Mathematica Compendium

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Chapter 1

Propositional Logic

Propositional logic, also called Boolean logic, is one of the foundational branches of mathematics and, thus, the first chapter of this book. In propositional logic, we use symbolic variables to represent the logic, and we can use any symbol to represent a proposition, such as p, q, r, s, t, etc. Propositions can be either *true* (1, T) or *false* (0, F), but they cannot be both.

Remark: In programming languages like C, any number greater than 0 is treated as T.

1.1 Predicates

A *predicate* is a statement which contains variables linked by some property stated in the language of formal logic, and whose value can be true or false.

1.2 Predicate Substitution

Given a predicate P on the variable v, we can substitute every free instance of v with the term v to extend our predicate to another variable space.

$$[t/v]P$$

Types of substitution:

- *Full replacement*: $[y/x]\forall z((z = x) \wedge (y \in x))$, converted to $\forall z((z = y) \wedge (y \in y))$.
- *Partial replacement*: $[y/x]x = y \wedge \forall x(x \in y)$, the x of $\forall x(x \in y)$ is not a free variable, hence it can not be replaced completely, just $y = y \wedge \forall x(x \in y)$.

Example:

Prove $\forall a(a = a)$:

Let $P := x = a$ be a predicate in x and a. We apply:

$$[a/x](x = a) \text{ to } a = a, \text{ then by extension of our predicate } \forall a(a = a).$$

Prove $\forall a \forall b(a = b \implies b = a)$:

Let $P := x = a$ be a predicate in a and x. We apply:

$$[b/x](x = a) \text{ to } b = a, \text{ then by extension of our predicate } \forall a \forall (a = b \implies b = a)).$$

$\mathfrak{QED}$

Prove $\forall a(a = b \wedge b = c \implies c = a)$:

We know $a = b \wedge b = a$, let this be our predicate with x and apply the substitution $[c/b]$, which gives us $a = x$, $a = b$ and finally $a = c$.

$\mathfrak{QED}$

1.3 Logic Operators and Truth Tables

NOT ($\neg$, $\sim$)

p	$\neg q$
T	F
F	T

AND ($\wedge$)

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

OR ($\vee$)

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

IMPLIES ($\implies$)

p	q	$p \implies q$
T	T	T
T	F	F
F	T	T
F	F	T

IFF ($\iff$)

p	q	$p \iff q$
T	T	T
T	F	F
F	T	F
F	F	T

XOR ($\oplus$)

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

NOR ($\downarrow$)

p	q	$p \downarrow q$
T	T	F
T	F	F
F	T	F
F	F	T

NAND ($\uparrow$)

p	q	$p \uparrow q$
T	T	F
T	F	T
F	T	T
F	F	T

1.4 Implications

As a side note, for the *implication*

$$p \implies q$$

we can think about it as *if, then.* For the cases false, true; and false, false; we call them *vacuously* true. From a false premise, anything can follow.

Example:

Let p be "it rained" and q "the street is wet", with theirs respective negations "it did not rain", "the street is not wet".

Now, let us study each of the cases:

p	q	$p \implies q$
T	T	T
T	F	F
F	T	T
F	F	T

- p and q are both true: "It rained and the street is wet." This makes the implication $p \implies q$ true.
- p is true and q is false: "It rained but the street is not wet." This makes the implication $p \implies q$ false, since the stree should be wet.
- p is false and q is true: "It did not rain but the street is wet." This makes the implication $p \implies q$ true (vacuously true). An external factor could have been responsible for the street being wet.
- p and q are both false: "It did not rain and the street is not wet." This makes the implication $p \implies q$ true (vacuously true).

We simplify the possibilites inside this mini universe and the implication becomes very intuitive.

1.5 If And Only If

If and only if $\iff$ is equivalent to

$$(A \implies B) \wedge (B \implies A).$$

Which is justs its definition.

1.6 Tautology and Contradiction

Tautology: A logical formula that is always true.

Common tautologies

- $p \implies q \vee p$
- $q \implies p \vee q$
- $p \wedge q \implies p$
- $q \wedge q \implies q$
- $[(p \implies q) \wedge (q \implies r)] \implies (p \implies r)$

Contradiction: A formula that is always false.

1.7 Logical Equivalences

Two logical formulas P, Q are said to be *logically equivalent* if they have the same truth value in every possible scenario. We can write this as $P \equiv Q$ or $P \iff Q$.

1.7.1 Laws of Logical Equivalences

Commutative Laws

$$p \wedge q \iff q \wedge p \qquad p \vee q \iff q \vee p$$

Associative Laws

$$(p \wedge q) \wedge r \iff p \wedge (q \wedge r) \qquad (p \vee q) \vee r \iff p \vee (q \vee r)$$

Distributive Laws

$$p \wedge (q \vee r) \iff (p \wedge q) \vee (p \wedge r) \qquad p \vee (q \wedge r) \iff (p \vee q) \wedge (p \vee r)$$

Identity Laws

$$p \wedge \mathrm{T} \iff p \qquad p \vee \mathrm{F} \iff p$$

Negation Laws

$$p \vee \sim p \iff \mathrm{T} \qquad p \wedge \sim p \iff \mathrm{F}$$

Double Negation Law

$$\sim (\sim p) \iff p$$

Idempotent Laws

$$p \wedge p \iff p \qquad p \vee p \iff p$$

Universal Bound Laws

$$p \vee \mathrm{T} \iff \mathrm{T} \qquad p \wedge \mathrm{F} \iff \mathrm{F}$$

De Morgan's Laws

$$\sim (p \wedge q) \iff (\sim p) \vee (\sim q) \qquad \sim (p \vee q) \iff (\sim p) \wedge (\sim q)$$

Absorption Laws

$$p \vee (p \wedge q) \iff p \qquad p \wedge (p \vee q) \iff p$$

Conditional Laws

$$(p \implies q) \iff (\sim p \vee q) \qquad \sim (p \implies q) \iff (p \wedge \sim q)$$

Complement Law

$$p \vee \neg p \iff \mathrm{T} \qquad p \wedge \neg p \iff \mathrm{F}$$

Biconditional Law

$$p \iff q \iff (p \implies q) \wedge (q \implies p)$$

Transitivity Law

$$(p \implies q) \wedge (q \implies r) \implies (p \Rightarrow r)$$

Indirect Proof (Contrapositive)

$$(p \implies q) \iff (\neg q \implies \neg p)$$

Disjunctive Syllogism (Disjunctive Exclusion)

$$p \nabla q \equiv (p \vee q) \wedge \neg p \implies q$$

$$p \nabla q \equiv (p \wedge q) \vee \neg (p \wedge q)$$

1.8 Truth Tables

Truth tables are a fundamental tool in logic that systematically show the truth value (true or false) of a compound statement for every possible combination of the truth values of its individual component statements.

Essentially, they lay out all the scenarios and the resulting truth of the overall logical expression. This helps determine if an argument is valid, if statements are logically equivalent, or the circumstances under which a complex statement is true or false.

Example:

p	q	$\neg p$	$\neg q$	$\neg p \implies q$	$(\neg p \implies q) \wedge \neg p$	$[(\neg p \implies q) \wedge \neg p] \Rightarrow q$
T	T	F	F	T	F	T
T	F	F	T	T	F	T
F	T	T	F	T	T	T
F	F	T	T	F	F	T

1.8.1 Filling a Truth Table

To fill a truth table for a logical expression with truth values (True or False), you follow a specific order for the input variables. This order ensures that all possible combinations of truth values for the variables are covered.

1. **List all possible combinations of truth values for the input variables**: If you have n variables, the number of rows in the truth table will be 2^n. Each variable can be either True (T) or False (F).
2. **Order of the input variables**:
   - Start by filling in the truth values for the first variable. It alternates between True and False every 2^{n-1} rows.
   - Then for the second variable, it alternates every 2^{n-2} rows, and so on.
   - In short: the first variable alternates every other row, the second variable every two rows, the third every four rows, and so on.

Example:

For 3 variables, there are $2^3 = 8$ possible combinations of truth values. The truth values are filled in the following order:

p	q	r	Expression Result
T	T	T	–
T	T	F	–
T	F	T	–
T	F	F	–
F	T	T	–
F	T	F	–
F	F	T	–
F	F	F	–

The pattern for filling the input truth values

- The first column p alternates every 4 rows: 'T, T, F, F, T, T, F, F'.
- The second column q alternates every 2 rows: 'T, T, F, F, T, T, F, F'.
- The third column r alternates every row: 'T, F, T, F, T, F, T, F'.

This ensures that all combinations of p, q, and r are covered, and you can then evaluate the logical expression for each combination.

Truth Table for the Expression $(p \wedge q) \vee q$

p	q	r	$(p \wedge q) \vee r$
T	T	T	T
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	F

1.8.2 Disjunctive Normal Form (DNF)

Disjunctive Normal Form (DNF) is a standard way of writing a logical expression as a *disjunction(OR)* of *conjunctions (ANDs)*. A DNF expression consists of a series of conjunctions of literals, where each conjunction is connected by disjunctions.

To find the DNF in a truth table take the rows of the final result where there are true statements and bind the propositions that generated it with an AND inside parenthesis. Repeat it with each of true rows and connect all parenthesis with OR's.

Example:

Consider the logical expression:

$$(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge \neg r).$$

This is in DNF because it is a disjunction (OR) of conjunctions (ANDs) of literals.

1.8.3 Conjunctive Normal Form (CNF)

Conjunctive Normal Form (CNF) is a standard way of writing a logical expression as a *conjunction (AND)* of *disjunctions (ORs)*. A CNF expression consists of a series of disjunctions of literals, where each disjunction is connected by conjunctions.

To find the CNF proceed just as the DNF but with the false rows and instead of ANDs inside the parenthesis use OR and connect the terms with OR. Also add a negation before each parenthesis.

Example:

Consider the logical expression:

$$\neg(p \vee q) \wedge \neg(\neg p \vee r) \wedge \neg(q \vee \neg r).$$

This is in CNF because it is a conjunction (AND) of disjunctions (ORs) of literals.

1.9 Basis

A *Basis* in the field of logic is a set of operations with which you can represent all possible logic operations.

Example: The set $\{\neg, \wedge\}$

1.10 Karnaugh Maps

Karnaugh Maps (K-Maps) are a graphical method used to simplify Boolean expressions. The main goal of a K-map is to group adjacent cells that contain 1's in order to simplify the expression. A K-map helps identify common terms, allowing the Boolean expression to be reduced to its simplest form.

1.10.1 Karnaugh Map for Two Variables

Consider the Boolean expression $(A \vee (B \wedge \neg A \wedge \neg B))$.

We first construct a K-map for two variables, A and B. The truth table for this expression gives the following values:

A	B	$(A \vee (B \wedge \neg A \wedge \neg B))$
0	0	0
0	1	1
1	0	1
1	1	1

The corresponding K-map is:

AB	00	01	11	10
Value	0	1	1	1

Here, we group the ones together to simplify the Boolean expression. The simplified expression is:

$$A \vee B.$$

1.10.2 Karnaugh Map for Three Variables

Now, let's consider the expression $\neg C$. This expression only depends on one variable, but for illustration, we will use a 3-variable K-map with variables A, B, and C.

The truth table for $\neg C$ is as follows:

A	B	C	$\neg C$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

The corresponding K-map for three variables A, B, and C is:

$AB\backslash C$	0	1
00	1	0
01	1	0
11	1	0
10	1	0

We see that the ones are grouped in a column, leading to the simplified Boolean expression:

$$\neg C.$$

1.10.3 Solving a Karnaugh Map

To solve a Karnaugh Map and simplify a Boolean expression, follow these steps:

1. **Determine the Number of Variables:** Decide how many variables are in the Boolean function. This determines the size of the K-Map:
   - 2 variables: 2×2.
   - 3 variables: 2×4.
   - 4 variables: 4×4.
   - etc.
2. **Fill in the K-Map:** Place 1's in the cells that correspond to the min-terms (where the function outputs 1). You may also include do not care conditions (usually denoted as X).
3. **Group the 1's:** Form groups (called *implicants*) of 1's. The groups must follow these rules:
   - Each group must contain $1, 2, 4, 8, \ldots$ (powers of 2) 1's.
   - Groups must be rectangular (e.g., 1×2, 2×2).
   - Groups can wrap around the edges of the K-Map.
   - Try to form the largest groups possible to simplify the expression.
   - Each 1 should be included in at least one group.
4. **Write the Simplified Expression:** For each group:
   - Identify the variables that are constant (either always 0 or always 1) across the group.
   - Write a product term (AND) using only the constant variables.
   - Combine all product terms with OR operations to get the final simplified SOP (Sum of Products) expression.

1.11 Open Sentences

When writting statements we normally do not write the statement with concrete values, but instead define it in terms of variables like x which is used to denote the members of a collection for which our statement holds or not holds. We denote this as an *open sentence*.

Example:

$$P(x): \text{ The sum of the angles of } x \text{ is } 180^\circ,\ \forall x \in S$$

where S is the set of the triangles for which P holds.

1.12 Mathematical Quantifiers

Mathematical quantifiers are symbols used as shorthand for common quantifying of elements when writting mathmatical statements.

1.12.1 Universal Quantifier $\forall$

Means **"for all"** or **"for every"**.

Negation: $\neg(\forall x)\, P(x) \equiv (\exists x)\, \neg P(x)$ "Not all" is the same as "There exists one that does not".

$$\neg(\forall x, P(x)) \equiv \exists x, \neg P(x)$$

Proof:

We want to prove that $\neg(\forall i \in I : E(i)) \iff (\exists i \in I : \neg E(i))$, given $I \neq \emptyset$

$$\begin{aligned}
\neg(\forall i \in I : E(i)) &\iff \neg(I = \{i \in I \mid E(i)\}) \\
&\iff I \neq \{i \in I \mid E(i)\} \\
&\iff \{i \in I \mid \neg E(i)\} \neq \emptyset \\
&\iff (\exists i \in I : \neg E(i)
\end{aligned}$$

$\mathfrak{QED}$

Example:

$\forall x \in \mathbb{R},\ x^2 \geq 0$ For all real numbers, the square is greater than or equal to zero.

Negated Example:

$\exists x \in \mathbb{R},\ x^2 < 0$ There exists a real number whose square is less than zero — this is false.

1.12.2 Existential Quantifier $\exists$

Means **"there exists at least one"**.

Negation: $\neg(\exists x)\, P(x) \equiv (\forall x)\, \neg P(x)$ "There does not exist" is the same as "For all, not".

Proof:

We want to prove that $\neg(\exists i \in I : E(i)) \iff (\forall i \in I : \neg E(i))$, given $I \neq \emptyset$

$$\begin{aligned}
\neg(\exists i \in I : E(i)) &\iff \neg(I = \{i \in I \mid E(i)\} \neq \emptyset) \\
&\iff I \neq \{i \in I \mid E(i)\} = \emptyset \\
&\iff \forall i \in I : \neg E(i)
\end{aligned}$$

$\mathfrak{QED}$

Example:

$\exists x \in \mathbb{N},\ x > 10$ There exists a natural number greater than 10.

Negated Example:

$\forall x \in \mathbb{N},\ x \leq 10$ All natural numbers are less than or equal to 10 — this is false.

1.12.3 Unique Existential Quantifier $\exists!$

Means **"there exists exactly one"**.

Negation: "Not exactly one" means

$$\neg(\exists! x)\, P(x) \equiv (\forall x)\, \neg P(x) \ \vee\ (\exists x_1 \neq x_2)\, P(x_1) \wedge P(x_2)$$

Either no such x exists, or more than one does.

Example:

$\exists! x \in \mathbb{R},\ x + 5 = 0$ There exists exactly one real number such that $x + 5 = 0$.

Negated Example:

$\exists x_1 \neq x_2 \in \mathbb{R},\ x_1^2 = 4 \wedge x_2^2 = 4$ There are multiple solutions to $x^2 = 4$.

1.12.4 Negations Remark

When dealing with proofs, it is not uncommon that we will need to negate an statement for the purpose of prooving a simpler equivalent statement. For example:

Note that $p \implies q$ is logically equivalent to $\neg p \vee q$. Hence, if would to prove its negation we can use the following equivalences:

$$\neg(p \implies q) \equiv \neg(\neg p \vee q) \equiv p \wedge \neg q.$$

1.12.5 Order Remark

When using quantifiers, *the order matters*, it is not same to write

$$\forall x \in M, \exists y \in N : E(x, y)$$

and

$$\exists y \in N, \forall x \in M : E(x, y)$$

Note that in the first example y can depend on x which means that for each x the may be different y; while in the second example we declare one or more y's which satisfy the condition for all x.

1.13 Equivalence Reformulations

An *equivalence reformulation* is when we trasnform a statement from one form to another which is *logically equivalent*, it has the values in a truth table.

$$A \iff B$$

We use them all the time during our proofs as the connectors when we want to rewrite a statement is form which more easier to work with.

Example:

$$|x| \leq y \iff (x \leq y \wedge -x \leq y) \iff -y \leq x \leq y$$

Chapter 2

Set Theory

Set theory is the foundation of mathematics alongside *logic*. The actual variant is founded on axioms, unlike its first appearance as **naive set theory**, which led to contradictions due to it allowing sets which are "an element of itself, if and only if, it is not an element of itself", the so-called *Russell's paradox*, but more on that later.

Note that I will start with sets as such instead of going through classes and then exporting some of their properties to sets. Classes will only be handled shortly with the axioms and some proofs, but some of their properties will be directly presented as set properties.

2.1 Set Definition and Symbology

To be concise, a *set* is a collection of other sets; yes, the definition is recursive. Once the foundation is set, we can abstract those nested sets as mathematical objects.

- $A \subseteq B$: Set A is a subset of B. This means every element of A is also in B.
- $A \subseteq B$ or $A = B$: This notation already includes the possibility that A equals B since a set is always a subset of itself.
- $A \cup B$: The union of sets A and B. It includes all elements that are in A, in B, or in both.
- $A \cap B = A \setminus (A \setminus B)$: The intersection of sets A and B. It includes only the elements that are in both sets.
- $A \setminus B$: The difference of sets. Elements in A that are not in B.
- A^c or $\overline{A}$: The complement of set A. All elements not in A, relative to a universal set.
- $|A|$: The cardinality of set A, which is the number of elements in the set.

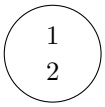
2.1.1 Venn Diagrams

1. $\emptyset$



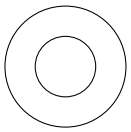
1. Empty Set

2. $\{1, 2\}$



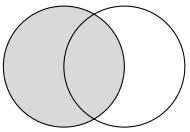
2. Set with Two Elements

3. $A \subseteq B$

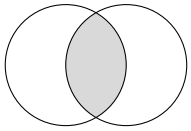


3. A is a subset of B

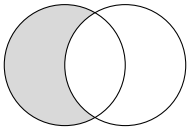
5. $A \cup B$



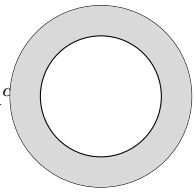
5. Union of A and B

6. $A \cap B$ 

6. Intersection of A and B

7. $A \setminus B$ 

7. A minus B

8. A^c 

8. Complement of A

Example:

Prove that $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$.

Proof:

$$\begin{aligned} x \in A \setminus (B \setminus C) &\iff x \in A \wedge x \notin (B \setminus C) \\ &\iff x \in A \wedge x \notin (x \in B \wedge x \notin C) \\ &\iff x \in A \wedge x \in (x \notin B \vee x \in C) \\ &\iff x \in A \wedge (x \notin B \vee x \in C) \\ &\iff (x \in A \wedge x \notin B) \vee (x \in A \wedge x \in C) \\ &\iff x \in (A \setminus B) \cup (A \cap C) \end{aligned}$$

$\mathfrak{QED}$

2.2 Short History Lesson

When set theory was invented back then by Cantor, he had no formal axioms but a set of basic 'common sense' rules he assumed when creating his theory. This choice lead to a famous paradox known as *Russell's Paradox*. The paradox states that: "Given a set S which is the set containing all sets which do not contain themselves. The S does not contain itself, but if it does not contain itself it must contain itself". In short $S = \{x : x \notin x\}$, this is a so-called *logical* paradox. This is of course a huge contradiction in the hearth of naive set theory. The choice to take a more e formal and axiomatic approach was to eliminate such fundamental faults.

As a short teasing, we will declare the *Axiom of Selection:* $\{x \in A : S(x)\}$ this means that we can construct a set of elements of an existing set as long as they satisfy a condition. Now the paradox of $x : x \notin x$ can not be state directly in our new system. We can emulate it by declaring

$$S = \{x \in A \mid x \notin x\}$$

By analyzing the expression we get that, if $S \notin S$ then $S \in S \wedge S \in A$ which will be a contradiction, therefore, $S \notin A$.

Continuing this small historic section, we will also need to limit the power of our condition of statements $S(x)$ to get rid of another type of paradoxes considered as *semantic*.

For example, we could define the following set:

$$\{x \in \mathbb{N} : \text{ x can be described in less than 20 words of the English language}\}$$

Note that this set would need to be finite because it there is only a limited amount of combination of words in the English language. But we could for example say: 'the natural number which can not be described in less than 20 words' this number. This is again an example of an element in in a set if and only if it is not in the set. To formalize our conditions we

2.3 Classes

One important distinction in the system of Neumann is the notion of the difference between a *set* and *class*.

Classes can not be contained inside *sets*, while sets can. Formally:

If $X \in Y$ for some class Y then X is a set. If $X \notin Y$ for any class Y then X is a class.

2.4 Axioms of Set Theory

I. *Axiom of Selection*: Given a class A, a class S can be build in the following manner $S = \{x \in A \mid P(x)\})$

II. *Axiom of Extensionality*: Let A and B be classes then $A = B \iff \forall X : A \in X \implies B \in X \wedge B \in X \implies A \in X$

III. *Empty-Set Axiom*: $\emptyset$ is a set.

IV. *Axiom of Class Construction*: Let $P(x)$ be a statement about x which can be expressed in terms of $\in, \wedge, \vee, \neg, \implies, \exists, \forall$ with variables $x, y, z, \dots$ then there exists a class C whose elements consist of all the elements x which satisfy $P(x)$. $\{x : P(x)\}$

V. *Axiom of Subset*: Every subclass of a set is a set.

VI. *Axiom of Paring*: Is a and b are sets, then $\{a, b\}$ is a sett.

VII. *Axiom of Union*: If A is a set of sets, then $\bigcup_{a \in A} a$ is a set. $\forall a \exists b \forall c(c \in a \implies \forall d(d \in c \implies d \in d))$.

VIII. *Axiom of the Power Set*: If A is a set then the set of all the subsets of the original set, $\mathcal{P}(A)$, is a set. $\forall a \exists b \forall c(\forall d(f \in c \implies d \in a) \implies d \in b)$.

IX. *Axiom of Foundation / Regularity*: If A is any set, there is an element $a \in A$ such that $a \cap A = \emptyset$. Alternatively, there exists not such set, which is formed from a chain like: $a_1 \in a_2 \in a_3 \in \dots$. $\forall a(\exists b(b \in a) \implies \exists c(c \in a \wedge \forall d(d \in c \implies d \notin a)))$

X. *Axiom of Replacement*: If A is a set and $f : A \to B$ is a surjective function, then B is a set. Let P be a predicate on a, b, c and let d, e, f, g not be yet introduced,

$$\begin{aligned}\forall a \exists b((\forall c((c \in a \wedge \exists d P) \implies \exists e(e \exists (e \in b \wedge [e/d]P)) \\ \wedge \forall f(f \in b \implies \exists g(g \in a \wedge [g/c][f/d]P)))\end{aligned}$$

XI. *Axiom of Infinity*: An infinity set exists. $\exists a(\exists b(b \in a) \wedge \forall c(c \in a \implies \exists d(d \in a \wedge c \in d)))$.

XII. *Axiom of Choice*: Given a non-empty set, $A = \{A_1, A_2, \dots\}$, there is a set containing exactly one element from each A_i.

$$\begin{aligned}\forall a(\forall b(b \in a \implies (\exists c(c \in b) \\ \wedge \forall d((d \in a \wedge \exists e(e \in b \wedge e \in d)) \implies b = d))) \\ \implies \exists f \forall g(g \in a \implies \exists h((h \in g \wedge h \in f) \\ \wedge \forall i((i \in g \wedge i \in f) \implies i = h))))\end{aligned}$$

Simplified

$$F : A \to \bigcup_{X \in A} X, \quad F(X) \in X.$$

2.4.1 Proofs Using the Axioms

- $A = A$ and also $A = B \implies B = A$: The statements is trivial by the first axiom
- $A \subseteq B \land B \subseteq A \implies A = B$: $(\forall x \in A \implies x \in B) \land (\forall x \in B \implies x \in A)$ Let us suppose for the sake of contradiction that $A \neq B$ then $\exists x \in A : x \notin B$ or the other way around. Note that this statement contradicts our initial conditions. Thus, the statement is true.
- $A \subseteq B \land B \subseteq C \implies A \subseteq C$: $(\forall x \in A \implies x \in B) \land (\forall x \in B \implies x \in C)$ By a similar manner , let us assume that A is not a subset of C this means that for some x the condition holds that $x \in A \land x \notin B$ because else it would be in C. Note that because $A \subseteq B$ such an x can not exist due to our initial condition and also $B \subseteq A$ for that case. Thus $A \subseteq C$.
- $\emptyset \subseteq A$: By definition $x \in \emptyset \implies x \in A$, by contrapositive $x \notin A \implies xnotint\emptyset$. This is true because of the contrapositive being true.
- Given a proper class $A \subseteq B$ prove that B is a proper class and that the union of two proper classes is also a proper class.

  Given $A \subseteq B$, if B would be not be a proper class then by the property which states that every subclass of a class is a set we would have a contradiction. Therefore, B is a proper class.

  $A \cap C$ were A and C are proper classes, gives us that their union is also a proper class due to $A, C \subseteq A \cup C$
- $\mathcal{P}(A) \nsubseteq A$: Let $x \in \mathcal{P}(a)$, then $\forall x \in \mathcal{P}(A) \implies x \in A$. By definition, $A \in \mathcal{P}(A)$, but $A \notin A$; hence we have contradiction and the power set of a set is not a subset of the original set.t.

$\mathfrak{QED}$

2.4.2 Consequences of the Axiom of Replacement

- Every singleton is a set.
- If A is in a one-to-one correspondence with B, then B is a set.
- If $\{A_i\}_{i \in I}$ is an indexed family of sets and I is a set, then $\{A_i \mid i \in I\}$ is a set.

2.5 Subclasses/Subsets

A subset/subclass A of the set/class B is a set which obeys the following property:

$$A \subseteq B \iff \forall x \in A, x \in A \implies x \in B$$

2.6 The Universal Class

By the axiom of construction we define $\mathcal{U}$ with $P(x)$ is $x = x$ as the *universal class* of all elements.

It also as the following propertyy

$$A \subseteq \mathcal{U},$$

which gives us the following: $A \cup \mathcal{U} = \mathcal{U}$ and $A \cap \mathcal{U} = A$ for any class or set A.

2.7 The Empty Class

The *empty class* is class with no elements. .

$$\emptyset = \{x \mid x \neq x\}$$

We can not use existence quantifiers $\exists$ with this class. Every statement with $\forall x \in \emptyset$ is true, because there are no counterexamples. This leads to interesting equalities like $\mathcal{U} = \emptyset$.

We can prove that the empty class exist by letting $\emptyset = \{x \mid x \neq x\}$ this is a perfectly valid statement.t.

Now we need to prove that it is unique, and this is given by the axiom of extensionality.

$\mathfrak{QED}$

2.8 The Empty Set

The *empty set* is the set that contains no elements, denoted as $\emptyset$. It is the unique set with zero elements. It inherits the properties of its class counterpart.

It also has the property that $\emptyset \subseteq A$ for any class or set A.

2.9 Set Complement Proof

Given $A, B \subseteq M$ then $A \subseteq B \implies B^c \subseteq A^c$).

For some $x \in B^c$ we know that $x \notin B$. If $x \in A$ then $x \in B$ which is a contradiction. Hence, $x \notin A$ therefore, for every $x \in B^c$ we conclude $B^c \subseteq A^c$.).

$\mathfrak{QED}$

2.10 Proof the Power Set of a Set Is Not an Element of the Set

Let X be a set and $Y = \{x \in X \mid x \notin x\}$.

This is a subset of X by definition and implies $Y \in \mathcal{P}(X)$.

Assume $\mathcal{P}(X) \subseteq X$ then $Y \subseteq X, Y \in X$ but

$$Y \in Y \implies Y \notin Y \implies Y \in Y \text{ which is a contradiction}$$

Because our assumption was proven false, then $\mathcal{P}(X)$ is not a subset of X.

$\mathfrak{QED}$

2.11 Proof of the Union of Subsets

Let A and B be sets then:

- $A \subseteq B$ if and only if $A \cup B = B$.
- $A \subseteq B$ if and only if $A \cap B = A$.

Proof:

For $A \subseteq B \iff A \cup B = B$ we know that $x \in A \implies x \in B$ Thenn

$$\begin{aligned} A \cup B &\iff x \in A \vee x \in B \quad \text{as subset then} \\ &\iff x \in B \vee x \in B \\ &\iff x \in B = B \end{aligned}$$

The other direction will be skipped due to it being very similar to the $\Rightarrow$ direction.

Similarly, for $A \subseteq B \iff A \cap B = A$ We know that for some $(y \in B \land y \notin A) \notin A \cap B$ because of the definition of the subset. Then this tells us that $\forall x \in A \cap B : x \in A$ thus $A \cap B = A$).

$\mathfrak{QED}$

2.12 Cut of Set and Complement

$A \cap B^c = \emptyset \iff A \subseteq B$

$\Rightarrow$:

$$\begin{aligned} A \cap B^c = \emptyset &\implies \nexists x \in A \land x \nexists x \in B^c \\ &\implies \forall x \in A : x \in B \\ &\iff A \subseteq B \end{aligned}$$

$\Leftarrow$: Let us assume that A is not a subset of B then

$$\begin{aligned} \exists x \in A \land x \notin B &\implies x \in A \land x \in B^c \\ &\implies x \in A \cap B^c \implies A \cap B^c \neq \emptyset \end{aligned}$$

$\mathfrak{QED}$

2.13 Properties of Set Differences

Given the sets A, B and C then the following properties apply:

1. $A \setminus B \subseteq A$.
2. $(A \setminus B) \cap B = \emptyset$.
3. $(A \setminus B) \cup B = A \cup B$.
4. $A \setminus (A \setminus B) = A \cap B$.
5. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
6. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Proof V:

Let $x \in A \setminus (B \cup C)$, then $x \in A$ and $x \notin B$ **and** $x \notin C$, we may think to use or in this case, but we need both conditions to hold, hence $x \in (A \setminus B) \cap (A \setminus C)$, and therefore

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

The other direction is trivial once we already broke down the definition of the sets with logic operators.

$\mathfrak{QED}$

2.14 The Singleton and Doubleton

We define the *singleton* as

$$\{a\} := \{x \mid x = a\},$$

and the *doubleton* as

$$\{a, b\} := \{x \mid x = a \lor x = b\},$$

2.14.1 Properties of Singletons and Doubletons

- $\{a\} = \{b\} \iff a = b$
- $\{a, b\} = \{a\} \iff a = b$
- $\{a, b\} = \{x, y\} \iff (a = x \wedge b = y) \vee (a = y \wedge b = x)$

Proofs:

$\{a\} = \{b\} \iff a = b$, by the axiom of the equality of sets if

$$\exists x \in \{a\} : x \notin \{b\} \implies \{a\} \neq \{b\},$$

which is a contradiction to our condition thus $a = b$. Similarly, we can use the same argument for the other direction of the proof.

$\mathfrak{QED}$

$\{a, b\} = \{a\} \iff a = b$ can be proven using the definition and the equivalence $P \vee P = P$.

$$\{a, b\} := \{x \mid x = a \vee x = a\} \implies \{x \mid x = a\}$$

$\mathfrak{QED}$

$\{a, b\} = \{x, y\} \iff (a = x \wedge b = y) \vee (a = y \wedge b = x)$, we need to first consider the case $a = b$. In that case our problem simplifies to $\{a\} = \{x, y\}$ which by the axiom guarantees the equality for $\{a\} = \{x\} \vee \{a\} = \{y\}$ because otherwise we will arrive at a contradiction.

For the case $a \neq b$ we have the following cases $(x = a \wedge y = b) \vee (x = b \wedge y = a)$, from which, again by the use of the axiom of the equality we can discard the cases where the equality does not hold.

$\mathfrak{QED}$

2.15 Ordered Pair

Let A and B be two non-empty sets and $m \in A, n \in B$. We define

$$(m, n) := \{\{m, n\}, m\}$$

and

$$(n, m) := \{\{m, n\}, n\}$$

as ordered pairs which are elements of $A \times B$ and $B \times A$ respectively.

2.16 The Cartesian Product

The Cartesian product of two sets A and B, written $A \times B$, is the set of all ordered pairs in which the first element belongs to A and the second belongs to B:

$$A \times B = \{(a, b) : a \in A, \wedge\ b \in B\}$$

Example:

Table 2.1: Cartesian Product of $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$

	$b \in B$		
$A \times B$	4	5	6
$a \in A$			
1	$(1, 4)$	$(1, 5)$	$(1, 6)$
2	$(2, 4)$	$(2, 5)$	$(2, 6)$
3	$(3, 4)$	$(3, 5)$	$(3, 6)$

The general Cartesian product of n sets can be written as:

$$X_{i=1}^{n+1} A_i = (X_{i=1}^{n} A_i) \times A_{n+1} \quad \text{with} \quad X_{i=1}^{1} A_i = A_1$$
$$\text{When } A_i = M \text{ for all } i:$$
$$M^n := M \times M \times \cdots \times M = X_{i=1}^{n} M \quad \text{with} \quad M^1 = M$$

Distributive Law for the Cartesian Product

Let $A, A' \subseteq X$ and $B, B' \subseteq Y$. Then

$$(A \times B) \cap (A' \times B') = (A \cap A') \times (B \cap B').$$

We prove the equality by showing both inclusions.

1. Direction: $(A \times B) \cap (A' \times B') \subseteq (A \cap A') \times (B \cap B')$.

$$\begin{aligned}
&\text{Let } (x, y) \in (A \times B) \cap (A' \times B'). \\
&\iff (x, y) \in A \times B \text{ and } (x, y) \in A' \times B' \\
&\iff x \in A,\ y \in B \text{ and } x \in A',\ y \in B' \\
&\iff x \in A \cap A' \text{ and } y \in B \cap B' \\
&\iff (x, y) \in (A \cap A') \times (B \cap B').
\end{aligned}$$

2. Direction: $(A \cap A') \times (B \cap B') \subseteq (A \times B) \cap (A' \times B')$.

$$\begin{aligned}
&\text{Let } (x, y) \in (A \cap A') \times (B \cap B'). \\
&\iff x \in A \cap A' \text{ and } y \in B \cap B' \\
&\iff x \in A,\ x \in A' \text{ and } y \in B,\ y \in B' \\
&\iff (x, y) \in A \times B \text{ and } (x, y) \in A' \times B' \\
&\iff (x, y) \in (A \times B) \cap (A' \times B').
\end{aligned}$$

Thus, both inclusions hold and the equality follows.

QED

2.16.1 Proof That the Cartesian Product Is a Set

We need to prove that $A \times B \subseteq \mathcal{P}(A \cup B)$. By the definition of the ordered pair we know that for $x \in A \cup B \implies \{x\} \subseteq A \cup B$ the same for $x, y \in A \cup B$ which implies $\{x, y\} \subseteq A \cup B$. By our axiom of union we know that $\{\{x\}, \{x, y\}\}$ is a set, and this is a subset of $\mathcal{P}(\mathcal{P}(A \cup B))$ for any x and y. Hence

$$(x, y) \subseteq A \times B \subseteq \mathcal{P}(\mathcal{P}(A \cup B))$$

QED

2.16.2 The Empty Set of the Cartesian Product

$A \times B = \emptyset \iff A = \emptyset \vee B = \emptyset$. We will prove the negation of the statement, which is logically equivalent.

$$A \times B \neq \emptyset \iff A \neq \emptyset \wedge B \neq \emptyset$$

$\Rightarrow$:

$$\begin{aligned} A \times B \neq \emptyset &\implies \exists (x, y) \in A \times B : x \in A \wedge y \in B \\ &\implies \exists x \in A \wedge \exists y \in B \\ &\implies A \neq \emptyset \wedge B \neq \emptyset \end{aligned}$$

$\Leftarrow$:

$$\begin{aligned} A \neq \emptyset \wedge B \neq \emptyset &\implies \exists x \in A \wedge \exists y \in B \\ &\implies A \times B \neq \emptyset \end{aligned}$$

$\mathfrak{QED}$

2.17 Laws of Set Algebra

Let X be the universal set and $A, B, C \subseteq X$.

- $\emptyset \subseteq A$
- $A \subseteq B \iff A \cap B = A \iff A \cup B = B \iff X \setminus B \subseteq X \setminus A \iff B \subseteq A$
- $A \cup B = B \cup A$
- $A \cap B = B \cap A$
- $(A \cup B) \cup C = A \cup (B \cup C)$
- $(A \cap B) \cap C = A \cap (B \cap C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cup A = A \quad \text{and} \quad A \cap A = A$
- $A \setminus B = A \cap (X \setminus B) = A \cap \overline{B}$
- $B = \overline{A} \iff (A \cup B = X \wedge A \cap B = \emptyset)$
- $\overline{A} \cap \overline{B} = \overline{A \cup B}$
- $\overline{A} \cup \overline{B} = \overline{A \cap B}$
- $\overline{\overline{A}} = A$
- $(A \setminus B) \cup (B \setminus A) = A \Delta B$
- $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
- $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

2.17.1 Proof of the Distributive Law

$$
\begin{aligned}
&\text{Let } x \in A \cap (B \cup C).\\
&\iff x \in A \text{ and } x \in B \cup C\\
&\iff x \in A \text{ and } (x \in B \text{ or } x \in C)\\
&\iff (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)\\
&\iff x \in A \cap B \text{ or } x \in A \cap C\\
&\iff x \in (A \cap B) \cup (A \cap C).
\end{aligned}
$$

Thus,

$$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C).$$

$$
\begin{aligned}
&\text{Conversely, let } x \in (A \cap B) \cup (A \cap C).\\
&\iff x \in A \cap B \text{ or } x \in A \cap C\\
&\iff (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)\\
&\iff x \in A \text{ and } (x \in B \text{ or } x \in C)\\
&\iff x \in A \text{ and } x \in B \cup C\\
&\iff x \in A \cap (B \cup C).
\end{aligned}
$$

Thus,

$$(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C).$$

QED

2.17.2 Proof of De Morgan's Law for Sets and Logic

The complement of $A \cup B$ is $\overline{(A \cup B)}$, and the law on disjoint decomposition states:

$$B = \overline{A} \iff (A \cup B = X) \wedge (A \cap B = \emptyset)$$

So define $\overline{C} := A \cup B$ and $D := \overline{A} \cap \overline{B}$, and use Law (11) to show the disjoint decomposition:

$$D = C \iff A \cap B = A \cup B$$

To show:

$$D \cup C = X \iff (\overline{A} \cap \overline{B}) \cup (A \cup B) = X$$

$$
\begin{aligned}
(\overline{A} \cap \overline{B}) \cup (A \cup B) &= (\overline{A} \cup A \cup B) \cap (\overline{B} \cup A \cup B) \quad \text{(Law (8))}\\
&= (X \cup B) \cap (X \cup A)\\
&= X \cap X\\
&= X
\end{aligned}
$$

To show:

$$D \cap C = \emptyset \iff (\overline{A} \cap \overline{B}) \cap (A \cup B) = \emptyset$$

$$\begin{aligned}(\overline{A} \cap \overline{B}) \cap (A \cup B) &= (A \cap B \cap A) \cup (A \cap B \cap B) \quad \text{(Law (7))} \\ &= (\overline{A} \cap A \cap \overline{B}) \cup (\overline{A} \cap \overline{B} \cap B) \\ &= (\emptyset \cap \overline{B}) \cup (\overline{A} \cap \emptyset) \\ &= \emptyset \cup \emptyset \\ &= \emptyset\end{aligned}$$

ΩΕΔ

2.18 Indexed Sets

Let X be a set, and $A_i \subseteq X$ for all $i \in J$, where I is the *index set*.

If $I = \{1, 2, \ldots, n\}$, then we define

$$\{A_i\}_{i \in I},$$

as the set of all classes/sets for which an index in I exists.

We also can define the graph of the sets with their respective indexes:

$$G_{A_i} = A_i \times I = \{(i, A_i) \mid i \in I \wedge A_i \in \{A_i\}_{i \in I}\}$$

2.18.1 Generalized Union, Intersection and Cartesian Product for Indexed Sets

For a class of classes or sets of sets with an index set I we define the generalized *union*, *intersection* and *Cartesian product* as:

$$\bigcup_{i=1}^{n} A_i := A_1 \cup A_2 \cup \cdots \cup A_n = \{x \mid \exists i \in I, (x \in A_i)\}$$

$$\bigcap_{i=1}^{n} A_i := A_1 \cap A_2 \cap \cdots \cap A_n = \{x \mid \forall i \in I, (x \in A_i)\}$$

$$X_{i=1}^{n} A_i = \{(a_1, \ldots, a_n) \mid a_i \in A_i\}$$

2.19 Pairwise Disjoint

If I is any set, then $(A_i)_{i \in I}$ are *pairwise disjoint* if and only if:

$$\forall i_1, i_2 \in I,\ i_1 \neq i_2 \implies A_{i_1} \cap A_{i_2} = \emptyset$$

Example:

Prove that given $M \neq \emptyset$ then for all $T \subseteq M$ the following holds

$$\bigcap_{t \in T} \overline{\{t\}} = \overline{T}.$$

Proof:

$$x \in \bigcap_{t \in T} \overline{\{t\}} \iff x \in \overline{\{t\}}, \forall \overline{\{t\}}$$

Hence, $x \notin T$. This equivalent to saying $x \in \overline{T}$.

$\mathfrak{QED}$

2.20 Disjoint Decomposition

If I is any set, then $(A_i)_{i\in I}$ forms a *disjoint decomposition* of X if and only if:

$$(A_i)_{i\in J} \text{ are pairwise disjoint and } \bigcup_{i\in J} A_i = X$$

2.21 Union and Interscion for Indexed Sets

Let $A_i, B_j \subseteq X$ for $i \in I$ and $j \in J$. Then the following holds:

- *De Morgan's Laws:*

$$\overline{\bigcap_{i\in I} A_i} = \bigcup_{i\in I} \overline{A_i} \quad \text{and} \quad \overline{\bigcup_{i\in I} A_i} = \bigcap_{i\in I} \overline{A_i}$$

- *Generalized Distributive Laws:*

$$\left(\bigcap_{i\in I} A_i\right) \cup \left(\bigcap_{j\in J} B_j\right) = \bigcap_{i,j} (A_i \cup B_j) \quad \text{with} \quad \bigcap_{i,j} = \bigcap_{(i,j)\in I\times J}$$

$$\left(\bigcup_{i\in I} A_i\right) \cap \left(\bigcup_{j\in J} B_j\right) = \bigcup_{i,j} (A_i \cap B_j) \quad \text{with} \quad \bigcup_{i,j} = \bigcup_{(i,j)\in I\times J}$$

Here, $I = \{1, 2, 3, \ldots, n\}$ and $J = \{1, 2, 3, \ldots, m\}$. Then:

$$\begin{aligned} I \times J &= \{(i, j) \mid 1 \leq i \leq n, 1 \leq j \leq m, i, j \in \mathbb{N}\} \\ &= \{(1,1), (1,2), \ldots, (1,m), (2,1), (2,2), \ldots, (2,m), \ldots, (n,1), (n,2), \ldots, (n,m)\} \end{aligned}$$

2.22 Cardinality

The cardinality of a set is the number of elements in that set.

Let A and B be finite sets with $|A| = n$, $|B| = m$, and let X be the finite universal set. Then the following holds:

2.22.1 Cardinality Via Maps

Given $f : A \to B$, where A, B have a finte number of elements, for all elements of the image of A, we can add the cardinalities of the fibers

$$|A| = \sum_{b\in B} |f^{-1}(b)|,$$

which will give us the total cardinality. We use this for the following proofs.

2.23 Cardinality Laws

Cardinality of a Set

$$A = (A \cap B) \cup (A \setminus B), \quad |A| = |A \cap B| + |A \setminus B|$$

Proof:

Cardinality of the Complements

$$|A| = |X \setminus A| = |X| - |A|$$
$$A \setminus B = A \cap (X \setminus B), \quad |A \setminus B| = |A| - |A \cap B|$$

Proof:

Cardinality of the Cartesian Product

$$|A \times B| = |A| \cdot |B| = n \cdot m$$

Proof:

Inclusion-Exclusion Formula for Two Disjoint Sets

$$|A \cup B| = |A| + |B| = n + m$$

Proof:

2.23.1 Inclusion-Exclusion Formula for Two Non-Disjoint Sets

Let $|A \cap B| = k$, then:

$$|A \cup B| = |A| + |B| - |A \cap B| = n + m - k \quad \text{(since we do not count the intersection twice)}$$

2.23.2 Inclusion-Exclusion Formula for Three Non-Disjoint Sets

$$\begin{aligned} |A \cup B \cup C| &= |(A \cup B) \cup C| \\ &= |A \cup B| + |C| - |(A \cup B) \cap C| \\ &= |A| + |B| - |A \cap B| + |C| - |(A \cap C) \cup (B \cap C)| \\ &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \end{aligned}$$

2.23.3 General Formula for the Cardinality of the Union of Sets

$$\left| \bigcup_{i=1}^{n} M_i \right| = \sum_{I \subseteq \{1,\dots,n\}, I \neq \emptyset} (-1)^{|I|-1} \left| \bigcap_{i \in I} M_i \right|$$

2.24 Finite Definition

A set A is *finite*, if there is a number $\mathbb{Z}_{\geq 0}$ and a bijection

$$\xi : \hat{n} \to M, \quad \hat{n} = \{0,\ 1,\ 2,\ \cdots,\ n\}.$$

is such a bijection does not exists, then is it *non-finite, infinite.*

2.25 Countable and Non-Countable

A set A is *infinite countable* or just *countable* if there exists a bijection from $\mathbb{N}$ to A. If A has infinitely many elements and there is no such bijection, we call it *non-countable*.

2.26 Countable Union

Given $A_n, n \in \mathbb{N}$ countable sets, then

$$M := \bigcup_{n \in \mathbb{N}} A_n$$

is also countable.

2.27 Countability of Subsets

Given $A \subseteq B$, where B is countable, then A is also countable. The other case is that if A is non-countable, then B must be non-countable as well.

2.27.1 Countability of the Rational Numbers

We show that the set of rational numbers $\mathbb{Q}$ is countable.

Every rational number can be expressed as

$$q = \frac{p}{q}, \quad p \in \mathbb{Z},\ q \in \mathbb{N},\ q \neq 0.$$

Thus,

$$\mathbb{Q} = \left\{ \frac{p}{q} \;\middle|\; p \in \mathbb{Z},\ q \in \mathbb{N} \right\}.$$

The Cartesian product $\mathbb{Z} \times \mathbb{N}$ is countable because both $\mathbb{Z}$ and $\mathbb{N}$ are countable, and a countable product of countable sets is countable.

Hence, there exists an enumeration $(p_i, q_i)_{i \in \mathbb{N}}$ of all pairs (p, q) with $q \neq 0$. Defining

$$f(i) = \frac{p_i}{q_i},$$

we obtain a surjective mapping $f : \mathbb{N} \to \mathbb{Q}$.

Therefore, the set of rational numbers $\mathbb{Q}$ is countable.

$\mathfrak{QED}$

2.28 Cantor's Diagonal Argument: The Real Numbers Are Uncountable

We show that the set of real numbers in the interval $(0, 1) \subseteq \mathbb{R}$ is uncountable.

Assume, for contradiction, that $(0, 1)$ is countable. Then there exists a surjective mapping

$$f : \mathbb{N} \to (0, 1),$$

and thus all real numbers in $(0, 1)$ can be listed as

$$f(1) = 0.a_{11}a_{12}a_{13}\ldots, \quad f(2) = 0.a_{21}a_{22}a_{23}\ldots, \quad f(3) = 0.a_{31}a_{32}a_{33}\ldots, \quad \text{and so on.}$$

We now construct a new real number $r \in (0,1)$ by defining its decimal digits as

$$r = 0.b_1b_2b_3\ldots$$

where

$$b_i = \begin{cases} 5, & \text{if } a_{ii} \neq 5, \\ 6, & \text{if } a_{ii} = 5. \end{cases}$$

By construction, r differs from each $f(i)$ in at least the i-th decimal place. Hence, $r \neq f(i)$ for all $i \in \mathbb{N}$, contradicting the assumption that f is surjective.

Therefore, the interval $(0,1)$, and thus the set of real numbers $\mathbb{R}$, is uncountable.

$\mathfrak{QED}$

2.29 The Power Set

The *power set* of a set is the set of all subsets of a given set.

$$\mathcal{P}(x) := \{M : M \subseteq X\}$$

Its cardinality is 2^n with n being the number of elements in the original set X.

Example:

Prove that

$$M \subseteq N \iff \mathcal{P}(M) \subseteq \mathcal{P}(N)$$

Proof:

$\Rightarrow$: Let $M = N$, then because they have the same elements they are going to have the same subsets.

$\Leftarrow$: Let us take the contraposite statement, i.e., assume that $M \nsubseteq N$.

Then there exists an element $x \in M$ such that $x \notin N$. Consequently, the singleton set $\{x\} \subseteq \mathcal{P}(M)$ but $\{x\} \nsubseteq \mathcal{P}(N)$, which contradicts the assumption that $\mathcal{P}(M) \subseteq \mathcal{P}(N)$. Hence, $M \subseteq N$.

$\mathfrak{QED}$

2.30 Set Systems

Let A be a non-empty set. A subset $\mathscr{F} \subseteq \mathcal{P}(A)$ is called a *set system* or *family of sets* of A.

2.30.1 Intersection and Union of Set Systems

We can define the *intersection* as

$$\bigcap_{F \in \mathscr{F}} F := \{x \in A \mid \forall F \in \mathscr{F} \implies x \in F\}$$

and the *union* as

$$\bigcup_{F \in \mathscr{F}} F := \{x \in A \mid \exists F \in \mathscr{F} \implies x \in F\}.$$

2.30.2 The Intersection of the Empty Set Is the Universal Class

Proof:

The set intersection of the empty-set gives us the set of all elements that are in every set of the empty-set. Recall our definition

$$\bigcap \mathscr{F} := \{x \mid \forall F, F \in \mathscr{F} \implies x \in F\}$$

For our case

$$\bigcap \emptyset = \{x \mid \forall F, F \in \emptyset \implies x \in F\}.$$

We can see that due to the implication our left side is always false which leads our right side to be always true which means that the statement is valid for all x. Resulting in

$$\bigcap \emptyset = \mathcal{U}$$

For the union

$$\bigcup \emptyset = \{x \mid \exists F \in \emptyset, x \in F\}$$

The condition is always false; therefore the union has to be the empty set.

$\mathfrak{QED}$

2.30.3 Subset-Property of Set Systems

Let $A \in \mathcal{B}$ then $A \subseteq \bigcup \mathcal{B} \wedge \bigcap \mathcal{B} \subseteq A$.

Proof:

$A \in \mathcal{B}$ gives us two cases for $A \subseteq \bigcup \mathcal{B}$.

Case $A = \bigcup \mathcal{B}$: this case is given when A is the only set in $\mathcal{B}$ and the equality holds, because otherwise we will have a contradiction.

Case $A \subset \bigcup \mathcal{B}$: Assume that $\exists x \in A : x \notin \bigcup \mathcal{B} \implies A \notin B$, and therefore we have a contradiction. Hence, A is a proper subset of the union of the members of B.

For our second statement $\bigcap \mathcal{B} \subseteq A$, we have again the two cases of

Case $\bigcap \mathcal{B} = A$ which is true if A is already the intersection of all members and due to $A \cap A = A$ it is proven true.

For the case $\bigcap \mathcal{B} \subset A$, we assume that the opposite is the case which implies $\exists b \in \bigcap \mathcal{B} : b \notin A$, which is a big contradiction to the definition of $\bigcap \mathcal{B}$. Therefore, the intersection must be a proper subset of A.

$\mathfrak{QED}$

One application of this property is, for example: Given $\{A_i\}_{i \in I}$, prove that $\bigcap A_i$ is a set.

We know that $\bigcap A_i \subseteq A_i$ for some $i \in I$. Hence, it is a subset which means by the axiom of subsets of classes and sets that it has to be a set.

$\mathfrak{QED}$

2.31 Fundamental Set Functions

2.31.1 Inclusion Function

Let $A \subseteq B$. We define the function

$$E_B(x) : A \to B, \quad x \mapsto x, \quad \forall x \in B$$

as the *inclusion function* of A in B.

2.31.2 Partition Function

If P is a partition of M, then each element of M belongs to exactly one $X \in P$.

$$f_p : M \to P : m \mapsto X \in P \text{ with } m \in X,$$

is a map, called *natural map* of P.

2.31.3 Characteristic Function

Given two sets A and B, we define the *characteristic function* of B as:

$$C_B(x) : A \to \{0,1\}, x \mapsto \begin{cases} 0 & x \notin B \\ 1 & x \in B \end{cases}$$

2.32 Partitions

Let A be a set. A *partition* of A is a set of non-empty subsets in $\mathcal{P}(A)$ with the following properties:

$$\forall i, j \in I A_i \cap A_j = \emptyset \quad \text{for} \quad i \neq j$$

and

$$\bigcup_{i \in I} A_i = A \quad \text{or equivalently} \quad \bigcup_{i \in I} A_i \subseteq A$$

Example:

Prove that $B_n = \{m \in \mathbb{Z} \mid \exists q, m = n + 5q\}$ is a partition of $\mathbb{Z}$.

First, we prove the disjoint property, which means $m \in B_x \wedge m \in B_y \iff x = y$.

For the union, we need to prove that every B_n is not empty, which is very easy by letting $m = n+5\cdot 0 \implies m = n$; then combining all of them with the union will give us the entire set of integers, because each of the sets B_n has at least the number n in it, which can be any integer.

$\mathfrak{QED}$

2.32.1 Equivalence Classes

Given a partition P of a set M, the elements of P are called *classes*. If X is a class, then its members are called *representatives* of the class.

A subset of M with an element of each of the classes of M is called the *representative set* or *traversal*.

The function $v : P \to M$ with $v(X) \in X$ for all $X \in P$ is called *representative function*.

2.32.2 Equivalence Relation From a Partition

Given the partition S of the set A, then

$$R_S = \{(a,b) \mid (a,b) \in A \times A, \exists X \in S, a,b \in X\}$$

is an equivalence relation.

Proof:

Reflexivity: $a \in A = \bigcup S \implies a \in X \in S \implies (a,a) \in R_S$.

Symmetry: $a,b \in A \bigcup S \implies a,b \in X \in S \implies (a,b),(b,a) \in R_S$.

Transitivity: $(a,b),(b,c) \in R_S \implies a,b \in X \in S \wedge b,y \in Y \in S \implies b \in X \cap Y$ by definition $X \cap Y \neq \emptyset \iff X = Y$; therefore, $a,c \in X, (a,c) \in R_S$.

$\mathfrak{QED}$

2.32.3 Relation From the Relation of the Quotient Space and Partition From the Quotient Space

Given R on A, $R_{A/R} = R$ with

$$R_{A/R} = \{(a,b) \mid (a,b) \in A \times A, \exists X \in A/R, a,b \in X\}.$$

And for S a partition of A, then $A/R_S = S$.

Proof:

For the first statement we need to prove the condition of an equivalence relation:

Reflexivity: $(a,b) \in R_{A/R} \implies a,b \in [x]_R \in A/R \implies a \sim x \wedge b \sim x \wedge a \sim b \wedge a \sim a \implies (a,a) \in R_{A/R}$.

Symmetry: $(a,b),(b,a) \in R_{A/R} \implies a,b \in [x]_R \in A/R \implies a \sim x \wedge b \sim x \wedge a \sim b \wedge b \sim a \implies (a,b),(b,a) \in R_{A/R}$.

Transitivity: $(a,b),(b,c) \in R_{A/R} \implies a,b \in [x]_R \in A/R \wedge b,c \in [y]_R \in A/R \implies a \sim x \wedge b \sim x \wedge b \sim y \wedge a \sim y \implies (a,b),(b,c),(a,c) \in R_{A/R}$.

Thus, $R_{A/R} = R$. For the other direction $(a,b) \in R \implies b \in [a]_R \in A/R \implies aR_{A/R}b)$.

For the second part: $A/R_S = S$:

Let S be a partition of A, and $X \in A/R_S$. For $X = [a]_{R_S} = \{x \in A \mid aR_Sx\}$. For a and b to be in a relation they need to be part of the same element $Y \in S$. We need to prove that $Y = [a]_{R_S}$.

This is proven directly by the very definition of the relation R_S which implies that all elements belong to the same set Y. For the other direction if $a,b \in Y \implies aE_Sb \implies b \in [a]_{R_S} \in A/R_S$.

$\mathfrak{QED}$

2.32.4 The Quotient Space Satisfies the Partition Properties

Given A/R, it satisfies the properties of a partition.

Proof:

For $X,Y \in A/R$, we know by the one of the properties of equivalence classes that $[a]_R \cap [b]_R \neq \emptyset \iff [a]_R = [b]_R$. Therefore, the first condition is satisfied.

To prove: $\bigcup_{[a]_R \in A/R} [a]_R = A$.

$x \in \bigcup A/R \implies x \in [a]_R \in A/R$, this implies that $x \in A$ because $\nexists a \in A, a \notin (x,y) \in R$. This was also already proven in the section subsection dedicated to the quotient space.

$\mathfrak{QED}$

2.32.5 Properties of Partitions

- Every equivalence relation corresponds to a partition:

$$J : \{R : R \text{ is an equivalence relation on } X\} \to \{F : F \text{ is a partition of } X\}$$

  where $J(R) := X/R$ is a bijection.

- If F is a partition of X, then we can define an equivalence relation R_F by:

$$R_F := \{(x, y) \in X \times X : \exists M \in F \text{ such that } x, y \in M\}$$

  Then R_F is an equivalence relation on X.

- Two equivalence classes are either disjunctive or equal.

Proof: Equivalence Relations Correspond to Partitions

We want to prove that there exists a one-to-one correspondence between equivalence relations and partitions. We first start by start proving that from a family of equivalence classes we get a partition.

Let $\mathscr{C}$ be the set of all the equivalence classes for an equivalence relation on A, and C_x be some of those sets of equivalence classes.

We must prove that each of its equivalence classes are disjoint if not equal. By the definition if $\exists y \in [a]_\sim \wedge y \in [b]_\sim$, then this would imply that y is related to both a and b which means that they are in the same equivalence class, because otherwise $a \sim y \wedge b \sim y \iff \neg(a \sim b)$, which is a contradiction. Hence, equivalence classes are either equal or disjoint.

Now, for the union. We can state that $\bigcap [a]_\sim \subseteq A$ due to all of its elements forming at least a proper subset of A. The equality, can be shown via the definition of the equivalence class which states that each element in A in has its own equivalence class.

Analogously, we can prove it the other way around. By having disjoint sets, we know that they have to be in different equivalence classes $A_i \cap A_j = \emptyset \implies \forall x \in A_i, x \neg(\sim) y, \forall y \in A_j$.

$\mathfrak{QED}$

2.32.6 Equivalence Relations and Partitions

Let M be a set.

1. If $\sim$ is an equivalence relation on M, then the equivalence classes defined by

$$[m_0] := [m_0]_\sim := \{\, m \in M \mid m \sim m_0 \,\}$$

   for $m_0 \in M$ form a *partition* of M. This partition is usually denoted by $M/\sim$.

2. If P is a partition of M, then the relation $\sim_P$ defined by

$$m \sim_P n \quad \text{if and only if there exists } X \in P \text{ such that } m, n \in X,$$

   is an *equivalence relation.*

3. The following holds:

$$M/\sim_P = P, \quad \text{for all partitions } P \text{ of } M,$$

   and

$$\sim_{(M/\sim)} = \sim, \text{ for every equivalence relation } \sim \text{ on } M.$$

2.33 Equivalence Relations Under Functions

2.33.1 Pre-Image

Let $f : A \to B$ and Gbe an equivalence relation in B. The *pre-image* of G in A is defined as

$$f^{-1}(G) = \{(x, y) \mid (f(x), f(y)) \in G\},$$

and it is also an equivalence relation.

2.33.2 Restriction

Similarly, we can also define the *restriction* for when $B \subseteq A$:

$$G_{[B]} = \{(x, y) \mid x \in B \land y \in B \land (x, y) \in G\}$$

2.33.3 Refinement

Let G and H be equivalence relations in A. We call G a *refinement* of H if $G \subseteq H$; we also say that G is *finer* than H, and that H is *coarser* than G.

2.33.4 Quotient Space for Refinements

Let G and H be equivalence relations on a set A and let G be a refinement of H. The *quotient* of H by G usually denoted as H/G, is a relation in A/G defined as follows:

$$H/G = \{([x]_G, [y]_G) \mid (x, y) \in H\}$$

In simpler words, it is a set of the pairs consisting of the equivalence classes of a pair $(x, y) \in H$.

2.33.5 The Quotient Is an Equivalence Relation

The Quotient is an Equivalence Relation.

Proof:

Reflexivity: For $([x]_G, [y]_G) \in H/G$ we know claim that $[x]_G \sim [x]_G$. By the reflexivity of H we know that $x \sim x$, which means that $(x, x) \in H$; and therefore, $[x]_G \sim [x]_G$.

Symmetry: For $([x]_G, [y]_G) \in H/G$ we know claim that $([x]_G \sim [y]_G) \land ([y]_G \sim [x]_G)$. Analogously to our last property, if $(x \sim y \land (y \sim x)$ were not true then H would not be an equivalence relation which leads to a contradiction of our initial conditions. Hence, the symmetric property is present.

Transitivity: Lastly, the transitivity we again use the same argument as before using our base elements and pairs from H to arrive at a contradiction, which hich guarantees us that their equivalence class counterpart are also part of an equivalence relation..

$\mathfrak{QED}$

2.33.6 Equivalence Classes Under Refinements

Let G and H be equivalence relations in A, suppose $G \subseteq H$. Then $z \in [x]_H \implies [z]_G \subseteq [x]_H$.

Proof:

$z \in [x]_H$ means that $(z, x) \in H$, then

$$\begin{aligned} y \in [z]_G \implies (y, z) \in G \subseteq H &\implies (y, x) \in H \\ &\implies y \in [x]_H \end{aligned}$$

Thus, $[z]_G \subseteq [x]_H$.

$\mathfrak{QED}$

This can also be extended for all elements $x \in A$, as $[x]_G \subseteq [x]_H$.

2.34 Equivalence Relations and Surjective Mappings

- Let M be a set and $\sim$ an equivalence relation on M. Then there exists a set P and a surjective mapping $\Gamma : M \to P$ such that $\sim = \sim_\Gamma$. Hence, every equivalence relation can be represented as the equality of images under a surjective mapping.
- If $\Gamma : M \to N$ is a surjective mapping and $P := M/\sim_\Gamma$ is the partition corresponding to equality of images, then the function
$$f : P \to N, \quad [m]_{\sim_\Gamma} \mapsto \Gamma(m)$$
is a well-defined bijection.
- If $\Gamma_1 : M \to P_1$ and $\Gamma_2 : M \to P_2$ are two surjective mappings, then $\sim_{\Gamma_1} = \sim_{\Gamma_2}$ if and only if there exists a bijection $f : P_1 \to P_2$ with $\Gamma_2 = f \circ \Gamma_1$

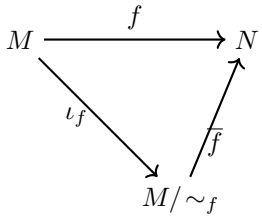
2.35 Homomorphism Theorem for Sets

Given $f : M \to N$ with $M/\sim_f = \{f^{-1}(\{n\}) \mid m \in f(M)\}$

By factorizing f as $f = \overline{f} \circ \iota_f$ with ι_f surjective and $\overline{f}$ injective, we obtain the following:

- The mapping $\iota_f : M \to M/\sim_f$ is the natural map of the partition $M/\sim_f$ induced by f.
- The mapping $\overline{f} : M/\sim_f \to N : f^{-1}(\{f(m)\}) \mapsto f(m)$ is well-defined.

The homomorphism theorem for Sets can be visualized through the following commutative diagram:



This is saying us to first form collections of elements and then use our second function to associate that connections to their r respective images.

2.36 Family of Subsets Operations

Let $\mathscr{F}$ be a system of sets (a family of subsets) on the set X. We define:

$$\bigcup_{M \in F} M := \bigcup F := \{x \in X : \text{there exists } M \in F \text{ such that } x \in M\},$$

$$\bigcap_{M \in F} M := \bigcap F := \{x \in X : x \in M \text{ for all } M \in F\}.$$

2.37 Product of Classes

Given a family of classes $\{A_i\}_{i \in I}$ we define the *product of classes* as

$$\prod A_i = \{f : I \to A \mid f(i) \in A_i \wedge i \in I\}$$

Is the set of all functions from I to $\bigcup A_i$.

2.37.1 Properties Products of Classes

- $(\prod_{i\in I} A_i) \cap \left(\prod_{j\in J} B_j\right) = \prod_{(i,j)\in I\times J}(A_i \cap B_i)$
- $(\prod_{i\in I} A_i) \cup \left(\prod_{j\in J} B_j\right) = \prod_{(i,j)\in I\times J}(A_i \cup B_i)$

2.38 Pigeon Principle

For $g : A \to B$ and $|A| = n + 1$, $|B| = n$ then there exists $b \in B$ such that $|g^{-1}(\{b\})| \geq 2$.

In other words if you have n boxes and $n + 1$ objects, then at least one of those boxes will have more than 2 elements.t.

2.39 The Cantor Set

In the following, we want to study the Cantor set $\mathcal{C} \subseteq \mathbb{R}$.

First, we describe its construction informally: We start with the unit interval $A_0 := [0, 1]$ and remove from it the "middle third" (where the boundary points are not removed). The result A_1 thus consists of the sub-intervals $\left[0, \frac{1}{3}\right]$ and $\left[\frac{2}{3}, 1\right]$. Now we remove from each of these two sub-intervals the middle third and continue in this manner.

For fixed but arbitrary $n \in \mathbb{N}_0$, A_n thus consists of 2^n pairwise disjoint closed sub-intervals of length 3^{-n}, and A_{n+1} arises from A_n by removing the middle third from each of these sub-intervals. What "remains" is the Cantor set.

Now mathematically precise: For $n \in \mathbb{N}_0$, let

$$A_{n+1} := \frac{A_n}{3} \cup (\frac{2 + A_n}{3}), \text{ with } A_0 = [0, 1]$$

Regarding the notation - for $X \subseteq \mathbb{R}$ and $a, t \in \mathbb{R}$, we define

$$aX := \{ax : x \in X\}$$

$$t + X := \{t + x : x \in X\}$$

Then $\mathcal{C} := \bigcap_{n=0}^{\infty} A_n$.

For the intuition, imagine our fraction squishing all the numbers and then adding two to a copy of those numbers. Then repeating the process multiple times.

Properties:

- The Cantor Set is uncountable
- The Cantor Set has a length of 0
- The set can be interpreted as all infinite sequences of zeroes and ones, when handling left and right as 0 and one.

Proof I:

We give a proof by contradiction.

Assume the Cantor set is countable. Then there exists a surjective function

$$c : \mathbb{N} \to \mathcal{C}.$$

After the first step of the Cantor construction we have

$$A_1 = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1].$$

Since

$$c(1) \in \mathcal{C} \subset A_1,$$

the point $c(1)$ lies in exactly one of the two disjoint intervals. We choose the other interval and call it I_1.

Thus:

- $I_1 \subset A_1$,
- I_1 is closed,
- $c(1) \notin I_1$.

For the inductive step, assume there exists a closed interval

$$I_n \subset A_n \quad \text{with} \quad c(n) \notin I_n.$$

When passing from A_n to A_{n+1}, the middle third of each interval is removed. Therefore, I_n splits into exactly two disjoint closed sub-intervals of A_{n+1}.

Since $c(n+1)$ can lie in only one of these two intervals, we choose the other and call it I_{n+1}.

Thus:

- $I_{n+1} \subsetneq I_n$,
- I_{n+1} is closed and non-empty,
- $c(n+1) \notin I_{n+1}$.

By construction the nested sequence

$$I_1 \supsetneq I_2 \supsetneq I_3 \supsetneq \cdots$$

is obtained, and the lengths satisfy

$$|I_n| = 3^{-n}.$$

Thus, the lengths converge to 0.

Since the intervals are closed, bounded, nested, and their lengths tend to 0, the interval-nesting principle implies:

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

There exists exactly one point

$$x \in \bigcap_{n=1}^{\infty} I_n.$$

Since each $I_n \subset A_n$, it follows that

$$x \in \mathcal{C}.$$

Contradiction

Because c is surjective by assumption, there exists a $k \in \mathbb{N}$ such that

$$x = c(k).$$

However, by our construction I_k was chosen so that

$$c(k) \notin I_k,$$

while at the same time

$$x \in I_k.$$

This is a contradiction. Thus, the assumption that the Cantor set is countable leads to a contradiction. Therefore, the Cantor set has to be uncountable.

$\mathfrak{QED}$

Proof II:

This is not the most rigorous proof, but accomplishes its goal:

The original interval as length 1 and consist of one unit, the next $1 - \frac{1}{3}$ and consist of two units, this pattern with pattern with $2^n \frac{1}{3^n}$ continues forever thus,

$$L = 1 - \sum_{i=1}^{\infty} 2^n \frac{1}{3^n}$$

Note that the sum is a geometric series therefore,

$$L = 1 - 1 = 0$$

$\mathfrak{QED}$

2.40 Convexity

A set X is *convex* is $\|x - y\| \in X,\ \forall x, y \in X$.

Chapter 3

Relations, Maps and Functions

3.1 Relations

Given two classes A and B, a *relation* R is a subset of its *Cartesian product* $R \subseteq A \times B$.

Example:

Given $A = \{1, 2, 3\}$ find the elements of the relation R defined by $R = \{(a, b) \in A \times A \mid a < b\}$.

$$A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$$

The relation R would then be:

$$R = \{(1,2), (1,3), (2,3)\}$$

Common types of relations include:

- *Functions*: (special relations where each input has exactly one output).
- *Equivalence relations*: (reflexive, symmetric, and transitive).
- *Partial orders*: (reflexive, antisymmetric, and transitive).

3.1.1 Types of Relations

Let A be a set and X be a relation on A.

- *Reflexive:* $\forall a \in A : (a, a) \in X$ (or written as $a \sim a$)
- *Irreflexive:* $\forall a \in A : (a, a) \notin X$
- *Symmetric:* $\forall a, b \in A : (a, b) \in X \to (b, a) \in X$ (or written as $(a \sim b) \Rightarrow (b \sim a)$)
- *Antisymmetric:* $\forall a, b \in A : (a, b) \in X$ and $(b, a) \in X \implies a = b$
- *Transitive:* $\forall a, b, c \in A : (a, b) \in X$ and $(b, c) \in X \to (a, c) \in X$ (or written as $(a \sim b)$ and $(b \sim c) \Rightarrow (a \sim c)$)
- *Total:* $\forall a, b \in A : a \neq b \to (a, b) \in X$ or $(b, a) \in X$

3.2 Diagonal Graph

Given a relation, the *graph* I_A is defined as

$$\{(x, x) \mid x \in A\}$$

and is denoted as the *diagonal graph* of A.

3.2.1 Lemmas for Proofs

Let the graph G be a relation on A.

- G is symmetric if and only if $G = G^{-1}$.
- G is antisymmetric if and only if $G \cap G^{-1} \subseteq I_A$.
- G is transitive if and only if $G \circ G \subseteq G$.

3.3 Order Relation

A relation is called an *order relation* $R \subseteq A \times A$ if it is reflexive, antisymmetric and transitive.

3.3.1 Inverse Order Relation

Given an order relation G on A, then G^{-1} is also an order relation on A.

3.3.2 Strict Order Relation

A relation is called a *strict order relation* $R \subseteq A \times A$ if it is irreflexive, antisymmetric and transitive.

3.3.3 Inverse Strict Order Relation

Given a strict order relation G on A, then G^{-1} is also a strict order relation on A.

3.4 Equivalence Relation

An *equivalence relation* is a relation $R \subseteq A \times A$ that is **symmetric**, **transitive** and **reflexive**.

Example:

$$R := \{(a, b) \in \mathbb{N} \times \mathbb{N} : a = b\}$$

3.4.1 Properties of Equivalence Relations

Given the equivalence relations G and H, the following properties apply:

- $G \circ H$ is an E.R. if and only if $G \circ H = H \circ G$.
- $G \cup H$ is an E.R. if and only if $G \circ H \subseteq G \cup H \wedge H \circ G \subseteq G \cup H$.

3.4.2 Inverse Equivalence Relation

Given an equivalence relation G on A, then G^{-1} is also an equivalence relation on A.

3.5 Equivalence Class

Let $\sim$ be an equivalence relation on a set A and $a \in A$, the *equivalence class* of a, denoted by

$$[a]_\sim := \{y \in A \mid (a, y) \in \sim\}$$

is the set of all elements in A that are related to a.

3.5.1 Lemmas About Equivalence Classes

For $a, b \in A$ then

- $aEb \iff [a]_E = [b]_E$
- $a \neg Eb, (a,b) \notin E \iff [a]_E \cap [b]_E = \emptyset$

Proof I:

$aEb, x \in [a]_E$ tells us that also $xEb \implies x \in [b]_E$. For the other direction using the reflexive property we get $[a]_E = [b]_E \implies aEa \implies a \in [a]_E = [b]_E$.

$\mathfrak{QED}$

Proof II:

$x \in [a]_E \cap [b]_E \implies xEa, xEb, aEx \implies aEb$, a contradiction. Now, for $[a]_E \cap [b]_E = \emptyset$ we can say $aEb \implies a \in [a]_E, a \in [b]_E$, another contradiction.

$\mathfrak{QED}$

3.6 Quotient Space

Let $\sim$ be an equivalence relation on a set A. The *quotient space* of A by $\sim$, denoted by $A/\sim$ (or sometimes A/R), is the set of all distinct equivalence classes of elements in A. Formally, it is defined as:

$$A/\sim := \{[x]_\sim \mid x \in A\}$$

The quotient space $A/\sim$ is a partition of the original set A into disjoint equivalence classes. Each element of the quotient space is an equivalence class $[x]$, which itself is a subset of A.

3.6.1 The Union of the Quotient Space

Given A/R, then $\bigcup A/R = A$.

Proof:

Suppose $a_0 \in A \land a_0 \notin [a]_R, \forall [a]_R \in A/R$. If R is an equivalence relation then $a_0 \in A$ with R implies that $(a_0, a_0) \in R$. Therefore, the only way for a_0 to not belong to any equivalence class is if R were not an equivalence relation, which contradicts our initial condition; hence, we know that the union of the equivalence classes given an equivalence relation must be equal to the original set.

$\mathfrak{QED}$

3.7 Maps

A *map* for two sets A and B is defined as a subset of $A \times B$, and it is denoted by

$$f : A \to B \subseteq \{(a,b) \mid (a,b) \in A \times B\}$$

where

$$\forall a \in A, !\exists b \in B : (a,b) \in f.$$

This is informally known as the vertical line test.

3.7.1 The Set of All Maps

For the sets M and N, we define the set of all maps $f : M \to N$ as

$$N^M := \{f \mid f : M \to N\}$$

3.7.2 Equality Of Maps

Given two maps $f : A \to B$ and $g : M \to N$. They are equal if $A = M$ and $B = N$.

3.7.3 The Characteristic Map

Given $T \subseteq M$ we define the *characteristic map* as

$$\mathcal{X}_T : M \to \{0,1\} : m \to \begin{cases} 1, & m \in T \\ 0, & m \notin T \end{cases}$$

This map tells us if an element is in the subset. Also, each map $\alpha : M \to \{0,1\}$ defines a unique subset T of M,

$$T := M\{|\ \alpha(m) = 1\}$$

Because of that fact, we can use this map an the concept of a subset interchangeably.

Proof:

We want to show $\alpha = \mathcal{X}_T$. Therefore, we need to show that both have the same domain rand image.

For the domain, $M \subseteq M$, $\forall m \in M : m \in M$ which is trivial.

Now we need to prove that the images are the same; let $m \in T \wedge m \in M$, we know that for $\alpha(m) = 1$. Also if $m \notin T$, then $\alpha(m) = 0$, which implies $m \not\in nT \subseteq M$. This is the definition of the characteristic map.

$\mathfrak{QED}$

3.8 Functions

A subset $f \subseteq X \times Y$ is called a *function* if:

$$\forall x \in X : \exists y \in Y : f(x) = y$$

and

$$(x,y) \in f \text{ and } (x,y') \in f \implies y = y'.$$

That is, for **each** $x \in X$, there exists at most one $y \in Y$ such that $(x,y) \in f$.

When writting proofs, it is important to take into consideration that the whole set X is in use by the function. This guarantees the existance of $f(x$.

3.9 The Graph

We can also consider the graph as a set of ordered pairs x, y where $y = f(x)$.

$$\text{graph}(f) := \{(x, f(x)) \mid x \in X, f(x) \in Y\} = G$$

3.9.1 The Inverse Graph

Let G be a graph. We define

$$G^{-1} := \{(x,y) \mid (y,x) \in G\}$$

as *inverse graph* of G.

3.9.2 The Composite Graph

Let G and H be graphs. We define

$$G \circ H := \{(x, y) \mid \exists z \in (x, z) \in H \wedge (z, y) \in G\}$$

as the *composite graph* of G and H.

- $G \circ (H \circ J) = (G \circ H) \circ J$.
- $(G^{-1})^{-1} = G$.
- $(G \circ H)^{-1} = H^{-1} \circ G^{-1}$.

Proof I:

$(G \circ H) \circ J) = G \circ (H \circ J)$

$$G \circ H = \{(z, y) \mid \exists w \in (z, w) \in H \wedge (w, y) \in G\}$$

Proceeding

$$(G \circ H) \circ J = \{(x, y) \mid \exists z, w \text{ s.t. } (x, z) \in J \wedge (z, w) \in H \wedge (w, y) \in G\}.$$

For $G \circ (H \circ J)$

$$H \circ J = \{(x, w) \mid \exists z \in (x, z) \in J \wedge (z, w) \in H\}.$$

Proceeding

$$G \circ (H \circ J) = \{(x, y) \mid \exists z, w \in (x, z) \in J \wedge (z, w) \in H \wedge (w, y) \in G\}$$

$\mathfrak{QED}$

Proof II:

$(G^{-1})^{-1} = G$

$$\begin{aligned}(x, y) \in (G^{-1})^{-1} &\iff (y, x) \in G^{-1} \\ &\iff (x, y) \in (G^{-1})^{-1}\end{aligned}$$

$\mathfrak{QED}$

Proof III:

$(G \circ H)^{-1} = H^{-1} \circ G^{-1}$.

$$\begin{aligned}(G \circ H)^{-1} &= \{(x, y) \mid \exists z \in (y, z) \in H \wedge (z, x) \in G\} \\ &\iff (y, z) \in H \wedge (z, x) \in G \\ &\iff (z, y) \in H^{-1} \wedge (x, z) \in G^{-1}\end{aligned}$$

$\mathfrak{QED}$

3.9.3 Functional Graph

A functional graph is a graph that satisfies the second constraint of functions. Therefore, it corresponds to a function. Hence, all of our graph properties are applicable to functions.

3.10 Domain

The *domain* of a relation R from set A to set B is the set of all elements in A that are related to at least one element in B. Formally, if $R \subseteq A \times B$ is a relation, then the domain of R is defined as:

$$\text{dom}(R) = \{a \in A \mid \exists b \in B \text{ such that } (a, b) \in R\}$$

In other words, the domain consists of all the input values that appear in the ordered pairs of the relation. For example, if $R = \{(1, 4), (2, 5), (3, 4), (2, 6)\}$, then $\text{Dom}(R) = \{1, 2, 3\}$.

3.11 The Identity

$$\text{id}(f) := idx := \{(x, x) \mid x \in X\}$$

3.12 Image

The *image* of a relation R from set A to set B is the set of all elements in B that are related to at least one element in A. Formally, if $R \subseteq A \times B$ is a relation, then the image of R is defined as:

$$\text{img}(R[A]) = \{y \in \text{ ran}(R) \mid \exists a \in A \text{ such that } xRy\} \subseteq B$$

In other words, the image consists of all the output values that appear in the ordered pairs of the relation. For example, if $R = \{(1, 4), (2, 5), (3, 4), (2, 6)\}$, then $\text{img}(R) = \{4, 5, 6\}$.

3.13 The Range

The *range* of an function is the goal-set of the function. Note that this is **not** the image of the function.

3.13.1 Proof That the Domain and Image Are Sets

Given a graph G which is a set. Prove that the domain and image are also sets.

$$\forall (x, y) \in G : (x, y) = \{x, \{x, y\}\}$$

Thus, the union of all the elements in G has a structure of the form

$$\bigcup G = \{x, y, z, \dots, \{x, y\}, \{x, z\}, \{y, z\}, \dots\}$$

By iterating one more time we get the union of the union

$$\bigcup\bigcup G = \{x, y, z, \dots\}$$

Hence, by the definition of the domain and the image of the graph we can directly see that they are subsets of the union of the union of the graph G, and thus they are sets.

$\mathfrak{QED}$

3.14 The Fiber

We donote a *fiber* for $b \in B$ as

$$f^{-1}(\{b\}) := \{a \in A \mid f(a) = b\}.$$

3.15 The Pre-Image

Given a map $f : AtoB$, we denote the set

$$f^{-1}(B) := \{a \in A \mid f(a) \in B\} = \bigcup_{b \in B} f^{-1}(\{b\})$$

as the *pre-image* of B.

3.16 Composition of Maps

The composition of two functions is the operation of applying one function to the result of another function. If we have functions $f : A \to B$ and $g : B \to C$, then the composition of g and f, denoted as $g \circ f$ (read as g composed with f), is a function from A to C defined by:

$$(g \circ f)(a) = g(f(a)) \quad \text{for all } a \in A$$

The composition applies f first, then applies g to the result. Note that the co-domain of f must match the domain of g for the composition to be defined.

The formal definition is

$$g \circ f := \{(s, u) \in S \times U \mid \exists t \in T \text{ such that } (s, t) \in f \wedge (t, u) \in g\}$$

Proof:

We want to prove that this is actually a map, for that we need $g \circ f \subseteq A \times C$ which is trivial for $A \neq \emptyset \neq B$. Now we need to prove the existence and uniqueness.

For f

$$a \in A, \implies f(a) = b \in B.$$

And by our Cartesian product $g(b) \in C$, which guarantees the existence, for the uniqueness:

$$(a, b_1) \in f, (a, b_2) \in f : f(a) = b_1,\ f(a) = b_2$$

Therefore, $g(b_1) \in g, g(b_2) \in g$, because g is a map $b_1 = b_2$.

$\mathfrak{QED}$

Example:

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = x + 3$. Then:

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 3$$

$$(f \circ g)(x) = f(g(x)) = f(x + 3) = (x + 3)^2 = x^2 + 6x + 9$$

Note that $g \circ f \neq f \circ g$ in general, which shows that function composition is not commutative.

3.16.1 Associativity of the Composition

Given the maps $f : A \to B$, $g : B \to C$, and $h : C \to D$, the composition of functions is associative, i.e.,

$$h \circ (g \circ f) = (h \circ g) \circ f$$

under the conditions $f(A) \subseteq B$ and $g(B) \subseteq C$.

Proof:

For any $a \in A$, we know that $f(a) \in B$, hence by the map property, $\exists c \in C,\ (f(a), c) \in g$, continuing because $c = g(f(a)) \in C$ there exits $d \in D$ such that $(c, d) \in h$, and hence $h(g(f(a))) = d$ and we have a proper map.

Therefore, $(a, d) \in (h \circ g) \circ f$ and $(a, d) \in h \circ (g \circ f)$, proving the associativity.

$\mathfrak{QED}$

3.16.2 Injectivity and Surjectivity Under Composition

Let $f : A \to B$ and $g : B \to C$, then the following holds:

- Is $g \circ f$ injective, then f is injective.
- Is $g \circ f$ surjective, then g is surjective.

Proof I:

$g \circ f$ is injective therefore for $a_1, a_2 \in A$,

$$g(f(a_1) = g(f(a_2)) \implies f(a_1) = f(a_2) \implies a_1 = a_2$$

$\mathfrak{QED}$

Proof II:

$g \circ f$ is surjective therefore, for all $c \in C$, $\exists a \in A$ such that $g(f(a)) = c$. This also implies that $\exists f(a) \in B : g(b) \in C$.

$\mathfrak{QED}$

3.16.3 Domain and Range Under Composition Corollary

Let G and H be graphs, then $\text{img}(H) \subseteq \text{dom}(G) \implies \text{dom}(G \circ H) = \text{dom}(H)$

Proof:

Note that for

$$\forall y \in \text{img}(H) : y \in \text{dom}(G).$$

Also, when constructing $G \circ H$ we get a map from $\text{dom}(H)$ to the $\text{img}(G)$, because for all of our $y \in \text{img}(H)$ we can always find a pair $(y, w) \in \text{img}(G)$ due to the structure of our graph. Hence, the domain of our composite graph is $\text{dom}(H)$.

$\mathfrak{QED}$

3.17 Types of Functions

3.17.1 Injective Functions

An injective function (also called a one-to-one function) is a function that maps distinct elements from the domain to distinct elements in the co-domain.

Formally, a function $f : A \to B$ is injective if for all $a_1, a_2 \in A$:

$$a_1 \neq a_2 \rightarrow f(a_1) \neq f(a_2)$$

Equivalently, using the contrapositive:

$$f(a_1) = f(a_2) \rightarrow a_1 = a_2$$

Example:

Prove that $\mathrm{Id}_X : X \rightarrow X$ is an injective function.

Proof:

Let $x_1, x_2 \in X$ such that $\mathrm{Id}_X(x_1) = \mathrm{Id}_X(x_2)$. Then,

$$x_1 = \mathrm{Id}_X(x_1) = \mathrm{Id}_X(x_2) = x_2$$

Hence, Id_X is injective.

$\mathfrak{QED}$

3.17.2 Injection

Given $T \subseteq M$, a subset of M, the *injection* from T to M defined as

$$\iota_T : T \rightarrow M : t \mapsto t$$

is an *injective function.*

3.17.3 Surjective Functions

A *surjective function* (also called an onto function) is a function whose image equals its co-domain, meaning that every element in the co-domain has at least one pre-image in the domain.

Formally, a function $f : A \rightarrow B$ is surjective if:

$$\forall b \in B, \exists a \in A \text{ such that } f(a) = b$$

To prove that a function is surjective, one must show that given any element $b \in B$, there exists an element $a \in A$ such that $f(a) = b$. When a concrete function definiton like $f(x) = 2x$ is given we just need to solve for x, and show that it is an element of the domain. For the case where we are dealing with an more abstract function, we need to use the **function definition** to guarantee the existence of elements in the domain.

Example I:

Prove that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x$ is surjective.

Proof:

Let $b \in B$ the $b = 2x$, then we can solve for x giving us $x = \frac{b}{2}$. Note that $\frac{b}{2} \in \mathbb{R}$, hence we have shown that for any $b \in B$, there exists an $x \in \mathbb{R}$ such that $f(x) = b$. Therefore, f is surjective.

$\mathfrak{QED}$

Example II:

Given the characteristic function $\mathcal{X}_T : X \rightarrow \{0, 1\}$ defined by

$$\mathcal{X}_T(x) = \begin{cases} 1 & \text{if } x \in T \\ 0 & \text{if } x \notin T \end{cases}$$

where $T \subseteq X$.

Prove that $\mathcal{X}_T$ is surjective.

Proof:

Let $b \in \{0, 1\}$. We need to show that there exists an $x \in X$ such that $\mathcal{X}_T(x) = b$.

If $b = 1$, then choose $x \in T$. By definition, $\mathcal{X}_T(x) = 1$.

If $b = 0$, then choose $x \in X \setminus T$. By definition, $\mathcal{X}_T(x) = 0$.

Hence, for every $b \in \{0, 1\}$, there exists an $x \in X$ such that $\mathcal{X}_T(x) = b$. Therefore, $\mathcal{X}_T$ is surjective.

$\mathfrak{QED}$

3.17.4 Bijective Functions

A *bijective function* (also called a one-to-one correspondence) is a function that is both injective and surjective. In other words, every element in the co-domain is mapped to by exactly one element in the domain.

Formally, a function $f : A \to B$ is bijective if it is both:

- *Injective*: $\forall a_1, a_2 \in A, a_1 \neq a_2 \to f(a_1) \neq f(a_2)$
- *Surjective*: $\forall b \in B, \exists a \in A$ such that $f(a) = b$

Bijective functions establish a perfect pairing between elements of the domain and co-domain, where each element in the domain corresponds to exactly one element in the co-domain, and vice versa. A bijection allows us to define an inverse function $f^{-1} : B \to A$.

3.18 Cardinality of Injective and Surjective Functions

Given the finite sets A, B with the map $f : A \to B$, the following holds:

- If f is injective, then $|A| \leq |B|$.
- If f is surjective, then $|A| \geq |B|$.
- If f is bijective, then $|A| = |B|$.

Proof I:

f is injective therefore, $|f^{-1}(b)| \leq 1$, $\forall b \in B$ hence,

$$|A| = \sum_{b \in B} |f^{-1}(b)| \leq \sum_{b \in B} 1 = |B|.$$

$\mathfrak{QED}$

Proof II:

f is surjective $|f^{-1}(b)| \geq 1$, $\forall b \in B$ therefore,

$$|A| = \sum_{b \in B} |f^{-1}(b)| \geq \sum_{b \in B} 1 = |B|.$$

$\mathfrak{QED}$

Proof III:

Proven by using I and II.

$\mathfrak{QED}$

3.19 Propositions on Images and Pre-Images Under Set Operations

Let $f : X \to Y$ be a function.

3.19.1 Union and Cut Sets

- For subsets $A_1, A_2 \subseteq X$, we have:

$$f(A_1 \cup A_2) = f(A_1) \cup f(A_2) \quad \text{and} \quad f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$$

- For subsets $B_1, B_2 \subseteq Y$, we have:

$$f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2) \quad \text{and} \quad f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2)$$

Proof:

We prove the second part of 1 and the first part of 2.

Let $y \in f(A_1 \cap A_2)$. Then there exists $x \in A_1 \cap A_2$ such that $f(x) = y$. Since $x \in A_1$ and $x \in A_2$, it follows that $y \in f(A_1)$ and $y \in f(A_2)$, hence $y \in f(A_1) \cap f(A_2)$. Therefore, every element of $f(A_1 \cap A_2)$ is also an element of $f(A_1) \cap f(A_2)$, so:

$$f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$$

The reason we do not have equality is because:

$$\begin{aligned} y \in f(A_1) \cap f(A_2) &\iff \exists f^{-1}(y) \in A_1 \wedge f^{-1}(y) \in A_2 \\ &= x_1 \in A_1 \wedge x_2 \in A_2 \end{aligned}$$

Note, that if $x_1 = x_2$ we have equality, but if $x_1 \neq x_2$ then $y \notin f(A_1 \cap A_2)$. Hence we have a propert subset.

$\mathfrak{QED}$

Let $x \in f^{-1}(B_1 \cup B_2)$. Then $f(x) \in B_1 \cup B_2$, which means $f(x) \in B_1$ or $f(x) \in B_2$. Thus, $x \in f^{-1}(B_1)$ or $x \in f^{-1}(B_2)$, which implies:

$$x \in f^{-1}(B_1) \cup f^{-1}(B_2)$$

Hence, both sets contain the same elements and are therefore equal:

$$f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2)$$

$\mathfrak{QED}$

3.19.2 Union and Cut of the Whole Domain and Range

Let $f : X \to Y$ be a function.

- Let $\mathcal{F}$ be a collection of subsets of X. Then:

$$f\left(\bigcup_{A \in \mathcal{F}} A\right) = \bigcup_{A \in \mathcal{F}} f(A) \quad \text{and} \quad f\left(\bigcap_{A \in \mathcal{F}} A\right) \subseteq \bigcap_{A \in \mathcal{F}} f(A)$$

- Let $\mathcal{G}$ be a collection of subsets of Y. Then:

$$f^{-1}\left(\bigcup_{B\in\mathcal{G}} B\right) = \bigcup_{B\in\mathcal{G}} f^{-1}(B) \quad \text{and} \quad f^{-1}\left(\bigcap_{B\in\mathcal{G}} B\right) = \bigcap_{B\in\mathcal{G}} f^{-1}(B)$$

Proof (partial):

We show the first statement of part (ii); the rest follows analogously.

Let $x \in f^{-1}\left(\bigcup_{B\in\mathcal{G}} B\right)$. Then:

$$f(x) \in \bigcup_{B\in\mathcal{G}} B \quad \Leftrightarrow \quad \exists B \in \mathcal{G} \text{ such that } f(x) \in B \quad \Leftrightarrow \quad \exists B \in \mathcal{G} \text{ such that } x \in f^{-1}(B)$$

Hence:

$$x \in \bigcup_{B\in\mathcal{G}} f^{-1}(B)$$

It follows that:

$$f^{-1}\left(\bigcup_{B\in\mathcal{G}} B\right) = \bigcup_{B\in\mathcal{G}} f^{-1}(B)$$

$\mathfrak{QED}$

Example:

We define $A = \{1, 0\}$ and $B = \{2\}$ and $f(1) = 3$, $f(2) = 3$, $f(0) = 1$. Then

$$f(A \cap B) = f(\emptyset) = \emptyset$$

And

$$f(A) \cap f(B) = 3$$

3.20 Bijection of the Power Set and 2^A

We denote 2^A as the set of all functions from A to a set of two elements. The statement goes as follows: there is a bijection between the power set $\mathcal{P}(A)$ and 2^A.

Proof:

Let $\gamma : \mathcal{P}(A) \to 2^A$. If $B \in \mathcal{P}$ let

$$\gamma(B) = C_B, \forall B \in \mathcal{P}(A).$$

By this definition we get the properties of a function directly. Now we have to prove that the function is a one-to-one correspondence.

For the injectivity, we already know that for $\gamma(B) = \gamma(C) \implies B = C$, since otherwise it would be the case that we have two elements in the power set which are identical.

For the surjectivity, if $f \in 2^A$, and we let $B = f^{-1}(0)$, then $f = C_B = \gamma(B)$. Thus, γ satisfies the surjectivity.

$\mathfrak{QED}$

3.21 Restriction Function

Let G be a graph and $B \subseteq \text{dom}(G)$ then we denote the *restriction* of G as:

$$G_{[B]} = \{(x, y) \mid (x, y) \in G \wedge x \in B\}.$$

The range of this restricted map can be denoted as:

$$G(B) = \{y \mid x \in \text{dom}(B) \wedge)(x, y) \in G\}$$

3.22 Even Function

A real function f is *even* if

- $\forall x \in \text{Dom}(f) \exists -x \in \text{Dom}(f)$
- $\forall x \in \text{Dom}(f)$ the following is true $f(x) = f(-x)$

3.23 Odd Function

A real function f is *odd* if

- $\forall x \in \text{Dom}(f) \exists -x \in \text{Dom}(f)$
- $\forall x \in \text{Dom}(f)$ the following is true $f(x) = -f(x)$

3.24 Inverse of a Function

The *inverse* of a function $f : A \to B$ is a function $f^{-1} : B \to A$ that reverses the operation of f. That is, if f maps an element $a \in A$ to an element $b \in B$, then the inverse function f^{-1} maps b back to a.

Formally, a function $f : A \to B$ has an inverse $f^{-1} : B \to A$ if and only if f is bijective (both injective and surjective). The inverse function satisfies the following properties:

$$f^{-1}(f(a)) = a \quad \text{for all } a \in A,$$

$$f(f^{-1}(b)) = b \quad \text{for all } b \in B.$$

In other words, composing a function with its inverse yields the identity function. That is:

$$f^{-1} \circ f = id_A \quad \text{and} \quad f \circ f^{-1} = id_B$$

where id_A and id_B are the identity functions on sets A and B, respectively.

Proof:

$\Rightarrow$: direction is trivial, due to the definition of a bijective function.

$\Leftarrow$: Our function has the following properties:

$$f^{-1}(f(a)) = a \quad \text{for all } a \in A,$$

$$f(f^{-1}(b)) = b \quad \text{for all } b \in B.$$

The first condition gives us the injectivity due to the constraint that for all $b \in B$ there exists exactly one $a \in A$ such that $f(a) = b$. Else, the condition would not be obeyed.

The second one gives us the surjectivity of f by providing a definition for each of the $b \in B$. Otherwise, our condition would be violated.

The uniqueness is guaranteed by

$$f^{-1}(b) \neq g^{-1}(b)$$

This gives us a contradiction if they were to have the same properties but were different.

$\mathfrak{QED}$

3.24.1 Steps to Find the Inverse of a Function

To find the inverse of a function $f(x)$, follow these steps:

1. Replace $f(x)$ with y: $y = f(x)$
2. Interchange the variables x and y: $x = f(y)$
3. Solve for y in terms of x: $y = f^{-1}(x)$
4. Verify that the resulting function is indeed the inverse by checking that $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$

Example:

Let's apply the steps above to find the inverse of $f(x) = 2x + 3$.

Step 1: Replace $f(x)$ with y

$$y = 2x + 3$$

Step 2: Interchange the variables x and y

$$x = 2y + 3$$

Step 3: Solve for y in terms of x

$$\begin{aligned} x &= 2y + 3 \\ x - 3 &= 2y \\ \frac{x-3}{2} &= y \end{aligned}$$

So, the inverse function is:

$$f^{-1}(x) = \frac{x-3}{2}$$

Step 4: Verify that $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$

Let's verify $f^{-1}(f(x)) = x$:

$$\begin{aligned} f^{-1}(f(x)) &= f^{-1}(2x+3) \\ &= \frac{(2x+3)-3}{2} \\ &= \frac{2x}{2} \\ &= x \end{aligned}$$

And let's verify $f(f^{-1}(x)) = x$:

$$\begin{aligned} f(f^{-1}(x)) &= f\left(\frac{x-3}{2}\right) \\ &= 2\left(\frac{x-3}{2}\right)+3 \\ &= (x-3)+3 \\ &= x \end{aligned}$$

Since both compositions yield the identity function, $f^{-1}(x) = \frac{x-3}{2}$ is indeed the inverse of $f(x) = 2x+3$.

3.24.2 Properties of the Inverse Function

Let $f : X \to Y$ be a function.

- Assume $x \sim y$ if $f(x) = f(y)$; then $\sim$ is an equivalence relation.
- Consider the quotient set $X_f := X/\sim$, where $\sim$ is the equivalence relation defined by $x \sim x' \iff f(x) = f(x')$. Let $q_f : X \to X_f$ be the canonical projection defined by $q_f(x) = [x]_\sim$, and let $\iota_f : f(X) \to Y$ be the inclusion map, $y \mapsto y$. Then the function

  $$\hat{f} : X_f \to f(X), \quad [x]_\sim \mapsto f(x)$$

  is a bijection, and the original map f can be written as the composition

  $$f = \iota_f \circ \hat{f} \circ q_f.$$

  Here, q_f is surjective, $\hat{f}$ is bijective, and ι_f is injective. This yields the following commutative diagram:

  $$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow{\scriptstyle q_f} & & \uparrow{\scriptstyle \iota_f} \\ X_f & \xrightarrow{\hat{f}} & f(X) \end{array}$$

3.24.3 Left and Right Inverses

Given $f : A \to B$ and $g : B \to A$ and $\gamma : B \to A$, then

- f is *injective* if and only if $g \circ f = \mathrm{Id}_A$. Each g is surjective and called the *left inverse* of f.
- f is *surjective* if and only if $f \circ \gamma = \mathrm{Id}_B$. Each γ is injective and called the *right inverse* of f.

Proof I:

$\Leftarrow$: We assume $g \circ f = \mathrm{Id}_A$. Let $f(a) = f(b)$ for $a, b \in A$, now we apply our composite function $g(f(a)) = g(f(b)) \implies a = b$, since $g \circ f = \mathrm{Id}_A$. Therefore, f is injective since we just wrote the defintion of it.

$\Rightarrow$: Let f be injective; now let us define

$$g : \begin{cases} a, & f(a) = b \\ a_0, & b \notin \mathrm{img}(f) \end{cases}$$

For $a \in \mathrm{dom}(f)$, $g \circ f = \mathrm{Id}_A$ and g is also surjective since for all $b \in B$ there exists a pre-image which is not empty.

$\mathfrak{QED}$

Proof II:

$\Rightarrow$: A $f \circ \gamma = \mathrm{Id}_B$; now let $b \in B$, then $f(\gamma(b)) = b \implies b \in \mathrm{img}(f) \implies$ f is surjective.

$\Leftarrow$: Let f be surjective, this means that for all $|f^{-1}(\{b\})| \neq 0$, now let us define

$$\gamma : \begin{cases} a, & |f^{-1}(\{b\})| = 1 \\ \text{choose}(f^{-1}(\{b\})), & \text{else} \end{cases}$$

Note that $f \circ \gamma = \mathrm{Id}_B$ since there is always at least one a such that we return to B, and for the case there are multiple we just choose one from the lists. Also note that $a \neq \text{choose}(f^{-1}(\{b\}))$, since this would mean that f is not a map.

$\mathfrak{QED}$

Example:

Given $f : M \to N$ having a right inverse $\gamma : N \to M$ guarantees the existence of the solution of $f(m) = n$. Set $m = \gamma(n)$ then $f(m) = f(\gamma(n)) = f \circ \gamma = \mathrm{Id}_N(n) = n$.

3.24.4 Equivalent Definitions of the Inverse Function

Given the map $f : A \to B$, both statements are equivalent

1. $f^{-1} := \{(b, a) \in B \times A \mid (a, b) \in f\}$ is a map from $B \to A$.
2. $\forall b \in B$, $|f^{-1}(b)| = 1$.

Proof:

$\Rightarrow$: We assume that 1. is true, hence $\forall b \in B, !\exists a \in A : f^{-1}(b) = a$, in other words

$$f^{-1}(\{b\}) = \{f^{-1}(a)\}.$$

This know that $f^{-1}(\{b\}) = \{a\} \subseteq \{f^{-1}(a)\}$ and the other way around, hence, we are done.

$\Leftarrow$: For the other direction, if the fiber only has one element, then f^{-1} is a map.

$\mathfrak{QED}$

3.24.5 Inverse Definition over Left and Right Inversess

A function is *bijective* (invertible) is and only if it has a left and right inverse.

Proof::

$\Rightarrow$: Our function is bijective, and we know that by the equivalences that it has to have both a left and a right inverse.

$\Leftarrow$: If a function has both a left and a right inverse, it must be bijective. The existence of a left inverse ensures injectivity, and the existence of a right inverse ensures surjectivity. Therefore, the function is bijective.

$\mathfrak{QED}$

3.25 Transformations of a Function

Transformations modify the appearance of a function's graph without altering its basic shape. Here, we examine how different algebraic changes to a function $f(x)$ affect its graph:

Vertical Translation: $f(x) + a$

- Shifts the graph *upward* if $a > 0$, and *downward* if $a < 0$.
- Each point on the graph moves vertically by a units.

Horizontal Translation: $f(x + a)$

- Shifts the graph *left* if $a > 0$, and *right* if $a < 0$.
- This is opposite of what might be expected: adding to x shifts the graph in the negative direction.

Vertical Scaling (Stretch/Compression): $af(x)$

- If $|a| > 1$: the graph is *stretched* vertically (taller and narrower).
- If $0 < |a| < 1$: the graph is *compressed* vertically (shorter and wider).
- If $a < 0$: includes a reflection across the *x-axis*.

Horizontal Scaling (Stretch/Compression): $f(ax)$

- If $|a| > 1$: the graph is *compressed* horizontally (narrower).
- If $0 < |a| < 1$: the graph is *stretched* horizontally (wider).
- If $a < 0$: includes a reflection across the *y-axis*.

Reflection across the x-axis: $-f(x)$

- Flips the graph upside-down over the x-axis.
- Each point (x, y) becomes $(x, -y)$.

Reflection across the y-axis: $f(-x)$

- Flips the graph left-to-right over the y-axis.
- Each point (x, y) becomes $(-x, y)$.

3.26 Counting Functions From $A \to B$

Given $f : A \to B$ with $|A| = n$ and $|B| = m$, we can count the number of different types of functions from A to B:

- *Total functions:* There are m^n total functions from A to B. Each of the n elements in A can be mapped to any of the m elements in B.
- *Injective functions:* If $n \leq m$, the number of injective functions is given by $m \times (m-1) \times (m-2) \times \ldots \times (m-n+1)$. This is because the first element in A has m choices, the second has $m-1$ choices, and so on.
- *Surjective functions:* The number of surjective functions can be calculated using the principle of inclusion-exclusion:

$$\text{Number of surjective functions} = \sum_{k=0}^{m} (-1)^k \binom{m}{k} (m-k)^n$$

  This formula counts all functions and subtracts those that miss at least one element in B.
- *Bijective functions:* If $n = m$, the number of bijective functions (one-to-one and onto) is $n!$ (n factorial), which counts all possible permutations of the n elements in A to the n elements in B.

3.27 Counting Relations

Given two sets A and B with $|A| = n$ and $|B| = m$, we can count the number of different types of relations from A to B:

- *Total relations:* There are $2^{n \times m}$ total relations from A to B. Each of the $n \times m$ possible ordered pairs (a, b) can either be included in or excluded from the relation. This comes from the cardinality of the power set of the Cartesian product of $A \times B$.

- *Reflexive relations:* If $A = B$, the number of reflexive relations is 2^{n^2-n}. This is because the n diagonal pairs (a, a) must be included, while the remaining $n^2 - n$ pairs can be chosen freely.
- *Symmetric relations:* If $A = B$, the number of symmetric relations is $2^{\frac{n(n+1)}{2}}$. This counts the diagonal pairs and half of the off-diagonal pairs, as each off-diagonal pair (a, b) must be paired with (b, a).
- *Antisymmetric relations:* If $A = B$, the number of antisymmetric relations is $2^{n^2-\frac{n(n-1)}{2}}$. This counts all pairs except for those that would violate antisymmetry.
- *Transitive relations:* Counting transitive relations is more complex and does not have a simple closed-form formula. It often requires combinatorial methods or recursive counting techniques.

Chapter 4

Mathematical Proofs

In this chapter, I will provide both definitions and examples of different types of proofs for the reader to gain an overview of how to use the different proving methods.

4.1 Nomenclature

In this book I do not really used a lot of the formal mathematical nomenclature found in academic texts like: theorem, lemma, etc. with the exception of the QED symbol to denote the end of a proof and really important theorems like the Fundamental Theorem of Calculus. Yet, it is important to understand these formal terms.

- *Theorem*: A "big" result that is considered important and is usually proven using a formal argument.
- *Lemma*: A "small" result that is usually proven to help in proving a theorem.
- *Corollary*: A result that follows directly from a theorem or a lemma.
- *Proposition*: A result that is considered less important than a theorem but still worth stating formally.

4.2 Vacouly True

As a reminder, a stement $p \implies q$ is *vacuously true* if p is false. In some cases we can arrive at interesing results due to this phenomenon.

Example:

Take the definition of the intersection of a set system

$$\bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}.$$

Now let $I = \emptyset$. Then the intersection over an empty index set is

$$\bigcap_{i \in \emptyset} A_i = \{x \mid x \in A_i \text{ for all } i \in \emptyset\}.$$

Since there are no elements $i \in \emptyset$, the condition "for all $i \in \emptyset$" is vacuously true for any x. Therefore,

$$\bigcap_{i \in \emptyset} A_i = \{x \mid \text{true}\} = \text{the universal set}.$$

$\mathfrak{QED}$

4.3 Trivial Proofs

A proof is *trivial* if for

$$\forall x \in X, P(x) \implies Q(x)$$

we can find a simpler statement $\forall x \in X, Q(x)$ which is true. In other words, an implication is trivially true when the conclusion of the implicationis always true.

4.4 Proof by Direct Argument (Direct Proof)

Assume A to prove B.

$$A \implies B$$

Example I:

For any integer n, if n is even, then n^2 is even. We will prove this theorem by direct argument.

Assume n is an even integer. Then we can write $n = 2k$ for some integer k.

Now, we compute n^2:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

This shows that n^2 is even, as it can be expressed as $2m$ where $m = 2k^2$ is an integer. Therefore, we conclude that if n is even, then n^2 is even.

$\mathfrak{QED}$

Example:

Let $z \in \mathbb{Z}$ be odd. It can be written as the difference of the squares of $a, b \in \mathbb{N}_0$. First some examples:

$$1 = 1^2 - 0^2, \quad 3 = 2^2 - 1^2, \quad 5 = 3^2 - 2^2, \quad 7 = 4^2 - 3^2, \quad \ldots$$

We know that any odd integer has the form

$$z = 2m \pm 1 \quad \text{for some } m \in \mathbb{Z}.$$

and we have noticed the pattern $b = a - 1$. Now let us assume that every odd integer can be written as $a^2 - b^2$.

$$\begin{aligned} z &= a^2 - b^2 \\ &= a^2 - (a-1)^2 \\ &= a^2 - (a^2 - 2a + 1) \\ &= 2a - 1 \in \mathbb{Z}. \end{aligned}$$

$\mathfrak{QED}$

4.5 Proof by Contrapositive

The contrapositive of an implication is logically equivalent to the original implication:

$$A \implies B \iff \neg B \implies \neg A.$$

Example:

If n is an integer such that n^2 is odd, then n is odd.

We will prove this theorem by contrapositive. The contrapositive of the statement is: If n is an integer such that n is even, then n^2 is even.

Assume n is even. Then we can write $n = 2k$ for some integer k.

Now, we compute n^2:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

This shows that n^2 is even.

Since the contrapositive statement is true, the original statement, "If n^2 is odd, then n is odd," is also true.

$\mathfrak{QED}$

4.6 Proof by Contradiction

Assume the negation of A to derive a false statement.

$$A \implies B \iff \neg A \implies F$$

Example I:

If n is an integer such that n^2 is even, then n is even.

We will prove this theorem by contradiction. Assume that n is an integer such that n^2 is even, but n is odd. Then we can write $n = 2k + 1$ for some integer k.

Now, we compute n^2:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

This shows that n^2 is odd, which contradicts our assumption that n^2 is even. Therefore, our assumption that n is odd must be false, and thus, n must be even.

$\mathfrak{QED}$

Example II:

We will prove that there are infinitely many prime numbers by assuming that there is a biggest prime number p_n. Let us then create a new number

$$p_1 \cdot \cdots \cdot p_n + 1 = a$$

Note that a like any other integer has to be a multiple of some prime which means $p_k c = a$. If we divide both sides by p_k we get

$$(p_1 \cdot \cdots \cdot p_{k-1} \cdot p_{k+1} \cdot \cdots \cdot p_n) + \frac{1}{p_k} = c.$$

On the left we have a rational whereas on the right we have an integer which is contradiction. Therefore, there are infinitely many prime numbers.

$\mathfrak{QED}$

Example III:

Suppose, for contradiction, that $\sqrt{2}$ is rational. Then there exist integers p and q with $q \neq 0$, $\gcd(p, q) = 1$, and

$$\sqrt{2} = \frac{p}{q}.$$

Squaring both sides gives

$$2 = \frac{p^2}{q^2} \implies p^2 = 2q^2.$$

Thus, p^2 is even, so p is even. Write $p = 2k$ for some integer k. Then

$$(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies q^2 = 2k^2,$$

so q^2 is even and hence q is even. Therefore, both p and q are even, contradicting $\gcd(p, q) = 1$. This contradiction shows $\sqrt{2}$ is irrational.

$\mathfrak{QED}$

4.7 If and only Proofs

Given two statements A and B, the statement $A \iff B$ means that A is true if and only if B is true. This is equivalent to proving both

$$A \iff B \equiv A \implies B \text{ and } B \implies A.$$

Note that we can use techniques like: direct, contrapositive and contradiction proofs for both directions of the implication.

Example

Let x be a odd integer. We want to prove that x^2 is also odd if and only if x is odd.

Proof:

($\Rightarrow$) Assume x is odd. Then we can write $x = 2k + 1$ for some integer k. Now, we compute x^2:

$$x^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which shows that x^2 is odd.

($\Leftarrow$) Assume x^2 is odd. Suppose, for contradiction, that x is even. Then $x = 2k$ for some integer k, and

$$x^2 = (2k)^2 = 4k^2 = 2(2k^2),$$

which is even, contradicting our assumption that x^2 is odd. Therefore, x must be odd.

$\mathfrak{QED}$

4.8 Proof by Full Induction

For all $n \in \mathbb{N}$, $A(n)$ is a statement. Then the following holds

- *Base case*: Is $A(1)$ true and,
- *Inductive Hypothesis*: We assume that $A(n)$ is true for some arbitrary $n \in \mathbb{N}$.
- *Inductive step*: $A(n) \implies A(n+1)$ for all $n \in \mathbb{N}$

Then $A(n)$ is true for all $n \in \mathbb{N}$.

Proof:

Let

$$M := \{n \in \mathbb{N} \mid A(n) \text{ holds } \}.$$

We need to show that $M = \mathbb{N}$.

In our first step we assume that $A(1)$ is true, then we know $1 \in M$. We need to prove the inductive step, which is done by using the follwing axiom $1 \in M \implies 1' \in M$, hence by the last Peano axiom

$$M = \mathbb{N}.$$

$\mathfrak{QED}$

Remark: Most induction proofs follow the same structure during the inductive step in which we can just substitute one part of the expression with our inductive hypothesis, **but** sometimes this wield yield no progress and way need to use extra information and also consider other manipulations to our expresssion to derive at a useful conclusion; like for example the *Bernoulli inequality*.

Another trick is to start from the final result we want to get and then work backwards to arrive at some conclusion.

Example:

For all $n \in \mathbb{N}$, the sum of the first n positive integers is given by:

$$\sum_{i=1}^{n} i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

We will prove this theorem by induction on n.

Base Case: For $n = 1$:

$$\sum_{i=1}^{n} i = 1 = \frac{1(1+1)}{2}.$$

The base case holds.

Inductive Hypothesis: Let

$$\sum_{i=1}^{n} i = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

be true.

Inductive Step: Let us prove the stement for $n + 1$

$$\sum_{i=1}^{n+1} i = 1 + 2 + 3 + \cdots + n + 1 = \frac{(n+1)(n+2)}{2}.$$

By the inductive hypothesis, we have:

$$\begin{aligned}\sum_{i=1}^{n+1} i &= \sum_{i=1}^{n} i + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n(n+1) + 2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2}\end{aligned}$$

Thus, we have prooven the statement for $n+1$, completing the inductive step. By the principle of mathematical induction, the statement holds for all $n \in \mathbb{N}$.

$\mathfrak{QED}$

4.9 Strong Induction

Strong induction is a form of mathematical induction where instead of assuming $A(n)$ is true in our inductive hypothesis, we assume $A(j)$ is true for $1 \leq j \leq n$.

The inductive step: Show $[A(1) \wedge A(2) \ldots A(n)] \implies A(n+1)$.

Sometimes, we will prove specific values are true for $1 \leq j \leq n$, and sometimes we just assume that $A(j)$ is true for $1 \leq j \leq n$.

Example:

Prove that any integer ≥ 1 can be written as the product of primes.

Let $A(n)$ be that if n is an integer greater than 1, then n can be written as the product of primes.

Basis: $A(2)$ is true because 2 is prime.

Inductive hypothesis: $A(j)$ for $2 \leq j \leq n$ is true. **Important!**

Consider the cases where $k+1$ is prime or composite.

The first case is true.

For the case where $k+1$ is not prime, we have $k+1 = ab$ with $2 \leq a \leq b < k+1$, and for both a and b we know that they can be written as the product of primes; thus, $k+1$ can be written as the product of primes.

$\mathfrak{QED}$

4.10 Proof by Cases (Exhaustion)

In *exhaustion*, we test all possible cases and their respective branches. This kinds of proofs tend to be lengthy and tedious, especially when the number of cases is large.

Example:

The only integer solutions to the equation $x^2 + y^2 = 1$ are $(0,1), (1,0), (0,-1), (-1,0)$.

We will prove this theorem by exhaustion. We will check all possible integer values of x and y such that $x^2 + y^2 = 1$.

The possible integer values for x and y are $-1, 0, 1$. We will check each case:

If $x = 0$:

Then $y^2 = 1$ gives $y = 1$ or $y = -1$.

Solutions: $(0,1),(0,-1)$.

If $x = 1$:

Then $y^2 = 0$ gives $y = 0$.

Solution: $(1,0)$.

If $x = -1$:

Then $y^2 = 0$ gives $y = 0$.

Solution: $(-1,0)$.

Thus, the only integer solutions to the equation are:

$$(0,1),(1,0),(0,-1),(-1,0).$$

QED

4.11 Proof by Counter-Example

With a *counter-example*, it is enough to find one case in which a statement is not true to prove the statement false, and in some cases to prove that the opposite is true.

Example:

The statement "all prime numbers are odd" is false.

To prove this theorem, we will provide a counterexample.

The number 2 is a prime number, as its only divisors are 1 and 2. However, 2 is even, which contradicts the statement that all prime numbers are odd.

Therefore, the statement "all prime numbers are odd" is false.

QED

4.12 Mathematical Writting

As mathematicians we must not only obey the respective language of our field as well as the grammar and punctuation rules of our writing tongue, but also it is essential to follow conventions and formulate our sentences to be as clear and free from ambiguity as possible.

For example, look at the following sentences:

- "The set A is empty." is good in contrast to "A is empty", what is A
- "Let $x \in \mathbb{R}$ be such that $x^2 = 4$." is good in contrast to "$x^2 = 4$", what is x?

Other rules which apply are

- Punctuation at the end of equations which are part of a sentence.
- Do not mix different notations for the same mathematical object. This is confusing.
- Use the first person plural "we" when writing mathematical proofs, as it is the convention in mathematical writing. This creates a connection between the author and the reader, making the exposition more engaging and easier to follow.
- Use the active voice.
- Use "thus", "hence", "therefore" to indicate logical conclusions.

4.13 Combining Techniques

For more complex theorems somtimes a single proof technique is not enough, and we may need to combine multiple of them.

Common patterns include:

- Combining direct proof and proof by contradiction: Sometimes we can start with a direct proof and then use contradiction to handle a specific case.
- Combining direct proof and proof by induction: We may use a direct argument for the base case and then apply induction for the general case.
- Combining proof by contrapositive and proof by contradiction: In some situations, proving the contrapositive can simplify the argument, and contradiction can be used for the remaining cases.
- Contraposite for if and only if proofs where a direction is easier to prove by considering the contrapositive. This can simplify the argument for one direction, while the other direction may be handled directly or by another technique.
- Negation of the if and only if statements.

Chapter 5

Order

5.1 Partially Ordered Classes

A *partially ordered class* is a pair of objects $\langle A, R\rangle$ where A is a class and R is an order relation in A. Remember that an order relation is one which is: reflexive, antisymmetric, and transitive.

5.2 Lexicographical Ordering

Let A and B be partially ordered classes; by the *lexicographical ordering* of $A \times B$ we mean the following order relation in $A \times B$: If $(a_1, b_1), (a_2, b_2) \in A \times B$ and $(a_1, b_1) \leq (a_2, b_2)$ if and only if

$$a_1 < a_2$$

or

$$(a_1 = a_2) \wedge (b_1 \leq b_2).$$

5.3 Antilexicographical Ordering

Let A and B be partially ordered classes; by the *antilexicographical ordering* of $A \times B$ we mean the following order relation in $A \times B$: If $(a_1, b_1), (a_2, b_2) \in A \times B$ and $(a_1, b_1) \leq (a_2, b_2)$ if and only if

$$b_1 < b_2$$

or

$$(b_1 = b_2) \wedge (a_1 \leq a_2).$$

5.4 Comparable

Let A be a partially ordered class. Two elements $x, y \in A$ are considered *comparable* if either $x \leq y$ or $y \leq x$; otherwise, they are said to be *incomparable*.

5.5 Fully Ordered Subclass

Let A be a partially ordered class, and let $B \subseteq A$. If two elements of B are comparable, then we call B a *fully ordered subclass of* A, or *linearly ordered subclass of* A, or more commonly, a *chain* of A. If every two elements of A are comparable, then A is also considered as such.

Example:

Let C and D be chains of A and B, respectively; prove that $A \times B$ lex. sorted $\implies$ $C \times D$ is chain of $A \times B$.

Proof:

By the contrapositive, if $C \times D$ is not a chain of $A \times B$, then there is some element (c, d) that cannot be compared, but $(c, d) \in A \times B$, so $A \times B$ cannot be lexicographically sorted.

$\mathfrak{QED}$

5.6 Initial Segments

Let A be a partially ordered class and suppose $a \in A$. The *initial segment of A determined by a* is the class S_a, defined as follows:

$$S_a = \{x \in A \mid x < a\}.$$

Initial segments are transitive, which means that if we have an initial segment of an initial segment, then the last one is also an initial segment of the parent class.

An Initial Segment of an Initial Segment of a Class Is an Initial Segment of the Same Class

Proof:

Let $P = \{x \in A \mid x < a\}$ for some $a \in A$ and let $Q = \{p \in P \mid p < c\}$ for some $c \in P$. By definition $Q \subseteq A$. Because $c \in A$ it is clear that Q is an initial segment of A.

$\mathfrak{QED}$

5.7 Cuts

If A is a partially ordered class, then a *cut* of A is a pair (L, U) of nonempty subclasses of A with the following properties:

- $L \cup U = \emptyset \wedge L \cup U = A$.
- If $x \in L \wedge y \leq x \implies y \in L$.
- If $x \in U \wedge y \geq x \implies y \in U$.

The last two conditions simply tell us that we can compare its elements.

Example:

Given the partially ordered classes A and B, let $A \times B$ be ordered antilexicographically. Prove that (L, U) is a cut of $B \implies (A \times L, A \times U)$ is a cut of $A \times B$.

Proof:

Let (L, U) be a cut of B and define

$$L' = A \times L, \qquad U' = A \times U.$$

We show (L', U') is a cut of $A \times B$ under the antilexicographical order.

Nonempty, disjoint, and covering:

Suppose $L' \cap U' \neq \emptyset$. Then,

$$\exists (x, y) \in L' \cap U' : x \in A \wedge (y \in L \vee y \in U).$$

Hence, $L \cap U \neq \emptyset$, a contradiction. Therefore, they must be disjoint.

For the covering, we use the distributive property of the Cartesian product under set union, $A \times (L \cup U) = A \times L \cup A \times U = A \times B$.

Downward closure of L':

Let $(a_1, b_1) \in A \times L$ and $(a_2, b_2) \leq (a_1, b_1)$.

By antilexicographical order,

$$b_2 < b_1 \text{ or } (b_2 = b_1 \wedge a_2 \leq a_1).$$

Since $b_1 \in L$ and L is downward closed in B, we have $b_2 \in L$. Hence $(a_2, b_2) \in A \times L$.

Upward closure of U':

Let $(a_1, b_1) \in A \times U$ and $(a_1, b_1) \leq (a_2, b_2)$. Then

$$b_1 < b_2 \quad \text{or} \quad (b_1 = b_2 \wedge a_1 \leq a_2).$$

Since $b_1 \in U$ and U is upward closed in B, we have $b_2 \in U$. Thus $(a_2, b_2) \in A \times U$.

Therefore $(A \times L, A \times U)$ is a cut of $A \times B$.

$\mathfrak{QED}$

5.8 Order Preserving Functions

Let A and B be partially ordered classes; a function $f : A \to B$ is said to be *increasing* or *decreasing* if it satisfies the following condition: For every two elements $x, y \in A$,

$$x \leq y \implies f(x) \leq f(y).$$

5.8.1 Isomorphism

If f is bijective and $x \leq y \iff f(x) \leq f(y)$ for every two elements $x \in A, y \in B$, we call it an *isomorphism*.

5.8.2 Strictly Increasing Function

We consider f strictly increasing if it satisfies that:

$$x < y \implies f(x) < f(y).$$

Example:

Prove that if $f : A \to B$ is an injective, increasing function, then it is strictly increasing.

Proof:

Suppose it is not strictly increasing. The negation is $x < y \implies f(x) \geq f(y)$.

Now we have that $x \leq y \implies f(x) \geq f(y) \wedge x \leq y \implies f(x) \leq f(y) \iff f(x) = f(y)$ for some $x, y \in A$, hence we have a contradiction to the injectivity.

$\mathfrak{QED}$

5.9 Order by Inverses

Let A and B be partially ordered classes and let $f : A \to B$ be a bijective function. Then f is an isomorphism if and only if $f : A \to B$ and $f^{-1} : B \to A$ are increasing functions.

Proof:

$$f(x) \leq f(y) \implies f^{-1}(f(x)) \leq f^{-1}(f(y)) \implies x \leq y$$

By the isomorphic property we get the increasing property for f and also for f^{-1} because else for $f(x) \leq f(y) : y \leq x$ violates our initial condition that $x \leq y$.

$$x \leq y \implies f(x) \leq f(y) \implies f^{-1}(f(x)) \leq f^{-1}(f(y))$$

Thus, f and f^{-1} are increasing functions.

$\mathfrak{QED}$

5.10 Common Isomorphism on Partially Ordered Classes

Let A, B and C be partially ordered classes.

- $I_A : A \to A$ with $I_A(x) = x, \quad x \in A$ is an isomorphism.
- If $f : A \to B$ is an isomorphism, then $f^{-1} : B \to A$ is an isomorphism.
- If $f : A \to B$ and $g : B \to C$ are isomorphisms, then $g \circ f : A \to C$ is an isomorphism.

Proof I:

Let $I_A = f$, for $x, y \in A$ we have that $x \leq y$ gives us $x < y \iff f(x) < f(y)$ and $x = y \iff f(x) = f(y)$.

$\mathfrak{QED}$

Proof II:

In one of our previous results we got that f^{-1} is increasing if and only f is an isomorphism. We prove that f^{-1} is also one by assuming it is not, then $\exists y \in B : f(x) \neq y, \forall x \in A$ which is a contradiction.

$\mathfrak{QED}$

Proof III:

We keep in mind the bijectivity of f and g and state the following

$$x \leq y \iff f(x) \leq f(y) \iff g(f(x)) \iff (f(y))$$

$\mathfrak{QED}$

Example:

Given the isomorphism $f : A \to B$, if (L, U) is a cut of A, then $(f(L), f(U))$ is a cut of B.

Proof:

Suppose $(f(L), f(U))$ is not a cut, then $y \in f(L) \cap f(U)$, then $y \in f^{-1}(f(L)) \cap f^{-1}(f(U))$ but this is a contradiction. Therefore, $(f(L), f(U))$ must be disjoint.

For the covering, we use a similar argument $f(L) \cup f(U) \neq B$, then f would not be bijective, hence, another contradiction.

For the downward and upward closure, we know that our function targets the whole set B and by its isomorphic properties it respects the order. Therefore, these properties are also satisfied.

$\mathfrak{QED}$

5.11 Isomorphic Definition

Let A and B be partially ordered classes. If there exists an order preserving isomorphism between the two classes, we say that A is *isomorphic* with B.

Note that the isomorphic property allows us to define an equivalence relation between classes.

5.12 Convex

Let A be a partially ordered class and $C \subseteq A$. If for $a \in C$ and $b \in C$ the following holds $a \leq x \leq b, \forall x \in C$, then we consider C *convex*.

Convex Properties

Let A and B be partially ordered classes, then

- If $C \subseteq B$ is convex and $f : A \to B$ is an increasing function, then $f^{-1}(C)$ is also convex.
- If H is the functional equivalence relation, and $f : A \to B$ is an increasing function, then all equivalence classes of A/H are convex subsets of A.

Proof I:

Given $f(a), f(b) \in C$, then using the properties of the increasing functions we get $f^{-1}(f(a)) \leq f^{-1}(f(x)) \leq f^{-1}(f(b))$ which gives us $a \leq x \leq b$ which is the definition of a convex class.

$\mathfrak{QED}$

Proof II:

Given the equivalence relation given by the function f we have that for every member of an arbitrary equivalence class $f(a) = f(x) = f(b)$ can be written as $f(a) \leq f(x) \leq f(b)$, hence we get the definition of convex classes.

$\mathfrak{QED}$

5.13 Maximal Element

An element $m \in A$ is considered a *maximal element* of A under the partial order relation R if $\forall x \in A$

$$x \leq m$$

and if

$$x \geq m \implies x = m.$$

Or equivalently m under the relation rule does not relate to anything except itself:

$$\forall (m, y) \in R \implies m = y \text{ or equivalently } (x, m) \in R, \forall x \in A.$$

5.14 Minimal Element

An element $m \in A$ is considered a *minimal element* of A under the partial order relation R if $\forall x \in A$

$$x \geq m$$

and if

$$x \leq m \implies x = m.$$

Or m under the relation rule does not relate to anything except itself:

$$\forall (x, m) \in R \implies m = x \text{ or equivalently } (m, x) \in R, \forall x \in A.$$

5.15 Upper Bound

Let $B \subseteq A$. An *upper bound* of B in A is an element $a \in A$ such that $a \geq x, \forall x \in B$. Equivalently

$$\forall x \in B, (x, a) \in R.$$

5.15.1 Set of Upper Bounds

Let $B \subset A$, we define the function v as

$$v(B) := \{a \in A \mid a \geq x, \forall x \in B\}.$$

5.16 Lower Bound

Let $B \subseteq A$. A *lower bound* of B in A is an element $a \in A$ such that $a \leq x, \forall x \in B$. Equivalently

$$\forall x \in B, (a, x) \in R.$$

5.16.1 Set of Lower Bounds

Let $B \subset A$, we define the function λ as

$$\lambda(B) := \{a \in A \mid a \leq x, \forall x \in B\}.$$

5.17 Maximum

An element x in a partially ordered class A is a *maximum* if

$$(a, x) \in R, \forall a \in A.$$

5.18 Minimum

An element x in a partially ordered class A is a *minimum* if

$$(x, a) \in R, \forall a \in A.$$

5.19 Supremum and Infimum

Given a partially ordered class $B \subseteq A$, the least upper bound is called the *supremum* $s \in A$ and defined as:

$$\forall b \in B, (b, s) \in R \text{ and } \forall a \in A, [(\forall b \in B, (b, a) \in R) \implies (s, a) \in R],$$

and the greatest lower bound is its *infimum* $i \in A$ is defined as:

$$\forall b \in B, (i, b) \in R \text{ and } \forall a \in A[(\forall b \in B, (a, b) \in R) \implies (a, i) \in R].$$

5.20 Duality

Let $\mathscr{S}$ be a statement about partially ordered classes with $\geq, \leq$. If we replace the occurrences of $\geq, \leq$ with their contrary, then the resulting statement $\mathscr{S}'$ is called the *dual* of $\mathscr{S}$.

We know that given an order relation G on A, then G^{-1} is also an order relation on A. For the dual this means that for any true statement on G the dual of the statement is also true for G^{-1}.

5.21 Greatest and Least Elements Under Inclusion

If A has a greatest element a, and B has a greatest element b, and $B \subseteq A$, then

$$b \leq a.$$

Proof:

$$a \geq x, \forall x \in A \implies a \geq b, \forall b \in B \implies a \geq b$$

$\mathfrak{QED}$

Dual: If A has a least element a, and B has a least element b, and $B \subseteq A$, then

$$b \geq a.$$

5.21.1 Upper and Lower Bounds Under Inclusion

Let B and C be subclasses of A. If $B \subseteq C$, then

$$v(C) \subseteq v(B).$$

Proof:

$x \in v(C) \implies x \geq y,\ \forall y \in C$ because $y \in B \implies b \in C$, then $x \geq y, \forall y \in B \implies x \in v(B)$.

The opposite is not always true, due to the fact that if $z \in v(B)$, such that $z \in C$, but $z \notin v(C)$, which is possible if z is not an upper bound of C, then the theorem does not hold.

$\mathfrak{QED}$

Dual: If $B \subseteq C$, then

$$\lambda(C) \subseteq \lambda(B).$$

5.21.2 Suprema and Infima Under Inclusion

Let B and C be subclasses of A, and suppose that B and C each has a sup in A. If $B \subseteq C$, then

$$\sup B \leq \sup C.$$

Proof:

By the dual of the greatest element under inclusion applied to $v(C) \subseteq v(B)$,

$$\sup(B) \leq \sup(C).$$

$\mathfrak{QED}$

Dual: If B and C each has an inf in A, and if $B \subseteq C$, then

$$\inf B \geq \inf C.$$

5.21.3 Relation Between a Set and Bounds of Bounds

Let B be a subclass of A. Then

$$B \subseteq v(\lambda(B)).$$

Proof: Suppose $x \in B$; for each $y \in \lambda(B)$, $y \leq x$; hence $x \in v(\lambda(B))$.

$\mathfrak{QED}$

Dual:

$$B \subseteq \lambda(v(B)).$$

5.22 Infimum via Lambda Function

Let $B \subseteq A$, then

$$\sup(\lambda(B)) \in A \implies \inf(B) \in A \wedge \inf(B) = \sup(\lambda(B))$$

Dual: $\inf(v(B)) \in A \implies \sup(B) \in A \wedge \sup(B) = \inf(v(B))$.

Proof:

Let $s = \sup(\lambda(B))$, $\forall b \in B$, $s \leq b$; hence, s is a lower bound of B. Suppose $\inf(B) \neq s$, then $s > \inf(B) \vee s < \inf(B)$.

Case $s > \inf(B)$: this leads to a contradiction because if $s \in A$ and s is a lower bound of B, but it is greater than its infimum, then $\inf(B)$ cannot be the greatest lower bound of B, which does not make sense by the definition.

Case $s < \inf(B)$: $\inf(B)$ is a lower bound of B; therefore, $\inf(B) \in \lambda(B)$. If $s < \inf(B)$, then s cannot be the supremum of $\lambda(B)$ since there exists a greater element in it; hence $s = \inf(B)$.

$\mathfrak{QED}$

5.23 Example Exercises

1. Prove that given $B \subseteq A$, $\sup(B) = b \in A \implies \lambda(v(B)) \cap v(B) = \{b\}$.

Proof:

By definition $b \in v(B)$. Also, by the definition of $\lambda(v(B))$:

$$\{a \in A \mid x \leq b, \forall b \in v(B)\};$$

we can guarantee that $b \in \lambda(v(B))$. It remains to prove that the intersection only consists of b.

Suppose that there also exists another x in the intersection such that $x \neq b$, then $x \in v(B) \wedge x \in \lambda(v(B))$.

If $x \in v(B)$ and $x \neq b$, then $x < b$ or $x > b$ which when handled separately lead to the following conclusion:

If $x < b$, then b is not the least upper bound of B which is a contradiction to the definition of the supremum. Hence, by the $\wedge$ in our original statement, we arrive at the conclusion that such an x cannot exist.

$\mathfrak{QED}$

2. Let $B \subseteq A$, $C \subseteq A$, then $\sup(B) = \sup(C) \implies v(B) = v(C)$.

Proof:

By definition for $\sup(B) \geq b, \forall b \in B$, and also $\sup(B) \in v(B)$, but because it is the least upper bound we can write

$$v(B) = \{a \in A \mid a \geq \sup(B)\}.$$

Therefore, if $\sup(B) = \sup(C) = s$, then $v(B) = \{a \in A \mid a \geq s\}$, and because they are shared, then $v(C) = \{a \in A \mid a \geq s\}$.

$\mathfrak{QED}$

3. Suppose $B \subseteq A$; prove that $v(B) = v(\lambda(v(B)))$.

For $x \in v(B)$ holds that $x \geq b, \forall b \in B$, then for $y \in \lambda(v(B))$ holds $y \leq z, \forall z \in v(B)$ which implies that $y \leq b, \forall b \in B$. Finally, if $y \leq b, b \in B$, then for $w \in v(\lambda(v(B)))$ it must be true that $w \geq y, \forall y \in \lambda(v(B))$, which by transitivity implies that $w \geq b, \forall b \in B$; hence, $v(B) = v(\lambda(v(B)))$.

$\mathfrak{QED}$

5.24 Bounded

Let A be a partially ordered class. If A has both a lower and an upper bound, then it is *bounded*.

5.25 Conditionally Complete

Let A be a partially ordered class. If every non-empty subclass of A that is bounded has a sup, then A is said to be *conditionally complete*.

5.26 Lattices

Let A be a partially ordered class. If every doubleton $\{x, y\}$ in A has a sup and inf then A is called a *lattice*.

When dealing with lattices, we denote $\sup(\{x, y\}) := x \vee y$ and $\inf(\{x, y\}) := x \wedge y$.

5.26.1 Properties of Lattices

For $a, b \in A$ the following properties apply

1. $a \leq a \vee b$ and $b \leq a \vee b$
2. $a \geq a \wedge b$ and $b \geq a \wedge b$
3. If $a \leq c$ and $b \leq c$, then $a \vee b \leq c$
4. If $a \geq c$ and $b \geq c$, then $a \wedge b \geq c$
5. $a \vee a = a$ and $a \wedge a = a$
6. $a \vee b = b \vee a$ and $a \wedge b = b \wedge a$
7. $(a \vee b) \vee c = a \vee (b \vee c)$ and $(a \wedge b) \wedge c = a \vee (b \wedge c)$
8. $(a \vee b) \wedge a = a$ and $(a \wedge b) \vee a = a$

Proof 3:

To prove: $a \vee a = a$ and $a \wedge a = a$.

$a \vee a \geq x, \forall x \in \{a\}$. Using the definition of the infimum we can also prove the statement for infimum.

The other statements are easily proved by cases.

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5.27 Definition of Less Than

Let A be a *lattice*; we define the symbol $\leq$ as:

$$x \leq y \iff x \vee y = y$$

Example I:

Given a lattice A with $\leq$, prove that it is an order relation.

Reflexivity: $(a, a) \in R$ because $a \leq a$ implies $a \vee a = a$.

Antisymmetry: $a \leq b \implies a \vee b = b$ if by supposition $b \leq a$, then $b \vee a = a$, but our property tells us that $a \vee b = b \vee a$; therefore, $a = b$.

Transitivity: $(a, b), (b, z) \in R \implies (a, z) \in R$, if $a \vee b = b$ and $b \vee z = z$, then $a \leq b$ and $a \leq z$, hence; $a \leq b \vee z = z \implies (a, z) \in R$.

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Example II:

Given $a \leq b$ and $c \leq d$, prove that $a \vee c \leq b \vee d$.

Proceeding by cases, note that: $a \geq c$ delivers $a \vee b \vee d$. If $b \geq c$, then $a \leq b$; if $d \geq b$ then $a \leq d$ by transitivity. Similarly, for the case $a \vee c = c$, then if $b \geq d$ gives us $c \leq b$; else if $b \leq d$ then $c \leq b \leq d$.

ΩΕΔ

5.27.1 Distributive Laws for Lattices

- $x \wedge (y \vee z) \geq (x \wedge y) \vee (x \wedge z)$.
- $(x \vee y) \wedge z \leq (x \vee y) \wedge (x \vee z)$.

5.28 Sublattices

Let A be a lattice, and let B be a subclass of A. If

$$x \in B \text{ and } y \in B \implies x \vee y \in B \text{ and } x \wedge y \in B,$$

then B is called a *sublattice* of A.

5.29 Boolean Algebra

A *Boolean algebra* is defined to be a lattice A with the following additional properties:

1. $\exists 0 \in A$ and $1 \in A$ such that $x \vee 0 = x$ and $x \wedge 1 = x$.
2. $\forall x \in A, \exists \overline{x} \in A : x \wedge \overline{x} = 0$ and $x \vee \overline{x} = 1$.
3. $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$ and $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ Let A be a partially ordered class; A is called a *complete lattice* if every subclass of A has a sup. Alternatively, if every subclass of A has an inf.

In other words, 1 is the biggest and 0 the smallest element in A.

5.29.1 Distributive Lattices

A *distributive lattice* is a Boolean algebra.

5.29.2 Fully Ordered Classes

5.30 Principle of the Well-Ordering

Every non-empty set of natural numbers has a minimum.

5.31 Order in Fields

An *ordered field* is a field $\mathbb{F}$ equipped with a total order $<$ that is compatible with the field operations $+$ and $\cdot$. That is, the order satisfies both algebraic and ordering properties.

5.31.1 Order Axioms for Fields

A field F is called an *ordered field* if it satisfies the following properties for all $a, b, c \in F$:

I. *Trichotomy:* Exactly one of the following holds:

$$a < b, \quad a = b, \quad a > b$$

II. *Transitivity:* If $a < b$ and $b < c$, then $a < c$.

III. *Additive Compatibility:* If $a < b$, then $a + c < b + c$.

IV. *Multiplicative Compatibility:* If $0 < a$ and $0 < b$, then $0 < ab$.

These properties ensure that arithmetic operations respect the order structure.

5.31.2 Consequences of the Order Axioms

- $a < b \implies -b < -a$
- $a < b \land c < 0 \implies ac > bc$
- $a < 0, b < 0 \implies a + b < 0$ the contrary applies to the multiplication $ab > 0$.
- $a > 0, b > 0 \implies a + b > 0$
- Squares are always non-negative: $a^2 \geq 0$.
- The order is total: any two elements are comparable.

Examples of ordered fields:

- $\mathbb{Q}$: Rational numbers with the usual order.
- $\mathbb{R}$: Real numbers with the usual order.
- $\mathbb{C}$: Complex numbers are not an ordered field, since $i^2 = -1 < 0$ would violate positivity of squares.

Chapter 6

The Natural Numbers

6.1 Definition as Sets

We define for the members of $\mathbb{N}$ the following properties:

- $0 := \emptyset$
- $1 := \{0\} = \{\emptyset, \{\emptyset\}\}$
- $2 := \{0, 1\} = \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$
- $n := \{n_1, \ldots, 0\}$

6.1.1 Addition

$$S(n) := n \cup n := n + 1$$

6.1.2 Inductive Sets

A set I is inductive if

$$0 \in I, x \in I \implies S(x) \in I.$$

6.1.3 Axiom of Infinity

An inductive set exists.

6.1.4 Definition of the Set of Natural Numbers

$$\mathbb{N} := \{x \mid x \in I, \text{ for every inductive set } \}.$$

Proof that $\mathbb{N}$ is inductive for every inductive set $I, \mathbb{N} \subseteq I$:

$\forall I, \emptyset \in I$, hence $0 \in \mathbb{N}$. For $n \in \mathbb{N}$, let I be inductive, then

$$S(n) \in I \therefore S(n) \in \mathbb{N}.$$

6.2 The Standard Order

We define $<$ for $\mathbb{N}$ as:

$$\forall m, n \in \mathbb{N}, m < n \iff m \in n.$$

Lemma

- $0 < n, \forall n \in \mathbb{N}$
- $\forall k, n \in \mathbb{N}, k < S(n) \iff k < n \vee k = n$

Proof:

Suppose for the sake of contradiction that $\exists n \in \mathbb{N}, n \neq 0 : n < 0$. By definition $n < 0 \iff n \in 0 = \emptyset$. Hence, we arrive at our desired contradiction.

For the second part of the lemma $k < S(n) \iff k < n \vee k = n$:

$$\begin{aligned} k < S(n) \implies k \in \{n\} \cup n &\iff k \in n \vee k = n \\ &\iff k < n \vee k = n \\ &\iff k < n \end{aligned}$$

$\mathfrak{QED}$

6.3 The Peano Axioms

We will define the set of natural numbers, $\mathbb{N}$, via the following 9 axioms. These axioms are known as the *Peano Axioms*. The first 4 axioms define equality on the set $\mathbb{N}$.

I. *Axiom of Reflexivity*: For every $x \in \mathbb{N}$, we have $x = x$.

II. *Axiom of Symmetry*: For every $x, y \in \mathbb{N}$, if $x = y$ then $y = x$.

III. *Axiom of Transitivity*: For every $x, y, z \in \mathbb{N}$, if $x = y$ and $y = z$ then $x = z$.

IV. *Axiom of the Closure of Equality*: For all x, y, if $x \in \mathbb{N}$ and $x = y$, then $y \in \mathbb{N}$.

The remaining 5 axioms define the structure of $\mathbb{N}$:

V. $1 \in \mathbb{N}$

VI. If $x \in \mathbb{N}$, then the successor $S(x) \in \mathbb{N}$.

VII. There is no $x \in \mathbb{N}$ such that $S(x) = 1$.

VIII. For all $x, y \in \mathbb{N}$, if $S(x) = S(y)$, then $x = y$.

IX. Let $P(x)$ be a statement about the natural number x. If:

- $P(1)$ is true, and
- for all $n \in \mathbb{N}$, if $P(n)$ is true, then $P(S(n))$ is also true,

then $P(x)$ is true for all $x \in \mathbb{N}$. (Mathematical Induction)

As shorthand, we denote:

$$S(1) = 2, \quad S(S(1)) = 3, \quad S(S(S(1))) = 4, \quad \text{and so on.}$$

6.4 Propositions and Proofs

Proposition 1: $n \neq m \implies S(n) \neq S(m)$

Proof:

We will prove this by contradiction. Assume $n \neq m$ and $S(n) = S(m)$. By Axiom 8, we have $n = m$, which is a contradiction. Therefore, $S(n) \neq S(m)$.

6.4.1 Proposition 2: For any $n \in \mathbb{N},\ n \neq S(n)$

Proof:

$M = \{n \in \mathbb{N} \| n \neq S(n)\}$

By $1 \neq S(n)$ for any $n \in \mathbb{N}$, this implies that 1 is part of the set M. This implies that $S(n) \neq S(S(n)) \implies S(n) \in M$. By Axiom 9, $M = \mathbb{N}$.

6.4.2 Proposition 3: $n \neq 1\ \exists m \in \mathbb{N} \mid n = S(m)$

Proof:

$$M = \{1\} \cup \{n \in \mathbb{N} | \text{Proposition 3 is true}\}$$

We know that 1 is in the set M. And by Proposition 1 we know that

$$S(n) = S(S(m)) \implies S(n) \in M,$$

and by Axiom 9 we know that $M = \mathbb{N}$.

6.5 Definition of Addition in $\mathbb{N}$

For any pair n, m $\in \mathbb{N}$, there is a unique way to define

$$\text{Add}(n, m) = n + m.$$

1. **Base Case:** $n + 1 = S(n)$.
2. **Inductive Step:** $n + S(m) = S(n + m) \iff S(n + m)$.

Uniqueness:

Suppose $A\,B$ satisfy our conditions. Fix n and then let $M = \{m \in \mathbb{N} | A(n, m) = B(n, m)\}$. Then

$$A(n, 1) = S(n) = B(n, 1) \implies 1 \in M$$

$$m \in M \implies A(n, m) = B(n, m) \implies A(n, S(m)) = S(A(n, m)) = S(B(n, m)) = B(n, S(m))$$

$$\implies A(n, S(m)) = B(n, S(m))$$

By Axiom 9, we know that $M = \mathbb{N}$ and $A = B$.

Construction: For $n = 1$, define $A(n, m) = S(m)$.

1. $A(n, 1) = S(n) = S(1)$
2. $A(n, S(m)) = S(A(n, m)) = S(S(m))$

Define: $A(S(n), S(m)) = S(A(n, m))$

1. $A(S(n), 1) = S(A(n, 1)) = S(S(n))$
2. $A(S(n), S(m)) = S(A(n, m)) = S(S(m))$

Commutativity of Addition:

The proposition says $n + m = m + n$ for any $n, m \in \mathbb{N}$. We will prove this by induction on m. Fix n and consider $M = \{n \in \mathbb{N} | A(n, m) = A(n, m)\}$.

Now recall that $A(n, 1) = S(n)$.

For $n = 1$:

$$A(n, 1) = S(1) = A(1, n) \implies 1 \in \mathbb{N}$$

Also, $A(n, k) = 1 + k \implies 1 + m = S(m) \implies 1 + m = m + 1 \implies 1 \in \mathbb{N}$.

Suppose $n \in \mathbb{N} \implies n + m = m + n$ or $A(n, m) = A(m, n)$.

By construction $A(S(n), m) = S(A(n, m))$ and by definition $A(S(n), m) = S(A(n, m)) = A(m, S(n)) = S(n) + m = m + S(n) \implies S(n) \in M \implies$ by induction $M = \mathbb{N}$.

Chapter 7

The Archimedean Principle

If a and b are real numbers with $a > 0$, then there exists $n \in \mathbb{N}$ such that $na > b$.

Equivalently, for any real number $x \in \mathbb{R}$, there exists a natural number $n \in \mathbb{N}$ such that $n > x = \frac{b}{a}$.

Proof:

We prove the Archimedean Principle by contradiction.

Assume that there exists some real number $x \in \mathbb{R}$ such that $n \leq x$ for all $n \in \mathbb{N}$. That is, x is an upper bound for the set $\mathbb{N} \subset \mathbb{R}$, backed up by the Completeness Axiom of the real numbers.

Let $s = \sup(\mathbb{N})$ be the least upper bound of $\mathbb{N}$. Then,

$$s - 1 < s \implies \exists m \in \mathbb{N} \text{ s. t. } m > s - 1.$$

But, if we add 1 to both sides, we get

$$s < m + 1 \in \mathbb{N}.$$

This is a contradiction to the assumption that s is the supremum of the Naturals. Hence, there is no real number x which is greater than all natural numbers.

$\mathfrak{QED}$

7.1 Equivalent Formulation

For any $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$.

Applications:

1. **Density of Rational Numbers:** The Archimedean Principle helps in proving that between any two real numbers, there exists a rational number.
2. **Limits and Infinitesimals:** It ensures that sequences like $\left\{\frac{1}{n}\right\}$ converge to 0, foundational in real analysis and calculus.
3. **Bounding Functions:** It is used in analysis to show that functions do not grow faster than natural numbers in certain contexts.
4. **Non-Existence of Infinitely Small Numbers:** The principle implies that real numbers do not contain infinitesimals (nonzero numbers smaller than all $\frac{1}{n}$), distinguishing $\mathbb{R}$ from non-standard number systems.

Example:

Prove that $\forall \varepsilon > 0, \exists N \in \mathbb{N} : \frac{1}{n} < \varepsilon, \forall n \geq N$.

We know that

$$\begin{aligned} \frac{1}{N} &< \varepsilon \\ \varepsilon > \frac{1}{N} &\geq \frac{1}{n} \text{ by transitivity} \\ \varepsilon &> \frac{1}{n}. \end{aligned}$$

We also get $N > \frac{1}{\varepsilon}$.

And by the Archimedean Principle, we let $b = 1$, $a = \varepsilon$ and $n = N$. Then by transitivity we get our proof.

QED

Chapter 8

Arithmetic

8.1 Divisor

Let $a, b \in \mathbb{Z}$. We say that a *divides* b if

$$a \mid b \iff \exists k \in \mathbb{Z} : b = ak.$$

8.2 Even and Odd Numbers

A natural number a is *even* if $a = 2k$ with $k \in \mathbb{N}$.

A natural number b is *odd* if $b = 2q - 1$ with $q \in \mathbb{N}$.

8.3 Perfect Squares

A natural number a is a *perfect square* if $\sqrt{a} \in \mathbb{N}$.

8.4 Division Algorithm

Given any two integers a and b with $a > 0$, there exist unique integers q and r such that $b = qa + r$ with $0 \geq r < a$.

Proof:

Let $S = \{b - xa \mid b - xa \geq 0, x \in \mathbb{Z}\}$.

By the well-ordering principle of the natural numbers, we know that our set has a minimum r.

First, we prove $0 \leq r$.

If $b \geq 0$, then let $x = 0 \implies b \in S$. For $b < 0$, let $x = b$. Thus,

$$b - ba = b(1 - a) \implies b - ba \in S.$$

Now, to prove that $r < a$, assume $r \geq a$, but then

$$\begin{aligned} b - qa &\geq a \\ b - qa - a &\geq 0 \\ b - a(q + 1) &\geq 0 \\ b - a(q + 1) \geq 0 &\implies b - a(q + 1) \in S. \end{aligned}$$

And $b - a(q+1) < b - aq = r \implies r < a$. Therefore, we have a contradiction.

Now, we will prove the uniqueness.

Assume $pq + r = a = pq' + r'$ with $q \neq q'$ and $r \neq r'$.

$$r - r' = a(q - q') = 0$$

Using the property of inverses of $\mathbb{Z} \setminus \{0\}$ as a field, we know that $r = r'$ and $q - q'$.

$\mathfrak{QED}$

8.5 Prime Numbers

A *prime number* $p \in \mathbb{N} \setminus \{1\}$ fulfills the following condition:

I. p is only divisible by itself and 1.

8.6 Composite Number

A number $z \in \mathbb{Z}$ is *composite* if

$$z = xy, \quad 1 < x, y < z.$$

8.7 Pythagorean Triplets

Pythagorean triplets are 3 natural numbers such that

$$a^2 + b^2 = c^2.$$

8.8 Division

Given $a, b \in \mathbb{Z}$. There exist unique $q, r \in \mathbb{Z}$ such that

$$a = qb + r \quad 0 \leq r < |b|.$$

8.9 Greatest Common Divisor

For two integers $a, b \in \mathbb{Z}$ with $(a, b) \neq (0, 0)$, the positive integer t satisfying:

- $t \mid a$ and $t \mid b$.
- For all $c \in \mathbb{Z}$ with $c \mid a$ and $c \mid b$, it follows that $c \mid t$.

We call it the *greatest common divisor* (gcd) of a and b, denoted as $t = \gcd(a, b)$.

If $t = 1$, then a and b are said to be *coprime* or *relatively prime*.

8.10 The Euclidean Algorithm

If $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$.

Input: $a, b \in \mathbb{Z}$, $a \neq 0 \neq b$.

Output: The greatest common divisor t of a and b.

Algorithm:

1. If $a = 0$, then return $|b|$. If $b = 0$, return $|a|$.

2. Set $r_1 := a$; $r_2 := b$; $k := 2$.
3. As long as $r_k \neq 0$, define $r_{k+1} := r_{k-1} \bmod r_k$.
4. As soon as $r_k = 0$, return r_{k-1}.

In simpler terms, as long as the remainder is not zero, we keep dividing the previous remainder by the current remainder and updating the remainders. When we reach a remainder of zero, the previous remainder is the gcd.

Proof:

Notice that the algorithm terminates because the sequence given r_n is monotone decreasing due to $0 \leq r < b$ and the well-ordering principle of the natural numbers.

For the correctness: given is $a = bq + r$ to show that $\gcd(a, b) = \gcd(b, r)$. This is like saying:

$$\begin{aligned} \gcd(a, b) &= gcd(b, r) \\ \gcd(b, r) &= \gcd(r, r_1) \\ \gcd(r, r_1) &= \gcd(r_1, r_2) \\ &\vdots \\ \gcd(r_{n-1}, r_n) &= \gcd(r_n, 0) = r_n. \end{aligned}$$

Let d be a common divisor of a, b. Then $d \mid a$ and $d \mid b$, so $d \mid ((a - bq) = r) \implies d \mid r$.

Let e be a common divisor of b, r. Then $e \mid b$ and $e \mid r$, so $e \mid ((bq + r) = a) \implies e \mid a$.

Thus, the set of common divisors of a, b is the same as the set of common divisors of b, r, so their greatest common divisors are equal.

$\mathfrak{QED}$

Example:

Find $\gcd(252, 198)$:

$$\begin{aligned} r_{n-1} &= q_{n-1} r_n + r_{n+1} \\ 252 &= 1 \cdot 198 + 54 \\ 198 &= 3 \cdot 54 + 36 \\ 54 &= 1 \cdot 36 + 18 \\ 36 &= 2 \cdot 18 + 0 \end{aligned}$$

So, $\gcd(252, 198) = 18$.

8.10.1 Euclidean Algorithm as Linear Map

Given $E := \{(a, b)^T \mid a, b \in \mathbb{Z}, b \neq 0\}$ and

$$\varepsilon : E \to \mathbb{Z}^{2\times 1} : \begin{bmatrix} a \\ b \end{bmatrix} \mapsto \begin{bmatrix} b \\ a \mod b \end{bmatrix}$$

Then the Euclidean algorithm can be described as repeated application of ε until the second component is zero. The first component at that point is the gcd.

$$\varepsilon^{k-1} \begin{bmatrix} a \\ b \end{bmatrix} = \varepsilon \circ \cdots \circ \varepsilon \begin{bmatrix} t \\ 0 \end{bmatrix}$$

$t = \gcd(a, b)$.

Also, we can express ε as a matrix multiplication:

$$\varepsilon \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -q \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ r \end{bmatrix}$$

with $a = qb + r$ and $q \in \mathbb{Z}, 0 \leq q < |b|$.

8.10.2 Bezout Identity

This is one of the most important properties of the *gcd*. Given $a, b \in \mathbb{Z}$, not both zero, there exist integers $\alpha, \beta \in \mathbb{Z}$ such that

$$\gcd(a, b) = a\alpha + b\beta$$

This equation comes directly from *Euclid's Algorithm*. For the intuition, when writing your iterations you can manipulate them to arrive at the solution.

Proof:

Given

$$M := \{a\alpha + b\beta \mid a\alpha + b\beta > 0 \wedge \alpha \in \mathbb{Z} \wedge \beta \in \mathbb{Z}\}$$

M is not empty because $\alpha, \beta \in M$, and it has a minimum by the well-ordering principle of the natural numbers which we will call c.

Then, let $k \in M$. Then there exist $q \in \mathbb{N}_0$ and $r \in \mathbb{N}_0$ with $r < c$ such that

$$k = qc + r$$

because k and c are elements of M we should be able to find r_1, r_2, s_1, s_2 such that

$$k = r_1 a + s_1 b \quad c = r_2 a + s_2 b$$

Now let us

$$\begin{aligned} k = r_1 a + s_1 b &= qc + r \\ r_1 a + s_1 b &= q(r_2 a + s_2 b) + r \\ r &= r_1 a + s_1 b - q(r_2 a + s_2 b) \\ r &= r_1 a + s_1 b - q r_2 a - q s_2 b \\ r &= (r_1 - q r_2) a + (s_1 - q s_2) b \end{aligned}$$

$r \geq 0$ and $r < c$, but if $r \neq 0$ then $r \in M$ would be the minimum which is a contradiction to the assumption that c is the minimum, also $r = 0$, $k = qc$. Thus, *every element of M is a multiple of c*. Therefore, a, b are also divisible by c.

Now we need to show that c is in fact the greatest common divisor. $c = r_2 a + s_2 b$

$$c = a\alpha_0 + b\beta_0$$

Because $c \mid a \wedge c \mid b$ we know that $c \mid a\alpha_0 + b\beta_0$. We also know that $c \mid \gcd(a, b)$ and thus, $|c| = |g = \gcd(a, b|$, but because both are positive $c = g$.

$\mathfrak{QED}$

8.11 Prime Field

Given $p \in \mathbb{N}, p > 1$ is prime, then $\mathbb{Z}/p\mathbb{Z}$ is a field with p elements. It can also be noted $\mathbb{F}_p$.

Proof:

We need to show that for $a + p\mathbb{Z} \neq p\mathbb{Z}$ there exists some $b + p\mathbb{Z}$ such that $(a + p\mathbb{Z}))(b + p\mathbb{Z}) = 1 + p\mathbb{Z}$.

For $a \notin p\mathbb{Z}$ we know that $gcd(a, p) = 1$, thus we do not have zero divisors.

8.11.1 Implication of Prime Numbers

For $a, b \in \mathbb{Z}$ and a prime number p

$$p \mid ab \implies (p \mid a) \vee (p \mid b).$$

Proof:

"⇒": Let p be a prime and $a, b \in \mathbb{Z}$ with $p \mid ab$. Assume $p \nmid a$. Then there exist $\alpha, \beta \in \mathbb{Z}$ with $\alpha p + \beta a = 1$. Since $p \mid ab$, it follows that $p \mid \alpha pb + \beta ab = (\alpha p + \beta a)b = b$.

"⇐": Let $c \in \mathbb{N}$ be a divisor of p. Then there exists $d \in \mathbb{N}$ with $p = cd$. In particular $1 \leq c \leq p$ and $1 \leq d \leq p$ and $p \mid cd$. By our assumption it follows that $p \mid c$ or $p \mid d$, i.e. $p \leq c$ or $p \leq d$. Together with $c \leq p$ and $d \leq p$ this forces $c = 1$ or $c = p$. Hence, p has no nontrivial divisors and is prime.

$\mathfrak{QED}$

8.12 Fundamental Theorem of Arithmetic

Every integer greater than 1 can be written in the form

$$p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k},$$

where $n_i \geq 0$ and the p_i are distinct primes. The factorization is unique, except possibly for the order of the factors.

Example:

$$4312 = 2 \cdot 2156 = 2 \cdot 2 \cdot 1078 = 2 \cdot 2 \cdot 2 \cdot 539 = 2 \cdot 2 \cdot 2 \cdot 7 \cdot 77 = 2 \cdot 2 \cdot 2 \cdot 7 \cdot 7 \cdot 11$$

That is,

$$4312 = 2^3 \cdot 7^2 \cdot 11$$

8.12.1 Least Common Multiple

The least common multiple of a and b, denoted $[a, b]$, is the smallest positive integer that is a multiple of both a and b.

Example:

$$[6, 4] = 12, \quad [33, 15] = 165$$

8.12.2 Multiplication Property of the Least Common Multiple

$$[a, b] \cdot \gcd(a, b) = ab$$

Proof:

Let

$$a = p_1 \cdots p_l q_1 \cdots q_m, \quad b = q_1 \cdots q_m r_1 \cdots r_n,$$

then

$$\begin{aligned} \gcd(a, b) &= q_1 \cdots q_m \\ [a, b] &= p_1 \cdots p_l q_1 \cdots q_m r_1 \cdots r_n \\ ab &= p_1 \cdots p_l q_1^2 \cdots q_m^2 r_1 \cdots r_n. \end{aligned}$$

So

$$[a, b] \cdot \gcd(a, b) = ab.$$

Example:

$$\gcd(36, 90) = 18, \quad [36, 90] = 180, \quad 36 \cdot 90 = 32400 = 18 \cdot 180$$

8.13 Goldbach's Conjecture

Every even integer greater than 4 can be written as the sum of two primes.

8.14 Prime Factorization

To find the *prime factorization* of a number a, the algorithm goes as follows:

- Let n be the first prime number.
- Try dividing a by n. If $n \mid a$ note the number and set $a = a \div n$ to repeat this step. If n did not divide a set n to the next prime number and repeat this step.
- When 1 was reached using the previous step then the algorithm terminates and the noted numbers are our prime factorization of a.

8.15 GCD via Prime Factorization

The gcd of two or more numbers can also be found with the following algorithm.

- Find the prime factorization of all target numbers.
- Note the numbers which appear in both prime factorizations, always taking the one with the least exponent.
- Multiply the noted prime factors together.

8.16 Modular Arithmetic

We define an equivalence relation

$$a \equiv b \mod n \iff n|(a-b) \wedge a, b \in \mathbb{Z}.$$

or equivlently

$$a \equiv b \mod n \iff a - b = nk \wedge a, b, k \in \mathbb{Z}.$$

This means that both have the **same remainder** when divided by n.

8.17 Residual Classes and Residual Class Rings

Because we have an equivalence relation, we can classify the partitions of our set $\mathbb{Z}$ into classes, which we will call *residual classes*.

Example:

$$[13]_7 + \{m \in \mathbb{Z} \mid m \equiv 13 \mod 7\}$$

The set of all residual classes can be written as $\mathbb{Z}/n\mathbb{Z}$, where n is the modulus.

8.18 Properties

- $$(a+b) \mod n = (a \mod n + b \mod n) \mod n$$
- $$(a*b) \mod n = (a \mod n * b \mod n) \mod n$$
- $$(a^m) \mod n = ((a \mod n)^n) \mod n$$
- $$\left(\frac{a}{b}\right) \mod n \neq \frac{a \mod n}{b \mod n}$$

8.19 General Division Rules

Sum Rules

If the number a can be expressed as a sum and b is a common divisor of all the terms, then b is also a divisor of a.

Example:

$742 = 700 + 42$. Then 7, as a common divisor of 700 and 42, is a divisor of 742.

Product Rule

If the number a can be written as a product of two factors and b is a divisor of one of these factors, then b is also a divisor of a.

Example:

$1500 = 15 \cdot 100$. Then 3 and 5 are divisors of 15, and 2, 4, 5, 10, 25, and 50 are divisors of 100—thus, all are divisors of 1500.

Divisor Pair Rules

If the number a has b as a divisor, then $\frac{a}{b}$ is also a divisor of a.

Example:

3 is a divisor of 315. Then $105 = \frac{315}{3}$ is also a divisor of 315.

Divisor Rule

If the number a has b as a divisor, then every divisor of b is also a divisor of a.

Example:

45 is a divisor of 450. Then 3, 5, 9, and 15 (divisors of 45) are also divisors of 450.

Difference Rule

If the number a can be expressed as a difference and b is a common divisor of the minuend and subtrahend, then b is also a divisor of a.

Example:

$686 = 700 - 14$. Then 7, as a common divisor of 700 and 14, is a divisor of 686.

Relative Prime Rule

If the number a has two relatively prime numbers b and c as divisors, then their product $b \cdot c$ is also a divisor of a.

Example:

1540 has the relatively prime numbers 7 and 11 as divisors. Then $77 = 7 \cdot 11$ is also a divisor of 1540.

8.20 Using Modular Arithmetic to Find the n-th Digit of a Large Number

What is the last digit of the number 13^{2023}?

Note that $13 \mod 10 = 3$ and $13^2 \mod 10 = ((13 \mod 10)(13 \mod 10)) \mod 10 = 9$.

We can continue this process, and we will notice a pattern. After some iterations, we will know that

$$(13^{2020}13) \mod 10 = 7.$$

8.21 Gaussian Brackets

The Gaussian brackets refer to two closely related functions used to round real numbers to integers in specific ways:

8.21.1 Floor Function

The floor function, denoted by $\lfloor x \rfloor$, gives the greatest integer less than or equal to a real number x.

Example:

$\lfloor 3.7 \rfloor = 3$, $\lfloor -1.2 \rfloor = -2$

8.21.2 Ceiling Function

The ceiling function, denoted by $\lceil x \rceil$, gives the smallest integer greater than or equal to a real number x.

Example:

$\lceil 3.7 \rceil = 4$, $\lceil -1.2 \rceil = -1$

These functions are particularly useful in number theory, computer science, and rounding operations, where precise integer bounds of real numbers are required.

8.22 Residual Class Ring Modulo n

Given $n \in \mathbb{N}$, we define the relation $\sim_n$ for $\mathbb{Z}$.

$$a \sim b \iff n \mid (a - b).$$

Also, a/n and b/n have the same remainder.

8.22.1 Equivalence Class of $\mathbb{Z}$

For $i, n \in \mathbb{Z}$

$$\overline{i} := i + n\mathbb{Z} = \{i + na \mid a \in \mathbb{Z}\}$$

and

$$\mathbb{Z}_N = \mathbb{Z}/n\mathbb{Z} = \{\overline{0} := n\mathbb{Z}, \overline{1} := 1 + n\mathbb{Z}, \ldots, \overline{n-1} := n - 1 + n\mathbb{Z}\},$$

is the set of all equivalence classes for $\mathbb{Z}$ with respect to the relation $k \sim l \iff n \mid (k - l)$. We define the addition and multiplication for $\mathbb{Z}/n\mathbb{Z}$ using the representatives:

$$(a + n\mathbb{Z}) + (b + n\mathbb{Z}) := (a + b) + n\mathbb{Z}$$

and

$$(a + n\mathbb{Z}) \cdot (b + n\mathbb{Z}) := ab + n\mathbb{Z}$$

For all $a, b \in \mathbb{Z}$. Therefore,

$$\overline{a} + \overline{b} = \overline{a + b}, \quad \overline{a} \cdot \overline{b} = \overline{ab}.$$

Chapter 9

Real Numbers

9.1 Axiom Definition

A way of defining the *real numbers* is via the *Cauchy sequences*. Another way is via the axioms of a field with the order axioms and the *completeness axiom*.

The following axioms apply for any $a, b, c \in \mathbb{R}$:

- *Associativity:* $(a \odot b) \odot c = a \odot (b \odot c)$ for all $a, b, c \in S$.
- *Commutativity:* $a \odot b = b \odot a$ for all $a, b \in S$.
- *Identity Element:* There exists $e \in S$ such that $a \odot e = e \odot a = a$ for all $a \in S$.
- *Inverse Element:* For every $a \in S$, there exists $a^{-1} \in S$ such that $a \odot a^{-1} = a^{-1} \odot a = e$.
- *Distributivity:* $a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c)$ and/or $(b \odot c) \oplus a = (b \odot a) \oplus (c \odot a)$.

We also refresh the order axioms:

I. *Trichotomy:* Exactly one of the following holds:

$$a < b, \quad a = b, \quad a > b$$

II. *Transitivity:* If $a < b$ and $b < c$, then $a < c$.

III. *Additive Compatibility:* If $a < b$, then $a + c < b + c$.

IV. *Multiplicative Compatibility:* If $0 < a$ and $0 < b$, then $0 < ab$.

9.1.1 Properties Derived from the Axioms

- The neutral and inverse elements of the addition and multiplication are well defined.
- $a + b = c$ and $a + c = 0 \implies b = c$
- $a \neq 0$, $ab = 1$ and $ac = 1 \implies b = c$
- $a \cdot 0 = 0$
- $-a = (-1)a$
- $-(-a) = a$
- $(-1)^2 = 1$
- $\left(a^{-1}\right)^{-1} = a, \forall \in \mathbb{R}, a \neq 0$

Some proofs of these properties can be found in the section about the algebraic structures.

9.2 Existence of Solutions

Given $a, b \in \mathbb{R}$, then

$$!\exists x \in \mathbb{R}, \text{ such that, } a + x = b$$

Also, if $a \neq 0$

$$!\exists x \in \mathbb{R}, \text{ such that, } ax = b$$

Proof:

For the first proposition, let $x = -a + b$

$$\begin{aligned} a + x &= a - a + b \\ &= b \end{aligned}$$

For the second, let $x = a^{-1}b$

$$\begin{aligned} ax &= aa^{-1}b \\ &= 1 \cdot b \end{aligned}$$

$\mathfrak{QED}$

9.3 The Completeness Axiom

All non-empty upper-bounded subsets of the real numbers have a *supremum* $s \in \mathbb{R}$.

9.4 Minima in the Real Numbers

As a consequence of the *completeness axiom*, each non-empty lower-bounded subset of real numbers has a minimum.

Proof:

The main idea is that the supremum times minus 1 of a set is the minimum of another set.

We define $-A := \{-a \mid a \in A\}$, then $-A \neq \emptyset$ and is upper-bounded. By definition s is less or greater than all the elements of the set A.

$$\begin{aligned} s \leq a, \forall a \in A \\ -s \geq -a, \forall a \in A \end{aligned}$$

Thus, $-s = \sup -A$.

Now we want to prove that $-\sup -A$ is first, the lower bound of A and second, it is the greatest upper bound.

For the first part

$$\begin{aligned} \sup -A \geq -a \forall a \in A \\ -\sup -A \leq a \forall a \in A \end{aligned}$$

Thus, $-\sup -A$ is in fact a lower bound of A. For the second part let us assume that there exists some element s such that $s > -\sup -A$ and $s \leq a, \forall a \in A$, then

$$-s < \sup -A \text{ and } -s \geq -a, \forall a \in A$$

This is a contradiction because we have an upper bound, which is not an upper bound. Therefore, $-\sup -A$ is the infimum of A.

$\mathfrak{QED}$

9.5 Approximation of the Supremum

Given a non-empty, upper-bounded set $A \subseteq \mathbb{R}$ and $s = \sup A$. Then there is for each $y < s$ an $x \in A$, such that

$$y < x \leq s$$

An analogous theorem can be written for the minimum as well.

Proof:

If $x \in A$, then the proof is completed. For the case $s \notin A$, if there existed some $y^* \in \mathbb{R}$ where $y^* < s$ and $y^* \geq x$ for all $x \in A$, then we still have a contradiction to the assumption that s is the supremum of A.

$\mathfrak{QED}$

9.6 Supremum of the Naturals

$\mathbb{N}$ is not upper-bounded.

Proof:

Assume $a = \sup \mathbb{N}$. Since $\mathbb{N} \subset \mathbb{R}$ is not empty, it has an upper-bound by the completeness axiom.

By the definition of the supremum, $a - 1 \in \mathbb{N}$ is not an upper-bound of $\mathbb{N}$, therefore $a - 1 < k, k \in \mathbb{N}$.

Then, the following also applies

$$a < k + 1.$$

This is a contradiction to the assumption that a is our supremum, thus our assumption has been proven false.

9.7 The Properties of Archimedes and Eudoxus

Archimedes:

For all pairs of positive real numbers x, y, there exists some $n \in \mathbb{N}$ such that:

$$nx > y.$$

Eudoxus:

For all positive real numbers x, there exists an $n \in \mathbb{N}$ such that

$$\frac{1}{n} < x.$$

Proof Eudoxus:

$x \in \mathbb{R}_{>0} \implies \frac{1}{x} \in \mathbb{R}_{>0}$. By the Archimedean property, $\exists n \in \mathbb{Z}_{0>}, \frac{1}{x} < n$. Equivalently $\frac{1}{n} < x$.

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9.8 Density of the Rationals in the Reals

For each pair $a, b \in \mathbb{R}$ there exists some $q \in \mathbb{Q}$ such that

$$a < q < b.$$

Proof:

Let $q = \frac{m}{n}, m \in \mathbb{Z}, n \in \mathbb{N}$ to prove $a < \frac{m}{n} < b$ equivalently $an < m < bn$.

For $b - a > 0$, we know $\frac{1}{n} < b - a$ or $a < b - \frac{1}{n}$, by the Archimedean property.

Next take the smallest $m \in \mathbb{Z}$ such that $m - 1 \leq na < m$ or $a < \frac{m}{n}$. This gives us the first half of our theorem.

Finally, for the second half

$$\begin{aligned} m - 1 &\leq na \\ &= na + 1 \\ &< n(b - \frac{1}{n}) + 1 \text{ using our result from the Archimedean property} \\ &= nb \end{aligned}$$

This tells us that $m < nb \implies \frac{m}{n} < b$.

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9.9 Existence of Square Roots

For each $x \in [0, \infty], !\exists r \in [0, \infty)$ s. d. $r^2 = x$. Also, $\sqrt{x} := r$ and we call $x \mapsto \sqrt{x}$ the *root function*.

Proof:

Let $T = \{x \in [0, \infty) \mid x < y\}$ and let us define $a = y^2$ as the supremum of T.

To prove that the equality holds we will prove that the other order cases can not be true.

Assume $a < y$, because $a - y < 0$, we know that $\frac{1}{n} < a - y$ and $a + \frac{1}{n} < y$. Thus, $a + \frac{1}{n} \in T$, but because a is the supremum, this is a contradiction because our supremum is not the smallest upper-bound.

For the second case, assume $a > y$, by applying the same reasoning with $a - \frac{1}{n}$, we will arrive at a similar contradiction.

Assume $a > y$, then $a - y > 0 \implies \frac{1}{n} < a - y, \implies a - \frac{1}{n} < y$

But then $a - \frac{1}{n}$ is an upper bound and this contradicts our assumption. Thus, by elimination $a^2 = y$.

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9.10 Irrational Numbers

$$\mathbb{R} \setminus \mathbb{Q} \text{ or } \mathbb{Q}^C := \{x \in \mathbb{R} \mid x \notin \mathbb{Q}\}$$

9.11 Absolute Value for Real Numbers

The *absolute* is defined as

$$|x| := \sqrt{x \cdot x}.$$

9.11.1 Distance

- $|x| \geq 0, \forall x \in \mathbb{R}$
- $|x| = 0, \implies x = 0$
- $|\alpha x| = |\alpha||x|, \forall x, \alpha \in \mathbb{R}$
- $|x + y| \leq |x| + |y| \forall x, y \in \mathbb{R}$

We define the *distance* between two real numbers as

$$d(x, y) = |x - y|$$

with the following properties:

- $d(x, y) \geq 0, \forall x, y \in \mathbb{R}$
- $d(x, y) = 0, \iff x = y$
- $d(x, y) = d(y, x)$
- $d(x, y) \geq d(x, y) + d(z, y), \forall x, y \in \mathbb{R}$

9.12 Nested Interval Property

For each $n \in \mathbb{N}$, assume we are given a closed interval $I_n = [a, b] = \{x \in \mathbb{R} \mid a_n \leq x \leq b_n\}$. Also assume each I_n contains I_{n+1}. Then the nested sequence of closed intervals $I_1 \supseteq I_2 \supseteq I_3 \supseteq \ldots$ has a non-empty intersection.

Proof:

By definition every I_n is non-empty, thus the sequences $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ are well-defined. Also, every set $A = \{a_n : n \in \mathbb{N}\}$ is bounded above by the corresponding b_n, and this is guaranteed by the axiom of completeness.

Let $x = \sup A$. Then consider $I_n = [a_n, b_n]$.

$$a_n \leq x \leq b_n, \forall n \in \mathbb{N} \implies x \in I_n$$

Thus, $x \in \bigcap_{n=1}^{\infty} I_n$ and the intersection is non-empty.

$\mathfrak{QED}$

9.13 Definition as Cauchy Sequences

Let $\mathbb{F}$ be an ordered field. On the set

$$\text{ch}(\mathbb{F}) := \{x : \mathbb{N} \to \mathbb{F} \mid x \text{ is a Cauchy sequence}\}$$

and on the set

$$c(\mathbb{F}) := \{x : \mathbb{N} \to \mathbb{F} \mid x \text{ is a convergent sequence}\}$$

we can define an addition and a multiplication as follows:

If $x = (x_n)_{n\in\mathbb{N}}$ and $y = (y_n)_{n\in\mathbb{N}}$ are Cauchy sequences (respectively, convergent sequences), then their sum is defined as

$$x + y := (x_n)_{n\in\mathbb{N}} + (y_n)_{n\in\mathbb{N}} := (x_n + y_n)_{n\in\mathbb{N}}$$

and their product is defined as

$$x \cdot y := (x_n)_{n\in\mathbb{N}} \cdot (y_n)_{n\in\mathbb{N}} := (x_n \cdot y_n)_{n\in\mathbb{N}}$$

The sum and product satisfy all field axioms except for the existence of the multiplicative inverse. The zero element is $0_{\mathbb{N}} = (0, 0, \ldots)$, the unit element is $1_{\mathbb{N}} = (1, 1, \ldots)$, and the additive inverse of $x = (x_n)_{n\in\mathbb{N}}$ is $-x = (-x_n)_{n\in\mathbb{N}}$. We demonstrate the distributive law as an example:

Let $x = (x_n)_{n\in\mathbb{N}}, y = (y_n)_{n\in\mathbb{N}}, z = (z_n)_{n\in\mathbb{N}}$ be Cauchy sequences (convergent sequences). Then we have:

$$\begin{aligned} x(y+z) &= (x_n)_{n\in\mathbb{N}} \cdot \big((y_n)_{n\in\mathbb{N}} + (z_n)_{n\in\mathbb{N}}\big) \\ &= (x_n)_{n\in\mathbb{N}} \cdot (y_n + z_n)_{n\in\mathbb{N}} \\ &= (x_n(y_n + z_n))_{n\in\mathbb{N}} \\ &= (x_n y_n + x_n z_n)_{n\in\mathbb{N}} \\ &= (x_n y_n)_{n\in\mathbb{N}} + (x_n z_n)_{n\in\mathbb{N}} \\ &= xy + xz. \end{aligned}$$

We now aim to construct the ordered field $\mathbb{R}$ of the real numbers; it will have the following properties:

- There exists an injective mapping $j : \mathbb{Q} \to \mathbb{R}$ which respects addition, multiplication, and order, such that the following holds:

  For all $z, w \in \mathbb{R}$ with $z < w$, there exists an $x \in \mathbb{Q}$ such that

  $$z < j(x) < w$$

- Every Cauchy sequence in $\mathbb{R}$ converges.

Via j, we identify $\mathbb{Q}$ with $j(\mathbb{Q})$ and consider $\mathbb{Q}$ as a subset of $\mathbb{R}$. In $\mathbb{R}$, the following will additionally hold:

- For all $y > 0$ and $n \in \mathbb{N}$, the equation $x^n = y$ has a solution.
- Every bounded above subset of $\mathbb{R}$ has a supremum.

We define the following relation on the set $\mathrm{ch}(\mathbb{Q})$ of all Cauchy sequences in $\mathbb{Q}$:

$$x \sim y \quad \text{if and only if} \quad x - y \text{ is a null sequence}$$

That is, $(x_n)_{n\in\mathbb{N}} \sim (y_n)_{n\in\mathbb{N}}$ if and only if

$$x_n - y_n \to 0 \quad (n \to \infty)$$

9.13.1 Definition

The set

$$\mathbb{R} := \{[x]_\sim : x \in \mathrm{ch}(\mathbb{Q})\}$$

is called the set of real numbers.

Analogous to the construction of the rational numbers, the real numbers consist of equivalence classes. Roughly speaking, an equivalence class consists of those Cauchy sequences in $\mathbb{Q}$ that exhibit the same limit behavior.

Equipped with the addition

$$+ : \mathbb{R} \times \mathbb{R} \to \mathbb{R}, \quad [x], [y] \mapsto [x] + [y] := [x + y]$$

and the multiplication

$$\cdot : \mathbb{R} \times \mathbb{R} \to \mathbb{R}, \quad [x], [y] \mapsto [x] \cdot [y] := [xy].$$

$\mathbb{R}$ is a field. The zero element is $[0_{\mathbb{N}}]$, and the unit element is $[1_{\mathbb{N}}]$.

Chapter 10

Complex Numbers

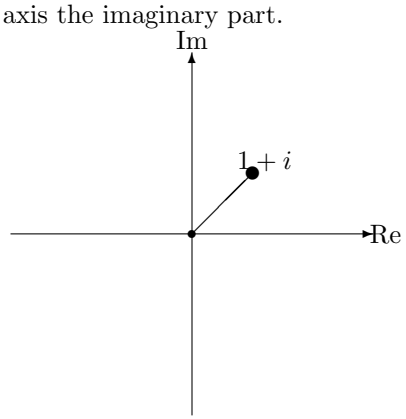
A *complex number* is a number of the form

$$z = a + bi,$$

where $a, b \in \mathbb{R}$, and i is the imaginary unit defined by $i^2 = -1$. The set of all complex numbers is denoted by $\mathbb{C}$.

10.1 The Complex Plane

Complex numbers can be represented graphically in the *complex plane*, where the horizontal axis represents the real part and the vertical axis the imaginary part.



The point $1 + i$ is located at (1,1), showing 1 unit on the real axis and 1 unit on the imaginary axis.

10.2 Conjugate of a Complex Number

The *conjugate* of a complex number $z = a + bi$ is denoted $\overline{z}$ and is defined as

$$\overline{z} = a - bi.$$

Geometrically, it reflects the point z across the real axis in the complex plane. Conjugates are useful in division and in finding the modulus, since

$$z \cdot \overline{z} = a^2 + b^2 = |z|^2.$$

10.3 Operations in Cartesian Coordinates

Let $z_1 = a + bi$ and $z_2 = c + di$ be two complex numbers.

Addition:

$$z_1 + z_2 = (a + c) + (b + d)i$$

Multiplication:

$$z_1 \cdot z_2 = (ac - bd) + (ad + bc)i$$

Quotient:

$$\frac{z_1}{z_2} = \frac{(a + bi)}{(c + di)}\frac{(c - di)}{(c - di)} = \frac{(a + bi)(c - di)}{c^2 + d^2} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2}$$

10.4 Polar Coordinates

A complex number can also be expressed in polar form as:

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta},$$

where

$$r = |z| = \sqrt{a^2 + b^2} \quad \text{(modulus)}$$
$$\theta = \arg(z) = \tan^{-1}\left(\frac{b}{a}\right) \quad \text{(argument)}$$

The value of θ depends on the quadrant where the complex number lies:

- Quadrant I: $a > 0, b > 0$; use $\tan^{-1}(b/a)$
- Quadrant II: $a < 0, b > 0$; add π to $\tan^{-1}(b/a)$
- Quadrant III: $a < 0, b < 0$; add π to $\tan^{-1}(b/a)$
- Quadrant IV: $a > 0, b < 0$; use $\tan^{-1}(b/a)$

10.4.1 Multiplication and Division in Polar Coordinates

Given:

$$z_1 = r_1 e^{i\theta_1}, \quad z_2 = r_2 e^{i\theta_2},$$

Multiplication:

$$z_1 \cdot z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

This polar form is especially useful in simplifying powers and roots of complex numbers using De Moivre's Theorem.

10.5 Exponentiation and Roots (De Moivre's Theorem)

Let $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$ be a complex number in polar form.

Exponentiation

To raise z to the power $n \in \mathbb{N}$, we use De Moivre's Theorem:

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta)) = r^n e^{in\theta}$$

Roots of Complex Numbers

To find the nth roots of a complex number $z = re^{i\theta}$, we use the formula:

$$z^{1/n} = r^{1/n}\left(\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right), \quad k = 0, 1, \ldots, n-1$$

This yields n distinct roots, each separated by an angle of $\frac{2\pi}{n}$ in the complex plane.

Example:

Solve $z^4 = 1 + \sqrt{3}i$.

Step 1: Convert RHS to polar form:

Let $w = 1 + \sqrt{3}i$. Real part: $a = 1$, Imaginary part: $b = \sqrt{3}$.

$$r = |w| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$$
$$\theta = \arg(w) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$$

So,

$$w = 2\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right).$$

Step 2: Solve $z^4 = w \to z = w^{1/4}$:

Using the root formula:

$$z_k = 2^{1/4}\left(\cos\left(\frac{\pi + 2k\pi}{12}\right) + i\sin\left(\frac{\pi + 2k\pi}{12}\right)\right), \quad k = 0, 1, 2, 3.$$

So the four roots are:

$$z_0 = 2^{1/4}\left(\cos\left(\frac{\pi}{12}\right) + i\sin\left(\frac{\pi}{12}\right)\right)$$
$$z_1 = 2^{1/4}\left(\cos\left(\frac{5\pi}{12}\right) + i\sin\left(\frac{5\pi}{12}\right)\right)$$
$$z_2 = 2^{1/4}\left(\cos\left(\frac{9\pi}{12}\right) + i\sin\left(\frac{9\pi}{12}\right)\right) = 2^{1/4}\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right)$$
$$z_3 = 2^{1/4}\left(\cos\left(\frac{13\pi}{12}\right) + i\sin\left(\frac{13\pi}{12}\right)\right)$$

These represent the four complex 4th roots of $1 + \sqrt{3}i$, equally spaced around the circle of radius $2^{1/4}$ in the complex plane.

10.6 Solving Equations with Complex Numbers

Solving equations in $\mathbb{C}$ can involve various forms. Here are the most common cases:

10.6.1 Linear Equations

Solve for z in $az + b = 0$, where $a, b \in \mathbb{C}$, $a \neq 0$:

$$z = -\frac{b}{a}$$

10.6.2 Equations Involving the Conjugate

Solve for z in equations like $z + \overline{z} = 4$. Let $z = x + iy$, then $\overline{z} = x - iy$. So:

$$z + \overline{z} = 2x \quad \rightarrow \quad x = 2 \quad \Rightarrow \quad z = 2 + iy$$

The imaginary part remains free unless further constraints are given.

10.6.3 Modulus Equations

Solve $|z| = r$. Let $z = x + iy$, then:

$$\sqrt{x^2 + y^2} = r \rightarrow x^2 + y^2 = r^2$$

This is a circle of radius r centered at the origin in the complex plane.

10.6.4 Equations Involving $z \cdot \overline{z}$

Recall $z \cdot \overline{z} = |z|^2$. For example, solve:

$$z \cdot \overline{z} = 9 \rightarrow |z| = 3$$

Again, a circle in the complex plane of radius 3.

10.6.5 Quadratic Equations

Complex roots occur naturally. For example:

$$z^2 + 1 = 0 \rightarrow z^2 = -1 \Rightarrow z = \pm i$$

10.6.6 General Polynomial Equations

Use De Moivre's Theorem or polar form.

Example 1:

$$z^n = w \rightarrow z_k = \sqrt[n]{|w|} \cdot e^{i\left(\frac{\arg(w) + 2k\pi}{n}\right)}, \quad k = 0, 1, \ldots, n-1$$

Example 2:

Solve:

$$\left(\frac{2+3i}{1+i}+\frac{4+5i}{2-2i}\right)\hat{z}=\frac{i+2}{i}$$
$$\left(\frac{-3-i}{4}\right)\hat{z}=\frac{i+2}{i}$$
$$\hat{z}=\frac{i+2}{i}:\frac{-3-i}{4}$$

Example 3:

Solve:

$$z-3i+(2-i)\hat{z}+2=0$$

In this case we let $z=x+iy$ and $\hat{z}=x-iy$.

$$\begin{aligned}
z &= 3i-(2-i)\hat{z}-2\\
x+iy &= 3i-(2-i)(x-iy)-2\\
x+iy &= 3i-[2x-2yi-xi+yi^2]-2\\
x+yi &= 3i-2+2x+2yi+xi+y\\
x+yi &= (y-2-2x)+i(3+2y+x)\\
x &= y-2-2x \;\; y=3+2y+x\\
x &= \frac{y-2}{3} \;\; y=3+2y+\frac{y-2}{3}=-7\\
x &= \frac{-7-2}{3}=-3\\
z &= -3-7i \;\; \hat{z}=-3+7i
\end{aligned}$$

10.7 The Complex Logarithm

The *logarithm* of a complex number is multivalued due to the periodic nature of the complex exponential.

Let $z=re^{i\theta}$ with $r>0$, $\theta\in\mathbb{R}$. Then:

$$\log z=\ln r+i(\theta+2\pi k),\quad k\in\mathbb{Z}$$

Here:

- $\ln r$ is the natural (real) logarithm of the modulus.
- θ is the principal argument $\arg(z)\in(-\pi,\pi]$.
- The term $2\pi k$ accounts for the infinitely many branches of the logarithm in $\mathbb{C}$.

10.7.1 Principal Value

The *principal value* of the complex logarithm is often written:

$$\operatorname{Log} z=\ln|z|+i\operatorname{Arg}(z),\quad \text{where } \operatorname{Arg}(z)\in(-\pi,\pi]$$

Example:

Let $z=-1$. Then:

$$|z| = 1, \quad \arg(z) = \pi, \quad \to \log(-1) = i(\pi + 2\pi k), \quad k \in \mathbb{Z}$$

$$\to \operatorname{Log}(-1) = i\pi$$

The multivalued nature of $\log z$ is crucial in advanced complex analysis, especially in defining analytic continuations and branch cuts.

10.8 Complex Exponents

$$a^x \approx 1\Big(a + \alpha\frac{x}{N}\Big)^N \to e^z := \lim_{N\to\text{inf}}\Big(1 + \frac{z}{N}\Big)^N$$

The process above is called *linearization of the exponential function* by zooming $\alpha \approx \frac{dy}{d}$.

For $e^{ci} = \cos\theta + i\sin\theta$, every exponentiation of a complex number is a rotation in the complex plane.

$$e^{ic} = \lim_{N\to\text{inf}}\left(1 + \frac{ic}{N}\right)^N$$

Now imagine that in a sector of a circumference you put triangles one above the other with a base of length one, a height of $\frac{C}{N}$, and an angle of δ.

$$\tan\delta \approx \delta \text{ for } \delta \ll 1$$

$$1 + \frac{ci}{N} = 1\angle\frac{c}{N}, N \gg 1$$

$$e^{ic} = \lim_{N\to\text{inf}}\Big(1 + \frac{c}{N}\Big)^N \to e^{ci} = 1\angle c = \cos c + i\sin c$$

10.9 Euler's Formula Proof

We know that

$$e^{i\pi} = -1 \text{ and } e^{i\theta} = |r|(\cos\theta + i\sin\theta)$$

$$e^z = \lim_{n\to\infty}\Big(1 + \frac{z}{n}\Big)^n \implies e^{i\pi} = \lim_{n\to\infty}\left(1 + \frac{i\pi}{n}\right)^n = -1$$

$$\implies \lim_{n\to\infty}|r_n| = 1 \quad \lim_{n\to\infty}\theta = 0+$$

Now we can demonstrate the formula.

$$|r_n| = \Big(1 + \Big|\frac{z}{n}\Big|^n\Big) \implies \left(\sqrt{1 + \frac{\pi^2}{n^2}}\right)$$

$$\theta = \sum_{k=1}^{n} n\arctan\frac{\pi}{2} = n\arctan\frac{\pi}{n}$$

$$\lim_{n\to\infty}\left(\sqrt{1 + \frac{\pi^2}{n^2}}\right)^n = \lim_{n\to\infty}\left(1 + \frac{\pi^2}{2n}\right)^{\frac{n}{2}} = \lim_{n\to\infty} e^{\ln\left(1+\frac{\pi^2}{2}\right)\frac{n}{2}} = e^0 = 1$$

$$\lim_{n\to\infty}\theta = \lim_{n\to\infty} n\arctan\frac{\pi}{n} = \lim_{n\to\infty} n^{-1}\arctan\frac{\pi}{n} = 0.$$

Thus, for $e^{i\pi} = 1$ for $x = \pi\forall x \in \lim r_n(x) = 1$ and $\lim\theta(x) = x$.

$\mathfrak{QED}$

10.10 The Fundamental Theorem of Algebra

Given any positive integer $n \geq 1$ and any choice of complex numbers $a_0, a_1, \ldots, a_n$, such that $a_n \neq 0$, the polynomial equation

$$a_n z^n + \cdots + a_1 z + a_0 = 0$$

has at least one solution $z \in \mathbb{C}$.

10.11 Complex Numbers as Matrices

Given a complex number $z = a + bi$, we can represent it as a real 2x2 matrix:

$$z \leftrightarrow \begin{bmatrix} a & -b \\ b & a \end{bmatrix},$$

with the linear function $f : M_2(\mathbb{R}) \to \mathbb{C}$ defined by

$$f\left(\begin{bmatrix} a & -b \\ b & a \end{bmatrix}\right) = a + bi$$

Taking the determinant of this matrix gives us the modulus squared of the complex number:

$$\det\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = a^2 + b^2 = |z|^2$$

Note that this matrix representation preserves addition and multiplication:

$$\begin{bmatrix} a_1 & -b_1 \\ b_1 & a_1 \end{bmatrix} + \begin{bmatrix} a_2 & -b_2 \\ b_2 & a_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & -(b_1 + b_2) \\ b_1 + b_2 & a_1 + a_2 \end{bmatrix}$$

$$\begin{bmatrix} a_1 & -b_1 \\ b_1 & a_1 \end{bmatrix} \cdot \begin{bmatrix} a_2 & -b_2 \\ b_2 & a_2 \end{bmatrix} = \begin{bmatrix} a_1a_2 - b_1b_2 & -(a_1b_2 + b_1a_2) \\ a_1b_2 + b_1a_2 & a_1a_2 - b_1b_2 \end{bmatrix}$$

Hence,

$$M_2(\mathbb{R}) \supseteq \left\{\begin{bmatrix} a & -b \\ b & a \end{bmatrix}\right\} \cong \mathbb{C}.$$

Chapter 11

Topology

In this chapter, we introduce essential vocabulary used in *topology*. Each term is accompanied by a brief explanation and its formal mathematical definition.

11.1 Introduction to Topological Nomenclature

11.1.1 Tological Space (Informal)

A *topological space* is a mathematical space whose elements are called points and which obeys certain axioms. The formal definition will come later, but for the sake of this chapter, we will introduce it later.

11.1.2 Open Ball $B_\varepsilon(x)$

For a metric space (X, d), the *open ball* centered at $x \in X$ with radius $\varepsilon > 0$ is defined as:

$$B_\varepsilon(x) := \{y \in X \mid d(x, y) < \varepsilon\} = (x - \varepsilon, x + \varepsilon).$$

It represents the set of all points within distance ε from x, excluding the boundary.

11.1.3 Open Set

A subset $U \subseteq X$ of a topological space is called *open* if for every point $x \in U$, there exists an $\varepsilon > 0$ such that the open ball $B_\varepsilon(x) \subseteq U$.

Intuitively, an open set contains none of its boundary points and every point has some wiggle room around it.

Example:

We will prove that $(0, 1)$ is an open set.

This means $x_0 > 0$ and $1 - x_0 > 0$. First we define

$$\varepsilon := \frac{1}{2} \min x_0, 1 - x_0;$$

then from $|y - x_0| \leq \varepsilon$ we get

$$y \geq x_0 |y - x_0| \geq x_0 - \varepsilon \geq \frac{1}{2} x_0 > 0,$$
$$y \leq x_0 |y - x_0| \leq x_0 + \varepsilon \leq x_0 + \frac{1}{2}(1 - x_0) = \frac{1}{2} + \frac{1}{2} x_0 < 1.$$

$\mathfrak{QED}$

11.1.4 The Interior of a Set

The *interior* denoted as A° of a set A is the set of all inner points of A.

11.1.5 Closed Set

A subset $A \subseteq X$ is called *closed* if its complement $X \setminus A$ is open. Equivalently, A contains all its limit points. That is, A is closed if it includes its boundary.

Example: $[0, 1]$

11.1.6 Boundary Point

A point $x \in X$ is a *boundary point* of a set $A \subseteq X$ if every open ball around x contains both points in A and in $X \setminus A$. The set of all boundary points is called the *boundary* of A, denoted ∂A.

11.1.7 Accumulation Point / Limit Point

A point $x \in X$ is an *accumulation point* of a set $A \subseteq X$ if every open ball $B_\varepsilon(x)$ contains a point of $A \setminus \{x\}$. In other words, points of A cluster arbitrarily close to x, even if $x \notin A$.

11.1.8 Isolated Point

A point $x \in A$ is an *isolated point* if there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \cap A = \{x\}$. That is, x stands alone in A without other points of A nearby.

Example:

-1 is an isolated point of $-1 \cup (0, 1)$.

11.1.9 Boundary

The set $A \cup \{$ all boundary points $\}$ is called the *boundary* of A. It is denoted as $\overline{A}$.

11.1.10 Compact Set

A set $K \subseteq X$ is *compact* if every open cover of K has a finite sub-cover. In $\mathbb{R}^n$, this is equivalent to K being closed and bounded (by the Heine–Borel theorem).

11.1.11 Dense Set

A subset $D \subseteq X$ is *dense* in X if every point $x \in X$ is either in D or is a limit point of D. Equivalently, the closure of D is X, i.e., $\overline{D} = X$.

Chapter 12

Inequalities

Inequalities play a big role in all of mathematics, but especially at the start of real analysis it is of major importance to get familiar with them.

12.1 The Absolute Value

The *absolute value* of a real number x, denoted by $|x|$, is defined as

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

12.1.1 Properties of Absolute Value

- $|x| \geq 0$
- $|x| = 0 \iff x = 0$
- $|-x| = |x|$
- $|xy| = |x| \cdot |y|$
- $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}, \quad y \neq 0$

12.1.2 Interval Property of the Absolute Value

Let $a, b, x \in \mathbb{R}$ and assume that $b \geq 0$.

$$|x - a| \leq b \implies a - b \leq x \leq a + b$$

Proof:

Assume that $|x - a| \leq b$. Now we have two cases.

If $x - a \geq 0$, then $0 \leq x - a \leq b$, which implies $a \leq x \leq a + b$. Hence,

$$a - b \leq x \leq a + b.$$

If $x - a < 0$, then $|x - a| = -(x - a) = -x + a$, which implies $0 \leq -x + a \leq b$, and hence $a - b \leq x \leq a$. Therefore, combining both cases, we get the desired interval property.

$$a - b \leq x \leq a + b.$$

$\mathfrak{QED}$

12.2 Triangle Inequality

Let $x, y \in \mathbb{R}$. The following inequalities hold:

- $|x + y| \leq |x| + |y|$
- $|x - y| \geq |x| - |y|$
- $||x| - |y|| \leq |x - y|$
- $xy \leq |x| \cdot |y|$

Proof of the triangle inequality:

Let $x \geq 0$ and $y \geq 0$, then $|x + y| = x + y$ and $|x| = x, |y| = y$. Therefore,

$$|x + y| = x + y = |x| + |y|.$$

If $x < 0$ and $y < 0$, then $|x + y| = -(x + y) = -x - y$ and $|x| = -x, |y| = -y$. Therefore,

$$|x + y| = -x - y = |x| + |y|.$$

If $x \geq 0$ and $y < 0$, then $|x + y| = x + (-y) = x - y$ and $|x| = x, -|y| = y$. Because

$$-|y| < |y| \implies x - y < x + |y| = |x| + |y|.$$

Similarly, for the case $x < 0$ and $y \geq 0$, then $|x + y| = -x + y$ and $|x| = -x, |y| = y$. Because

$$-|x| < |x| \implies -x + y < |x| + |y|.$$

QED

12.3 Reciprocal Inequality

$$a < b < c \implies \frac{1}{a} > \frac{1}{b} > \frac{1}{c}$$

12.4 Bernoulli Inequality

Given $x \in \mathbb{R}, x \geq -1, x \neq 0$, then for $n \in \mathbb{N}, n \geq 1$, we have

$$(1 + x)^n \geq 1 + nx$$

This inequality is known as the *Bernoulli inequality* and it is useful for proving that some functions are stricly greater than other on some intervals of the real line, among other applications.

Proof:

Suppose $a \geq -1$.

Let our *base case* be $n = 1$. Then $(1 + x)^1 = 1 + x$, which satisfies the inequality $1 + x \geq 1 + x$.

Induction hypothesis: suppose the inequality holds for some $n = k$, i.e.,

$$(1 + x)^k \geq 1 + kx$$

Induction Step: We want to prove that $(1 + x)^{k+1} \geq 1 + (k + 1)x$.

$$\begin{aligned}(1+x)^{k+1} &= (1+x)^k(1+x)\\ &\geq (1+xn)(1+x)\\ &= 1+x+xn+nx^2\\ &= 1+(k+1)x+nx^2\\ &\geq 1+(k+1)x\end{aligned}$$

In this induction proof we could not just use the induction hypothesis directly and manipulate our expression; we had to multiply both sides by $1+x$ and then expand and simplify to show that the inequality still holds for $n = k+1$.

ΩΕΔ

12.5 Inequalities for Real Numbers

For $x, y \in \mathbb{R}$ the following holds:

$$x < y + \varepsilon, \forall \varepsilon > 0 \iff x \leq y$$

and

$$|x| < \varepsilon, \forall \varepsilon > 0 \iff x = 0.$$

Proof:

For the first, if $x > y$, then I can choose some epsilon only at some point where $y + \varepsilon > x$; therefore, not all epsilons satisfy the inequality.

For the second one, if $x \neq 0$, then $|x| \neq 0 \wedge |x| > 0$; therefore, the only valid option is for x to be zero.

ΩΕΔ

12.5.1 Elementary Young Inequalities

For all $x, y \in \mathbb{R}$, the following holds:

$$xy \leq \frac{1}{2}x^2 + \frac{1}{2}y^2.$$

And this implies:

$$\sqrt{xy} \leq \frac{1}{2}x + \frac{1}{2}y, \quad \forall x, y \geq 0$$

and

$$xy \leq \frac{\varepsilon}{2}x^2 + \frac{1}{2\varepsilon}y^2, \quad \forall x, y \in \mathbb{R}, \varepsilon > 0.$$

12.6 Inequalities With and Without Absolute Value

Inequalities are like normal *equalities*, but with ranges of solutions instead of one particular solution. The inequalities without absolute value are straightforward to solve, just like equations, with the particularity that when dividing both sides by a negative number, the relation $<, >, \geq, \leq$ has to be converted to its opposite.

The case with absolute value is similar, but here we have to account for different cases depending on the expression inside the $|\circ|$.

Example:

Solve $|x+1| \leq 10 + |2x-6|$.

Main cases:

$$F_1 : x + 1 \geq 0 \implies x \geq -1$$

$$F_2 : x + 1 < 0 \implies x < -1$$

Subcases:

$$F_{1,1} : 2x - 6 \geq 0 \implies x \geq 3$$

$$\begin{aligned} x + 1 &\geq 10 + 2x - 6 \\ x &\leq 5 \\ \mathcal{L} &= [3; \infty) \end{aligned}$$

$$F_{1,2} : 2x - 6 < 0 \implies x < 3$$

$$\begin{aligned} x + 1 &\geq 10 - (2x - 6) \\ x &\geq 5 \\ \mathcal{L} &= (-\infty; 3] \end{aligned}$$

$$F_{2,1} : 2x - 6 \geq 0 \implies x \geq 3$$

$$\begin{aligned} -(x + 1) &\geq 10 + 2x - 6 \\ x &\leq \frac{-5}{3} \\ \mathcal{L} &= \emptyset \end{aligned}$$

$$F_{2,2} : 2x - 6 < 0 \implies x < 3$$

$$\begin{aligned} -(x + 1) &\geq 10 - (2x - 6) \\ x &\leq 17 \\ \mathcal{L} &= (-\infty, 3) \end{aligned}$$

Union of the solutions:

$$\mathcal{L} = (-\infty; 3] \cup [3; \infty) \cup \emptyset \cup (-\infty, 3) = (-\infty, \infty)$$

Chapter 13

Means

13.1 Important Means

13.1.1 Arithmetic Mean

The *arithmetic mean* of n numbers $x_1, x_2, \ldots, x_n$ is given by:

$$A = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

13.1.2 Geometric Mean

The *geometric mean* of n numbers $x_1, x_2, \ldots, x_n$ is given by:

$$G = \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$$

13.1.3 Harmonic Mean

The *harmonic mean* of n numbers $x_1, x_2, \ldots, x_n$ is given by:

$$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$$

13.1.4 Quadratic Mean

The *quadratic mean* of n numbers $x_1, x_2, \ldots, x_n$ is given by:

$$Q = \sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}}$$

13.2 Proof of the Arithmetic Mean-Geometric Mean Inequality

Let $a_1, a_2, \ldots, a_n > 0$. We will prove by induction that:

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n},$$

with equality if and only if $a_1 = a_2 = \cdots = a_n$.

Base Case: $n = 2$

We want to prove:

$$\frac{a_1 + a_2}{2} \geq \sqrt{a_1 a_2}$$

Let $a_1, a_2 > 0$. Then, by the identity

$$\left(\frac{a_1 - a_2}{2}\right)^2 \geq 0,$$

we get

$$\frac{a_1^2 - 2a_1a_2 + a_2^2}{4} \geq 0 \Rightarrow a_1^2 + a_2^2 \geq 2a_1a_2.$$

So,

$$(a_1 + a_2)^2 \geq 4a_1a_2 \Rightarrow \left(\frac{a_1 + a_2}{2}\right)^2 \geq a_1a_2,$$

and taking square roots gives the desired result:

$$\frac{a_1 + a_2}{2} \geq \sqrt{a_1 a_2}.$$

For $n \geq 2$:

$$\begin{aligned}
A_{n+1} &:= \left(\sum_{i=1}^{n+1} a_i\right)/(n+1) = \frac{a_1 + a_2 + \cdots + a_n + a_{n+1}}{n+1} \\
G_{n+1} &:= \sqrt[n+1]{a_1a_2\cdots a_n a_{n+1}} = \sqrt[n+1]{(a_1a_2\cdots a_n)a_{n+1}} \\
A_{n-1}^{n+1} &:= \left(\frac{a_1 + a_2 + \cdots + a_{n+1}}{n+1}\right)^{n-1} = A_{n+1}^{n-1} \\
G_{n+1}^{n+1} &:= \left(\sqrt[n+1]{a_1a_2\cdots a_n a_{n+1}}\right)^{n+1} = (a_1a_2\cdots a_n)a_{n+1} = \sqrt[n]{(a_1a_2\cdots a_n)^n}\, a_{n+1}^{n+1} = G_n^n a_{n+1}^{n+1}.
\end{aligned}$$

Then

$$G_{n+1}^{n+1} A_{n+1}^{n-1} = G_n^n a_{n+1}^{n+1} A_{n+1}^{n-1} \leq A_n^n A_{n+1}^{n-1}.$$

This comes from:

$$\begin{aligned}
G_n &\leq A_n \\
G_n^n &\leq A_n^n \\
G_n^n a_{n+1} &\leq A_n^n a_{n+1} \\
G_n^n a_{n+1}^{n+1} A_{n+1}^{n-1} \leq A_n^n A_{n+1}^{n-1} &= \left(A_n^n (a_{n+1}A_{n+1})^{n-1}\right)^{\frac{n}{n}} \\
&\leq A_n^n \left(\frac{a_{n+1} + A_{n+1} + \cdots A_{n+1}}{n}\right)^n \\
&A_n^n \left(\frac{a_{n+1} + (n-1)A_{n+1}}{n}\right)^n \\
&\left(A_n \frac{a_{n+1} + (n-1)A_{n+1}}{n}\right)^n.
\end{aligned}$$

Note that:

$$\left(A_n \frac{a_{n+1}+(n-1)A_{n+1}}{n}\right) \to \left(\sqrt{A_n \frac{a_{n+1}+(n-1)A_{n+1}}{n}}\right)^2 \leq \left(\frac{A_n + \frac{a_{n+1}+(n-1)A_{n+1}}{n}}{2}\right)^{2n}.$$

Now, with power of n, we have

$$\begin{aligned}\left(A_n + \frac{a_{n+1}+(n-1)A_{n+1}}{n}\right)^n &\leq \left(\frac{A_n + \frac{a_{n+1}+(n-1)A_{n+1}}{n}}{2}\right)^{2n} \\ &= \left(\frac{A_n}{2} + \frac{a_{n+1}+(n-1)A_{n+1}}{2n}\right)^{2n} \\ &= \left(\frac{A_n n}{2n} + \frac{a_{n+1}+(n-1)A_{n+1}}{2n}\right)^{2n} \\ &= \left(\frac{A_n n + a_{n+1}+(n-1)A_{n+1}}{2n}\right)^{2n} \\ &= \left(\frac{A_{n+1}(n+1)+(n-1)A_{n+1}}{2n}\right)^{2n} \\ &= \left(\frac{2nA_{n+1}}{2n}\right)^{2n} = (A_{n+1})^{2n}.\end{aligned}$$

Now, we have proven that $G_{n+1}^{n+1}A_{n+1}^{n-1} \leq (A_{n+1})^{2n}$. By dividing both sides by A_{n+1}^{n-1}, we get

$$G_{n+1}^{n+1} \leq A_{n+1}^{n+1}.$$

$\mathfrak{QED}$

13.3 Proof of the Harmonic Mean-Geometric Mean Inequality

Let $a_1, a_2, \ldots, a_n > 0$. We will prove the inequality.

We know that $G_n \leq A_n$, which implies

$$\begin{aligned} G_n &\leq A_n \\ \sqrt[n]{x_1 \ldots x_2 \cdots x_n} &\leq \frac{x_1 + x_2 + \cdots + x_n}{n} \\ \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}} &\leq \sqrt[n]{x_1 \cdot x_2 \cdots x_n}. \end{aligned}$$

This concludes the proof of the harmonic mean-geometric mean inequality.

$\mathfrak{QED}$

Chapter 14

Solving Polynomial Equations

In this section, we discuss the solution formulas for polynomial equations of degrees 2 and 3: the PQ formula, the ABC formula, and the Cubic formula. We also derive each of them step by step. There will be no mention of the formula for polynomials of degree 4 since it is not really used in worksheets.

14.1 The PQ Formula

The PQ formula

$$x_{1,2} = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}$$

solves quadratic equations of the form:

$$x^2 + px + q = 0.$$

Proof:

To derive the PQ formula, we complete the square

$$\begin{aligned} x^2 + px + q &= 0 \\ x^2 + px &= -q \\ x^2 + px + \left(\frac{p}{2}\right)^2 &= -q + \left(\frac{p}{2}\right)^2 \\ \left(x + \frac{p}{2}\right)^2 &= \left(\frac{p}{2}\right)^2 - q \\ x + \frac{p}{2} &= \pm\sqrt{\left(\frac{p}{2}\right)^2 - q} \\ x &= -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}. \end{aligned}$$

14.2 The Quadratic Formula

The general quadratic equation is:

$$ax^2 + bx + c = 0 \quad \text{with } a \neq 0$$

which can be solved by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Proof:

We normalize the equation by dividing through by a and complete the square:

$$\begin{aligned}
ax^2 + bx + c &= 0 \\
x^2 + \frac{b}{a}x + \frac{c}{a} &= 0 \\
x^2 + \frac{b}{a}x &= -\frac{c}{a} \\
x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 &= -\frac{c}{a} + \left(\frac{b}{2a}\right)^2 \\
\left(x + \frac{b}{2a}\right)^2 &= \frac{b^2 - 4ac}{4a^2} \\
x + \frac{b}{2a} &= \pm\frac{\sqrt{b^2 - 4ac}}{2a} \\
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\end{aligned}$$

14.3 The Cubic Formula

To solve a general cubic equation:

$$ax^3 + bx^2 + cx + d = 0,$$

we use the following formula known as *Cardano's formula for depressed cubics.*

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}$$

Proof:

We first need to transform our cubic polynomial into its depressed cubic using a substitution.

Step 1: Depress the cubic

Let $x = t - \frac{b}{3a}$, then the equation becomes:

$$t^3 + pt + q = 0$$

with:

$$p = \frac{3ac - b^2}{3a^2}, \quad q = \frac{2b^3 - 9abc + 27a^2d}{27a^3}$$

Step 2: Solve the depressed cubic using Cardano's method

Assume a solution of the form

$$t = u + v.$$

Then substitute and simplify

$$(u+v)^3 + p(u+v) + q = 0.$$

Expanding and setting

$$u^3 + v^3 + (3uv + p)(u+v) + q = 0.$$

To eliminate the $(u+v)$ term, set:

$$3uv + p = 0 \quad \Rightarrow \quad uv = -\frac{p}{3}.$$

Now

$$u^3 + v^3 = -q.$$

Let

$$u^3 = A, \quad v^3 = B \quad \Rightarrow \quad A + B = -q, \quad AB = -\frac{p^3}{27}.$$

These are the roots of the quadratic

$$z^2 + qz - \frac{p^3}{27} = 0;$$

solve for A and B, then take cube roots to get u and v. The final solution is:

$$x = u + v$$

Finally,

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}$$

This formula gives one real root. The other roots (if real) can be found using trigonometric or complex methods depending on the discriminant.

Chapter 15

The Binomial Coefficient

In this short section, we will cover the definition and properties of the *binomial coefficient*. This object can represent the number of combinations in which we can take k from n distinct items.

It is defined as:

$$\binom{n}{k} := \frac{n!}{k!(n-k)!},$$

or as a product:

$$\prod_{j=1}^{n} \frac{n-k+j}{j},$$

with n and k as natural numbers (for the moment).

15.1 Properties of the Binomial Coefficient

- $\binom{n}{k} = \binom{n}{n-k}$
- $\binom{n}{k-1} = \binom{n}{k}\frac{n-k}{k+1}$
- $\sum_{k=0}^{n}\binom{n}{k} = 2^n$
- $\sum_{k=0}^{n}\binom{m}{k} = \binom{n+1}{m+1}$
- $\binom{n}{0} = 1 = \binom{n}{n}$
- $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$
- $\binom{n}{k} = (-1)^k\binom{k-n-1}{k}$
- *Pascals Identity*: $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$

For

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

we think of it as: "number of k-elements subsets of $S = \{0, 1, \ldots, n\}$" is equal to "number of k-elements subsets of S with 0" plus "number of k-elements subset of S without 0"

Proof of Pascal's Indentity:

$$\begin{aligned}
\binom{n}{k} + \binom{n}{k-1} &= \binom{n+1}{k} \\
\frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-(k-1))!} &= \frac{(n+1)!}{k!(n+1-k)!} \\
\frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} &= \frac{(n+1)!}{k!(n+1-k)!} \\
\frac{(n+1-k)n!}{k!(n+1-k)!} + \frac{kn!}{k!(n-k+1)!} &= \frac{(n+1)!}{k!(n+1-k)!} \\
\frac{(n+1-k)n! + kn!}{k!(n+1-k)!} &= \frac{(n+1)!}{k!(n+1-k)!} \\
\frac{n!(n+1-k+k)}{k!(n+1-k)!} &= \frac{(n+1)!}{k!(n+1-k)!} \\
\frac{n!(n+1)}{k!(n+1-k)!} &= \frac{(n+1)!}{k!(n+1-k)!} \\
\frac{(n+1)!}{k!(n+1-k)!} &= \frac{(n+1)!}{k!(n+1-k)!}
\end{aligned}$$

$\mathfrak{QED}$

15.2 Proof that the Binomial Coefficient is a Natural Number

We will prove this via induction for all $n \geq k$.

Base case: $k = 0$.

$$\binom{0}{0} = 1, \quad \binom{n+1}{n+1} = 1, \binom{n}{0} = 1$$

Induction hypothesis:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \binom{n+1}{k} - \binom{n}{k-1} \in \mathbb{N}$$

Induction step:

We want to prove

$$\binom{n+1}{k} \in \mathbb{N}$$

We know that

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

Note that we can manipulate this expression so that

$$\binom{n}{k} = \binom{n+1}{k} - \binom{n}{k-1}$$

$\mathfrak{QED}$

15.3 The Binomial Theorem

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

The *binomial*, and later multinomial theorem looks out of place at first, but if we first take a look at the sum without the sum without the binomial coefficient, we will notice that $a^{n-k}b^k$ is just a patter in which the power of x decreases and the one of y increases. Finally the binomial coefficients just tells us "how many ways are there to get the specific $a^{n-k}b^k$"

Proof:

We will prove that the equality holds for any n.

For n, we have $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k$.

For $n+1$, we get:

$$(a+b)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k}b^k.$$

Now, let us manipulate the expression to demonstrate our theorem.

Assume that the statement is true for some integer $n \geq 0$.

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k.$$

Now, for some greater n,

$$(a+b)^{n+1} = (a+b)(a+b)^n.$$

Using the induction hypothesis:

$$(a+b)^{n+1} = (a+b)\sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k.$$

We distribute:

$$(a+b)^{n+1} = \sum_{k=0}^{n} \binom{n}{k} a^{\,n+1-k}b^{\,k} + \sum_{k=0}^{n} \binom{n}{k} a^{\,n-k}b^{\,k+1}.$$

In the second sum, change the index of summation by setting $j = k+1$, then rename j back to k:

$$\sum_{k=0}^{n} \binom{n}{k} a^{\,n-k}b^{\,k+1} = \sum_{k=1}^{n+1} \binom{n}{k-1} a^{\,n+1-k}b^{\,k}.$$

Now, combine both sums:

$$(a+b)^{n+1} = \sum_{k=0}^{n+1} \left(\binom{n}{k} + \binom{n}{k-1} \right) a^{n+1-k}b^k,$$

where we use the convention $\binom{n}{k} = 0$ for $k < 0$ or $k > n$.

By *Pascal's Identity*:

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}.$$

Therefore,

$$(a+b)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k} b^{k}.$$

$\mathfrak{QED}$

Example:

We will expand the binomial $(x+y)^3$ using the binomial theorem with some extra information to make it easier to understand.

First,

$$(x+y)^3 = (x+y)(x+y)(x+y) = \sum_{k=0}^{3} \binom{3}{k} x^{3-k} y^{k} = \binom{3}{0} x^3 + \binom{3}{1} x^2 y + \binom{3}{2} x y^2 + \binom{3}{3} y^3.$$

$k=0$	xxx	$\binom{3}{0} x^{3-0} y^0$
$k=1$	$xxy + xyx + yxx$	$\binom{3}{1} x^{3-1} y^1$
$k=2$	$xyy + yxy + yyx$	$\binom{3}{2} x^{3-2} y^2$
$k=3$	yyy	$\binom{3}{3} x^{3-3} y^3$

15.4 Pascal's Triangle

This is a triangle built by the recursion $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$.

$n=0$						1						
$n=1$					1		1					
$n=2$				1		2		1				
$n=3$			1		3		3		1			
$n=4$		1		4		6		4		1		
$n=5$	1		5		10		10		5		1	
$n=6$	1		6		15		20		15		6		1

Here is the triangle, but with the binomial coefficients:

$n=0$						$\binom{0}{0}$						
$n=1$					$\binom{1}{0}$		$\binom{1}{1}$					
$n=2$				$\binom{2}{0}$		$\binom{2}{1}$		$\binom{2}{2}$				
$n=3$			$\binom{3}{0}$		$\binom{3}{1}$		$\binom{3}{2}$		$\binom{3}{3}$			

It can be used with the Binomial Theorem to get the coefficients of a polynomial of degree n.

Example:

$$(a+b)^2 = 1a^2b^0 + 2a^1b^1 + 1a^0b^2,$$

or

$$(a-b)^2 = 1a^2b^0 - 2a^1b^1 + 1a^0b^2.$$

Chapter 16

Proportionality and the Rule of Three

This is handled in most schools, but since it tends to be forgotten and can be applied in real life, here it is.

16.1 Proportionality

Two quantities are said to be *proportional* if their ratio remains constant.

16.1.1 Direct Proportionality

Two quantities a and b are in *direct proportion* if:

$$\frac{a}{b} = k \quad \Rightarrow \quad a = k \cdot b,$$

where k is the constant of proportionality.

Example:

If 2 pencils cost 1, then 4 pencils cost 2. The ratio is constant: $\frac{2}{1} = \frac{4}{2}$.

16.1.2 Inverse Proportionality

Two quantities a and b are in *inverse proportion* if their product is constant:

$$a \cdot b = k$$

Example:

If 4 workers finish a job in 6 hours, then 2 workers would need 12 hours:

$$4 \cdot 6 = 2 \cdot 12 = 24$$

16.2 The Rule of Three (Simple)

The *Rule of Three* is a method used to find a fourth value when three values are known and a proportional relationship is assumed.

16.2.1 Direct Rule of Three (Simple)

Given: $a : b = c : x$, solve for x:

$$x = \frac{b \cdot c}{a}$$

Example:

If 3 apples cost 6, how much do 5 apples cost?

$$x = \frac{6 \cdot 5}{3} = 10$$

16.2.2 Inverse Rule of Three (Simple)

If the relationship is inverse:

$$a : b = x : c \quad \Rightarrow \quad x = \frac{a \cdot c}{b}$$

Example:

If 5 people finish a task in 8 hours, how long will 10 people need?

$$x = \frac{5 \cdot 8}{10} = 4$$

16.3 The Rule of Three (Compound)

The *Compound Rule of Three* (or *composed rule of three*) involves more than two variables.

Example:

If 4 machines produce 120 items in 5 hours, how many items will 6 machines produce in 8 hours?

Step 1: Set up the proportional relationships:

$$\text{Items} \propto \text{Machines} \quad \text{(direct)} \text{Items} \propto \text{Time} \quad \text{(direct)}$$

Step 2: Adjust the quantities:

$$\begin{aligned} \text{Initial: } & 4 \text{ machines, } 5 \text{ hrs } \rightarrow 120 \text{ items} \\ \text{New: } & 6 \text{ machines, } 8 \text{ hrs } \rightarrow x \text{ items} \end{aligned}$$

Step 3: Apply proportionality:

$$x = 120 \cdot \frac{6}{4} \cdot \frac{8}{5} = 120 \cdot 1.5 \cdot 1.6 = 288$$

Answer: 288 items.

Summary Table

Type	Formula
Direct Proportion	$x = \frac{b \cdot c}{a}$
Inverse Proportion	$x = \frac{a \cdot c}{b}$
Compound Rule of Three	Multiply by all direct ratios and divide by all inverse ratios

Chapter 17

Factorization

Factorization is the process of writing a mathematical expression as a product of its factors. This is a fundamental technique which can help in all kinds of situations.

17.1 Common Factorization Techniques

17.1.1 Common Factor

Factor out the greatest common divisor (GCD) of all terms.

Example:

$$6x^2 + 9x = 3x(2x + 3)$$

17.1.2 Difference of Squares

$$a^2 - b^2 = (a - b)(a + b)$$

17.1.3 Sum of Cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

17.1.4 Difference of Cubes

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Example:

$$x^3 + 8 = (x + 2)(x^2 - 2x + 4)$$

17.1.5 Trinomial: Special Case $x^2 + bx + c$

This is the case where $a = 1$. Find two numbers whose product is c and whose sum is b.

Example:

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

17.1.6 Trinomial: General Form $ax^2 + bx + c$

As in the previous case, we are going to use a similar procedure, but we start by multiplying and dividing by $\frac{a}{a}$, and then proceed with the rest. Look at the example:

$$\begin{aligned} 3x^2 - 5x - 2 &= \frac{3(3x^2 - 5x - 2)}{3} \\ &= \frac{(3x)^2 - 5(3x) - 6}{3} \\ &= \frac{(3x - 6)(3x + 1)}{3} \\ &= (x - 2)(3x + 1). \end{aligned}$$

17.1.7 Perfect Square Trinomial

These follow the identities:

$$(a + b)^2 = a^2 + 2ab + b^2,$$

$$(a - b)^2 = a^2 - 2ab + b^2.$$

Example:

$$x^2 + 6x + 9 = (x + 3)^2$$

17.1.8 Rationalization of Radicals

Rationalizing removes radicals from the denominator.

Example (Single Radical):

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Example (Binomial):

$$\frac{1}{\sqrt{3} + 1} = \frac{\sqrt{3} - 1}{(\sqrt{3} + 1)(\sqrt{3} - 1)} = \frac{\sqrt{3} - 1}{2}$$

17.2 Substitution

Substitute a more complex expression with a single variable, factor, then back-substitute.

Example:

Given $x^4 + 2x^2 + 1$. Let $y = x^2$, then

$$y^2 + 2y + 1 = (y + 1)^2 = (x^2 + 1)^2.$$

This one is extremly important in the analysis.

17.3 Horner's Method

This method is used to divide a polynomial by a binomial of the form $(x - r)$.

We can apply this method by following these steps:

1. Write the coefficients of the polynomial $P_1(x)$.
2. Find the factors of the term with x^0.
3. Bring down the first coefficient.
4. Multiply it by r, and add to the next coefficient. Repeat until you reach the term with x^0.
5. If the final addition gives zero, then you have found a zero of the polynomial. Now, it can be written in the following form: $P_2(x)(x-r)$, with $P(x)$ being the new polynomial whose coefficients were computed during the process to get the zero.

Example:

Given $P_1(x) = x^3 - 6x^2 + 11x - 6$.

The factors of -6 are $\{\pm 1, \pm 2, \pm 3, \pm 6\}$. We use $r = 1$:

$$\begin{array}{c|cccc} 1 & 1 & -6 & 11 & -6 \\ & & 1 & -5 & 6 \\ \hline & 1 & -5 & 6 & 0 \end{array} \implies P_2(x) = (x^2 - 5x + 6)(x - 1)$$

Now, by repeating the process with $r = 2$:

$$\begin{array}{c|ccc} 2 & 1 & -5 & 6 \\ & & 2 & -6 \\ \hline & 1 & -3 & 0 \end{array} \implies P_3(x) = (x - 3)(x - 2)(x - 1)$$

The final result is $(x - 3)(x - 2)(x - 1)$.

17.4 Long Division of Polynomials

The *long division for polynomials* works exactly as the division algorithm for real numbers, but we use powers of x and the scalars of $\mathbb{F}$ for the multiplication step.

Example:

$$\begin{array}{r|l} & x^2 + 2x + 2 \\ \hline x - 1 & x^3 + x^2 - 1 \\ & -x^3 + x^2 \\ \hline & 2x^2 \\ & -2x^2 + 2x \\ \hline & 2x - 1 \\ & -2x + 2 \\ \hline & 1 \end{array}$$

with the result: $x^2 + 2x + 2 + \frac{1}{x-1}$.

Chapter 18

Recursion

Recursion is a term used to refer to something defined in terms of itself. An example is the factorial: $n! = (n-1)!$, with $0! = 1$ being the base case where the recursion stops.

18.1 From Recursive to Closed Form

Often, a sequence defined recursively can be rewritten in an explicit or closed form.

Recursive form defines a_n based on previous terms (e.g., $a_n = a_{n-1} + 2$).

Closed form expresses a_n directly in terms of n, with no reference to previous terms.

18.2 Linear Recursion

Linear recursion has the form:

$$\begin{aligned} a_n &= \lambda a_{n-1} + c \\ a_1 = \lambda a_0 + c &\implies a_2 = \lambda(\lambda a_0 + c) + c \\ \implies a_n &= \lambda^n a_0 + (1 + \lambda + \lambda^2 + \cdots + \lambda^{n-1})c. \end{aligned}$$

Therefore,

$$a_n = \lambda^n a_0 + \frac{\lambda^n - 1}{\lambda - 1} c.$$

18.2.1 General Method for Linear Recursion

1. Calculate some terms and observe the difference or quotient between them. Maybe the pattern is clear, and you can build a geometric or arithmetic sequence directly.
2. Identify the type (e.g., linear, homogeneous, constant coefficients).
3. Solve the characteristic equation (if linear).
4. Use initial conditions to determine constants.

18.3 Recursive Inequalities

Recursive inequalities bound a sequence rather than define it exactly.

Example:

Suppose

$$a_{n+1} \leq \frac{1}{2}a_n + 3.$$

To analyze

- Guess a bound (e.g., show $a_n \leq M$ for all n).
- Use induction or iteration to justify convergence or boundedness.
- Compare with a simpler sequence (e.g., geometric series).

These techniques are common in analysis, approximation, and numerical algorithms.

18.4 Closed Form of the Fibonacci Sequence

The Fibonacci sequence is defined recursively:

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2}.$$

The characteristic equation is:

$$x^2 - x - 1 = 0 \Rightarrow x = \varphi = \frac{1+\sqrt{5}}{2}, \quad \psi = \frac{1-\sqrt{5}}{2}.$$

The closed form (Binet's Formula) is:

$$F_n = \frac{1}{\sqrt{5}}\left(\varphi^n - \psi^n\right).$$

- $\varphi \approx 1.618$ is the golden ratio.
- This formula demonstrates exponential growth of F_n.

This formula can be derived using the steps of the chapter about **generating functions**.

Chapter 19

Sums and Products

These notations are essential for working with sequences, series, and algebraic expressions. In mathematics, the sum and product notations are compact ways to represent repeated addition and multiplication, respectively.

19.1 Sum Notation $\sum$

The summation symbol $\sum$ represents the addition of a sequence of terms:

$$\sum_{i=m}^{n} a_i = a_m + a_{m+1} + \cdots + a_n$$

Here, i is the index of summation, m is the lower bound, and n is the upper bound.

Linearity:

$$\sum_{i=m}^{n} (a_i + b_i) = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i$$

$$\sum_{i=m}^{n} c \cdot a_i = c \cdot \sum_{i=m}^{n} a_i$$

Splitting:

$$\sum_{i=m}^{n} a_i = \sum_{i=m}^{k} a_i + \sum_{i=k+1}^{n} a_i \quad (m \leq k < n)$$

19.1.1 Change of Index

Let $j = i + k$, then:

$$\sum_{i=m}^{n} a_i = \sum_{j=m+k}^{n+k} a_{j-k}$$

Example:

$$\sum_{i=1}^{4} a_i = \sum_{j=2}^{5} a_{j-1}$$

19.1.2 Power Sums and Their Formulas

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$
$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$
$$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$
$$\sum_{i=1}^{n} i^k = (\text{Higher-order polynomial in } n)$$

19.1.3 Derivation of $\sum_{i=1}^{n} i^2$

We use the method of finite differences or induction. Assume a quadratic form:

$$\sum_{i=1}^{n} i^2 = An^3 + Bn^2 + Cn$$

Plug in small values of n (e.g., 1, 2, 3), solve the system of equations to find:

$$A = \frac{1}{3}, \quad B = \frac{1}{2}, \quad C = \frac{1}{6} \Rightarrow \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

19.1.4 Telescoping Sum

A telescoping sum is a sum in which intermediate terms cancel out, leaving only the first and last terms.

$$\sum_{k=m}^{n} (a_k - a_{k+1}) = a_m - a_{n+1}$$

Example:

$$\sum_{i=1}^{n} \left(\frac{1}{i} - \frac{1}{i+1}\right) = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{n+1}$$

19.2 Geometric Series

19.2.1 Finite Geometric Series

For a geometric sequence $a, ar, ar^2, \ldots, ar^n$:

$$\sum_{i=0}^{n} ar^i = a \cdot \frac{1 - r^{n+1}}{1 - r}, \quad r \neq 1$$

19.2.2 Infinite Geometric Series

If $|r| < 1$, then:

$$\sum_{i=0}^{\infty} ar^i = \frac{a}{1 - r}$$

19.2.3 Proof of the Geometric Series

Let $S_n = \sum_{k=0}^{n} aq^k$ with $|q| < 1$.

Then

$$\begin{aligned} S_n &= a + -aq^{n+1} \\ S_n - \sum_{k=0}^{n} aq^{k+1} &= a - aq^{n+1} \\ S_n - q\left(\sum_{k=0}^{n} aq^k\right) &= a - aq^{n+1} \\ S_n - qS_n &= a - aq^{n+1} \\ S_n(1-q) &= a - aq^{n+1} \\ S_n &= a\frac{1-q^{n+1}}{1-q} \end{aligned}$$

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19.3 Product Notation $\prod$

The product notation $\prod$ represents repeated multiplication:

$$\prod_{i=m}^{n} a_i = a_m \cdot a_{m+1} \cdot \cdots \cdot a_n$$

Multiplication:

$$\prod_{i=m}^{n} (a_i \cdot b_i) = \left(\prod_{i=m}^{n} a_i\right) \cdot \left(\prod_{i=m}^{n} b_i\right)$$

Power Rule:

$$\prod_{i=m}^{n} a^k = a^{k(n-m+1)}$$

19.3.1 Change of Index

Let $j = i + k$, then:

$$\prod_{i=m}^{n} a_i = \prod_{j=m+k}^{n+k} a_{j-k}$$

Example:

$$\prod_{i=1}^{3} a_i = \prod_{j=2}^{4} a_{j-1}$$

19.3.2 Telescoping Product

A telescoping product occurs when consecutive terms simplify or cancel each other.

$$\prod_{k=m}^{n} \frac{a_k}{a_{k+1}} = \frac{a_m}{a_{n+1}}$$

Example:

$$\prod_{i=1}^{n} \frac{i}{i+1} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{n}{n+1} = \frac{1}{n+1}$$

Chapter 20

Logarithms

The *logarithm* of a number x with base b is the exponent to which the base must be raised to produce that number. It is denoted as:

$$\log_b(x) = y \iff b^y = x,$$

where $b > 0$, $b \neq 1$, and $x > 0$.

$$\log_2(8) = 3 \quad \text{because} \quad 2^3 = 8.$$

We can also approximate a logarithm by dividing our target x by the base b repeatedly until we get 1 or less.

20.1 Properties of Logarithms

- *Product Rule:* $\log_b(xy) = \log_b(x) + \log_b(y)$
- *Quotient Rule:* $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$
- *Power Rule:* $\log_b(x^k) = k \cdot \log_b(x)$
- *Logarithm of the Base:* $\log_b(b) = 1$
- *Logarithm of One:* $\log_b(1) = 0$
- *Change of Base for Powers:* $\log_{b^k}(x^w) = \frac{w}{k} \cdot \log_b(x)$
- *Logarithm of the Reciprocal:* $\log_b\left(\frac{1}{x}\right) = -\log_b(x)$
- *Logarithm of a Power of the Base:* $\log_b(b^x) = x$
- *Exponential and Natural Logarithm:* $e^{\ln(x)} = x$.
- *Natural Logarithm of a Fraction:* If $0 < a < 1$, then $\ln(a)$ is a negative number.
- *Monotonicity:* $\log(x) < \log(y)$ for $x < y$.
- *Logarithm Inequality:* $\log(x+1) \leq x$ for $x > -1$.
- *Logarithm Growth:* For all $c > 0$, there is some M such that $\log(x) < cx$ for all $x \in (0, \infty), x > M$.

20.1.1 Fundamental Identity of Logarithms

$$b^{\log_b(x)} = x$$

20.1.2 Change of Base Formula

$$\log_b(x) = \frac{\log_k(x)}{\log_k(b)},$$

where k is any positive number different from 1.

Proof:

$$\begin{aligned} b^a &= x \\ \log_k(b^a) &= \log_k(x) \\ a\log_k(b) &= \log_k(x) \\ a &= \frac{\log_k(x)}{\log_k(b)} \\ \log_b(x) &= \frac{\log_k(x)}{\log_k(b)} \end{aligned}$$

QED

20.1.3 The Chain Rule

$$\log_y(a)\log_a(b) = \log_y(b)$$

Let

$$y^w = a, \quad a^x = b, \quad y^z = b.$$

Then

$$\begin{aligned} y^z &= a^x \\ y^z &= (y^w)^x \\ y^z &= y^{wx} \\ z &= wx \iff \log_y(b) = \log_y(a)\log_a(b) \end{aligned}$$

QED

Proof:

20.2 The Derivative of the Natural Logarithm

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Let $f(x) = \ln(x)$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\ln(x+h) - \ln(x)}{h} \\ &= f'(x) = \lim_{h\to 0} \frac{\ln\left(\frac{x+h}{x}\right)}{h} \\ &= f'(x) = \lim_{h\to 0} \frac{\ln\left(1+\frac{h}{x}\right)}{h} \\ &= f'(x) = \lim_{h\to 0} \ln\left(1+\frac{h}{x}\right)^{\frac{1}{h}} \end{aligned}$$

Now, let $n = \frac{h}{x}$.

$$\begin{aligned} f'(x) &= \lim_{n\to 0} \ln(1+n)^{\frac{x}{h}\frac{1}{x}} \\ f'(x) &= \lim_{n\to 0} \frac{1}{x} \ln(1+n)^{\frac{1}{n}} \\ f'(x) &= \frac{1}{x} \ln\left(\lim_{n\to 0} (1+n)^{\frac{1}{n}}\right) \\ f'(x) &= \frac{1}{x} \end{aligned}$$

Therefore, the derivative of the natural logarithm is:

$$\frac{d}{dx}\ln(x) = \frac{1}{x}.$$

QED

20.3 Proofs of the Properties

20.3.1 Product

Let $a^n a^m = a^{n+m}$, and $a^n = x$, and $a^m = y$, then

$$\log_a x = n \text{ and } \log_a y = m \to \log_a(xy) = n + m$$

$$\log_a(xy) = \log_a x + \log_a y$$

QED

20.3.2 Quotient

Let $\frac{a^n}{a^m} = a^{n-m}$, and $a^n = x$, and $a^m = y$, then as in the previous proof,

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

QED

20.3.3 Power

Let $(a^n)^m = a^{nm}$, and $a^n = x$, and $a^{nm} = y$.

$$\log_x y = m \text{ and } \log_a x = n \text{ thus, } \log_a y = (\log_a x)m$$

QED

20.3.4 Power Rule: General Case

Let $(a^n)^m = b^x$, then $\log_{a^n} b^x = m$. Now, let us manipulate the original expression:

$$\begin{aligned} \log_a (a^n)^m &= \log_a b^x \\ nm \log_a a &= x \log_a b \\ m = \frac{x}{n} \log_a b &= \log_{a^n} b^x \end{aligned}$$

QED

Chapter 21

Fractions, Roots, and Exponents

This is a brief chapter to review the properties of fractions, roots, and exponents. It is not a complex chapter, but it is useful to include it in the compendium.

The following statements apply for all $a, b \in \mathbb{R}$.

21.1 Properties of Fractions

- *Addition*: $\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$.
- *Multiplication*: $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$.
- *Division*: $\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$.
- *Reciprocal*: $\frac{a}{b} = \frac{1}{\frac{b}{a}}$.
- *Equality*: $\frac{a}{b} = \frac{c}{d} \iff ad = bc$.
- *Simplification*: $\frac{ac}{bc} = \frac{a}{b}$.

21.2 Properties of Roots and Exponents

- *Multiplication*: $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$.
- *Quotient*: $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$.
- *Nested Roots*: $\sqrt[n]{\sqrt[m]{a}} = \sqrt[n]{\sqrt[m]{a}}$.
- *Factor Outside Root*: $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.
- *Multiplication*: $a^m \cdot a^n = a^{m+n}$.
- *Division*: $a^m \div a^n = a^{m-n}$.
- *Power of a Power*: $(a^m)^n = a^{mn}$.
- *Negative Exponent*: $a^{-n} = \frac{1}{a^n}$.
- *Fractional Exponent*: $a^{\frac{m}{n}} = \sqrt[n]{a^m}$.
- *Square Root*: $a^{\frac{1}{2}} = \sqrt{a}$.
- *Cube Root*: $a^{\frac{1}{3}} = \sqrt[3]{a}$.
- *Nth Root*: $a^{\frac{1}{n}} = \sqrt[n]{a}$.
- *Zero Exponent*: $a^0 = 1$.
- *Identity Exponent*: $a^1 = a$.

Chapter 22

Generating Functions

Given a sequence $\{a_n\}_{n\geq 0}$, its *generating function* is defined as:

$$F(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Here, x is an indeterminate, and the series is usually treated formally, meaning we do not worry about convergence unless the context requires it.

22.1 Connection to Subsets

There is a deep similarity between generating functions and the process of forming subsets. Consider the generating function for the set $\{1, x\}$:

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

This expansion encodes all subsets of an n-element set, where each term x^k corresponds to selecting a subset of size k. The coefficient $\binom{n}{k}$ gives the number of such subsets.

In this way, generating functions can be viewed as a powerful algebraic analogue of subset selection and counting, with the exponents representing sizes or weights, and the coefficients representing counts.

Generating functions serve as a bridge between algebra and combinatorics, allowing abstract operations to yield concrete combinatorial results.

22.2 Closed Form of the Fibonacci Sequence

Let us use generating functions to find a closed form for the Fibonacci numbers.

We know that $F_0 = 1$ and $F_1 = 1$, and for $n \geq 2$, $F_n = F_{n-1} + F_{n-2}$.

Now, let us use the following function:

$$F(x) = \sum_{n=0}^{\infty} F_n x^n,$$

which is going to have the Fibonacci numbers as the coefficients of the power series.

$$1 + x + 2x^2 + 3x^3 + 5x^4 + \ldots,$$

but before that, we have to look at when our recursive formula is defined.

In this case, $n = 2$. Thus, we have to write our expression in the following way:

$$\begin{aligned} F(x) &= 1 + x + \sum_{n=2}^{\infty} F_n x^n \\ &= 1 + x + \sum_{n=2}^{\infty} F_{n-1} x^n + \sum_{n=2}^{\infty} F_{n-2} x^n. \end{aligned}$$

Here, we are going to adjust our expression by making our n be equal

$$\begin{aligned} F(x) &= 1 + x + \sum_{n=2}^{\infty} F_{n-1} x^n + \sum_{n=2}^{\infty} F_{n-2} x^n \\ F(x) &= 1 + x + x \left(\sum_{n=2}^{\infty} F_{n-1} x^{n-1} \right) + x^2 \left(\sum_{n=2}^{\infty} F_{n-2} x^{n-2} \right). \end{aligned}$$

Now, we are going to reset our sum to start from zero by adding a smart zero to our term with $n - 1$. This allows us to shift the sum to start from 0, because when $n = 1$, $F_{n-1}x^{n-1}$ turns into $F_0 x^0$.

$$\begin{aligned} F(x) &= 1 + x + x \left(-F_0 x^0 + F_0 x^0 + \sum_{n=2}^{\infty} F_{n-1} x^{n-1} \right) + x^2 \left(\sum_{n=2}^{\infty} F_n x^n \right) \\ F(x) &= 1 + x + x \left(-F_0 x^0 + \sum_{n=1}^{\infty} F_{n-1} x^{n-1} \right) + x^2 \left(\sum_{n=2}^{\infty} F_n x^n \right) \\ F(x) &= 1 + x + x \left(\sum_{n=0}^{\infty} F_n x^n \right) + x^2 \left(\sum_{n=2}^{\infty} F_n x^n \right) \\ F(x) &= 1 + x + xF(x) - x + x^2 F(x), \end{aligned}$$

which, at the end, simplifies to:

$$F(x) = \frac{1}{1 - x - x^2}.$$

If you start with $F_0 = 0$ then the result is

$$F(x) = \frac{x}{1 - x - x^2}.$$

Now, we can use partial fraction decomposition to make this expression even more explicit by using the following expression, where α and β are zeros of the polynomial in the denominator.

$$\frac{1}{(1 - \alpha x)(1 - \beta x)}.$$

The zeros of the polynomial are:

$$x_1 = \frac{1 + \sqrt{5}}{2} \quad \text{and} \quad x_2 = \frac{1 - \sqrt{5}}{2}.$$

Now, we can write our expression as:

$$F(x) = \frac{1}{1 - x - x^2} = \frac{1}{\left(1 - x \cdot \frac{1+\sqrt{5}}{2}\right)\left(1 - x \cdot \frac{1-\sqrt{5}}{2}\right)},$$

which, after performing the partial fraction decomposition, leaves us with:

$$F(x) = \left(\left(\frac{1}{\sqrt{5}}(1 + \sqrt{5})\right)^n - \left(\frac{1}{\sqrt{5}}(1 - \sqrt{5})\right)^n\right).$$

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Chapter 23

Relational Algebra

Relational algebra is a *formal, procedural query language* for the relational model. It provides a set of operators that take relations (tables) as input and produce relations as output.

23.1 Relations in Set Theory

In mathematics, a *relation* is a set formed from the Cartesian product of two sets. It contains ordered pairs (a, b).

23.2 The Cartesian Product for Relation Algebra

The Cartesian Product for our entities is defined as:

$$S \times T := \bigcup_{(s_1,\ldots,s_n)\in S} \left[\bigcup_{(t_1,\ldots,t_k)\in T} \{(s_1,\ldots,s_n,t_1,\ldots,t_k)\} \right]$$

Simplified: each element of S is combined with each element of T. The size of this special product is $|S \times T| = |S| \cdot |T|$.

In relational algebra, relations represent tables and their rows are the tuples.

23.3 Axioms of Relational Algebra

Relational Algebra is based on *set theory* and satisfies the following axioms:

- *Closure*: Every operation produces a relation.
- *Set Semantics*: Relations do not contain duplicate tuples.
- *Order Irrelevance*: The order of tuples and attributes does not matter.
- *Composability*: Operations can be nested arbitrarily.

Example:

If $\sigma_{id>1}(A)$ returns a relation, it can be used as input to another operator such as projection.

23.4 Relations in Relational Algebra

In database theory, a *relation* is a finite set of tuples over a fixed set of attributes.

Formally:

$$R \subseteq D_1 \times D_2 \times \cdots \times D_n$$

Example:

Relation $A(id, value)$ is a set of tuples:

$$A = \{(1, X), (2, Y)\}$$

23.5 Identifier and Schema

We define:

$$\text{ident}(R) = \{a_i\}_i \quad \text{and} \quad \text{schema}(R) = \{[a_1 : T_1, \ldots, a_n : T_n]\}.$$

ident returns the names of the attributes of the relation, while schema returns the attributes and their corresponding types.

23.6 Domain of an Attribute

The *domain* of an attribute a in a relation R is the set of all possible values that a can take in R.

23.7 Relational Algebra Operations

23.7.1 Cartesian Product for Relations

The *Cartesian product* combines every tuple of one relation with every tuple of another. This is also called the *cross join* in some implementations.

$$S \times T = \bigcup_{(s_1,\ldots,s_n)\in S} \left[\bigcup_{(t_1,\ldots,t_n)\in T} \{(s_1, \ldots, s_n, t_1, \ldots, t_k)\} \right]$$

With size $|S \times T| = |S| \times |T|$.

Example:

```
SELECT * FROM T, S;
-- or equivalently:
SELECT * FROM T CROSS JOIN S;
```

23.7.2 Union ($\cup$)

Combines tuples from two relations with the *same schema* (same columns).

$$A \cup B$$

For the union of relations they have to be *union compatible* which means:

- They have the same number of elements in the tuple.
- The relations need to have the same columns.

Example:

```
SELECT * FROM T UNION SELECT * FROM S;
```

23.7.3 Set Difference ($\setminus$)

Returns tuples in one relation but not in another.

$$A \setminus B$$

Example:

```
SELECT * FROM T EXCEPT SELECT * FROM S;
-- or in some SQL dialects:
SELECT * FROM T MINUS SELECT * FROM S;
```

23.7.4 Intersection $\cap$

Returns tuples common to both relations.

$$A \cap B = A \setminus (A \setminus B)$$

Example:

```
SELECT * FROM T INTERSECT SELECT * FROM S;
```

23.7.5 Selection σ

Filters tuples based on a predicate.

$$\sigma_\theta(S) = \{s \in S \mid s \text{ satisfies } \Theta\}$$

Example:

```
SELECT * FROM T WHERE a > 10;
```

23.7.6 Projection π

Selects specific attributes.

$$\prod_{a_{i1},\ldots,a_{ik}} (S)\{(a_{i1}, \ldots, a_{ik}) \mid a \in S\}$$

This gives us the same tuples of the original relation but only with the columns we selected.

It can also be noted as π.

Example:

```
SELECT a, b FROM T;
```

23.7.7 Symmetric Difference $\triangle$

Tuples in either relation but not in both.

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

Example:

```
(SELECT * FROM T EXCEPT SELECT * FROM S) UNION (SELECT * FROM S EXCEPT SELECT * FROM T);
```

23.7.8 Rename ρ

Renames relations or attributes.

$$\rho_{A'(key,data)}(A)$$

Example:

```
SELECT a AS x, b AS y, c AS z FROM T;
-- or rename the table:
SELECT * FROM T AS T_new;
```

23.7.9 with

with allows us to store products of specific operations with a special name which can be used later in other operations.

23.7.10 Join $\bowtie$

A *theta/inner join* is the Cartesian product plus a selection based on a condition θ.

$$S \bowtie_\theta T = \sigma_\theta(S \times T)$$

Example:

```
SELECT * FROM T INNER JOIN S ON T.a = S.d;
-- or with theta condition:
SELECT * FROM T INNER JOIN S ON T.a < S.d;
```

23.7.11 Equi-Join

An *equi-join* is a join where the join condition consists only of equality comparisons between attributes of the two relations.

$$R \bowtie_{R.a_1=S.b_1 \wedge \cdots \wedge R.a_k=S.b_k} S.$$

The name and domain of the attributes must be the same.

$$R \bowtie_{R.a_1=S.b_1 \wedge \cdots \wedge R.a_k=S.b_k} S = \sigma_{R.a_1=S.b_1 \wedge \cdots \wedge R.a_k=S.b_k}(R \times S)$$

Example:

```
SELECT * FROM T JOIN S ON T.a = S.d AND T.b = S.e;
```

23.7.12 Natural Join

It is a special case of the equi-join which can only be performed if there is a common attribute.

$$R \bowtie S$$

Let $C = \text{ident}(R) \cap \text{ident}(S)$ be the common attributes.

$$R \bowtie S = \pi_{\text{ident}(R) \cup \text{ident}(S)} \left(\sigma_{\bigwedge_{c \in C} R.c=S.c}(R \times S) \right)$$

Example:

```
-- assumes T and S have common column names
SELECT * FROM T NATURAL JOIN S;
```

23.7.13 Semi-Join

It is a case where we only care about the attribute names but not their values.

$$R \ltimes S$$

$$R \ltimes S = \pi_{\text{ident}(R)}(R \bowtie S)$$

Or alternatively:

$$R \ltimes S = \pi_{\text{ident}(R)} \left(\sigma_{\bigwedge_{c \in C} R.c = S.c}(R \times S) \right)$$

where $C = \text{ident}(R) \cap \text{ident}(S)$.

Example:

```
SELECT T.* FROM T WHERE EXISTS (SELECT 1 FROM S WHERE T.a = S.d);
-- or using IN:
SELECT * FROM T WHERE a IN (SELECT d FROM S);
```

23.7.14 Anti Semi-Join

In this case, it works very similar to the semi join, but we take the attributes which do not participate in the natural join.

$$R \triangleright S$$

$$R \triangleright S = R \setminus (R \ltimes S) = R \setminus \pi_{\text{ident}(R)}(R \bowtie S)$$

Example:

```
SELECT T.* FROM T WHERE NOT EXISTS (SELECT 1 FROM S WHERE T.a = S.d);
-- or using NOT IN:
SELECT * FROM T WHERE a NOT IN (SELECT d FROM S);
```

23.7.15 Outer Join

The *outer join* preserves non-matching tuples.

- *Left Outer Join*: It returns all tuples from the left relation with not matches on the right.
- *Right Outer Join*: It returns all tuples from the right relation with not matches on the left.
- *Full Outer Join*: It returns all tuples from both relations which are not in the inner join.

Example:

```
-- Left Outer Join:
SELECT * FROM T LEFT OUTER JOIN S ON T.a = S.d;

-- Right Outer Join:
SELECT * FROM T RIGHT OUTER JOIN S ON T.a = S.d;

-- Full Outer Join:
SELECT * FROM T FULL OUTER JOIN S ON T.a = S.d;
```

23.7.16 Division ÷

Used to express *"for all"* queries. It can only be performed when the relation on the right is a proper subset of the left relation.

Let

- $R(X, Y)$
- $S(Y)$

Then:

$$R \div S = \{x \mid \forall y \in S : (x, y) \in R\}$$

$$R \div S = \pi_X(R) \setminus \pi_X\left((\pi_X(R) \times S) \setminus R\right),$$

where $X = \text{ident}(R) \setminus \text{ident}(S)$.

Example Concept:

If $A(id, value)$ contains all values in $B(value)$, division returns the corresponding ID's.

```
-- Find tuples in T that match all tuples in S
SELECT DISTINCT T.a FROM T
WHERE NOT EXISTS (
SELECT S.d FROM S
WHERE NOT EXISTS (
    SELECT * FROM T AS T2
    WHERE T2.a = T.a AND T2.b = S.d
)
);
```

23.7.17 Assignment (=)

This is just using a placeholder to store the result of a relational algebra operation.

Example:

```
-- Create a view or temporary table
CREATE VIEW Result AS SELECT * FROM T WHERE a > 5;
-- or
CREATE TEMPORARY TABLE Result AS SELECT * FROM T WHERE a > 5;
```

23.8 Types of Keys

A *key* is an attribute or combination of attributes (composite key) which uniquely identifies an entity.

- *Candidate Key:* The term refers to the attributes which are suited for being the key.
- *Irreducible:* A key has to be as atomic as possible.
- *Super key:* It consists of a candidates key's attributes + potentially some extra attributes.
- *Alternate Key:* This is an alternative primary key. This means that the UNIQUE constraints will be imposed on the attribute/s.
- *Surrogate Key:* This key has only meaning inside the database. Example: ID → Person's Name.
- *Foreign Key* This key is an attribute of one table which references another table.

23.9 Relation Schema

The $(\mathcal{R}, \tau)$ represents a *relation schema*. We can constrain our relations to obey certain rules. Thus, limiting our space. Here, $\mathcal{R}$ is the set of all relations of n elements and τ is the set of n-tuple (attributes).

The rules we use to limit the space of allowed relations are called *restrictions*.

This is the analog of using `CREATE TABLE` in SQL. We are creating rules for the tuples inside the relation.

Also, this is just the definition of the name of the relations and the types of its attributes.

23.10 Constraint

We denote that $d[x]$ is a function from $x \to R^+$, where R^+ is the relation R without the attribute x. This is called the *constraint*.

23.11 Functional Dependencies

A *functional relation* is dependence between one or more attributes commonly noted $A \to B$.

Formally, given $\alpha \subseteq \text{ident}(R)$ and $\beta \subseteq \text{ident}(R)$ and $D \subseteq \text{dom}(R)$ $\alpha \to \beta$, β is dependent on α if

$$\forall d_1, d_2 \in D : d_1[\alpha] = d_2[\alpha] \implies d_1[\beta] = d_2[\beta].$$

In simple terms the notation $X \to Y$ means "we know Y if we know X".

A complete dependence is when $A \to B$ is when we can not further reduce the tuple A so that we still get B.

23.12 Armstrong Axioms

For a relation $R = \{a_1, \ldots, a_n\}$ with the schema $\alpha, \beta, \gamma \subseteq \text{ident}(R)$.

The Armstrong axioms are given:

I. *Reflexivity:* for all $\beta \subseteq \alpha$ holds $\alpha \to \beta$

II. *Transitivity:* $\alpha \to \beta \land \beta \to \gamma \implies \alpha \to \gamma$

III. *Connectivity:* $\alpha \to \beta \implies \alpha\gamma \to \beta\gamma$

IV. *Union:* $\alpha \to \beta \land \alpha \to \gamma \implies \alpha \to \beta\gamma$

V. *Decomposition:* $\alpha \to \beta\gamma \implies \alpha \to \beta \land \alpha \to \gamma$

VI. *Pseudotransivity:* $\alpha \to \beta \land \beta\gamma \to \delta \implies \alpha\gamma \to \delta$

These are not strictly axioms, but they are called that way.

23.13 Attribute Closure

The *attribute closure* $AC(FD, x)$is the set of all attributes reachable via functional dependency including the starting point.

Example:

$R = \{A, B, C, D\}$ with $FD = \{A \to B, B \to C, AB \to CD\}$.

- $AC(FD, A) = \{A, AB, ABC, ABCD\} = ABCD$
- $AC(FD, B) = \{B, BC\} = BC$

23.14 Canonical Cover

We consider two sets of functional dependencies if and only if their attribute closures FD^+ are identical.

For a schema R with a set of functional dependencies FD we denote FC^c as *the canonical cover* of FD if the following criteria are satisfied:

- $FD^c \cong FD$, $FD^{c+} = FD^+$
- For all $\alpha \to \beta \in FD^c$ are minimal in the sense that

  $$\forall a \in \alpha : FD^c \setminus (\alpha \to \beta) \cup (\alpha \setminus a \to \beta) \neq FD^c$$

  $$\forall b \in \beta : FD^c \setminus (\alpha \to \beta) \cup (\alpha \to \beta \setminus b) \neq FD^c$$

- All left side attributes appear only one time

The following steps are always performed for the whole set of functional dependencies one after the other.

23.14.1 Left-Reduction

For all $\alpha \to \beta \in FD$ check if there is a redundant $a \in \alpha$. $\beta \subseteq AC(FD, \alpha \setminus a)$. If that is the case then replace $\alpha \to \beta$ with $\alpha \setminus a \to \beta$.

After that

23.14.2 Right-Reduction

For all $\alpha \to \beta \in FD$ check if there is a redundant $b \in \beta$. $\beta \subseteq AC(FD \setminus \{\alpha \to \beta\} \cup \{\alpha \to \beta \setminus b, \alpha\})$. If that is the case then replace $\alpha \to \beta$ with $\alpha \to \beta \setminus b$.

Then

23.14.3 Elimination

Remove $\alpha \to \emptyset$.

And finally

23.14.4 Union

If $\alpha \to \beta_1$ and $\alpha \to \beta_2$ then combine them into $\alpha \to \beta_1, \beta_2$.

Example:

Given are $R = \{A, B, C, D, E, F\}, FD = \{ABC \to DEF, A \to BC, D \to EF\}$.

Left-Reduction:

First iteration, $ABC \to DEF$

- Is A redundant? No: $AC(F, BC) = BC$
- Is B redundant? Yes: $AC(F, AC) = AC \implies ABC \implies ABCDEF$. Thus, we have $AC \to DEF, A \to BC, D \to EF$.
- Is C redundant? Yes: $AC(F, A) = A \implies ABC \implies ABCDEF$. Thus, we have $A \to DEF, A \to BC, D \to EF$.

Right-Reduction

Using $\{A \to DEF, A \to BC, D \to EF\}$ we proceed with the first dependency $A \to DEF$.

- Is D redundant? No, without it, we can not reach EF.
- Is E redundant? Yes: $AC(A \to DF, A \to BC, A \to EF, A) = A \implies ADF \implies ADEFBC$. Thus, we have $F = \{A \to DF, A \to BC, D \to EF\}$.
- Is F redundant? Yes: $AC(A \to D, A \to BC, D \to EF, A) = A \implies ABC \implies ABCDEF$. Thus, we have $F = \{A \to D, A \to BC, D \to EF\}$.

Elimination and Union

$$FC^c = \{A \to BCD, D \to EF\}$$

Important notes:

- By the left-reduction we only care about fd's which have at least two attributes on the right side.
- By the right-reduction we only care about the fd's with one attribute at the right.

23.14.5 How to Find a Candidate Key

Given a set of attributes $\{A, B, C, D\}$ with the following functional dependencies:

$$A \to B, C \to B, A \to D$$

The set is already left-reduced.

- $AC(F, A) = ABD$
- $AC(F, C) = CB$

We get $A \to BD, C \to B$.

For the right reduction:

- Is B redundant? Yes

Finally, we get our candidate key $AC \to ABCD$.

23.14.6 Losslessness and Maintenance of Dependency

Given a relation which needs to be split into more relations:

- *Losslessness:* After the split, the information in the original relation needs to be accessible via joins.
- *Maintenance of Dependency:* The original functional dependencies do not change after the split.

23.15 Normal Forms

A *normalized* table are protected from data inconsistencies and make the tables more readable.

23.15.1 I Normal Form

- No row order for information.
- No Mixing data types in the same column.
- No Tables without a primary key.
- No Repeating groups.

This normal is mostly given in relational algebra problems.

23.15.2 II Normal Form:

Each non-key attribute must depend on the entire primary key.

The relation is in the second normal form if it is already in the first normal form and each **non-key** attribute is functionally dependent on the **candidate key**.

Example I:

$R = \{A, B, C, D\}, \quad FD = \{A \to BC, B \to A\}$ not in 2NF due to D not being present.

Example II:

$R = \{A, B, C, D\}, \quad FD = \{A \to B, B \to A, AD \to C\}$

Candidates AD, BD, not prime attribute C. Thus, it obeys the 2NF.

23.15.3 III Normal Form

A schema R is in 3NF if for all $\alpha \to \beta \in FD$ at least one of the following statements is true:

- $\alpha \to \beta$ is trivial $\beta \subseteq \alpha$
- β has only prime attributes
- α is a superkey

The relation is in the third normal form if it is already in the second normal form and each *non-key* attribute is not dependent on other *non-key* attribute.

23.15.4 Boyce-Codd III Normal Form

A schema R is in BCNF if for all $\alpha \to \beta \in FD$ at least one of the following statements is true:

- $\alpha \to \beta$ is trivial $\beta \subseteq \alpha$
- α is a superkey

Every attribute in a table should depend on the key, the whole key, and nothing but the key.

Example I

$R = \{A, B, C, D\}, \quad FD = \{A \to BC, C \to D\}$, not in 3NF due to D being dependent on C.

23.15.5 IV Normal Form

Multivalued dependencies in a table must be multivalued dependencies on the key.

23.15.6 V Normal Form

A table in the fifth normal form cannot be described as the logical result of joining some other tables.

23.16 Synthesis Algorithm

The *synthesis algorithm* transforms a relation schema into a set of schemas that are in Third Normal Form (3NF), while ensuring *lossless joins* and *dependency preservation*. The algorithm proceeds as follows:

- Compute the *canonical cover* F_c of the given set of functional dependencies F:
  - Decompose all dependencies so that each has a single attribute on the right-hand side.
  - Remove extraneous attributes from the left-hand sides.
  - Eliminate redundant dependencies.
- For each functional dependency $\alpha \to \beta \in F_c$, construct a relation schema

$$R_k = \alpha \cup \beta.$$

  All dependencies whose attributes are contained in R_k are associated with this schema.
- If none of the generated schemas contains a candidate key of the original relation, create an additional schema whose attributes form a candidate key.
- If a schema R_k is completely contained in another schema R_m, remove R_k, since it is redundant.

Example:

Given

$$R = \{A, B, C, D, E\}$$

with

$$F = \{ADE \to BE,\ B \to ACD,\ D \to AC,\ ACE \to BD,\ C \to B\}.$$

Step 1: Right-hand side decomposition

$$\begin{aligned} &ADE \to B, \quad ADE \to E, \quad B \to A, \quad B \to C, \quad B \to D, \\ &D \to A, \quad D \to C, \quad ACE \to B, \quad ACE \to D, \quad C \to B \end{aligned}$$

Step 2: Left reduction

- $ADE \to B$: A and E are extraneous since $D^+ = \{D, B, A, C, E\}$, hence $D \to B$.
- $ADE \to E$ is trivial and removed.
- $ACE \to B$: A and E are extraneous, yielding $C \to B$.
- $ACE \to D$: A and E are extraneous, yielding $C \to D$.
- All other dependencies already have minimal left-hand sides.

Current set:

$$\{D \to B,\ B \to A,\ B \to C,\ B \to D,\ D \to A,\ D \to C,\ C \to B,\ C \to D\}$$

Step 3: Remove redundant dependencies

- $D \to A$ is redundant since $D \to C$ and $C \to B$ and $B \to A$.
- $B \to C$ is redundant since $B \to D$ and $D \to C$.
- $C \to D$ is redundant since $C \to B$ and $B \to D$.

Thus, the canonical cover is:

$$F_c = \{B \to A,\ B \to D,\ D \to C,\ C \to B\}$$

Step 4: Construct relation schemas

$$\begin{aligned} R_1 &= \{B, A\}, \\ R_2 &= \{B, D\}, \\ R_3 &= \{D, C\}, \\ R_4 &= \{C, B\} \end{aligned}$$

Step 5: Remove redundant schemas

None of these schemas is contained in another, so all are kept.

Step 6: Test for candidate keys

For the original relation $R = \{A, B, C, D, E\}$, attribute E does not appear on any right-hand side and must be part of every key.

$$(BE)^+ = \{A, B, C, D, E\}$$

so $\{B, E\}$ is a candidate key.

Since no schema contains $\{B, E\}$, we add:

$$R_5 = \{B, E\}$$

Final 3NF Decomposition:

$$\{\{B, A\},\ \{B, D\},\ \{D, C\},\ \{C, B\},\ \{B, E\}\}$$

23.17 Decomposition Algorithm

The *decomposition algorithm* transforms a relation schema into Boyce-Codd Normal Form (BCNF) by repeatedly decomposing relations that violate the BCNF condition. A relation schema R is in BCNF if for every non-trivial functional dependency $\alpha \to \beta$ on R, α is a superkey of R.

- If R is already in BCNF, stop
- Otherwise, choose a functional dependency $\alpha \to \beta$ that violates BCNF, meaning α is not a superkey of R
- Decompose R into two schemas

$$R_1 = \alpha \cup \beta \quad \text{and} \quad R_2 = R \setminus (\beta \setminus \alpha)$$

- Apply the algorithm recursively to R_1 and R_2 until all resulting schemas are in BCNF

Example:

Given

$$R = \{A, B, C, D\},$$

with

$$F = \{A \to B,\ B \to C,\ C \to D\}.$$

Step 1: Check BCNF

$A \to B$ violates BCNF since A is not a superkey of R

Step 2: First decomposition

Using $A \to B$, decompose into

$$R_1 = \{A, B\} \quad \text{and} \quad R_2 = \{A, C, D\}$$

Step 3: Check BCNF of R_1

In $R_1(A, B)$, $A \to B$ holds and A is a key, so R_1 is in BCNF

Step 4: Check BCNF of R_2

From $B \to C$ and $C \to D$, we obtain $A \to C$ and $A \to D$, but A is not a key of R_2, so BCNF is violated by $A \to C$

Step 5: Second decomposition

Using $A \to C$, decompose R_2 into

$$R_3 = \{A, C\} \quad \text{and} \quad R_4 = \{A, D\}$$

Step 6: Check BCNF of remaining schemas

In $R_3(A, C)$, A is a key, so R_3 is in BCNF

In $R_4(A, D)$, $A \to D$ holds and A is a key, so R_4 is in BCNF

Final BCNF Decomposition:

$$\{\{A, B\},\ \{A, C\},\ \{A, D\}\}$$

23.17.1 Optimization

The following equivalences hold for relational algebra expressions:

- $\sigma_{C_1 \wedge C_2 \wedge \cdots \wedge C_n} \cong \sigma_{C_1}(\sigma_{C_2}(\ldots(\sigma_{C_n}(R))\ldots))$
- $\sigma_{C_1}(\sigma_{C_2}(R)) \cong \sigma_{C_1}(\sigma_{C_2}(R))$
- If $L_1 \subseteq L_2 \subseteq \cdots \subseteq L_n$ then

$$\pi_{L_1}(\pi_{L_2}(\ldots(\pi_{L_n}(R))\ldots)) \cong \pi_{L_1}(R)$$

- $\pi_{A_1,\ldots,A_n}(\sigma_C(R)) \cong \sigma_C(\pi_{A_1,\ldots,A_n}(R))$
- $\times, \cup, \cap$ and $\bowtie$ are commutative
- Pushing Selections, select as early as possible $\sigma_C\,(R \bowtie_{C_J} S) \cong \sigma_C(R) \bowtie_{C_J} S$

  Special Case:

$$\sigma_{C_1 \wedge C_2}\,(R \bowtie_{C_J} S) \cong \sigma_{C_1}(R) \bowtie_{C_J} \sigma_{C_2}(S)$$

- $\pi_L(R \bowtie_{C_J} S) \cong \pi_{A_1,\ldots,A_n}(R) \bowtie_{C_J} \pi_{b_1,\ldots,b_n}(S)$
- $\times, \cup, \cap$ and $\bowtie$ are associative
- σ is distributive with $\cup, \cap, \setminus$

$$\sigma_C(R \Theta S) = \sigma_C(R) \Theta \sigma_C(S)$$

- π is distributive with $\cup$

$$\pi_C(R \cup S) = \pi_C(R) \cup \pi_C(S)$$

- De Morgan's Laws for Joins with selections

$$\neg(A \wedge B) = \neg A \vee \neg B \quad \neg(A \vee B) = \neg A \wedge \neg B$$

- $\sigma_C(R \times S) \cong R \bowtie_C S$

Chapter 24

Geometry

24.1 Formulas for Area and Perimeter of Geometric Figures

- **Square**
  - Area: $A = a^2$
  - Perimeter: $P = 4a$
- **Rectangle**
  - Area: $A = l \cdot w$
  - Perimeter: $P = 2(l + w)$
- **Circle**
  - Area: $A = \pi r^2$
  - Circumference: $C = 2\pi r$
- **Triangle (General)**
  - Area: $A = \frac{1}{2}b \cdot h$
  - Perimeter: $P = a + b + c$
- **Equilateral Triangle**
  - Area: $A = \frac{\sqrt{3}}{4}a^2$
  - Perimeter: $P = 3a$
- **Isosceles Triangle**
  - Area: $A = \frac{b}{4}\sqrt{4a^2 - b^2}$
- **Trapezoid**
  - Area: $A = \frac{1}{2}(a + b)h$
  - Perimeter: $P = a + b + c + d$
- **Parallelogram**
  - Area: $A = b \cdot h$
  - Perimeter: $P = 2(a + b)$

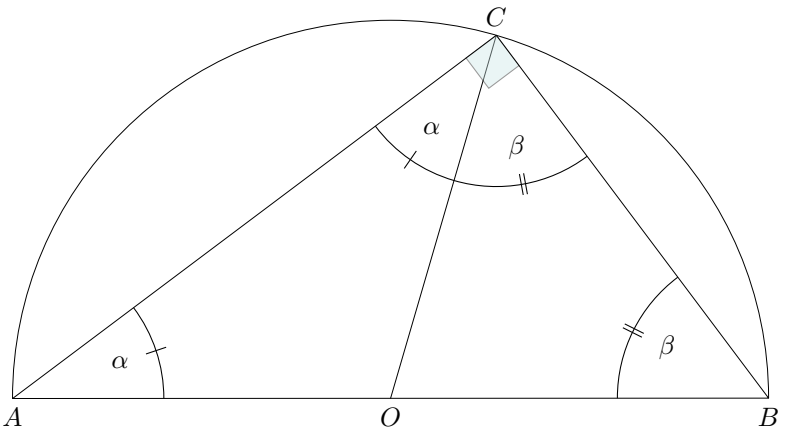
24.2 Formulas for Area and Volume of 3D Geometric Figures

- **Cube**
  - Surface Area: $A = 6a^2$
  - Volume: $V = a^3$
- **Cylinder**
  - Surface Area: $A = 2\pi r(h + r)$
  - Volume: $V = \pi r^2 h$
- **Cone**
  - Surface Area: $A = \pi r(r + l)$ with l = slant height
  - Volume: $V = \frac{1}{3}\pi r^2 h$
- **Sphere**
  - Surface Area: $A = 4\pi r^2$
  - Volume: $V = \frac{4}{3}\pi r^3$
- **Square Pyramid**
  - Surface Area: $A = a^2 + 2a \cdot l$
  - Volume: $V = \frac{1}{3}a^2 h$
- **Triangular Pyramid (Tetrahedron)**
  - Volume: $V = \frac{1}{3}A_b \cdot h$
- **Prism (General)**
  - Surface Area: $A = 2A_b + P_b \cdot h$
  - Volume: $V = A_b \cdot h$

24.3 Thales' Theorem

Thales' theorem states that if A, B, and C are points on a circle where the line AB is the diameter, then the angle $\angle\alpha + \beta$ is a right angle.

$\angle\alpha + \beta = 90°$ if AB is a diameter of the circle



Proof:

Note that we have two isosceles triangles where the sides AO, AC, and OC all have the same length, this being the radius of the circle.

Because they are isosceles, two of their angles must be equal, and the sum of the angles of each triangle has to add up to 180°. Therefore,

$$\begin{aligned} \alpha + \alpha + \beta + \beta &= 180 \\ 2\alpha + 2\beta &= 180 \\ \alpha + \beta &= 90 \end{aligned}$$

ΩΕΔ

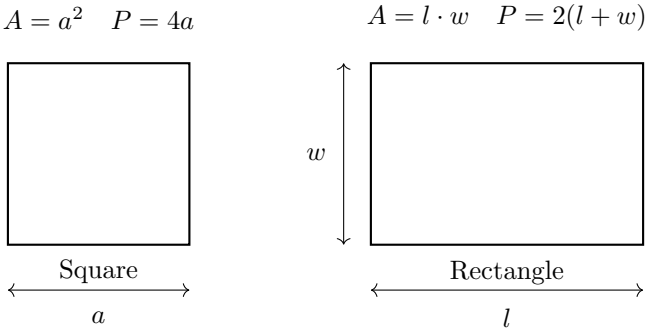
24.4 Conversion Between Radians and Degrees

To convert between radians and degrees, use the following formulas:

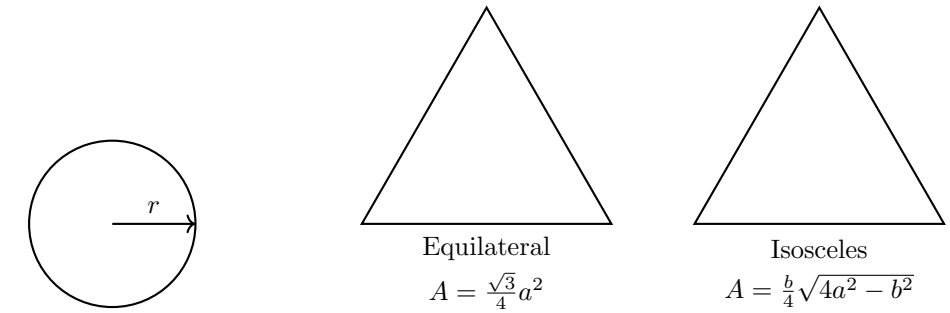
$$\text{Degrees} = \text{Radians} \times \left(\frac{180^\circ}{\pi}\right)$$
$$\text{Radians} = \text{Degrees} \times \left(\frac{\pi}{180^\circ}\right)$$

24.5 Geometric Figures With Formulas (2D)

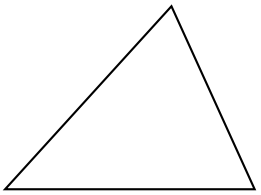
Square and Rectangle



Circle and Triangle Types



Scalene Triangle, Trapezoid, Parallelogram



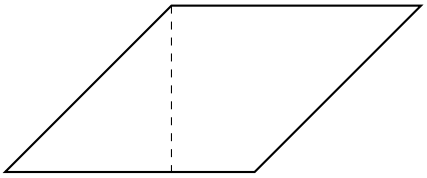
Scalene Triangle

$A = \frac{1}{2}b \cdot h$



Trapezoid

$A = \frac{1}{2}(a + b)h$

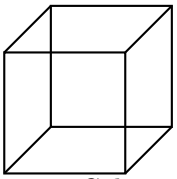


Parallelogram

$A = b \cdot h \quad P = 2(a + b)$

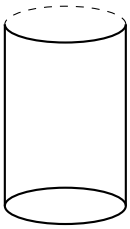
24.6 3D Geometric Figures With Formulas

24.6.1 Cube and Cylinder



Cube

$A = 6a^2 \quad V = a^3$

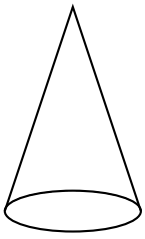


Cylinder

$A = 2\pi r(h + r)$

$V = \pi r^2 h$

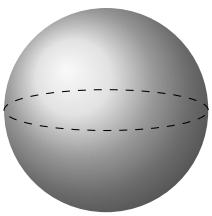
24.6.2 Cone and Sphere



Cone

$A = \pi r(r + l)$

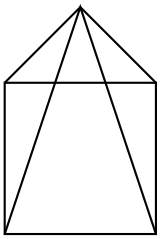
$V = \frac{1}{3}\pi r^2 h$



Sphere

$A = 4\pi r^2 \quad V = \frac{4}{3}\pi r^3$

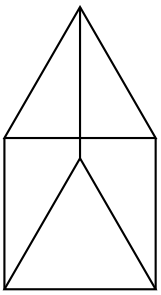
24.6.3 Pyramid and Prism



Square Pyramid

$A = a^2 + 2al$

$V = \frac{1}{3}a^2h$



Triangular Prism

$A = 2A_b + P_bh$

$V = A_b \cdot h$

24.7 Axioms of Euclidean Geometry

Euclidean geometry is founded on a set of basic assumptions known as *axioms* or postulates, originally formulated by the Greek mathematician Euclid in his work *Elements*. These axioms form the logical basis for plane geometry and include the following:

I. *Line Postulate:* A straight line segment can be drawn joining any two points.

II. *Extension Postulate:* Any straight line segment can be extended indefinitely in a straight line.

III. *Circle Postulate:* Given any straight line segment, a circle can be drawn having the segment as radius and one endpoint as center.

IV. *Right Angle Postulate:* All right angles are congruent.

V. *Parallel Postulate:* If a line segment intersects two straight lines and makes the interior angles on the same side less than two right angles, then the two lines, if extended indefinitely, meet on that side.

These five postulates define the framework of classical geometry in a flat (Euclidean) space. Notably, the fifth postulate—the parallel postulate—has been the subject of much scrutiny and led to the development of non-Euclidean geometries when it was altered or replaced.

24.8 Types of Angles and Angle Relationships

An *angle* is the distance between two rays or intersecting lines.

- If an angle is less than 90°, it is an *acute* angle.
- If it is exactly 90°, it is a *right* angle.
- If it is more than 90° but less than 180°, it is called an *obtuse* angle.

Now let us look at other types of angles.

24.8.1 Vertical Angles

When two lines intersect, they form two pairs of opposite (or vertical) angles. Vertical angles are always congruent. That is, if two angles are vertical, their measures are equal:

$$\angle A = \angle B$$

For example, if two lines intersect and form angles of 60° and 120°, the angles opposite each other are both 60° and both 120°.

24.8.2 Adjacent Angles

Adjacent angles are two angles that share a common side and a common vertex, and do not overlap. When two lines intersect, each pair of adjacent angles forms a straight angle (or a linear pair), summing to 180°:

$$\angle 1 + \angle 2 = 180^\circ$$

This is known as the Linear Pair Postulate.

24.8.3 Complementary Angles

Complementary angles are two angles whose measures add up to 90°:

$$\angle X + \angle Y = 90^\circ$$

Complementary angles do not have to be adjacent, but if two intersecting lines form a right angle, the two acute angles adjacent to the right angle may be complementary.

24.8.4 Angles Generated by Transversals

When a *transversal* crosses two or more lines, it forms several pairs of angles with specific relationships. Some key types include:

- *Corresponding Angles:* Angles in the same relative position at each intersection. If the lines are parallel, corresponding angles are congruent:

$$\angle 1 \cong \angle 2$$

- *Alternate Interior Angles:* Angles that lie between the two lines but on opposite sides of the transversal. They are congruent if the lines are parallel:

$$\angle 3 \cong \angle 4$$

- *Alternate Exterior Angles:* Angles outside the two lines and on opposite sides of the transversal. Also, congruent when the lines are parallel:

$$\angle 5 \cong \angle 6$$

- *Consecutive (Same-Side) Interior Angles:* Angles on the same side of the transversal and inside the parallel lines. These are supplementary:

$$\angle 7 + \angle 8 = 180^\circ$$

24.9 Triangle Similarity and Congruence

Triangles can be compared using criteria of similarity or congruence, depending on whether their shapes are the same or both shape and size are identical.

24.9.1 Triangle Congruence

Two triangles are *congruent* if all corresponding sides and angles are equal. This means the triangles are identical in shape and size. Common criteria for triangle congruence include:

- *SSS (Side-Side-Side):* All three pairs of corresponding sides are equal.
- *SAS (Side-Angle-Side):* Two pairs of sides and the angle between them are equal.
- *ASA (Angle-Side-Angle):* Two pairs of angles and the included side are equal.
- *AAS (Angle-Angle-Side):* Two angles and a non-included side are equal.
- *HL (Hypotenuse-Leg):* For right triangles, if the hypotenuse and one leg are equal.

24.9.2 Triangle Similarity

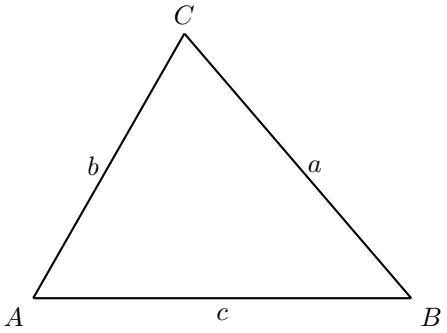
Two triangles are *similar* if their corresponding angles are equal and their corresponding sides are in proportion. This means they have the same shape but not necessarily the same size. Criteria for similarity include:

- *AA (Angle-Angle):* Two pairs of corresponding angles are equal.
- *SAS (Side-Angle-Side):* One pair of corresponding angles is equal, and the sides including those angles are in proportion.
- *SSS (Side-Side-Side):* All three pairs of corresponding sides are in proportion.

24.10 Visualizations of Triangle Similarity and Congruence

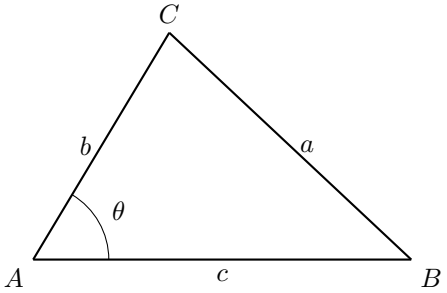
24.10.1 SSS (Side-Side-Side)

A triangle is uniquely determined if all three sides are known.



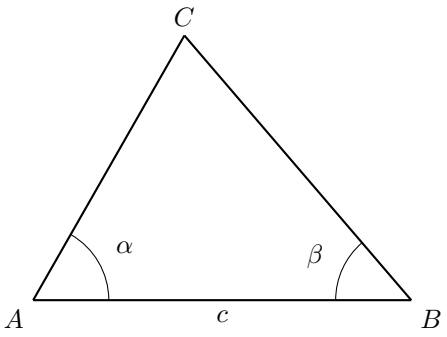
24.10.2 SAS (Side-Angle-Side)

A triangle is uniquely determined if two sides and the included angle are known.



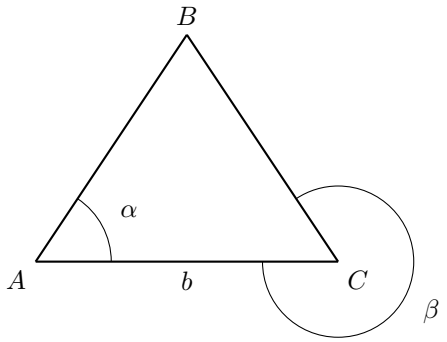
24.10.3 ASA (Angle-Side-Angle)

A triangle is uniquely determined if two angles and the included side are known.



24.10.4 AAS (Angle-Angle-Side)

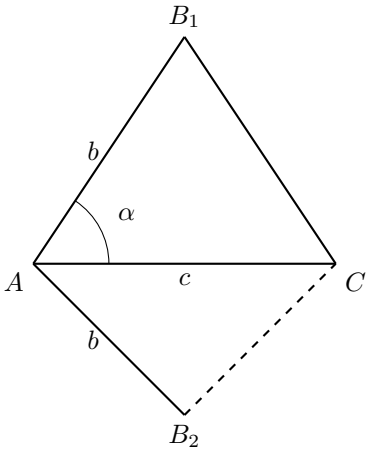
A triangle is uniquely determined if two angles and a non-included side are known.



24.10.5 SSA (Side-Side-Angle)

Two sides and a non-included angle do *not always* determine a unique triangle. This case is ambiguous and can result in:

- One triangle (when the opposite side is long enough),
- Two triangles (ambiguous case),
- No triangle (when the side is too short).



24.11 Triangle Medians, Angle Bisectors, and Altitudes

In any triangle, there are special line segments that connect vertices to specific points on the opposite sides. These segments play key roles in triangle geometry and have unique properties.

Median

A *median* of a triangle is a line segment that connects a vertex to the midpoint of the opposite side. Every triangle has three medians, and they intersect at a single point called the *centroid*, which is the triangle's center of mass. The centroid divides each median in a ratio of $2:1$, with the longer part closer to the vertex.

Angle Bisector

An *angle bisector* is a line segment that divides an angle of the triangle into two equal parts and extends to the opposite side. All three angle bisectors intersect at a point called the *incenter*, which is the center of the triangle's incircle (the largest circle that fits inside the triangle and touches all three sides).

Altitude

An *altitude* of a triangle is a perpendicular segment from a vertex to the line containing the opposite side (not necessarily within the triangle). Altitudes intersect at a point called the *orthocenter*. Unlike medians and bisectors, altitudes can fall outside the triangle in obtuse triangles.

Centroid $\rightarrow$ Intersection of medians

Incenter $\rightarrow$ Intersection of angle bisectors

Orthocenter $\rightarrow$ Intersection of altitudes

24.12 The Midsegment Theorem

The *Midsegment Theorem* states that the segment connecting the midpoints of two sides of a triangle is:

- *Parallel* to the third side of the triangle
- *Half the length* of the third side

Let $\triangle ABC$ be a triangle, and let D and E be the midpoints of sides AB and AC, respectively. Then the segment $\overline{DE}$ is the *midsegment*, and we have:

$$\overline{DE} \parallel \overline{BC} \quad \text{and} \quad DE = \frac{1}{2}BC$$

This theorem is useful in coordinate geometry, triangle proofs, and for simplifying complex geometric relationships within triangles.

24.13 Intercept Theorems

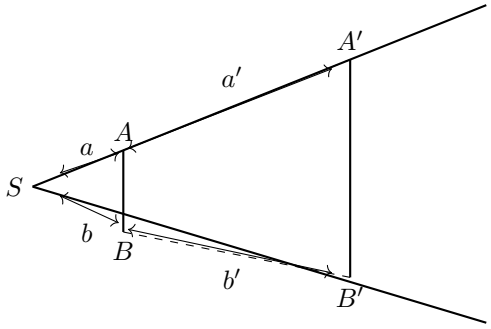
The *intercept theorems* describe relationships between segment lengths when two rays from a point intersect two parallel lines. They are based on similar triangles and allow us to calculate unknown lengths using proportions.

24.13.1 First Intercept Theorem

If two rays start from a common point and are intersected by two parallel lines, then:

$$\frac{a}{a'} = \frac{b}{b'},$$

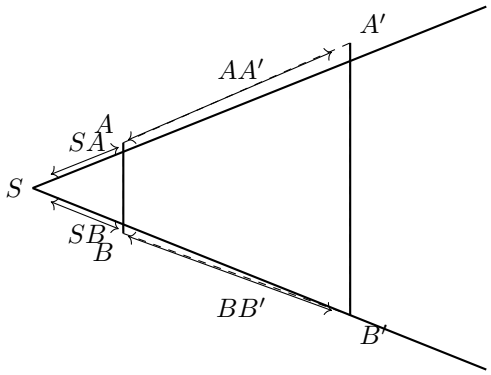
where a and b are segments on one ray, and a', b' are the corresponding segments on the other ray.



24.13.2 Second Intercept Theorem (General Form)

If a ray intersects two parallel lines, the segments from the origin point to the lines are in the same ratio as the segments along the parallels:

$$\frac{SA}{SA'} = \frac{SB}{SB'} \quad \text{and} \quad \frac{AB}{A'B'} = \frac{SA}{SA'}$$



24.14 Proof of the Pythagorean Theorem

We want to prove that $c^2 = a^2 + b^2$ is a true statement.

Let us build a square with area c^2, take our original triangle, and place it on the sides of the square.

This will create a square of side $(a + b)$, which is bigger than the inner square c^2.

Now let us find the area c^2 using what we know about this shape.

$$\begin{aligned}
(a+b)^2 &= 4(\frac{ab}{2}) + c^2 \\
(a+b)^2 &= 2ab + c^2 \\
a^2 + 2ab + b^2 &= 2ab + c^2 \\
a^2 + b^2 &= c^2
\end{aligned}$$

QED

Chapter 25

Trigonometry

In this chapter, we will cover a variety of trigonometric functions and their properties. *Trigonometry* is widely used in different areas of mathematics; hence, this chapter provides the most important facts.

25.1 Trigonometric Functions (Triangle Definition)

- $\sin(x) = \frac{1}{\csc(x)}$
- $\cos(x) = \frac{1}{\sec(x)}$
- $\tan(x) = \frac{\sin(x)}{\cos(x)} = \frac{1}{\cot(x)}$
- $\csc(x) = \frac{1}{\sin(x)}$
- $\sec(x) = \frac{1}{\cos(x)}$
- $\cot(x) = \frac{\cos(x)}{\sin(x)} = \frac{1}{\tan(x)}$

25.2 SOH-CAH-TOA

- $\sin(x) = \frac{\text{opposite}}{\text{hypotenuse}}$
- $\cos(x) = \frac{\text{adjacent}}{\text{hypotenuse}}$
- $\tan(x) = \frac{\text{opposite}}{\text{adjacent}}$

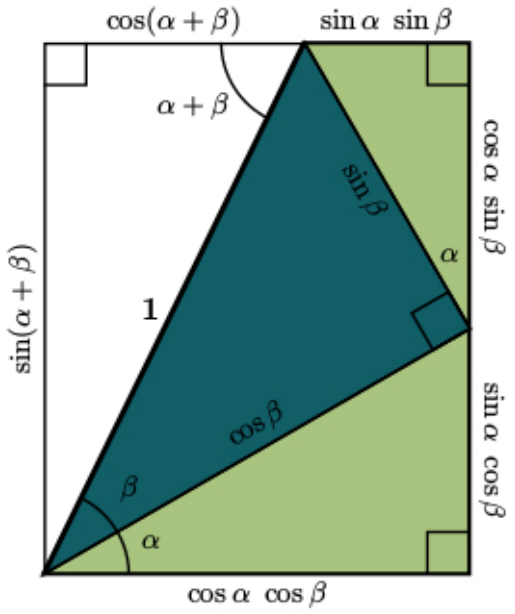
25.3 Pythagorean Identities

- $\sin^2(x) + \cos^2(x) = 1$
- $1 + \tan^2(x) = \sec^2(x)$
- $1 + \cot^2(x) = \csc^2(x)$

25.4 Sum and Difference Formulas

- $\sin(a + b) = \sin(a)\cos(b) + \cos(a)\sin(b)$
- $\sin(a - b) = \sin(a)\cos(b) - \cos(a)\sin(b)$
- $\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$
- $\cos(a - b) = \cos(a)\cos(b) + \sin(a)\sin(b)$
- $\tan(a + b) = \frac{\tan(a)+\tan(b)}{1-\tan(a)\tan(b)}$

- $\tan(a-b) = \frac{\tan(a)-\tan(b)}{1+\tan(a)\tan(b)}$



$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$
$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

25.5 Double Angle Formulas

- $\sin(2x) = 2\sin(x)\cos(x)$
- $\cos(2x) = \cos^2(x) - \sin^2(x) = 2\cos^2(x) - 1 = 1 - 2\sin^2(x)$
- $\tan(2x) = \frac{2\tan(x)}{1-\tan^2(x)}$
- $\csc(2x) = \frac{2\csc(x)\sec(x)}{1-\tan^2(x)}$
- $\sec(2x) = \frac{1+\tan^2(x)}{2\tan(x)}$
- $\cot(2x) = \frac{1-\tan^2(x)}{2\tan(x)}$

25.6 Half Angle Formulas

- $\sin\left(\frac{x}{2}\right) = \pm\sqrt{\frac{1-\cos(x)}{2}}$
- $\cos\left(\frac{x}{2}\right) = \pm\sqrt{\frac{1+\cos(x)}{2}}$
- $\tan\left(\frac{x}{2}\right) = \frac{\sin(x)}{1+\cos(x)} = \frac{1-\cos(x)}{\sin(x)} = \frac{\tan(x)}{1+\tan^2(x)}$
- $\csc\left(\frac{x}{2}\right) = \frac{1}{\sin\left(\frac{x}{2}\right)} = \pm\sqrt{\frac{2}{1-\cos(x)}}$
- $\sec\left(\frac{x}{2}\right) = \frac{1}{\cos\left(\frac{x}{2}\right)} = \pm\sqrt{\frac{2}{1+\cos(x)}}$

- $\cot\left(\frac{x}{2}\right) = \frac{1}{\tan\left(\frac{x}{2}\right)} = \frac{1+\cos(x)}{\sin(x)} = \frac{1-\tan^2(x)}{2\tan(x)}$

25.7 Product to Sum Formulas

- $\sin(a)\sin(b) = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$
- $\cos(a)\cos(b) = \frac{1}{2}[\cos(a-b) + \cos(a+b)]$
- $\sin(a)\cos(b) = \frac{1}{2}[\sin(a+b) + \sin(a-b)]$
- $\cos(a)\sin(b) = \frac{1}{2}[\sin(a+b) - \sin(a-b)]$

25.8 Power Reducing Formulas

- $\sin^2(x) = \frac{1-\cos(2x)}{2}$
- $\cos^2(x) = \frac{1+\cos(2x)}{2}$
- $\tan^2(x) = \frac{1-\cos(2x)}{1+\cos(2x)}$
- $\csc^2(x) = \frac{1}{\sin^2(x)} = \frac{2}{1-\cos(2x)}$
- $\sec^2(x) = \frac{1}{\cos^2(x)} = \frac{2}{1+\cos(2x)}$
- $\cot^2(x) = \frac{1}{\tan^2(x)} = \frac{1+\cos(2x)}{1-\cos(2x)}$

25.9 Even and Odd Functions

- $\sin(-x) = -\sin(x)$
- $\cos(-x) = \cos(x)$
- $\tan(-x) = -\tan(x)$
- $\csc(-x) = -\csc(x)$
- $\sec(-x) = \sec(x)$
- $\cot(-x) = -\cot(x)$

25.9.1 Graph Identity

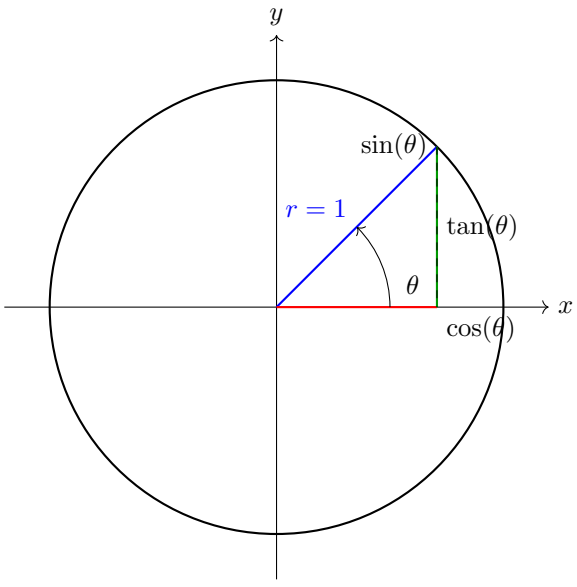
$\cos(x) = \sin(x + \frac{\pi}{2})$

25.10 Trigonometric Values

θ (radians)	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
0	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	undefined
$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{3}}$
π	0	-1	0
$\frac{7\pi}{6}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	1
$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$
$\frac{3\pi}{2}$	-1	0	undefined
$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\sqrt{3}$
$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-1
$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{3}}$
2π	0	1	0

25.11 Unit Circle and Trigonometric Functions

The *unit circle* is a circle with a radius of 1 centered at the origin. Trigonometric functions like $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$ can be defined based on the coordinates of a point on the circle corresponding to angle θ.



25.11.1 Definitions of the Trigonometric Functions on the Unit Circle

- $\cos(\theta)$ is the *x-coordinate* of the point on the circle.
- $\sin(\theta)$ is the *y-coordinate*.

- $\tan(\theta) = \dfrac{\sin(\theta)}{\cos(\theta)}$ is the ratio of the opposite side to the adjacent side in the triangle.

25.12 Trigonometric Rules

- $\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$
- $c^2 = a^2 + b^2 - 2ab\cos\gamma$
- $\frac{a-b}{a+b} = \frac{\tan\frac{1}{2}(\alpha-\beta)}{\tan\frac{1}{2}(\alpha+\beta)}$

25.13 Proof of the Cosine Rule

The *Cosine Rule* relates the lengths of the sides of a triangle to the cosine of one of its angles. For any triangle ABC with sides $a = BC$, $b = AC$, and $c = AB$, and angle $\gamma = \angle ACB$, the cosine rule states:

$$c^2 = a^2 + b^2 - 2ab\cos(\gamma)$$

Proof:

Consider triangle ABC, and place it on the coordinate plane such that point C is at the origin $(0, 0)$, point B is at $(a, 0)$, and point A lies somewhere in the plane. Let angle $\gamma = \angle ACB$.

Let the coordinates of point A be $(b\cos\gamma, b\sin\gamma)$, since $b = AC$ and we're using polar coordinates to define its location relative to point C.

Using the distance formula to find side $c = AB$, we get:

$$\begin{aligned} c^2 &= (a - b\cos\gamma)^2 + (0 - b\sin\gamma)^2 \\ &= a^2 - 2ab\cos\gamma + b^2\cos^2\gamma + b^2\sin^2\gamma \\ &= a^2 + b^2(\cos^2\gamma + \sin^2\gamma) - 2ab\cos\gamma \\ &= a^2 + b^2 - 2ab\cos\gamma \end{aligned}$$

Since $\cos^2\gamma + \sin^2\gamma = 1$.

Thus, we have proven that:

$$c^2 = a^2 + b^2 - 2ab\cos\gamma.$$

$\mathfrak{QED}$

Chapter 26

Linear Systems of Equations

A *linear system of equations* is a set of equations that can be expressed in the form:

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n &= b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n &= b_2 \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= b_m, \end{aligned}$$

where $a_{i,j}$ are the coefficients of the variables x_j, and b_i are the constants on the right-hand side of the equations. The goal is to find the values of $x_1, x_2, \ldots, x_n$ that satisfy all equations simultaneously.

More formally, given M, N with $f : M \to N$, for each $n \in N$ the corresponding *linear system of equations* is given by

$$f(m) = n,$$

with the solution set

$$f^{-1}(\{n\}) = \{m \in M \mid f(m) = n\}.$$

There are the following types of solutions depending on the nature of f:

- *At least one solution* if f is surjective.
- *One solution and no more* if f is injective/surjective.

26.1 Linear System as Matrices

A linear system can be represented in matrix form as:

$$Ax = b$$

where A is the coefficient matrix, x is the vector of variables, and b is the vector of constants. The coefficient matrix A is an $m \times n$ matrix, where m is the number of equations and n is the number of variables. The vector x is an $n \times 1$ column vector, and the vector b is an $m \times 1$ column vector.

26.2 Solution Set of a Linear System

The solution set $L(A, b)$ of the system of equations associated with (A, b) is defined as

$$L(A, b) := \{x \in \mathbb{F}^n \mid Ax = b\}.$$

26.3 Reduced Echelon Form

For a matrix $A \in \mathbb{F}^{m \times n}$ with rows $Z_1, \ldots, Z_n$, we define for $i \in \underline{\mathrm{m}}$ the i-th *echelon index* $Ei_i(A)$ as

$$Ei := \min\{j \in \underline{\mathrm{n}} \mid A_{i,j} \neq 0\}.$$

We define the *echelon form* as the sequence

$$Ei(A) := (Ei_1(A), Ei_2(A), \ldots, Ei_s(A)).$$

If the index of all $Ei_i(A)$ coincides with the column number, then we call it the *strict echelon form*.

26.4 Equality of the Rank $\mathrm{rank}(A) = \mathrm{rank}_S(A)$

We have $\mathrm{rank}(A) = \mathrm{rank}_S(A)$.

Proof:

For $x = (x_1, \ldots, x_n)^T \in \mathbb{F}^n$, it holds that

$$Ax = \sum_{i=1}^{n} x_i a_i.$$

It follows that

$$\mathrm{rank}(A) = \mathrm{rank}(L_A) = \dim(\mathrm{img}(L_A)) = \dim\left(\{L_A(x) \mid x \in \mathbb{F}^n\}\right) = \dim\left(\{Ax \mid x \in \mathbb{F}^n\}\right)$$

$$= \dim\left\{\sum_{i=1}^{n} x_i a_i \,\middle|\, x_i \in K\right\} = \dim(L(a_1, \ldots, a_n)) = \mathrm{rank}_S(A).$$

26.5 Solvability of a Linear System $Ax = b$

The linear system $Ax = b$ is solvable if and only if

$$\mathrm{rank}(a_1, \ldots, a_n) = \mathrm{rank}(a_1, \ldots, a_n, b).$$

More concisely, we write

$$\mathrm{rank}(A) = \mathrm{rank}(A, b).$$

Proof:

We have

$$Ax = (a_1, \ldots, a_n)x = x_1 a_1 + \cdots + x_n a_n = b.$$

Therefore, a solution vector x exists if and only if

$$b \in L(a_1, \ldots, a_n) \quad \iff \quad \operatorname{rank}(A) = \operatorname{rank}(A, b),$$

where $L(a_1, \ldots, a_n)$ denotes the linear span of $a_1, \ldots, a_n$.

26.6 Solution Set of the Linear System $Ax = b$

Let $x_s \in \mathbb{F}^n$ be a solution of $Ax = b$. Then the solution set is given by

$$L(A, b) = x_s + \ker(A) = \{x_s + x \mid x \in \ker(A)\}.$$

Proof:

$$x \in \ker(A) \iff Ax = 0 \iff A(x_s + x) = Ax_s + Ax = Ax_s = b.$$

26.7 Free Variables

In a system of linear equations, a *free variable* is a variable whose column in the coefficient matrix does not correspond to a pivot column after row reduction. Free variables are not directly solved by any equation in the system; instead, they can take any real value, and the values of the leading (or dependent) variables are expressed in terms of them.

Free variables arise naturally in systems with infinitely many solutions and play a key role in describing the solution set. Given a system with p equations and q unknowns, if $q > p$ then there are $q - p$ free variables.

Example:

Consider the system

$$\begin{cases} x_1 + 2x_2 - x_3 = 4 \\ 3x_1 + 6x_2 - 3x_3 = 12 \end{cases}$$

which row reduces to

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Here, the pivot is in the first column corresponding to x_1, while x_2 and x_3 are free variables. Assigning parameters to x_2 and x_3, the general solution can be expressed in terms of these free variables.

26.8 Gaussian Elimination

Gaussian elimination is a method for solving linear systems by transforming the system into an upper triangular form. The steps involved in Gaussian elimination are:

1. *Forward elimination*: Transform the system into an upper triangular form by eliminating the variables from the equations.
2. *Back substitution*: Solve for the variables starting from the last equation and substituting back into the previous equations.

The forward elimination process involves performing row operations on the augmented matrix $[A|b]$ to create zeros below the diagonal. The row operations include:

- Swapping two rows.
- Multiplying a row by a non-zero scalar.

- Adding or subtracting a multiple of one row from another row.

These steps are defined as linear maps applied to a system like $f(m) = n$ to $(g \circ f)(m) = g(n)$. We need to be able to find the respective inverses to get to the result $(g \circ f)^{-1}(g(n)) = n$. This is accomplished via the *elementary matrices*.

Once the matrix is in upper triangular form, back substitution is used to find the values of the variables. The last equation gives the value of the last variable, which can then be substituted into the previous equations to find the other variables.

26.8.1 Procedure

Starting with $Ax = b$

$$\left[\begin{array}{cccc|c} a_{1,1} & a_{1,2} & \cdots & a_{1,n} & b_1 \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} & b_m \end{array}\right]$$

For eliminating the elements underneath a pivot element we apply

$$\text{Row}_n - \frac{a_{n,i}}{a_{n-1,i}}\text{Row}_{n-1},$$

where n is the index of the target row and i is the index of the column.

We denote the coefficient $\frac{a_{n,i}}{a_{n-1,i}} = l_{n,i}$.

This results in

$$\left[\begin{array}{cccc|c} a_{1,1} & a_{1,2} & \cdots & a_{1,n} & b_1 \\ 0 & a_{2,2} - a_{1,2}\frac{a_{2,1}}{a_{1,1}} & \cdots & a_{2,n} - a_{1,n}\frac{a_{2,1}}{a_{1,1}} & b_2 - b_1\frac{a_{2,1}}{a_{1,1}} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \lambda & \gamma \end{array}\right]$$

With λ and γ being complex sums of the results of the other manipulations.

26.9 Invariability of the Solution Set

Let $f : M \to N$ and $n \in N$. For every bijective map $g : N \to N$, the following holds for the fiber of f over n:

$$f^{-1}(\{n\}) = (g \circ f)^{-1}(\{g(n)\}).$$

Proof:

The first inclusion does not use the bijectivity of g:

Let $m \in f^{-1}(\{n\})$, i.e. $f(m) = n$. Then $g(f(m)) = g(n)$, i.e. $m \in (g \circ f)^{-1}(\{g(n)\})$. For the reverse implication one needs a left inverse of g, which of course exists because g is bijective.

$\mathfrak{QED}$

26.10 Gauss-Jordan Elimination

Gauss-Jordan elimination is an extension of Gaussian elimination that transforms the matrix into reduced row echelon form (RREF). In RREF, each leading entry in a row is 1, and all entries above and below the leading entry are zeros. The steps involved in Gauss-Jordan elimination are:

1. Forward elimination: Transform the system into an upper triangular form.
2. Back substitution: Transform the upper triangular matrix into RREF by eliminating the entries above the leading 1s.
3. Solve for the variables directly from the RREF matrix.

The Gauss-Jordan elimination method is particularly useful for finding the inverse of a matrix, as it can be applied to the augmented matrix $[A|I]$, where I is the identity matrix. If the left side of the augmented matrix becomes I, then the right side will be the inverse of A.

Example:

Solve the following system of equations using Gaussian elimination:

$$\begin{aligned} 2x + 3y + z &= 1 \\ 4x + y - z &= 2 \\ -2x + y + 3z &= 3 \end{aligned}$$

Solution: The augmented matrix for the system is:

$$\left[\begin{array}{ccc|c} 2 & 3 & 1 & 1 \\ 4 & 1 & -1 & 2 \\ -2 & 1 & 3 & 3 \end{array}\right]$$

Performing row operations to eliminate the variables, we can transform the matrix into upper triangular form:

$$\left[\begin{array}{ccc|c} 1 & \frac{3}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & -5 & -3 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

Now, we can perform back substitution to find the values of x, y, and z:

$$\begin{aligned} z &= 1 \\ -5y - 3z = 0 &\implies y = -\frac{3}{5} \\ 2x + 3y + z = 1 &\implies x = \frac{1}{5} \end{aligned}$$

Thus, the solution to the system is:

$$\begin{aligned} x &= \frac{1}{5} \\ y &= -\frac{3}{5} \\ z &= 1 \end{aligned}$$

26.11 Homogeneous Linear Equations

A linear equation is said to be *homogeneous* if its constant term is zero. That is, it can be written in the form:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$$

Such equations always have at least the trivial solution $x_1 = x_2 = \cdots = x_n = 0$.

26.12 Particular Solution

A *particular solution* to a linear system of equations is a specific solution that satisfies the system. It can be found using various methods, including substitution, elimination, or matrix methods. A particular solution is not unique; there may be multiple particular solutions depending on the system. A particular solution can be found by substituting specific values for the variables and solving for the remaining variables. For example, in the system:

$$\begin{aligned} 2x + 3y &= 5 \\ 4x - y &= 1 \end{aligned}$$

we can substitute $x = 1$ into the first equation to find y:

$$\begin{aligned} 2(1) + 3y &= 5 \\ 3y &= 3 \\ y &= 1 \end{aligned}$$

Thus, $(x, y) = (1, 1)$ is a particular solution to the system. However, this is not the only solution; other values of x may yield different values of y.

26.13 General = Particular + Homogeneous

The *general solution* of a linear system of equations is the complete set of solutions that satisfy the system. It can be expressed as the sum of a particular solution and the general solution of the associated homogeneous system.

The general solution can be written as:

$$x = x_p + x_h$$

Where x_p is a particular solution to the non-homogeneous system, and x_h is the general solution to the homogeneous system. The homogeneous system is obtained by setting the right-hand side of the equations to zero:

$$Ax = 0.$$

The theorem says:

Any linear system's solution set has the form:

$$\left\{ \vec{p} + c_1\vec{\beta}_1 + \cdots + c_k\vec{\beta}_k \;\middle|\; c_1, \ldots, c_k \in \mathbb{R} \right\},$$

where $\vec{p}$ is a particular solution to the system, and the vectors $\vec{\beta}_1, \ldots, \vec{\beta}_k$ form a basis of the solution space to the corresponding homogeneous system. The number k equals the number of *free variables* the system has after applying Gaussian elimination.

Proof:

We want to prove that every linear system of equations $Ax = b$ which has at least one solution fulfills the following properties:

1. $L = \{Ax = 0\} \neq \emptyset$
2. The general solution is linear combination of the particular solutions plus the homogeneous solution.

For the first property, the solution of $Ax = 0$ is $\ker(A)$, more precisely $\ker(\phi_A)$, and thus it is not empty because $0 \in \ker(A)$.

The second proposition follows because the addition of the zero vector to a linear combination of other vectors just gives you the original combination by the axioms of the vector spaces.

$\mathfrak{QED}$

26.14 Linear Combination Lemma

Any linear combination of linear combinations is a linear combination.

26.15 Example: Gaussian Elimination With 3 Equations and 4 Unknowns

Consider the following system of linear equations:

$$\begin{aligned} x_1 + 2x_2 + x_3 + x_4 &= 4 \\ 2x_1 + 5x_2 + x_3 + 3x_4 &= 10 \\ x_1 + 3x_2 + 2x_3 + 2x_4 &= 7 \end{aligned}$$

Step 1: Augmented Matrix

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 4 \\ 2 & 5 & 1 & 3 & 10 \\ 1 & 3 & 2 & 2 & 7 \end{bmatrix}$$

Step 2: Eliminate below pivot in column 1

- Row 2 = Row 2 − 2 × Row 1
- Row 3 = Row 3 − Row 1

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 4 \\ 0 & 1 & -1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 3 \end{bmatrix}$$

Step 3: Eliminate below pivot in column 2

- Row 3 = Row 3 − Row 2

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 4 \\ 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 2 & 0 & 1 \end{bmatrix}$$

Step 4: Back Substitution

From Row 3:

$$2x_3 = 1 \implies x_3 = \frac{1}{2}$$

From Row 2:

$$x_2 - x_3 + x_4 = 2 \implies x_2 = 2 + x_3 - x_4 = 2 + \frac{1}{2} - x_4 = \frac{5}{2} - x_4$$

From Row 1:

$$x_1 + 2x_2 + x_3 + x_4 = 4 \implies x_1 = 4 - 2x_2 - x_3 - x_4$$

Substitute:

$$x_1 = 4 - 2\left(\frac{5}{2} - x_4\right) - \frac{1}{2} - x_4 = 4 - 5 + 2x_4 - \frac{1}{2} - x_4 = -1 - \frac{1}{2} + x_4 = -\frac{3}{2} + x_4$$

General Solution

Let $x_4 = t$ (free variable), then:

$$\begin{aligned} x_1 &= -\frac{3}{2} + t \\ x_2 &= \frac{5}{2} - t \\ x_3 &= \frac{1}{2} \\ x_4 &= t \end{aligned} \qquad \text{with } t \in \mathbb{R}$$

Solution Set:

$$\left\{ \begin{bmatrix} -\frac{3}{2} \\ \frac{5}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \middle| \, t \in \mathbb{R} \right\}$$

26.16 The Determinant, the Cross Product and the Solutions of Linear Systems of Equations

A linear system of three equations has the following properties:

- There is a unique solution if the determinant of the coefficient matrix is non-zero.

$$\langle (a \times b), c \rangle = \det(a, b, c) \neq 0$$

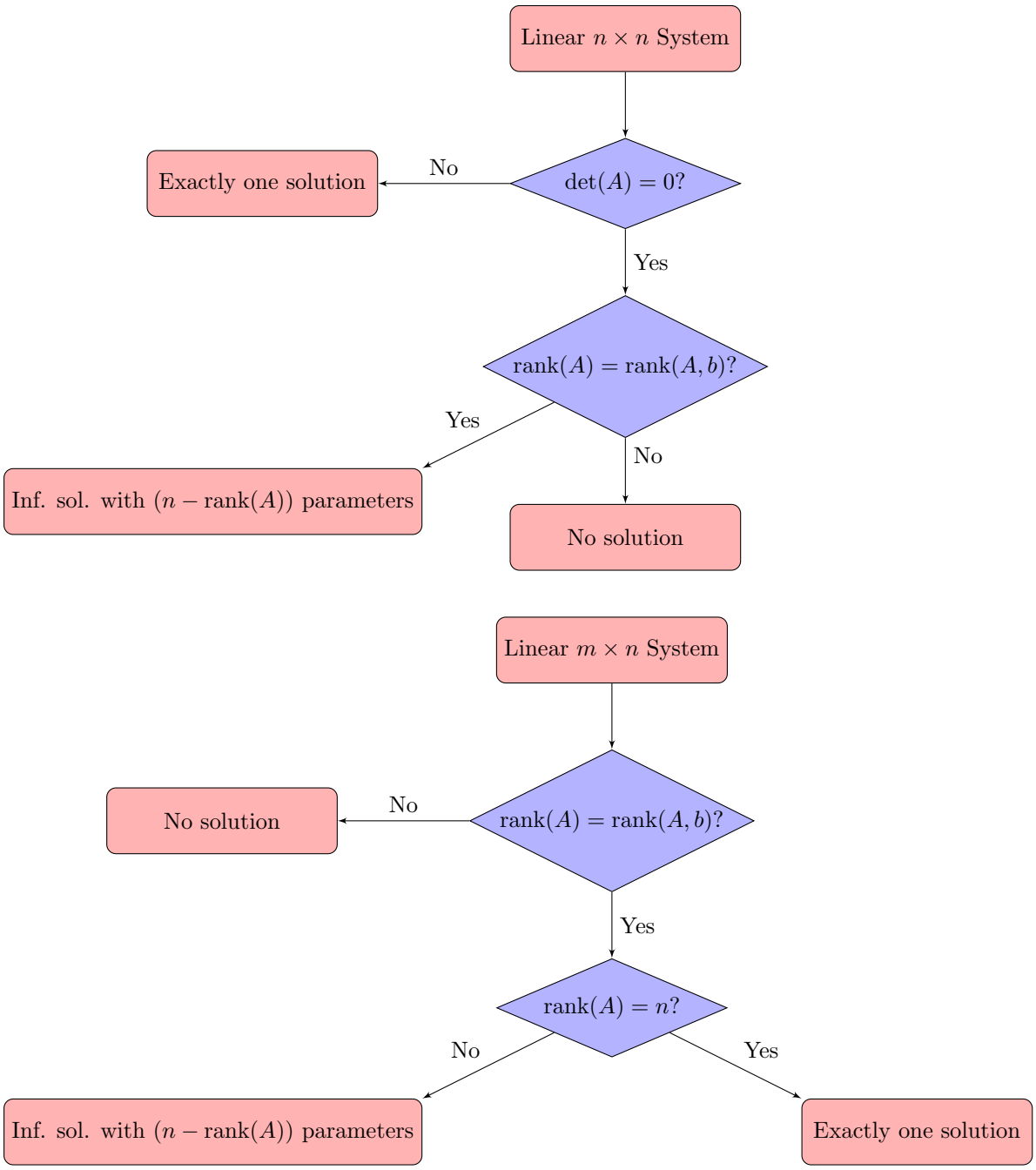
- There are infinitely many solutions if the determinant of the coefficient matrix is zero.

$$\langle (a \times b), c \rangle = \det(a, b, c) = 0$$

- There is no solution if the determinant of the coefficient matrix is zero and the system is inconsistent.

$$\langle (a \times b), c \rangle = 0$$

26.17 Chart for the Number of Solutions



26.18 Proof of the Gaussian Elimination

The row operations do not change the solution set.

Proof:

A linear system of equations of the form $Ax = b$, $A \in \mathbb{F}^{m \times n}$ with some solution x can be multiplied by some elementary matrix C_1, C_2, C_3 which correspond to a row operation $CAx = Cb$.

The equality is kept and therefore, $L(A, b) \subset L(CA, Cb)$. If $x \in L(CA, Cb)$ then

$$CAx = Cb \implies C^{-1}CAx = C^{-1}Cb \implies Ax = b$$

Because of the invertibility of elementary matrices we can be sure that $L(CA, Cb) \subset L(A, b)$.

Now for the case that $L(A,b) = \emptyset$. If we assume that $L(CA,Cb) \neq \emptyset$, then there would be some $x \in L(CA,Cb)$ with $CAx = Cb$. By multiplying both sides by C^{-1} we get $Ax = b$ and therefore, $L(A,b) \neq \emptyset$, which is a contradiction to our original assumption $L(A,b) = \emptyset$.

$\mathfrak{QED}$

Now we only have to prove that each regular matrix can be transformed in the reduced echelon form or even row reduced echelon form. This is the equivalent of saying that the row operations do not change the rank of the matrix.

Proof:

If C is an elementary matrix and $A \in \mathbb{F}^{m\times n}$ some matrix. The statement $\text{rank}(CA) = \text{rank}(A)$ is true because of the invertibility of the elementary matrices and the isomorphism in vector spaces.

Given $A = (a_1, a_2, \ldots, a_n)$. The exchange of two columns preserves $L := L(a_1, \ldots, a_n)$, and also the $\text{rank}_s(A)$. Concerning the scaling of a column by some λ, $L := L(a_1, \ldots, \lambda a_k, \ldots, a_n)$ also preserves $dim(L) = \text{rank}_s(A)$. And finally, the addition $L = L(a_1, \ldots, a_i + \lambda a_k, \ldots, a_n)$ also preserves $\text{rank}_s(A) = \text{rank}(A)$. Thus, $\text{rank}(CA) = \text{rank}(A)$.

$\mathfrak{QED}$

The only part left is to prove that if A has an inverse, then with use of the row operations we can transform it in to a matrix with an upper triangular with also no zero element on the main diagonal.

Proof:

First we eliminate all the elements under the main diagonal for all columns until the column $k-1$. With this, we have created a new matrix $A^(\ell) = C_\ell \cdot \vdots \cdot \dot{C}_1 A$ which looks like:

$$A^{(\ell)} = \begin{bmatrix} a_{1,1}^{(\ell)} & \cdots & a_{1,k-1}^{(\ell)} & a_{1,k}^{(\ell)} & a_{1,k+1}^{(\ell)} & \cdots & a_{1,n}^{(\ell)} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & a_{k-1,k-1}^{(\ell)} & a_{k-1,k}^{(\ell)} & a_{k-1,k+1}^{(\ell)} & \cdots & a_{k-1,n}^{(\ell)} \\ 0 & \cdots & 0 & a_{k,k}^{(\ell)} & a_{k,k+1}^{(\ell)} & \cdots & a_{k,n}^{(\ell)} \\ 0 & \cdots & 0 & \vdots & a_{k+1,k+1}^{(\ell)} & \cdots & a_{k+1,n}^{(\ell)} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{n,k}^{(\ell)} & \cdots & \cdots & a_{n,n}^{(\ell)} \end{bmatrix}$$

Because of $\text{rank}(A) = \text{rank}(A^{(\ell)})$, $a_{i,i}^{(\ell)} \neq 0$ for $1 \leq i < k$. All of these elements will not change for the rest of the Gaussian Elimination. Now we want to build another matrix with row operations $A^{(\mu)}$ where $a_{k,k}^{(\mu)} \neq 0$ and $a_{i,k}^{(\mu)} = 0$ for $i > k$. If $a_{k,k}^{(\ell)} \neq 0$ then (i,k)-th element will become 0 by subtracting $\frac{a_{i,k}^{(\ell)}}{a_{k,k}^{(\ell)}}$ time the k-th row of the i-th row. So can the k-th column be transformed with a series of row operations to the desired form. But if $a_{k,k}^{(\ell)} = 0$, then you have to swap it with another row under it. This is the critical point of the algorithm: What if there is no row, also $a_{i,k}^{(\ell)} = 0$ for $i \geq k$? Then $A^{(\ell)}$ would be:

$$A^{(\ell)} = \begin{bmatrix} a_{1,1}^{(\ell)} & \cdots & a_{1,k-1}^{(\ell)} & a_{1,k}^{(\ell)} & a_{1,k+1}^{(\ell)} & \cdots & a_{1,n}^{(\ell)} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & a_{k-1,k-1}^{(\ell)} & a_{k-1,k}^{(\ell)} & a_{k-1,k+1}^{(\ell)} & \cdots & a_{k-1,n}^{(\ell)} \\ 0 & \cdots & 0 & 0 & a_{k,k+1}^{(\ell)} & \cdots & a_{k,n}^{(\ell)} \\ 0 & \cdots & 0 & \vdots & a_{k+1,k+1}^{(\ell)} & \cdots & a_{k+1,n}^{(\ell)} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & \cdots & a_{n,n}^{(\ell)} \end{bmatrix}$$

But, that would imply that the k-th column is a linear combination of the other $k-1$ columns, and then $\text{rank}(A^{(\ell)}) \leq n-1$ which is a contradiction to the assumption $\text{rank}(A^{(\ell)}) = n$.

$\mathfrak{QED}$

26.19 Alternative Method for Finding the Rank of a Matrix

For a matrix $A \in \mathbb{F}^{(m\times n)}$. The biggest submatrix of size $r \times r$ with a non-zero determinant gives us the *rank* of the matrix. $\operatorname{rank}(A) = r$.

26.20 Cramer's Rule

For a square matrix $A = (a_1, \ldots, a_n)$ and $x, b \in \mathbb{F}^n$ with $Ax = b$ and $\det A \neq 0$, then

$$A_i = (a_1, \ldots, a_{i-1}, b, a_{i+1}, \ldots, a_n), i = 1, \ldots, n.$$

The solution i is then

$$x_i = \frac{\det(A_i)}{\det(A)}$$

Proof:

Given a matrix $X_i \in \mathbb{F}^{n\times n}$ by

$$\begin{bmatrix} 1 & & \cdots & x_1 & \cdots \\ \vdots & & \ddots & \vdots & 0 \\ & & x_{i-1} & & \\ & 0 & x_i & & \\ & & x_{i+1} & & \\ & & \vdots & \ddots & \\ & & x_n & & 1 \end{bmatrix}$$

$X_i = (e_1, \ldots, e_{i-1}, e_i, e_{i+1}, \ldots, e_n)$. Applying the Laplace formula n times along the first row shows that $|X_i| = x_i$.

$$\begin{aligned} AX_i &= A(e_1, \ldots, e_{i-1}, e_i, e_{i+1}, \ldots, e_n) \\ AX_i &= A(a_1, \ldots, a_{i-1}, Ax, a_{i+1}, \ldots, a_n) \\ AX_i &= A(a_1, \ldots, a_{i-1}, b, a_{i+1}, \ldots, a_n) = A_i \end{aligned}$$

This implies $|A_i| = |AX_i| = |A||X_i|$ therefore, $x_i = \frac{|A_i|}{\det(A)}$.

$\mathfrak{QED}$

Another way of thinking about this in a more geometric way is to first think of the determinant as the area/volume/etc. that is spanned by the basis vector and the stretching factor of the transformation $\det(A)$. Now visualize your matrix A as a linear transformation with a non-zero determinant and think about $\det(A_i)$ as the area/volume/etc. which is spanned by the basis vector but with one of the vectors substituted by b.

Now in two dimensions, think of the area spanned by the basis vectors $A = xy$. The signed area of the original parallelogram gets stretched by $\det(A)$, which gives us $Area = \det(A)y \implies y = \frac{Area}{\det(A)}$. This also works for x.

26.21 LU Decomposition

Given a regular system $Ax = b$ with $a_{k,k}^{(k-1)} \neq 0$, then the Gaussian elimination delivers a so-called *LU-Decomposition*

$$A = LU$$

of A, where L is a lower triangular matrix with main diagonal consisting only of ones and U being an upper triangular matrix. $a_{k,k}^{(k-1)} \neq 0$ denotes the elements of the main diagonal for the $k-1$ matrix generated after applying a Gauss operation.

26.21.1 Derivation

Given our matrix A, we want to eliminate the coefficients underneath the main diagonal. This is accomplished by applying an elementary matrix of the form:

$$F = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & \lambda & 1 & \\ & & & & \ddots \end{bmatrix}$$

Hence, to get to the row-echelon-form we apply a series of such F matrices, where for the elimination of the k-th row we use the *Frobenius* matrix

$$F_k = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & -l_{k+1,k} & 1 & \\ & & -l_{n,k} & & \ddots \end{bmatrix}$$

where $-l_{i,k} = \frac{a_{i,k}^{(k-1)}}{a_{k,k}^{k-1}}$.

For example:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix}, a_{1,1} \neq 0$$

To eliminate $a_{2,1}$ we need to subtract $\frac{a_{2,1}}{a_{1,1}}$ times the first row.

$$F = \begin{bmatrix} 1 & 0 \\ -\frac{a_{2,1}}{a_{1,1}} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -l_{2,1} & 1 \end{bmatrix}$$

The multiplication results in

$$FA = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} - \frac{a_{1,1}a_{2,1}}{a_{1,1}} & a_{2,2} - \frac{a_{2,1}}{a_{1,1}} \end{bmatrix} = \begin{bmatrix} a_{1,1} & a_{1,2} \\ 0 & a_{2,2} - \frac{a_{2,1}}{a_{1,1}} \end{bmatrix}$$

Note that the main diagonal does not have to be normalized.

By repeating this process we get

$$U = A^{(n-1)} = F_{n-1} \cdot \cdots \cdot F_1 A$$

and

$$y = b^{(n-1)} = F_{n-1} \cdot \dots \cdot F_1 b$$

Due to all F_k being regular, we get that $Ax = b$ and $Ux = y$ are equivalent. Hence, by applying the inverses

$$A = F_1^{n-1} \cdot \dots \cdot F_{n-1} U$$

This chain of the inverses is called

$$L = \begin{bmatrix} 1 & & & & \\ l_{2,1} & \ddots & & & \\ \vdots & & 1 & & \\ & & & 1 & \\ l_{n,1} & & \cdots & l_{n,n-1} & 1 \end{bmatrix}$$

The individual members are

$$F_k^{-1} = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & l_{k+1,k} & 1 & \\ & & l_{n,k} & & 1 \end{bmatrix}$$

Therefore $A = LU$ and $Ly = b$.

26.21.2 Algorithm

Overview:

1. Get L and U via Gaussian Elimination.
2. Solve $Ly = b$.
3. Solve $Ux = y$.

Concrete algorithm:

Start with your $n \times n$ matrix and perform the following steps across all of the entries in the main diagonal and its corresponding sub-square-matrix.

1. Take the current diagonal element and proceed to divide all entries in the same column underneath by it. Do not forget that we may be working within a sub-matrix.
2. The current row of our matrix is going to be subtracted from all of the rows underneath inside our sub-matrix. The factor we are going to use for our row at each subtraction is going to be the first entry of the current column inside our sub-matrix.
3. We reduce the scope of our matrix and repeat the steps above until we have a 1 by 1 matrix.
4. After the previous process, the resulting matrix will be divided into two matrices. L consisting of the elements underneath the main diagonal with it being all ones and above all zeros; and U being the rest, the upper triangular matrix.

Example:

Given

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 0 \\ 2 & 7 & 1 \end{bmatrix}, b = \begin{bmatrix} 7 \\ 12 \\ 19 \end{bmatrix} \text{ and } b' = \begin{bmatrix} 3 \\ 4 \\ 9 \end{bmatrix}$$

solve using the LU-decomposition.

First iteration:

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 0 \\ 2 & 7 & 1 \end{bmatrix} \rightarrow_{\text{division by 2}} \begin{bmatrix} 2 & 1 & 1 \\ 2 & 4 & 0 \\ 1 & 7 & 1 \end{bmatrix}$$

Now we subtract $II - 2I$ and $III - 1I$:

$$\begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & -2 \\ 1 & 6 & 0 \end{bmatrix}$$

Second iteration:

$$\begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & -2 \\ 1 & 6 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & -2 \\ 1 & 3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & -2 \\ 1 & 3 & 6 \end{bmatrix}$$

Hence

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 6 \end{bmatrix}$$

We solve $Ly = b$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix} y = \begin{bmatrix} 7 \\ 12 \\ 19 \end{bmatrix} \iff y = \begin{bmatrix} 7 \\ -2 \\ 18 \end{bmatrix}$$

Then we solve $Ux = y$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 6 \end{bmatrix} x = \begin{bmatrix} 7 \\ -2 \\ 18 \end{bmatrix} \iff x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

26.21.3 Extended Algorithm for Any Regular Matrix

We want to expand the algorithm to be compatible with the swap rows operations. Given a regular matrix $A^{(k-1)} = F_{k-1} \cdot \cdots \cdot F_1 A$, where after Gaussian operations we get $a_{k,k}^{(k-1)} = 0$, then there exists in the k-th row underneath at least one $a_{i,k}^{(k-1)} \neq 0, i > k$, so that we can swap the rows.

Proof:

Let us suppose that there does not exist such an element; then the k-th column must be a linear combination of the other columns because there is a free variable, which contradicts our initial assumption. Note that due to F_i also being regular, the multiplication of $A^{(k-1)}$ must also be regular. Therefore, it is always possible to swap rows.

$\mathfrak{QED}$

26.21.4 Derivation of the Algorithm

We start with the same setup as before, but now we also have

$$P = \begin{bmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 0 & & 1 & & & \\ & & & 1 & & & & \\ & & \vdots & & \ddots & \vdots & & \\ & & 1 & & & 0 & & \\ & & & & & & 1 & \\ & & & & & & & \ddots \\ & & & & & & & & 1 \end{bmatrix}$$

with the following properties: $P = P^T$ and $PP = I$, which means $P = P^{-1}$.

Our U and L becomes

$$U = F_{n-1}P_{n-1} \cdot \cdots \cdot F_1 P_1 A$$

$$\hat{L} = (F_{n-1}P_{n-1} \cdot \cdots \cdot F_1 P_{-1})^{-1}$$

And $U = \hat{L}^{-1}A$, the question is how to get $\hat{L}$ to always be a lower triangular matrix.

For that we use the inverse properties of permutation matrices to multiply by the identity matrix

$$I = P_1 P_2 \cdot \cdots \cdot P_{n-1} P_{n-1} \cdot \cdots \cdot P_2 P_1$$

by associativity.

We get

$$\begin{aligned} U &= F_{n-1}P_{n-1} \cdot \cdots \cdot F_1 P_1 A \\ &= F_{n-1}P_{n-1} \cdot \cdots \cdot F_1 P_1 \cdot (P_1 P_2 \cdot \cdots \cdot P_{n-1} P_{n-1} \cdot \cdots \cdot P_2 P_1) A \\ &= (F_{n-1}P_{n-1} \cdot \cdots \cdot F_1 P_1 \cdot P_1 P_2 \cdot \cdots \cdot P_{n-1}) P_{n-1} \cdot \cdots \cdot P_2 P_1 A \\ &= (F_{n-1}P_{n-1} \cdot \cdots \cdot F_1 P_1 \cdot P_1 P_2 \cdot \cdots \cdot P_{n-1}) P A \\ &= L^{-1} P A \end{aligned}$$

Where

$$\begin{aligned} L &= (F_{n-1}P_{n-1} \cdot \cdots \cdot F_1 P_1 \cdot P_1 P_2 \cdot \cdots \cdot P_{n-1})^{-1} \\ &= P_{n-2} \cdot \cdots \cdot P_2 F_1^{-1} P_2 F_2^{-1} \cdot \cdots \cdot P_{n-1} F_{n-1}^{-1} \end{aligned}$$

Now L is a lower triangular matrix, because of the following inductive proof:

Let $L_1 = F_1^{-1}$, $L_k = P_k L_{k-1} P_k F_k^{-1}$ for $k = 2, \ldots, n-1$. That is, $L_{n-1} = L$.

We show that all L_k are lower triangular matrices.

Base Case: For $k = 1$

$$L_1 = \begin{bmatrix} 1 & & & & & \\ l_{2,1} & \ddots & & & & \\ \vdots & & 1 & & & \\ & & & 1 & & \\ l-n,1 & & & & \ddots & \end{bmatrix}$$

Inductive Step:

For $L_2 = P_2 L_1 P_2 F_2^{-1}$:

In $P_2 L_1$, the second and j-th rows are changed

$$P_2 L_1 = \begin{bmatrix} 1 & & & & & \\ l_{21} & \ddots & & 1 & & \\ l_{j,1} & & 1 & & & \\ & \vdots & & 1 & & \\ l-n,1 & & & & \ddots & \end{bmatrix}$$

Then in $P_2 L_1 P_2$ the change gets undone; and finally, when multiplied by

$$F_2^{-1} = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & l_{3,2} & 1 & & \\ & \vdots & & 1 & \\ & l_{n,2} & & & \ddots \end{bmatrix}$$

we get

$$L_2 = P_2 L_1 P_2 F_2^{-1} = \begin{bmatrix} 1 & & & & \\ l_{2,1} & \ddots & & & \\ \vdots & l_{3,2} & 1 & & \\ & \vdots & & 1 & \\ l_{n,1} & l_{n,2} & & & \ddots \end{bmatrix}$$

Hence, by induction L_k is always a lower triangular matrix.

$\mathfrak{QED}$

Formally we state that given a regular matrix $A \in \mathbb{R}^{n \times n}$, there are also permutations $P_k, k = 1, \dots n-1$ such that $P = P_{n-1} \cdot \dots \cdot P_1$ for which

$$PA = LU$$

where L and U are lower and upper triangular matrices, respectively.

26.21.5 Extended LU-Decomposition Algorithm

The only important change for the algorithm is that for each row swaps to our matrix $A^{(i)}$ we perform the same on the matrix $P^{(i)}$. The process for the Gaussian elimination remains the same.

Overview:

1. Decompose A into L, U and P.
2. Solve $Ly = Pb$. First permute b just like A, and then solve $Ly = Pv$.
3. Solve $Ux = y$.

Example:

Find the L, U and P for

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 2 & 1 & 2 & 0 \\ 4 & 4 & 0 & 0 \\ 2 & 3 & 1 & 0 \end{bmatrix}$$

We now use a vector p of size $n \times 1$, where the j-th entry p_j is read as: switch row j with p_j.

We swap rows 1 and 2, giving us $p_1 = 2$:

$$\begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 4 & 4 & 0 & 0 \\ 2 & 3 & 1 & 0 \end{bmatrix}$$

Proceeding with the algorithm as for the simple LU-decomposition

$$\begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 4 & 4 & 0 & 0 \\ 2 & 3 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 2 & -4 & 0 \\ 1 & 2 & -1 & 0 \end{bmatrix} (p_2 = 3) \rightarrow \begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & -4 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 2 & -1 & 0 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & -4 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & -4 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 3 & 0 \end{bmatrix} (p_3 = 4) \rightarrow \begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & -4 & 0 \\ 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Hence

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 2 & -4 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad v = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

Now the solving part is equal for both procedures:

First $Ly = Pb$, where Pb is the permuted vector using p.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} y = \begin{bmatrix} 5 \\ 4 \\ 5 \\ 3 \end{bmatrix} \iff y = \begin{bmatrix} 5 \\ -6 \\ 6 \\ 3 \end{bmatrix}$$

Finally, $Ux = y$:

$$\begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 2 & -4 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 5 \\ -6 \\ 6 \\ 3 \end{bmatrix} \iff x = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

26.21.6 Greatest Pivot Strategy

As a side note, to mitigate the rounding errors in scientific computing, the swap row strategy is modified to always take the row with the biggest element in the current column, before performing the division step. In such a case we swap the entire row and not just the row inside our sub-matrix.

We also need to keep track of the swaps, which is accomplished by storing a vector $x \in \mathbb{R}^n$ where x_i is the position of the i-th row.

26.22 Cholesky Decomposition

This is a special case of the LU-Decomposition. Given $A \in \mathbb{R}^{n \times n}$ a symmetric, positive definite matrix it can be decomposed as

$$A = \tilde{L} D \tilde{L}^T,$$

where $\tilde{L}$ is a lower triangular matrix with 1's on the main diagonal and D a diagonal matrix with positive diagonal elements. (*Rational Cholesky Decomposition*).

Or $A = GG^T$, where G is a lower triangular matrix. (*Cholesky Decomposition*).

26.22.1 Algorithm for the Cholesky Decomposition

1. Find the LU-decomposition of A.
2. D is given by the entries in the diagonal of U.
3. Determine V by taking L^T, replacing its main diagonal with the diagonal of U and then dividing each row by the pivot elements of D. For symmetric matrices $V = L^T$.
4. Find the root of D by splitting it into two diagonal matrices which consist of the square roots of the elements in the main diagonal. Take one of them and call it S.
5. Find $G = LS$.
6. Finally $A = GG^T$.

Example:

Find the Cholesky decomposition of

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -2 \\ 1 & -2 & 14 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -2 \\ 1 & -2 & 14 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -3 \\ 1 & -3 & 13 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -3 \\ 1 & -3 & 13 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -3 \\ 1 & -3 & 4 \end{bmatrix}$$

We get

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & -3 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 4 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Then, we get

$$S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad V = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{bmatrix}$$

And finally,

$$Q = LS = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & -3 & 2 \end{bmatrix}$$

We got $A = QQ^T$.

Chapter 27

Over/Underdetermined Systems of Equations

Let us start with a little motivation for this chapter. Imagine that you have collected some data points, and you want to relate them via a linear function $Ax = (\alpha, \beta, \dots)^T$. This would probably not work because it will be impossible to connect all the points, but you could make an approximation which is the line $\alpha x_1 + \beta x_2 + \dots$ that is the closest to all the points in your data set. Thus, we will find an approximation b that is close to the image of A ($\text{img}(A)$).

For this purpose, we will use a theorem that says that the best approximation is the orthogonal projection of $p_A(b)$. This projection can be found if there exists an orthonormal basis of the subspace we want to project our vector onto.

27.1 Normal Equations

Given a projection $p_A(b)$ onto the subspace U generated by $A = (a_1, \dots, a_n)$ then there exists an $x \in \mathbb{R}^n$ such that

$$p_A(b) = \sum_{k=1}^{n} x_k a_k = Ax.$$

Also, because of the orthogonal property,

$$\begin{aligned} b - p_A(b) \perp U &\implies b - Ax \in U^\perp \\ &\implies b - Ax \perp a_k \quad \forall k \\ &\implies \langle a_k, b - Ax \rangle = 0 \quad \forall k \\ &\iff A^T(b - Ax) = \vec{0} \\ &\implies A^T b = Ax. \end{aligned}$$

This equation $A^T b = Ax$ is called the *normal equation*.

A *normal equation* of a matrix $A \in \mathbb{R}^{(m \times n)}$ has a solution. If $\text{rank}(A) = n$ then

$$x = (A^T A)^{-1} A^T b$$

is a unique solution to the system. The term $(A^T A)^{-1} A^T$ is called the *Generalized Inverse* of A.

Proof:

The statement is true because of the existence of the orthogonal projection, which gives us a concrete basis. Also, the fact that $rg(A) = n$ implies that, and also because of the invertibility of $(A^T A)^{-1}$.

QED

27.2 The Least Squares Method

Consider a linear system of equations $Ax = b$ with $A \in \mathbb{R}^{(m \times n)}$, $b \in \mathbb{F}^m$ and $m \geq n$. For the case $\text{rank}(A) = n$ we have

$$x_s = (A^T A)^{-1} A^T b$$

and

$$\|b - Ax_s\| = \min_{z \in \mathbb{R}^n} \|b - Az\| .$$

Here the vector z is called *approximation solution via the method of the least squares.*

Proof:

We know that because of $\text{rank}(A) = n$ our solution is unique, and that $p_A(b) = Ax$ is our orthogonal projection. We can use the Pythagorean theorem to obtain

$$\begin{aligned} \|b - Az\|^2 &= \|(b - p_A(b)) + (p + p_A(b))\|^2 \\ &= \|(b - p_A(b))\|^2 + \|(p + p_A(b))\|^2 \\ &\geq \|(b - p_A(b))\|^2 \\ &= \|b - Ax\|^2 . \end{aligned}$$

Thus, $p_A(b)$ gives us the smallest distance to the $\text{img}(A)$.

QED

Example I:

$$x = 1, x = -1$$

$$\|Ax - b\|^2 = (x+1)^2 + (x-1)^2 = 2x^2 + 2$$

Then $\|Ax - b\|^2$ gives us the smallest distance for $x = 0$.

Example II:

We can also take the last example and solve it with an alternative method.

$$A = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, x = (x), b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Let us use the normal equations

$$A^T Ax = A^T b$$

$$(1,1) \begin{bmatrix} 1 \\ 1 \end{bmatrix} x = (1,1) \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\implies 2x = 0$$

Example 3:

Given are

$$x_1 = 1, x_2 = 2, x_1 + x_2 = -1$$

with the corresponding matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

and solution vector

$$b = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

Now let us build the normal equation

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \vec{x} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Now with Gaussian elimination we get $x = (-\frac{1}{3}, \frac{2}{3})^T$.

Finally, we can use $\vec{x}$ to solve $Ax - b$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \frac{4}{3} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

And therefore, $\|Ax - b\|^2 = \frac{48}{9}$.

27.3 Underdetermined Systems of Equations

Consider a linear system $Ax = b$ such that $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$ and $m \leq n$. For $\text{rank}(A) = m$ we get the solution vector

$$x_s = A^T (AA^T)^{-1} b.$$

Also, x_s is the solution with the smallest length.

$$\|x_s\| = min_{Ax=b} \|x\|,$$

and x_s is orthogonal to the homogeneous system $Az = 0$, which means

$$x_s \perp ker(A).$$

Finally, we call for $\text{rank}(A) = m$, $A^T(AA^T)^{-1}$ the *Generalized Inverse* of A.

Proof:

First, we show the invertibility of AA^T because this implies that x_s is defined.

Given $x \in \ker(AA^T)$. This implies that

$$AA^T x = 0$$

which implies

$$x^T AA^T x = 0$$

because of $x^T A = A^T x$ and this gives us

$$\left\|A^T x\right\| = 0$$

therefore, $x \in kern(A^T)$.

Now because of $\text{rank}(A^T) = \text{rank}(A) = m$ we get $\dim(\ker(A)^T) = 0$, thus $x = \vec{0}$. Because of that $\ker(AA^T) = 0$ and $\text{rank}(AA^T) = m$ thus, the invertibility is proven.

x_s is a solution of $Ax = b$ because of

$$Ax_s = A(A^T(AA^T)^{-1}b) = AA^T(AA^T)^{-1}b = b.$$

$x_s \perp \ker(A)$ because for $z \in \ker(A)$

$$\langle z, x_s\rangle = z^T A^T(AA^T)^{-1}b = \left\langle Az, (AA^T)^{-1}b\right\rangle = 0$$

due to $Az = 0$.

Finally, thanks to the general solution theorem we know that $x = x_s + z$. Using the Pythagorean theorem and taking into consideration $x_s \perp \ker(A)$, we obtain

$$\|x\|^2 = \|x\|^2 + \|z\|^2 \geq \|x_s\|^2 .$$

Example:

We have the equation $2x_1 + 2x_2 - x_3 = 6$.

In this case, the solution builds a plane in $\mathbb{R}^3$, thus the solution is the normal vector that cuts the plane. We build the Hessian normal form.

$$\frac{1}{3}(2x_1 + 2x_2 - x_3) = 2$$

whose solution is $x = \frac{2}{3}(2, 2, -1)^T$.

Now with our method: $A = (2, 2, -1)$, $x = (x_1, x_2, x_3)$, $b = 6$

$$x = A^T(AA^T)^{-1}b = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \left((2, 2, -1) \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \right)^{-1} 6$$

$$x = \frac{2}{3}(2, 2, -1)^T$$

Chapter 28

Analytical Geometry

In this chapter, we will cover the topics for the *geometry* of $\mathbb{R}^2$ and $\mathbb{R}^3$, and higher dimensional spaces.

28.1 Vectors and Points

In analytical geometry, points and vectors are the basic elements.

A *point* in $\mathbb{R}^3$ is represented as $\vec{P} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, or as $(x_1, x_2, \ldots, x_n)$.

A vector is an object with direction and magnitude (in this case), also represented as $\vec{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$, or as $[x_1, x_2, \ldots, x_n]^T$. Sometimes, when it is clear that we are talking about vectors we can omite the arrow on top. We will do this for the rest of the book.

28.1.1 Vector Addition and Scalar Multiplication

Given two vectors a and b:

$$a \pm b = \begin{bmatrix} a_x \\ a_y \end{bmatrix} \pm \begin{bmatrix} b_x \\ b_y \end{bmatrix} = \begin{bmatrix} a_x \pm b_x \\ a_y \pm b_y \end{bmatrix}$$

For a scalar λ and vector a:

$$\lambda a = \lambda \begin{bmatrix} a_x \\ a_y \end{bmatrix} = \begin{bmatrix} \lambda a_x \\ \lambda a_y \end{bmatrix}$$

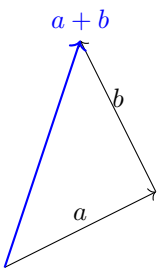


Figure 28.1: Vector addition: $a + b$

28.2 The Dot Product

The *dot product* of vectors a and b is a function $\langle x, y\rangle := V \times V \to \mathbb{F}$. $\forall\, a, b \in V$ the following applies

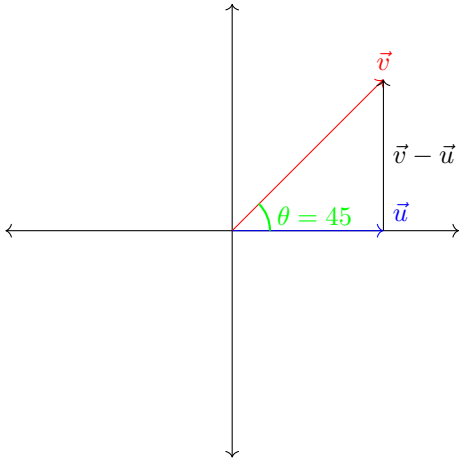
- $\langle a, b\rangle = \langle b, a\rangle$.
- $\langle a, b + c\rangle = \langle a, b\rangle + \langle a, c\rangle$.
- $\langle a, \lambda b\rangle = \lambda \langle a, b\rangle$
- $\langle a, b\rangle = 0 \Leftrightarrow a = \vec{0} \vee b = \vec{0}$

Formula:

$$\langle a, b\rangle = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \sum_{i=1}^{n} a_i b_i$$

Proof:

Let us look at the following triangle:



We can use the cosine law: $\|v - u\|^2 = \|v\|^2 + \|u\|^2 - 2\,\|u\|\,\|v\|\cos(\theta)$.

Now we expand:

$$\|u\|^2 = u_1^2 + u_2^2 \quad \text{and} \quad \|v\|^2 = v_1^2 + v_2^2$$

Then: $\|v - u\|^2 = (v_1 - u_1)^2 + (v_2 - u_2)^2$ and by re-grouping after expansion we get

$$\langle v, u\rangle = v_1 u_1 + v_2 u_2 = \|v\|\,\|u\|\cos(\theta)$$

$\mathfrak{QED}$

If we manipulate this expression, we can also prove the angle formulas. With that information, we can prove that

$$\cos\frac{\pi}{2} = \frac{\langle v, u\rangle}{\|v\|\,\|u\|} = 0$$

This holds only if the dot product is equal to zero, so we have also proved that only orthogonal vectors have a dot product of zero.

28.2.1 Orthogonal Vectors

Two vectors a and b are orthogonal if:

$$\langle a, b\rangle = 0$$

28.3 The Orthogonal Projection

The *orthogonal projection of vector* a onto vector b is given by:

$$\text{Proj}_b(a) = \frac{\langle a, b\rangle}{\|b\|^2} b$$

Proof:

We know that

$$p \| q \implies p = \alpha q \quad \forall \alpha \in \mathbb{F} \text{ and } \forall\ p, q \in V$$

Then let us define the orthogonal projection of vector a on vector b as p. We want to find α such that $p = \alpha q$ where $q = b$. We also know that the vector $a - p$ is orthogonal to q:

$$\begin{aligned} \langle a - p, q\rangle = 0 &\implies \langle a - \alpha q, q\rangle = 0 \\ &\implies \langle a, q\rangle - \alpha \langle q, q\rangle = 0 \quad \text{by the properties of the inner product} \\ \alpha &= \frac{\langle a, q\rangle}{\langle q, q\rangle} \end{aligned}$$

Therefore, the orthogonal projection of vector a on b is given by multiplying b by the scalar α.

QED

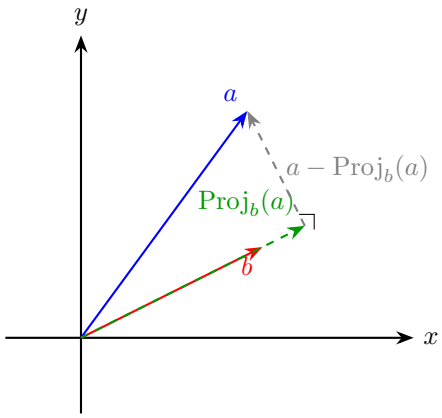


Figure 28.2: Orthogonal projection of a onto b

28.3.1 Vector Projections in Data Science

In data science, vector projections are used in various applications such as dimensionality reduction, feature extraction, and data visualization. For example, in Principal Component Analysis (PCA), data points are projected onto the directions of maximum variance to reduce the dimensionality of the dataset while preserving as much information as possible.

We find a unit vector in the direction of interest (e.g., a principal component) u_i onto which we want to project our data points x_j. Note we can find multiple such vectors $u_1, u_2, \ldots, u_k$ to project onto a lower-dimensional subspace.

$$\text{Proj}_u(v) = \langle v, u\rangle\, u = ku$$

Proof:

Let us start with the derivation. We have a vector v, and we want to project it onto a unit vector u. Then

$$u = \frac{v}{\|v\|}$$

and

$$p = ku \quad \text{and } d = v - p$$

also $\langle d, u \rangle = 0$

Analog to the derivation of the formula of the orthogonal projection we get

$$d = v - ku \implies \langle v - ku, u \rangle = 0 \implies \langle v, u \rangle - k\langle u, u \rangle = 0$$

Notice that $\langle u, u \rangle = 1$ because u is a unit vector. Therefore,

$$k = \langle v, u \rangle$$

Thus, the projection of v onto the unit vector u is given by:

$$\text{Proj}_u(v) = \langle v, u \rangle u = ku$$

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28.4 Dot Product as Projection

The *dot product* of two vectors can be interpreted as projecting one vector onto the other and then multiplying the length of the projecting vector by the magnitude of the other, for real vectors.

This comes from the fact that taking a dot product of real vectors is the same as multiplying one of them with the linear transformation formed by the transpose version of the other vector.

$$\langle a, b \rangle = a^T b$$

The transformation will take a vector from our space and take it to the number line, and it is also linear. In our case,

$$\text{Proj} = \begin{bmatrix} u_{x_1} & u_{x_2} & \cdots & u_{x_n} \end{bmatrix}$$

is defined this way because, when we project our basis vector b_i onto the line our vector $\vec{u}$ lies in, by the symmetry we will get the same value as if we were projecting $\vec{u}$ onto the line b_i lives in.

28.5 Metrics for Vectors

The metric of a vector a is described by different norms:

It also has the following properties:

- $\|a\| \geq 0$ and $\|a\| = 0 \Leftrightarrow a = \vec{0}$.
- $\|\lambda a\| = |\lambda| \, \|a\|$.
- $\|a + b\| \leq \|a\| + \|b\|$ (*Triangle inequality*).
- $\|a + b\|^2 = \|a\|^2 + \|b\|^2 + 2 \langle a, b \rangle$ (*Pythagorean theorem*).

These are the most common metrics, although we will be using primarily the Euclidean metric:

- *Euclidean metric*:
$$\|a\| = \sqrt{\langle a, a\rangle} = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}.$$
- *Manhattan metric*:
$$\|a\|_1 = |a_1| + |a_2| + \cdots + |a_n|.$$
- *Maximum metric*:
$$\|a\|_\infty = \max(|a_1|, |a_2|, \ldots, |a_n|).$$

28.6 Equation of a Line

A *line* is defined by a point $\vec{P}$ and a direction vector $\vec{v}$:

$$\vec{r}(t) = \vec{P} + t\vec{v}, \quad t \in \mathbb{R}$$

28.7 Equation of a Plane in 2D

A *plane* is defined by a point $\vec{P}$ and a normal vector $\vec{n}$:

$$P : \vec{x} = \vec{s} + \lambda_1 \vec{v_1} + \lambda_2 \vec{v_2}$$

28.8 Normal Vectors and Hyperplanes

A *hyperplane* in n dimensions is given by the equation

$$\langle w, x\rangle = 0.$$

Therefore, it is determined by all the vectors which are *normal* to the plane.

28.9 Angle Relations

The *angle* θ between two vectors is:

$$\cos(\theta) = \frac{\langle a, b\rangle}{\|a\|\,\|b\|}, \quad \theta = \arccos\left(\frac{\langle a, b\rangle}{\|a\|\,\|b\|}\right)$$

Line-Line Angle

Use direction vectors $\vec{v}_1$ and $\vec{v}_2$:

$$\theta = \arccos\left(\frac{\langle \vec{v}_1, \vec{v}_2\rangle}{\|\vec{v}_1\|\,\|\vec{v}_2\|}\right)$$

Line-Plane Angle

Let $\vec{v}$ be the line's direction and $\vec{n}$ the plane's normal:

$$\theta = \arcsin\left(\frac{\langle \vec{v}, \vec{n}\rangle}{\|\vec{v}\|\,\|\vec{n}\|}\right)$$

Plane-Plane Angle

Angle between planes is angle between their normals:

$$\theta = \arccos\left(\frac{\langle \vec{n}_1, \vec{n}_2 \rangle}{\|\vec{n}_1\| \, \|\vec{n}_2\|}\right)$$

28.9.1 Line Relations

Two lines can be:

- *Identical*: same direction vector and point
- *Parallel*: direction vectors are proportional
- *Intersecting*: one solution for t_1, t_2 such that $\vec{r}_1(t_1) = \vec{r}_2(t_2)$
- *Skew*: not parallel, do not intersect

To find the relation:

1. Check if direction vectors are scalar multiples $\Rightarrow$ parallel
2. Solve $\vec{P}_1 + t\vec{v}_1 = \vec{P}_2 + s\vec{v}_2$ for t and s:
   - Solution exists $\Rightarrow$ intersect
   - No solution $\Rightarrow$ skew
3. If same point and direction vector $\Rightarrow$ identical

28.10 Normalization of a Vector

To normalize a vector a, we divide it by its length:

$$\hat{a} = \frac{a}{\|a\|} = \frac{\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}}{\sqrt{a_x^2 + a_y^2 + a_z^2}} = \begin{bmatrix} \frac{a_x}{\|a\|} \\ \frac{a_y}{\|a\|} \\ \frac{a_z}{\|a\|} \end{bmatrix}$$

28.11 The Cross Product

The *cross product* of two vectors a and b in $\mathbb{R}^3$ is defined as:

$$a \times b = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \times \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

or

$$\det\left(\begin{bmatrix} \imath & a_1 & b_1 \\ \jmath & a_2 & b_2 \\ \hat{k} & a_3 & b_3 \end{bmatrix}\right) = \imath(a_2 b_3 - a_3 b_2) - \jmath(a_1 b_3 - a_3 b_1) + \hat{k}(a_1 b_2 - a_2 b_1)$$

28.11.1 Properties of the Cross Product

- $a \times b = -(b \times a)$
- $a \times (b + c) = a \times b + a \times c$
- $(\lambda_1 a) \times (\lambda_2 b) = \lambda_1 \lambda_2 (a \times b)$
- $\|a \times b\| = \|a\| \, \|b\| \sin(\theta)$

- $\langle a, (b \times \vec{c}) \rangle = 0$ (*scalar triple product*).
- $a \times b = \vec{0} \Leftrightarrow a = \lambda b$ for some $\lambda \in \mathbb{R}$ (*parallel vectors*).
- $a \times b = \vec{0} \Leftrightarrow a = \vec{0} \vee b = \vec{0}$ (*zero vector*).

The cross product is not defined in $\mathbb{R}^2$. The cross product is *not commutative*, but it is *associative*:

$$(a \times b) \times c = a \times (b \times c).$$

The cross product is *distributive over vector addition*:

$$a \times (b + c) = a \times b + a \times c$$

The cross product is *anti-commutative*:

$$a \times b = -(b \times a)$$

The cross product is *not associative*:

$$(a \times b) \times \vec{c} \neq a \times (b \times \vec{c})$$

The cross product is *not distributive over scalar multiplication*:

$$\lambda(a \times b) \neq (\lambda a) \times b$$

The length of the cross product in $\mathbb{R}^3$ is the area of the parallelogram spanned by the two vectors.

28.11.2 Determinant Trick

For two vectors a and b in $\mathbb{C}^3$ we can compute the cross product by using the *determinant trick* and treating the first row as basis vectors.

$$a \times b = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$$

28.11.3 Connection to the Volume of the Parallelepiped

We want to define a linear transformation f which relates the volume of the parallelepiped formed by three vectors, the determinant, and the dot product of an input vector of the transformation with some vector p.

$$f\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \det\left(\begin{bmatrix} x & v_1 & w_1 \\ y & v_2 & w_2 \\ z & v_3 & w_3 \end{bmatrix}\right) = \left\langle \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}, \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right\rangle$$

This gives us another version of the determinant trick seen previously, and we know that this gives us a vector perpendicular to v and w. The motivation behind this weird formula is that if you compute both sides of the equation you will get the coordinates p from the determinant trick.

This whole equality can be related to the geometric meaning of the cross product. Remember that the dot product of two vectors can be interpreted as projecting one onto another and then multiplying the length of the projecting vector by the magnitude of the other. This is exactly the same as projecting our input vector onto a line which is perpendicular to the parallelogram and then multiplying it by the area, which will give us the volume.

28.11.4 Derivation of the Cross Product Formula

We define n as a vector perpendicular to v and w. Therefore,

$$\langle v, n\rangle = 0 \quad \text{and} \quad \langle w, n\rangle = 0$$

Writing this out in components, let $v = [v_1, v_2, v_3]$, $w = [w_1, w_2, w_3]$, and $n = [n_1, n_2, n_3]$. Then:

$$v_1 n_1 + v_2 n_2 + v_3 n_3 = 0$$

and

$$w_1 n_1 + w_2 n_2 + w_3 n_3 = 0$$

Now we multiply the first equation by $-w_1$, the second by v_1, and add them

$$n_2(w_2 v_1 - v_2 w_1) + n_3(w_3 v_1 - v_3 w_1) = 0$$

We know that the equation $pc_1 + qc_2 = 0$ has the results $c_1 = -q$ and $c_2 = p$. Thus,

$$n_2 = v_3 w_1 + w_3 v_1$$

$$n_3 = v_1 w_2 - w_1 v_2$$

Finally, by substituting into the original equation and solving we get

$$n_1 = v_2 w_3 - v_3 w_2$$

Thus, we have obtained the formula for the cross product.

$\mathfrak{QED}$

28.12 Orthogonal Vectors in $\mathbb{R}^2$ $\mathbb{R}^3$

In this small section I will show you how to find orthogonal vectors to a vector x in both $\mathbb{R}^2$ and $\mathbb{R}^3$.

28.12.1 Orthogonal Vectors in $\mathbb{R}^2$

1. Interchange the components
2. Change the sign of one component

28.12.2 Orthogonal Vectors in $\mathbb{R}^3$

1. Interchange two components
2. The one that was not changed, set to zero
3. Change the sign of first component

Example:

$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad b = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

28.13 Hessian Normal Form

The *Hessian normal form* is a way of expressing the equation of a plane in three-dimensional space using a normalized normal vector. It is particularly useful in computational geometry and physics, where signed distances from points to planes are important.

28.13.1 Geometric Interpretation

The Hessian normal form represents a plane by specifying:

- A unit normal vector $\vec{n} = (a, b, c)$ to the plane,
- and the shortest distance d from the origin to the plane.

This form is derived by normalizing the general plane equation. A plane in 3D can be written as:

$$ax + by + cz + d = 0,$$

where (a, b, c) is a normal vector to the plane and d is the dot product of the normal vector with a point p. If we divide all terms by $\sqrt{a^2 + b^2 + c^2}$, we normalize the normal vector:

$$\frac{a}{\sqrt{a^2 + b^2 + c^2}}x + \frac{b}{\sqrt{a^2 + b^2 + c^2}}y + \frac{c}{\sqrt{a^2 + b^2 + c^2}}z + \frac{d}{\sqrt{a^2 + b^2 + c^2}} = 0.$$

Let:

$$n = \left(\frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}}\right), \quad d' = \frac{d}{\sqrt{a^2 + b^2 + c^2}},$$

then the equation becomes:

$$\langle n, r + d' \rangle = 0,$$

which is the Hessian normal form. Here, $r = (x, y, z)$ is any point on the plane, and n is the unit normal.

28.13.2 Signed Distance to the Plane

This form allows easy calculation of the signed distance from any point p to the plane:

$$\text{distance} = \langle n, p \rangle + d',$$

which is positive if p lies on the same side of the plane as the normal vector.

28.13.3 Derivation Illustration

To visualize the derivation, imagine a plane with unit normal vector n, and a point P in space. The shortest distance from P to the plane is the projection of the vector $p - q$ onto n, where q is any point on the plane. This leads to:

$$\text{distance} = \langle (p - q), n \rangle .$$

This gives the signed distance formula and thus, motivates the Hessian form.

28.13.4 Converting From the Parametric Form to the Hessian Normal Form

Steps:

1. Find the normal vector $\vec{n}$ of the plane
2. Normalize the normal vector
3. Find the distance d from the origin to the plane
4. Write the Hessian normal form

28.13.5 Converting From the Hessian Normal Form to the Parametric Form

Steps:

1. Find a point on the plane
2. Find two direction vectors in the plane
3. Write the parametric form

28.14 Properties of Lines and Planes

- Two planes are parallel if their normal vectors are scalar multiples of each other.
- A line and a plane are parallel if the direction vector of the line is orthogonal to the normal vector of the plane.
- A line intersects a plane if there exists a point on the line that satisfies the equation of the plane.
- Two planes intersect in a line if their normal vectors are not parallel.
- Three planes can intersect in a point, a line, or not at all.
- If we have a line G and a point on the line, for every vector n that is orthogonal to the direction vector of the line: $x \in G \iff \langle x, n\rangle$
- If p and q are two points in the line G with a normal vector then $\langle p, n\rangle = \langle q, n\rangle$
- Let E be a plane with the origin p and the direction vectors $\vec{v}$ and w, then there exists a normal vector and $x \in E \iff \langle x, n\rangle = \langle p, n\rangle$

28.15 Convert Normal Vector in Two Direction Vectors

1. Given the normal vector $n = [a, b, c]$, interchange a and b and multiply b by -1.
2. Set the other component to 0. This gives you the first direction vector $v = [-b, a, 0]$.
3. Take the original normal vector n and interchange a and c and multiply c by -1.
4. Set the other component to 0. This gives you the second direction vector $\vec{w} = [-c, 0, a]$.

28.16 Intersection Between Line and Plane

To find the *intersection between a line and a plane*, we can use the following steps:

1. Write the parametric form of the line: $\vec{r}(t) = \vec{P} + t\vec{v}$, where $\vec{P}$ is a point on the line and $\vec{v}$ is the direction vector.
2. Write the equation of the plane in Hessian normal form: $\langle \vec{n}, \vec{x} - \vec{P}\rangle = 0$, where $\vec{n}$ is the normal vector and $\vec{P}$ is a point on the plane.
3. Substitute the parametric form of the line into the equation of the plane.
4. Solve for t to find the intersection point.
5. Substitute t back into the parametric form of the line to find the intersection point.

Example:

Given the line:

$$g(t) = \begin{bmatrix}0\\0\\1\end{bmatrix} + t \begin{bmatrix}1\\1\\1\end{bmatrix}$$

and the plane:

$$E(u,m) = \begin{bmatrix}0\\1\\2\end{bmatrix} + u \begin{bmatrix}1\\0\\1\end{bmatrix} + m \begin{bmatrix}1\\1\\0\end{bmatrix},$$

we want to find the intersection point.

Step 1: Determine the normal vector of the plane using the cross product of the two direction vectors:

$$\vec{v}_1 = \begin{bmatrix}1\\0\\1\end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix}1\\1\\0\end{bmatrix}$$

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{bmatrix}1\\0\\1\end{bmatrix} \times \begin{bmatrix}1\\1\\0\end{bmatrix} = \begin{bmatrix}-1\\1\\1\end{bmatrix}$$

Step 2: Use the normal vector and a point on the plane to write the plane equation:

$$\langle \vec{n}, \vec{x} - \vec{Q} \rangle = 0, \quad \vec{Q} = \begin{bmatrix}0\\1\\2\end{bmatrix}$$

Step 3: Plug the line into the plane equation:

$$\vec{x}(t) = \begin{bmatrix}0\\0\\1\end{bmatrix} + t \begin{bmatrix}1\\1\\1\end{bmatrix} \Rightarrow \vec{x}(t) - \vec{Q} = \begin{bmatrix}t\\t-1\\t-1\end{bmatrix}$$

Now compute the dot product:

$$\langle \vec{n}, \vec{x}(t) - \vec{Q} \rangle = \begin{bmatrix}-1\\1\\1\end{bmatrix} \cdot \begin{bmatrix}t\\t-1\\t-1\end{bmatrix} = -t + (t-1) + (t-1) = t - 2$$

Step 4: Solve for t:

$$t - 2 = 0 \Rightarrow t = 2$$

Step 5: Substitute $t = 2$ into the line:

$$\vec{x}(2) = \begin{bmatrix}0\\0\\1\end{bmatrix} + 2 \begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}2\\2\\3\end{bmatrix}$$

Result: The line intersects the plane at the point

$$\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

Proof:

$$E : \langle x, n \rangle = \langle p, n \rangle$$
$$G : x = p + t \cdot v$$
$$\langle x, n \rangle = c$$
$$\langle p + t \cdot v, n \rangle = c$$
$$\langle p, n \rangle + t \cdot \langle v, n \rangle = c$$
$$t = \frac{c - \langle p, n \rangle}{\langle v, n \rangle}$$

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28.17 Distances Between Points, Lines and Planes

28.17.1 Distance Between Two Points

The distance between two points $\vec{P_1}$ and $\vec{P_2}$ in $\mathbb{R}^n$ is given by:

$$d(\vec{P_1}, \vec{P_2}) = \left\| \vec{P_1} - \vec{P_2} \right\| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + \cdots + (z_1 - z_2)^2}$$

where $\vec{P_1} = (x_1, y_1, \ldots, z_1)$ and $\vec{P_2} = (x_2, y_2, \ldots, z_2)$.

28.17.2 Distance Between a Point and a Hyperplane

The distance between a point $\vec{P}$ and a hyperplane defined by the equation $\langle \vec{n}, \vec{x} - \vec{P_0} \rangle = 0$ is given by:

$$d(\vec{P}, \text{hyperplane}) = \frac{|\langle \vec{n}, \vec{P} - \vec{P_0} \rangle|}{\|\vec{n}\|}$$

where $\vec{P_0}$ is a point on the hyperplane and $\vec{n}$ is the normal vector of the hyperplane.

28.17.3 Distance Between Two Lines

The distance between two lines in $\mathbb{R}^3$ can be calculated using the formula:

$$d = \frac{|\langle \vec{v_1} \times \vec{v_2}, \vec{P_2} - \vec{P_1} \rangle|}{\|\vec{v_1} \times \vec{v_2}\|}$$

where $\vec{P_1}$ and $\vec{P_2}$ are points on the two lines, and $\vec{v_1}$ and $\vec{v_2}$ are the direction vectors of the lines.

28.17.4 Distance Between a Point and a Line

The distance between a point $\vec{P}$ and a line defined by the parametric equation $\vec{r}(t) = \vec{P_0} + t\vec{v}$ is given by:

$$d(\vec{P}, \text{line}) = \frac{\left\| \vec{v} \times (\vec{P} - \vec{P_0}) \right\|}{\|\vec{v}\|}$$

where $\vec{P_0}$ is a point on the line and $\vec{v}$ is the direction vector of the line.

28.17.5 Distance Between Two Planes

The distance between two parallel planes defined by the equations $\langle \vec{n}, \vec{x} - \vec{P_1} \rangle = 0$ and $\langle \vec{n}, \vec{x} - \vec{P_2} \rangle = 0$ is given by:

$$d = \frac{|\langle \vec{n}, \vec{P_2} - \vec{P_1} \rangle|}{\|\vec{n}\|}$$

where $\vec{P_1}$ and $\vec{P_2}$ are points on the two planes, and $\vec{n}$ is the normal vector of the planes.

28.17.6 Distance Between a Point and a Plane

The distance between a point $\vec{P}$ and a plane defined by the equation $\langle \vec{n}, \vec{x} - \vec{P_0} \rangle = 0$ is given by:

$$d(\vec{P}, \text{plane}) = \frac{|\langle \vec{n}, \vec{P} - \vec{P_0} \rangle|}{\|\vec{n}\|}$$

where $\vec{P_0}$ is a point on the plane and $\vec{n}$ is the normal vector of the plane.

Example:

Distance Between Two Skew Lines

To find the shortest distance between two skew lines, we use the formula:

$$\text{distance} = \frac{|\langle (\vec{P_2} - \vec{P_1}), (\vec{v_1} \times \vec{v_2}) \rangle|}{\|\vec{v_1} \times \vec{v_2}\|}$$

where:

- $\vec{P_1}$ and $\vec{P_2}$ are points on each line.
- $\vec{v_1}$ and $\vec{v_2}$ are the direction vectors.
- $\vec{v_1} \times \vec{v_2}$ is the cross product of the direction vectors.

Given:

$$g_1 : \vec{r}_1(a) = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} + a \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad g_2 : \vec{r}_2(b) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + b \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Step 1: Set

$$\vec{P_1} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, \quad \vec{v_1} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\vec{P_2} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v_2} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Step 2: Compute the vector between base points:

$$\vec{P_2} - \vec{P_1} = \begin{bmatrix} 1-2 \\ 2-2 \\ 3-2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Step 3: Compute the cross product:

$$\vec{v}_1 \times \vec{v}_2 = \begin{bmatrix}1\\1\\0\end{bmatrix} \times \begin{bmatrix}3\\2\\1\end{bmatrix} = \begin{bmatrix}(1)(1)-(1)(2)\\(1)(3)-(0)(1)\\(0)(2)-(1)(3)\end{bmatrix} = \begin{bmatrix}-1\\3\\-3\end{bmatrix}$$

Step 4: Compute scalar triple product:

$$\langle(\vec{P}_2 - \vec{P}_1), (\vec{v}_1 \times \vec{v}_2)\rangle = \langle \begin{bmatrix}-1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\3\\-3\end{bmatrix}\rangle = (-1)(-1) + (0)(3) + (1)(-3) = 1 + 0 - 3 = -2 \Rightarrow |\ldots| = 2$$

Step 5: Magnitude of the cross product:

$$\|\vec{v}_1\vec{v}_2\| = \sqrt{(-1)^2 + 3^2 + (-3)^2} = \sqrt{1+9+9} = \sqrt{19}$$

Final Answer:

$$\text{distance} = \frac{2}{\sqrt{19}} \approx 0.458$$

$$\text{Shortest distance between the lines is } \frac{2}{\sqrt{19}}$$

28.18 Foot of the Perpendicular and Mirror Point

28.18.1 Foot of the Perpendicular

The *foot of the perpendicular* from a point $\vec{P}$ to a line (or plane) is the point on the line (or plane) where the perpendicular from $\vec{P}$ meets it.

Line case:

Given a line in parametric form:

$$g : \vec{r}(t) = a + t\vec{v}$$

and a point $\vec{P}$ not on the line, the foot of the perpendicular $\vec{F}$ satisfies:

$$(\vec{P} - \vec{F}) \perp \vec{v} \quad \Rightarrow \quad (\vec{P} - (a + t\vec{v})) \cdot \vec{v} = 0$$

Solve this inner product for t, then compute:

$$\vec{F} = a + t\vec{v}$$

Plane case:

Given a plane in normal form:

$$\langle \vec{n}, \vec{x} - \vec{Q} \rangle = 0$$

then the foot of the perpendicular from point $\vec{P}$ to the plane is:

$$\vec{F} = \vec{P} - ((\vec{P} - \vec{Q}) \cdot \vec{n}) \cdot \vec{n}$$

28.18.2 Mirror Point

The *mirror point* (or reflected point) of $\vec{P}$ across a line or plane is the point $\vec{P}'$ such that the midpoint between $\vec{P}$ and $\vec{P}'$ is the foot of the perpendicular.

Formula:

$$\vec{P}' = 2\vec{F} - \vec{P}$$

Where $\vec{F}$ is the foot of the perpendicular from $\vec{P}$ to the line or plane.

Chapter 29

Algebraic Structures

Algebraic structures are mathematical systems consisting of a set equipped with one or more operations that satisfy certain axioms.

29.1 Internal and External Operations

An *internal composition law* is a binary operation that takes two elements from a set and returns another element in the same set. Formally, for a set S and operation $\circ$, we have:

$$\circ : S \times S \to S$$

An *external composition law* involves a second set acting on the structure, such as scalar multiplication in vector spaces:

$$\circ : \mathbb{F} \times V \to V$$

where $\mathbb{F}$ is a field and V is a vector space.

29.1.1 Properties of Operations

Let $\circ$ and $\circledast$ be binary operations on a set S. The most important properties include:

- *Associativity:* $(a \circ b) \circ c = a \circ (b \circ c)$ for all $a, b, c \in S$.
- *Commutativity:* $a \circ b = b \circ a$ for all $a, b \in S$.
- *Identity Element:* There exists $e \in S$ such that $a \circ e = e \circ a = a$ for all $a \in S$.
- *Inverse Element:* For every $a \in S$, there exists $a^{-1} \in S$ such that $a \circ a^{-1} = a^{-1} \circ a = e$.
- *Distributivity:* $a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c)$ and/or $(b \oplus c) \odot a = (b \odot a) \oplus (c \odot a)$.

29.2 Homomorphisms and Isomorphisms

Let $(G, \oplus)$ and $(H, \oplus')$ be two algebraic structures.

A *homomorphism* is a function $\varphi : G \to H$ such that:

$$\varphi(a \oplus b) = \varphi(a) \oplus \varphi(b), \quad \forall a, b \in G$$

An *isomorphism* is a bijective homomorphism. If such a map exists, we say the structures are *isomorphic*, written as $G \cong H$.

29.3 Lattice

A *lattice* $(L, \wedge, \vee)$ is an algebraic structure with two binary, associative, and commutative operations, which satisfy the following properties:

$$a \vee (a \wedge b) = a$$

and

$$a \wedge (a \vee b) = a$$

for all $a, b \in L$.

29.4 Semigroup $(S, \oplus)$

A *semigroup* is a set S equipped with a binary operation $\oplus$ that is *associative*. This means:

$$(a \oplus b) \oplus c = a \oplus (b \oplus c) \quad \text{for all } a, b, c \in S.$$

29.5 Monoid $(M, \oplus)$

A *monoid* builds on a semigroup by adding an *identity element* e such that:

$$a \oplus e = e \oplus a = a \quad \text{for all } a \in M.$$

This structure is useful when an operation must have a "do nothing" element, like 0 for addition or 1 for multiplication.

29.6 Group $(G, \oplus)$

A *group* is a monoid where every element has an *inverse*:

$$\text{For each } a \in G, \text{ there exists } a^{-1} \in G \text{ such that } a \oplus a^{-1} = a^{-1} \oplus a = e.$$

29.7 Abelian Group $(A, \oplus)$

An *Abelian group* (or commutative group) is a group where the operation is *commutative*:

$$a \oplus b = b \oplus a \quad \text{for all } a, b \in A.$$

29.8 Ring $(R, \oplus, \odot)$

A *ring* is a set with two operations:

- $(R, \oplus)$ is an Abelian group.
- $(R, \odot)$ is a semigroup (associative multiplication).
- Multiplication distributes over addition:

$$a \odot (b \oplus c) = a \odot b \oplus a \odot c.$$

29.9 Commutative Ring $(R, \oplus, \odot)$

A *commutative ring* is a ring where multiplication is also *commutative*:

$$a \odot b = b \odot a \quad \text{for all } a, b \in R.$$

This is the kind of ring most often encountered in basic algebra, such as the integers $\mathbb{Z}$.

29.10 Field $(\mathbb{F}, \oplus, \odot)$

A *field* is a commutative ring where every nonzero element has a *multiplicative inverse*:

$$(\mathbb{F} \setminus \{0\}, \odot) \text{ is an Abelian group.}$$

Fields like $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ allow division (except by zero), enabling the full range of arithmetic operations.

29.10.1 Field Proofs

Uniqueness of Differences

There exists exactly one $x \in \mathbb{F}$ such that $a + x = b$, $x = b + (-a)$ also written $x = b - a$.

Let $x, x' \in \mathbb{F}$ such that the property is satified then,

$$x = b - a \wedge x' = b - a \implies x =' x.$$

$\mathfrak{QED}$

A special case of this property is when $b = a$, which gives us that

$$a + x = a \wedge a + x' = a \iff a + 0 = a \wedge a - 0 = a \implies 0 = -0.$$

Distribution of Negative Signs

For any $a, b \in \mathbb{F}$ the following holds $-(a + b) = (-a) + (-b)$.

Note that $-(a + b)$ satisfies

$$(a + b) - (a + b) = 0$$

Now note that

$$\begin{aligned}(a + b) - a - b &= (b + a) - a - b \\ &= b + (a - a) - b \\ &= b - b = 0\end{aligned}$$

Hence, $-a - b$ also satisfies the same difference. By our previous propert we know that difference are unique hence, $-(a + b) = (-a) + (-b)$.

$\mathfrak{QED}$

Inverses Are Well-Defined:

We know $a + -a = 0$. Now let us say $-a' \neq -a$ and $a - a' = 0$.

$$\begin{aligned} a + -a &= a - a' \quad | -a \\ -a &= -a' \end{aligned}$$

This contradicts our initial assumption; therefore, they are the same. The proof is analogous for the multiplication:

$$\begin{aligned} a \cdot a^{-1} &= a \cdot a'^{-1} \quad | *a^{-1} \\ a^{-1} &= a'^{-1} \end{aligned}$$

QED

Neutral Element for the Addition Is Well-Defined:

We know $a + 0 = a$. Now let us say $0' \neq 0$ and $a + 0' = a$.

$$\begin{aligned} a + 0 = a + (a - a) &= a + 0' \\ a - a + a &= a + 0' \\ a &= a + 0' \quad | -a \\ 0 &= 0' \end{aligned}$$

There is a contradiction to the assumption that there is more than one neutral element for the addition. Thus, it is well-defined.

Neutral Element for the Multiplication is well-defined: This proof is very similar to the previous one; thus, it will be skipped.

QED

Minus Times Minus Is Plus:

We will prove that $ab = (-a)(-b)$ for any two elements of a field.

$$\begin{aligned} ab &= (-a)(-b) \\ ab + (-a)b &= (-a)(-b) + (-a)b \\ b(a + (-a)) &= (-a)((-b) + b) = 0 \end{aligned}$$

From that point, for the right side we can do the following:

$$\begin{aligned} (-a)((-b) + b) &= 0 \\ (-a)(-b) + (-a)b &= 0 \\ (-a)(-b) &= -(-a)b \end{aligned}$$

Then, by looking at the left side of the original equality:

$$\begin{aligned} b(a + (-a)) &= 0 \\ ab + (-a)b &= 0 \\ ab &= -(-a)b \end{aligned}$$

Thus, we have obtained our equality $ab = (-a)(-b)$.

$\mathfrak{QED}$

29.11 Vector Space $(V, \oplus, \cdot)$

A *vector space* over a field $\mathbb{F}$ consists of:

- An Abelian group $(V, \oplus)$ for vector addition.
- A scalar multiplication $\cdot : \mathbb{F} \times V \to V$ satisfying:
  - Distributivity over vector addition: $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$
  - Distributivity over field addition: $(a + b) \cdot v = a \cdot v + b \cdot v$
  - Associativity: $a \cdot (b \cdot v) = (ab) \cdot v$
  - Identity: $1 \cdot v = v$

Vector spaces provide the foundation for linear algebra, where scalars from a field combine with vectors from a set.

Chapter 30

Polynomial Rings

Given a field $\mathbb{F}$ with $\mathbb{F}^{\mathbb{Z}_{\geq 0}}$, we define the multiplication

$$(a_0, a_1, \dots) \cdot (b_0, b_1, \dots) := (c_0, c_1, \dots)$$

where

$$c_0 := a_0 b_0, \quad c_1 := a_0 b_1 + a_1 b_0, \quad c_2 := a_0 b_2 + a_1 b_1 + a_2 b_0$$

In general, for $n \geq 0$,

$$c_n := \sum_{i=0}^{n} a_i b_{n-i} = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0$$

and the addition as

$$(a_i)_{i \in \mathbb{N}} + (a_i)_{i \in \mathbb{N}} = (a_i + b_i)_{i \in \mathbb{N}}.$$

This $(\mathbb{F}^{\mathbb{Z}_{\geq 0}}, +, \cdot)$ is also noted as $\mathbb{F}[[\mathbb{F}]]$ and called the *power-ring* over $\mathbb{F}$. Also, instead of $(a_0, a_1, \dots)$, we write

$$f = \sum_{i=0}^{\infty} a_i X^i$$

for $X := (0, 1, 0, 0, \dots)$. We also call this ring the *formal power series* over $\mathbb{F}$.

The members $f = (a_0, a_1, \dots,) \in \mathbb{F}[[X]]$ are called *polynomials*.

For the finite case, we write

$$f = \sum_{i=0}^{n} a_i X^i$$

The set of all polynomials reachable via the multiplication vector-space properties of $\mathbb{F}[[X]]$ is called the *polynomial ring* $\mathbb{F}[X]$. The field before the $[X]$ tells us which elements are allowed as coefficients.

Example:

For $f := (1, 2, 3, 4, \dots) = 1 + 2X + 3X^2 + 4X^3$

$$f \cdot 4 = (4,\ 1,\ 5,\ 2,\ 0,\ 0,\ \dots) = 4 + 1X + 5X^2 + 2X^3$$

$$f \cdot 3X = (0,\, 3,\, 6,\, 2,\, 5,\, 0,\, 0,\, \ldots) = 3X + 6X^2 + 2X^3 + 5X^4$$

$$f \cdot 2X^2 = (0,\, 0,\, 2,\, 4,\, 6,\, 1,\, 0,\, 0,\, \ldots) = 2X^2 + 4X^3 + 6X^4 + X^5$$

$$f \cdot X^3 = (0,\, 0,\, 0,\, 1,\, 2,\, 3,\, 4,\, 0,\, 0,\, \ldots) = X^3 + 2X^4 + 3X^5 + 4X^6$$

30.1 Degree of a Polynomial

The *degree* of a polynomial is given by the n in $\sum_{i=0}^{n} a_i X^i$ with $f \neq \vec{0}$. Also, note that we define $\deg(0) := -\infty$.

Degree Addition

For $f, g \in \mathbb{F}[X] \setminus \{0\}$, the following holds

$$\deg(f + g) \leq \max(\deg(f), \deg(g)).$$

Degree Multiplication

For $f, g \in \mathbb{F}[X] \setminus \{0\}$, the following holds

$$\deg(fg) \leq \max(\deg(f) + \deg(g)).$$

30.1.1 Properties of the Polynomial Ring

Neutral Element

$$1 := (1, 0, 0, \ldots) \in \mathbb{F}[X]$$

is the neutral element of the multiplication in $\mathbb{F}[[X]]$ and $\mathbb{F}[X]$.

X_a

$$X_a = (0, a_0, a_1, \ldots) \quad \forall a \in \mathbb{F}[[X]].$$

Multiplication

$$X^i X^j = X^{i+j}, \text{ with } X^0 := 1.$$

Identity

Given $f \in \mathbb{F}[X]$ with degree n then

$$f = a_0 + a_1 X + \cdots + a_n X^n$$

gives a clear identification of $\mathbb{F}$ with

$$\{0\} \cup \{f \in \mathbb{F}[X] \mid \deg(f) = 0\}.$$

Vector space

$$\mathbb{F}[X]_{\deg<n} := \{0\} \cup \{f \in \mathbb{F}[X] \mid \deg(f) < n\}$$

is a vector space.

30.2 Field Algebra

Given a ring R with a one that is also a vector space over the field $\mathbb{F}$, we call R an *associative algebra* with the one if and only if

$$a(rs) = (ar)s = r(as)$$

for all $r, s \in R$ and $a \in \mathbb{F}$.

30.3 Division for Polynomials

Given the field $\mathbb{F}$, then

- $\mathbb{F}[X]$ is a commutative ring with one and exactly one commutative $\mathbb{F}$-algebra with 1.
- For $f, g \in \mathbb{F}[X]$ with $g \neq 0$, there are well-defined $q, r \in \mathbb{F}[X]$ with

$$f = qg + r \text{ and } \deg(r) < \deg(g) \vee r = 0.$$

Proof:

For the first lemma, to show that the multiplication is commutative, we just use the commutativity of our field $\mathbb{F}$.

The distributive property for $f, g, h \in \mathbb{F}[X]$ and the scalar multiplication are also easy to show using our definitions. We will show the associativity.

We will use induction: If $\deg(h) = 0$, then $h \in \mathbb{F}$ and the associativity holds because of the associativity for scalars.

Let us assume that it holds for $\deg(h) \leq n$. If $\deg(h) > 0$, then

$$h = \hat{h} + a_{n+1}X^{n+1} \text{ with } \hat{h} \in \mathbb{F}[X]$$

is a polynomial of degree at most n or 0. Due to that fact, $(fg)\hat{h} = f(g\hat{h})$.

Induction step:

$$\begin{aligned}(fg)h &= (fg)(\hat{h} + a_{n+1}X^{n+1}) \\ &= fg\hat{h} + fga_{n+1}X^{n+1}) \\ &= fg\hat{h} + f(ga_{n+1)}X^{n+1}) \\ &= f(g\hat{h} + ga_{n+1}X^{n+1})) \\ &= f(g(\hat{h} + a_{n+1}X^{n+1}))) \\ &= f(gh)\end{aligned}$$

$\mathfrak{QED}$

As for the second lemma, given $\deg(f) = m, \deg(g) = n, f = \sum_{i=0}^{m} a_i X^i, g = \sum_{i=0}^{n} b_i X^i$. If $f = 0$ then $q = 0$ and $r = 0$. If $m < n$ we are done, because $q = 0, r = f$.

For the case $m \geq n$, we will show the existence of q, r by induction on m. For $m = 0$ and $n = 0$, $f = a_0$, $g = b_0$. Set $q = \frac{a_0}{q_0}$ and $r = 0$, and we get $f = qg + r$.

Let $m > 0$, and assume that for each polynomial h with $\deg(h) \leq m-1$ there exist polynomials $\hat{q}, \hat{r}$ such that $\deg(\hat{r}) < \deg(g)$ and $h = \hat{q}g + \hat{r}$.

Define $\hat{f} := f - \frac{a_m}{b_n}X^{m-n}g$. Then $\deg(\hat{f}) \leq m - 1$. There also exist $\hat{q}, \hat{r}$ with $\deg(\hat{r}) < \deg(g)$ and $\hat{f} = \hat{q}g + \hat{r}$. Then

$$f = \hat{f} + \frac{a_m}{b_n} X^{m-n} g = \hat{q}g + \hat{r} + \frac{a_m}{b_n} X^{m-n} g = \left(\hat{q} + \frac{a_m}{b_n} X^{m-n} \right) g + \hat{r}.$$

Hence, with $q = \hat{q} + \frac{a_m}{b_n} X^{m-n}$ and $r = \hat{r}$, existence is shown.

For uniqueness: Let $f = qg + r$ and $f = \hat{q}g + \hat{r}$ with $\deg(r) < \deg(g)$ and $\deg(\hat{r}) < \deg(g)$. Then it follows that

$$r - \hat{r} = (q - \hat{q})g.$$

If $q - \hat{q} \neq 0$, then $\deg(r - \hat{r}) \geq \deg(g)$, which is a contradiction. Thus, $q = \hat{q}$ and $r = \hat{r}$.

ΩΕΔ

30.4 Greatest Common Divisor for Polynomials

For two polynomials $f, g \in \mathbb{F}[X]$ with $(f, g) \neq (0, 0)$, the polynomial $t \in \mathbb{F}[X]$ with

- $t \mid f$ and $t \mid g$
- For all $c \in \mathbb{F}[X]$ with $c \mid f$ and $c \mid g$ holds $c \mid t$

is considered the *greatest common divisor* of f and g. If $t = 1$, then f, g are *relatively prime*.

30.5 More Properties of Polynomials

- In $\mathbb{Z}$ one has $|ab| = |a||b|$; in $\mathbb{F}[X]$ one has $\deg(fg) = \deg(f) + \deg(g)$.
- The gcd of two polynomials $f, g \in \mathbb{F}[X]$ with $(f, g) \neq (0, 0)$ is unique only up to factors in $\mathbb{F}[X]^*$, that is, up to factors in $\mathbb{F}[X]$ of degree 0. Notation: $t = \gcd(f, g)$ actually also denotes every multiple at with $a \in \mathbb{F} \setminus \{0\}$ as a gcd of f and g.
- In both rings there is a division with remainder, and therefore $\mathbb{F}[X]$ also has a Euclidean algorithm for computing the gcd and an extended Euclidean algorithm for determining Bézout coefficients. Here the degree of a polynomial plays the role of the absolute value in $\mathbb{Z}$. We give the first part of the algorithm again.
- Furthermore, one also has the analogue of prime numbers in $\mathbb{F}[X]$, namely irreducible polynomials, and correspondingly a unique prime factorization for polynomials.

30.6 Euclidean Algorithm for Polynomials

Input: $f, g \in \mathbb{F}[X]$, $(f, g) \neq (0, 0)$.

Output: The greatest common divisor t of f and g.

Algorithm:

1. If $f = 0$, then return g. If $g = 0$, return f.
2. Set $r_1 := f$, $r_2 := g$, $k := 2$.
3. As long as $r_k \neq 0$, define $r_{k+1} := r_k$.
4. As soon as $\deg(r_k) = 0$, return r_{k-1}.

30.7 Reducible and Irreducible

A polynomial $p \in \mathbb{F}[X] \setminus \{0\}$ is called *irreducible* if p is not a unit (that is, $p \notin \mathbb{F}[X]^*$) and if for $f, g \in \mathbb{F}[X]$ the following holds: $p = f \cdot g \Rightarrow f \in \mathbb{F}[X]^* \vee g \in \mathbb{F}[X]^*$. An element $h \in \mathbb{F}[X] \setminus \{0\}$ for which $h = f \cdot g$ holds with $f, g \in \mathbb{F}[X] \setminus \mathbb{F}[X]^*$ is called *reducible*.

30.8 Normalized

A polynomial $f \in \mathbb{F}[X] \setminus \{0\}$ with $f = \sum_{i=0}^{n} a_i X^i$ is considered *normalized* if $a_n = 1$.

30.9 Factorization Property

Each polynomial $f \in \mathbb{F}[X] \setminus \{0\}$ has, up to the order of the factors, a unique factorization into a normalized irreducible form and a unit.

Chapter 31

Vector Spaces

Given a field $\mathbb{F}$, a *vector space* is a set $(V, \mathbb{F}, \oplus, \odot)$ with two operations, vector addition and scalar multiplication, such that:

I. $(a \oplus b)\vec{v} = a\vec{v} \oplus b\vec{v}, \forall a, b \in \mathbb{F} \wedge \vec{v} \in V$

II. $(\vec{v} \oplus \vec{u})a = a\vec{v} \oplus a\vec{u}, \forall a \in \mathbb{F} \wedge \vec{v}, \vec{u} \in V$

III. $(ab) \odot \vec{v} = a(b \odot \vec{v}), \forall a, b \in \mathbb{F} \wedge \vec{v} \in V$

IV. $1 \odot \vec{v} = \vec{v}, \forall \vec{v} \in V$

The scalar multiplication is defined as

$$\mathbb{F} \times V \to V : (a, v) \mapsto a \odot v.$$

Also, using these axioms, we get

- $\vec{v} \odot 0 = \vec{0}$
- $(-1) \odot \vec{v} = -1 \odot \vec{v}$

Proof:

For the first property, $0\vec{v} = (0 + 0)\vec{v} = 0\vec{v} + 0\vec{v}$, then

$$0\vec{v} = 0\vec{v} + 0\vec{v} \implies \vec{0} = 0\vec{v}.$$

For the second property,

$$\begin{aligned}(-1)\vec{v} &= (1 - 2)\vec{v} \\ (-1)\vec{v} &= \vec{v} - 2\vec{v} \\ (-1)\vec{v} &= -\vec{v}\end{aligned}$$

$\mathfrak{QED}$

Examples:

1. The set of all n-tuples of real numbers $\mathbb{R}^n$ is a vector space over the field of real numbers $\mathbb{R}$.
2. The set of all polynomials of degree less than or equal to n is a vector space over the field of real numbers $\mathbb{R}$.
3. The set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$ is a vector space over the field of real numbers $\mathbb{R}$.
4. The set of all $m \times n$ matrices with real entries is a vector space over the field of real numbers $\mathbb{R}$.
5. The set of all linear maps is also a vector space.

31.1 Subspaces

A subset W of a vector space V is a *subspace* of V if:

I. The zero vector $\vec{0} \in W$.

II. For all $\vec{u}, \vec{v} \in W$, $\vec{u} + \vec{v} \in W$.

III. For all $a \in F$ and $\vec{v} \in W$, $a\vec{v} \in W$.

If W is a subspace of V, we write $W \subseteq V$.

A subset W of V is a subspace if and only if $W \neq \emptyset$ and, for all $v, w \in W$ and all $a \in \mathbb{F}$, $v + aw \in W$.

Proof:

$\Rightarrow$: $\vec{0} \in W$ thus $W \neq \emptyset$. Also, by the axioms $v + aw \in W$.

$\Leftarrow$: $W \neq \emptyset \implies \exists w \in W$ because $\mathbb{F}$ is a field; then $-1 \in \mathbb{F}$ and therefore, $w - w = 0 \in W$. For $v + aw \in W$ we choose $a = 1$ or $v = 0$, and we see that the addition and multiplication are also defined for our vector space.

31.1.1 Properties of the Subspaces

- The intersection of two or more subspaces $\bigcap_{W \in V} W$ is a subspace.
- The union of two subspaces is not necessarily a subspace.
- The sum of two subspaces U and W is defined as:

$$U + W = \{\vec{u} + \vec{w} : \vec{u} \in U, \vec{w} \in W\}$$

  The sum of two subspaces is a subspace.

31.1.2 Sum of Subspaces

- The *direct sum* of two subspaces U and W is defined as:

$$U \oplus W = \{\vec{u} + \vec{w} : \vec{u} \in U, \vec{w} \in W\}$$

  The direct sum of two subspaces is a subspace.
- The direct sum of two subspaces is the smallest subspace containing both subspaces, such that $U \cap W = \{\vec{0}\}$.
- The direct sum of two subspaces is denoted by $U \oplus W$.

The sum of two subspaces is the smallest subspace containing both subspaces.

31.2 Vector Space Product

The *product* $\langle M \rangle$ of $M \subseteq V$ is the intersection of all subspaces of V which contain M:

$$\langle M \rangle := \bigcap_{W \subseteq V_{M \subseteq W}} W.$$

31.3 Linear Combinations

A *linear combination* of vectors $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ in a vector space V is an expression of the form:

$$a_1\vec{v}_1 + a_2\vec{v}_2 + \cdots + a_n\vec{v}_n$$

where $a_1, a_2, \ldots, a_n$ are scalars from the field F.

31.4 The Span

The set of all linear combinations of a set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is called the *span* of those vectors, denoted by $\text{span}(\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n)$. The span of a set of vectors is a subspace of the vector space V.

- The span of a set of vectors is the smallest subspace containing those vectors.
- The set of all linear combinations of a set of vectors $B = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is given by $\mathcal{L}(B)$.

31.5 Linear Independence

A set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ in a vector space V is said to be *linearly independent* if the only solution to the equation:

$$\lambda_1\vec{v}_1 + \lambda_2\vec{v}_2 + \cdots + \lambda_n\vec{v}_n = 0$$

or

$$\sum_{i=1}^{n} \lambda_i\vec{v}_i = 0$$

is $a_1 = a_2 = \cdots = a_n = 0$. If there exists a non-trivial solution to this equation, then the set of vectors is said to be linearly dependent. A set of vectors is linearly independent if and only if the only linear combination of those vectors that equals the zero vector is the trivial combination where all coefficients are zero.

31.5.1 Properties of the Linear Independence

- A set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is linearly independent if and only if the only linear combination of those vectors that equals the zero vector is the trivial combination where all coefficients are zero.
- If a set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is linearly independent, then any subset of that set is also linearly independent.
- If a set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is linearly dependent, then at least one vector in that set can be expressed as a linear combination of the others.

31.6 The Basis

A *basis* of a vector space V is a set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ that is linearly independent and spans the vector space V.

31.6.1 How to Find the Basis of a Set of Vectors

To find the basis of a set of vectors, we can use the following steps:

1. Write the vectors as columns of a matrix.
2. Row-reduce the matrix to echelon form.
3. The non-zero rows of the echelon form matrix correspond to the basis of the vector space spanned by the original set of vectors.

The number of non-zero rows in the echelon form matrix is equal to the dimension of the vector space spanned by the original set of vectors.

The basis of a vector space is not unique. Different bases can span the same vector space.

31.6.2 Basis Isomorphism

Given $B = (v_1, \ldots, v_n)$. B is a basis of V if and only if

$$\lambda = \lambda_B : \mathbb{F}^n \to V : (a_1, \ldots, a_n) \mapsto \sum_{i=1}^{n} a_i v_i$$

is an isomorphism. This means that each vector in V can be uniquely represented as a linear combination of the basis vectors.

31.7 Dimension

The *dimension* of a vector space V is the number of vectors in a basis of V.

- The dimension of a vector space is denoted by $\dim(V)$.
- The dimension of a vector space can be finite or infinite.
- If the dimension of a vector space is finite, then it is said to be *finite-dimensional*.
- If the dimension of a vector space is infinite, then it is said to be *infinite-dimensional*.

31.8 Basis Extension Theorem

Let V be a vector space over a field K, and let

$$v_1, \ldots, v_r, \quad w_1, \ldots, w_s \in V.$$

Suppose that $(v_1, \ldots, v_r)$ is a linearly independent tuple and that

$$\operatorname{span}(v_1, \ldots, v_r, w_1, \ldots, w_s) = V.$$

Then it is possible to extend $(v_1, \ldots, v_r)$ to a basis of V by possibly adding suitable vectors from the set $\{w_1, \ldots, w_s\}$.

Proof:

If $\operatorname{span}(v_1, \ldots, v_r) = V$, the statement is obvious. So assume

$$\operatorname{span}(v_1, \ldots, v_r) \neq V.$$

Then there exists at least one w_i such that $w_i \notin \operatorname{span}(v_1, \ldots, v_r)$; otherwise, if all $w_i \in \operatorname{span}(v_1, \ldots, v_r)$, then

$$\operatorname{span}(v_1, \ldots, v_r, w_1, \ldots, w_s) = \operatorname{span}(v_1, \ldots, v_r) = V,$$

which contradicts our assumption that $\operatorname{span}(v_1, \ldots, v_r) \neq V$.

The tuple $(w_i, v_1, \ldots, v_r)$ is linearly independent, because of

$$\sum_{j=1}^{r} \lambda_j v_j + \lambda w_i = 0$$

It follows that $\lambda = 0$ (since $w_i \notin \operatorname{span}(v_1, \ldots, v_r)$), and then also $\lambda_j = 0$ for all j because the v_j are linearly independent.

Possibly, $(w_i, v_1, \ldots, v_r)$ is still not a basis of V. Then we repeat the previous step and keep adding further w_i until the tuple extends $(v_1, \ldots, v_r)$ to a basis of V. This process terminates after finitely many steps, since

$$\text{span}(v_1, \ldots, v_r, w_1, \ldots, w_s) = V.$$

$\mathfrak{QED}$

Every finitely generated vector space V has a basis.

31.8.1 Epimorphism of Finite Dimensional Vector Spaces

A finite dimensional vector space is an *epimorphic* image of $Field^n$ where n is the number of dimensions.

Proof:

Given $\{v_1, \ldots, v_n\} \subseteq V$ a generating system of V and $X = (v_1, \ldots, v_n) \in V^n$

$$\lambda : K^n \to V : a \mapsto a_1 X_1 + \cdots + a_n X_n$$

31.8.2 Exchange Lemma

Let $(v_1, \ldots, v_n)$ be a basis of V and $(w_1, \ldots, w_m)$ be a generating system of V. Then, for every v_i, there exists a w_j such that if we replace v_i by w_j in the tuple $(v_1, \ldots, v_n)$, it still forms a basis of V. We can exchange $m - n$ vectors in this way.

Proof I:

Let $(v_1, \ldots, v_n)$ and $(w_1, \ldots, w_m)$ be two bases of V. Suppose we remove v_i from the first basis. The truncated tuple $(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n)$ satisfies

$$\text{span}(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n) \neq V,$$

because if $\text{span}(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n) = V$, then v_i would lie in the span of the remaining vectors and could be written as a linear combination of them. This would contradict the assumption that $(v_1, \ldots, v_n)$ is linearly independent and a basis of V.

By the Basis Extension Theorem, we can extend the truncated tuple $(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n)$ to a basis of V by adding vectors from $(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n, w_1, \ldots, w_m)$. Therefore, by the Basis Extension Theorem, there exists a w_j such that

$$w_j \notin \text{span}(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n),$$

and the tuple $(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n, w_j)$ is linearly independent.

If this tuple does not form a basis, we can again apply the Basis Extension Theorem and add one of the vectors $v_1, \ldots, v_n$ to complete the basis. Clearly, the only possibility is to add v_i, but this would imply that the tuple $(v_1, \ldots, v_n, w_j)$ is not a basis, as w_j would then be linearly dependent on the other vectors. Therefore, $(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n, w_j)$ must form a basis of V.

$\mathfrak{QED}$

Proof II:

Let $A = (v_1, \ldots, v_n)$ and $B = (w_1, \ldots, w_m)$ be two bases of V. We prove the statement by induction on m.

For $m = 1$ $A \cup B = (v_1, \ldots, v_n, w_1)$ is linearly dependent because A is a basis of V. Thus, there exist $\lambda_1, \ldots, \lambda_n = w_1$.

Choose $\lambda_j \neq 0$ and define

$$v_j = -\frac{1}{\lambda_j}(\lambda_1 v_1 + \cdots + \lambda_{j-1} v_{j-1} + \lambda_{j+1} v_{j+1} + \cdots + \lambda_n v_n - w_1)$$

If we remove v_j from A and add w_1 we get a basis of V. This is true because let

$$\tilde{\lambda_1} v_1 + \cdots + \tilde{\lambda_{j-1}} v_{j-1} + \tilde{\lambda_j} w_1 + \tilde{\lambda_{j+1}} v_{j+1} + \cdots + \tilde{\lambda_n} v_n = 0$$

Assume $\tilde{\lambda_j} \neq 0$ then we can express w_1 as a linear combination of the other vectors. This is a contradiction because this would mean that we can write w_1 as another linear combination of the vectors in A which is not possible because B is a basis of V.

If we repeat this process $n - m$ times, we get the desired result.

$\mathfrak{QED}$

31.9 Dimension as a Sum of the Dimension of Subspaces

Let U and W be two subspaces of a vector space V. Then the dimension of the sum of the two subspaces is given by:

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

31.10 Linear Independence of Polynomials

Let P_n be the vector space of polynomials of degree at most n. The set of polynomials $\{1, x, x^2, \ldots, x^n\}$ is a basis for P_n. The dimension of P_n is $n + 1$.

- The set of polynomials $\{1, x, x^2, \ldots, x^n\}$ is linearly independent.
- The set of polynomials $\{1, x, x^2, \ldots, x^n\}$ spans the vector space P_n.
- The dimension of P_n is $n + 1$.

To prove that the set of polynomials $\{1, x, x^2, \ldots, x^n\}$ is linearly independent, we can use the following steps:

1. Assume that there exists a linear combination of the polynomials that equals zero:

   $$a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n = 0,$$

   where $a_0, a_1, \ldots, a_n$ are scalars.
2. Since the left-hand side is a polynomial of degree at most n, it can only be equal to zero if all coefficients are zero.
3. Therefore, we have $a_0 = a_1 = \cdots = a_n = 0$, which proves that the set of polynomials $\{1, x, x^2, \ldots, x^n\}$ is linearly independent.

So you only have to prove that the set of coefficient vectors is linearly independent.

31.11 Interpolation Polynomial

Given the $n + 1$ points (x_k, y_k), with $0 \leq k \leq n$ and all x_k distinct, there exists exactly one polynomial $p_n \in P_n$ such that $y_k = p_n(x_k)$ for all $0 \leq k \leq n$. This polynomial is called the interpolation polynomial.

Proof:

The uniqueness follows immediately. We prove the existence by induction on n. For $n = 0$, choose $p_0(x) = y_0$.

Now assume the statement is true for $n-1$. Let the polynomial p_{n-1}interpolate the points $(x_0, y_0), \ldots, (x_{n-1}, y_{n-1})$.

Define

$$p_n(x) = p_{n-1}(x) + q(x),$$

where

$$q(x) = \frac{(x - x_0)(x - x_1)\cdots(x - x_{n-1})}{(x_n - x_0)(x_n - x_1)\cdots(x_n - x_{n-1})}(y_n - p_{n-1}(x_n)).$$

We have $q \in P_n$, and it follows that $p_n \in P_n$. Furthermore, $q(x_k) = 0$ for $k \leq n - 1$ because a linear factor in the numerator always vanishes at x_k. Therefore, $p_n(x_k) = y_k$ for $k \leq n - 1$. Additionally, we have

$$q(x_n) = y_n - p_{n-1}(x_n),$$

so that $p_n(x_n) = y_n$.

$\mathfrak{QED}$

Example:

Consider the three points $(-2, 1)$, $(-1, -1)$, and $(1, 1)$. These points uniquely define an interpolating parabola p_2. This parabola can be determined using the definition of p_n. For hand calculations and a few points to interpolate, the following approach is also useful. The general form of the polynomial is

$$p_2(x) = ax^2 + bx + c.$$

Substituting the three points into this form gives the system of equations:

$$\begin{aligned} 1 &= a + b + c \quad \text{(from the point (1, 1))} \\ -1 &= a - b + c \quad \text{(from the point (-1, -1))} \\ 1 &= 4a - 2b + c \quad \text{(from the point (-2, 1))} \end{aligned}$$

This leads to the system of equations:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 4 & -2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Solving this system gives $a = 1$, $b = 1$, and $c = -1$, so the interpolation polynomial is

$$p_2(x) = x^2 + x - 1.$$

Chapter 32

Outer and Inner Product; Euclidean and Unitary Space

32.1 Outer Product

The *outer product* of two vectors $a \in \mathbb{F}^m$ and $b \in \mathbb{F}^n$ is defined as the matrix

$$a \otimes b := ab^T = \begin{bmatrix} a_1b_1 & a_1b_2 & \dots & a_1b_n \\ a_2b_1 & a_2b_2 & \dots & a_2b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_mb_1 & a_mb_2 & \dots & a_mb_n \end{bmatrix} \in \mathbb{F}^{m \times n}.$$

32.2 Inner Product

Let V be a vector space over a field $\mathbb{F}$. A mapping $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ is called an *inner product*, if the following conditions are satisfied:

Symmetry

For all $a, b \in V$:

$$\langle a, b \rangle = \begin{cases} \langle b, a \rangle & \text{if } K = \mathbb{R}, \\ \overline{\langle b, a \rangle} & \text{if } K = \mathbb{C}. \end{cases}$$

Linearity in the First Argument

For all $a, b, c \in V$:

$$\langle a, b + c \rangle = \langle a, b \rangle + \langle a, c \rangle$$

and

$$\langle a + b, c \rangle = \langle a, c \rangle + \langle b, c \rangle .$$

Homogeneity in the First Argument

For all $\alpha \in \mathbb{F}$, we have:

$$\langle \alpha a, b \rangle = \alpha \langle a, b \rangle = \begin{cases} \langle a, \alpha b \rangle & \text{if } \mathbb{F} = \mathbb{R}, \\ \langle a, \alpha b \rangle & \text{if } \mathbb{F} = \mathbb{C}. \end{cases}$$

Positive Definiteness

For all $a \in V \setminus \{0\}$:

$$\langle a, a \rangle > 0,$$

and

$$\langle 0, 0 \rangle = 0.$$

It can also be interpreted as a linear transformation that takes a vector and maps it to a real number via matrix-vector multiplication. Also, a more practical way of thinking about the *dot product* is the question: how much are two vectors pointing in the same direction? when dealing with analytical geometry.

32.2.1 Trigonometric Definition

Think about the orthogonal projection as a triangle. Now, remember that to find the adjacent side x you need to take $x = \cos(\theta)h$, which is the value α in our projection formula $p_a(b) = \frac{\langle a,b \rangle}{\langle a,a \rangle} \alpha a$.

Here we add $\|a\| \, \|b\|$ to make this function linear.

$$\langle a, b \rangle = \|a\| \, \|b\| \cos(\theta).$$

32.3 Standard Inner Product for Complex Numbers

Let $a = (a_i)_{i=1}^n$ and $b = (b_i)_{i=1}^n$ be vectors in $\mathbb{C}^n$. The *standard inner product* is defined by

$$\langle a, b \rangle := \sum_{i=1}^{n} a_i \overline{b_i} = a^T \overline{b}$$

and by the *Hermitian inner product*

$$\langle a, b \rangle := \overline{a^T} b.$$

Here the *symmetry* is given by $\langle a, b \rangle = \overline{\langle b, a \rangle}$.

32.4 Inner Product on $C[a, b]$

Let $f, g \in C[a, b]$. The inner product on $C[a, b]$ is defined by

$$\langle f, g \rangle := \int_a^b f(x) \cdot g(x) \, dx.$$

32.5 Euclidean and Unitary Vector Spaces

A real vector space equipped with an inner product is called a *Euclidean vector space*, while a complex vector space with an inner product is called a *unitary vector space*.

32.6 Norms in Vector Spaces

Let V be a K-vector space and $a, b \in V$. A function $\|\cdot\| : V \to \mathbb{R}$ is called a norm if and only if the following conditions hold:

- $\|a\| \in \mathbb{R}$,
- $\|a\| \geq 0$,
- $\|a\| = 0 \iff a = 0$,
- $\forall \lambda \in K,\ \|\lambda a\| = |\lambda| \, \|a\|$,
- $\|a + b\| \leq \|a\| + \|b\|$.

32.6.1 Induced Norm by a Inner Product

As in the special case $V = \mathbb{R}^n$, an inner product *induces* a norm. In a unitary (or Euclidean) space, the inner product induces a (standard) norm defined by

$$\|\cdot\| = \sqrt{\langle \cdot, \cdot \rangle}.$$

32.7 Cauchy-Schwarz Inequality in Unitary Vector Spaces

In all unitary vector spaces V, the *Cauchy-Schwarz inequality* holds:

$$|\langle a, b \rangle| \leq \|a\| \, \|b\| \quad \forall a, b \in V.$$

32.8 Proof of the Triangle Inequality

Both sides of the triangle inequality are real and, in particular, non-negative. Therefore, it is sufficient to prove that the squares of both sides satisfy the desired inequality, i.e., we need to show:

$$\langle a + b, a + b \rangle \leq (\|a\| + \|b\|)^2.$$

First, we expand the left-hand side:

$$\langle a + b, a + b \rangle = \langle a, a \rangle + \langle a, b \rangle + \langle b, a \rangle + \langle b, b \rangle .$$

Since $\langle b, a \rangle = \langle a, b \rangle$, we have:

$$\langle a, b \rangle + \langle b, a \rangle = 2 \operatorname{Re} \langle a, b \rangle .$$

Now, we know that the absolute value of a complex number is always greater than or equal to its real part, so:

$$2 \operatorname{Re} \langle a, b \rangle \leq 2 |\langle a, b \rangle| .$$

Using the Cauchy-Schwarz inequality, we can further bound this by:

$$2 \operatorname{Re} \langle a, b \rangle \leq 2 \, \|a\| \, \|b\| .$$

Thus, we have:

$$\langle a + b, a + b \rangle \leq \langle a, a \rangle + 2 \, \|a\| \, \|b\| + \langle b, b \rangle .$$

Using the definition of the norm, $\|a\|^2 = \langle a, a\rangle$ and $\|b\|^2 = \langle b, b\rangle$, we obtain:

$$\langle a+b, a+b\rangle \leq \|a\|^2 + 2\|a\|\|b\| + \|b\|^2.$$

This is exactly the expansion of $(\|a\| + \|b\|)^2$, which completes the proof.

$\mathfrak{QED}$

32.9 Definiteness of Inner Products

The concept of the *inner product* plays a major role in linear algebra, and there are various manifestations of this element. We are going to see how inner products are constructed based on the properties we are interested in, those being linearity for the columns, symmetry, and positive definiteness.

Let us start with a matrix $A \in \mathbb{R}^{n\times n}$ and a map $(\cdot,\cdot)_A : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, more specifically $(x, y)_A := \langle x, Ay\rangle$ using the standard Euclidean inner product.

For our first property, the linearity, it is granted for every matrix $A = (a_1, a_2, \ldots, a_n)$ because of the rules of matrix-vector multiplication. For example, for the basis vectors e_i, e_j we get

$$(e_i, e_j)_A = \langle e_i, Ae_j\rangle = e_i^T Ae_j = e_i^T a_j = a_{ij}.$$

Therefore, $(e_i, e_j)_A = (e_j, e_i)_A \iff a_{ij} = a_{ji}$. This means that A has to be symmetric. As a side note, the choice for the Euclidean inner product for the structure of our map is optional. Any kind of inner product would also do the job. This implies that each inner product can be described via a matrix.

32.10 Definiteness and Quadratic Forms

Given a symmetric $A \in \mathbb{R}^{n\times n}$ then:

1. The map $x \to \langle x, Ax\rangle$ is called the *Quadratic Form*
2. A is *positive definite* if $\langle x, Ax\rangle > 0 \forall x \in \mathbb{R}^n \setminus \{0\}$.
3. A is *negative definite* if $\langle x, Ax\rangle < 0 \forall x \in \mathbb{R}^n \setminus \{0\}$.
4. A is *positive semidefinite* if $\langle x, Ax\rangle \geq 0 \forall x \in \mathbb{R}^n \setminus \{0\}$.
5. A is *negative semidefinite* if $\langle x, Ax\rangle \leq 0 \forall x \in \mathbb{R}^n \setminus \{0\}$
6. A is *indefinite* if $\exists x, y \in \mathbb{R}^n \setminus \{0\} : \langle x, Ax\rangle < 0 \wedge \langle y, Ay\rangle > 0$.

$(\cdot,\cdot)_A$ is only an inner product if A is symmetric and positive definite.

Example:

Given is the term $x_1^2 - x_3^2 + x_1x_4$, show that it is a quadratic form.

First write the formula for the quadratic form $\langle x, Ax\rangle$ where A is a real symmetric matrix

$$\left\langle \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & q \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right\rangle$$

$$\left\langle \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \begin{bmatrix} ax_1 + bx_2 + cx_3 + dx_4 \\ ex_1 + fx_2 + gx_3 + hx_4 \\ ix_1 + jx_2 + kx_3 + lx_4 \\ mx_1 + nx_2 + ox_3 + px_4 \end{bmatrix} \right\rangle$$

$$x_1(ax_1+bx_2+cx_3+dx_4)+x_2(ex_1+fx_2+gx_3+hx_4)+x_3(ix_1+jx_2+kx_3+lx_4)+x_4(mx_1+nx_2+ox_3+px_4)$$

Now we have to compare the coefficients with the original expression $x_1^2 + (0)x_2^2 - x_3^2 + x_1x_4$.

$$\begin{aligned} &ax_1^2 + bx_1x_2 + cx_1x_3 + dx_4x_4 + \\ &ex_2x_1 + fx_2^2 + gx_2x_3 + hx_2x_4 + \\ &ix_3x_1 + jx_3x_2 + kx_3^2 + lx_3x_4) + \\ &mx_4x_1 + nx_4x_2 + ox_4x_3 + px_4^2 \end{aligned}$$

Note that the only quadratic terms present in the expression are x_1^2 and $-x_3^2$ while the others are zero. This means that they have to disappear in the main diagonal. Also, the only term that x_1 is also multiplying is x_4, therefore all the others have to be zero, like all other non-quadratic terms which are not present. Thus

$$\begin{bmatrix} 1 & 0 & 0 & d \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ m & 0 & 0 & 0 \end{bmatrix}$$

Now the question is what d and m are, because A has to be symmetric, but they cannot be 1 because then we would have $2x_1x_4$. Because of that, we will take the only available option $\frac{1}{2}$.

$$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$$

And now we are done.

32.11 Positive Definite Matrices

A matrix $A \in \mathbb{R}^{n\times n}$ is called *positive definite* if it is Hermitian and if the quadratic form

$$\langle x, Ax\rangle > 0 = x^T Ax > 0,\ \forall x \in \mathbb{R}^n,\ x \neq 0.$$

32.11.1 Properties of Positive Definite Matrices

- All eigenvalues of A are positive.
- If A is symmetric, then A is positive definite if and only if all eigenvalues of A are positive.
- A is invertible and A^{-1} is also positive definite.
- The determinant of A is positive.
- The main diagonal entries of A are all positive.
- The quadratic form $\langle x, Ax\rangle$ is strictly convex.

32.11.2 Condition for Positive Definiteness

A is positive definite if and only if one of the following conditions is satisfied:

$$A_s = \frac{1}{2}\left(A + A^T\right)$$

is also positive definite.

Proof:

If A is positive definite, then:

$$\langle x, Ax\rangle = \langle A_x, x\rangle = (Ax)^T x = x^T A^T x.$$

Therefore,

$$x^T Ax = x^T (A + A^T)x = 2x^T A_s x.$$

$\mathfrak{QED}$

32.12 Minors

For a matrix $A \in \mathbb{R}^{n\times n}$, consider the leading $k \times k$-submatrices $A_k = (a_{ij})_{i,j=1}^{k}$ of A. Then $D_k = \det(A_k)$ is called a *main minor* or *minor* of A.

32.12.1 Minors and Definiteness

Let A be a real, square, symmetric matrix. Then:

- A is positive definite if all main minors are positive.
- A is negative definite if all main minors are negative.
- A is positive semidefinite if all main minors are greater than or equal to zero but not all zero.
- A is negative semidefinite if all main minors are less than or equal to zero but not all zero.
- A is indefinite if all main minors are equal to zero.

32.12.2 Connection to Symmetric Matrices

For a real symmetric matrix A, the following statements are equivalent.

- A is positive definite.
- All eigenvalues of A are positive.
- All main minors of A are positive.

Continuing, A is positive semidefinite if all eigenvalues of A are not negative and the main minors are all not positive.

Example: Symmetry and Positive Definiteness of BAB^T and B^TAB

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix and $B \in \mathbb{R}^{m\times n}$ be a real matrix. We study the symmetry and positive definiteness of the matrices $C = BAB^T \in \mathbb{R}^{m\times m}$ and $D = B^TAB \in \mathbb{R}^{n\times n}$.

(i) Symmetry of BAB^T:

$$(BAB^T)^T = (B^T)^T A^T B^T = BAB^T,$$

since $(B^T)^T = B$ and $A^T = A$. Thus, BAB^T is symmetric.

(ii) Symmetry of B^TAB:

$$(B^TAB)^T = B^T A^T B = B^TAB,$$

since A is symmetric. Hence, B^TAB is symmetric as well.

To determine when BAB^T and B^TAB are positive definite, we use the quadratic form criterion.

(i) Positive definiteness of $C = BAB^T$:

Let $x \in \mathbb{R}^m$, $x \neq 0$. Consider:

$$x^T Cx = x^T BAB^T x.$$

Let $y = B^T x \in \mathbb{R}^n$. Then:

$$x^T BAB^T x = y^T Ay.$$

Since A is positive definite, $y^T Ay > 0$ for all $y \neq 0$. Therefore, $x^T Cx > 0$ if and only if $B^T x \neq 0$ for all $x \neq 0$, which holds if and only if $\ker(B^T) = \{0\}$, or equivalently, B has full column rank.

(ii) Positive definiteness of $D = B^T AB$:

Let $z \in \mathbb{R}^n$, $z \neq 0$. Then:

$$z^T Dz = z^T B^T ABz = (Bz)^T A(Bz).$$

Let $w = Bz$. Since A is positive definite, $w^T Aw > 0$ if $w \neq 0$. Thus, D is positive definite if and only if $Bz \neq 0$ for all $z \neq 0$, i.e., if B has full rank (i.e., $\ker(B) = \{0\}$).

In summary:

- BAB^T is positive definite if B has full column rank.
- $B^T AB$ is positive definite if B has full rank.

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32.13 Sign of the D_k

For the main minors D_k of a negative definite matrix it holds that

$$(-1)^k D_k = (-1)^k \det(A_k) = \det(-A_k) > 0,$$

because $-A$ is positive definite.

32.14 Negative Quadratic Form

If A is negative definite, then

$$-\langle x, Ax\rangle = \langle x - Ax\rangle > 0.$$

32.15 Theorems for Non-Square Matrices

32.15.1 Theorem I

Given $m \geq n$ and $A \in \mathbb{R}^{m\times n}$ with $rg(A) = n$. Then $A^T A$ is positive definite. For $rg(A) < n$, $A^T A$ is positive semidefinite.

Proof:

Because $(A^T A)^T = A^T A$, $A^T A$ is symmetric. Therefore, it holds that

$$\left\langle x, A^T Ax\right\rangle = x^T A^T Ax = (Ax)^T Ax = \langle Ax, Ax\rangle = \|Ax\|^2 \geq 0.$$

Also, $A^T A$ is positive semidefinite, and it is also invertible for $rg(A) = n$. If there were an $x \neq 0$ such that $\|Ax\| = 0$, this would imply that $Ax = 0$, and also $A^T(Ax) = 0$, therefore $x \in ker(A^T A)$. And $A^T A$ would not be invertible. Finally, $\left\langle x, A^T Ax\right\rangle > 0 \forall x \neq 0$, and $A^T A$ is symmetric and positive definite.

32.16 Theorem II

Given $m \leq n$ and $A \in \mathbb{R}^{m \times n}$. Then $A^T A$ is positive semidefinite and even positive definite if $rg(A) = m$.

Chapter 33

Orthogonality and Inner Products

33.1 Orthogonality and Projection

Let $a, b \in V$. The vectors a and b are *orthogonal* to each other if

$$\langle a, b \rangle = 0.$$

This is written as $a \perp b$.

Also, the Pythagorean Theorem holds

$$\|a + b\|^2 = \|a\|^2 + \|b\|^2,$$

for $a \perp b$ in all unitary vector spaces.

For the orthogonal projection $p_b(a)$ of a vector a onto b, with $b \neq 0$, the formula in every unitary vector space is:

$$p_b(a) = \frac{\langle a, b \rangle}{\langle b, b \rangle} b.$$

33.2 Induced Metric

Given $(V, \|\cdot\|)$ a metric space, there is an inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{F}$ such that $\|x\|^2 = \langle x, x \rangle$ if and only if the *parallelepiped equation* holds.

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|), \ \forall x, y \in V.$$

In such a case, the inner product is given by

$$\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2).$$

Proof:

$\Rightarrow$: If $\|x\|^2 = \langle x, x \rangle$, then

$$\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= \langle x+y, x+y\rangle + \langle x-y, x-y\rangle \\
&= 2(\langle x,x\rangle + \langle y,y\rangle) \\
\langle x, x+y\rangle + \langle y, x+y\rangle + \langle x, x-y\rangle - \langle y, x-y\rangle &= 2(\langle x,x\rangle + \langle y,y\rangle) \\
\langle x,x\rangle + \langle x,y\rangle + \langle y,y\rangle + \langle x,y\rangle + \langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle &= 2(\langle x,x\rangle + \langle y,y\rangle) \\
2\langle x,x\rangle + 2\langle y,y\rangle + &= 2(\langle x,x\rangle + \langle y,y\rangle)
\end{aligned}$$

$\Leftarrow$: Assume the parallelepiped equation holds, then we want to show that the equation

$$f(x,y) = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2)$$

satisfies the properties of an inner product:

- $f(x,y) \geq 0$ trivial.
- $f(x,y) = 0 \iff x = y = 0$ is also trivial and
- $f(x,y) = f(y,x)$ is also trivial by our definition.
- $f(\alpha x + y, z) = \alpha f(x,y) + f(y,z)$ must be shown in a two-step process.

First, we show that $f(x+y,z) = f(x,z) + f(y,z)$:

$$\begin{aligned}
4(f(x,y) + f(z,y)) &= \|x+y\|^2 - \|x-y\|^2 + \|z+y\|^2 - \|z-y\|^2 \\
&= \|x+y\|^2 + \|z\|^2 \\
&\quad - \|x-y\|^2 - \|z\|^2 \\
&\quad + \|z+y\|^2 + \|x\|^2 \\
&\quad - \|z-y\|^2 - \|x\|^2
\end{aligned}$$

Hence, by adding this clever zero and using the parallelepiped equation we get

$$\begin{aligned}
2(f(x,y) + f(z,y)) &= \|x+y+z\|^2 + \|x+y-z\|^2 \\
&\quad - \|x-y+z\|^2 - \|x-y-z\|^2 \\
&\quad + \|z+y+x\|^2 + \|z+y-x\|^2 \\
&\quad - \|z-y+x\|^2 - \|z-y-x\|^2 \\
&= 2(\|x+z+y\|^2 - \|x+z-y\|^2) \\
&= 2f(x+z,y)
\end{aligned}$$

Therefore, $f(x+z,y) = f(x,y) + f(z,y)$.

Now we need to prove that $f(\alpha x, y) = \alpha f(x,y)$:

For the naturals, starting with $n = 2$: $f(2x,y) = f(x+x,y) = f(x,y) + f(x,y) = 2f(x,y)$, hence using induction for $n \in \mathbb{N}$ with base case 2 and summation, the proof is concluded.

For the rationals, $r \in \mathbb{Q}$ with $r = \frac{m}{n}$, we have that

$$f(rx,y) = f\left(\frac{mx}{n}, y\right) = mf\left(\frac{1}{n}x, y\right) = \frac{m}{n}nf\left(\frac{1}{n}x, y\right) = \frac{m}{n}f\left(x,y\right).$$

Finally, for the reals: each real number can be approximated as the limit of an infinite sequence of rationals.

For $\alpha \in \mathbb{R}$ there is $r_K \in \mathbb{Q}$ such that $\lim_{k\to\infty} r_k = \alpha$.

Given that f is continuous, we can then use the following

$$f\left(\lim_{k\to\infty} r_K x, y\right) = \lim_{k\to\infty} f(r_K x, y) = \lim_{k\to\infty} r_k f(x,y) = \alpha f(x,y).$$

$\mathfrak{QED}$

33.3 Euclidean Inner Product as Sum of Metrics

Given our Euclidean inner product for any two vectors in $\mathbb{R}^n$, the following holds:

$$\langle x_i, x_j\rangle = \frac{1}{2}(\|x_i\|^2 + \|x_j\|^2 - \|x_i - x_j\|^2).$$

Proof:

$$\begin{aligned}
\|x_i\|^2 + \|x_j\|^2 - \|x_i - x_j\|^2 &= \langle x_i, x_i\rangle + \langle x_j, x_j\rangle - \langle x_i - x_j, x_i - x_j\rangle \\
&= \sum (x_i^{(k)})^2 + \sum (x_i^{(k)})^2 - \sum (x_i^{(k)} - x_j^{(k)})^2 \\
&= \sum (x_i^{(k)})^2 + \sum (x_i^{(k)})^2 - \left[\sum (x_i^{(k)})^2 - 2\sum x_i^{(k)} x_j^{(k)} + \sum (x_j^{(k)})^2\right] \\
&= \sum (x_i^{(k)})^2 + \sum (x_i^{(k)})^2 - \sum (x_i^{(k)})^2 + 2\sum x_i^{(k)} x_j^{(k)} - \sum (x_j^{(k)})^2 \\
&= 2\sum x_i^{(k)} x_j^{(k)}
\end{aligned}$$

Now by substituting into the original equality

$$\begin{aligned}
\langle x_i, x_j\rangle &= \frac{1}{2}(\|x_i\|^2 + \|x_j\|^2 - \|x_i - x_j\|^2) \\
&= \frac{1}{2}\left(2\sum x_i^{(k)} x_j^{(k)}\right) \\
&= \sum x_i^{(k)} x_j^{(k)} = \langle x_i, x_j\rangle
\end{aligned}$$

$\mathfrak{QED}$

33.4 Orthogonal Projection and the Orthogonal Complement

Let U be a finitely generated subspace of V and $a \in V$. A vector $p_U(a) \in U$ is called the *orthogonal projection* of a onto U if

$$a - p_U(a) \perp u \quad \forall u \in U.$$

The question arises about well-definedness, i.e., whether such a vector $p_U(a)$ always exists and if it is unique. The following concept helps in the discussion of uniqueness.

For $M \subseteq V$, the *orthogonal complement* of M is defined as

$$M^\perp = \{v \in V \mid v \perp u \forall u \in M\}.$$

- $M^\perp$ is a subspace of V.
- Let U be a subspace of V. Then, we have $U \cap U^\perp = \{0\}$.

Proof:

We need to check the closure property. For $u \in M$, $x, y \in M^\perp$, and $\lambda \in \mathbb{R}$, we have:

$$\langle x + y, u \rangle = \langle x, u \rangle + \langle y, u \rangle = 0$$

$$\langle \lambda x, u \rangle = \lambda \langle x, u \rangle = 0.$$

Let $a \in U \cap U^\perp$. Then, we have $\langle a, u \rangle = 0$ for all $u \in U$, because $a \in U^\perp$, and in particular, $\langle a, a \rangle = 0$ since $a \in U$, which implies $a = 0$.

$\mathfrak{QED}$

33.5 Orthogonality of the Basis to the Complement

Let U be as before, and let $(u_1, \ldots, u_m)$ be a basis of U. For $v \in V$, we have:

$$v \in U^\perp \quad \text{if and only if} \quad \langle v, u_i \rangle = 0 \quad \forall 1 \leq i \leq m.$$

33.6 How to Find the Orthogonal Complement

Let U be a finitely generated subspace of V with basis $(u_1, \ldots, u_m)$. To find the orthogonal complement $U^\perp$, we can solve the system of equations:

$$\langle v, u_i \rangle = 0 \quad \forall 1 \leq i \leq m.$$

This system can be expressed in matrix form as $A \cdot v = 0$, where A is the matrix whose rows are the vectors u_i and v is the vector we want to find in $U^\perp$.

33.7 Orthogonal and Orthonormal Systems

Let $B = (v_1, \ldots, v_m)$ be an m-tuple of vectors in $V \setminus \{0\}$.

- B is called an orthogonal system in V if all the vectors v_i are pairwise orthogonal.
- An orthogonal system is called an orthonormal system if, in addition, $\|v_i\| = 1$ for all $i = 1, \ldots, m$.
- An orthogonal system that forms a basis of V is called an orthogonal basis of V.
- An orthonormal system that forms a basis of V is called an orthonormal basis of V.

Notation:

OG-System OG-Basis ON-System ON-Basis

Using the Kronecker delta symbol $\delta_{i,j}$:

$$\delta_{i,j} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j, \end{cases}$$

we have $\langle v_i, v_j \rangle = \delta_{i,j}$ for any orthonormal system.

33.8 Writing a Vector in Terms of the Orthogonal Basis

Example I:

The vectors

$$a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

are orthogonal and normalized, but do not form a basis. Thus, they constitute an *orthonormal system*.

Example II:

The vectors

$$a_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \quad a_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

form a basis, are orthogonal and normalized. Therefore, they form an *orthonormal basis*.

33.9 Linear Independence of Orthogonal Systems

An orthogonal system $(v_1, \ldots, v_n)$ is linearly independent.

Proof:

Let $\sum_{i=1}^{n} \lambda_i v_i = 0$. Then, for any v_j, it follows that

$$\left\langle \sum_{i=1}^{n} \lambda_i v_i, v_j \right\rangle = \sum_{i=1}^{n} \lambda_i \langle v_i, v_j \rangle = 0.$$

Due to the orthogonality of the system, all terms vanish except $\lambda_j \langle v_j, v_j \rangle$, so

$$\lambda_j \langle v_j, v_j \rangle = 0.$$

Since $v_j \neq 0$ in any orthogonal system, we have $\langle v_j, v_j \rangle = \|v_j\|^2 > 0$, implying $\lambda_j = 0$ for all $1 \leq j \leq n$.

$\mathfrak{QED}$

33.10 Representation With Respect to an Orthogonal Basis

Let $B = (v_1, \ldots, v_n)$ be an orthogonal basis of a vector space V. Then for every $v \in V$, we have:

$$v = \sum_{k=1}^{n} \frac{\langle v, v_k \rangle}{\langle v_k, v_k \rangle} v_k,$$

that is, the coordinates of v with respect to the basis B are given by

$$\left(\frac{\langle v, v_k \rangle}{\|v_k\|^2} \right)_{1 \leq k \leq n}^{T}.$$

33.10.1 Orthogonal Projection and Direct Sum Decomposition

Let $B = (v_1, \ldots, v_m)$ be an orthogonal system in V, and let $U = L(B)$ be the subspace of V spanned by B.

- For every $v \in V$, the orthogonal projection of v onto U is given by:

$$p_U(v) = \sum_{i=1}^{m} \frac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle} v_i.$$

- Every vector $v \in V$ can be uniquely written as the sum $v = p_U(v) + w$, with $w \in U^{\perp}$. In this case, we have:

$$w = v - p_U(v).$$

- $V = U \oplus U^{\perp}$, which means that every vector in V can be uniquely decomposed into a sum of a vector from U and a vector from $U^{\perp}$.
- If $\dim(V) = n$, then:

$$\dim(U) + \dim(U^{\perp}) = n \quad \text{for every subspace } U.$$

33.11 The Gram-Schmidt Process

The *Gram-Schmidt process* is a method for orthonormalizing a set of vectors in an inner product space. Given a finite set of linearly independent vectors $(v_1, \ldots, v_n)$, the process generates an orthonormal basis $(u_1, \ldots, u_n)$ as follows:

1. Set $u_1 = \frac{v_1}{\|v_1\|}$.
2. For $k = 2, \ldots, n$:
   (a) Set $w_k = v_k - \sum_{j=1}^{k-1} \langle v_k, u_j \rangle u_j$.
   (b) Set $u_k = \frac{w_k}{\|w_k\|}$.
3. The resulting set $(u_1, \ldots, u_n)$ is an orthonormal basis of the subspace spanned by $(v_1, \ldots, v_n)$.

The Gram-Schmidt process can be applied to any finite set of linearly independent vectors in an inner product space, and it is particularly useful for constructing orthonormal bases in Euclidean spaces.

Proof:

Let W be a non-zero finite-dimensional subspace of an inner product space and let $(u_1, \ldots, u_n)$ be a basis for W. We will show that an *orthonormal basis* exists.

Let $v_1 = u_1$, then compute the vector u_2 orthogonal to the space W_1 spanned by v_1

$$v_2 = u_2 - proj_{w_1} u_2 = u_2 - \frac{\langle u_2, v_1 \rangle}{\|v_1\|} u_2$$

Then normalize the vector.

We can continue this approach of projecting the vector u_i orthogonal to the space W_i spanned by all the previously computed basis vectors v_i and then normalize it until we get our orthonormal basis. Note that, because our previous vectors formed a basis, we can be sure that during this process no zero vector will be generated, because then we would have some u_i that is a linear combination of the other vectors. Thus, this would be a contradiction.

$\mathfrak{QED}$

Example:

Let $v_1 = (1,0,0)$, $v_2 = (1,1,0)$, and $v_3 = (1,1,1)$. The Gram-Schmidt process yields:

$$u_1 = \frac{v_1}{\|v_1\|} = (1,0,0), \quad u_2 = \frac{v_2 - \langle v_2, u_1\rangle\, u_1}{\|v_2 - \langle v_2, u_1\rangle\, u_1\|} = (0,1,0)\,,$$

$$u_3 = \frac{v_3 - \langle v_3, u_1\rangle\, u_1 - \langle v_3, u_2\rangle\, u_2}{\|v_3 - \langle v_3, u_1\rangle\, u_1 - \langle v_3, u_2\rangle\, u_2\|} = (0,0,1)\,.$$

Thus, the orthonormal basis is $(1,0,0), (0,1,0), (0,0,1)$.

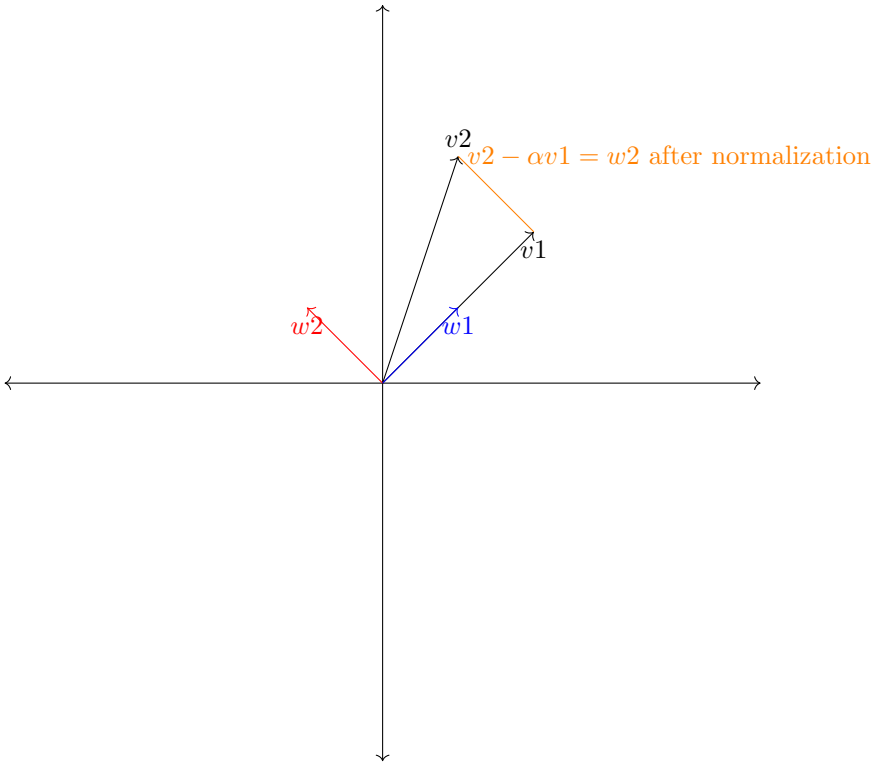
33.12 Existence of Orthonormal Bases and Orthogonal Projections

- Every finitely generated unitary vector space has an orthonormal basis.
- This is in general false for vector spaces that are not finitely generated.
- We now provide the existence proof of the orthogonal projection.
- Let V be a unitary vector space and U a finitely generated subspace. Then for every $v \in V$, the orthogonal projection $p_U(v)$ of v onto U exists.
- Let V be a finitely generated unitary vector space and U any subspace. Then we have $V = U \oplus U^\perp$, and

$$\dim(V) = \dim(U) + \dim(U^\perp).$$

- Every hyperplane in $\mathbb{R}^n$ admits a normal form; the normal vector is unique up to scalar multiplication.
- Let V be as above and let $v_1, \ldots, v_m \in V$. If it is possible to construct orthonormal vectors $w_1, \ldots, w_m$ from them using the Gram-Schmidt process, then $(v_1, \ldots, v_m)$ are linearly independent.

33.13 Visualization of the Gram-Schmidt Process in 2 Dimension



33.14 Best Approximation

Let V be a unitary vector space and U a finitely generated subspace of V. A vector $v* \in U$ is called the best approximation in U to v if:

$$\|v* - v\| = \inf_{x \in U} \|x - v\|$$

also

$$\|v* - P_u(v)\| = \inf_{u \in U} \|u - v\|.$$

Proof:

Let $x \in V$, v be some point and $v* = \text{proj}_U v$. Then

$$v - x = v - v* + v* - x$$

where $v - v* \in V^\perp$ and $v* - x \in V$. By the Pythagorean Theorem

$$\|x - v\|^2 = \|v* - v\|^2 + \|v* - x\|^2$$

Therefore

$$\|v* - v\|^2 \leq \|v* - x\|^2$$

QED

Chapter 34

Matrices

In this chapter, we explore the fundamental concepts of *matrices*, their operations, and important properties that form the foundation of linear algebra.

34.1 Definition of a Matrix

A *matrix* is a rectangular array of numbers, symbols, or expressions arranged in rows and columns (informally). An $m \times n$ matrix A consists of mn elements a_{ij} where $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$.

Formally,

$$A : m \times n \to \mathbb{F} : (i, j) \mapsto a_{i,j}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The set of all $m \times n$ matrices with real entries is denoted by $\mathbb{R}^{m \times n}$. Special cases include:

- Square matrix: matrix with the same number of rows and columns ($m = n$)
- Column vector: $m \times 1$ matrix
- Row vector: $1 \times n$ matrix
- Identity matrix I_n: An $n \times n$ matrix with ones on the main diagonal and zeros elsewhere
- Zero matrix: A matrix where all entries are zero

34.2 Matrix Addition and Subtraction

Matrix addition and *subtraction* are defined for matrices of the same dimensions.

$$c_{ij} = a_{ij} \pm b_{ij} \quad \text{for all } i = 1, 2, \ldots, m \text{ and } j = 1, 2, \ldots, n$$

Matrix addition/subtraction satisfies the following properties:

34.2.1 Properties of Matrix Addition and Subtraction

- *Commutativity*: $A \pm B = B \pm A$.
- *Associativity*: $(A \pm B) \pm C = A \pm (B \pm C)$.
- *Identity element*: $A \pm O = A$.
- *Inverse element*: $A \pm (-A) = O$.

Here O is the zero matrix.

34.3 Matrix Multiplication

Matrix multiplication is defined between matrices where the number of columns in the first matrix equals the number of rows in the second matrix.

For $A \in \mathbb{C}^{m \times p}$ and $B \in \mathbb{C}^{p \times n}$, their product $C = AB \in \mathbb{C}^{m \times n}$ is defined as:

$$c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{ip} b_{pj}$$

Matrix multiplication is defined this way so that we can think of each operation as the dot product of two vectors.

An example of this is thinking about the dot product as the total price of a purchase in a store. The total price is the sum of the amounts, which are given by the number of items per price. Now, to extend this idea to matrices, we think about the number of items as a vector of size i and the prices of each item as another vector p_1 with dot product $\langle i, p_1 \rangle$. If we want to buy other items with different prices, but in the same amounts given by the vector i, we get a matrix $A = (p_1, \ldots, p_n)$. Now it makes sense that, if we take three different arrays of products, we get a vector of three components, which represent each of the total amounts to pay.

Thus, each operation in matrix multiplication is taking the dot product of the *transpose* of the current row with the column vector of the other matrix.

$$A \times \vec{v} = \begin{bmatrix} \langle a_1^T, v \rangle \\ \langle a_2^T, v \rangle \\ \vdots \\ \langle a_n^T, v \rangle \end{bmatrix}$$

As for the case with multiple columns:

$$A \times B = \begin{bmatrix} \langle a_1^T, b_j \rangle & \ldots & \langle a_1^T, b_n \rangle \\ \vdots & \ldots & \vdots \\ \langle a_n^T, b_j \rangle & \ldots & \langle a_n^T, b_n \rangle \end{bmatrix}$$

Example:

Consider the matrices A and B given by:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 0 & -2 \end{bmatrix} \in \mathbb{R}^{2 \times 3} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ -1 & 3 \\ 4 & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 2}$$

To compute the product $C = AB \in \mathbb{R}^{2 \times 2}$, we calculate each entry c_{ij} using the formula:

$$c_{ij} = \sum_{k=1}^{3} a_{ik} b_{kj}$$

Let's calculate each entry of C:

$$\begin{aligned} c_{11} &= a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} \\ &= 2 \cdot 1 + 3 \cdot (-1) + 1 \cdot 4 \\ &= 2 - 3 + 4 = 3 \end{aligned}$$

$$\begin{aligned} c_{12} &= a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ &= 2 \cdot 2 + 3 \cdot 3 + 1 \cdot 0 \\ &= 4 + 9 + 0 = 13 \end{aligned}$$

$$\begin{aligned} c_{21} &= a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} \\ &= 1 \cdot 1 + 0 \cdot (-1) + (-2) \cdot 4 \\ &= 1 + 0 - 8 = -7 \end{aligned}$$

$$\begin{aligned} c_{22} &= a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \\ &= 1 \cdot 2 + 0 \cdot 3 + (-2) \cdot 0 \\ &= 2 + 0 + 0 = 2 \end{aligned}$$

Therefore, the product $C = AB$ is:

$$C = AB = \begin{bmatrix} 3 & 13 \\ -7 & 2 \end{bmatrix}$$

Let's verify that matrix multiplication is not generally commutative by attempting to compute BA:

Since B is a 3×2 matrix and A is a 2×3 matrix, the product BA would be a 3×3 matrix. However, this calculation cannot be performed since the number of columns in B (which is 2) does not equal the number of rows in A (which is 2). T hus, BA is undefined, demonstrating that matrix multiplication is not always commutative.

34.3.1 Properties of Matrix Multiplication

Matrix multiplication satisfies the following properties:

- *Associativity*: $A(BC) = (AB)C$.
- *Left distributivity*: $A(B + C) = AB + AC$.
- *Right distributivity*: $(A + B)C = AC + BC$.
- *Identity*: $AI_n = A \quad \text{and} \quad I_m A = A$.

Note that matrix multiplication is generally not commutative, i.e., $AB \neq BA$ in most cases.

34.4 Outer Product for Matrix Multiplication

The *outer product* of two vectors $a \in \mathbb{F}^m$ and $b \in \mathbb{F}^n$ is defined as the matrix

$$a \otimes b := ab^T = \begin{bmatrix} a_1b_1 & a_1b_2 & \dots & a_1b_n \\ a_2b_1 & a_2b_2 & \dots & a_2b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_mb_1 & a_mb_2 & \dots & a_mb_n \end{bmatrix} \in \mathbb{F}^{m \times n}.$$

For the case where $A \in \mathbb{F}^{n\times m}$ and $B \in \mathbb{F}^{m\times n}$:

$$A \times B = \sum_{k=1}^{n} a_k \otimes b_k = \sum_{k=1}^{n} a_k b_k^T$$

where a_k is the k-th column of A and b_k is the k-th row of B.

34.5 The Transpose of a Matrix

The *transpose* of a matrix $A \in \mathbb{R}^{m\times n}$, denoted $A^T \in \mathbb{R}^{n\times m}$, is obtained by interchanging rows and columns:

$$(A^T)_{ij} = a_{ji} \quad \text{for all } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m$$

34.5.1 Properties of the Transpose

- *Transpose of a transpose*: $(A^T)^T = A$.
- *Transpose of a sum*: $(A + B)^T = A^T + B^T$.
- *Transpose of a product*: $(AB)^T = B^T A^T$.
- *Transpose of a scalar multiple*: $(\alpha A)^T = \alpha A^T$ for any scalar α.
- *Transpose of an inverse*: $(A^{-1})^T = (A^T)^{-1}$ (assuming A is invertible).
- *Rank of the transpose*: $\text{rank}(A) = \text{rank}(A^T)$.
- *Determinant of the transpose*: $\det(A) = \det(A^T)$.
- *Eigenvalues of the transpose*: $\text{eig}(A) = \text{eig}(A^T)$, where eig() are the eigenvalues.

The transpose can also help us interpret the dot product of two vectors being 0 as a matrix multiplication. Here we are given $A = (a_1, \ldots, a_n)$ with Ax as some vector, b some other vector, and a_k some row of A.

$$\langle a_k, b - Ax\rangle = 0 \quad \forall k \iff A^T(b - Ax) = \vec{0}$$

Proof That the Inverse of the Transpose Is the Transpose of the Inverse

To prove: $(A^{-1})^T = (A^T)^{-1}$

$$\begin{aligned}(A^{-1})^T A^T &= (A^T)^{-1} A^T \\ (AA^{-1})^T &= (A^T)^{-1} A^T \\ I^T &= I\end{aligned}$$

$\mathfrak{QED}$

34.6 The Equivalence of Matrices

Two matrices A and B are said to be *equivalent* if one can be transformed into the other through a finite sequence of elementary row operations. We write $A \sim B$ to denote this equivalence.

The elementary row operations are:

- Interchanging two rows: $R_i \leftrightarrow R_j$.
- Multiplying a row by a non-zero scalar: $R_i \mapsto \alpha R_i$ where $\alpha \neq 0$.
- Adding a multiple of one row to another: $R_i \mapsto R_i + \alpha R_j$ where $i \neq j$.

Matrix equivalence is an equivalence relation, satisfying reflexivity, symmetry, and transitivity. Equivalent matrices represent the same linear system in different bases.

34.6.1 Row Echelon Form (REF)

A matrix is in *row echelon form* if:

- All rows consisting entirely of zeros are at the bottom of the matrix.
- The leading entry (first non-zero element) of each non-zero row is to the right of the leading entry of the row above it.
- All entries in a column below a leading entry are zeros.

34.6.2 Reduced Row Echelon Form (RREF)

A matrix is in *reduced row echelon form* if:

- It is in row echelon form.
- Each leading entry is 1.
- Each leading entry is the only non-zero entry in its column.

The Gauss-Jordan elimination algorithm proceeds as follows:

- Start with the leftmost non-zero column.
- Find the pivot (non-zero element) in this column. If necessary, swap rows to move a non-zero element to the pivot position.
- Divide the pivot row by the pivot value to make the pivot equal to 1.
- Eliminate all other entries in the pivot column by subtracting appropriate multiples of the pivot row.
- Cover the pivot row and column, and repeat steps 1-4 on the submatrix until all rows are processed.
- For RREF, eliminate all entries above each pivot as well.

Example:

We'll use Gauss-Jordan elimination to solve the linear system:

$$\begin{aligned} 2x + y - z &= 8 \\ -3x - y + 2z &= -11 \\ x + y + z &= 3 \end{aligned}$$

First, we set up the augmented matrix:

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ 1 & 1 & 1 & 3 \end{array}\right]$$

Now we apply Gauss-Jordan elimination to transform this into reduced row echelon form:

Step 1: We'll choose the first element in the first row as our pivot. Let's first swap row 1 and row 3 to get a simpler pivot:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ -3 & -1 & 2 & -11 \\ 2 & 1 & -1 & 8 \end{array}\right]$$

Step 2: Eliminate the first elements in rows 2 and 3:

Row 2 + 3 × Row 1:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 2 & 5 & -2 \\ 2 & 1 & -1 & 8 \end{array}\right]$$

Row 3 - 2 × Row 1:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 2 & 5 & -2 \\ 0 & -1 & -3 & 2 \end{array}\right]$$

Step 3: Make the pivot in row 2 equal to 1 by dividing the entire row by 2:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & \frac{5}{2} & -1 \\ 0 & -1 & -3 & 2 \end{array}\right]$$

Step 4: Eliminate the second element in rows 1 and 3:

Row 1 - Row 2:

$$\left[\begin{array}{ccc|c} 1 & 0 & -\frac{3}{2} & 4 \\ 0 & 1 & \frac{5}{2} & -1 \\ 0 & -1 & -3 & 2 \end{array}\right]$$

Row 3 + Row 2:

$$\left[\begin{array}{ccc|c} 1 & 0 & -\frac{3}{2} & 4 \\ 0 & 1 & \frac{5}{2} & -1 \\ 0 & 0 & -\frac{1}{2} & 1 \end{array}\right]$$

Step 5: Make the pivot in row 3 equal to 1 by multiplying the entire row by -2:

$$\left[\begin{array}{ccc|c} 1 & 0 & -\frac{3}{2} & 4 \\ 0 & 1 & \frac{5}{2} & -1 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

Step 6: Eliminate the third element in rows 1 and 2:

Row 1 + $\frac{3}{2}$× Row 3:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & \frac{5}{2} & -1 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

Row 2 - $\frac{5}{2}$× Row 3:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1+5=4 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

The matrix is now in reduced row echelon form:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

This corresponds to the system:

$$x = 1$$
$$y = 4$$
$$z = -2$$

Therefore, the solution to the original system is $x = 1$, $y = 4$, and $z = -2$.

34.7 The Inverse of a Matrix

For a square matrix $A \in \mathbb{R}^{n \times n}$, the *inverse* matrix A^{-1} (if it exists) satisfies:

$$AA^{-1} = A^{-1}A = I_n.$$

34.7.1 Properties of the Inverse

- *Inverse of the inverse*: $(A^{-1})^{-1} = A$.
- *Inverse of the transpose*: $(A^T)^{-1} = (A^{-1})^T$.
- *Inverse of a product*: $(AB)^{-1} = B^{-1}A^{-1}$.
- *Determinant of the inverse*: $\det(A^{-1}) = \frac{1}{\det(A)}$.
- *Scalars under the inverse*: $(cA)^{-1} = \frac{1}{c}A^{-1}$.
- *Rank of the inverse*: $\text{rank}(A^{-1}) = \text{rank}(A)$.
- *Eigenvalues of the inverse*: $\text{eig}(A^{-1}) = \text{eig}(A)^{-1}$.

34.7.2 Finding the Inverse

There are several methods to find the inverse of a matrix:

Gauss-Jordan Method

Form the augmented matrix $[A|I_n]$ and apply Gauss-Jordan elimination to transform it into $[I_n|A^{-1}]$:

1. Create the augmented matrix $[A|I_n]$
2. Apply row operations to transform the left side into I_n
3. The right side will be A^{-1}

Adjoint Method

For an $n \times n$ matrix A:

$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A)$$

Where $\text{adj}(A)$ is the adjoint (or adjugate) of A, defined as the transpose of the co-factor matrix.

A matrix is invertible if and only if its determinant is non-zero. Such matrices are called non-singular or regular matrices.

Example:

Let's find the inverse of the matrix:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

We form the augmented matrix $[A|I_3]$:

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Now we apply Gauss-Jordan elimination:

Step 1: Make the first pivot equal to 1 by dividing the first row by 2:

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Step 2: Eliminate the first element in rows 2 and 3:

Row 2 - 3 $\times$ Row 1:

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Row 3 - 2 $\times$ Row 1:

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

Step 3: Make the second pivot equal to 1 by multiplying the second row by 2:

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -1 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

Step 4: Eliminate the second element in row 1 and the third element in row 2:

Row 1 - $\frac{1}{2}\times$ Row 2:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & -1 & 0 \\ 0 & 1 & -1 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

Row 2 + Row 3:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & -1 & 0 \\ 0 & 1 & 0 & -4 & 2 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

Step 5: Eliminate the third element in row 1:

Row 1 - Row 3:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -1 \\ 0 & 1 & 0 & -4 & 2 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

The right side of the augmented matrix now gives us A^{-1}:

$$A^{-1} = \begin{bmatrix} 3 & -1 & -1 \\ -4 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

To verify, we can check that $AA^{-1} = I_3$:

$$\begin{aligned} AA^{-1} &= \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & -1 & -1 \\ -4 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \cdot 3 + 1 \cdot (-4) + 1 \cdot (-1) & 2 \cdot (-1) + 1 \cdot 2 + 1 \cdot 0 & 2 \cdot (-1) + 1 \cdot 1 + 1 \cdot 1 \\ 3 \cdot 3 + 2 \cdot (-4) + 1 \cdot (-1) & 3 \cdot (-1) + 2 \cdot 2 + 1 \cdot 0 & 3 \cdot (-1) + 2 \cdot 1 + 1 \cdot 1 \\ 2 \cdot 3 + 1 \cdot (-4) + 2 \cdot (-1) & 2 \cdot (-1) + 1 \cdot 2 + 2 \cdot 0 & 2 \cdot (-1) + 1 \cdot 1 + 2 \cdot 1 \end{bmatrix} \\ &= \begin{bmatrix} 6 - 4 - 1 & -2 + 2 + 0 & -2 + 1 + 1 \\ 9 - 8 - 1 & -3 + 4 + 0 & -3 + 2 + 1 \\ 6 - 4 - 2 & -2 + 2 + 0 & -2 + 1 + 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

This confirms that we have correctly found the inverse of matrix A.

34.8 How to Isolate Matrices in Equations

Like in other types of equations, we can isolate a matrix by performing the proper operations.

- $A \pm B = C \implies A = C \mp B$
- $AB = C \implies ABB^{-1} = CB^{-1} \implies A = CB^{-1}$
- $AB = C \implies A^{-1}AB = A^{-1}C \implies B = A^{-1}C$

34.9 The Rank of a Matrix

The *rank* of a matrix A, denoted $\text{rank}(A)$ or $\text{rg}(A)$, is the dimension of the column space (or equivalently, the row space) of A.

Equivalent definitions of rank include:

- The maximum number of linearly independent columns of A
- The maximum number of linearly independent rows of A
- The order of the largest non-zero minor of A
- The number of non-zero rows in any row echelon form of A

34.9.1 Finding the Rank

To find the rank of a matrix:

1. Transform the matrix into row echelon form using Gauss-Jordan elimination
2. Count the number of non-zero rows in the resulting matrix

34.9.2 Properties of Rank

- *Upper bound of the rank*: $\text{rank}(A) \leq \min(m, n)$ for $A \in \mathbb{R}^{m \times n}$.
- *Rank of the transpose*: $\text{rank}(A^T) = \text{rank}(A)$.
- *Rank of a product*: $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.
- *Rank of a sum*: $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$.

For a square matrix $A \in \mathbb{R}^{n \times n}$, the following are equivalent:

- A is invertible.
- $\text{rank}(A) = n$.
- $\det(A) \neq 0$.
- The columns of A are linearly independent.
- The rows of A are linearly independent.
- $Ax = 0$ has only the trivial solution $x = 0$.

34.10 Column Space

The column space of A, denoted $\text{Col}(A)$, is the span of the columns of A:

$$\text{Col}(A) = \{\vec{y} \in \mathbb{R}^m : \vec{y} = A\vec{x} \text{ for some } \vec{x} \in \mathbb{R}^n\}.$$

This is also called the range or image of the linear transformation represented by A. The dimension of the column space equals the rank of A.

34.11 Row Space

The *row space* of A, denoted $\text{Row}(A)$, is the span of the rows of A. It is also perpendicular to the *null space*.

$$\text{Row}(A) = \text{Col}(A^T)$$

The dimension of the *row space* also equals the rank of A.

34.11.1 Null Space

The *null space* (or *kernel*) of A, denoted $\text{Null}(A)$ or $\ker(A)$ is the set of all vectors that A maps to zero:

$$\text{Null}(A) = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\}$$

The dimension of the null space is called the nullity of A, denoted $\text{nullity}(A)$.

34.11.2 Left Null Space

The *left null space* of A is the null space of A^T:

$$\text{Null}(A^T) = \{\vec{y} \in \mathbb{R}^m : A^T\vec{y} = \vec{0}\} = \{\vec{y} \in \mathbb{R}^m : \vec{y}^T A = \vec{0}^T\}$$

The Rank-Nullity Theorem connects these spaces:

$$\text{rank}(A) + \text{nullity}(A) = n$$

To find a basis for these spaces:

- *Column space*: Take the linearly independent columns of A.
- *Row space*: Take the non-zero rows from any row echelon form of A.
- *Null space*: Solve the homogeneous system $A\vec{x} = \vec{0}$ and express the general solution in terms of free variables.

34.12 Linear Maps as Matrices

Linear maps between vector spaces V and W can be represented by matrices, in the following way:

$$A\vec{x} = \sum_{i=1}^{n} x_i \vec{a}_i$$

where $\vec{a}_i$ are images of the basis vectors of V as columns of A and $\vec{x} = (x_1, \ldots, x_n)^T$.

34.12.1 Matrix Representation of Linear Maps on Basis Vectors

Let V and W be two $\mathbb{F}$-vector spaces, $\beta = \{v_1, \ldots, v_n\}$ a basis of V and $\gamma = \{w_1, \ldots, w_m\}$ a basis of W, and let $f : V \to W$ be linear. Then there exists a unique matrix $A = (a_{ij}) \in \mathbb{F}^{m \times n}$ with

$$f(v_j) = \sum_{i=1}^{m} a_{ij} w_i \quad \forall j = 1, \ldots, n.$$

Example:

Let $V = \mathbb{R}^2$, $W = \mathbb{R}^3$, and define a linear map $f : V \to W$ by:

$$f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + 2y \\ 3x + y \\ 4y \end{bmatrix}$$

We aim to find the matrix representation $[M]_\beta^\gamma(f) \in \mathbb{R}^{3\times 2}$ of f, with respect to the standard bases:

$$\beta = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \subseteq V$$
$$\gamma = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \subseteq W$$

Step 1: Apply f to the basis vectors of V

$$f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 + 2 \cdot 0 \\ 3 \cdot 1 + 0 \\ 4 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$$

$$f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 + 2 \\ 0 + 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$$

Step 2: Form the matrix $[M]_\beta^\gamma(f)$

The columns of the matrix are the coordinate vectors of $f(v_1)$ and $f(v_2)$ in basis γ:

$$[M]_{\beta}^{\gamma}(f) = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 0 & 4 \end{bmatrix}$$

Step 3: Verify the matrix representation

We check that:

$$f(v_j) = \sum_{i=1}^{3} a_{ij} w_i \quad \text{for } j = 1, 2$$

$$f(v_1) = 1w_1 + 3w_2 + 0w_3$$
$$f(v_2) = 2w_1 + 1w_2 + 4w_3$$

This confirms the matrix representation is correct.

34.13 Matrices With Respect to a Basis

Expressing a linear map in terms of a different basis means that, given two vector spaces V and W with the bases $\beta = \{v_1, \ldots, v_n\}$ and $\gamma = \{w_1, \ldots, w_n\}$ respectively, the map $f : V \to W$ with a matrix representation A whose columns are vectors in the basis β can be expressed in terms of the basis γ using the following steps:

1. Determine the images of the basis vectors of V under f, which will give us the matrix A_β, if not already given.
2. For every basis vector w_i of W, apply $A_\beta w_i$. This is going to give you a vector k_i.
3. Express each k_i as a linear combination of the basis vectors of W, which means finding coefficients λ_k such that $\sum_{k=i}^{n} \lambda_k w_k = k_j$. This can be found using Gaussian elimination.
4. Form the matrix A_γ using each collection of coefficients λ_k as columns, where each column corresponds to the image of a basis vector v_j of V.

Another, much faster method to express a linear map in terms of another basis is to use the change of basis formula. If φ_γ is the change of basis matrix from γ to the canonical basis formed by the basis vector of W, and φ_γ^{-1} is the change of basis matrix from the canonical basis to γ, then:

$$A_\gamma = \varphi_\gamma^{-1} A_\beta \varphi_\gamma$$

The intuition behind it is that in the previous method we are solving the system of equations $A_\gamma \vec{x}_i = A_\beta w_i$ for each basis vector w_i in γ, which is equivalent to $\vec{x}_i = \varphi_\gamma^{-1} A_\beta w_i$ when using Gaussian elimination. If you then put all $\vec{x}_i$ in order in a matrix then you get A_γ.

34.14 Example of a Complete Exercise

1. Determination of the kernel
2. Determination of the dimension of the kernel
3. Determination of the rank (dimension formula)
4. Determination of the image

Example:

Given

$$f\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}2x_1 + x_2\\x_1 - x_2 + x_3\\4x_1 - x_2 + 2x_3\end{bmatrix}.$$

It should be shown that f is linear, and $\ker(f)$, $\mathrm{img}(f)$ and their dimensions should be determined.

A direct proof of linearity is easily possible. Instead, we give the transformation matrix A. The images of the (canonical) basis vectors are

$$f\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}2\\1\\4\end{bmatrix}, \quad f\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}1\\-1\\-1\end{bmatrix}, \quad f\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\1\\2\end{bmatrix}.$$

We obtain

$$A = \begin{bmatrix}2 & 1 & 0\\1 & -1 & 1\\4 & -1 & 2\end{bmatrix}.$$

But now it must be shown that indeed $f(x) = Ax \quad \forall x \in \mathbb{R}^3$ holds, by, for example, calculating both Ax and $f(x)$ for a general x and showing equality: Here, with $x = (x_1, x_2, x_3)^T$

$$A \cdot x = \begin{bmatrix}2 & 1 & 0\\1 & -1 & 1\\4 & -1 & 2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}2x_1 + x_2\\x_1 - x_2 + x_3\\4x_1 - x_2 + 2x_3\end{bmatrix}.$$

This obviously corresponds to $f(x)$, so that by Theorem 4.36, the map f is linear. To determine the kernel, one has to solve the system of linear equations

$$\begin{array}{cccc}2 & 1 & 0 & 0\\1 & -1 & 1 & 0\\4 & -1 & 2 & 0\end{array}$$

Which corresponds to the equation $Ax = f(x) = 0$. Gaussian elimination yields $x_3 = \lambda'$; $x_2 = \frac{2}{3} \cdot \lambda'$; $x_1 = -\frac{1}{3} \cdot \lambda'$, thus, with $\lambda = \frac{1}{3}\lambda'$:

$$\ker(f) = \left\{ x = \lambda \begin{bmatrix}-1\\2\\3\end{bmatrix} \middle| \lambda \in \mathbb{R} \right\}$$

It follows that $\dim(\ker(f)) = 1$ and because of $\dim(V) = 3$ from the dimension formula $\dim(\mathrm{img}(f)) = 2$.

$\mathrm{img}(f)$ corresponds to the linear span of the columns of the matrix. One chooses consequently $\dim(\mathrm{img}(f))$ column vectors, e.g., the first ones, and tests if they are linearly independent. In the concrete case, this is obvious, because the second column is not a multiple of the first. It follows therefore,

$$\mathrm{img}(f) = \left\{ x \middle| x = \lambda \begin{bmatrix}2\\1\\4\end{bmatrix} + \mu \begin{bmatrix}1\\-1\\-1\end{bmatrix}, \quad \lambda, \mu \in \mathbb{R} \right\}$$

34.15 Diagonal and Triangular Matrices

A matrix is called *Diagonal* if the only non-zero elements are the elements on the main diagonal. A matrix is *triangular* if all elements under or above the main diagonal are zero.

34.15.1 Properties of Diagonal and Triangular Matrices

- *Determinant*: $\det(A) = \prod_{i=1}^{n} a_{ii}$.
- *Eigenvalues*: $\text{eig}(A) = \{a_{1,1}, a_{2,2}, \ldots, a_{n,n}\}$.

34.16 Symmetric Matrix

A matrix $A \in \mathbb{R}^{n \times n}$ is symmetric if

$$A^T = A.$$

34.16.1 Properties of Symmetric Matrices

- Given $A \in \mathbb{R}^{n \times m}$, AA^T and $A^T A$ are symmetric matrices.
- For all symmetric matrices $x^T Ax > 0$.
- All eigenvalues are real.
- The eigenvectors are orthogonal.

Proof 1:

Let $A \in \mathbb{R}^{n \times m}$, by the definition of matrix multiplication $AA^T \in \mathbb{R}^{n \times n}$ and $A^T A \in \mathbb{R}^{m \times m}$, hence a transpose can be taken.

Then

$$(AA^T)^T = A^{T^T} A^T = AA^T, \quad (A^T A)^T = A^T A$$

$\mathfrak{QED}$

34.17 Derivative of a Matrix

Given a matrix $A \in \mathbb{C}^{n \times n}$ and a vector $x \in \mathbb{C}^n$ we can find the derivatives of

$$\frac{\partial (Ax)}{\partial x} = A$$

and

$$\frac{\partial (x^T Ax)}{\partial x} = (A + A^T)x$$

Proof:

For Ax, we have that

$$Ax = \begin{bmatrix} \sum_{i=1}^{n} a_{1i} x_i \\ \sum_{i=1}^{n} a_{2i} x_i \\ \vdots \\ \sum_{i=1}^{n} a_{ni} x_i \end{bmatrix} = \begin{bmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_n(x) \end{bmatrix} \in \mathbb{C}^n$$

Hence, the derivative of Ax with respect to x is given by the matrix A, since when taking the partial derivatives of each $f_i(x)$ with respect to each x_j, we obtain the entries of A.

$$\frac{\partial f_i(x)}{\partial x_j} = a_{ij}$$

Hence,

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \dots & a_{n,n} \end{bmatrix}$$

For $x^T Ax$, we take our vector

$$x^T Ax = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} \sum_{i=1}^{n} a_{1,i}x_i \\ \sum_{i=1}^{n} a_{2,i}x_i \\ \vdots \\ \sum_{i=1}^{n} a_{n,i}x_i \end{bmatrix} = x_1 \sum_{i=1}^{n} a_{1,i}x_i + x_2 \sum_{i=1}^{n} a_{2,i}x_i + \cdots + x_n \sum_{i=1}^{n} a_{n,i}x_i$$

This can be rewritten as

$$\sum_{i=1}^{n} a_{1,i}x_1x_i + \sum_{i=1}^{n} a_{2,i}x_2x_i + \cdots + \sum_{i=1}^{n} a_{n,i}x_nx_i$$

which can be expressed as a double summation:

$$f(x) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{i,j}x_ix_j.$$

When taking the derivative of $f(x)$ with respect to x_k, we have 4 cases to consider:

- $i = k$, $j = k$: is $x_kx_ka_{k,k}$ and the derivative with respect to x_k is $2x_ka_{k,k}$.
- $i = k$, $j \neq k$: is $x_kx_ja_{k,j}$ and the derivative with respect to x_k is $a_{k,j}x_j$.
- $i \neq k$, $j = k$: is $x_ix_ka_{i,k}$ and the derivative with respect to x_k is $a_{i,k}x_i$.
- $i \neq k$, $j \neq k$: is $x_ix_ja_{i,j}$ and does not depend on x_k, so the derivative is 0.

This gives us the following form for the components of our gradient:

$$\frac{\partial f(x)}{\partial x_k} = 2x_ka_{k,k} + \sum_{i\neq k}^{n}(a_{i,k} + a_{k,i})x_i.$$

When written in matrix form, this yields

$$(A + A^T)x$$

because the diagonal elements are counted twice and the off-diagonal elements are added in the form $a_{i,j} + a_{j,i}$. If A is symmetric, then $A = A^T$ and we have

$$2Ax.$$

$\mathfrak{QED}$

Example:

For $n = 2$:

$$x^T Ax = x_1^2a_{1,1} + x_1x_2a_{1,2} + x_1x_2a_{2,1} + x_2^2a_{2,2}$$

Differentiate and get:

$$\nabla x^T Ax = \begin{bmatrix} 2x_1 a_{1,1} + x_2(a_{1,2} + a_{2,1}) \\ x_1(a_{1,2} + a_{2,1}) + 2x_2 a_{2,2} \end{bmatrix} = \begin{bmatrix} 2a_{1,1} & (a_{1,2} + a_{2,1}) \\ (a_{1,2} + a_{2,1}) & 2a_{2,2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Chapter 35

Metrics

35.1 Metrics for Vectors

The metric of a vector a is described by different norms:

It also has the following properties:

- $\|a\| \geq 0$ and $\|a\| = 0 \Leftrightarrow a = 0$
- $\|\lambda a\| = |\lambda| \, \|a\|$
- $\|a + b\| \leq \|a\| + \|b\|$ (Triangle inequality)
- $\|a + b\|^2 = \|a\|^2 + \|b\|^2 + 2\langle a, b\rangle$ (Pythagorean theorem)

These are the most common metrics, although we will primarily be using the Euclidean metric:

- *Euclidean metric*:
$$\|a\| = \sqrt{\langle a, a\rangle} = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$$
- *Manhattan metric*:
$$\|a\|_1 = |a_1| + |a_2| + \cdots + |a_n|$$
- *Maximum metric*:
$$\|a\|_\infty = \max(|a_1|, |a_2|, \ldots, |a_n|)$$

35.2 Metric for Matrices

Given a metric $\|\cdot\|$ over $\mathbb{R}^n$, $A \in \mathbb{R}^{n\times n}$, we define

$$\|A\| := \sup_{x\neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\| \,,$$

as the *matrix metric* for $\mathbb{R}^{n\times n}$.

- This norm is dependent on the chosen vector metric, hence it is called an *induced matrix metric*.
- The vector metric and the induced metric norm are compatible, which means:
$$\|Ax\|_{\text{vector}} \leq \|A\|_{\text{matrix}} \cdot \|x\|_{\text{vector}} \,.$$
- For matrix metrics the following holds:
$$\|AB\| \leq \|A\| \cdot \|B\| \,.$$
- For $\|\mathbb{R}^{n\times n}\|$, all matrix metrics are equivalent.

35.2.1 Spectral Metric

For $\|\cdot\|_2$ there is the so-called *spectral metric*.

$$\|A\|_2 = \sqrt{\rho(A^T A)},$$

where the *spectral radius* $\rho(B)$ is the eigenvalue of B with the greatest absolute value.

35.2.2 Row Metric

For $\|\cdot\|_\infty$ there is the *row-sum metric*

$$\|A\|_\infty = \max_{i=1,\ldots,n} \sum_{j=1}^{n} |a_{ij}|.$$

35.2.3 Column Metric

For $\|\cdot\|_1$ we get the *column-sum metric*

$$\|A\|_1 = \max_{j=1,\ldots,n} \sum_{i=1}^{n} |a_{ij}|.$$

35.2.4 Frobenius Metric

The *Frobenius metric* given by

$$\|A\|_F = \sqrt{\mathrm{dig}(A^T A)}$$

is not induced, but the following holds

$$\|AB\|_F \leq \|A\|_F \cdot \|B\|_F,$$

and, due to

$$\|Ax\|_2 \leq \|A\|_F \cdot \|x\|_2, \forall x$$

it is compatible with the vector metric $\|\cdot\|_2$. More precisely, $\|A\|_F$ is an upper bound for $\|A\|_2$.

Chapter 36

Linear Maps

A *linear map* $f : V \to W$ is a function that satisfies the following properties:

- $f(v_1 + v_2) = f(v_1) + f(v_2)$ for all $v_1, v_2 \in V$.
- $f(\lambda v) = \lambda f(v)$ for all $v \in V$ and $\lambda \in \mathbb{F}$.
- $f(0) = 0$.

This kind of function is also called a *homomorphism*. If $V = W$, it is called an *endomorphism*.

36.1 Types of Linear Maps

- *Injective or Monomorphism* (One-to-One): A linear map $f : V \to W$ is injective if $f(v_1) = f(v_2)$ implies $v_1 = v_2$.
- *Surjective or Epimorphism* (Onto): A linear map $f : V \to W$ is surjective if for every $w \in W$, there exists a $v \in V$ such that $f(v) = w$.
- *Bijective or Isomorphism*: A linear map $f : V \to W$ is bijective if it is both injective and surjective.

Two vector spaces are *isomorphic* if there exists a bijective linear map between them. In this case, we can say that the two vector spaces are *isomorphic*, and we write $V \cong W$.

36.2 Properties of Linear Maps

- The composition of two linear maps is a linear map.
- The inverse of a bijective linear map is also a linear map.
- The zero map $f : V \to W$ defined by $f(v) = 0$ for all $v \in V$ is a linear map.
- The identity map $\mathrm{id}_V : V \to V$ defined by $\mathrm{id}_V(v) = v$ for all $v \in V$ is a linear map.
- The sum of two linear maps $f : V \to W$ and $g : V \to W$ is a linear map defined by $(f + g)(v) = f(v) + g(v)$.
- The scalar multiplication of a linear map $f : V \to W$ by a scalar c is a linear map defined by $(cf)(v) = c(f(v))$.
- The composition of linear maps is associative, i.e., $(f \circ g) \circ h = f \circ (g \circ h)$.
- The composition of linear maps is distributive over addition, i.e., $f \circ (g + h) = f \circ g + f \circ h$.
- The composition of linear maps is compatible with scalar multiplication, i.e., $(cf) \circ g = c(f \circ g)$.
- Linear maps form a vector space over the field of scalars.

$$(Hom(V, W), +, \cdot)$$

36.3 Composition as Matrix Multiplication

Given $A \in \mathbb{F}^{m\times n}, B \in \mathbb{F}^{n\times l}$ and $\phi_A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{m\times 1}$ and $\phi_B : \mathbb{F}^{l\times 1} \to \mathbb{F}^{n\times 1}$ be the induced linear maps, then

- $$\phi_A \circ \phi_B \text{ is a linear map}$$
- $$\phi_A \circ \phi_B = \phi_{AB}$$

Proof I:

$$\begin{aligned}(\varphi_A \circ \varphi_B)(au + bv) &= \varphi_A(\varphi_B(au + bv)) \\ &= \varphi_A(a\,\varphi_B(u) + b\,\varphi_B(v)) \\ &= a\,\varphi_A(\varphi_B(u)) + b\,\varphi_A(\varphi_B(v)) \\ &= a(\varphi_A \circ \varphi_B)(u) + b(\varphi_A \circ \varphi_B)(v)\end{aligned}$$

$\mathfrak{QED}$

Proof II:

$$\begin{aligned}(\varphi_A \circ \varphi_B)(x) &= \varphi_A(\varphi_B(x)) \\ &= \varphi_A(Bx) \\ &= A(Bx) \\ &= (AB)x \\ &= \varphi_{AB}(x)\end{aligned}$$

$\mathfrak{QED}$

36.4 Injectivity and Surjectivity

Let $A \in \mathbb{F}^{m\times n}$. Then:

- φ_A is injective if and only if the fiber of φ_A over the zero column $0_m \in \mathbb{F}^{m\times 1}$ consists only of the zero column $0_n \in \mathbb{F}^{n\times 1}$.
- φ_A is surjective if and only if every column in $\mathbb{F}^{m\times 1}$ can be written as a linear combination of the columns of A.

36.5 Linear Extension Theorem

Let $\beta = \{v_1, \ldots, v_n\}$ be a basis of the finite dimensional vector space V and $w_1, \ldots, w_n$ be arbitrary vectors in W. Then $!\exists L$ with $L : V \to W$ such that $T(v_i) = w_i$.

Proof:

Define T as follows: Whenever $a = \lambda_1 v_1 + \cdots + \lambda_n v_n$ then $T(a) = \lambda_1 w_1 + \cdots + \lambda_n w_n$.

We know it is unique because for $\lambda_1, \ldots, \lambda_n \neq \gamma_1, \ldots, \gamma_n$ if we assume $a - b = \vec{0}$ then:

$$a - b = (\lambda_1 - \gamma_1)v_1 + \cdots + (\lambda_n - \gamma_n)v_n = \vec{0}$$

If and only if $\lambda_1, \ldots, \lambda_n = \gamma_1, \ldots, \gamma_n$. Thus, it works uniquely for each vector.

For the linearity, using the definition of linearity and T this part becomes trivial.

Finally, for the uniqueness of T itself, let us assume there is a map U which $U(v_i) = T(v_i)$. Because we can write a vector a linear combination we will arrive at the fact that

$$U(x) = \lambda_1 U(v_1) + \cdots + \lambda_n U(v_n) = \lambda_1 T(v_1) + \cdots + \lambda_n T(v_n)$$

Therefore, $U = T$.

$\mathfrak{QED}$

36.6 The Kernel of a Linear Map

The *kernel* of a linear map $f : V \to W$ is the set of all vectors in V that are mapped to the zero vector in W:

$$\ker(f) = \{v \in V \mid f(v) = 0\}$$

and also $\ker(f) \subseteq V$.

To compute the *kernel* of the linear map, we just solve the homogeneous system of equations $Ax = \vec{0}$ where A is the matrix representation of f. The number of free variables is the $\dim(\ker(f))$.

Proof:

$\ker(f) \neq \emptyset$ because $0 \in \ker(f)$ (at least zero should be in the kernel by the definition of the vector space and linear maps). Also, for $v, w \in \ker(A)$ we know that $f(av + bw) = af(v) + bf(w) = a0 + b0 = 0$.

$\mathfrak{QED}$

36.7 The Nullity

The *nullity* of a linear map $f : V \to W$ is the dimension of its kernel:

$$\text{nullity}(f) = \dim(\ker(f))$$

36.8 The Image of a Linear Map

The *image* of a linear map $f : V \to W$ is the set of all vectors in W that can be expressed as $f(v)$ for some $v \in V$:

$$\text{img}(f) = \{w \in W \mid w = f(v) \text{ for some } v \in V\}.$$

Also, $\text{img}(f) \subseteq W$.

To find $\dim(\text{im}(f))$, compute the rank of the matrix representing f. This is the number of pivot columns in the row-reduced form of A.

Proof:

$f(0) \in \text{img}(f) \implies \text{img}(f) \neq \emptyset$. Therefore, for $v_1, v_2 \in V, \exists w_1, w_2 \in \text{img}(f)$ with $w_1 = f(v_1)$ and $w_2 = f(v_2)$ due to

$$aw_1 + bw_2 = af(v_1) = bf(v_2) = f(av_1 + bv_2) \in \text{img}(f).$$

$\mathfrak{QED}$

36.9 The Rank of a Linear Map

The *rank* of a linear map $f : V \to W$ is the dimension of its image.

$$\text{rank}(f) = \dim(\text{img}(f))$$

36.10 The Nullity of a Linear Map

The *nullity* of a linear map $f : V \to W$ is the dimension of its kernel:

$$\text{nullity}(f) = \dim(\ker(f)).$$

36.11 Proof of the Injectivity of the Kernel

Suppose we have a linear map f and two vectors v_1 and v_2 which are not equal. We are going to assume that they both map to the $\vec{0}$; therefore,

$$f(v_1) = f(v_2) = \vec{0}.$$

Because of the linearity of the map we can write

$$f(v_1 - v_2) = f(v_1) - f(v_2) = \vec{0} - \vec{0} = \vec{0}.$$

And that contradicts the assumption that v_1 and v_2 are not equal. Therefore, the kernel of a linear map is injective.

36.12 Dimension Formula for Linear Mappings

Let $f : V \to W$ be linear and $\dim(V) = n$. Then

$$\dim(\ker(f)) + \text{rank}(f) = n.$$

Remember that $\ker(f)$ forms a subspace of V and therefore, $\dim(\ker(f)) := r \leq n$. We extend an arbitrary basis $(v_1, \ldots, v_r)$ of $\ker(f)$ to a basis $(v_1, \ldots, v_r, v_{r+1}, \ldots, v_n)$ of V. Setting $w_{r+i} = f(v_{r+i})$ for $i = 1, \ldots, n-r$, we have $\forall v \in V$:

$$\begin{aligned} f(v) &= f(\lambda_1 v_1 + \cdots + \lambda_r v_r + \lambda_{r+1} v_{r+1} + \cdots + \lambda_n v_n) \\ &= \lambda_1 \underbrace{f(v_1)}_{=0} + \cdots + \lambda_r \underbrace{f(v_r)}_{=0} + \lambda_{r+1} f(v_{r+1}) + \cdots + \lambda_n f(v_n) \\ &= \lambda_{r+1} f(v_{r+1}) + \cdots + \lambda_n f(v_n) \\ &= \lambda_{r+1} w_{r+1} + \cdots + \lambda_n w_n \end{aligned}$$

Thus, $\text{img}(f) = \text{span}(w_{r+1}, \ldots, w_n)$. We now show that $w_{r+1}, \ldots, w_n$ are linearly independent. Let

$$\lambda_{r+1} w_{r+1} + \cdots + \lambda_n w_n = 0.$$

From

$$0 = \lambda_{r+1} w_{r+1} + \cdots + \lambda_n w_n = f(\lambda_{r+1} v_{r+1} + \cdots + \lambda_n v_n),$$

it follows that

$$\lambda_{r+1}v_{r+1} + \cdots + \lambda_n v_n \in \ker(f).$$

Therefore,

$$\lambda_{r+1}v_{r+1} + \cdots + \lambda_n v_n = \lambda_1 v_1 + \cdots + \lambda_r v_r$$

for some $\lambda_1, \ldots, \lambda_r$. Since $v_1, \ldots, v_n$ are linearly independent, we have $\lambda_1 = \cdots = \lambda_n = 0$. Thus, the vectors $w_{r+1}, \ldots, w_n$ are linearly independent.

It follows that $\dim(\text{img}(f)) = \text{rank}(f) = n - r$ and therefore,

$$\dim(\ker(f)) + \dim(\text{img}(f)) = \dim(\ker(f)) + \text{rank}(f) = r + n - r = n$$

$\mathfrak{QED}$

36.13 Linear Maps and Matrices

Let V and W be finite-dimensional vector spaces over the same field $\mathbb{F}$. If $\dim V = n$ and $\dim W = m$, then a linear map $f : V \to W$ can be represented by an $m \times n$ matrix. The action of the linear map on a vector can be expressed as:

$$f(x) = Ax,$$

where A is the matrix representation of f and v is the vector represented in a column format. The columns of the matrix A are the images of the basis vectors of V under the linear map f.

This fact comes from f being linear which means that

$$f(x) = x_1 f(e_1) + \cdots + x_n f(e_n)$$

with e_i being a basis vector. And from that expression we can derive $(f(e_1), \ldots, f(e_n))x$.

We can state that matrices *induce* linear maps, which we denote as φ_A.

36.14 How to Identify the Type of Linear Map

We want to summarize how to identify the type of linear map given $f : V \to W$,

- *Injective*: A linear map f is injective if and only if

$$\dim(\ker(f)) = 0.$$

- *Surjective*: The map f is surjective if every element of W is the image of some element in V. This occurs when

$$\dim(\text{img}(f)) = \dim(W).$$

  That is, the image of f consists of the linearly independent vectors of f.

- *Bijective*: A linear map is bijective if it is both injective and surjective. Therefore, it must satisfy both conditions:

$$\dim(\ker(f)) = 0 \quad \text{and} \quad \dim(\text{im}(f)) = \dim(W).$$

- *Neither*: If neither of the above conditions is satisfied, the map is neither injective nor surjective.

Example:

Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \end{bmatrix}.$$

To identify the type of this linear map:

- Compute $\dim(\ker(f))$: By reducing the matrix A to row echelon form, we find that $\dim(\ker(f)) = 1$. This means f is not injective.
- Compute $\dim(\text{im}(f))$: The image corresponds to the column space of A. Since there are 2 linearly independent rows in the row echelon form, we have $\dim(\text{im}(f)) = 2$, which equals $\dim(W) = 2$. Therefore, f is surjective.

Thus, f is surjective but not injective.

Chapter 37

The Change of Basis

Imagine you have a vector $\vec{\lambda}$ in a vector space V with basis B_v, and you also have another basis B_w for the same vector space.

The basis B_w is written in terms of the basis B_v as usual. The question is, given a vector written in the basis B_v, how is this vector expressed in the basis B_w? Note, that simply taking $B_w\vec{v}$ will not give you the correct answer, as it will give another vector.

The key to getting the correct answer is to frame the question as: which vector $\vec{\mu}$, when the transformation B_w is applied to it, will give us the vector $\vec{v}$? To answer this question, we need to take the inverse of B_w and apply it to $\vec{\lambda}$. This will give us the vector $\vec{\mu}$, which is the representation of $\vec{v}$ in the basis B_w, since when we apply B_w to $\vec{\mu}$, we will get back the vector $\vec{\lambda}$.

Using the information above, we can write the following formula for translating between bases:

$$B_v\vec{\lambda}_v = B_w\vec{\mu}_w.$$

37.1 Coordinate Mapping

Let V be a $\mathbb{F}$-vector space with a basis $\beta = \{v_1, \ldots, v_n\}$. Then there exists exactly one isomorphism

$$\varphi_\beta : \mathbb{F}^n \to V$$

such that

$$\varphi_\beta(e_i) = v_i, \quad \text{for } 1 \leq i \leq n,$$

which is called the *coordinate mapping* of V with respect to the basis β.

37.2 Coordinates With Respect to a Basis

The isomorphism φ_β from the previous theorem is called the *coordinate mapping*, and for $v \in V$, we define

$$(\vec{v})_\beta := \varphi_\beta^{-1}(v) \in \mathbb{F}^n$$

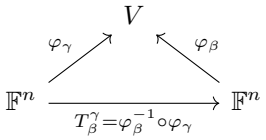
as the *coordinates of v with respect to β*.

37.3 Translation of Coordinates

We define the *Translation* of a vector to another basis as:

$$T_\beta^\gamma = \varphi_\beta^{-1} \circ \varphi_\gamma$$

where φ_β and φ_γ are the matrices made of the basis vectors of the bases β and γ with respect to the standard basis, respectively.



37.4 Change of Basis Theorem

Let $v \in V$ be arbitrary, with $(\vec{v})_\gamma = (x_1, \ldots, x_n)^T$ and $(\vec{v})_\beta = (y_1, \ldots, y_n)^T$. Then the following holds:

$$(\vec{v})_\beta = \varphi_\beta^{-1} \circ \varphi_\gamma (\vec{v})_\gamma.$$

In vector notation with the translation:

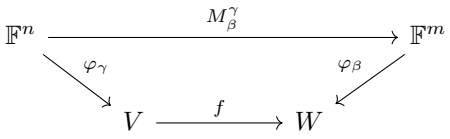
$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = T_\beta^A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

37.5 Change of Basis in Linear Maps

Let V and W be finitely generated $\mathbb{F}$-vector spaces with bases γ and β, respectively, and let $f \in \mathrm{Hom}(V, W)$. Then we have

$$M_\beta^\gamma(f) = \varphi_\beta^{-1} \circ f \circ \varphi_\gamma.$$

Thus, $M_\beta^\gamma(f)$ tells you to take a vector in the basis γ, translate it to the canonical basis, apply the linear map f, and then convert the result to the basis β.



37.6 Transformations Involving Multiple Bases

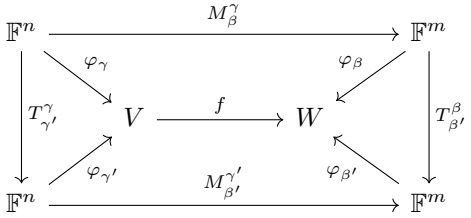
Let V and W be finitely generated vector spaces with bases γ and γ' for V, and β and β' for W, respectively. Furthermore, let $f : V \to W$ be a linear map. Then the following holds:

$$M_{\beta'}^{\gamma'}(f) = T_{\beta'}^{\beta} \cdot M_\beta^\gamma(f) \cdot \left(T_{\gamma'}^{\gamma}\right)^{-1}$$

More explicitly, we can write this as:

$$\begin{aligned} M_{\beta'}^{\gamma'}(f) &= T_{\beta'}^{\beta} \cdot M_{\beta}^{\gamma}(f) \cdot \left(T_{\gamma'}^{\gamma}\right)^{-1} \\ &= (\varphi_{\beta'}^{-1} \circ \varphi_{\beta}) \circ (\varphi_{\beta}^{-1} \circ f \circ \varphi_{\gamma}) \circ (\varphi_{\gamma'}^{-1} \circ \varphi_{\gamma})^{-1}. \\ &= (\varphi_{\beta'}^{-1} \circ \varphi_{\beta}) \circ (\varphi_{\beta}^{-1} \circ f \circ \varphi_{\gamma}) \circ (\varphi_{\gamma}^{-1} \circ \varphi_{\gamma'}). \end{aligned}$$

We can visualize this as a commutative diagram where, when we take an inverse, we reverse the direction of the arrows:



Chapter 38

The Determinant of a Matrix

The *determinant* of a matrix $A \in \mathbb{R}^{n \times n}$, denoted $\det(A)$ or $|A|$, is a scalar value that provides important information about the matrix, including whether it is invertible and the volume scaling factor of the linear transformation represented by A.

38.1 Determinant Formulas for $\mathbb{R}^{2\times2}, \mathbb{R}^{3\times3}$

The determinant of a 2×2 matrix is given by:

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

For a 3×3 matrix, the determinant can be computed using the rule of Sarrus or the co-factor expansion:

$$\det(A) = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

The determinant of larger matrices can be computed using co-factor expansion along any row or column:

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(A_{ij}),$$

where A_{ij} is the $(n-1) \times (n-1)$ sub-matrix obtained by deleting the i-th row and j-th column of A.

38.2 Properties of the Determinant

- $\det(A) = 0$ if and only if A is singular (not invertible).
- $\det(AB) = \det(A) \cdot \det(B)$ for any square matrices A and B of the same size.
- $\det(A^T) = \det(A)$.
- If a row (or column) of A is multiplied by a scalar α, then $\det(A)$ is multiplied by α.
- If two rows (or columns) of A are swapped, then $\det(A)$ changes sign.

$$\det(v_1, v_2, v_3, \ldots, v_n) = -\det(v2, v3, v1, \ldots, v_n)$$

- If a row (or column) of A is added to another row (or column), then $\det(A)$ remains unchanged.
- If one of the columns is a linear combination of the others, then $\det(A) = 0$.

- The determinant of the identity matrix I_n is 1.
- The determinant can be split into the sum of more determinants:

$$\det(a, b, c + d) = \det(a, b, c) + \det(a, b, d)$$

- The determinant of a diagonal matrix is the product of its diagonal entries.
- For elementary matrices $\det(C1) = 1$, $\det(C2) = -1$, $\det(C3) = \lambda$.
- For $A \in \mathbb{R}^{n\times n}$, $\det(\lambda A) = \lambda^n \det(A)$.

38.2.1 Proof of the Multiplication of Determinants

For $A, B \in \mathbb{F}^{n\times n}$, it holds that

$$\det(AB) = \det(A)\det(B).$$

Proof:

If $\det(A) = 0$ or $\det(B) = 0$, then $\operatorname{rank}(L_A) < n$ or $\operatorname{rank}(L_B) < n$.

Thus, $\operatorname{rank}(L_{AB}) < n$, and the statement is trivial.

So let us assume that A and B are invertible, i.e., $\operatorname{rank}(A) = \operatorname{rank}(B) = n$.

We now assume that in such a case, a matrix can always be transformed into reduced row echelon form using the row operations from Gaussian elimination.

Then, $A = C_k \cdot \dots \cdot C_1$ for certain elementary matrices C_i, and therefore:

$$\begin{aligned}
\det(AB) &= \det(C_k \cdot \dots \cdot C_1 B) \\
&= \det(C_k)\det(C_{k-1} \cdot \dots \cdot C_1 B) \\
&\;\;\vdots \\
&= \det(C_k) \cdots \det(C_1)\det(B) \\
&= \det(C_k \cdot \dots \cdot C_1)\det(B) \\
&= \det(A)\det(B).
\end{aligned}$$

Therefore, also if A is invertible, then

$$\det(A^{-1}) = (\det(A))^{-1}.$$

It holds that

$$\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det(E) = 1.$$

QED

38.3 The Leibniz Formula

Let $A = (a_{ij}) \in \mathbb{F}^{n\times n}$ be a square matrix over a field K. The determinant of A is defined via the *Leibniz formula* as:

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)},$$

where:

- S_n is the set of all permutations of $\{1, 2, \ldots, n\}$.
- $\operatorname{sgn}(\sigma)$ is the sign of the permutation σ, equal to $+1$ for even permutations and -1 for odd ones.

Derivation

Step 1: Expansion using multilinearity

Each column vector a_j of the matrix A can be written as a linear combination of the standard basis vectors $e_1, \ldots, e_n$:

$$a_j = \sum_{i=1}^{n} a_{ij} e_i.$$

Using the multilinearity of the determinant function (i.e., linearity in each column), we expand:

$$\det(a_1, a_2, \ldots, a_n) = \det\left(\sum_{i_1} a_{i_1 1} e_{i_1}, \sum_{i_2} a_{i_2 2} e_{i_2}, \ldots, \sum_{i_n} a_{i_n n} e_{i_n}\right).$$

By multilinearity, this becomes a sum over all n-tuples $(i_1, i_2, \ldots, i_n)$:

$$= \sum_{i_1, i_2, \ldots, i_n = 1}^{n} a_{i_1 1} a_{i_2 2} \cdots a_{i_n n} \cdot \det(e_{i_1}, e_{i_2}, \ldots, e_{i_n}).$$

Step 2: Eliminate repeated indices

If two indices $i_j = i_k$ for $j \neq k$, then the corresponding determinant is zero (because the determinant is alternating and thus zero for repeated columns). Therefore, only terms with pairwise distinct indices remain. These correspond to permutations of $\{1, 2, \ldots, n\}$.

Let $\sigma \in S_n$ be such a permutation. Then we can write:

$$\det(A) = \sum_{\sigma \in S_n} a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n} \cdot \det(e_{\sigma(1)}, e_{\sigma(2)}, \ldots, e_{\sigma(n)}).$$

Step 3: Sign of the permutation

Since the determinant of $(e_{\sigma(1)}, \ldots, e_{\sigma(n)})$ is $\operatorname{sgn}(\sigma)$, we obtain:

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n}.$$

Equivalently, by re-indexing:

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot \prod_{i=1}^{n} a_{i\sigma(i)}.$$

This formula expresses the determinant as a sum over all permutations of the indices $\{1, \ldots, n\}$, where each term is a product of matrix entries taken from different rows and columns, weighted by the sign of the corresponding permutation.

38.4 Laplace's Method (Co-Factor Expansion)

The determinant of an $n \times n$ matrix $A = [a_{ij}]$ can be computed using the *Laplace expansion*, also known as *co-factor expansion*. This method expands the determinant along a specific row or column.

Expansion Along Row i:

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(M_{ij})$$

Expansion Along Column j:

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det(M_{ij})$$

where:

- a_{ij} is the element in the i-th row and j-th column of matrix A.
- M_{ij} is the *minor* of a_{ij}, i.e., the determinant of the submatrix formed by removing the i-th row and j-th column from A.
- $(-1)^{i+j}$ is the *sign factor*, determining the sign of each co-factor term.

Steps:

1. **Choose a Row or Column:** Select any row or column of the matrix. It's often easiest to choose one with many zeros.
2. **For Each Element:** For each element, a_{ij}, in the chosen row or column:

   Find the Minor, M_{ij}: The minor M_{ij} is the determinant of the sub-matrix formed by deleting the i-th row and the j-th column of the original matrix.

   Find the Co-factor, C_{ij}: The co-factor C_{ij} is the minor multiplied by a sign factor:

   $$C_{ij} = (-1)^{i+j} M_{ij}$$

   The term $(-1)^{i+j}$ gives a checkerboard pattern of signs:

   $$\begin{bmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

   Calculate the Determinant: The determinant of the matrix, A, is the sum of the products of the elements in the chosen row or column and their corresponding co-factors.
3. **Expansion along the i-th row:**

   $$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} = a_{i1} C_{i1} + a_{i2} C_{i2} + \cdots + a_{in} C_{in}$$

   Expansion along the j-th column:

   $$\det(A) = \sum_{i=1}^{n} a_{ij} C_{ij} = a_{1j} C_{1j} + a_{2j} C_{2j} + \cdots + a_{nj} C_{nj}$$

Both expansions give the same result.

Example (3×3 Matrix):

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Expanding along the first row:

1. a_{11}: $M_{11} = \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$, $C_{11} = +M_{11}$
2. a_{12}: $M_{12} = \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}$, $C_{12} = -M_{12}$
3. a_{13}: $M_{13} = \det \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$, $C_{13} = +M_{13}$

Therefore,

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

Example of Determinant Calculation

Let's calculate the determinant of the matrix:

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

Using the rule of Sarrus for 3×3 matrices:

$$\begin{aligned} \det(A) &= 2 \cdot 0 \cdot 1 + 1 \cdot 2 \cdot 3 + 3 \cdot 1 \cdot 1 - (3 \cdot 0 \cdot 0 + 1 \cdot 2 \cdot 2 + 2 \cdot 1 \cdot 1) \\ &= 0 + 6 + 3 - (0 + 4 + 2) \\ &= 9 - 6 = 3 \end{aligned}$$

Thus, the determinant of matrix A is $\det(A) = 3$.

Now consider the following 4×4 matrix:

$$A = \begin{bmatrix} 2 & 1 & 3 & 2 \\ 4 & 0 & -1 & 3 \\ -2 & 3 & 1 & 5 \\ 1 & -1 & 0 & 2 \end{bmatrix}$$

For a 4×4 matrix, we can use Laplace Method along the first row:

$$\begin{aligned} \det(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + a_{14}C_{14} \\ &= 2 \cdot \det \begin{bmatrix} 0 & -1 & 3 \\ 3 & 1 & 5 \\ -1 & 0 & 2 \end{bmatrix} - 1 \cdot \det \begin{bmatrix} 4 & -1 & 3 \\ -2 & 1 & 5 \\ 1 & 0 & 2 \end{bmatrix} \\ &\quad + 3 \cdot \det \begin{bmatrix} 4 & 0 & 3 \\ -2 & 3 & 5 \\ 1 & -1 & 2 \end{bmatrix} - 2 \cdot \det \begin{bmatrix} 4 & 0 & -1 \\ -2 & 3 & 1 \\ 1 & -1 & 0 \end{bmatrix} \end{aligned}$$

$$\det(A) = 145$$

38.5 Determinant of the Reduced Echelon Form

The determinant has another trait that states that if your matrix A is *REF*, then its determinant is given by the product of the main diagonal.

$$\det(A) = \prod_{i=1}^{n} a_{i,i}$$

38.6 Gaussian Method for Finding Determinants

Another method for finding determinants uses the row operations we already know plus column operations, but with the catch that for the column case we have to take into consideration how operating with columns affects the determinant, in contrast to the row operations. The exception is scaling a row by some λ; in this case, we would need to undo the change by multiplying the determinant by $\frac{1}{\lambda}$.

Example:

We are given the matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

We apply Gaussian elimination:

Swap $R_2 \leftrightarrow R_3$:

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Now the matrix is upper triangular. The determinant is the product of diagonal elements times -1 for one row swap:

$$\det(A) = (-1)^1 \cdot (1) \cdot (-1) \cdot (-1) \cdot (3) = -3$$

$$\det(A) = -3$$

Chapter 39

Elementary Matrix

We are going to define 3 matrices that can be obtained from the Elementary Row Operations on the *Identity Matrix* and call them *Elementary Matrices*.

39.1 C1: Row Addition

$$C_1 := \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & \lambda & 1 & \\ & & & & \ddots \end{bmatrix} \in \mathbb{F}^{n\times n} \quad \text{or} \quad C_1 := \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & \lambda & \\ & & & 1 & \\ & & & & \ddots \end{bmatrix} \in \mathbb{F}^{n\times n}$$

39.2 C2: Row Swap

$$C_2 := \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 0 & 1 & & \\ & & 1 & 0 & & \\ & & & & \ddots & \\ & & & & & 1 \end{bmatrix} \in \mathbb{F}^{n\times n}$$

39.3 C3: Row Scaling

$$C_3 := \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix} \in \mathbb{F}^{n\times n}$$

Each of them also encodes the row operations from which they came. So now Gaussian elimination can be encoded as matrix multiplication.

Formally,

$$C_1 := \mathrm{Add}_m(i,j,\lambda) := I_m + ae_if_j : m \times m \to \mathbb{F} : (s,t) \mapsto \begin{cases} 1, & s = t \\ \lambda, & (s,t) = (i,j) \\ 0, & \text{else} \end{cases}$$

$$C_2 := \text{Swap}_m(i,j) := \\ (e_1,\ldots,e_{i-1},e_j,e_{j+1},\ldots,e_{j-1},e_i,e_{j+1},\ldots,e_m) : m \times m \to \mathbb{F} : (s,t) \mapsto \\ \begin{cases} 1, & s = t \notin \{i,j\} \\ 1, & (s,t) \in \{(i,j),(j,i)\} \\ 0, & \text{else} \end{cases}$$

$$C_3 := \text{Mul}_m(i,\lambda) := I_m + (a-1)e_i f_j : m \times m \to \mathbb{F} : (s,t) \mapsto \begin{cases} 1, & s = t \neq 1 \\ \lambda, & s = t = i \\ 0, & \text{else} \end{cases}$$

39.4 Invertibility

Because every row operation is *invertible*, we can now also claim that elementary matrices are invertible and that they also represent that inverted row operation.

Formally,

- $\text{Add}_m(i,j,a)$ has the inverse $\text{Add}_m(i,j,-a)$.
- $\text{Mul}_m(i,\lambda)$ has the inverse $\text{Mul}_m(i,\lambda^-1)$.
- $\text{Swap}_m(i,j)$ has itself as the inverse.

39.5 Determinants of Elementary Matrices

The determinants corresponding to each matrix are:

$$\det(C1) = 1$$
$$\det(C2) = -1$$
$$\det(C3) = \lambda$$

Chapter 40

Orthogonal Linear Maps

In this chapter we are going to take a look at a special type of linear map that is called *orthogonal* and at the term *isometry*. It will be useful to remember some concepts about the dot product, such as $\langle x, x\rangle = x^T \cdot x = \|x\|^2$.

40.1 Isometry

Given two vector spaces $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$, a map $f : V \to W$ is called *isometric* if and only if $\|x - y\|_V = \|f(x) - f(y)\|_W$.

It also has the following traits:

1. *Length-Preserving*: $\|x\|_V = \|f(x)\|_W \quad \forall x \in V$.
2. *Angle-Preserving*: $\angle(x, y) = \angle(f(x), f(y)) \quad \forall x, y \in V$.
3. *Linear*: $f(x + \lambda y) = f(x) + \lambda f(y)$.

40.2 Unit-Sphere

For $(V, \mathbb{R})$ with metric $\|\cdot\|$ we denote

$$S = \{x \mid x \in V : \|x\| = 1\}$$

as the *unit-sphere*.

40.3 Rotation Matrices

40.3.1 The Rotation Matrix

This matrix is a common example of an isometric linear map, because of the way it transforms the space.

For the $2D$ case we have

$$\begin{bmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{bmatrix}.$$

The derivation comes from the addition theorems for $\sin(\phi)$ and $\cos(\phi)$.

Given two vectors $\vec{v}$ and $\vec{g}$ whose angles are α and $\theta = \alpha + \beta$, we can rotate the vector $\vec{v}$ by some β to transform it into $\vec{g}$.

$$x = r\cos(\alpha)$$
$$y = r\sin(\alpha)$$

Then

$$x' = r'\cos(\alpha + \beta)$$
$$y' = r'\sin(\alpha + \beta)$$

And therefore

$$x' = r'\cos(\alpha)\cos(\beta) - r'\sin(\alpha)\sin(\beta)$$
$$y' = r'\cos(\alpha)\sin(\beta) + r'\cos(\beta)\sin(\alpha)$$

Because a rotation does not change the magnitude, we can say that $r = r'$. Thus

$$x' = r\cos(\alpha)\cos(\beta) - r\sin(\alpha)\sin(\beta)$$
$$y' = r\cos(\alpha)\sin(\beta) + r\cos(\beta)\sin(\alpha)$$

Now notice that we have our x and y inside the expression

$$x' = x\cos(\beta) - y\sin(\beta)$$
$$y' = x\sin(\beta) + y\cos(\beta)$$

which gives us

$$\begin{bmatrix}\cos(\phi) & -\sin(\phi)\\ \sin(\phi) & \cos(\phi)\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}$$

$\mathfrak{QED}$

40.3.2 Rotation Matrices for $\mathbb{R}^3$

Note that a rotation is determined by two axes; hence, we get 3 different matrices depending on the two axes we need to rotate.

R(y,z):

$$R_x(\phi) = \begin{bmatrix}1 & 0 & 0\\ 0 & \cos(\phi) & -\sin(\phi)\\ 0 & \sin(\phi) & \cos(\phi)\end{bmatrix}$$

R(x,z):

$$R_y(\phi) = \begin{bmatrix}1 & 0 & \sin(\phi)\\ 0 & 1 & 0\\ -\sin(\phi) & 0 & \cos(\phi)\end{bmatrix}$$

R(x,y):

$$R_z(\phi) = \begin{bmatrix}\cos(\phi) & -\sin(\phi) & 0\\ \sin(\phi) & \cos(\phi) & 0\\ 0 & 0 & 1\end{bmatrix}$$

40.4 Orthogonal Matrices

A matrix $A \in \mathbb{R}^{n\times n}$ is *orthogonal* if its column vectors form an orthonormal basis.

The set of all *orthogonal* matrices is $O(n)$.

40.4.1 Properties of Orthogonal Matrices

1. $A \in O(n)$.
2. If A has an inverse then $A^{-1} = A^T$.
3. $A^T \in O(n)$.
4. $\|Q\|_2 = \sup_{x\neq 0} \frac{\|Qx\|_2}{\|x\|_2} = 1$.

Proof 1-2:

$$\begin{aligned} A^T = A^{-1} &\iff A^T A = E \\ &\iff (A^T A)_{ij} = \langle a_i, a_j\rangle = \delta_{ij} 1 \leq i,j \leq n \\ &\iff a_i \perp a_j, i \neq j, \text{ and } \|a_i\| = 1, 1 \leq i \leq n \\ &\iff A \in O(n) \end{aligned}$$

Proof 1-3:

$$\begin{aligned} A \in O(n) &\iff \\ &\iff A^{-1} = A^T \\ &\iff (A^T)^{-1} = (A^T)^T \\ &\iff (A^1)^T = A \\ &\iff A^T \in O(n) \end{aligned}$$

Example of a proof

Prove that H_n is *orthogonal* for all $\vec{u} \in \mathbb{R}^n \backslash \{0\}$.

$$H_n = I_n - 2\frac{uu^T}{u^T u}$$

I_n is the identity matrix.

Proof:

$$\begin{aligned} (H_n)^2 &= I \\ &= \left(I_n - 2\frac{uu^T}{u^T u}\right)^2 \\ &= I_n^2 - 4\frac{uu^T}{u^T u} + 4\left(\frac{uu^T}{u^T u}\right)^2 \\ &= I_n - 4\frac{uu^T}{u^T u} + 4\frac{uu^T uu^T}{(u^T u)^2} \\ &= I_n - 4\frac{uu^T}{u^T u} + 4\frac{u(u^T u)u^T}{(u^T u)^2} \\ &= I_n - 4\frac{uu^T}{u^T u} + 4\frac{uu^T}{u^T u} \\ &= I_n \end{aligned}$$

QED

40.5 The Determinant of Orthogonal Matrices

The *determinant* of this kind of matrix can only be either positive 1 or negative 1.

Proof:

We know that $\det(I) = I$, therefore

$$\begin{aligned}\det(I) &= \det(AA^{-1}) \\ &= \det(AA^T) \\ &= \det(A)\det(A^T) \\ &= (\det(A))^2 \\ &= \|\det(A)\| = 1\end{aligned}$$

QED

40.6 The Orthogonal Group

$O(n)$ with the operation matrix multiplication forms the *orthogonal group*.

40.7 The Inner Product

Given $f : \mathbb{R}^n \to \mathbb{R}^n$, these two statements are equivalent:

$$\langle f(x), f(y)\rangle = \langle x, y\rangle \quad \forall x, y \in \mathbb{R}^n$$

and

f preserves the angles and length. f is also called *orthogonal*.

Proof $\Rightarrow$:

$$\|x\|^2 = \langle x, x\rangle = \langle f(x), f(x)\rangle = \|f(x)\|^2.$$

This implies the length preservation. The isometry gives us:

$$\begin{aligned}\|f(x) - f(y)\|^2 &= \langle f(x) - f(y), f(x) - f(y)\rangle \\ &= \langle f(x), f(x)\rangle - 2\langle f(x), f(y)\rangle + \langle f(y), f(y)\rangle \\ &= \langle x, x\rangle - 2\langle x, y\rangle + \langle y, y\rangle \\ &= \langle x - y, x - y\rangle = \|x - y\|^2\end{aligned}$$

Then

$$\begin{aligned}\angle(f(x), f(y)) &= \arccos\left(\frac{\langle f(x), f(y)\rangle}{\|f(x)\| \cdot \|f(y)\|}\right) \\ &= \arccos\left(\frac{\langle x, y\rangle}{\|x\| \cdot \|y\|}\right) \\ &= \angle(x, y)\end{aligned}$$

QED

Proof ⇐:

With $\alpha = \angle(x, y) = \angle(f(x), f(y))$, it holds:

$$\langle f(x), f(y)\rangle = \cos(\alpha) \cdot \|f(x)\| \cdot \|f(y)\| = \cos(\alpha) \cdot \|x\| \cdot \|y\| = \langle x, y\rangle .$$

QED

40.8 Orthonormal Matrices

A matrix $Q \in \mathbb{R}^{n\times n}$ is *orthonormal* if for each of its column vectors q_j, they are pairwise orthonormal.

$$(q_i, q_j) = \delta_{i,j} = \begin{cases} 1 & i = j \\ 0 & \text{else} \end{cases}$$

40.8.1 Properties of Orthonormal Matrices

If $Q, \tilde{Q} \in \mathbb{R}n \times n$ are orthonormal matrices, the following applies:

1. $QQ^T = Q^TQ = I$, i.e. $Q^{-1} = Q^T$.
2. $\kappa_2(Q) = 1$.
3. $Q\tilde{Q}$ is an orthonormal matrix.

Proof 1:

When applying the matrix product, we note that our repeated inner product is applied to each of the flipped columns. Since by our definition

$$(q_i, q_j) = \delta_{i,j} = \begin{cases} 1 & i = j \\ 0 & \text{else} \end{cases}$$

We get a matrix with 1's at the entries $QQ^t_{i,j}$ if $i = j$ and 0 otherwise. In other words, the identity.

QED

Proof 2:

For all $x \in \mathbb{R}^n$ it holds that

$$\|Qx\|_2^2 = (Qx)^T(Qx) = x^TQ^TQx = x^Tx = \|x\|_2^2.$$

Hence, orthonormal matrices preserve the length. Therefore $\|Q\|_2 = 1$, and analogously, the same holds for Q^{-1}, hence

$$\kappa_2(Q) = \|Q\|_2\|Q^{-1}\| = 1.$$

QED

Proof 3:

Applying our first property, we need to show that $(Q\tilde{Q})^T = (Q\tilde{Q})^{-1}$.

$$(Q\tilde{Q})^T(\tilde{Q}Q) = Q^T\tilde{Q}^T(\tilde{Q}Q) = Q^TIQ = Q^TQ = I$$

Therefore the columns are pairwise orthonormal.

QED

40.9 Theorem of the Matrix Representation

The mapping $f : \mathbb{R}^n \to \mathbb{R}^n$ is orthogonal if and only if the associated matrix representation Q (with respect to the standard basis) is an orthogonal matrix, i.e., $Q \in O(n)$.

Proof $\Rightarrow$:

f is linear, and thus has a matrix representation Q. It follows that

$$\langle Qx, Qy\rangle = \left\langle Q^\top Qx, y\right\rangle = \langle x, y\rangle \quad \forall x, y \in \mathbb{R}^n,$$

thus,

$$\left\langle Q^\top Qx - x, y\right\rangle = 0 \quad \forall x, y.$$

Choosing $y = Q^\top Qx - x$, we get

$$\langle Q^\top Qx - x, Q^\top Qx - x\rangle = 0 \quad \forall x,$$

and hence $(Q^\top Q - I)x = 0 \quad \forall x$. It follows that $Q^\top Q = I$, and by Theorem 7.3, $Q \in O(n)$.

$\mathfrak{QED}$

Proof $\Leftarrow$:

Follows by the same calculation as above.

$\mathfrak{QED}$

40.10 Change of Basis

Let f be orthogonal. Then the matrix representation of f with respect to any orthonormal basis is orthogonal.

Proof:

Let E be the standard basis and B an orthonormal basis. Let $Q = M_E^E(f)$ and $S = T_E^B$ be the change-of-basis matrix from B to E. Since the columns of S are the basis vectors of B, we have $S \in O(n)$.

$$M_B^B(f) = S^{-1}QS \in O(n),$$

due to the group property of $O(n)$.

$\mathfrak{QED}$

40.11 QR-Decomposition

A use case of orthogonal matrices is the *QR-Decomposition*. The goal is that given a system $Ax = b$ with a matrix $A \in \mathbb{R}^{m\times n}$,

$$A = \begin{bmatrix}\vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n\end{bmatrix}$$

$m \geq n$ and $\operatorname{rank}(A) = n$ we can find two matrices Q and R such that $A = QR$ with $R \in \mathbb{R}^{n\times n}$ and $Q \in \mathbb{R}^{m\times n}$.

R is an upper triangular matrix and Q is orthogonal for the columns.

We know that the best approximation by the method of the least squares is given by

$$x_s = (A^T A)^{-1} A^T b$$

Because of the orthogonality of the columns $\text{rank}(Q) = \text{rank}(A)$, therefore, it can be inverted. Hence, we get:

$$\begin{aligned} x_s &= (A^T A)^{-1} A^T b \\ &= ((QR)^T (QR))^{-1} (QR)^t b \\ &= (R^T Q^T QR)^{-1} (QR)^t b \\ &= (R^T R)^{-1} R^T Q^T b \\ &= R^{-1} (R^T)^{-1} R^T Q^T b \\ &= R^{-1} Q^T b \\ Rx_s &= Q^T b \end{aligned}$$

Thus, we can solve the linear system by finding Q which is matrix with an orthonormal basis of the vector space which can be found via the Gram-Schmidt algorithm. The matrix R consists of the matrix of the inner products our A matrix and Q

$$R = \begin{bmatrix} \langle q_1, \vec{a}_1 \rangle & \langle q_1, \vec{a}_2 \rangle & \cdots & \langle q_1, \vec{a}_n \rangle \\ 0 & \langle q_2, \vec{a}_2 \rangle & \cdots & \langle q_2, \vec{a}_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \langle q_n, \vec{a}_n \rangle \end{bmatrix}$$

Or even simpler $R = Q^T A$.

Example:

Find the *QR-Decomposition* of the matrix A given by:

$$\begin{bmatrix} 1 & 3 \\ 1 & -1 \end{bmatrix}$$

First, we find the orthonormal basis of the columns of our matrix A

$$q_1 = \frac{a_1}{\|a_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$r_2 = a_2 - \langle a_2, q_1 \rangle q_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \langle \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rangle \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$q_2 = \frac{1}{\sqrt{8}} \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{8}} \\ \frac{1}{\sqrt{2}} & -\frac{2}{\sqrt{8}} \end{bmatrix}$$

$$R = \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{8} \end{bmatrix}$$

Chapter 41

Eigenvectors and Eigenvalues

Eigenvector:

An eigenvector of a square matrix A is a non-zero vector $\vec{v}$ that, when multiplied by A, results in a vector that is a scalar multiple of itself. In other words, the direction of the vector $\vec{v}$ remains unchanged when the linear transformation represented by A is applied to it.

Eigenvalue:

The scalar multiple, denoted by λ, is called the eigenvalue associated with the eigenvector v. It represents the factor by which the eigenvector is scaled when transformed by the matrix A.

Mathematically, the relationship between a square matrix A, an eigenvector $\vec{v}$, and its corresponding eigenvalue λ is expressed by the following equation:

$$A\vec{v} = \lambda\vec{v}$$

where $A = f : V \to V$ is an *endomorphism*, $\vec{v} \neq \vec{0}$ and $\lambda \in \mathbb{C}$.

The way we find them is by manipulating the equation above:

$$\begin{aligned} A\vec{v} &= \lambda\vec{v} \\ A\vec{v} - \lambda\vec{v} &= \vec{0} \\ A\vec{v} - \lambda I\vec{v} &= \vec{0} \\ (A - \lambda I)\vec{v} &= \vec{0} \end{aligned}$$

Here, note that if $(A - \lambda I)$ were invertible then we could multiply both sides by its inverse and then get $\vec{v} = 0$. But we do not want that, therefore $(A\vec{v} - \lambda I)$ must be a linear transformation whose determinant is 0, which means that it cannot be invertible.

$$\begin{aligned} (A - \lambda I)\vec{v} &= \vec{0} \\ \det(A - \lambda I) &= 0 \end{aligned}$$

41.1 Characteristic Polynomial

The polynomial which we get from the determinant $\det(A - \lambda I) = 0$ is called the *characteristic polynomial of A*.

$$\mathcal{X}_A = \det(A - \lambda I) = 0$$

41.2 Eigenspace

The *Eigenspace* of an eigenvalue is the set of solutions of the homogeneous system $A - \lambda I$.

$$\text{Eig}(f; \lambda) = ker(A - \lambda I)$$

41.3 Theorems of Eigenvalues and Eigenvectors

41.3.1 Theorem I

Given an eigenvalue λ of f and $v_1, \ldots, v_n$ being the eigenvectors of f with λ, then $v \in L(v_1, \ldots, v_k) \backslash \{0\}$ is an eigenvector of f with λ.

41.3.2 Theorem II

For $\lambda \in \mathbb{C}$, $Eig(f; \lambda) := \{c \in V | f(v) = \lambda v\}$ is the *Eigenspace* of f with λ, and it is a subspace of V.

41.3.3 Theorem III

For $\lambda \neq \gamma \quad Eig(f; \lambda) \cap Eig(f; \gamma) = \{0\}$.

41.4 Linear Independence of Eigenvectors With Distinct Eigenvalues

Eigenvectors with different eigenvalues are linearly independent.

Proof:

If $\alpha v + \beta w = 0$, where v and w are eigenvectors of $A \in \mathbb{C}^{n \times n}$, then it must be true that

$$\alpha \lambda v + \beta \lambda w = 0$$
$$\alpha \mu v + \beta \mu w = 0$$

With $\lambda \neq \mu$ being eigenvalues of A.

Also, we have that

$$0 = A(\alpha v + \beta w) = \alpha \lambda v + \beta \mu w$$

By subtracting this equation from the previous two equations, we get

$$0 = \alpha(\lambda - \mu)v$$
$$0 = \beta(\mu - \lambda)w$$

But since $\mu \neq \lambda$ and $v, w \neq \vec{0}$ it must be that $\alpha = \beta = 0$. Thus, they are linearly independent.

$\mathfrak{QED}$

41.5 The Determinant via Eigenvalues

Given the following eigenvalues $\lambda_1, \ldots, \lambda_n$, then the determinant of the matrix A is given by

$$\det(A) = \prod_{i=1}^{n} \lambda_i$$

Proof:

Given the eigenvalues $\lambda_1, \ldots, \lambda_n$ are the roots of the characteristic polynomial

$$\begin{aligned}\det(A - \lambda I) = p(\lambda) &= (-1)^n(\lambda - \lambda_1)(\lambda - \lambda_2)\cdots \\ &= (-1)(\lambda - \lambda_1)(-1)(\lambda - \lambda_2)\cdots \\ &= (\lambda - \lambda_1)(\lambda - \lambda_2)\cdots\end{aligned}$$

Now, by setting $\lambda = 0$ we get

$$\det(A) = \prod_{i=1}^{n} \lambda_i$$

Or by using the definition of the diagonal matrix:

$$\begin{aligned}A &= XDX^{-1} \\ \det(A) &= \det(XDX^{-1}) \\ \det(A) &= \det(X)\det(D)\det(X^{-1}) \\ \det(A) &= \det(D) = \prod_{i=1}^{n} \lambda_i\end{aligned}$$

$\mathfrak{QED}$

41.6 Eigenvalues Under Isometric Maps

For every linear isometric map f with some eigenvalue λ, it holds that

$$\|x\| = \|f(x)\| = \|\lambda x\| = \|\lambda\|\|x\|$$

Also, $\lambda = 1$.

An example of this are rotation matrices, which map to the same space and preserve the length, and whose eigenvalues are complex with a modulus of one.

41.7 Eigenvalues as Powers

For some eigenvector x with an eigenvalue λ of A, it holds that

$$A^k x = \lambda^k x$$

41.8 How to Find the Eigenvectors and Eigenvalues

To find the eigenvalues and eigenvectors of a square matrix A we follow the next steps

1. **Form the characteristic equation:** Rewrite the equation $A\vec{v} = \lambda\vec{v}$ as $(A - \lambda I)\vec{v} = \vec{0}$, where λI is the identity matrix times λ because this matrix encodes the multiplication by some scalar.

   To have a non-trivial solution 0 for $\vec{v}$, the matrix $(A - \lambda I)$ must be singular as stated before, which means its determinant must be zero because otherwise $\vec{v}$ would be 0. Thus, we have the characteristic equation:

$$\det(A - \lambda I) = 0$$

2. **2. Solve for the eigenvalues:** Solve the characteristic equation for λ. The solutions $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of the matrix A.
3. **3. Find the eigenvectors:** For each eigenvalue λ_i, substitute it back into the equation $(A - \lambda_i I)\vec{v} = \vec{0}$ and solve for the vector $\vec{v}$. The non-zero solutions for $\vec{v}$ are the eigenvectors corresponding to the eigenvalue λ_i. Note that we are going to get infinitely many solutions; thus, we have to write our vector with dependence on the free parameters.

Example:

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$$

Let us build the characteristic equation and solve it

$$\det \begin{bmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{bmatrix} = 0$$

$$\begin{aligned} (1-\lambda)^2 - 4 &= 0 \\ 1 - 2\lambda + \lambda^2 - 4 &= 0 \\ \lambda^2 - 2\lambda - 3 &= 0 \\ (\lambda - 3)(\lambda + 1) &= 0 \end{aligned}$$

We have got the eigenvalues $\lambda_1 = 3$ and $\lambda_2 = -1$. Now let us continue by substituting our eigenvalues into the original equation.

$$\begin{aligned} \begin{bmatrix} 1-\lambda_2 & 1 \\ 4 & 1-\lambda-2 \end{bmatrix} &= \vec{0} \\ \begin{bmatrix} 1-2 & 1 \\ 4 & 1-2 \end{bmatrix} &= \vec{0} \\ \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} &= \vec{0} \end{aligned}$$

Using Gaussian elimination we get:

$$\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$$

Thus, let $v_2 = -2$ and $v_1 = \frac{2}{2} = 1$. This gives us the first eigenvector and all of its scalar versions.

$$e_1 = \left\{ c \begin{bmatrix} 1 \\ -2 \end{bmatrix} | c \in \mathbb{R} \right\}$$

The process for the other eigenvalue is exactly the same.

41.9 Complex Eigenvalues

This is a special case of *eigenvalues*. They appear for example in rotation matrices.

Example:

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\det\left(\begin{bmatrix} -\lambda & 1 \\ -1 & -\lambda \end{bmatrix}\right) = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

Now the corresponding eigenvectors.

For $\lambda = i$

$$\begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix} \to \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$$

Therefore, our first eigenvector is

$$c\begin{bmatrix} -i \\ 1 \end{bmatrix}$$

For $\lambda = -i$

$$\begin{bmatrix} i & 1 \\ -1 & i \end{bmatrix} \to \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}$$

Therefore, the second eigenvector is

$$c\begin{bmatrix} i \\ 1 \end{bmatrix}$$

41.10 Diagonal Matrices

A *diagonal matrix* is a matrix of dimensions $n \times n$ where only the entries of the main diagonal are not zero. This kind of matrix makes a lot of the computations in linear algebra trivial. Therefore, if our matrix is not diagonal, we are interested in finding a way of transforming the original matrix into a diagonal matrix.

A matrix is diagonalisable if and only if its eigenvectors are linearly independent.

41.10.1 Complex Diagonalization

Given a matrix $A \in \mathbb{C}^{n\times n}$ with n distinct eigenvalues, then A is diagonalisable. But this does not guarantee the diagonalisability for $\mathbb{R}$.

41.10.2 Geometric and Algebraic Multiplicity

Given a matrix $A \in \mathbb{C}^{n\times n}$ with λ_i as an eigenvalue:

The multiple occurrence of the zero λ_i in $\mathcal{X}_A$ is called *algebraic multiplicity* $\text{alg}(\lambda)_i$.

Continuing, $\text{geo}(\lambda_i) := \dim(\text{Eig}(A;\lambda_i))$ is the *geometrical multiplicity* of λ_i i.e. the number of linearly independent eigenvectors associated with λ_i.

Also, A is diagonalisable if and only if the *geometric multiplicity* of every eigenvalue is equal to its *algebraic multiplicity*.

Example:

This example shows a matrix which is not diagonalisable:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

The corresponding multiplicities are:

$$\begin{aligned} \mathcal{X} = \det(A - \lambda I) &= 0 \\ (1 - \lambda)^2 &= 0 \end{aligned}$$

With eigenvalue $\lambda = 1$.

This results in $\text{alg}(1) = 2$ and the $\text{geo}(1) = 1$ because of

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

More precisely $\dim(\text{Eig}(A; 1)) = 1$.

Example With Complex Eigenvalues

Given is the rotation matrix

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

With the characteristic polynomial

$$\begin{aligned} \mathcal{X} &= (\cos(\theta) - \lambda)^2 + \sin^2(\theta) = 0 \\ &= \cos^2(\theta) - 2\cos(\theta)\lambda + \lambda^2 + \sin^2(\theta) = 0 \\ &= \lambda^2 - 2\cos(\theta)\lambda + 1 = 0 \end{aligned}$$

Using the quadratic formula gives us

$$\begin{aligned} \lambda_{1,2} &= \cos(\theta) \pm \sqrt{\cos^2(\theta) - 1} \\ &= \cos(\theta) \pm \sqrt{-\sin^2(\theta)} \\ &= \cos(\theta) \pm i\sin(\theta) = e^{\pm i\theta} \end{aligned}$$

Therefore, our eigenvalues are complex. Now, is the matrix diagonalisable? The answer is yes.

Because $(\cos(\theta) - \lambda)^2$ gives us $\text{alg}(\lambda) = 2$.

$$\text{alg}(e^{i\theta}) = 2, \quad \text{alg}(e^{-i\theta}) = 2$$

And

$$\text{geo}(e^{i\theta}) = 2, \quad \text{geo}(e^{-i\theta}) = 2$$

For $\lambda = e^{i\theta}$ we get two linearly independent vectors

$$\begin{bmatrix} \cos(\theta) - \cos(\theta) - i\sin(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) - \cos(\theta) - i\sin(\theta) \end{bmatrix}$$

$$\begin{bmatrix} -i\sin(\theta) & -\sin(\theta) \\ \sin(\theta) & -i\sin(\theta) \end{bmatrix}$$

For $\lambda = e^{-i\theta}$ we also get two linearly independent vectors

$$\begin{bmatrix} \cos(\theta) - \cos(\theta) + i\sin(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) - \cos(\theta) + i\sin(\theta) \end{bmatrix}$$

$$\begin{bmatrix} i\sin(\theta) & -\sin(\theta) \\ \sin(\theta) & i\sin(\theta) \end{bmatrix}$$

41.11 Eigenvalues of Real Symmetric Matrices

Eigenvalues of *real symmetric matrices* are *real*, and so are their respective eigenvectors.

Proof:

By definition, we have that $Ax = \lambda x$. Then let us multiply both sides by $\bar{x}^T$. Note that because x is a real vector, the conjugate does not change anything.

$$\begin{aligned} Ax &= \lambda x \\ \bar{x}^T Ax &= \bar{x}^T \lambda x \\ \bar{x}^T Ax &= \lambda \bar{x}^T x \\ \bar{x}^T Ax &= \lambda \langle \bar{x} x \rangle \\ \bar{x}^T Ax &= \lambda \left\| x \right\|^2 \\ \frac{\bar{x}^T Ax}{\left\| x \right\|^2} &= \lambda \end{aligned}$$

With this we know that the denominator is a real number thus, if we prove that the numerator is also real we are done. For this we will use the properties of the conjugate.

$$\begin{aligned} \bar{x}^T Ax &= \overline{\bar{x}^T Ax} \\ &= \overline{\bar{x}^T}\,\overline{Ax} \\ &= x^T \overline{Ax} \\ &= (\overline{Ax})^T x \\ &= (\bar{A}\,\bar{x})^T x \\ &= \bar{x}^T \bar{A}^T x \end{aligned}$$

Notice that we ended just where we started. Therefore, the conjugate had no effect on the whole term, thus it is real-valued.

$\mathfrak{QED}$

41.12 Orthogonality of Eigenvectors

Given a symmetric matrix $A \in \mathbb{R}^{n\times n}$ with $\lambda \neq \mu$ as two eigenvalues of A with eigenvectors v and w. Then v and w are *orthogonal* to each other.

Proof:

$$\langle w, Av\rangle = w^T Av = (A^T w)^T v = (Aw)^T v = \langle Aw, v\rangle$$

Therefore, $\lambda \langle v, w\rangle = \langle Av, w\rangle = \langle v, Aw\rangle = \mu \langle v, w\rangle$. Because $\mu \neq \lambda$ and $\lambda - \mu \langle v, w\rangle = 0$, the vectors must be orthogonal.

$\mathfrak{QED}$

41.13 Eigenvalue Decomposition of Symmetric Matrices

Given a symmetric matrix $A \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then, because of the orthogonality of its eigenvectors and the fact that they can be normalized, we can write the matrix Q as an orthogonal matrix, and therefore

$$A = QDQ^T$$

41.14 Matrix Diagonalization

Diagonalizing a matrix involves finding a diagonal matrix that is similar to the given matrix. A square matrix A is diagonalisable if there exists an invertible matrix X made of eigenvectors of A such that $X^{-1}AX = D$, where D is a diagonal matrix made of eigenvalues.

The order of the eigenvalues and eigenvectors matters. If you choose $\lambda_1, \lambda_2, \ldots, \lambda_n$, then the order of the eigenvectors $v_1, v_2, \ldots, v_n$ must also correspond. This process only works if A has unique eigenvalues or, if they are duplicates, they have to be linearly independent.

The process of diagonalisation is as follows:

1. **1. Find the eigenvalues and eigenvectors of A:**

   Form the characteristic polynomial.

   Compute the determinant.

   Find the roots of the polynomial.

2. **2. Form the matrix X:** Create a matrix X whose columns are the linearly independent eigenvectors of A.

3. **3. Form the diagonal matrix D:** Create a diagonal matrix D whose diagonal entries are the eigenvalues of A, corresponding to the order of the eigenvectors in P. That is, if the i-th column of P is the eigenvector corresponding to the eigenvalue λ_i, then the i-th diagonal entry of D is λ_i.

4. **4. Verify the diagonalisation:** Check that $X^{-1}AX = D$.

A matrix $A \in \mathbb{C}^{n\times n}$ is diagonalisable if and only if it has n *linearly independent eigenvectors.*

Example:

Consider the matrix

$$A = \begin{bmatrix} -3 & -4 \\ 5 & 6 \end{bmatrix}$$

The characteristic equation is

$$\det(A-\lambda I)=\det\begin{bmatrix}-3-\lambda & -4\\ 5 & 6-\lambda\end{bmatrix}=(-3-\lambda)(6-\lambda)+20=\lambda^2-3\lambda+2=0$$

$$(\lambda-1)(\lambda-2)$$

Solving for λ, we get $\lambda_1=1$ and $\lambda_2=2$.

For $\lambda_1=1$:

$$(A-I)\vec{v}=\begin{bmatrix}-4 & -4\\ 5 & 5\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\left[\begin{array}{cc|c}-4 & -4 & 0\\ 5 & 5 & 0\end{array}\right]\Rightarrow\left[\begin{array}{cc|c}1 & 1 & 0\\ 5 & 5 & 0\end{array}\right]\Rightarrow\left[\begin{array}{cc|c}1 & 1 & 0\\ 0 & 0 & 0\end{array}\right]$$

Our free variable is y therefore

$$\vec{v}=\left\{y\begin{bmatrix}-1\\ 1\end{bmatrix}\mid y\in\mathbb{R}\right\}.$$

Now choose $y=1$ which gives us $x=-1$. Now we get the vector

$$\vec{x}=\begin{bmatrix}-1\\ 1\end{bmatrix}.$$

For $\lambda_2=2$ the process is similar, so we are going to skip it. We get the vector

$$\vec{y}=\begin{bmatrix}-\frac{4}{5}\\ 1\end{bmatrix}.$$

Now we will complete the process by writing $D, X,$ and X^{-1}.

$$D=\begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}\ X=\begin{bmatrix}-1 & -\frac{4}{5}\\ 1 & 1\end{bmatrix}\ X^{-1}=\begin{bmatrix}-1 & \frac{4}{5}\\ -1 & -1\end{bmatrix}.$$

41.15 Normal Matrices

A matrix $A\in\mathbb{C}^{n\times n}$ is considered *normal* if and only if

$$A\overline{A^T}=\overline{A^T}A.$$

For real matrices, this condition simplifies to $AA^T=A^TA$.

Also, with the euclidean inner product in $\mathbb{C}^n$, there exists an orthonormal basis of eigenvectors, and therefore A is diagonalisable.

This implies that

1. All real symmetric matrices are diagonalisable, and their eigenvalues are real.
2. All real antisymmetric matrices $A^T=-A$ are diagonalisable, and their eigenvalues are purely imaginary or 0.
3. All orthogonal matrices are diagonalisable.

41.16 Singular Value Decomposition (SVD)

The *Singular Value Decomposition (SVD)* is a fundamental matrix factorization technique in linear algebra. It expresses any real matrix $A \in \mathbb{R}^{m \times n}$ as a product of three specific matrices:

$$A = U\Sigma V^T$$

where:

- $U \in \mathbb{R}^{m \times m}$ is an orthogonal matrix whose columns are called the *left singular vectors* of A. A rotation to align with the eigenvectors.
- $V \in \mathbb{R}^{n \times n}$ is an orthogonal matrix whose columns are the *right singular vectors* of A. A stretching with dimension reduction/expansion.
- $\Sigma \in \mathbb{R}^{m \times n}$ is a diagonal matrix with non-negative entries $\sigma_i \geq 0$ on the diagonal, called the *singular values* of A. Another rotation to align with the original basis vectors.

Let $A \in \mathbb{R}^{m \times n}$ be a real square matrix with SVD given by:

$$A = U\Sigma V^T.$$

We show that:

1. The columns of U are eigenvectors of AA^T,
2. The columns of V are eigenvectors of A^TA,
3. The eigenvalues of both AA^T and A^TA are the squares of the singular values σ_i^2.

Proof:

For AA^T, we compute:

$$AA^T = (U\Sigma V^T)(V\Sigma^T U^T) = U\Sigma V^T V\Sigma^T U^T = U\Sigma\Sigma^T U^T = U\Sigma^2 U^T$$

Since $\Sigma\Sigma^T$ is diagonal with entries σ_i^2, this is an eigenvalue decomposition of AA^T. Thus, the columns of U are eigenvectors of AA^T, and the corresponding eigenvalues are σ_i^2.

For A^TA, we have:

$$A^TA = (V\Sigma^T U^T)(U\Sigma V^T) = V\Sigma^T U^T U\Sigma V^T = V\Sigma^T\Sigma V^T = V\Sigma^2 V^T$$

Again, $\Sigma^T\Sigma$ is diagonal with entries σ_i^2, so the matrix A^TA is diagonalized by V, and the columns of V are eigenvectors of A^TA, with eigenvalues σ_i^2.

$\mathfrak{QED}$

41.16.1 Finding the SVD of a Matrix

To find the SVD of a matrix $A \in \mathbb{C}^{m \times n}$ we follow these steps:

1. Compute A^TA and AA^T and their respective eigenvalues and eigenvectors.
2. Form $V \in \mathbb{C}^{n \times n}$ with the eigenvectors of A^TA as columns.
3. Form $U \in \mathbb{C}^{m \times m}$ with the eigenvectors of AA^T as columns.
4. Form $\Sigma \in \mathbb{C}^{m \times n}$ with the square root of the eigenvalues of AA^T as columns.

41.17 Similar Matrices

Given $A \in \mathbb{C}^{n\times n}$ and $X \in \mathbb{C}^{n\times n}$ then if

$$A = XDX^{-1} \iff X^{-1}AX = D$$

we denote XDX^{-1} as *similar* to A.

41.17.1 Case I

A is diagonalisable with eigenvalues and eigenvectors.

$$S^{-1}AS = D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \ddots & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & & \lambda_n \end{bmatrix}$$

41.17.2 Case II

Any complex eigenvalue must come in complex conjugate pairs if A is real valued.

$$\lambda_1 = a + ib,\ \lambda_2 = a - bi$$

$$S^{-1}AS = D = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \to e^{At} = \begin{bmatrix} \cos(bt) & \sin(bt) \\ -\sin(bt) & \cos(bt) \end{bmatrix} e^{at}$$

41.17.3 Case III

Repeated eigenvalues and not enough vectors in the null-space.

$$S^{-1}AS = D = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$$

41.18 Equality of Eigenvalues

Given $A \in \mathbb{C}^{n\times n}$ and $X \in \mathbb{C}^{n\times n}$, both regular, then XAX^{-1} has the same eigenvalues as A.

Proof:

$$\begin{aligned} \det(A - \lambda I) &= \det(XAX^{-1} - \lambda I) \\ &= \det(XAX^{-1} - \lambda XIX^{-1}) \\ &= \det(X[A - \lambda I]X^{-1}) \\ &= \det(X)\det(A - \lambda I)\det(X^{-1}) \\ &= \det(A - \lambda I) \\ &= \mathcal{X}_A \end{aligned}$$

Thus, similar matrices have the same **eigenvalues with their correspondent multiplicity**.

$\mathfrak{QED}$

41.19 Jordan Normal Form

Let $A \in \mathbb{C}^{n\times n}$, it is diagonalisable if

$$X^{-1}AX = D \iff A = XDX^{-1}$$

where D is a diagonal matrix.

Generalizing, we can also find a matrix $J \in \mathbb{C}^{n\times n}$ in the *Jordan normal form* such that $A = SJS^{-1}$. The matrix J is then considered the most not diagonalisable matrix similar to our matrix A. This comes in handy when we have repeated eigenvalues with enough eigenvectors.

$$A = XJX^{-1}, J = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_n \end{bmatrix}$$

Whereas J is a *block-diagonal matrix*

$$J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \ddots & \ddots & \\ & & \lambda_i & 1 \\ & & & \lambda_i \end{bmatrix}$$

For each of these blocks, the one corresponding eigenvalue is repeated across the diagonal.

If A has exactly n linearly independent eigenvectors, then J is just D.

41.19.1 Algorithm for Jordan Blocks

- Determine the eigenvalues of A and their algebraic multiplicities.
- For each eigenvalue, determine the size of the largest Jordan block by finding the smallest k such that $\text{nullity}((A-\lambda I)^k) = \text{nullity}((A-\lambda I)^{k+1})$.
- Construct the Jordan normal form J by placing the Jordan blocks corresponding to each eigenvalue along the diagonal, with the size of each block determined in the previous step.

Example:

Given the matrix

$$A = \begin{bmatrix} 5 & 4 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 3 \end{bmatrix}$$

The characteristic polynomial is

$$\mathcal{X}_A = (5-\lambda)(1-\lambda)(3-\lambda)$$

Delivering the eigenvalues $\lambda_1 = 5$, $\lambda_2 = 1$, and $\lambda_3 = 3$ with algebraic multiplicity of 1 for each.

Hence the Jordan normal form of A is

$$J = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

If we had a matrix with repeated eigenvalues and not enough eigenvectors, for example

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

The characteristic polynomial is

$$\mathcal{X}_A = (2 - \lambda)^2$$

Delivering the eigenvalue $\lambda = 2$ with algebraic multiplicity of 2. However, there is only one linearly independent eigenvector, so the Jordan normal form of A is

$$J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Note that the Jordan block has a size equal to the algebraic multiplicity of the eigenvalue, and the superdiagonal entry is 1, indicating that there is a single Jordan block for the eigenvalue 2.

41.20 The Hermitian Transpose

For a matrix $A \in \mathbb{C}^{n \times m}$, the *hermitian transpose* A^H is the transpose of the conjugate matrix of the matrix A.

$$A^H = \bar{A}^T$$

41.21 Unitary Matrices

Unitary Matrices are matrices whose columns form orthonormal vectors and

$$U^H = U^{-1}.$$

41.22 Unitary Decomposition (Schur's Theorem)

For each matrix $A \in \mathbb{C}^{n \times n}$, there exists a unitary matrix $U \in \mathbb{C}^{n \times n}$ such that

$$A = UTU^H, \quad T = \begin{bmatrix} \lambda_1 & t_{1,2} & \cdots & t_{1,n} \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & t_{n-1,n} \\ 0 & \cdots & 0 & \lambda_{n,n} \end{bmatrix}.$$

As a reminder $U^H = \overline{U^T}$.

41.23 Unitary Diagonalisation

For a matrix $A \in \mathbb{C}^{n \times n}$, it is considered *unitary diagonalisable* if there exists a matrix $U \in \mathbb{C}^{n \times n}$ such that

$$A = U\Lambda U^H, \ \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_{n,n} \end{bmatrix}.$$

41.24 Normal Matrices

A matrix $A \in \mathbb{C}^{n\times n}$ is considered *normal* if $AA^H = A^H A$.

41.24.1 Diagonalisability Criteria Based on Normal Matrices

A matrix $A \in \mathbb{C}^{n\times n}$ is unitary diagonalisable if and only if it is normal.

Proof:

A is unitary diagonalisable $\Rightarrow$ A is normal:

Then there exists a unitary matrix U such that $A = U\Lambda U^H$. Therefore,

$$A^H A = (U\Lambda U^H)^H (U\Lambda U^H) = U\Lambda^H U^H U\Lambda U^H = U\Lambda^H \Lambda U^H$$

and

$$AA^H = (U\Lambda U^H)(U\Lambda U^H)^H = U\Lambda U^H U\Lambda^H U^H = U\Lambda\Lambda^H U^H.$$

Hence, $AA^H = A^H A$, so A is normal.

A is normal $\Rightarrow$ A is unitary diagonalisable:

By Schur's theorem, there exists a unitary matrix U such that $A = UTU^H$, where T is upper triangular.

Since A is normal, we have:

$$AA^H = UTU^H(UT^H U^H) = UTT^H U^H$$

and

$$A^H A = (UTU^H)^H (UTU^H) = UT^H U^H UTU^H = UT^H TU^H.$$

Since $AA^H = A^H A$, it follows that $TT^H = T^H T$. This implies that T is a normal upper triangular matrix, which must be diagonal. Therefore, $T = \Lambda$ is diagonal, and we have $A = U\Lambda U^H$, showing that A is unitary diagonalisable.

$\mathfrak{QED}$

Example

The matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$$

is normal and unitary diagonalisable. We can verify that $AA^H = A^H A$ and then find a unitary matrix U such that $A = U\Lambda U^H$, where Λ is a diagonal matrix containing the eigenvalues of A.

$$\Lambda = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -\sqrt{3}i & 0 \\ 0 & 0 & \sqrt{3}i \end{bmatrix} \quad U = \begin{bmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{6} - \frac{i}{2} & -\frac{\sqrt{3}}{6} + \frac{i}{2} \\ \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{6} + \frac{i}{2} & -\frac{\sqrt{3}}{6} - \frac{i}{2} \\ \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{-1+i\sqrt{3}}{2\sqrt{3}} \end{bmatrix}$$

41.25 Properties of Unitary Diagonalisable Matrices

For matrices $A \in \mathbb{C}^{n\times n}$ that are unitary diagonalisable $A = U\Lambda U^H$, the following properties hold:

- For all eigenvalues λ_i its geometric and algebraic multiplicity are the same.
- The column vectors of U form an orthonormal basis of $\mathbb{C}^n$. This means that for a vector x

$$x = \sum_{i=1}^{n} x_i u_i, \quad x_i = \langle u_i, x\rangle_2 = u_i^H x$$

  and

$$Ax = \sum_{i=1}^{n} x_i A u_i = \sum_{i=1}^{n} x_i \lambda_i u_i$$

- For any vector $x, y \in \mathbb{C}^n$ we have that:

$$\langle x, y\rangle_2 = x^H y = \sum_{i=1}^{n} \bar{x_i} y_i, \quad \|x\|_2^2 = \sum_{i=1}^{n} |x_i|^2$$

41.25.1 Hermitian Matrices

A matrix $M \in \mathbb{C}^{n\times m}$ with the following property:

$$M^H = M$$

is considered *hermitian*.

41.25.2 Properties of Hermitian Matrices

- All eigenvalues of a hermitian matrix are real.
- Hermitian matrices are normal and therefore unitary diagonalisable.
- Symmetric matrices are a special case of hermitian matrices where all entries are real.

41.26 Real Eigenvalues Via Hermitian Matrices

Given a normal matrix $A \in \mathbb{C}^{n\times n}$, all eigenvalues of A are real if and only if A is hermitian.

Proof:

$\Rightarrow$: A is normal which means that it is unitary diagonalisable. Therefore, there exists a unitary matrix U such that $A = U\Lambda U^H$, where Λ is a diagonal matrix containing the eigenvalues of A.

Since all eigenvalues are real, Λ is a real diagonal matrix. Thus, $A^H = (U\Lambda U^H)^H = U\Lambda^H U^H = U\Lambda U^H = A$, showing that A is hermitian.

$\Leftarrow$: If A is hermitian, we know that $A^H = A$. Since A is also normal, it is unitary diagonalisable. Therefore, there exists a unitary matrix U such that $A = U\Lambda U^H$, where Λ is a diagonal matrix containing the eigenvalues of A.

$$A^H = (U\Lambda U^H)^H = U\Lambda^H U^H = U\Lambda U^H = A$$

Since, $\Lambda^H = \Lambda$, follows that Λ is a real diagonal matrix, and thus all eigenvalues of A are real.

$\mathfrak{QED}$

41.27 Gershgorin's Theorem

Given $A \in \mathbb{C}^{n\times n}$ and

$$K = \bigcup_{i=1}^{n} K_i, \quad K_i = \{z \in \mathbb{C} : |z - a_{i,i}| \leq R_i\}, \quad R_i = \sum_{j\neq i} |a_{i,j}|$$

Then, all eigenvalues of A are contained in K, $\lambda_i \in K\ \forall i$. The diagonal element is the middle point of the circle and the sum the radius. If all eigenvalues are real, then we get intervals.

Example:

Given the matrix

$$A = \begin{bmatrix} 4 & -2 & 3 \\ 3 & 2 & -2 \\ 2 & -1 & 3 \end{bmatrix}$$

We get the following circles:

- $K_1 = \{z \in \mathbb{C} : |z - 4| \leq 5\} = K(4, 5)$.
- $K_2 = \{z \in \mathbb{C} : |z - 2| \leq 5\} = K(2, 5)$.
- $K_3 = \{z \in \mathbb{C} : |z - 3| \leq 3\} = K(3, 3)$.

Which means that all eigenvalues of A are contained in the union of these circles, $K = K_1 \cup K_2 \cup K_3$.

If we also approximate the eigenvalues of A^T, then we get the following circles:

- $K_1'(4, 5)$.
- $K_2'(2, 3)$.
- $K_3'(3, 5)$.

With this information we can conclude that the eigenvalues have to be in the circle $K(2, 3)$, because it is the smallest circle (interval).

41.28 Approximation of the Differences of Eigenvalues

Given a symmetric matrix $A \in \mathbb{R}^{n\times n}$ with distinct eigenvalues and $c \in \mathbb{R}$ and $x \in \mathbb{R}^n \setminus \{\vec{0}\}$, then

$$\min_{i=1,\ldots,n} |\lambda_i - c| \leq \frac{\|Ax - cx\|}{x} = \frac{\|d\|}{\|x\|}$$

Proof:

Because of the distinct eigenvalues of A, we know that there exists an eigenbasis such that we can write x as

$$\sum_{i=1}^{n} \alpha_i v_i$$

We can write

$$Ax = \sum_{i=1}^{n} \alpha_i A v_i$$

Because of the eigenbasis we get

$$Ax = \sum_{i=1}^{n} \alpha_i \lambda_i v_i$$

Resulting in

$$Ax - cx = \sum_{i=1}^{n} \alpha_i \lambda_i v_i - \sum_{i=1}^{n} \alpha_i c v_i$$

Which results in

$$\sum_{i=1}^{n} \alpha_i v_i (\lambda_i - c) = d$$

Using the inner product

$$\|d\|_2^2 = \sum_{i=1}^{n} \alpha_i^2 (\lambda_i - c)^2$$

Turning our focus to the minimal of the distance $\min_{i=1,\dots,n} |\lambda_i - c|$, let us define m as the number associated with the minimum.

This gives us the following:

$$\begin{aligned} \sum_{i=1}^{n} \alpha_i^2 (\lambda_i - c)^2 &\geq m^2 \sum_{i=1}^{n} \alpha_i^2 \\ \frac{\|d\|^2}{\|x\|^2} &\geq m^2 \\ \frac{\|d\|}{\|x\|} &\geq \min_{i=1,\dots,n} |\lambda_i - c| \end{aligned}$$

$\mathfrak{QED}$

41.29 Eigenvalue of Block Matrices

Given a block matrix

$$A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix} \in C^{n \times n}, \quad B \in C^{p \times p}, \quad C \in C^{p \times (n-p)}, \quad D \in C^{(n-p) \times (n-p)}$$

Its eigenvalues are the eigenvalues of B and D.

Proof:

This fact comes from the determinant of a block matrix

$$\det(A - \lambda I) = \det(B - \lambda I) \cdot \det(D - \lambda I)$$

$\mathfrak{QED}$

41.30 Roth's Removal Rule

The matrices

$$A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix} \quad \tilde{A} = \begin{bmatrix} B & 0 \\ 0 & D \end{bmatrix}$$

are similar

$$A = V\tilde{A}V^{-1}$$

if and only if the *Sylvester equation* has a solution.

$$BX - XD = C$$

In such a case, the matrix V is given by

$$V = \begin{bmatrix} I & X \\ 0 & I \end{bmatrix}, \quad V^{-1} = \begin{bmatrix} I & -X \\ 0 & I \end{bmatrix}$$

41.31 Eigenvalues/Vectors of Matrices Representing Complex Numbers

Given a complex number $z = a + ib$ we can represent it as a matrix

$$A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

Its eigenvalues are z and $\bar{z}$ and the corresponding eigenvectors are

$$v_z = \begin{bmatrix} \frac{a}{b} - \frac{a-ib}{b} \\ 1 \end{bmatrix}, \quad v_{\bar{z}} = \begin{bmatrix} \frac{a}{b} - \frac{a+ib}{b} \\ 1 \end{bmatrix}$$

41.32 Real Jordan Form

For every matrix $A \in \mathbb{R}^{n\times n}$ there exists a regular matrix $V \in \mathbb{R}^{n\times n}$ such that

$$A = VJV^{-1}, \quad J = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_k \end{bmatrix} \quad J_i = \begin{bmatrix} \Lambda_i & I & & \\ & \ddots & \ddots & \\ & & \Lambda_i & I \\ & & & \Lambda_i \end{bmatrix}$$

For $\lambda_i \in \mathbb{R}$ is $\Lambda_i = \lambda_i$ else

$$\Lambda_i = \begin{bmatrix} a_i & b_i \\ -b_i & a_i \end{bmatrix}$$

where $\lambda_i = a_i \pm jb_i \in \mathbb{C}$ are the complex conjugate eigenvalues of A.

41.33 Real Schur Form

For every matrix $A \in \mathbb{R}^{n\times n}$ there exists an orthogonal matrix $Q \in \mathbb{R}^{n\times n}$ such that

$$A = QTQ^T, \quad T = \begin{bmatrix} T_{1,1} & T_{1,2} & \cdots & T_{1,m} \\ 0 & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & T_{m-1,m} \\ 0 & 0 & 0 & T_{m,m} \end{bmatrix}$$

where $T_{i,i} \in \mathbb{R}$ or $T_{i,i} \in \mathbb{R}^{2\times 2}$.

41.34 Orthonormal Standard Form

A matrix $A \in \mathbb{R}^{2\times 2}$ with complex eigenvalues $\lambda = a + bi$, $\bar{\lambda} = a - bi$ is then transformable to the *orthonormal standard form*

$$T = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

if A is normal.

41.35 Criterion for Orthonormal Diagonalisation

A matrix $A \in \mathbb{R}^{n\times n}$ is orthonormal diagonalisable i.e

$$A = QDQ^T, \quad D = \begin{bmatrix} \Lambda_1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 0 \\ 0 & \cdots & 0 & \Lambda_n \end{bmatrix}, \quad \Lambda_i \in \mathbb{R} \text{ or } \Lambda_i = \begin{bmatrix} a_i & b_i \\ -b_i & a_i \end{bmatrix}$$

if and only if A is normal, i.e. $AA^T = A^TA$.

Example:

The matrix

$$A = \begin{bmatrix} 41 & 12 & 60 \\ 12 & 34 & 45 \\ -60 & 45 & 50 \end{bmatrix}$$

is normal; hence it is unitary diagonalisable $A = U\Lambda U^H$.

Its eigenvalues are $\lambda = 25$, $\lambda = 50 - 75i$ and $\lambda = 50 + 75i$.

$$\Lambda = \begin{bmatrix} 25 & 0 & 0 \\ 0 & 50 - 75i & 0 \\ 0 & 0 & 50 + 75i \end{bmatrix}$$

The eigenvectors will be skipped for the sake of brevity, but there are three of them.

We can diagonalize A as stated in the criterion for orthonormal diagonalisation.

$$\Lambda = \begin{bmatrix} 25 & 0 & 0 \\ 0 & 50 & -75 \\ 0 & 75 & 50 \end{bmatrix}$$

Note that the complex eigenvalues are represented as 2×2 blocks in the diagonal of Λ.

Then Q is given by the matrix of the corresponding eigenvectors of A and we can verify that $A = Q\Lambda Q^T$.

$$Q = \begin{bmatrix} \frac{3}{5} & 0 & \frac{4}{5} \\ -\frac{4}{5} & 0 & -\frac{3}{5} \\ 0 & 1 & 0 \end{bmatrix}$$

41.36 Normal Property of Symmetric Matrices

All real symmetric matrices are normal, and therefore they are orthonormal diagonalisable, and all of their eigenvalues are real by the properties of hermitian matrices.

Chapter 42

Dual Spaces and Dual Vectors

Given a $\mathbb{F}$-vector space V, we define V^*, the *dual space* of V, as the set of linear transformations $f : V \to K$. f is called the *linear functional*.

Note, the vectors inside the space V^* are called *dual vectors* or *covariant vectors*, while vectors in the space V are called *contravariant vectors*.

Example:

$V = \mathbb{R}^3$, with $f : V \to \mathbb{R}$ and $f(x, y, z) = x + y + z$

42.1 The Dual Basis

Given a finite dimensional $\mathbb{F}$-vector space V, let $\beta = \{v_1, \ldots, v_n\}$ be a basis of V, and let $f \in V^*$. The set of functionals $\beta^* = \{f_1, \ldots, f_n\}$ of V^*, defined by:

$$f_i(v_j) = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases} = \delta_{ij}$$

is called a *dual basis*. The definition above is saying that only one of the basis vectors of V is going to be associated with a function of the basis β.

Proof:

To prove a set of vectors is a basis, we have to prove that they are linearly independent and that they span the entire vector space.

Linear Independence:

We want that

$$\sum_{i=1}^{n} \lambda_i f_i(v) = 0 \text{ only if all } \lambda_i = 0 \text{ for all } v \in V$$

Now just choose n-times $v = v_j, 1 \leq j \leq n$ and you are going to realize that because $f_i(v_j) = 1$ only if $j = i$, you are going to get $\lambda_j f_j(v_j) + 0 + \cdots + 0 = 0$. Therefore, by repeating this process you will arrive at the conclusion that all lambdas have to be zero. Thus, the linear independence is shown.

Span:

Suppose $f \in V^*$ is arbitrary and $f = \sum_{i=1}^{n} \lambda_i f_i(v)$ for all $v \in V$. Using the previously collected knowledge, which tells us that for $v = v_j$ the linear combination of $\sum_{i=1}^{n} \lambda_i f_i(v_j)$ is

$$\sum_{i=1}^{n} \lambda_i f_i(v_j) = \lambda_j f_j(v_j) = \lambda_j$$

Now let us guess $f = f(v_1)f_1 + \cdots f(v_n)f_n = g$. If $f(v_i) = g(v_i)$ for all basis vectors of V, then $f = g$.

Note for

$$g(v_i) = f(v_1)f_1(v_i) + \cdots + f(v_i)f_i(v_i) + \cdots f(v_i)f_n(v_i)$$

All terms are zero except for $f(v_i)f_i(v_i)$ which equals $f(v_i) = g(v_i)$.

$\mathfrak{QED}$

42.1.1 How to Find a Dual Basis

We know that to find the coefficients of a linear combination $f = a_1f_1 + a_2f_2 + \cdots + a_nf_n$ in the space $V*$ with the basis $\beta = \{f_1, \ldots, f_n\}$ we just need to evaluate the function f at each of the basis vectors of the space V which are $\{v_1, \ldots, v_n\}$.

The process for solving this kind of problem can be summarized in the following steps using multiple systems of equations.

1. For each f_i repeat the following steps except for the last one.
2. Use the linearity property of vectors to write non-canonical basis vectors as a linear combination $f_i(\vec{k}) = \sum \lambda_k f_i(\vec{v}_k)$, do this for every vector in the basis of V with the current f_i. This will set up the left-hand side of the system of equations.
3. Use the definition of the dual basis for setting up the right-hand side of the linear system
4. Solve the system of equations
5. If an f was given, then use the found basis to write $f = \sum_{i=1}^{n} f(v_i)f_i$

Example:

Given the space $V = \mathbb{R}^n$ with the basis $\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}$ find $\beta^* = \{f_1, f_2\}$.

Because of linearity

$$\begin{aligned} f_1(2,1) &= 2f_1(\vec{e}_1) + 1f_1(\vec{e}_2) = 1 \\ f_1(3,1) &= 3f_1(\vec{e}_1) + 1f_1(\vec{e}_2) = 0 \end{aligned}$$

We get a linear system of equations. By solving the system, we get $f_1(1,0) = -1$ and $f_2(0,1) = 3$. Thus, $f_1(x,y) = x(-1) + y(3)$.

And for f_2 it is the same process

$$\begin{aligned} f_2(2,1) &= 2f_1(\vec{e}_1) + 1f_1(\vec{e}_2) = 0 \\ f_2(3,1) &= 3f_1(\vec{e}_1) + 1f_1(\vec{e}_2) = 1 \end{aligned}$$

This time we get $f_2 = 2x - 5y$. Thus, our basis is $\beta^* = \{-x + 3y, 2x - 5y\}$, and if we let f be $f = 2x - 5$, we get that it can be decomposed into:

$$\begin{aligned} f &= f(2,1)f_1 + f(3,1)f_2 \\ &= (-1)f_1 + 1f_2 \end{aligned}$$

42.2 Change of Basis in the Dual Space

Given some finite $\mathbb{F}$-vector space V with basis β and β'. Then the matrix for the *change of basis* $\beta^* \to \beta'^*$ is given by

$$(T^{-1})^T = (T^T)^{-1}$$

where $T = \beta^{-1}\beta'$.

Example:

Given are $\beta = \{(1,1);(-1,1)\}$ and $\beta' = \{(2,3),(1,1)\}$.

$$T = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}$$

Then

$$(T^T)^{-1} = \begin{pmatrix} 0 & 1 \\ 2 & -5 \end{pmatrix}$$

42.3 Dimension Theorem

If the dimension of V is finite, then V and V^* are *isomorphic*.

42.4 The Transpose of a Linear Transformation

Let V and W be finite dimensional vector spaces with the field K. Let $T : V \to W$ be a linear transformation. Now define

$$T^T : W^* \to V^*$$

with an input $g \in W^*$ where $g : W \to K$; and an output $T^T(g) \in V^*$ with $T^T(g) : V \to K$.

Therefore, $T^T(g)(\vec{x}) = g(T(\vec{x}))$ for all $\vec{x} \in V$.

42.5 Connection Between the Dual and the Transpose Matrix

Let $T : V \to W$ be a linear transformation, $\beta = \{v_1, \ldots, v_n\}$ be a basis of V, $\gamma = \{w_1, \ldots, w_n\}$ be a basis of W. The matrix of the change of the transformation is given by $A = [T]^{\gamma}_{\beta}$, which takes vectors from β to γ.

Then let $\beta^* = \{f_1, \ldots, f_n\}$ be the dual basis of V^* and $\gamma^* = \{g_1, \ldots, g_n\}$ the dual basis of W^*. Then

$$[T^T]^{\beta^*}_{\gamma^*} = A^T$$

Which takes functions from γ^* to β^*.

Proof:

Because $T(v_i) \in W$ we can write

$$T(v_i) = sum^n_{k=1} A_{ki} w_k$$

as a vector in terms of the basis γ.

For $T^T : W^* \to V^*$ we can write $T^T(g)$ as a linear combination of the basis vectors β^*

$$T^T(g_j) = \sum_{i=1}^{n} T^T(g_j)(v_i) f_i = \sum_{i=1}^{n} g(T(v_i)) f_i = \sum_{i=1}^{n} A_{ij} f_i$$

where $A = (T^T(g_1), \ldots, T^T(g_n)))$.

We have assumed that $T^T(g)(\vec{x}) = g(T(\vec{x}))$. Now let us apply this to any $v_j \in V$ and $g_i \in W^*$

$$\begin{aligned} T^T(g_j)(v_i) &= g_j(T(v_i)) \\ &= g_j\left(\sum_{k=1}^{n} A_{ki} w_k\right) = \sum_{k=1}^{n} A_{ki} g_j(w_k) \\ &= A_{ji} \end{aligned}$$

This sum is equal to A_{ji} when $g_j(w_j)$. Therefore, we have flipped the columns and rows. Thus, we have shown that

$$[T^T]_{\gamma^*}^{\beta^*} = A^T$$

Because we have shown it for the basis vectors, it gets extended to all of its linear combinations.

$\mathfrak{QED}$

Example:

Let $P_1\mathbb{R}$ be the space of polynomials of degree at most 1 and $W = \mathbb{R}^2$. Also, let $T : V \to W$ be defined by $T(p) = \begin{bmatrix} p(0) - 2p(1) \\ p(0) + p'(0) \end{bmatrix}$

For example $T(1 + 2x) = \begin{bmatrix} 1 - 2(1+2) \\ 1 + 2 \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$

If $f(a, b) = a - 2b \in W^*$ find $T^T(f)$.

The output of $T^T(f)$ is in V^* and because we are using the transpose we can write it as $T^T(f)(p) = f(T(p))$ and

$$\begin{aligned} f(a,b) &= a - 2b \text{ then} \\ f(T(p)) &= f((p(0) - 2p(1), p(0) + p'(0))) \\ &= (p(0) - 2p(1)) - 2(p(0) + p'(0)) \\ &= p(0) - 2p(1) - 2p(0) - 2p'(0) \\ T^T(f)(p) &= -p(0) - 2p(1) - 2p'(0) \end{aligned}$$

Now calculate the matrix of T and T^T. The basis of V is $\{1, x\}$ and the basis of W is $\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\}$.

First we know that $\beta^* = \{f_1, f_2\}$ and $\gamma^* = \{g_1, g_2\}$, and we remember that

$$f_i(v_j) = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases} = \delta_{ij}$$

To find the matrix of T^T, which is a 2×2 matrix

$$T^T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

We need to evaluate $T^T(g_1)$ and $T^T(g_2)$ and then write them in terms of β^*.

$$T^T(g_1) = af_1 + cf_2$$

Now we evaluate at the basis vectors of V

$$T^T(g_1)(1) = af_1(1) + cf_2(1) = a$$

Now we use the definition of T

$$\begin{aligned} g_1(T(1)) &= g_1((\begin{bmatrix} 1-2 \\ 1+0 \end{bmatrix}) = g_1(\begin{bmatrix} -1 \\ 1 \end{bmatrix}) \\ &= g_1(-1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 0 \\ 1 \end{bmatrix}) = -1 \end{aligned}$$

Thus, we have $a = -1$.

Now let's repeat this for x

$$T^T(g_1)(x) = af_1(1) + cf_2(x) = c$$

$$\begin{aligned} g_1(T(x)) &= g_1((\begin{bmatrix} 0-2 \\ 0+1 \end{bmatrix}) = g_1(\begin{bmatrix} -2 \\ 1 \end{bmatrix}) \\ &= g_1(-2\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 0 \\ 1 \end{bmatrix}) = -2 \end{aligned}$$

Thus, we have $c = -2$.

Now if you repeat this for g_2, you will get $b = 1$ and $d = 0$.

Thus, the matrix of T^T is

$$T^T = \begin{bmatrix} -1 & 1 \\ -2 & 0 \end{bmatrix}$$

And the matrix of T is calculated by applying $T(1)$ and $T(x)$ and then writing them in terms of the basis γ.

This gives us

$$T(1) = \begin{bmatrix} 1-2 \\ 1+0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Which is $-1\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

And

$$T(x) = \begin{bmatrix} 0-2 \\ 0+0 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

Which is $-2\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Thus, the matrix of T is

$$T = \begin{bmatrix} -1 & -2 \\ 1 & 0 \end{bmatrix}$$

42.6 The Dual of the Dual Space

Given a dual space $V^* = L(V, \mathbb{F})$ then

$$(V^*)^* = L(V^*, \mathbb{F})$$

We also define the linear functional $\varphi : V \to V^{**}$ where $\varphi(x) = \hat{x}$ and $\hat{x}(f) = f(x)$ for all $f \in V$.

42.6.1 Isomorphism of the Dual Space

If V is finite dimensional, then V^{**} and V are isomorphic.

42.6.2 Isomorphism From V to V^{**}

If V is finite dimensional, then the map $\phi : V \to V^{**}$ is an isomorphism.

42.7 Basis Theorem for the Dual Space

Every basis β^* of V^* is the dual basis of some basis β of V.

Proof:

Let $\beta^* = \{f_1, \ldots, f_n\}$ be a basis of V^*. We can define a basis β of V with the help of the following:

Let us find a basis $\beta^{**} = \{\mathcal{F}_1, \ldots, \mathcal{F}_n\}$ of V^{**}. The functional $\varphi : V \to V^{**}$ is an isomorphism and $\mathcal{F}_i = \hat{v}_i$ for some $v_i \in V$.

Let us choose a basis $\beta = \{v_1, \ldots, v_n\}$ of V and use β^*. By definition

$$f_i(v_j) = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases} = \delta_{ij}$$

Using the isomorphism φ we get

$$f_j(v_i) = \hat{v}_i(f_j) = \mathcal{F}_i(f_j) = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases} = \delta_{ij}$$

$\mathfrak{QED}$

42.7.1 Transpose of the Transpose Is T

We define $T^{TT} V^{**} \to W^{**}$ by

$$T^{TT}(\hat{x}) = \hat{x}(T^T)$$

and claim $T^{TT} = T$.

Proof:

In this proof we know that $T^T : W^* \to V^*$. Now let us declare our basis $\beta = \{v_1, \ldots, v_n\}$ for V, $\gamma = \{w_1, \ldots, w_n\}$ for W, $\beta^* = \{f_1, \ldots, f_n\}$ for V^*, $\gamma^* = \{g_1, \ldots, g_n\}$ for W^*, $\beta^{**} = \{\mathcal{F}_1, \ldots, \mathcal{F}_n\}$ for V^{**} and finally $\gamma^{**} = \{\mathcal{G}_1, \ldots, \mathcal{G}_n\}$ for W^**.

When we evaluate at $g : W \to \mathbb{F}$, we get

$$\begin{aligned} T^{TT}(\hat{x})(g) &= \hat{x}(T^T)(g) \\ &= \hat{x}(T^T(g)) \text{ by associativity} \\ &= T^T(g)(x) \text{ using the definition of } \hat{x} \\ &= g(T)(x) \text{ by associativity} \\ &= g(T(x)) \\ &= \hat{T(x)}(g) \end{aligned}$$

Here $\hat{T(x)} \in W^{**}$. Finally

$$\begin{aligned} T^{TT}(\hat{x})(g) &= \hat{T(x)}(g) \\ T^{TT}(\hat{x}) &= \hat{T(x)} \\ T^{TT} &= \hat{T} \\ T^{TT} &= T \text{ by ignoring the hats} \end{aligned}$$

They are not strictly the same, but because of the isomorphism we can set them equal.

$\mathfrak{QED}$

42.8 The Annihilator

We define the *Annihilator* of a subspace of V as

$$W^\circ = \{f \in V^* \mid f(w) = 0, \forall w \in W\}$$

We also get the following formula

$$\dim(W^\circ) + \dim(W) = \dim(V)$$

whose proof is similar to the one of the rank nullity theorem.

Proof:

Let $\{w_1, \ldots, w_{k-1}, w_k, w_{k+1}, \ldots, w_n\}$ be a basis of W with dimension k. Let us extend the basis to a basis of V by $\beta = \{w_1, \ldots, w_{k-1}, w_k, w_{k+1}, \ldots, w_n\}$ with dimension n.

Now let us claim that the vectors $f_{k+1}, \ldots, f_n$ are a basis of W°. This implies that for every linear combination of vectors from v_1 to v_k which equals w.

$$f_{k+1}(w) = \sum_{i=1}^{k} f_{k+1}(\lambda_i v_i) = 0$$

Therefore, $f_{k+1}, \ldots, f_n$ are in W°. Now we have to show that they are linearly independent.

For $f \in W^\circ$

$$f(x) = \sum_{i=0}^{n} f(v_i) f_i(x) = f(v_{k+1}) f_{k+1}(x) + \ldots f(v_n) f_n(x)$$

Therefore, we get that for some f in the annihilator we get a linear combination of our chosen basis.

$\mathfrak{QED}$

42.9 The Duality Argument

$$T \text{ is surjective} \iff T^T \text{ is injective}$$

Proof:

Let us assume the opposite

$$\neg T \text{ is surjective} \iff \neg T^T \text{ is injective}$$

If T is not surjective we can assume that the image of it is a strict subset of W which we will call W'. We can also define a function $g \in W^*$ which is not 0 everywhere except for the vectors in W'.

Then if we use the definition of T^T which tells us that when evaluating at g which is not the zero functional gives us

$$T^T(g)(x) = g(T(x)) = 0 \text{ because } T(x) \in W'$$

This concludes the proof in one way because we have found some linear combination for the zero vector different from the trivial solution.

For the other way of the if and only if, we know that T^T is not injective. This means that multiple vectors get transformed into the zero vector under T^T. Therefore, if suppose T is surjective and

$$\begin{aligned} T^T(g)(x) &= 0 \text{ for some } g \neq 0 \\ g(T(x)) &= 0, \quad \forall x \in V \end{aligned}$$

Then for $y = T(x)$ it is always 0, but because we have assumed that $g \neq 0$, we have a contradiction. T must not be surjective, and by getting rid of the negation on both sides we have proven the statement.

$\mathfrak{QED}$

42.10 The Rank of a Transpose Is Equal to the Rank of A

For this proof we will claim:

$$(T^T) = (\text{img}(T))^\circ.$$

The right part tells us that for some $g \in (T^T)$, $T^T(g) = f_0$. But because we can write

$$T^T(g)(x) = g(T(x)) = 0 \text{ where } T(x) \in \text{img}(T).$$

We know that g is in the annihilator of the image of T.

With this knowledge we can now prove $\text{rank}(A) = \text{rank}(A^T)$ using the rank nullity theorem

$$\begin{aligned} \dim((T^T)) + \dim(\text{img}(T^T)) &= \dim(W^*) = \dim(W) \\ \dim((\text{img}(T^\circ)) + \dim(\text{img}(T^T)) &= \dim(W) \end{aligned}$$

For $\dim(W)$ we get

$$\begin{aligned} \dim((\text{img}(T)^\circ)) + \dim(\text{img}(T)) &= \dim(W) \\ \dim((\text{img}(T)^\circ)) &= \dim(W) - \dim(\text{img}(T)) \end{aligned}$$

By substituting back into the original equation

$$\begin{aligned} \dim((\text{img}(T)^\circ)) + \dim(\text{img}(T^T)) &= \dim(W) \\ \dim(W) - \dim(\text{img}(T)) + \dim(\text{img}(T^T)) &= \dim(W) \\ \dim(\text{img}(T^T)) &= \dim(\text{img}(T)) \end{aligned}$$

$\mathfrak{QED}$

Chapter 43

Tensors

A *tensor* T of rank(k, l) is a multilinear map from k dual vectors and l vectors to the $\mathbb{F}$.

$$T : V^* \times \cdots \times V^* \times V \times \cdots \times V \to \mathbb{F}.$$

Example:

Suppose $k = 1, \quad l = 1$ and our field is $\mathbb{R}$. Then we have a tensor $T(\varphi, v) \to \mathbb{R}$, where $\varphi = \alpha f + \beta g, f, g \in V^*$ and $v = \gamma v_1 + \lambda v_2, v_1, v_2 \in V$ and $\alpha, \beta, \gamma, \lambda \in \mathbb{R}$. We get

$$T(\varphi, v) = T(\alpha f + \beta g, \gamma v_1 + \lambda v_2) = \alpha\gamma T(f, v_1) + \alpha\lambda T(f, v_2) + \beta\gamma T(g, v_1) + \beta\lambda T(g, v_2)$$

where each term is a real number.

More examples: a rank$(0, 1)$ tensor is a dual vector, a rank$(1, 0)$ tensor is a vector, and finally a scalar is rank$(0, 0)$.

Chapter 44

The Exponential Function and Relatives

The *exponential function* and its related functions, such as sine, cosine, and the hyperbolic functions, are fundamental in both pure and applied mathematics. This chapter explores these functions through their series expansions and interrelationships.

$$\exp(x) := e^x := \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n$$

44.1 Exponential Function as a Series

The *exponential function* e^x is defined as an infinite series:

$$e^x := \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

This series converges for all real and complex values of x.

44.2 Properties of the Exponential

- $\exp(x) < \exp(y), \forall x, y \in \mathbb{R} : x < y$
- $\exp(x) \geq 1 + x, \forall 1 + x$
- $|\exp(x) - 1| \leq \frac{|x|}{1-|x|}$ for all $x \in \mathbb{R}$ such that $|x| < 1$
- For a large enough x: $\exp(x) > x^n, \forall n \in \mathbb{N}$ and $x > 4n^2$

44.3 General Definition

For $a > 0$ and $x \in \mathbb{R}$, we define

$$a^x := \exp(x \ln(x)) = e^{x \ln(a)}.$$

44.4 Difference of Exponentials

For $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$a^n - b^n = (a-b)\sum_{k=1}^{n} a^{n-k}b^{k-1}.$$

44.5 Sine and Cosine Definitions With e

Using Euler's identities and Taylor series, *sine and cosine* can be defined in terms of exponential functions:

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}, \quad \cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

These definitions are consistent with their Taylor expansions:

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad \cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

44.6 Euler's Formula

One of the most elegant formulas in mathematics is *Euler's formula*:

$$e^{ix} = \cos(x) + i\sin(x)$$

This identity provides a deep connection between exponential and trigonometric functions and is valid for all real (and complex) x.

Special Case:

$$e^{i\pi} + 1 = 0$$

This is known as Euler's Identity and is often celebrated as one of the most beautiful equations in mathematics.

44.7 Reason for the Infinite Series of e (Differentiation)

The exponential series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

has the remarkable property that its derivative is the same function:

$$\frac{d}{dx}e^x = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!}$$

Now replace the index $m = n - 1$:

$$= \sum_{m=0}^{\infty} \frac{x^m}{m!} = e^x$$

This shows that e^x is the unique function (up to constant multiples) that is equal to its own derivative.

44.8 Hyperbolic Functions With e

The *hyperbolic functions* are analogs of the trigonometric functions but defined using the exponential function:

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$
$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$
$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

These functions satisfy identities similar to those of trigonometric functions, for example:

$$\cosh^2(x) - \sinh^2(x) = 1$$

They naturally arise in the study of differential equations, special relativity, and the geometry of hyperbolas.

44.9 Inverse of the Hyperbolic Trigonometric Functions

- $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$
- $\cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1})$
- $\tanh^{-1}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$
- $\coth^{-1}(x) = \frac{1}{2\ln\left(\frac{x+1}{x-1}\right)}$
- $\text{sech}^{-1} = \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right)$
- $\text{csch}^{-1} = \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right)$

Chapter 45

Growth Models

Growth models are fundamental to understanding population dynamics, resource consumption, and many natural and artificial processes. Two primary models of growth are the **exponential** and the **logistic** growth models. Each of these has discrete and continuous forms.

45.1 Exponential Growth

Exponential growth describes a process that increases at a rate proportional to its current value.

45.1.1 Discrete Exponential Growth

$$P_n = P_0 \cdot (1 + r)^n,$$

where:

- P_n is the population at time step n.
- P_0 is the initial population.
- r is the growth rate per time step.
- n is the number of time steps.

Example:

If $P_0 = 100$ and $r = 0.1$, after 5 steps:

$$P_5 = 100 \cdot (1 + 0.1)^5 = 100 \cdot 1.61051 \approx 161.05.$$

45.1.2 Continuous Exponential Growth

$$P(t) = P_0 \cdot e^{rt},$$

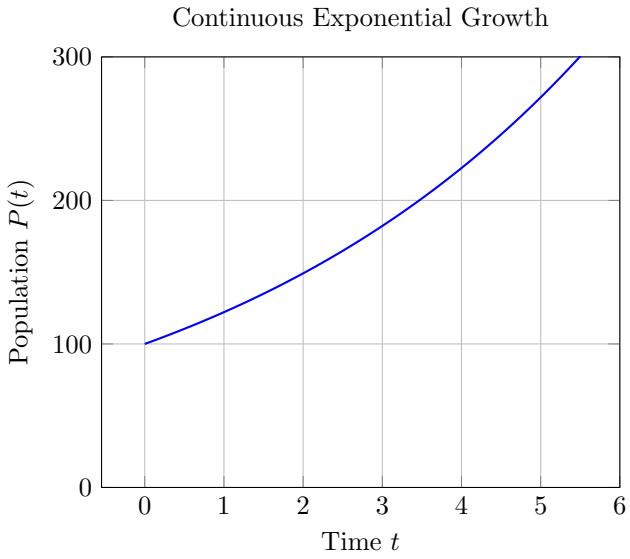
where:

- $P(t)$ is the population at time t.
- e is Euler's number (≈ 2.718).
- r is the continuous growth rate.

Example:

If $P_0 = 100$, $r = 0.1$, and $t = 5$:

$$P(5) = 100 \cdot e^{0.5} \approx 100 \cdot 1.64872 \approx 164.87.$$



45.2 Logistic Growth

Logistic growth occurs when a population grows rapidly at first and then slows as it approaches a maximum sustainable size (carrying capacity).

45.2.1 Discrete Logistic Growth

$$P_{n+1} = P_n + r \cdot P_n \cdot \left(1 - \frac{P_n}{K}\right),$$

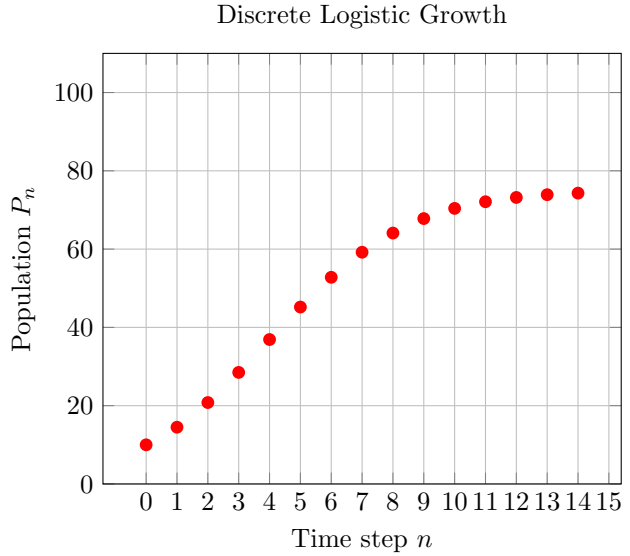
where:

- K is the carrying capacity
- r is the intrinsic growth rate

Example:

With $P_0 = 10$, $r = 0.5$, $K = 100$:

$$P_1 = 10 + 0.5 \cdot 10 \cdot \left(1 - \frac{10}{100}\right) = 10 + 0.5 \cdot 10 \cdot 0.9 = 10 + 4.5 = 14.5.$$



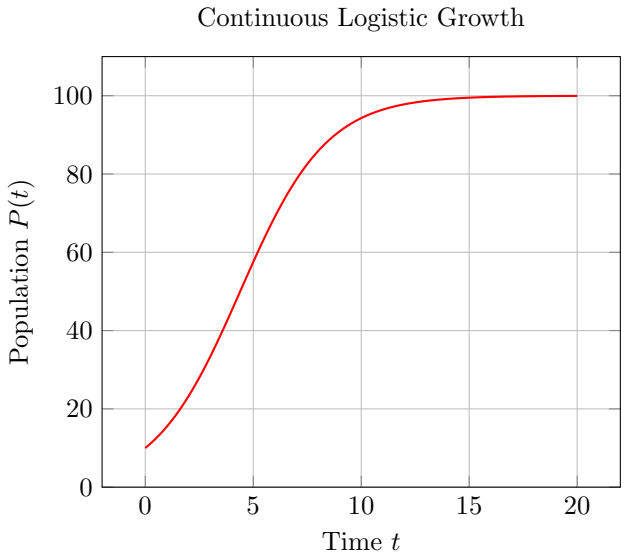
45.2.2 Continuous Logistic Growth

$$P(t) = \frac{K}{1 + \left(\frac{K - P_0}{P_0}\right) e^{-rt}}.$$

Example:

With $P_0 = 10$, $K = 100$, and $r = 0.5$, at $t = 5$:

$$P(5) = \frac{100}{1 + \left(\frac{90}{10}\right) e^{-0.5 \cdot 5}} = \frac{100}{1 + 9e^{-2.5}} \approx \frac{100}{1.7387} \approx 57.52.$$



Chapter 46

Asymptotes

An *asymptote* of a curve is a line that the curve approaches but never touches as the input or output grows infinitely large in magnitude. There are three main types of asymptotes: vertical, horizontal, and oblique (slant). Each one describes a different long-term behavior of a function.

46.1 Vertical Asymptotes

A *vertical asymptote* is a vertical line $x = a$ where the function grows without bound as x approaches a from the left or the right.

$$\lim_{x \to a^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \to a^+} f(x) = \pm\infty$$

46.1.1 How to Find Vertical Asymptotes

1. Find values of x that make the denominator of a rational function zero.
2. Eliminate removable discontinuities (common factors).
3. For each remaining zero in the denominator, check if the limit tends to infinity.

Example:

$$f(x) = \frac{1}{x-2} \quad \Rightarrow \quad x = 2 \text{ is a vertical asymptote.}$$

46.2 Horizontal Asymptotes

A *horizontal asymptote* is a horizontal line $y = L$ that the function approaches as x tends to infinity or negative infinity:

$$\lim_{x \to \pm\infty} f(x) = L$$

46.2.1 How to Find Horizontal Asymptotes (Rational Functions)

1. Compare the degrees of the numerator (n) and denominator (m):
   - If $n < m$: Horizontal asymptote at $y = 0$
   - If $n = m$: Horizontal asymptote at $y = \frac{\text{leading coefficient of numerator}}{\text{leading coefficient of denominator}}$
   - If $n > m$: No horizontal asymptote (check for slant)

Example:

$$f(x) = \frac{2x^2 + 3}{x^2 - 1} \quad \Rightarrow \quad y = \frac{2}{1} = 2 \text{ is the horizontal asymptote.}$$

46.3 Oblique (Slant) Asymptotes

An *oblique or slant asymptote* occurs when the degree of the numerator is exactly one more than the degree of the denominator.

$$\lim_{x \to \infty} [f(x) - (mx + b)] = 0$$

46.3.1 How to Find Oblique Asymptotes

1. Perform polynomial long division (or synthetic division) on the rational function.
2. The quotient (ignoring the remainder) is the equation of the slant asymptote.

Example:

$$f(x) = \frac{x^2 + 1}{x - 1}$$

Long division gives:

$$\frac{x^2 + 1}{x - 1} = x + 1 + \frac{2}{x - 1} \Rightarrow \text{Slant asymptote: } y = x + 1.$$

46.4 Summary Table

Type	Equation	Occurs When
Vertical	$x = a$	Denominator $\to 0$ (non-removable)
Horizontal	$y = L$	$\lim_{x \to \pm\infty} f(x) = L$
Oblique	$y = mx + b$	Degree numerator = degree denominator + 1

Chapter 47

Sequences and Series

A *sequence* is an ordered list of numbers, usually defined by a formula or a recurrence relation. It is one of the fundamental objects of study in mathematical analysis.

47.1 Arithmetic Sequence

An *arithmetic sequence* is a sequence where the difference between consecutive terms is constant.

$$a_n = a_1 + (n-1)d$$

- a_1 is the first term.
- d is the common difference $d = a_{n+1} - a_n$.

Example:

$$a_n = 3 + (n-1) \cdot 2 = 2n + 1 \quad \text{(Odd numbers)}$$

47.2 Geometric Sequence

A *geometric sequence* is a sequence where each term is obtained by multiplying the previous term by a fixed non-zero constant.

$$a_n = a_1 \cdot r^{n-1}$$

- a_1 is the first term.
- r is the common ratio $r = \frac{a_{n+1}}{a_n}$.

Example:

$$a_n = 2 \cdot 3^{n-1} = 2, 6, 18, 54, \ldots$$

47.3 Function Sequence

A *function sequence* is a sequence composed of functions and is denoted as:

$$(f)_n \text{with } x^t := \{f : D \to R |\ f \text{ is a function}\}$$

47.4 Boundness of a Sequence

To show the *boundness* of a sequence, we can use the definition of the *supremum* or *infimum* and approximations.

One useful definition is that a sequence is bounded below and above if

$$|a_n| \leq M, \forall n \geq 1$$

Example:

Given is $a_n = \frac{1}{2n+3}$

$$\left|\frac{1}{2n+3}\right| \leq \frac{1}{2n+3} < \frac{1}{2n} < \frac{1}{n} \leq 1$$

Because we have found a concrete bound, we can state that the sequence is bounded.

47.5 Convergence

A sequence (a_n) *converges* to a limit L if for every $\varepsilon > 0$, there exists an integer n_0 such that for all $n \geq n_0$:

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : \forall n \geq N |a_n - L| < \varepsilon$$

In this case, we write:

$$\lim_{n \to \infty} a_n = L$$

This also means that $\{a_n - a\}_{n=1}^{\infty}$ is a zero sequence.

47.5.1 Properties of Convergent Sequences

Given the convergent sequences $\{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty}$ with a and b as their respective limits, and a bounded sequence $\{c_n\}_{n=1}^{\infty}$, the following applies:

- $\{a_n\}_{n=1}^{\infty}$ is bounded
- $\{\alpha a_n + \beta b_n\}_{n=1}^{\infty}, \alpha, \beta \in \mathbb{R}$ converges towards $\alpha a + \beta b$
- $\{|a_n|\}_{n=1}^{\infty}$ converges towards $|a|$ and $\{\sqrt{a_n}\}_{n=1}^{\infty}$ towards $\sqrt{|a|}$
- $\{a_n \cdot b_n\}_{n=1}^{\infty}$ converges towards $a \cdot b$
- $\{a_n \cdot c_n\}_{n=1}^{\infty}$ is bounded
- If $a_n \neq 0, \forall n \in \mathbb{N}$ and $a \neq 0$ then $\lim_{n \to \infty} \frac{1}{a_n} = \frac{1}{a}$

47.6 The Limit of a Sequence Is Unique

Given a convergent sequence a_n, its limit is unique.

Proof:

Suppose that a_n has two different limits a and b. Let $\varepsilon = \frac{|a-b|}{2} > 0$.

Because we have distinct limits, we write

$$\forall \varepsilon > 0 \exists i \in \mathbb{N} : \forall i \geq N_1 |a_i - a| < \frac{|a-b|}{2}$$

and

$$\forall \varepsilon > 0 \exists j \in \mathbb{N} : \forall j \geq N_2 |a_j - b| < \frac{|a-b|}{2}$$

Let $M = \max N_1, N_2$. Then

$$|a-b| = |a - a_n + a_n - b| \leq |a_n - a| + |a_n - b| < 2\varepsilon = \varepsilon > 0$$

Thus, $|a-b| < |a-b|$, which is a contradiction. The limit must be unique.

$\mathfrak{QED}$

47.7 Convergence by Power of One Half

If for a sequence a_n the following holds, then it converges:

$$|a_n - a_{n+1}| \leq \frac{1}{2^n}$$

47.8 Boundness of Convergent Sequences

Let a_n be a convergent sequence; then a_n is bounded.

Proof:

Because a_n converges, it has a limit a. Now, let $\varepsilon = 1$, and let us use the definition of convergence.

$$\begin{aligned} -\varepsilon < |a_n - a| &< \varepsilon \\ -1 < a_n - a &< 1 \\ a - 1, a_n &< 1 + a \text{ for all } n > n_0 \end{aligned}$$

Up to this point, we have proven the boundness for all $n > n_0$. To guarantee that the sequence is also bounded for $1 \leq n \leq n_0$, we define

$$U := \max(x \mid x \in a_n, n \leq n_0, a+1)$$

and

$$L := \min(x \mid x \in a_n, n \leq n_0, a-1)$$

Thus, our sequence is bounded:

$$L \leq a_n \leq U$$

$\mathfrak{QED}$

47.9 Cauchy Sequence

A sequence (a_n) is a *Cauchy sequence* if for every $\varepsilon > 0$, there exists an integer n_0 such that for all $n, m \geq n_0$:

$$\forall \varepsilon > 0 \exists n, m, N \in \mathbb{N} : |a_n - a_m| < \varepsilon \text{ with } n, m \geq N$$

Every convergent sequence is a Cauchy sequence. In complete metric spaces (like $\mathbb{R}$), the converse also holds.

Example: Convergence Exercise

Let $a_i = \frac{1}{i}$ and $\varepsilon = 0.01$. We want to find i_0 such that:

$$\left|\frac{1}{n} - 0\right| < \varepsilon \implies \frac{1}{i} \leq \frac{1}{n} < \varepsilon \to \frac{1}{\varepsilon} < n$$

$$\Rightarrow \frac{1}{i} < 0.01 \Rightarrow i > 100 \Rightarrow i_0 = 101$$

47.10 Boundness of Cauchy Sequence

Every Cauchy sequence is *bounded*.

47.11 Proof That Every Convergent Sequence Is a Cauchy Sequence

Given a sequence a_n, it is convergent if and only if it is a Cauchy sequence.

$\rightarrow$: a_n is convergent; therefore, the following holds:

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : \forall n \geq N |a_n - L| < \varepsilon$$

Thus, we can use the triangle inequality for our definition of the Cauchy sequence and the following definition:

$$|a_n - a| < \frac{\varepsilon}{2}$$
$$|a_m - a| < \frac{\varepsilon}{2}$$

This is allowed because we know that a_n converges. We choose the max of each of the indices for which the inequality holds. Then

$$|a_n - a_m + a - a| \leq |a_n - a| + |a_m - a| < 2\frac{\varepsilon}{2} = \varepsilon$$

$\leftarrow$: a_n is a Cauchy sequence, thus:

$$\forall \varepsilon > 0 \exists n, m, N \in \mathbb{N} : |a_n - a_m| < \varepsilon \text{ with } n, m \geq N$$

a_n is bounded because otherwise $|a_n - a_m| > \varepsilon$ for $n, m \geq N_0$. Therefore, by the Bolzano-Weierstraß theorem it must converge.

$\mathfrak{QED}$

47.12 Cauchy Sequences Are Convergent

Given a Cauchy sequence a_n, it converges.

Proof:

We know that a_n is bounded, and by the Bolzano-Weierstraß theorem it has a convergent subsequence.

$$\forall \varepsilon > 0 \exists n, m, N_1 \in \mathbb{N} : |a_n - a_m| < \frac{\varepsilon}{2} \text{with } n, m \geq N_1$$

And

$$\forall \varepsilon > 0 \exists n, m, N_2 \in \mathbb{N} : |a_{n_k} - a| < \frac{\varepsilon}{2} \text{with } n \geq N_2$$

Then let $N = \max N_1, N_2$; therefore

$$|a_n - a_m| = |a_n + a_{n_{N+1}} - a_{n_{N+1}} - a_m| \leq |a_n - a_{n_{N+1}}| + |a_m - a_{n_{N+1}}| = 2\frac{\varepsilon}{2} = \varepsilon$$

Thus, it converges.

$\mathfrak{QED}$

47.13 Definition of ε-Neighborhood

The *ε-neighborhood* of a point $a \in \mathbb{R}$ is the set:

$$B_\varepsilon(a) = \{x \in \mathbb{R} \mid |x - a| < \varepsilon\}$$

This is the open interval $(a - \varepsilon, a + \varepsilon)$.

47.14 Supremum, Infimum, Maximum and Minimum

The *supremum* sup of a set $A \subseteq \mathbb{R}$ is the least upper bound: the smallest real number s such that $a \leq s$ for all $a \in A$.

$$x \leq s, \forall x \in A$$

$$x \leq s', \forall x \in A \implies s' \geq s$$

The *maximum* is the largest element in the set, if it exists.

The *minimum* is the smallest element in the set, if it exists.

Finally, we define $\sup\emptyset := -\infty$ and $\inf\emptyset := +\infty$.

Example I:

For $A = (0, 1)$:

- $\sup A = 1$ (not in A).
- $\inf A = 0$ (not in A).
- $\max A$ and $\min A$ do not exist.

Example II:

For $(0, 5)$ and $(0, 5]$, the supremum is 5.

Example III:

For the real numbers we can define $\sqrt{2}$ as the supremum of the set $\{x \in \mathbb{R} \mid x^2 < 2\}$.

The *infimum* inf of a set A is the greatest lower bound: the largest number t such that $a \geq t$ for all $a \in A$.

$$x \geq s, \forall x \in A$$

$$x \geq s', \forall x \in A \implies s' \leq s$$

Example IV:

Prove that $\sup\left\{\frac{n}{n+1} \mid n \in \mathbb{N}\right\} = 1$.

To prove this, we need to show that 1 is an upper bound, and then that for all $\varepsilon > 0$, $a - \varepsilon$ is not an upper bound.

$$\begin{aligned} 1 &\geq \frac{n}{n+1} \\ n+1 &\geq n \end{aligned}$$

Thus, 1 is in fact an upper bound.

For the second part, let $\varepsilon > 0$, and we want to prove that $\exists n \in \mathbb{N} : \frac{n}{n+1} > 1 - \varepsilon$.

$$\begin{aligned} \frac{n}{n+1} &> 1 - \varepsilon \\ \varepsilon &> 1 - \frac{n}{n+1} \\ \varepsilon &> \frac{n+1-n}{n+1} \\ \varepsilon &> \frac{1}{n+1} \\ n+1 &> \frac{1}{\varepsilon} \\ n &> \frac{1}{\varepsilon} - 1 \end{aligned}$$

By the Archimedean principle, we know that we can always find a larger natural number such that $n > \frac{1}{\varepsilon} - 1$

Now we can reverse the steps in the inequality, and we will get our original inequality.

$\mathfrak{QED}$

47.14.1 Alternative Characterization of the Supremum

Given a non-empty set $A \subseteq \mathbb{R}$. Then $s \in \mathbb{R}$ is a supremum of A if and only if

$$(\nexists s \in A, s < x) \wedge (s' \in \mathbb{R}, s' < s \implies !\exists x \in A, x > s')$$

47.15 Properties of Suprema and Infima

For non-empty sets $A, B \subseteq \mathbb{R}^+$:

- $\sup AB = \sup A \cdot \sup B$
- $\inf AB = \inf A \cdot \inf B$
- $\sup(A + B) = \sup A + \sup B$
- $\inf(A + B) = \inf A + \inf B$
- $\sup(A/B) = \sup(A)/\inf(B)$
- $\sup(-A) = -\inf A$
- $\inf(-A) = -\sup A$

For the sake of completeness, we will prove one of these properties:

Proof:

By definition, $\forall ab \in AB$ the following holds:

$$ab \leq \sup A \cdot \sup B \implies \sup AB \leq \sup A \cdot \sup B$$

Because $\sup A$ and $\sup B$ do not necessarily belong to A or B. With this, we already have half of the proof. Now we need to prove that $\sup AB \geq \sup A \cdot \sup B$.

Start with

$$ab \leq \sup AB$$
$$a \leq \frac{\sup AB}{b}$$

Therefore, $\frac{\sup AB}{b}$ is an upper bound of A. This means that:

$$\sup A \leq \frac{\sup AB}{b} \implies \sup A \cdot b \leq \sup AB$$

From this we can also derive that

$$\sup A \cdot b \leq \sup AB \implies b \leq \frac{\sup AB}{\sup A}$$

Thus, because $\frac{\sup AB}{\sup A}$ is an upper bound of B we have that:

$$\sup B \leq \frac{\sup AB}{\sup A} \implies \sup A \cdot \sup B \leq \sup AB$$

Combining both parts of the proof we have that:

$$\sup AB = \sup A \cdot \sup B$$

$\mathfrak{QED}$

The other proofs can be proven using similar arguments.

47.16 Limit and Accumulation Point

- A *limit* of a sequence (a_n) is the value the terms get arbitrarily close to as $n \to \infty$.
- An *accumulation point* (or limit point) of a set A is a point x such that every neighborhood of x contains infinitely many points of A.

Example:

The sequence $a_n = (-1)^n + \frac{1}{n}$ has two accumulation points: 1 and -1.

47.17 The Monotone Convergence Theorem

Let a_n be a monotone sequence. Then a_n converges if and only if it is bounded.

Proof:

a_n converges; thus, there exists a limit, and therefore the sequence is bounded.

The sequence a_n is bounded and monotone. Therefore, there exist both an infimum and a supremum. We will prove the case for the supremum:

$$a \geq a_i \forall a_i \in a_n \implies a - \varepsilon < a_i$$

Suppose a is the limit; then for all $i > n_0, n_0 \in \mathbb{N}$

$$\begin{aligned} |a_i - a| &< \varepsilon \\ |a - a_i| &< \varepsilon \\ |a - a_i| = a - a_i &< \varepsilon \\ a - \varepsilon &< a_i \end{aligned}$$

Therefore, the supremum is the limit.

$\mathfrak{QED}$

47.18 Monotony

We define a real-valued function f as *monotone increasing/decreasing*:

$$x < y \implies f(x) \leq f(y)$$

$$x < y \implies f(x) \geq f(y)$$

We call it *strictly monotone increasing or decreasing* if $f(x) < f(y)$ or $f(x) > f(y)$.

47.19 Monotonicity by Difference and Quotient

A sequence is considered *monotone* if, for all $n \in \mathbb{N}$, either of the following holds

$$a_{n+1} \geq a_n \text{ or } a_{n+1} \leq a_n$$

Monotonicity by Difference:

- If $a_{n+1} - a_n \geq 0$ for all n, then the sequence is non-decreasing.
- If $a_{n+1} - a_n < 0$, it is non-increasing.

Monotonicity by Quotient: (Typically for positive sequences)

- If $\frac{a_{n+1}}{a_n} \geq 1$, then the sequence is non-decreasing.
- If $\frac{a_{n+1}}{a_n} < 1$, it is non-increasing.

47.20 Divergence

A sequence *diverges* if it does not converge. That is, there is no real number L to which it converges. If it diverges towards some infinity, then:

$$\forall M \in \mathbb{R} \exists n, N \in \mathbb{N} : |a_n| \geq |M| \forall n \geq N$$

- A sequence can diverge to ∞ or $-\infty$
- A sequence can also oscillate without approaching any limit

Example:

The sequence $a_n = (-1)^n$ diverges since it oscillates between 1 and -1.

47.21 Subsequences and Their Properties

A *subsequence* of a sequence (a_n) is a sequence (a_{n_k}) where (n_k) is a strictly increasing sequence of natural numbers.

Properties:

- Every subsequence of a convergent sequence converges to the same limit.
- A sequence converges if and only if all of its subsequences converge to the same limit.
- A bounded sequence always has at least one convergent subsequence (Bolzano-Weierstraß theorem).
- $n_k \geq k$

47.21.1 Existence of the Accumulation Point as Limit

A sequence $\{x_n\}_{n=1}^{\infty}$ has x_0 as an accumulation point if and only if there exists a subsequence which converges towards x_0.

Proof:

Suppose that x_0 is an accumulation point; then $(x_0 - 1, x_0 + 1)$ contains a sequence member $x_{n_0}, n_0 \in \mathbb{N}$. Let

$$n_1 := n_0$$

For each $k \in \mathbb{N}$, there exists some $N(k) \in \mathbb{N}$ such that $N(k) > N(k-1)$ and

$$x_{N(k)} \in (x_0 - \frac{1}{k}, x_0 + \frac{1}{k})$$

Now we set

$$n_k = N(k)$$

For each subsequence $\{x_{n_k}\}_{k=1}^{\infty}$, the following holds

$$|x_{n_k} - x_0| \leq \frac{1}{k}$$

Therefore, the subsequence also converges towards x_0.

For the other direction of the proof, if the sequence $\{x_{n_k}\}_{k=1}^{\infty}$ converges towards x_0, then there exist a $\delta > 0$ and $K \in \mathbb{N}$ such that

$$x_{n_k} \in (x_0 - \delta, x_0 + \delta), \forall k \geq K$$

Therefore, each neighborhood contains infinitely many sequence members.

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47.21.2 Monotone Subsequence

Every sequence a_n has a monotone subsequence.

Proof:

Let a_n be a sequence, and let a_m be a peak if for all $a_i \in a_n, i > m, m, i \in \mathbb{N}$ the following holds: $a_i \leq a_m$.

Now consider the case that a_n has infinitely many peaks. Let a_{n_k} be the sequence of the k-th peak. This gives us that each peak is greater than or equal to the previous one; therefore, we have a monotone increasing subsequence. If we define the peaks with $a_i \geq a_m$, we would also get a monotone sequence, but a decreasing one instead.

For the case that there are only a finite number of peaks, let a_n be the last peak $a_i \geq a_m$. We can always find another a_{i+1} such that $a_{i+1} \geq a_n$; therefore, we can find a monotone increasing subsequence. Similarly, the decreasing case can be shown.

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47.21.3 Convergence Due to Subsequences

A sequence $\{a_n\}_{n=1}^{\infty}$ converges to a if and only if each of its subsequences also converges to a.

Proof:

Let a_{n_k} be a subsequence of a_n. Let $\varepsilon > 0$.

$$\exists n_0 \in \mathbb{N}, |a_n - a| < \varepsilon \forall n > n_0$$

We want to show

$$\exists n_1 \in \mathbb{N}, |a_{n_k} - a| < \varepsilon \forall k > n_1$$

with $n_k \geq k$.

Let $n_1 = n_0$ then if $k > n_1$, $k > n_0$ and $n_k \geq k > n_0$. Thus, for all $k > n_1$, $|a_{n_k} - a| < \varepsilon$.

For the other direction of the proof, assume every subsequence of a_n converges to a. Since a_n is itself a subsequence of a_n, it converges to a.

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47.22 Proof That -1^n Is Divergent

Let us assume that a limit exists; then

$$|-1^n - L| < \varepsilon \forall n \geq N$$

Let us choose $\varepsilon = \frac{1}{2}$.

Now our sequence bounces between -1 and 1, thus we can say that

$$|-1 - L| < \frac{1}{2}$$

$$|1 - L| < \frac{1}{2}$$

Now let us use the triangle inequality:

$$2 = |-1-L+1-L| \leq |-1+L| + |1-L| < 2\frac{1}{2} = 2$$

This is a contradiction. Thus, such a limit cannot exist.

QED

47.23 Bolzano-Weierstraß Theorem

Every bounded sequence in $\mathbb{R}$ has at least one convergent subsequence.

Equivalently, every bounded sequence of real numbers has at least one accumulation point.

Proof:

Let a_n be a bounded sequence, and let a_{n_k} be a monotone subsequence given by the monotone subsequence theorem. a_{n_k} is bounded because our original sequence is bounded.

Due to the monotone convergence theorem, we also know that the subsequence converges. Therefore, our theorem is proven.

QED

47.24 Series

A *series* is the sum of the terms of a sequence:

$$\sum_{n=1}^{\infty} a_n$$

We define the partial sums $S_n = \sum_{k=1}^{n} a_k$. If the sequence (S_n) converges to a limit S, then the series is said to converge:

$$\sum_{n=1}^{\infty} a_n = S$$

47.25 Power Series

A *power series* is a series of the form:

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n$$

where x_0 is the center of the expansion. Power series converge within a radius R:

$$R = \frac{1}{\limsup\limits_{n\to\infty} \sqrt[n]{|a_n|}}$$

47.26 Geometric Series

There are two types of *geometric series*.

1. Finite geometric series:

$$\sum_{n=0}^{r+1} aq^n = a \cdot \frac{1-q^{r+1}}{1-q} \quad \text{for } q \neq 1$$

2. Infinite geometric series:

$$\sum_{n=0}^{\infty} aq^n = \frac{a}{1-q} \quad \text{for } |q| < 1$$

The proof of this in the chapter *Sums and Products*.

47.26.1 Generalized Geomtric Series

Let $n \in \mathbb{N}_0$ and $a, b \in \mathbb{C}$ such that $a \neq b$, then

$$\sum_{k=0}^{n} a^k b^{n-k} = \frac{a^{n+1} - b^{n+1}}{a - b}.$$

47.27 Point-Wise Convergence

A sequence of functions (f_n) converges *point-wise* to a function f on a set A if:

$$\forall x \in A, \quad \lim_{n\to\infty} f_n(x) = f(x)$$

Point-wise convergence does not preserve properties like continuity or differentiability.

47.28 Uniform Convergence

A sequence of functions (f_n) converges *uniformly* to f on A if:

$$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N} \text{ such that } \forall n \geq n_0, \forall x \in A, \quad |f_n(x) - f(x)| < \varepsilon$$

Key Property: Uniform convergence preserves continuity, integration, and differentiation under certain conditions.

47.29 Cauchy Criterion for Series

A series $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there exists n_0 such that:

$$\left| \sum_{k=m}^{n} a_k \right| < \varepsilon \quad \forall n > m \geq n_0$$

47.30 Interval Nesting and the Interval Nesting Theorem

Let (I_n) be a sequence of closed intervals:

$$I_n = [a_n, b_n] \quad \text{with } I_{n+1} \subseteq I_n, \text{ and } \lim_{n\to\infty} (b_n - a_n) = 0$$

Then, the intersection contains exactly one point:

$$\bigcap_{n=1}^{\infty} I_n = \{x\}$$

This is useful in proving the existence of limits, roots, and fixed points.

47.31 Weierstraß Approximation Theorem

The *Weierstraß Approximation Theorem* states:

Every continuous function $f : [a, b] \to \mathbb{R}$ can be uniformly approximated by a polynomial $P(x)$, i.e., for every $\varepsilon > 0$ there exists a polynomial $P(x)$ such that:

$$|f(x) - P(x)| < \varepsilon \quad \forall x \in [a, b]$$

This theorem is foundational in numerical analysis and approximation theory.

47.32 Zero Sequence

A sequence (a_n) is called a *null sequence* or *zero sequence* if:

$$\lim_{n\to\infty} a_n = 0$$

Every zero sequence is convergent (to 0), and plays an essential role in convergence proofs and epsilon-delta arguments.

More formally, a sequence $\{a_n\}_{n=1}^{\infty}$ of real numbers is called a zero sequence if and only if for all $\varepsilon > 0$ there exists n_0 such that:

$$|a_{n_0}| < \varepsilon, \forall n \geq n_0 \in \mathbb{N}$$

Note that $\{|a_n|\}_{n=1}^{\infty}$ is only a zero sequence if $\{a_n\}_{n=1}^{\infty}$ is a zero sequence.

Example:

Is $a_n = \frac{1}{n}$ a zero sequence?

Choose $n_0 = \frac{1}{\varepsilon}$; then

$$n \geq \frac{1}{\varepsilon} \implies \frac{1}{n} \leq \varepsilon, \forall n \in \mathbb{N}, n \geq n_0$$

Thus, it is a zero sequence.

47.33 Comparison Criteria for Sequences

Given $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ with

$$|b_n| \leq |a_n|$$

For all $n \in \mathbb{N}$ with $n \geq n_0 \in \mathbb{N}$. If $a_n \to 0$ as $n \to \infty$, then $b_n \to 0$ as $n \to \infty$.

We also call a_n a majorant of b_n.

47.34 Properties of Zero Sequences

Given the zero sequences $\{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty}$ and the bounded sequence $\{c\}_{n=1}^{\infty}$, then

- $\{a_n\}_{n=1}^{\infty}$ is also bounded.
- $\{\alpha a_n + \beta b_n\}_{n=1}^{\infty}, \alpha, \beta \in \mathbb{R}$ is also a zero sequence.
- $\{\sqrt{|a_n|}\}_{n=1}^{\infty}$ is a zero sequence.
- $\{a_n \cdot c_n\}_{n=1}^{\infty}$ is a zero sequence.

47.35 The Monotony Principle

Every bounded monotonic sequence converges.

Proof:

Define $a := \sup\{a_n : n \in \mathbb{N}\}$. We know that our sequence is bounded, and in this case it has the property that $a_n \leq a$.

We use the definition of convergence:

$$\begin{aligned} |a_{n_0} - a| &< \varepsilon \\ a_{n_0} &> a - \varepsilon \end{aligned}$$

Using the approximation property of the supremum, we get

$$a_n \geq a_{n_0} > a - \varepsilon, \forall n \geq n_0$$

Therefore, we have for each $\varepsilon > 0$ an $n_0 \in \mathbb{N}$

$$a - \varepsilon \leq a_{n_0} \leq a \forall n \geq n_0$$

Thus, our theorem is true.

$\mathfrak{QED}$

- If (a_n) is non-decreasing and bounded above, then $\lim a_n$ exists and equals $\sup\{a_n\}$.
- If (a_n) is non-increasing and bounded below, then $\lim a_n = \inf\{a_n\}$.

To prove the convergence of a sequence via this principle, we follow the next steps:

1. Prove the monotonicity by induction
2. Calculate the limit
3. Prove the limitedness via induction
4. Write the conclusion

Example:

$a_{n+1} = \sqrt{2a_n - 3} + 1$

Let us assume that the sequence is decreasing, i.e., $a_{n+1} < a_n$:

$$\begin{aligned} a_{n+2} &< a_{n+1} \\ \sqrt{2a_{n+1} - 3} + 1 &< \sqrt{2a_n - 3} + 1 \\ \sqrt{2a_{n+1} - 3} &< \sqrt{2a_n - 3} \\ 2a_{n+1} - 3 &< 2a_n - 3 \\ a_{n+1} &< a_n \end{aligned}$$

$\mathfrak{QED}$

Now let us calculate the limit:

$$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \sqrt{2a_n - 3} + 1$$

Because of the recursion, we know that both a_{n+1} and a_n have the same value, which we call a.

$$a = \sqrt{2a - 3} + 1$$

$$a = 2$$

We only have to prove the limitedness now:

For $n = 0$, we have $a_1 = 4 > 2$; then let us assume that a never goes below 2.

$$a_{n+1} = \sqrt{2a_n - 3} + 1 \geq \sqrt{2(2) - 3} + 1 = 2$$

By setting $a_n = 2$, we get the limit again.

$\mathfrak{QED}$

47.36 Operations on Sequences

47.36.1 Sum of Sequences

The sum of two sequences (a_n) and (b_n) is a new sequence (c_n) where each term c_n is the sum of the corresponding terms of (a_n) and (b_n):

$$c_n = a_n + b_n, \quad \forall n \in \mathbb{N}.$$

If $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$, then the limit of the sum sequence is:

$$\lim_{n\to\infty} (a_n + b_n) = A + B.$$

47.36.2 Multiplication of Sequences

The product of two sequences (a_n) and (b_n) is a new sequence (d_n) where each term d_n is the product of the corresponding terms of (a_n) and (b_n):

$$d_n = a_n \cdot b_n, \quad \forall n \in \mathbb{N}.$$

If $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$, then the limit of the product sequence is:

$$\lim_{n\to\infty} (a_n \cdot b_n) = A \cdot B.$$

47.36.3 Absolute Value of a Sequence

The absolute value of a sequence (a_n) is a new sequence (e_n) where each term e_n is the absolute value of the corresponding term of (a_n):

$$e_n = |a_n|, \quad \forall n \in \mathbb{N}.$$

If $\lim_{n\to\infty} a_n = A$, then the limit of the absolute value sequence is:

$$\lim_{n\to\infty} |a_n| = |A|.$$

47.36.4 Conjugate of a Complex Sequence

If (a_n) is a sequence of complex numbers, where $a_n = x_n + iy_n$ with $x_n, y_n \in \mathbb{R}$, then the conjugate of (a_n) is a new sequence $(\overline{a_n})$ where each term $\overline{a_n}$ is the complex conjugate of a_n:

$$\overline{a_n} = x_n - iy_n, \quad \forall n \in \mathbb{N}.$$

If $\lim_{n\to\infty} a_n = A = X + iY$, then the limit of the conjugate sequence is:

$$\lim_{n\to\infty} \overline{a_n} = \overline{A} = X - iY.$$

47.36.5 Complex Limit (Real and Imaginary Parts)

For a sequence of complex numbers (a_n) where $a_n = x_n + iy_n$, the limit of the sequence $\lim_{n\to\infty} a_n = A = X + iY$ exists if and only if the limits of the real part sequence (x_n) and the imaginary part sequence (y_n) exist individually:

$$\lim_{n\to\infty} a_n = A \iff \left(\lim_{n\to\infty} \operatorname{Re}(a_n) = \operatorname{Re}(A) = X \quad \text{and} \quad \lim_{n\to\infty} \operatorname{img}(a_n) = \operatorname{img}(A) = Y \right).$$

This means we can analyze the convergence of a complex sequence by examining the convergence of its real and imaginary parts separately.

47.36.6 Asymptotic Equality

Two sequences $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ are said to be *asymptotically equal*, denoted by $a_n \sim b_n$, if

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 1,$$

provided that $b_n \neq 0$ for all sufficiently large n. Asymptotic equality implies that the sequences behave similarly as n approaches infinity.

47.37 Boundness of a Function

A real-valued function f is *bounded below* if

$$\forall x \in \mathcal{D}(f) f(x) \geq m$$

and *bounded above* if

$$\forall x \in \mathcal{D}(f) f(x) \leq m$$

47.38 Inverse Function Due to Monotony

Given a strictly monotonic function f. Then f has an inverse.

Proof:

Given $x_1 < x_2$ or $x_1 > x_2$, then due to the monotony

$$f(x_1) < f(x_2) \text{ or } f(x_1) > f(x_2)$$

Thus, $f(x_1) = f(x_2) \iff x_2 = x_1$.

$\mathfrak{QED}$

47.39 Domain Restriction

Given $f : A \to B$ and $C \subsetneq A$, then

$$g := f_{|C} : C \to B, x \mapsto f(x)$$

This is f with a *restricted* domain. The case with $A \subsetneq C$ is the *expansion*.

Chapter 48

Limits

Limits are a foundational concept in calculus and analysis. They describe the behavior of functions and sequences as the input approaches a particular value or infinity.

48.1 Definition of the Limit

Let f be a function defined near a point a. We say:

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that:

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

This is the precise ε-δ *definition* of a limit.

48.2 Alternative Definition

Given a neighborhood U of x_0, then $\dot{U} := U \setminus \{x_0\}$ is the *pointed neighborhood* of x_0. For $\delta > 0$, we define

$$\dot{U_\delta}(x_0) = (x_0 - \delta, x_0 + \delta) \setminus \{x_0\}$$

Using this, we can then define the limit as follows.

Given $f : D \to \mathbb{R}$ with $x_0 \in D$ being an accumulation point, L is the limit if for all $\varepsilon > 0$, there is a $\delta > 0$ such that

$$x \in D \cap \dot{U_\delta}(x_0) \text{ holds } |f(x) - L| < \varepsilon$$

48.3 Limit Calculation Rules

- *Sum Rule:* $\lim_{x \to a}(f(x) + g(x)) = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$.
- *Difference Rule:* $\lim_{x \to a}(f(x) - g(x)) = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$.
- *Product Rule:* $\lim_{x \to a}(f(x)g(x)) = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$.
- *Quotient Rule:* $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim f(x)}{\lim g(x)}$ if $\lim g(x) \neq 0$.
- *Power Rule:* $\lim_{x \to a} f(x)^n = (\lim_{x \to a} f(x))^n$.

- *Root Rule:* $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$ (when defined).
- *Logarithm Rule:* $\lim_{x\to a} \ln(f(x)) = \ln(\lim_{x\to a} f(x))$ (when defined).
- *e limit Rule:* $\lim_{x\to 0}(1+f(x))^{\frac{1}{f(x)}} = e$.

Proof of the Sum Rule:

For any $\varepsilon > 0,$ there exists a $\delta_1 > 0$ such that if $0 < |x-a| < \delta_1$ then $0 < |f(x) - L| < \varepsilon/2$. The same goes for $g(x)$ with its respective limit and delta δ_2. Now choose $\delta = \min\{\delta_1, \delta_2\}$ such that

$$|f(x) - L| < \frac{\varepsilon}{2}$$

$$|g(x) - K| < \frac{\varepsilon}{2}$$

Now, using the triangle inequality, we get

$$|f(x) + g(x) - (L+K)| \leq |f(x) - L| + |g(x) - K| < 2\frac{\varepsilon}{2} = \varepsilon$$

$\mathfrak{QED}$

Example:

Find the limit of $\lim_{x\to 0} \frac{e^x - 1}{x}$.

Apply a change of variable

$$z = e^x - 1 \iff z + 1 = e^x$$

Use the logarithm

$$z + 1 = e^x \iff \ln(z+1) = x$$

The trend for $x \to 0$ tells us that $z \to 0$ because $\lim_{x\to 0} z = \lim_{x\to 0} e^x - 1 = 0$.

Our limit is now

$$\lim_{z\to 0} \frac{z}{\ln(z+1)} = \lim_{z\to 0} \frac{1}{\frac{1}{z}\ln(z+1)} = \lim_{z\to 0} \frac{1}{\ln\left((z+1)^{\frac{1}{z}}\right)}$$

The limit of a logarithm is the limit of the logarithm, and we also use the quotient rule.

$$\frac{1}{\ln\left(\lim_{z\to 0}(z+1)^{\frac{1}{z}}\right) = \ln(e)} = \frac{1}{\ln e} = 1$$

48.4 Limits at Infinity

$$\lim_{n\to\infty} f(n) \iff \forall \varepsilon > 0, \exists N > 0 : \forall n > N : |a_n - L| < \varepsilon$$

with $a_n = f(n)$ as a sequence.

Some common limits at infinity are:

$$\lim_{n\to\infty} \frac{1}{n} = 0, \quad \lim_{n\to\infty} e^n = \infty, \quad \lim_{n\to\infty} \frac{1}{n^2} = 0$$

Example:

Prove that $\lim_{n\to\infty} \frac{2n+1}{3n+7} = \frac{2}{3}$.

We start with the definition. Let $\varepsilon > 0$ be given. We need to find $N > 0$ such that for all $n > N$:

$$\begin{aligned} \left|\frac{2n+1}{3n+7} - \frac{2}{3}\right| &< \varepsilon \\ \left|\frac{(6n+3)-(6n+14)}{9n+21}\right| &< \varepsilon \\ \left|\frac{-11}{9n+21}\right| \leq \frac{|-11|}{|9n|+|21|} = \frac{|-11|}{9n+21} &< \frac{11}{9n} < \varepsilon \end{aligned}$$

Thus, we get

$$n > \frac{11}{9\varepsilon}$$

By the Archimedean property, there exists an integer N such that

$$N > \frac{11}{9\varepsilon}$$

Then for all $n > N$, we have

$$\left|\frac{2n+1}{3n+7} - \frac{2}{3}\right| < \varepsilon \text{ for all } n > N > \frac{11}{9\varepsilon}$$

Thus, $\varepsilon > \frac{11}{9n}$, which satisfies the definition of the limit.

$\mathfrak{QED}$

48.5 Indeterminate Forms

Indeterminate forms arise in limits when the expression does not directly imply a limit value:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 1^\infty, \quad \infty^0, \quad 0^0$$

48.6 Limit of the n-th Root of n

We will show that $\lim_{n\to\infty} \sqrt[n]{n} = 1$.

Proof:

Let $n = (1 + (\sqrt{n} - 1))^n$. This is a valid substitution.

Now let us use the binomial theorem $(1 + (\sqrt{n} - 1))^n = (1 + b_n)^n$.

$$\begin{aligned} (1+b_n)^n &= \sum_{k=0}^{n} \binom{n}{k} b_n^k \\ &= 1 + b_n + \frac{n(n-1)}{2} b_n^2 + \cdots \\ &> \frac{n(n-1)}{2} b_n^2 \end{aligned}$$

Then

$$(1+b_n)^n = n > \frac{n(n-1)}{2}b_n^2$$
$$\sqrt{\frac{2}{n-1}} > b_n$$

Now if we take the limit

$$\lim_{n\to\infty}\sqrt{\frac{2}{n-1}} = 0 > \lim_{n\to\infty} b_n$$

Then

$$\lim_{n\to\infty} 1 + b_n = 1 + 0 = 1$$

QED

48.7 Limit of a Composite Function

If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to L} g(x) = g(L)$ (i.e., g is continuous at L), then:

$$\lim_{x\to a} g(f(x)) = g\left(\lim_{x\to a} f(x)\right)$$

48.8 Right and Left Limits

For $(a-\delta, a)$ and $(a, a+\delta)$

$$\lim_{x\to a^-} f(x) = L_-, \quad \lim_{x\to a^+} f(x) = L_+$$

The two-sided limit $\lim_{x\to a} f(x)$ exists if and only if $L_- = L_+$.

48.9 L'Hôpital's Rule

If $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ or $\pm\infty$, and $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists, then:

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$$

This rule is used to resolve indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$.

48.10 Limit of a Fraction

For expressions like:

$$\lim_{n\to\infty}\left(\frac{a}{b}\right)^n$$

- If $\frac{a}{b} < 1$: limit is 0
- If $\frac{a}{b} > 1$: limit diverges to ∞

48.11 Limit Due to Sequence

Given an accumulation point x_0 in the domain of f, then

$$\lim_{x \to x_0} f(x) = L$$

If and only if, for a sequence in the domain of f, the following is true

$$\lim_{n \to \infty} x_n = x_0$$

Proof:

$\Rightarrow$ Assume $\lim_{x \to x_0} f(x) = L$. Let a_n be a sequence in the domain with $a_n \to x_0$ and $a_n \neq x_0$, we must show that $f(a_n) \to L$.

Let $\varepsilon > 0$. There exists $\delta > 0$ such that if $0 < |x - x_0| < \delta$ then $|f(x) - L| < \varepsilon$. But since $a_n \to x_0$ then $\exists N : |a_n - x_0| < \varepsilon, \forall n > N$. Thus, for all $n > N$, $|f(a_n) - L| < \varepsilon$ and so $f(a_n) \to L$.

$\Leftarrow$ Assume that for every sequence a_n in the domain which converges to x_0, $f(a_n) \to L$.

Suppose for the sake of contradiction that $\lim_{x \to x_0} f(x) \neq L$.

Then, there exists $\varepsilon > 0$ such that for all $\delta > 0, \exists x \in \mathcal{D}, 0 < |x - x_0| < \delta$ where $|f(a_n) - L| \geq \varepsilon$. Therefore, by letting $\delta_n = \frac{1}{n}$, $\exists a_n \in \mathcal{D}$ satisfying $0 < |a_n - x_0| < \frac{1}{n}$ for which $|f(a_n) - L| \geq \varepsilon$.

Then by contradiction $f(a_n)$ does not converge to L.

Hence, it must be that $\lim_{x \to x_0} f(x) \neq L$.

$\mathfrak{QED}$

48.12 Limit of a Recursive Sequence

Let (a_n) be defined recursively:

$$a_{n+1} = f(a_n)$$

If (a_n) converges to L and f is continuous, then:

$$\lim_{n \to \infty} a_n = L \Rightarrow L = f(L)$$

This is used to find fixed points of recursive definitions.

48.13 Important Limits

- $\lim_{x \to 0} \frac{\sin x}{x} = 1$
- $\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
- $\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$
- $\lim_{n \to \infty} \frac{n!}{n^n} = 0$
- $\lim_{n \to \infty} \sqrt[n]{a} = 1$ (for $a > 0$)
- $\lim_{x \to 0} \frac{1}{\cos x} = 1$

48.14 Squeeze Theorem (Sandwich Theorem)

If $g(x) \leq f(x) \leq h(x)$ near a, and:

$$\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L \Rightarrow \lim_{x \to a} f(x) = L$$

Proof:

We know that $g(x) \leq f(x) \leq h(x)$; therefore, for $g(x)$ and $h(x)$, we can define the following inequalities.

$$|g(x)_{N_1} - L| < \varepsilon$$
$$|h(x)_{N_2} - L| < \varepsilon$$

Then let us define $n = \max(N_1, N_2)$. And we can also write

$$-\varepsilon < g(x)_n - L < \varepsilon$$
$$-\varepsilon < h(x)_n - L < \varepsilon$$

Now, we write the following, because we are assuming they all have the same limit:

$$-\varepsilon < g(x)_n - L \leq f(x)_n - L \leq h(x)_n - L < \varepsilon$$
$$-\varepsilon < f(x)_n - L < \varepsilon$$

from our original inequality. Thus, there has been no contradiction.

$\mathfrak{QED}$

Example:

$$\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right) = 0$$

48.15 The Number e as a Limit

The number e is defined as:

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

More generally:

$$\lim_{n \to \infty} \left(1 + \frac{a}{bn^d}\right)^{n^d} = e^{\frac{a}{b}}$$

Example I:

$$\lim_{x \to \infty} \left(\frac{x^2 + 1}{x^2 + 2}\right)^{x^2}$$

$$\lim_{x \to \infty} \left(\left(\frac{x^2 + 1}{x^2 + 2}\right)^{x}\right)^{x}$$

$$\lim_{x\to\infty}\left(\left(\frac{1+\frac{1}{x^2}}{1+\frac{2}{x^2}}\right)^x\right)^x$$

$$\lim_{x\to\infty}\left(\frac{e}{e^2}\right)^x=\left(\frac{1}{e}\right)^x=0$$

Example II:

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x^2}$$

$$\lim_{x\to\infty}\left(\left(1+\frac{1}{x}\right)^{x^2}\right)^{\frac{x}{x}}$$

$$\lim_{x\to\infty}\left(\left(1+\frac{1}{x}\right)^{x}\right)^{x}$$

$$\lim_{x\to\infty}\left(\lim_{n\to\infty}\left(1+\frac{1}{x}\right)^{x}\right)^{x}$$

$$\lim_{x\to\infty}(e)^x=\infty$$

48.16 Limits of arctan

- $\lim_{x\to\infty}\arctan x=\frac{\pi}{2}$
- $\lim_{x\to-\infty}\arctan x=-\frac{\pi}{2}$
- $\arctan x$ is continuous and differentiable everywhere, with a horizontal asymptote at $y=\pm\frac{\pi}{2}$

48.17 Bolzano-Weierstraß Theorem

Every bounded, convergent sequence in $\mathbb{C}$ has a convergent subsequence. More precisely, the sequence a has a biggest and smallest accumulation point, h^* and h_*.

$$\forall\varepsilon>0: h_*-\varepsilon<a_n<h^*+\varepsilon \text{ for almost all } b\in\mathbb{N}$$

$$h^*:=\lim_{n\to\infty}\sup a_n$$

$$h_*:=\lim_{n\to\infty}\inf a_n$$

48.18 Uniqueness of the Limit

Let $f: D\subset\mathbb{R}\to\mathbb{R}$, and suppose a is a limit point of D. Then: If $\lim_{x\to a}f(x)=L_1$ and $\lim_{x\to a}f(x)=L_2$, then $L_1=L_2$.

In other words, a function can have at most one limit at a given point. This is the *uniqueness of limits*.

Proof:

Assume $L_1\neq L_2$, and let $\varepsilon=\frac{|L_1-L_2|}{2}>0$.

- Since $\lim_{x \to a} f(x) = L_1$, there exists $\delta_1 > 0$ such that for all $x \in D$, $0 < |x - a| < \delta_1$ implies:

$$|f(x) - L_1| < \varepsilon$$

- Since $\lim_{x \to a} f(x) = L_2$, there exists $\delta_2 > 0$ such that for all $x \in D$, $0 < |x - a| < \delta_2$ implies:

$$|f(x) - L_2| < \varepsilon$$

Let $\delta = \min(\delta_1, \delta_2)$, and choose $x \in D$ such that $0 < |x - a| < \delta$.

Then:

$$|f(x) - L_1| < \varepsilon, \quad |f(x) - L_2| < \varepsilon$$

Using the triangle inequality:

$$|L_1 - L_2| = |L_1 - f(x) + f(x) - L_2| \leq |f(x) - L_1| + |f(x) - L_2| < \varepsilon + \varepsilon = 2\varepsilon$$

Thus,

$$|L_1 - L_2| < |L_1 - L_2|$$

This contradiction tells us that our assumption was false. Hence, $L_1 = L_2$. The limit is unique.

$\mathfrak{QED}$

48.19 Every Convergent Sequence Is a Cauchy Sequence

Let (a_n) be a sequence in $\mathbb{R}$. We say:

If (a_n) converges, then it is a Cauchy sequence.

A sequence (a_n) is a *Cauchy sequence* if:

$$\forall \varepsilon > 0, \ \exists N \in \mathbb{N} \text{ such that } \forall n, m \geq N, \quad |a_n - a_m| < \varepsilon$$

Proof:

Assume $\lim_{n \to \infty} a_n = L$. Then:

- For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that:

$$\forall n \geq N, \quad |a_n - L| < \frac{\varepsilon}{2}$$

- Now, for any $n, m \geq N$, use the triangle inequality:

$$|a_n - a_m| = |a_n - L + L - a_m| \leq |a_n - L| + |a_m - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Conclusion:

$|a_n - a_m| < \varepsilon$ for all $n, m \geq N \Rightarrow (a_n)$ is Cauchy

This completes the proof that every convergent sequence is Cauchy.

$\mathfrak{QED}$

48.20 Limsup and Liminf

Given a sequence $\{a_k\}_{k=1}^{\infty}$ and the infinite sequences x_n and y_n defined as

$$x_n := \sup\{a_k \mid k \geq n\} \quad y_n := \inf\{a_k \mid k \geq n\}$$

We define *limes superior* and *limes inferior* as

$$a^* = \lim_{n\to\infty} \sup a_n := \lim_{n\to\infty} x_n$$
$$a_* = \lim_{n\to\infty} \inf a_n := \lim_{n\to\infty} y_n$$

48.21 Limit by Bounds

If the limit superior and inferior are equal, then the limit exists.

Chapter 49

Convergence Criteria

In this chapter, we present various tools used to determine whether an infinite series or sequence converges.

49.1 Quotient (Ratio) Test

Let $\sum a_n$ be a series with positive terms, and suppose the limit

$$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

exists.

- If $L < 1$, the series converges absolutely.
- If $L > 1$ or $L = \infty$, the series diverges.
- If $L = 1$, the test is inconclusive.

Proof of the Ratio Test (for $L < 1$)

Let $r < 1$ such that $L < r < 1$. Then for sufficiently large n, we have:

$$\left| \frac{a_{n+1}}{a_n} \right| \leq r \Rightarrow a_{n+1} \leq r a_n$$

By induction:

$$a_{n+k} \leq a_n r^k$$

So,

$$\sum_{n=N}^{\infty} a_n \leq a_N \sum_{k=0}^{\infty} r^k = \frac{a_N}{1-r}$$

Hence, the series converges by comparison with a geometric series.

49.2 Zero Sequence Criterion

If a series $\sum a_n$ converges, then:

$$\lim_{n\to\infty} a_n = 0$$

The converse is not true — the harmonic series is a classic counterexample.

49.3 Majorant (Comparison Test)

If $0 \leq a_n \leq b_n$ for all n and $\sum b_n$ converges, then $\sum a_n$ converges.

49.4 Minorant

If $0 \leq b_n \leq a_n$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

49.5 Root Test

Let $\sum a_n$ be a series with non-negative terms, and suppose:

$$L = \limsup_{n \to \infty} \sqrt[n]{|a_n|}$$

- If $L < 1$, the series converges absolutely.
- If $L > 1$, the series diverges.
- If $L = 1$, the test is inconclusive.

Proof of the Root Test (for $L < 1$)

Let r be such that $L < r < 1$. Then for large n:

$$\sqrt[n]{|a_n|} \leq r \Rightarrow |a_n| \leq r^n$$

Since $\sum r^n$ converges, by comparison $\sum |a_n|$ converges absolutely.

49.6 Leibniz Criterion (Alternating Series Test)

Let $\sum (-1)^n a_n$ be a series with:

- a_n decreasing: $a_{n+1} \leq a_n$
- $\lim a_n = 0$

Then the series converges.

Example:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

49.7 Compactness Criterion

A sequence (x_n) in $\mathbb{R}^n$ has a convergent subsequence if and only if it is bounded. This is a consequence of the *Bolzano-Weierstraß theorem*, which is a form of sequential *compactness* in $\mathbb{R}^n$.

49.8 Cauchy Criterion I

A sequence (a_n) converges if and only if it is a *Cauchy sequence*:

$$\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall n, m \geq n_0 \Rightarrow |a_n - a_m| < \varepsilon$$

This is an internal (non-limit based) criterion for convergence, valid in complete metric spaces.

49.9 Cauchy Criterion II

A series $a_n = \sum_{i=0}^{k} x_i$ converges $\iff \forall d \in \mathbb{R} d > 0 \exists N \in \mathbb{N} \forall n, m > N : |\sum_{i=0}^{n} x_i + \sum_{i=0}^{m}|$

$$= \left| \sum_{i=0}^{n} x_i \right| < d$$

49.10 The Cauchy Product

Let $\sum a_n$, $\sum b_n$ be convergent series. The *Cauchy product* is:

$$\sum_{n=0}^{\infty} c_n, \quad \text{where } c_n = \sum_{k=0}^{n} a_k b_{n-k}$$

If $\sum a_n$ and $\sum b_n$ converge absolutely, then their Cauchy product converges absolutely and:

$$\sum c_n = \left(\sum a_n \right) \left(\sum b_n \right)$$

49.10.1 Cauchy Product Example

Prove that $e^x e^y = e^{x+y}$
Note that $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ and $e^y = \sum_{n=0}^{\infty} \frac{y^n}{n!}$.

Therefore,

$$e^x e^y = \sum_{n=0}^{\infty} \frac{x^n}{n!} \sum_{n=0}^{\infty} \frac{y^n}{n!}$$

And, because of the absolute convergence,

$$e^x e^y = \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{x^k}{n!} \frac{y^{n-k}}{(n-k)!}$$

Finally,

$$e^x e^y = \sum_{n=0}^{\infty} \sum_{k=0}^{n} x^k y^{n-k} \binom{n}{k} \frac{1}{n!} = \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!}$$

$\mathfrak{QED}$

49.11 Geometric Series

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad \text{for } |r| < 1$$

If $|r| \geq 1$, the series diverges.

49.12 Harmonic Series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

This series diverges, even though its terms approach zero.

Proof:

Group terms:

$$\left(\frac{1}{2}\right) + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots > \sum_{k=1}^{\infty} \frac{1}{2}$$

The lower bound is a divergent series, hence the *harmonic series* diverges.

Chapter 50

Convergence Radius

Consider a power series of the form

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n,$$

where c_n are the coefficients, x is the variable, and a is the center of the series. The radius of convergence, denoted by R, is a non-negative real number (or ∞) such that the series converges if $|x-a| < R$ and diverges if $|x-a| > R$. In other words, the interval of convergence is $(a-R, a+R)$, possibly including one or both endpoints.

$$R = \frac{1}{\limsup_{n\to\infty} \sqrt[n]{|a_n|}},$$

or

$$R = \limsup_{n\to\infty} \frac{a_n}{a_{n+1}}.$$

50.1 Test of the Borders

When the absolute value of the difference between x and a is equal to the radius of convergence (i.e., $|x-a| = R$), the convergence of the power series at the *borders* must be checked separately. This involves substituting the values $x = a \pm R$ into the series and determining whether the resulting series converges or diverges using other convergence tests .

50.2 Approximating the Error

When a power series is used to approximate a function, it is often necessary to estimate the *error* in the approximation. If the power series converges, the error can be made arbitrarily small by including a sufficient number of terms. For an alternating series that satisfies the conditions of the Alternating Series Test, the error in approximating the sum by the first n terms is bounded by the absolute value of the $(n+1)$-th term:

$$\left| \sum_{k=0}^{\infty} c_k (x-a)^k - \sum_{k=0}^{n} c_k (x-a)^k \right| \leq |c_{n+1}(x-a)^{n+1}|.$$

For other power series, techniques such as Taylor's Theorem with remainder can be used to estimate the error.

50.3 Complete Example

Find the convergence radius of

$$\sum_{n=1}^{\infty} \frac{x^n}{n2^n}$$

at $x = 0$.

Step 1: Extract the sequence a_n

The general form is

$$\sum_{0}^{\infty} a_n (x - x_0)^n,$$

which in our case is

$$= \sum_{n=1}^{\infty} x^n \frac{1}{n2^n},$$

with $\frac{1}{n2^n}$ as a_n.

Step 2: Compute the radius

$$\begin{aligned} R &= \frac{1}{\limsup_{n\to\infty} \sqrt[n]{\frac{1}{n2^n}}} \\ &= \frac{1}{\limsup_{n\to\infty} \frac{\sqrt[n]{1}}{\sqrt[n]{n}\,\sqrt[n]{2^n}}} \\ &= \frac{1}{\frac{1}{2}} = 2 \end{aligned}$$

Step 3: Check Borders

The borders to check are from $0 - 2$ to $0 + 2$.

For -2:

$$\sum_{n=1}^{\infty} \frac{(-2)^n}{n2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n (2)^n}{n2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n},$$

which converges by Leibniz.

For 2:

$$\sum_{n=1}^{\infty} \frac{(2)^n}{n2^n} = \sum_{n=1}^{\infty} \frac{1}{n},$$

which is the harmonic series; therefore, it diverges.

Step 4: Write the interval

The final interval of convergence is $[-2, 2)$.

Chapter 51

Calculating Series Values

51.1 Telescoping Sums and Partial Fraction Decomposition

Telescoping sums are series where most of the terms cancel out, leaving only a few terms. *Partial fraction decomposition* is a technique used to rewrite rational functions as a sum of simpler fractions, which can often lead to telescoping sums.

Example: $\sum_{k=1}^{\infty} \frac{1}{4k^2-1}$

Consider the infinite series

$$S = \sum_{k=1}^{\infty} \frac{1}{4k^2 - 1}.$$

We can decompose the fraction using partial fraction decomposition:

$$\frac{1}{4k^2 - 1} = \frac{1}{(2k-1)(2k+1)} = \frac{A}{2k-1} + \frac{B}{2k+1}.$$

Multiplying both sides by $(2k-1)(2k+1)$ gives

$$1 = A(2k+1) + B(2k-1).$$

To solve for A and B, we can use the following values of k:

- For $k = \frac{1}{2}$: $1 = A(1+1) + B(0) \implies A = \frac{1}{2}$.
- For $k = -\frac{1}{2}$: $1 = A(0) + B(-1-1) \implies B = -\frac{1}{2}$.

Thus, we have

$$\frac{1}{4k^2 - 1} = \frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right).$$

Now we can write the series as a telescoping sum:

$$S = \frac{1}{2}\sum_{k=1}^{\infty}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right).$$

Let S_n be the n-th partial sum:

$$\begin{aligned} S_n &= \frac{1}{2}\left[\left(\frac{1}{1}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{7}\right)+\cdots+\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)\right] \\ &= \frac{1}{2}\left[1-\frac{1}{2n+1}\right]. \end{aligned}$$

Taking the limit as n approaches infinity, we get

$$S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1}{2}\left(1-\frac{1}{2n+1}\right) = \frac{1}{2}(1-0) = \frac{1}{2}.$$

Therefore, the value of the series is $\frac{1}{2}$.

Example II:

Now, an example using the geometric series:

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{4}{3^n} &= \sum_{n=0}^{\infty} 4\left(\frac{1}{3}\right)^n \\ &= \frac{4}{1-\frac{1}{3}}. \end{aligned}$$

Chapter 52

Transforming Functions into Power Series

Many functions can be represented as *power series*, which is useful for various applications in calculus and analysis. This chapter discusses some common techniques for transforming functions into power series.

52.1 Common Cases

Here are some *common cases* where functions can be easily transformed into power series:

$\frac{1}{1-x}$

The function $\frac{1}{1-x}$ has the well-known geometric series representation:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1.$$

$\frac{1}{1-(x-x_0)^n}$

A slight generalization is:

$$\frac{1}{1-(x-x_0)^n} = \sum_{k=0}^{\infty} (x-x_0)^{nk}, \quad |x-x_0| < 1.$$

$\frac{1}{1-d(x-x_0)^n}$

For a constant d:

$$\frac{1}{1-d(x-x_0)^n} = \sum_{k=0}^{\infty} d^k (x-x_0)^{nk}, \quad |d(x-x_0)^n| < 1.$$

Example: $\frac{x+1}{3-x}$

Let's find the power series representation of the function $f(x) = \frac{x+1}{3-x}$.

52.1.1 Partial Fractions

First, we perform *partial fraction* decomposition. We can rewrite the function as:

$$\frac{x+1}{3-x} = \frac{-(3-x)+4}{3-x} = -1 + \frac{4}{3-x}.$$

52.1.2 Manipulating the Expression

Now, we *manipulate* the fraction to fit the form $\frac{1}{1-u}$:

$$\frac{4}{3-x} = \frac{4}{3(1-\frac{x}{3})} = \frac{4}{3} \cdot \frac{1}{1-\frac{x}{3}}.$$

52.1.3 Power Series Representation

Using the geometric series formula, we have

$$\frac{1}{1-\frac{x}{3}} = \sum_{n=0}^{\infty} \left(\frac{x}{3}\right)^n, \quad \left|\frac{x}{3}\right| < 1.$$

Thus,

$$\begin{aligned} \frac{x+1}{3-x} &= -1 + \frac{4}{3} \sum_{n=0}^{\infty} \left(\frac{x}{3}\right)^n \\ &= -1 + \frac{4}{3} \sum_{n=0}^{\infty} \frac{x^n}{3^n} \\ &= -1 + \sum_{n=0}^{\infty} \frac{4}{3^{n+1}} x^n, \quad |x| < 3. \end{aligned}$$

Therefore, the *power series representation* of the function is

$$\frac{x+1}{3-x} = -1 + \sum_{n=0}^{\infty} \frac{4}{3^{n+1}} x^n, \quad |x| < 3.$$

Chapter 53

Continuity

53.1 Definition of Continuity at a Point

A function f is *continuous* at a point x_0 if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$$

Equivalently,

$$\lim_{x \to x_0} f(x) = f(x_0)$$

In other words, a real-valued function f is continuous if it is continuous at all points in its domain.

53.1.1 Approach to Solve Problems

To solve continuity problems, the general approach is to:

1. Write down the definition.
2. Substitute the concrete function for f in the definition.
3. Modify the expression with the concrete f, $|f(x) - f(y)|$, in such a way that you get $|x - y|$ inside the expression.
4. Choose a δ if not given. On one hand, it is often $\delta = |x - y|$, and then substitute it into the expression. On the other hand, if δ is given, make an approximation which is greater than the current expression and then simplify. In both cases, after simplifying, you should get an ε if the function is continuous.

Example:

$\varepsilon - \delta$ Exercise for $f(x) = \frac{1}{x}$ at $x_0 > 0$

We want to find δ such that:

$$|x - x_0| < \delta \Rightarrow \left| \frac{1}{x} - \frac{1}{x_0} \right| < \varepsilon$$

Start by rewriting:

$$\left| \frac{1}{x} - \frac{1}{x_0} \right| = \left| \frac{x_0 - x}{x x_0} \right| = \frac{|x - x_0|}{|x||x_0|}$$

Assume:

$$x \in \left[\frac{x_0}{2}, \frac{3x_0}{2}\right] \Rightarrow |x| \geq \frac{x_0}{2}$$

Then:

$$\left|\frac{1}{x} - \frac{1}{x_0}\right| \leq \frac{|x - x_0|}{\frac{x_0}{2} \cdot x_0} = \frac{2}{x_0^2}|x - x_0|$$

Choose:

$$\delta = \min\left\{\frac{x_0}{2}, \frac{x_0^2 \varepsilon}{2}\right\}$$

Example:

Prove that

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

is continuous at $x_0 = 0$.

Recall, f is continuous at $x_0 = 0$ if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$$

We have $\varepsilon > 0$ and we want to find $\delta > 0$:

Let us assume $|x - 0| < \delta \iff \delta > |x|$, then

$$\begin{aligned} |f(x) - f(0)| = |f(x) - 0| &= |x^2 \sin\left(\frac{1}{x}\right)| \\ &\leq |x^2| \cdot \left|\sin\left(\frac{1}{x}\right)\right| \\ &\leq |x^2| \cdot 1 = x^2 < \varepsilon \\ &\Rightarrow x < \sqrt{\varepsilon} \end{aligned}$$

Choose $\delta = \sqrt{\varepsilon}$.

Let $\varepsilon > 0$ be given and choose $\delta = \sqrt{\varepsilon}$. Then for all x with $|x - 0| < \delta$ we have:

$$|f(x) - f(0)| < \varepsilon$$

QED

53.2 Characterization Using Sequences

Let $f : (a, b) \to \mathbb{R}$ be a function and $x_0 \in (a, b)$.

The function f is continuous at x_0 if and only if for every *sequence*

$$\{x_n\}_{n \in \mathbb{N}}$$

with $x_n \in (a, b)$ for all $n \in \mathbb{N}$ and $\lim_{n \to \infty} x_n = x_0$, it holds that

$$\lim_{n \to \infty} f(x_n) = f(x_0)$$

53.3 Left and Right Continuity

Let $f : D \to \mathbb{R}$ be a function and $x_0 \in D$.The function f is called *left-continuous* at x_0 if f is continuous on $D \cap (-\infty, x_0]$.

Analogously, f is *right-continuous* at x_0 if$f$ is continuous on $D \cap [x_0, \infty)$.

53.4 Rules of Continuity

Let f and g be continuous at x_0:

- $f + g$, $f - g$, $f \cdot g$ are continuous at x_0
- $\frac{f}{g}$ is continuous at x_0 if $g(x_0) \neq 0$
- Compositions: if g is continuous at x_0, and f is continuous at $g(x_0)$, then $f \circ g$ is continuous at x_0

53.5 Continuity and Compactness

A function on an interval $[a, b]$ which is continuous is *bounded*.

Proof:

If f is not bounded, then $\forall n \in \mathbb{N}, \exists x_n \in [a, b]$ such that

$$|f(x_n)| > n$$

By the Bolzano-Weierstrass theorem, there exists a convergent subsequence $\{x_{n_k}\} \subset [a, b]$ such that

$$\lim_{k\to\infty} x_{n_k} = x^* \in [a, b]$$

By continuity of f:

$$\lim_{k\to\infty} f(x_{n_k}) = f(x^*) = \infty$$

But by construction $|f(x_{n_k})| > n_k \to \infty$ as $k \to \infty$, which is a contradiction to $x^* \in \mathcal{D}(f)$.

$\mathfrak{QED}$

53.6 Extreme-Value Theorem

If $f : [a, b] \to \mathbb{R}$ is continuous on the closed interval $[a, b]$, then there exist $x_m, x_M \in [a, b]$ such that

$$f(x_m) = \inf_{[a,b]} f, \quad f(x_M) = \sup_{[a,b]} f$$

Proof:

Let $m := \inf_{[a,b]} f, \quad M := \sup_{[a,b]} f$. We know by the previous theorem that f is bounded, so $m, M \in \mathbb{R}$. By the Bolzano-Weierstrass theorem, there exist convergent subsequences $\{x_{n_k}\}, \{x_{m_k}\} \subset [a, b]$ such that

$$\lim_{k\to\infty} f(x_{n_k}) = m, \quad \lim_{k\to\infty} f(x_{m_k}) = M$$

Since $[a, b]$ is closed, the limits of these subsequences $x_m, x_M \in [a, b]$. Then by continuity of f:

$$f(x_m) = \lim_{k\to\infty} f(x_{n_k}) = m, \quad f(x_M) = \lim_{k\to\infty} f(x_{m_k}) = M$$

$\mathfrak{QED}$

53.7 Extreme Points

If for a set $A \subseteq \mathbb{R}$ with $f : A \to \mathbb{R}$ it holds that

$$f(x_m) = \inf_{x \in A} f(x) =: m \quad \text{or} \quad f(x_M) = \sup_{x \in A} f(x) =: M$$

then x_m is called a *point of minimum* and x_M a *point of maximum* of f on A. Also, $f(x_m)$ is called the *minimum* and $f(x_M)$ the *maximum* of f on A. These are the *extreme values*.

53.8 Middle Value Theorem

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) \neq f(b)$, then for every y_0 between $f(a)$ and $f(b)$ there exists $x_0 \in (a, b)$ such that $f(x_0) = y_0$.

53.9 Continuity of the Inverse by a Decreasing Function

Given a monotonically decreasing continuous function f on $[a, b]$ with $f(a) = c$ and $f(b) = d$, the inverse function $f^{-1} : [d, c] \to [a, b]$ is also continuous and monotonic.

Proof:

53.9.1 Continuity of the Root and Logarithm Function

$x \mapsto \sqrt[n]{x}$ is continuous on $[0, \infty)$.

$x \mapsto \log_n x$ is continuous on $(0, \infty)$.

53.10 Lipschitz Continuity

A function $f : D \subset \mathbb{R} \to \mathbb{R}$ is *Lipschitz continuous* if:

$$\exists L > 0 \text{ such that } |f(x) - f(y)| \leq L|x - y| \quad \forall x, y \in D$$

- Every Lipschitz continuous function is uniformly continuous.
- The smallest such L is called the *Lipschitz constant*.

53.11 Fixed Point

A *fixed point* of a function f is a point x^* such that:

$$f(x^*) = x^*$$

53.12 Fixed Point on Interval

Given a continuous function $f : [a, b] \to \mathbb{R}$ with $f([a, b]) \subseteq [a, b]$, there exists at least one fixed point $x^* \in [a, b]$ such that $f(x^*) = x^*$.

53.13 Banach Fixed Point Theorem

Let (X, d) be a complete metric space, and $f : X \to X$ a *contraction mapping*, i.e.,

$$\exists L < 1 \text{ such that } d(f(x), f(y)) \leq L \cdot d(x, y)$$

Then:

- f has a unique fixed point $x^* \in X$, such that $f(x^*) = x^*$
- Iterating $x_{n+1} = f(x_n)$ converges to x^*

Interpretation and Meaning

The Banach Fixed Point Theorem ensures:

- Existence and uniqueness of solutions (fixed points)
- Convergence of approximation by iteration
- Powerful tool in numerical methods and differential equations

53.13.1 A Priori and a Posteriori Approximations

A priori estimate:

$$|x_n - x^*| \leq \frac{L^n}{1-L}|x_1 - x_0| < \epsilon$$

A posteriori estimate:

$$|x_n - x^*| \leq \frac{L}{1-L}|x_n - x_{n-1}|\epsilon$$

53.13.2 Steps for a Fixed Point Exercise

Given $f : [a, b] \to [a, b]$, to prove existence and convergence:

1. Show monotonicity: f is increasing or decreasing.
2. Check interval preservation: $f([a, b]) \subseteq [a, b]$
3. Check Lipschitz continuity: Find $L < 1$
4. Fixed point iteration: Choose x_0, compute $x_{n+1} = f(x_n)$
5. Apply a priori or a posteriori bound

Example:

$f(x) = \frac{1}{x} + 3$ on $[2, 5]$

Step 1: Monotonicity

In this case, we will assume that it is monotonic.

Step 2: Domain check

$x \in [2, 5] \Rightarrow \frac{1}{x} \in [0.2, 0.5] \Rightarrow f(x) \in [3.2, 3.5] \subset [2, 5]$

Step 3: Lipschitz constant

$$f'(x) = -\frac{1}{x^2} \Rightarrow |f'(x)| \leq \frac{1}{2^2} = \frac{1}{4} < 1 \Rightarrow \text{Lipschitz with } L = \frac{1}{4} < 1$$

Step 4: Fixed point iteration

$$x_0 = 3, \quad x_1 = f(3) = \frac{1}{3} + 3 = \frac{10}{3}, \quad x_2 = f(x_1), \ldots$$

Step 5: Approximations

A priori

$$|x_n - x^*| \leq \frac{L^n}{1-L}|x_1 - x_0|$$

$$|x_n - x^*| \leq \frac{\frac{1}{4}^n}{1-\frac{1}{4}}|\frac{10}{3} - 3|$$

$$\begin{aligned}
|x_n - x^*| &\leq \frac{\left(\frac{1}{4}\right)^n}{1-\frac{1}{4}} \cdot \left|\frac{10}{3} - 3\right| \\
&= \frac{\left(\frac{1}{4}\right)^n}{\frac{3}{4}} \cdot \frac{1}{3} \\
&= \frac{4}{3} \cdot \left(\frac{1}{4}\right)^n \cdot \frac{1}{3} \\
&= \frac{4}{9} \cdot \left(\frac{1}{4}\right)^n
\end{aligned}$$

For $\varepsilon = 0.1$

$$\begin{aligned}
\frac{4}{9} \cdot \left(\frac{1}{4}\right)^n &< \varepsilon \\
\left(\frac{1}{4}\right)^n &< \frac{9}{4}\varepsilon \\
\ln\left(\left(\frac{1}{4}\right)^n\right) &< \ln\left(\frac{9}{4}\varepsilon\right) \\
n \cdot \ln\left(\frac{1}{4}\right) &< \ln\left(\frac{9}{4}\varepsilon\right) \\
n &> \frac{\ln\left(\frac{9}{4}\varepsilon\right)}{\ln\left(\frac{1}{4}\right)} \\
n &> \frac{\ln\left(\frac{9}{4}\varepsilon\right)}{-\ln 4}
\end{aligned}$$

A posteriori

$$|x_n - x^*| \leq \frac{L}{1-L}|x_n - x_{n-1}| = \frac{\frac{1}{4}}{1-\frac{1}{4}}|x_n - x_{n-1}| = \frac{1}{3}|x_n - x_{n-1}|$$

53.14 Uniform Continuity

A function $f : \mathcal{D} \subset \mathbb{R} \to \mathbb{R}$ is *uniformly continuous* if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon \quad \forall x, y \in \mathcal{D}$$

Example:

$f(x) = \frac{1}{x}$ is not uniformly continuous on $(0, 1)$

Let $\varepsilon = 1$. We want to find numbers in $(0, 1)$ for x, y such that $|x - y| < \delta$ but $|f(x) - f(y)| \geq 1$.

Choose $x \in (0, 1)$ with $x < \delta$ and let $y = \frac{x}{2}$. Then

$$|x - y| = |x - \frac{x}{2}| = \frac{x}{2} < \frac{\delta}{2} < \delta$$

But,

$$|f(x) - f(y)| = \left|\frac{1}{x} - \frac{1}{y}\right| = \left|\frac{1}{x} - \frac{2}{x}\right| = \left|-\frac{1}{x}\right| = \frac{1}{x} > 1$$

Since $\frac{1}{x} > 1$ for $x < 1$, we have a contradiction, and therefore f is not uniformly continuous on $(0, 1)$.

53.15 Boundedness by Uniform Continuity

If a function $f : (a, b) \to \mathbb{R}$ is uniformly continuous on the open interval (a, b), then f is bounded.

53.16 Cauchy Sequence by Uniform Continuity

Given $\mathcal{D} \subseteq \mathbb{R}$ and a uniformly continuous function $f : \mathcal{D} \to \mathbb{R}$, for every *Cauchy sequence* $\{x_n\} \subseteq \mathcal{D}$, the sequence $\{f(x_n)\}$ is also a Cauchy sequence.

Chapter 54

Differentiation

Differentiation is a core concept in calculus. It describes the rate at which a quantity changes and provides tools for analyzing and modeling change in real-world phenomena.

54.1 Definition of the Derivative

The *derivative* of a function f at a point x is defined as the limit:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

or

$$f'(x) = \lim_{h \to a} \frac{f(x) - f(a)}{x - a}$$

provided this limit exists.

Geometrically, $f'(a)$ is the slope of the tangent line to the graph of f at $x = a$.

54.2 Derivative Rules

Power Rule:

$$\frac{d}{dx} x^n = nx^{n-1}, \quad \text{for } n \in \mathbb{R}$$

Constant Rule:

$$\frac{d}{dx} c = 0, \quad \text{for constant } c$$

Constant Multiple Rule:

$$\frac{d}{dx} [c \cdot f(x)] = c \cdot f'(x)$$

Sum and Difference Rule:

$$\frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Product Rule:

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

Quotient Rule:

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

Chain Rule:

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$$

54.3 Trigonometric Derivatives

$$\begin{aligned}
\frac{d}{dx}\sin(x) &= \cos(x) \\
\frac{d}{dx}\cos(x) &= -\sin(x) \\
\frac{d}{dx}\tan(x) &= \sec^2(x) \\
\frac{d}{dx}\cot(x) &= -\csc^2(x) \\
\frac{d}{dx}\sec(x) &= \sec(x)\tan(x) \\
\frac{d}{dx}\csc(x) &= -\csc(x)\cot(x)
\end{aligned}$$

54.4 Derivatives of Inverse Trigonometric Functions

$$\begin{aligned}
\frac{d}{dx}\arcsin(x) &= \frac{1}{\sqrt{1-x^2}} \\
\frac{d}{dx}\arccos(x) &= -\frac{1}{\sqrt{1-x^2}} \\
\frac{d}{dx}\arctan(x) &= \frac{1}{1+x^2} \\
\frac{d}{dx}\operatorname{arccot}(x) &= -\frac{1}{1+x^2} \\
\frac{d}{dx}\operatorname{arcsec}(x) &= \frac{1}{|x|\sqrt{x^2-1}} \\
\frac{d}{dx}\operatorname{arccsc}(x) &= -\frac{1}{|x|\sqrt{x^2-1}}
\end{aligned}$$

54.5 The Differential

The *differential* dy of a function $y = f(x)$ is defined as:

$$dy = f'(x)\,dx.$$

This linear approximation estimates the change in y for a small change in x.

54.6 The Secant Equation and Graphical Meaning

The *secant line* through the points $(x_0, f(x_0))$ and $(x_0 + h, f(x_0 + h))$ has slope

$$\frac{f(x_0 + h) - f(x_0)}{h}.$$

As $h \to 0$, the secant line approaches the tangent line. Graphically, this means

$$\lim_{h \to 0} \text{slope of secant} = \text{slope of tangent} = f'(x) = \Delta_h.$$

The secant line equation is given by

$$L(x) = f(x) + \Delta_h f(x_0)(x - x_0).$$

54.7 Optimization via the Derivative

To find local or global extrema:

1. Compute $f'(x)$.
2. Solve $f'(x) = 0$ to find the critical points.
3. Use the first or second derivative test to classify them.
4. Evaluate the endpoints if optimizing over a closed interval.

Example I:

Maximize the area $A = x(10 - 2x)$:

$$A = 10x - 2x^2, \quad A' = 10 - 4x, \quad A' = 0 \Rightarrow x = 2.5.$$

Example II:

Given a number $c > 0$, we want to factor it into two positive numbers x and y such that their product is c and their sum $S = x + y$ is minimal.

Since $x \cdot y = c$, we can express one variable in terms of the other, e.g.,

$$y = \frac{c}{x}.$$

Substituting into the sum,

$$S(x) = x + \frac{c}{x}.$$

To minimize $S(x)$, we take the derivative:

$$S'(x) = 1 - \frac{c}{x^2}.$$

Setting the derivative to zero gives

$$1 - \frac{c}{x^2} = 0 \quad \Rightarrow \quad x^2 = c \quad \Rightarrow \quad x = \sqrt{c}.$$

Then

$$y = \frac{c}{x} = \sqrt{c}.$$

So the minimal sum occurs when $x = y = \sqrt{c}$, and the minimum value of the sum is

$$x + y = 2\sqrt{c}.$$

Let $c = 25$. Then the two positive factors that minimize the sum are

$$x = y = \sqrt{25} = 5,$$

and the minimal sum is

$$x + y = 5 + 5 = 10.$$

54.8 Power Rule Derivation

We derive the *power rule*

$$\frac{d}{dx}x^n = nx^{n-1}$$

for $n \in \mathbb{N}$ using the limit definition:

$$\frac{d}{dx}x^n = \lim_{h\to 0}\frac{(x+h)^n - x^n}{h}.$$

Using the Binomial Theorem,

$$(x+h)^n = x^n + nx^{n-1}h + \cdots + h^n \Rightarrow \text{only the linear term survives after dividing by } h.$$

54.9 Implicit Differentiation

Given

$$9x^2 + 4y^2 = 25,$$

differentiate both sides implicitly:

$$18x + 8y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{18x}{8y} = -\frac{9x}{4y}.$$

This method is used when y is not isolated and is defined implicitly.

54.10 Hermit Polynomials

A *Hermit polynomial* is defined as:

$$H_n(x) = (-1)^n e^{x^2}\frac{d^n}{dx^n}e^{-x^2}.$$

They appear when taking the multiple derivatives of e^{-x^2}.

Chapter 55

Integration

Integration is the reverse process of differentiation and is used to calculate areas, volumes, accumulated change, and more.

55.1 Riemann Integral

Let $f : [a, b] \to \mathbb{R}$ be a bounded function. We define the *definite integral* as:

$$\int_a^b f(x)\,dx = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(\xi_i)\Delta x_i$$

where $P = \{x_0, \ldots, x_n\}$ is a partition, $\xi_i \in [x_{i-1}, x_i]$, and $\Delta x_i = x_i - x_{i-1}$.

55.1.1 Upper and Lower Sums

Let f be bounded on $[a, b]$ and P a partition:

$$\underline{S}(f, P) = \sum_{i=1}^{n} m_i \Delta x_i, \quad \overline{S}(f, P) = \sum_{i=1}^{n} M_i \Delta x_i$$

with:

$$m_i = \inf_{x\in[x_{i-1},x_i]} f(x), \quad M_i = \sup_{x\in[x_{i-1},x_i]} f(x)$$

If $\sup \underline{S} = \inf \overline{S}$, then f is *Riemann integrable*.

55.2 Concrete and Non-Concrete Integrals

Concrete integrals can be explicitly evaluated as anti-derivatives with an explicit function. *Non-concrete* integrals cannot be integrated in elementary terms (e.g., $\int e^{-x^2} dx$).

55.3 Basic Rules of Integration

- $\int 0\,dx = C$.
- $\int c\,dx = cx$.
- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$.
- $\int \frac{1}{x}\,dx = \ln|x| + C$.

- $\int e^x\,dx = e^x + C$.
- $\int \sin x\,dx = -\cos x + C$.
- $\int \cos x\,dx = \sin x + C$.
- Linearity: $\int (af + bg)\,dx = a\int f\,dx + b\int g\,dx$.

55.4 Integration by Substitution

If $u = g(x)$ and $f = F'$, then:

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

Example:

$$\int \frac{1}{(x-1)^4}\,dx, \quad u = x-1 \Rightarrow \int \frac{1}{u^4}du = -\frac{1}{3u^3} + C = -\frac{1}{3(x-1)^3} + C$$

55.5 Integration by Parts

Integration by parts is the inverse process of the product and quotient rules of differentiation and is given by

$$\int u\,dv = uv - \int v\,du.$$

To compute, we use the abbreviation *ILATE* to find the order in which to choose u:

1. Inverse Trigonometric.
2. Logarithmic.
3. Algebraic.
4. Trigonometric.
5. Exponential.

Sometimes we encounter *cyclic integrals*, which cycle back to the original when applying our formula, e.g., $e^x \sin(x)$.

Proof:

$$\begin{aligned}
(f(x)g(x))' &= f(x)g'(x) + f'(x)g(x) \\
\int (f(x)g(x))'dx &= \int f(x)g'(x)dx + \int f'(x)g(x)dx \\
f(x)g(x) &= \int f(x)g'(x)dx + \int f'(x)g(x)dx \\
-\int f(x)g'(x)dx &= -f(x)g(x) + \int f'(x)g(x)dx \\
\int f(x)g'(x)dx &= f(x)g(x) - \int f'(x)g(x)dx \\
\int udv &= uv - \int vdu
\end{aligned}$$

$\mathfrak{QED}$

Example:

$\int e^x \sin(x)dx$

Let:

$$u = \sin(x),\ dv = e^x \Rightarrow du = \cos(x)dx,\ v = e^x$$

$$\int e^x \sin(x)dx = e^x \sin(x) - \int e^x \cos(x)\,dx$$

$$u = \cos(x),\ dv = e^x \Rightarrow du = -\sin(x),\ v = e^x$$

$$\begin{aligned}
\int e^x \sin(x)dx &= e^x \sin(x) - \left[\cos(x)e^x - \int \sin(x)e^x dx\right] \\
\int e^x \sin(x)dx &= e^x \sin(x) - \cos(x)e^x - \int \sin(x)e^x dx \\
2\int e^x \sin(x)dx &= e^x \sin(x) - \cos(x)e^x \\
\int e^x \sin(x)dx &= \frac{e^x \sin(x) + \cos(x)e^x + C}{2}
\end{aligned}$$

55.6 Integration by Partial Fractions

Applies to rational functions. Four cases:

55.6.1 Linear Factors

Denominator: $(x-a)(x-b)$.

Decomposition:

$$\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$$

55.6.2 Repeated Linear Factors

Denominator: $(x-a)^n$.

Decomposition:

$$\frac{P(x)}{(x-a)^n} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_n}{(x-a)^n}.$$

55.6.3 Irreducible Quadratic Factors

Denominator: (x^2+px+q), where x^2+px+q cannot be factored over the reals

Decomposition:

$$\frac{P(x)}{x^2+px+q} = \frac{Ax+B}{x^2+px+q}$$

55.6.4 Repeated Irreducible Quadratic Factors

Denominator: $(x^2 + px + q)^n$.

Decomposition:

$$\frac{P(x)}{(x^2+px+q)^n} = \frac{A_1x + B_1}{x^2+px+q} + \frac{A_2x+B_2}{(x^2+px+q)^2} + \cdots + \frac{A_nx+B_n}{(x^2+px+q)^n}$$

Example:

$$\int \frac{1}{x^2-1}dx = \int \left(\frac{1}{2(x-1)} - \frac{1}{2(x+1)}\right)dx = \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$$

55.7 Algebraic Manipulation

When dealing with certain integrals, it seems that it is not possible to apply common methods; for these cases, the solution tends to be closely tied to *algebraic manipulation*, of the kind of adding a clever zero or multiplying by an elegant one.

Example:

Find the integral of $f(x) = \frac{30+x^2}{1+x^2}$.

$$\begin{aligned}
\int \frac{30+x^2}{1+x^2}dx &= \int \frac{30}{1+x^2}dx + \frac{x^2}{1+x^2}dx \\
&= 30\int \frac{1}{1+x^2}dx + \int \frac{x^2}{1+x^2}dx \\
&= 30\arctan(x) + \int \frac{x^2-1+1}{1+x^2}dx \\
&= 30\arctan(x) + \int \frac{x^2+1}{1+x^2}dx + \int \frac{-1}{1+x^2}dx \\
&= 30\arctan(x) + \int 1dx - \arctan(x) + C \\
&= 29\arctan(x) + x + C
\end{aligned}$$

55.8 Trigonometric Substitution

Trigonometric substitution is useful for evaluating integrals involving square roots of quadratic expressions.

- $\sqrt{a^2-x^2}$: use $x = a\sin(\theta)$.
- $\sqrt{a^2+x^2}$: use $x = a\tan(\theta)$.
- $\sqrt{x^2-a^2}$: use $x = a\sec(\theta)$.

Example I:

$\int \frac{1}{\sqrt{4-x^2}}\,dx$

We use: $x = 2\sin(\theta) \Rightarrow dx = 2\cos(\theta)\,d\theta$.

$$\sqrt{4-x^2} = \sqrt{4-4\sin^2\theta} = \sqrt{4\cos^2(\theta)} = 2\cos(\theta)$$

Substitute:

$$\int \frac{1}{\sqrt{4-x^2}}\,dx = \int \frac{1}{2\cos(\theta)}\cdot 2\cos(\theta)\,d\theta = \int d\theta = \theta + C$$

Return to x using:

$$x = 2\sin(\theta) \Rightarrow \theta = \arcsin\left(\frac{x}{2}\right) \Rightarrow \int \frac{1}{\sqrt{4-x^2}}\,dx = \arcsin\left(\frac{x}{2}\right) + C$$

Example II:

$\int \frac{x^3}{\sqrt{x^2+4}}\,dx$

Use: $x = 2\tan(\theta) \Rightarrow dx = 2\sec^2(\theta)\,d\theta$

$$\sqrt{x^2+4} = \sqrt{4\tan^2(\theta)+4} = \sqrt{4\sec^2(\theta)} = 2\sec(\theta)$$

Substitute:

$$\int \frac{x^3}{\sqrt{x^2+4}}\,dx = \int \frac{(2\tan(\theta))^3}{2\sec(\theta)} \cdot 2\sec^2(\theta)\,d\theta = \int \frac{8\tan^3(\theta)}{\sec(\theta)} \cdot 2\sec^2(\theta)\,d\theta$$

Simplify:

$$8\int \tan^2(\theta)\sec(\theta)\tan(\theta)\,d\theta = 8\int (\sec^2(\theta)-1)\sec(\theta)\tan(\theta)\,d\theta$$

$$8\int (\sec^2(\theta)\sec(\theta) - \sec(\theta))\tan(\theta)\,d\theta$$

$$8\left(\int \sec^2(\theta)\sec(\theta)\tan(\theta)\,d\theta - \int \sec(\theta)\tan(\theta)\,d\theta\right)$$

$$8\left(\int \sec^2(\theta)\sec(\theta)\,d\theta - \int \sec(\theta)\tan(\theta)\,d\theta\right)$$

For the first one let $u = \sec^3(\theta)$ and $du = 3\sec^2(\theta)\,d\theta$

$$\int u du = \frac{\sec^3(\theta)}{3} + c$$

For the second one

$$\int \sec(\theta)\tan(\theta)\,d\theta = \sec(\theta) + c$$

Together

$$8\left(\frac{\sec^3(\theta)}{3} - \sec(\theta) + c\right)$$

Then solve the integral using trigonometric identities and reduction formulas. Return to x using: $\theta = \arctan\left(\frac{x}{2}\right)$

$$\tan(\theta) = \frac{x}{2} \;\; \sec(\theta) = \frac{\sqrt{x^2+4}}{2}$$

This leaves us with

$$\frac{8}{3}\left(\frac{\sqrt{x^2+4}}{2}\right)^3 - 8\left(\frac{\sqrt{x^2+4}}{2}\right) + c$$

Example III:

$\int \frac{x^3}{\sqrt{x^4-4}}\,dx$

Use: $x = 2\sec\theta,\ dx = 2\sec\theta\tan\theta\,d\theta$

Then

$$\int \frac{(2\sec\theta)^2 2\sec\theta\tan\theta\,d\theta}{2\tan\theta} = \int 8\sec^4\theta\,d\theta$$

$$8\int(\tan^2\theta+1)\sec^2\theta = 8\left(\int\tan^2\theta\sec^2\theta + \int\sec^2\theta\right)$$

Now let $u = \tan\theta$ and $du = \sec^2\theta\,d\theta$

$$8\left(\int u\,du + \int\sec^2\theta\,d\theta\right) = \frac{8}{3}\tan^3\theta + 8\tan\theta + c$$

Now we return to x via our triangle:

$$\tan(\theta) = \frac{\sqrt{x^2+4}}{2}$$

$$\frac{8}{3}\left(\frac{\sqrt{x^2+4}}{2}\right)^3 + 8\left(\frac{\sqrt{x^2+4}}{2}\right) + c$$

55.9 Average Value of a Function

$$\bar{f} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

55.10 Mean Value Theorem for Integrals

If f is continuous on $[a,b]$, then:

$$\exists c\in[a,b] \text{ such that } \int_a^b f(x)\,dx = f(c)(b-a)$$

55.11 Length, Area, Volume Formulas

Arc length:

$$L = \int_a^b \sqrt{1+f'(x)^2}\,dx$$

Area between curves:

$$A = \int_a^b |f(x)-g(x)|\,dx$$

Volume (disk):

$$V = \pi\int_a^b f(x)^2\,dx$$

Shell:

$$V = 2\pi \int_a^b x f(x) \sqrt{1 + f'(x)^2}\, dx$$

55.12 Differentiation of Integrals With Variable Bounds

If

$$F(x) = \int_{a(x)}^{b(x)} f(t)\, dt,$$

then

$$F'(x) = f(b(x)) \cdot b'(x) - f(a(x)) \cdot a'(x)$$

55.13 Parameter-Dependent Integrals

Let $F(p) = \int_a^b f(x,p)\, dx$. If f is continuous and differentiable in p, then:

$$\frac{d}{dp} \int_a^b f(x,p)\, dx = \int_a^b \frac{\partial f}{\partial p}(x,p)\, dx$$

55.14 Leibniz Rule for Differentiating Under the Integral Sign

$$\frac{\partial F}{\partial x} \int_{a(x)}^{b(x)} f(x,t)\, dx = f(x,b(x)) \cdot b'(x) - f(x,a(x)) \cdot a'(x) + \int_{a(x)}^{b(x)} \frac{\partial f}{\partial t}(x,t)\, dt$$

55.15 Improper Integrals With Infinite Bounds

$$\int_a^\infty f(x)\, dx = \lim_{R \to \infty} \int_a^R f(x)\, dx$$

The value when the bounds look like $\int_{-\infty}^{\infty} f(x)\, dx$ is called the *Cauchy-Main-Value*.

$$\int_{-\infty}^\infty f(x)\, dx = \lim_{R \to \infty} \int_{-R}^c f(x)\, dx + \int_c^R f(x)\, dx$$

55.16 Improper Integrals With Discontinuity

These integrals have some kind of asymptote between the bounds we want to evaluate. More concrete:

$$\int_a^b f(x)\, dx \text{ improper if } f \text{ has an infinite discontinuity on } [a,b]$$

To tackle this type of problems we just like before use a limit to approach the discontinuity without reaching it.

$$\int_a^b dx = \lim_{\varepsilon \to 0} \int_a^{c-\varepsilon} f(x) dx + \int_{c+\varepsilon}^b f(x) dx$$

Here c is the value of the discontinuity.

Example:

$$\int_0^1 \frac{1}{2\sqrt{x}}$$

Discontinuity at $c = 0$

$$\begin{aligned}\int_0^1 \frac{1}{2\sqrt{x}} &= \lim_{\varepsilon\to 0}\int_\varepsilon^1 \frac{1}{2\sqrt{x}}dx \\ &= \lim_{\varepsilon\to 0}\sqrt{x}|_\varepsilon^1 \\ &= 1 - \lim_{\varepsilon\to 0}\sqrt{\varepsilon} = 1\end{aligned}$$

55.17 Convergence Criteria for Improper Integrals

- *Majorant:* If $0 \le f(x) \le g(x)$ and $\int g(x)\,dx$ converges, then so does $\int f(x)\,dx$.
- *Minorant:* If $f(x) \ge g(x) \ge 0$ and $\int g(x)\,dx$ diverges, then $\int f(x)\,dx$ diverges.

55.18 Integral Test for Series

Let $f(n) = a_n$ where f is continuous, decreasing, and positive for $n \ge N$:

$$\sum_{n=N}^{\infty} a_n \text{ converges } \iff \int_N^\infty f(x)\,dx \text{ converges}$$

Example:

$\sum \frac{1}{n^p}$ converges $\iff p > 1$

55.19 Integrals with Absolute Value

When given an integral with an absolute value in it, we have to take a look at the function to see how it behaves when x is negative and then split the integral into two or more integrals where the border contain the critical point.

Example:

Integrate

$$\int_{-\infty}^{\infty} e^{-|x|}\,dx$$

This is function will be split at zero, and we will get

$$\begin{aligned}&\int_{-\infty}^{\infty} e^{-|x|}\,dx \\ &\lim_{R\to\infty}\int_{-R}^{0} e^x\,dx + \int_0^R e^{-x}\,dx \\ &\lim_{R\to\infty} 1 - e^{-R} + -e^{-R} + 1 = 2\end{aligned}$$

Chapter 56

Important Theorems of Single Variable Calculus

56.1 Intermediate Value Theorem

Let f be a continuous function on the closed interval $[a, b]$. If $f(a) \neq f(b)$ and N is any number between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that $f(c) = N$.

Proof:

Without loss of generality, assume $f(a) < N < f(b)$. Define:

$$S = \{x \in [a, b] \mid f(x) < N\}.$$

Since $a \in S$, S is non-empty and bounded above by b. Let $c = \sup S$. We want to show that $f(c) = N$ by satisfying the inequalities $f(c) \geq N$ and $f(c) \leq N$.

If $f(c) < N$, by the continuity of f, there exists $\delta > 0$ such that for all $x \in (c, c + \delta)$:

$$f(x) < N.$$

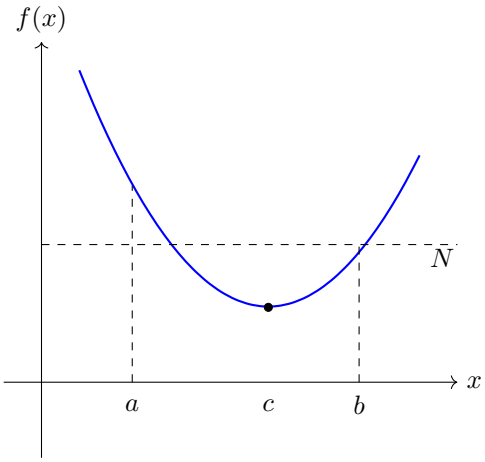
This contradicts the definition of c as the supremum of S.

If $f(c) > N$, again by the continuity of f, there exists $\delta > 0$ such that for all $x \in (c - \delta, c)$:

$$f(x) > N.$$

This contradicts the definition of c as the supremum of S. Hence, we must have $f(c) = N$.

$\mathfrak{QED}$



56.2 Extreme Value Theorem

If f is continuous on the closed interval $[a, b]$, then f attains a maximum and a minimum value on $[a, b]$. That is, there exist $c, d \in [a, b]$ such that:

$$f(c) \leq f(x) \leq f(d) \quad \text{for all } x \in [a, b].$$

Proof:

Since f is continuous on the compact interval $[a, b]$, the image $f([a, b]) \subset \mathbb{R}$ is also compact. A compact subset of $\mathbb{R}$ is closed and bounded, and hence attains its supremum and infimum.

Therefore, there exist $c, d \in [a, b]$ such that:

$$f(c) = \min\{f(x) : x \in [a, b]\}, \quad f(d) = \max\{f(x) : x \in [a, b]\}.$$

$\mathfrak{QED}$

56.3 Rolle's Theorem

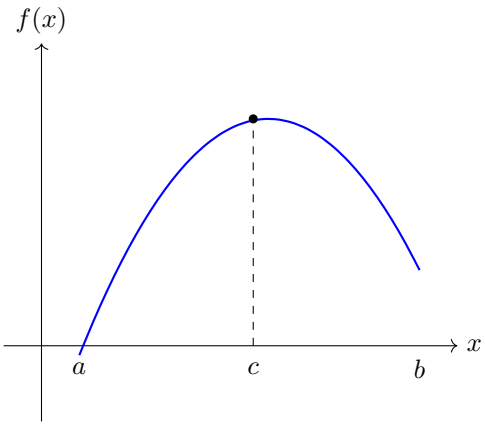
Let f be continuous on $[a, b]$, differentiable on (a, b), and suppose $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Proof:

By the Extreme Value Theorem, f attains a maximum or minimum at some point $c \in [a, b]$.

If $c \in (a, b)$, then by Fermat's theorem, $f'(c) = 0$. If the maximum or minimum occurs at a or b, since $f(a) = f(b)$, f must be constant, so $f'(x) = 0$ everywhere on (a, b), and in particular for some $c \in (a, b)$.

$\mathfrak{QED}$



56.4 Mean Value Theorem

Let f be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof:

Define the auxiliary function:

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a).$$

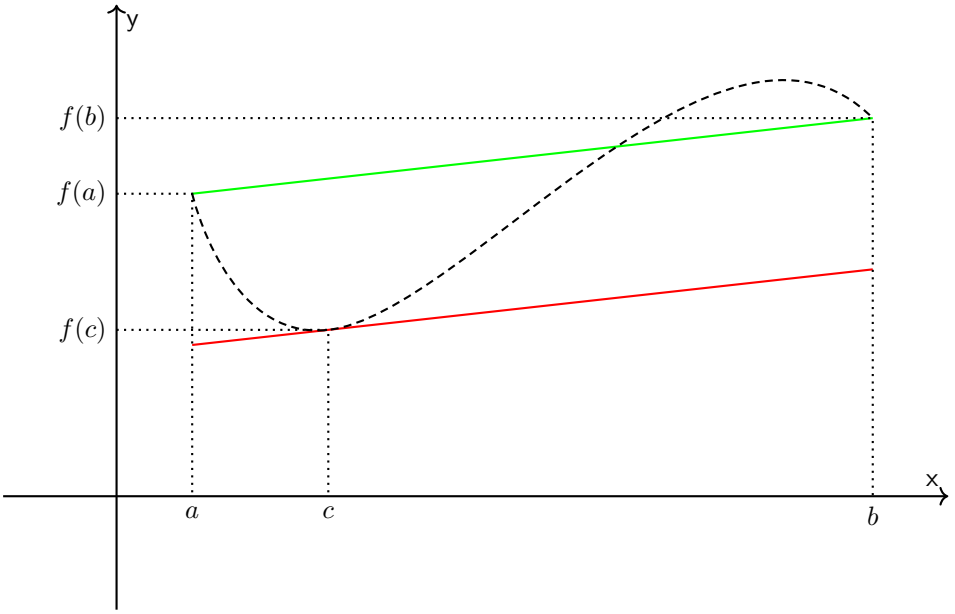
Then $g(a) = f(a)$, $g(b) = f(b) - m(b - a) = f(a) \Rightarrow g(a) = g(b)$.

Apply Rolle's theorem to $g(x)$: Since $g(a) = g(b)$ and g is continuous and differentiable, there exists $c \in (a, b)$ such that $g'(c) = 0$.

Then:

$$g'(x) = f'(x) - \left(\frac{f(b) - f(a)}{b - a} \right) \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}.$$

QED



Chapter 57

Coordinate Systems

Coordinate systems provide different ways of describing points in space, adapted to the geometry of the problem. Below are the most common systems used in multi-variable calculus.

57.1 Cartesian Coordinates

The *Cartesian coordinate system* uses mutually orthogonal axes. A point in $\mathbb{R}^2$ is given by:

$$(x, y) \quad \text{and in } \mathbb{R}^3: \quad (x, y, z).$$

This is the most natural system for rectangular domains and is the basis for most definitions in vector calculus. The coordinate axes are orthogonal and equidistant.

57.2 Polar Coordinates

In *polar coordinates*, a point $(x, y) \in \mathbb{R}^2$ is described by:

$$r \geq 0 \text{ (radius from origin)}, \quad \varphi \in [0, 2\pi) \text{ (angle from the } x\text{-axis)}$$

Transformations:

$$x = r\cos(\varphi), \quad y = r\sin(\varphi)$$

This comes from our right triangle with $r = 1$, where

$$\cos(\varphi) = \frac{x}{r}, \quad \sin(\varphi) = \frac{y}{r}$$

Inverse:

$$r = \sqrt{x^2 + y^2}, \quad \varphi = \arctan\left(\frac{y}{x}\right)$$

This system is useful for circular or radial symmetry. The Jacobian determinant is:

$$|J| = r\,dr\,d\theta$$

57.3 Cylindrical Coordinates

Cylindrical coordinates extend polar coordinates by adding a height z component:

$$(r, \varphi, z) \quad \text{where } r \geq 0,\ \varphi \in [0, 2\pi),\ z \in \mathbb{R}$$

Transformations:

$$x = r\cos(\varphi), \quad y = r\sin(\varphi), \quad z = z$$

Inverse:

$$r = \sqrt{x^2 + y^2}, \quad \varphi = \arctan\left(\frac{y}{x}\right), \quad z = z$$

This coordinate system is ideal for objects with rotational symmetry about the z-axis. The Jacobian determinant is:

$$|J| = r\, dr\, d\varphi\, dz$$

57.4 Spherical Coordinates

In *spherical coordinates*, a point $(x, y, z) \in \mathbb{R}^3$ is represented using:

$$\rho \geq 0 \text{ (radial distance)}, \quad \theta \in [0, \pi] \text{ (inclination)}, \quad \varphi \in [0, 2\pi) \text{ (azimuth)}$$

Transformations:

$$x = \rho\sin(\theta)\cos(\varphi), \quad y = \rho\sin(\theta)\sin(\varphi), \quad z = \rho\cos(\theta)$$

Inverse:

$$\rho = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \arccos\left(\frac{z}{\rho}\right), \quad \varphi = \arctan\left(\frac{y}{x}\right)$$

Spherical coordinates are useful for spherical symmetry (e.g., in gravitational or electric fields). The Jacobian determinant is:

$$|J| = \rho^2\sin(\theta)\, d\rho\, d\phi\, d\theta$$

Each coordinate system is tailored to specific geometries and greatly simplifies integrals when used appropriately. Coordinate transformations require careful application of the Jacobian determinant to account for stretching or distortion of space.

Chapter 58

Parameterized Curves

58.1 Definition

A *parameterized curve* is a mathematical object where each point on the curve is described as a function of one or more parameters, typically denoted by t. Formally, a parameterized curve in $\mathbb{R}^n$ is a continuous function:

$$\mathbf{r}(t) = (x(t), y(t), z(t), \ldots),$$

where t ranges over some interval $I \subseteq \mathbb{R}$, and each component function (e.g., $x(t)$, $y(t)$) is real-valued.

58.1.1 Purpose

The *purpose* of parameterizing a curve is to provide a flexible and detailed way of describing geometric objects, particularly when they cannot be easily expressed as a simple function $y = f(x)$. Parameterizations allow us to:

- Describe curves that loop, cross themselves, or have vertical segments.
- Compute quantities like arc length, tangent vectors, and curvature.
- Easily model motion along a curve, where t can represent time.

58.2 From Parametric to Explicit Form

This process is rather simple because, in the case where it is possible to change from one form to the other, we only need to write the parameter in terms of one of the variables and then substitute.

58.2.1 How to Parameterize a Function

To *parameterize* a curve or a function:

1. Introduce a new parameter t, often representing time or distance along the curve.
2. Express the coordinates (x, y) (or (x, y, z)) as functions of t.
3. Ensure that as t varies over a chosen interval, the set of points $(x(t), y(t))$ traces the desired curve.

Sometimes, parameterizations are natural (e.g., circles using trigonometric functions), while in other cases, one must carefully design a suitable parameterization based on the curve's properties.

Example:

Parameterizing a Circle:

Consider the unit circle defined by the equation:

$$x^2 + y^2 = 1$$

A natural parameterization uses the trigonometric functions sine and cosine:

$$x(t) = \cos(t), \quad y(t) = \sin(t),$$

for $t \in [0, 2\pi]$.

As t varies from 0 to 2π, the point $(x(t), y(t))$ traces out the entire circle exactly once.

58.3 Arclength and t Parameter

Recall that the *arclength* of a curve in 3 dimensions is

$$\int_a^b \sqrt{f'(t)^2 + g'(t)^2 + h'(t)^2}\, dt,$$

which is the norm of our tangent vector,

$$\int_a^b \sqrt{|\vec{v}(t)|}\, dt.$$

58.3.1 Arclength Parameter

This is given by

$$s(t) = \int_a^t \sqrt{|\vec{\tau}(t)|} dt,$$

where τ is the length parameter.

This parameter is intrinsic to the curve and represents one choice, because there is only one valid interpretation of the length. The downside is that this is hard to compute.

58.3.2 The t Parameter

For the t parameter, we can say that it is intrinsic to the object along the curve. It can be interpreted in multiple ways and therefore offers many choices, and it is easy to compute.

Chapter 59

Multi-Variable Functions

59.1 Open Sets

Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. The ε-neighborhood of a point $\vec{x}_0 \in \mathbb{R}^n$ is defined as:

$$U_\varepsilon(\vec{x}_0) := \{\vec{x} \in \mathbb{R}^n \mid \|\vec{x} - \vec{x}_0\| < \varepsilon\}$$

A point $\vec{x}_0 \in D \subseteq \mathbb{R}^n$ is called an *interior point* of D if there exists $\varepsilon > 0$ such that $U_\varepsilon(\vec{x}_0) \subseteq D$. A set D is called *open* if all its points are interior points.

Although ε-neighborhoods depend on the chosen norm, due to the equivalence of norms in $\mathbb{R}^n$, openness is norm-independent.

59.2 Closed Sets

Let $D \subseteq \mathbb{R}^n$ and $\|\cdot\|$ a norm. A point $\vec{x}_0$ is called an *accumulation point* (or limit point) of D if for every $\varepsilon > 0$, the neighborhood $U_\varepsilon(\vec{x}_0)$ contains a point $\vec{x} \neq \vec{x}_0$ such that $\vec{x} \in D$.

A set $D \subseteq \mathbb{R}^n$ is called *closed* if it contains all its accumulation points.

59.3 Boundedness and Order

A set $D \subseteq \mathbb{R}^n$ is called *bounded* if there exists a real number $M > 0$ such that:

$$\|\vec{x}\| < M \quad \text{for all } \vec{x} \in D$$

If no such bound M exists, the set is called *unbounded*.

59.4 Sequences

Let $(\vec{x}_n) \subseteq \mathbb{R}^n$ be a *sequence* of vectors. We say that $\vec{x}_n \to \vec{x} \in \mathbb{R}^n$ as $n \to \infty$ if for every $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that:

$$\|\vec{x}_n - \vec{x}\| < \varepsilon \quad \text{for all } n > n_0$$

The sequence $(\vec{x}_n)$ is called *convergent* if such $\vec{x}$ exists.

The sequence $(\vec{x}_n)$ is called a *Cauchy sequence* if for every $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that:

$$\|\vec{x}_n - \vec{x}_m\| < \varepsilon \quad \text{for all } n, m > n_0$$

Example I:

$$\lim_{n\to\infty} \vec{X}_n = \lim_{n\to\infty} \begin{bmatrix} \frac{1}{n} \\ 1+\frac{1}{n^2} \\ \frac{\sin n}{n} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

59.5 Different Norms

From linear algebra, it is known what a *Norm* is, along with its properties:

Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$. A function $\|\cdot\| : V \to \mathbb{R}$ is called a norm if for all $x, y \in V$ and all scalars α, the following properties hold:

1. *Non-negativity:* $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = 0$.
2. *Absolute homogeneity:* $\|\alpha x\| = |\alpha| \cdot \|x\|$.
3. *Triangle inequality:* $\|x + y\| \leq \|x\| + \|y\|$.

Commonly used norms in $\mathbb{R}^n$ (or $\mathbb{C}^n$) include:

- *Sum Norm* (also known as the ℓ_1 norm):

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$

- *Max Norm* (also known as the ℓ_∞ norm):

$$\|x\|_\infty = \max_{1\leq i\leq n} |x_i|$$

- *Euclidean Norm* (also known as the ℓ_2 norm):

$$\|x\|_2 = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2}$$

Example II:

Given is

$$\vec{X}_n = \begin{bmatrix} 5 - \frac{10}{n} \\ \frac{1}{n} \end{bmatrix}$$

We want to find n_0 for $\varepsilon = 0,1$.

$$\lim_{n\to\infty} \vec{X}_n = \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \vec{X}_0$$

Now use the definition of convergence and the *Euclidean Norm*:

$$\left\| \vec{X}_n - \vec{X}_0 \right\| < \varepsilon$$

$$\left\| \begin{bmatrix} -\frac{10}{n} \\ \frac{1}{n} \end{bmatrix} \right\|$$

$$\frac{\sqrt{101}}{n} < 0,1$$

$$n > 0, 1\sqrt{101} = 100, 5$$

Therefore, $n_0 = 101$.

59.5.1 Equivalence of Norms

In finite-dimensional vector spaces, all norms are equivalent. That is, if $\|\cdot\|_a$ and $\|\cdot\|_b$ are two norms on a finite-dimensional vector space V, then there exist constants $c, C > 0$ such that for all $x \in V$,

$$c\,\|x\|_a \leq \|x\|_b \leq C\,\|x\|_a$$

This implies that while different norms may measure length differently, they all induce the same topology and convergence behavior in finite-dimensional spaces.

59.5.2 Equivalence in the Reals

In $\mathbb{R}^n$, all norms are equivalent, and

$$\begin{aligned}
\|\vec{x}\|_\infty &\leq \|\vec{x}\|_2 \leq \sqrt{n}\,\|\vec{x}\|_\infty \\
\|\vec{x}\|_\infty &\leq \|\vec{x}\|_1 \leq n\,\|\vec{x}\|_\infty \\
\|\vec{x}\|_2 &\leq \|\vec{x}\|_1 \leq n\,\|\vec{x}\|_2
\end{aligned}$$

59.6 Accumulation Point

A vector $\vec{x}_0 \in \mathbb{R}^n$ is called an *accumulation point* of a set $D \subseteq \mathbb{R}^n$ if for every $\varepsilon > 0$, the neighborhood $U_\varepsilon(\vec{x}_0)$ contains a point $\vec{x} \in D \setminus \{\vec{x}_0\}$.

In the context of sequences, an accumulation point of a sequence $(\vec{x}_n)$ is a point $\vec{x} \in \mathbb{R}^n$ for which there exists a subsequence $(\vec{x}_{n_k})$ such that $\vec{x}_{n_k} \to \vec{x}$.

59.7 Bolzano–Weierstraß Theorem

Every infinite, bounded sequence in $\mathbb{R}^n$ has at least one accumulation point.

- Every bounded infinite sequence has at least one convergent subsequence.
- A bounded sequence is *convergent* if and only if it has exactly one accumulation point.
- Every Cauchy sequence in $\mathbb{R}^n$ is convergent (since $\mathbb{R}^n$ is complete).

59.8 Limits

A function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ has the *limit* $g \in \mathbb{R}$ at a point $\vec{X}_0 \in U$ if:

$$\lim_{\vec{X} \to \vec{X}_0} f(\vec{X}) = g$$

means that for every sequence $(\vec{X}_n) \subset U$ converging to $\vec{X}_0$, we have:

$$\lim_{n \to \infty} f(\vec{X}_n) = g$$

The convergence must hold along all possible paths to the point $\vec{X}_0$, making the multi-variable limit path-independent and unique (if it exists).

Example:

$$\vec{X}_n = \begin{bmatrix} \frac{1}{n} \\ 1 + \frac{1}{n^2} \\ \frac{\sin(n)}{n} \end{bmatrix}$$

The limit towards infinity is then

$$\lim_{n\to\infty} \begin{bmatrix} \frac{1}{n} \\ 1 + \frac{1}{n^2} \\ \frac{\sin(n)}{n} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

59.9 Partial Functions

For a function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$, a *partial function* is a restriction of f to a variable by fixing all others.

Examples in $\mathbb{R}^3$:

$$f(x_1, x_2, x_3) = 5x_1 + x_2x_3$$

Fixing values:

$$x_2 = 3, \quad x_3 = 5 \Rightarrow f_1(x_1) = 5x_1 + 15$$
$$x_1 = 2, \quad x_3 = 5 \Rightarrow f_2(x_2) = 10 + 5x_2x_1 = 2, \quad x_2 = 3 \Rightarrow f_3(x_3) = 10 + 3x_3$$

Each f_i reduces the multi-variable function to a single-variable slice (cross-section), allowing analysis along coordinate axes.

59.10 Continuity

Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$, and $\vec{X}_0 \in U$. The function f is said to be *continuous at* $\vec{X}_0$ if:

$$\lim_{\vec{X}\to\vec{X}_0} f(\vec{X}) = f(\vec{X}_0)$$

This means that for any sequence $(\vec{X}_n) \to \vec{X}_0$, we have:

$$\lim_{n\to\infty} f(\vec{X}_n) = f(\vec{X}_0)$$

To check if a multi-variable function is continuous, it is enough to prove that it is continuous at the origin:

$$\begin{aligned} \lim_{\vec{X}\to\vec{X}_0} f(\vec{X}) = f(\vec{X}_0) & \\ \iff & \lim_{\vec{X}\to\vec{X}_0} \left(f(\vec{X}) - f(\vec{X}_0) \right) = 0 \\ \iff & \lim_{\vec{h}\to 0} \left(f(\vec{X}_0 + \vec{h}) - f(\vec{X}_0) \right) = 0 \\ \iff & \lim_{\vec{h}\to 0} g(\vec{h}) = 0 \end{aligned}$$

If we are checking whether a function is continuous and get an *indeterminate form*, there are three main approaches.

1. *Factorizing*: Either normal factorization, adding a clever zero, or multiplying by a clever one.

2. *Change of Coordinates*: Sometimes changing the coordinate system allows us to simplify the expression.
3. *Different Paths*: Trying different paths can be pretty straightforward. If we get two different values for the same point across different paths, then the function is not continuous.

Example:

Given is $\frac{x^3+2yx^2+y^2x+2y^3}{x+2y}$ for $(x,y) \neq (0,0)$ and 0 for $(0,0)$.

$$\begin{aligned}\frac{x^3+2yx^2+y^2x+2y^3}{x+2y} &= \frac{x^2(x+2y)+y(x+2y)}{x+2y} \\ &= \frac{(x+2y)(x^2+y^2)}{x+2y} \\ &= x^2+y^2\end{aligned}$$

Now by taking the limit we see that

$$\lim_{(x,y)\to(0,0)} x^2+y^2 = 0$$

Thus, the function is continuous.

Example:

Check for continuity towards the origin of $f(x,y) = \frac{xy^2}{x^2+y^4}$.

For the path $x = y$ and $y = 0$:

$$f(x,0) = \frac{x0}{x^2+0} = 0$$

For the path $x = y^2$:

$$f(y^2,y) = \frac{y^4}{2y^4} = \frac{1}{2}$$

Because we got two different values, the function is not continuous.

59.10.1 Continuity in Polar Coordinates for 2D

Let $\vec{X} = (x,y)$ be written in polar coordinates:

$$x = r\cos\varphi, \quad y = r\sin\varphi$$

Then:

$$\vec{X}_n \to (0,0) \iff r_n \to 0$$

This alternative approach helps evaluate limits and continuity at the origin using radial convergence.

Example:

Given is

$$f(xy) = \begin{cases} \frac{xy}{x^2+y^2} \text{ for } (x,y) \neq (0,0) \\ 0 \text{ else} \end{cases}$$

Let $x = r\cos(\theta)$ and $y = r\sin(\theta)$

$$\begin{aligned}
&\lim_{r\to 0} \frac{r\cos(\theta) r\sin(\theta)}{r^2\cos^2(\theta) + r^2\sin^2(\theta)} \\
&= \lim_{r\to 0} \frac{r^2\cos(\theta)\sin(\theta)}{r^2(\cos^2(\theta) + \sin^2(\theta))} \\
&= \lim_{r\to 0} \frac{\cos(\theta)\sin(\theta)}{1}
\end{aligned}$$

Because of the dependence on θ, which for $\theta = \frac{\pi}{4}$ gives us $\frac{1}{2}$, this function is not continuous.

59.11 Substitution

When dealing with multi-variable functions we can sometimes use *substitution* for converting our function into a single variable function to use *L'Hospital's Rule*.

Example:

Given is $\frac{e^{x^2+y^2}-1}{x^2+y^2}$.

See that we have repeated terms we can substitute: $p^2 = x^2 + y^2$.

$$\frac{e^{p^2} - 1}{p^2}$$

Now differentiate:

$$\frac{2pe^{p^2}}{2p} = e^{p^2}$$

Now take the limit:

$$\lim_{p\to 0} e^{p^2} = 1$$

59.12 Uniform and Lipschitz Continuity

59.12.1 Uniform Continuity

A function $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ is called *uniformly continuous* on D if:

$$\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } \|\vec{X} - \vec{X_0}\| < \delta \Rightarrow |f(\vec{X}) - f(\vec{X_0})| < \varepsilon$$

The key feature is that δ depends only on ε, not on the points $\vec{X}, \vec{X_0}$ themselves. If f is continuous on a compact set (closed and bounded), then f is uniformly continuous.

59.12.2 Lipschitz Continuity

A function $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ is *Lipschitz continuous* if there exists a constant $L > 0$ such that:

$$|f(\vec{X}) - f(\vec{Y})| \leq L\|\vec{X} - \vec{Y}\| \quad \text{for all } \vec{X}, \vec{Y} \in D$$

If $L < 1$, then f is called a *contraction*.

59.13 Epsilon–Delta Criterion

Let $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$, and $\vec{X}_0 \in D$. The function f is continuous at $\vec{X}_0$ if:

$$\forall \varepsilon > 0 \; \exists \delta > 0 \text{ such that } \|\vec{X} - \vec{X}_0\| < \delta \Rightarrow |f(\vec{X}) - f(\vec{X}_0)| < \varepsilon$$

This generalizes the familiar *epsilon-delta criterion* from single-variable calculus to multi-variable settings using norms.

59.14 Fixpoints

A point $\vec{X}_0 \in D \subseteq \mathbb{R}^n$ is called a *fix-point* of a function $\vec{\varphi} : D \to \mathbb{R}^n$ if:

$$\vec{\varphi}(\vec{X}_0) = \vec{X}_0$$

That is, applying the function does not change the point — it maps to itself.

59.15 Banach Fixed Point Theorem

Let $\vec{\varphi} : D \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be a *contraction mapping*, i.e., there exists a constant $L < 1$ such that:

$$\|\vec{\varphi}(\vec{X}) - \vec{\varphi}(\vec{Y})\| \leq L\|\vec{X} - \vec{Y}\| \quad \text{for all } \vec{X}, \vec{Y} \in D$$

Then

- There exists a unique fix-point $\vec{X}^* \in D$ such that $\vec{\varphi}(\vec{X}^*) = \vec{X}^*$.
- Iteratively defined sequences $\vec{X}_{n+1} = \vec{\varphi}(\vec{X}_n)$ converge to the fix-point.

This result is fundamental in nonlinear analysis and iterative numerical methods.

59.16 Vector Fields

A *vector field* is a function $\vec{f} : \mathbb{R}^n \to \mathbb{R}^n$; in other words, simply a function with vectors as inputs and outputs.

$$\vec{f}(x_1, x_2, \ldots, x_n) = A_1(x_1, \ldots, x_n)\vec{e}_1 + \cdots + A_n(x_1, \ldots, x_n)\vec{e}_n$$

We call a vector field *continuous/differentiable* if all functions

$$A_1, A_2, \ldots, A_n$$

are continuous/differentiable.

As a side note, the *gradient* of a multi-variable function is a vector field.

Chapter 60

Multi-Variable Differentiation

60.1 Partial Derivatives

Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ and $\vec{X}_0 = (x_1^{(0)}, \ldots, x_n^{(0)}) \in U$. Then f is *partially differentiable with respect to* x_i at $\vec{X}_0$ if the limit:

$$\frac{\partial f}{\partial x_i}(\vec{X}_0) := \lim_{h \to 0} \frac{f(x_1^{(0)}, \ldots, x_i^{(0)} + h, \ldots, x_n^{(0)}) - f(\vec{X}_0)}{h}$$

exists. This derivative measures the rate of change of f in the direction of the x_i-axis while keeping all other variables fixed.

A function is *(partially) differentiable* at $\vec{X}_0$ if all partial derivatives $\frac{\partial f}{\partial x_i}(\vec{X}_0)$ exist.

60.2 Differentiability

We call a function f *differentiable* at $(x_1, x_2, \cdots, x_n)$ if and only if:

$$f(x_1 + \Delta x_1, x_2 + \Delta x_2, \ldots, x_n + \Delta x_n) - f(x_1, x_2, \ldots, x_n) =$$

$$\frac{\partial f(x_1, y_2, \ldots, x_n)}{\partial x_1}\Delta x_1 + \frac{\partial f(x_1, x_2, \ldots, x_n)}{\partial x_2}\Delta x_2 + \cdots + E_1(\Delta x_1) + E_2(\Delta x_n) + \cdots + E_n(\Delta x_n)$$

where $\lim_{\Delta x_1 \to 0} \frac{E_1(\Delta x_1)}{\Delta x_1} = \lim_{\Delta x_2 \to 0} \frac{E_2(\Delta x_2)}{\Delta x_2} = \cdots = 0$.

60.3 The Gradient

Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\vec{X}_0 = (x_1^{(0)}, \ldots, x_n^{(0)})$. The *gradient* of f at $\vec{X}_0$ is:

$$\nabla f(\vec{X}_0) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\vec{X}_0) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\vec{X}_0) \end{bmatrix}$$

The angle of the maximal slope is $\phi = 0$. This comes from the fact that the directional derivative, which is defined as

$$D_{\vec{v}} f(\vec{x}_0) = \langle \nabla f(\vec{x}_0), \vec{v} \rangle$$

which can be written as

$$\langle \nabla f, \vec{v} \rangle = \|\nabla f\| \, \|\vec{v}\| \cos(\phi)$$

tells us that our gradient is orthogonal to the tangent vector of the level curve, i.e., $\phi = \frac{\pi}{2}$, and the largest slope is when $\phi = 0$, which is the direction of the gradient.

Another way of thinking about this fact is using the definition as a projection of the dot product.

$$\text{proj}_{\nabla f}(\vec{v}) = \frac{\langle \nabla f, \vec{v} \rangle}{\|\nabla f\|^2} \nabla f$$

Now our directional derivative is the projection of the gradient onto the direction vector $\vec{v}$, and we want to maximize this projection. This happens when $\vec{v}$ is in the same direction as the gradient, i.e., $\vec{v} = \nabla f$. Hence, the direction of the maximal slope is in the direction of the gradient.

Formally, we have a vector d^* that points in the direction of the maximum slope, defined by

$$d^* = \arg_{\|d\|=1} \max D_{\vec{v}} f(\vec{x})$$

where $\vec{v}$ is a unit vector.

The directional derivative, written as the inner product with the gradient, gives us

$$d^* = \arg_{\|d\|=1} \max \langle \nabla f, \vec{v} \rangle$$

Then, using the definition of the dot product,

$$d^* = \arg_{\|d\|=1} \max \|\nabla f\| \, \|\vec{v}\| \cos(\phi)$$

Finally, because the magnitude of $\vec{v}$ is one, and to maximize cosine we set $\phi = \frac{\pi}{2}$,

$$d^* = \|\nabla f\|$$

Therefore, the magnitude of the gradient is the maximal slope.

60.4 Origin of the Formula

Suppose along the curve $r(t) = x_1(t)\hat{v_1} + x_2(t)\hat{v_2} + \cdots + x_n(t)\hat{v_n}$ that $f(x_1(t), \ldots, x_n(t)) = C$.

For a fixed t, we get a constant value C, and differentiating with respect to t gives us the following:

$$\frac{\partial}{\partial t} f(x_1(t), \ldots, x_n(t)) = \frac{\partial}{\partial t} C$$

$$\frac{\partial f}{\partial x_1} \frac{dx_1}{dt} + \cdots + \frac{\partial f}{\partial x_n} \frac{dx_n}{dt} = 0$$

We can write it as a dot product of two vectors:

$$\left\langle \left(\frac{\partial f}{\partial x_1} \hat{v_1} + \cdots + \frac{\partial f}{\partial x_n} \hat{v_n} \right), \left(\frac{dx_1}{dt} \hat{v_1} + \cdots + \frac{dx_n}{dt} \hat{v_n} \right) \right\rangle = 0$$

From the formula, we see that ∇f is the normal and $\frac{d\vec{r}}{dt}$ is the tangent vector of the curve.

$\mathfrak{QED}$

60.4.1 Gradient Operations

- $\nabla(f+g) = \nabla f + \nabla g$
- $\nabla(\lambda f) = \lambda \nabla f$ for $\lambda \in \mathbb{R}$
- $\nabla(fg) = f\nabla g + g\nabla f$

The $\frac{\text{grad } f}{\|\text{grad } f\|}$ points in the direction of the steepest increase of f, and $-\frac{\text{grad } f}{\|\text{grad } f\|}$ in the direction of the steepest decrease.

60.5 The Tangent Plane

The *tangent plane* to a differentiable surface $z = f(x,y)$ at the point (x_0, y_0) is the plane that best approximates the surface near that point. It is given by the linearization of f:

$$T(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$$

This plane touches the graph of the function at a single point and shares its slope in both x- and y-directions.

It can also be written in the parameterized form:

$$T(x,y) = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ f_x(x_0,y_0) \end{bmatrix} + \mu \begin{bmatrix} 0 \\ 1 \\ f_y(x_0,y_0) \end{bmatrix}$$

60.5.1 Generalization

$$\begin{aligned} T(\vec{X}) &= f(\vec{x_0}) + \sum_{i=1}^{n} f_{x_i}(\vec{x_0})(x_i - x_i^0) \\ &= f(\vec{x_0}) + \langle \nabla f, (\vec{x} - \vec{x_0}) \rangle \end{aligned}$$

Example: $f(x,y) = 4x^2y + xy^2 + 1$

Compute partial derivatives:

$$f_x(x,y) = 8xy + y^2, \quad f_y(x,y) = 4x^2 + 2xy$$

At the point $(1,1)$:

$$f(1,1) = 4(1)^2(1) + 1(1)^2 + 1 = 6 f_x(1,1) = 8(1)(1) + 1 = 9, \quad f_y(1,1) = 4(1)^2 + 2(1)(1) = 6$$

Thus, the tangent plane is:

$$T(x,y) = 6 + 9(x-1) + 6(y-1) \Rightarrow T(x,y) = 9x + 6y - 9$$

60.6 The Directional Derivative

Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\vec{x}_0$, and let $\vec{v} \in \mathbb{R}^n$ be a direction vector with $\|\vec{v}\| = 1$. The *directional derivative* of f at $\vec{x}_0$ in the direction $\vec{v}$ is defined as:

$$D_{\vec{v}} f(\vec{x}_0) := \lim_{h \to 0} \frac{f(\vec{x}_0 + h\vec{v}) - f(\vec{x}_0)}{h}$$

If f is differentiable at $\vec{x}_0$, then the directional derivative can be computed using the gradient:

$$D_{\vec{v}} f(\vec{x}_0) = \langle \nabla f(\vec{x}_0), \vec{v} \rangle$$

Schwarz Inequality (Cauchy–Schwarz):

$$|\langle \vec{a}, \vec{b} \rangle| \leq \|\vec{a}\| \cdot \left\|\vec{b}\right\|$$

This implies:

$$|D_{\vec{v}} f(\vec{x}_0)| \leq \|\nabla f(\vec{x}_0)\|$$

60.6.1 Derivation of the Directional Derivative Formula

Let $f(x, y)$ be differentiable, and let $\vec{v} = (e_1, e_2)$ be a direction vector. We aim to derive the formula for the directional derivative $D_{\vec{v}} f(x_0, y_0)$.

We begin by considering the tangent plane to the surface $z = f(x, y)$ at the point $(x_0, y_0, f(x_0, y_0))$. This plane is spanned by the vectors:

$$\vec{u}_1 = \begin{bmatrix} 1 \\ 0 \\ f_x(x_0, y_0) \end{bmatrix}, \quad \vec{u}_2 = \begin{bmatrix} 0 \\ 1 \\ f_y(x_0, y_0) \end{bmatrix}$$

These correspond to directional derivatives along the x- and y-axes, respectively.

Now consider the direction vector $\vec{v} = (e_1, e_2)$. Lift it into 3D space (onto the tangent plane) as:

$$\vec{u}_3 = \begin{bmatrix} e_1 \\ e_2 \\ D_{\vec{v}} f \end{bmatrix}$$

Since all three vectors lie in the same plane, the determinant of the matrix formed by these vectors as columns must vanish:

$$\det(\vec{u}_1, \vec{u}_2, \vec{u}_3) = 0$$

Explicitly:

$$\det \begin{bmatrix} 1 & 0 & e_1 \\ 0 & 1 & e_2 \\ f_x(x_0, y_0) & f_y(x_0, y_0) & D_{\vec{v}} f \end{bmatrix} = 0$$

Expanding the determinant gives:

$$f_x(x_0, y_0) \cdot e_2 - f_y(x_0, y_0) \cdot e_1 + D_{\vec{v}} f = 0$$

Solving for $D_{\vec{v}} f$, we obtain:

$$D_{\vec{v}} f = f_x(x_0, y_0) \cdot e_1 + f_y(x_0, y_0) \cdot e_2$$

Vector notation:

$$D_{\vec{v}} f = \langle \nabla f(x_0, y_0), \vec{v} \rangle$$

This is the desired result: the directional derivative equals the dot product of the gradient of f and the direction vector $\vec{v}$.

60.7 The Total Differential

If $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\vec{x}_0$, then the *total differential* of f at $\vec{x}_0$ is the linear approximation:

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\vec{x}_0)\, dx_i = \langle \nabla f(\vec{x}_0), d\vec{x} \rangle$$

This expresses how small changes in the input variables propagate into changes in the function values.

60.8 The Chain Rule

Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable, and suppose $\vec{x} = \vec{x}(t) \in \mathbb{R}^n$ is a differentiable path. Then:

$$\frac{d}{dt} f(\vec{x}(t)) = \left\langle \nabla f(\vec{x}(t)), \frac{d\vec{x}(t)}{dt} \right\rangle$$

This is the *chain rule* in vector form.

Or, to be more explicit, a small change in t induces a change in $x_1, x_2, \cdots$, and all these changes sum up to the total change in the original function.

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x_1}\frac{\partial x_1}{\partial t} + \cdots + \frac{\partial z}{\partial x_n}\frac{\partial x_n}{\partial t}$$

60.9 Absolute and Relative Error

Given an approximate value $\tilde{x}$ for the exact value x:

- *Absolute error:* $\Delta z_{\max} \leq |f_{x_1}||\Delta x_1| + \cdots + |f_{x_n}||\Delta x_n|$
- *Relative error:* $\varepsilon = \frac{\Delta z_{\max}}{z}$ with dependence on the error of the inputs $\frac{\Delta x}{x}$ etc.

In multi-variable contexts, similar formulas apply using vector norms.

Absolute Error Example

We consider the function:

$$z = \sqrt{x^2 + y^2}$$

with the measured values:

$$x = 4\,\text{cm}, \quad y = 3\,\text{cm}$$

Both x and y are measured with a precision of $\Delta x = \Delta y = 0{,}1\,\text{cm}$. The calculated length is $z = 5\,\text{cm}$. We now determine the maximum possible absolute error in z.

Step 1: Partial derivatives of z:

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}, \quad \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

Step 2: Evaluate at the given point:

$$\frac{\partial z}{\partial x}(4, 3) = \frac{4}{5}, \quad \frac{\partial z}{\partial y}(4, 3) = \frac{3}{5}$$

Step 3: Use the absolute error formula:

$$\Delta z \leq \left|\frac{\partial z}{\partial x}\right| \cdot \Delta x + \left|\frac{\partial z}{\partial y}\right| \cdot \Delta y$$

$$\Delta z \leq \frac{4}{5} \cdot 0{,}1 + \frac{3}{5} \cdot 0{,}1 = \frac{7}{5} \cdot 0{,}1 = 0{,}14\,\text{cm}$$

Conclusion:

The result for z is accurate to within 0,14 cm.

Relative Error Example

The relative error of a function $z = f(x_1, \ldots, x_n)$ is defined by:

$$\varepsilon = \frac{\Delta z_{\max}}{z}$$

Assuming z depends on variables with known relative measurement errors, and the function is differentiable, we estimate:

$$\varepsilon = \frac{\left|\frac{\partial f}{\partial x_1}\right| \cdot \Delta x_1 + \cdots + \left|\frac{\partial f}{\partial x_n}\right| \cdot \Delta x_n}{f(x_1, \ldots, x_n)}$$

or more directly using relative errors:

$$\varepsilon \approx \left|\frac{\Delta x_1}{x_1}\right| + \cdots + \left|\frac{\Delta x_n}{x_n}\right|$$

Example: Volume of a Cuboid

Let:

$$z = f(a, b, c) = a \cdot b \cdot c$$

Assume the side lengths $a = b = c$ are measured with a uniform relative error $\varepsilon_{\text{in}} = \frac{\Delta a}{a} = \frac{\Delta b}{b} = \frac{\Delta c}{c} = \varepsilon$.

Then the partial derivatives are:

$$\frac{\partial z}{\partial a} = bc, \quad \frac{\partial z}{\partial b} = ac, \quad \frac{\partial z}{\partial c} = ab$$

Applying the relative error formula:

$$\frac{\Delta z_{\max}}{z} \leq \frac{bc \cdot \Delta a + ac \cdot \Delta b + ab \cdot \Delta c}{abc} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c} = 3\varepsilon$$

Target accuracy:

If the final result z must be accurate to within $\varepsilon_z = 0{,}01$ (i.e., 1%), then we must satisfy:

$$3\varepsilon \leq 0{,}01 \Rightarrow \varepsilon \leq \frac{0{,}01}{3} = 0,0033 \quad (\text{or } 0,33\%)$$

Conclusion:

Each input measurement must be made with a maximum relative error of $0,33\%$ to ensure the output $z = abc$ is accurate to within 1 percent.

60.10 Implicit Differentiation

If a function $F(x, y) = 0$ defines y *implicitly* as a function of x, and F is differentiable, then:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

This generalizes to higher dimensions using the total differential and the implicit function theorem.

Example:

Given $x^2 + y^2 + z^2 = 1$, differentiate with respect to x. Regard z, the target function, as a function of the other variables, in this case $z(x, y)$.

$$\frac{d}{dx}x^2 + y^2 + z^2 - 1 = 0$$

$$2x + 2z\frac{dz}{dx} = 0$$

$$\frac{dz}{dx} = -\frac{x}{z}$$

Example:

Given the implicitly defined function:

$$yx^2 + \frac{x}{y} + x + y - 4 = 0$$

We are asked to compute the derivative $y'(1)$, that is, the derivative of y with respect to x at the point $(1, 1)$.

Step 1: Differentiate implicitly

Differentiate both sides of the equation with respect to x, treating $y = y(x)$ as a function of x:

$$\frac{d}{dx}(yx^2) + \frac{d}{dx}\left(\frac{x}{y}\right) + \frac{d}{dx}(x) + \frac{d}{dx}(y) - \frac{d}{dx}(4) = 0$$
$$y'x^2 + y \cdot 2x + \frac{y - xy'}{y^2} + 1 + y' = 0$$

Collect terms involving y'. Group all y' terms on one side:

$$y'x^2 - \frac{xy'}{y^2} + y' + 2xy + \frac{1}{y} + 1 = 0$$

Factor y':

$$y'\left(x^2 - \frac{x}{y^2} + 1\right) = -\left(2xy + \frac{1}{y} + 1\right)$$

Evaluate at $x = 1, y = 1$:

Substitute $x = 1$, $y = 1$:

$$\begin{aligned} y'(1-1+1) &= -(2+1+1) \\ y'(1) &= -4 \\ y' &= -4 \end{aligned}$$

Final Answer:

$$y'(1) = -4$$

60.11 Schwarz's Theorem (Clairaut's Theorem)

Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable. Then for any $i \neq j$, the mixed partial derivatives satisfy:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

That is, the order of partial differentiation does not matter if all second derivatives are continuous.

60.12 The Hessian

For a function $f : \mathbb{R}^n \to \mathbb{R}$, we define the *Hessian* as the matrix composed of its second derivatives.

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(x)}{\partial x_n \partial x_n} \end{bmatrix}$$

When analyzing the Hessian at critical points, we get:

- For the $\mathbb{R}^2$ case we use:

  If $H_f(x)$ is *positive definite* $f_{xx}(x_0, y_0) > 0$, $\det(H) > 0$, then (x_0, y_0) is a local minimum.

  If $H_f(x)$ is *negative definite* $f_{xx}(x_0, y_0) < 0$, $\det(H) > 0$, then it is a local maximum.

  If $H_f(x)$ has mixed signs, *indefinite*, $\det(H) < 0$, then it is a saddle point.

  If $\det(H) = 0$, then the next derivative decides.

- For the general case $\mathbb{R}^n$, we use the eigenvalues:

  Positive definite if $\lambda_i > 0$, $\forall \lambda_i$, which implies a local minimum.

  Negative definite if $\lambda_i < 0$, $\forall \lambda_i$, which implies a local maximum.

  Indefinite for mixed signs, which implies a saddle point.

60.13 The Jacobian

Let $\vec{f} : \mathbb{R}^n \to \mathbb{R}^m$, where

$$\vec{f}(\vec{x}) = \begin{bmatrix} f_1(x_1, \ldots, x_n) \\ \vdots \\ f_m(x_1, \ldots, x_n) \end{bmatrix}$$

The *Jacobian matrix* of $\vec{f}$ is the $m \times n$ matrix:

$$J_{\vec{f}}(\vec{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

In the case $n = m$, the *Jacobian determinant* $\det J_{\vec{f}}(\vec{x})$ describes local invertibility and orientation.

60.14 Taylor's Theorem

Let $f : \mathbb{R}^n \to \mathbb{R}$ be k-times continuously differentiable at $\vec{x}_0$. Then the *Taylor polynomial* of degree 2 around $\vec{x}_0$ is:

$$f(\vec{x}) \approx f(\vec{x}_0) + \langle \nabla f(\vec{x}_0), \vec{x} - \vec{x}_0 \rangle + \frac{1}{2}(\vec{x} - \vec{x}_0)^T H_f(\vec{x}_0)(\vec{x} - \vec{x}_0) + \cdots,$$

where $H_f(\vec{x}_0)$ is the *Hessian matrix* of second partial derivatives.

Example:

Let $f(x, y) = e^y \sin(x + 2y)$. We expand around the point $(a, b) = (0, 0)$.

First, compute the necessary derivatives at $(0, 0)$:

$$\begin{aligned} f(0,0) &= \sin(0) = 0 \\ f_x &= e^y \cos(x+2y) \quad \Rightarrow \quad f_x(0,0) = 1 \\ f_y &= 2ye^y \cos(x+2y) + e^y \sin(x+2y) \quad \Rightarrow \quad f_y(0,0) = 0 + 2 = 2 \\ f_{xx} &= -e^y \sin(x+2y) \quad \Rightarrow \quad f_{xx}(0,0) = 0 \\ f_{xy} = f_{yx} &= 2e^y \sin(x+2y) + e^y \cos(x+2y) \quad \Rightarrow \quad f_{xy}(0,0) = 3 \\ f_{yy} &= -3e^y \sin(x+2y) + 4e^y \cos(x+2y) \quad \Rightarrow \quad f_{yy}(0,0) = 0 + 4 = 4 \end{aligned}$$

Now, the second-order Taylor polynomial at $(0, 0)$ is:

$$\begin{aligned} f(x,y) &\approx 0 + 1 \cdot x + 2 \cdot y \\ &\quad + \frac{1}{2} \cdot 0 \cdot x^2 + 3 \cdot xy + \frac{1}{2} \cdot 4 \cdot y^2 \\ &= x + 2y + 3xy + 2y^2 \end{aligned}$$

60.15 The Tangent Vector

Let $\vec{X}(t) \in \mathbb{R}^n$ be a differentiable, parameterized curve. The *tangent vector* at the point $\vec{X}(t_0)$ is given by:

$$\vec{X}'(t_0) = \begin{bmatrix} x_1'(t_0) \\ x_2'(t_0) \\ \vdots \\ x_n'(t_0) \end{bmatrix}$$

This vector points in the direction in which the curve is moving at t_0, and its magnitude corresponds to the instantaneous speed. The unit tangent vector is:

$$\vec{X}(t) = \frac{\vec{X}'(t)}{\|\vec{X}'(t)\|}$$

The speed of the vector in $\mathbb{R}^2$ is given by:

$$\vec{X}(t) = \begin{bmatrix} t \\ f(t) \end{bmatrix}$$

and the tangent is:

$$T(t) = \begin{bmatrix} t \\ f(t) \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ f'(t) \end{bmatrix}$$

60.15.1 Derivation of the Tangent Vector Formula

Given a position vector function $\vec{r}(t)$, the tangent vector to the curve described by $\vec{r}(t)$ is obtained by taking the derivative of $\vec{r}(t)$ with respect to time. This derivative is defined as the following limit:

$$\frac{d\vec{r}}{dt} = \lim_{\Delta t \to 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t}$$

Assuming that $\vec{r}(t)$ is expressed in terms of the canonical basis vectors $\hat{\imath}, \hat{\jmath}, \hat{k}$, we can write:

$$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}$$

Then, the increment becomes:

$$\vec{r}(t + \Delta t) = x(t + \Delta t)\hat{\imath} + y(t + \Delta t)\hat{\jmath} + z(t + \Delta t)\hat{k}$$

So the difference in the numerator of the derivative is:

$$\begin{aligned} \vec{r}(t + \Delta t) - \vec{r}(t) = &[x(t + \Delta t) - x(t)]\,\hat{\imath} \\ &+ [y(t + \Delta t) - y(t)]\,\hat{\jmath} \\ &+ [z(t + \Delta t) - z(t)]\,\hat{k} \end{aligned}$$

Dividing by Δt and taking the limit:

$$\begin{aligned} \frac{d\vec{r}}{dt} &= \lim_{\Delta t \to 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t} \\ &= \left(\lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} \right) \hat{\imath} \\ &\quad + \left(\lim_{\Delta t \to 0} \frac{y(t + \Delta t) - y(t)}{\Delta t} \right) \hat{\jmath} \\ &\quad + \left(\lim_{\Delta t \to 0} \frac{z(t + \Delta t) - z(t)}{\Delta t} \right) \hat{k} \\ &= \frac{dx}{dt}\hat{\imath} + \frac{dy}{dt}\hat{\jmath} + \frac{dz}{dt}\hat{k} \end{aligned}$$

This vector $\frac{d\vec{r}}{dt}$ points in the direction of motion and is tangent to the curve at each point t.

Chapter 61

Multidimensional Integrals

61.1 Arclength of a Curve

We will use the famous technique of approximating the length/area/volume of something by dividing it into smaller, equal sections that are straight lines in this case. In other cases they will be prisms, squares, etc. Now, if we look at the difference between two points in our 3D space, for this example we notice that there are three differences: Δi, Δj, Δk. Now let us take the length of the vector $\sqrt{(\Delta i)^2 + (\Delta j)^2 + (\Delta k)^2}$.

The total *arclength* is going to be the sum of all of these sectors.

$$L = \sum_{i=1}^{n} \sqrt{(\Delta i)^2 + (\Delta j)^2 + (\Delta k)^2}$$

If we use a clever one $\left(\frac{\Delta t}{\Delta t}\right)^2$, we get

$$L = \sum_{i=1}^{n} \sqrt{\left(\frac{\Delta i}{\Delta t}\right)^2 + \left(\frac{\Delta j}{\Delta t}\right)^2 + \left(\frac{\Delta k}{\Delta t}\right)^2}\,\Delta t$$

Now, if we take the limit, we get

$$L = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{\left(\frac{\Delta i}{\Delta t}\right)^{2} + \left(\frac{\Delta j}{\Delta t}\right)^2 + \left(\frac{\Delta k}{\Delta t}\right)^2}\,\Delta t$$

which is

$$L = \int_a^b \sqrt{\left(\frac{\Delta i}{\Delta t}\right)^2 + \left(\frac{\Delta j}{\Delta t}\right)^2 + \left(\frac{\Delta k}{\Delta t}\right)^2}\,dt$$

$$L = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2}\,dt$$

For a differentiable function on the interval $t \in [a, b]$.

61.2 Line Integrals

61.3 Over Vector Field

Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a *vector field*, and let γ be a smooth curve parametrized by $\vec{r}(t)$, $t \in [a, b]$. The *line integral* of F along γ is:

$$\int_\gamma \left\langle \vec{F}(\vec{r}(t)), \frac{d\vec{r}(t)}{dt} \right\rangle dt = \int_a^b \left\langle \vec{F}(\vec{r}(t)), \frac{\vec{dr}}{dt} \right\rangle dt = \vec{F}(\vec{r}(b)) - \vec{F}(\vec{r}(a))$$

In the context of physics, this integral measures the work done by the field $\vec{F}$ along the path γ, such as force over distance.

In compact form:

$$\int_\gamma \langle \vec{F}, d\vec{r} \rangle$$

Example: Line Integral over a Vector Field

Let $f(x, y) = 2x + 2y$ be our vector field.

Let the curve c be the quarter-circle of radius 1 from $(1, 0)$ to $(0, 1)$, parametrized by:

$$\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} \cos(t) \\ \sin(t) \end{bmatrix} \quad t \in \left[0, \frac{\pi}{2}\right]$$

We compute the line integral of $\vec{F}$ along c:

$$\int_C \langle \vec{F}(\vec{r}(t)), \frac{d\vec{r}(t)}{dt} \rangle dt = \int_0^{\frac{\pi}{2}} \langle \vec{F}(\vec{r}(t)), \frac{d\vec{r}(t)}{dt} \rangle dt$$

First, compute $\vec{F}(\vec{r}(t))$:

$$\vec{F}(\vec{r}(t)) = \begin{bmatrix} 2\cos(t) \\ 2\sin(t) \end{bmatrix}$$

Then the tangent vector of $\vec{r}(t)$:

$$\frac{d\vec{r}(t)}{dt} = \begin{bmatrix} -\sin(t) \\ \cos(t) \end{bmatrix}$$

Now compute the inner product:

$$\langle \vec{F}(\vec{r}(t)), \frac{d\vec{r}(t)}{dt} \rangle = -2\cos(t)\sin(t) + \cos(t)2\sin(t) = 0$$

Therefore, the line integral is:

$$\begin{aligned} &\int_C \langle \vec{F}, \frac{d\vec{r}}{dt} \rangle dt \\ &= \int_0^{\frac{\pi}{2}} 0 dt \\ &= 0 \end{aligned}$$

The line integral of the gradient vector field $f(x, y) = 2x + 2y$ over this curve is zero.

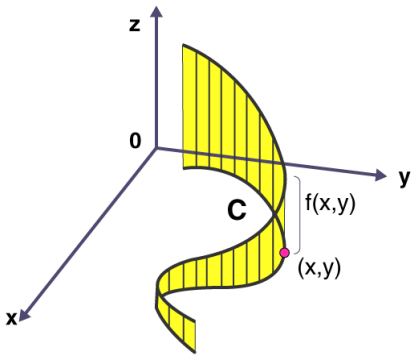
61.3.1 Over Scalar Field

Let $f(x_1, x_2, \ldots, x_n)$ be a scalar field defined on a smooth curve c, which is parametrized by a vector function.

$$\vec{r}(t) = (x_1(t), x_2(t), \ldots, x_n(t)), \quad t \in [a, b]$$

Then, the line integral of f over c is given by:

$$\int_C f(x_1, x_2, \ldots, x_n)\, ds = \int_a^b f(\vec{r}(t)) \left\| \frac{\vec{r}(t)}{dt} \right\| dt$$



This formula comes from approximating the area of the squares under the curve with height $f(x_k, y_k)$ and the arclength Δs_k. We will simplify to three dimensions for the sake of simplicity, but as seen before, this works for n dimensions. $A_k = f(x_k, y_k)\Delta s_k$, where $\Delta s_k = \sqrt{(\Delta s_k)^2 + (\Delta y_k)^2}$. Now, by using the formula for the arclength we already know, we get $\| \frac{\vec{r}(t)}{dt} \| dt = ds$.

Example: Scalar Field

Compute the line integral of $f(x, y) = x^2 + y^2$ over the straight line segment going from $(1, 1)$ to $(3, 3)$.

This time our curve is $y = x$, so the parametrized curve is $\begin{bmatrix} x(t) = t \\ y(t) = t \end{bmatrix}$. This leaves us with:

$$\int_1^3 2t^2\sqrt{2}dt = \frac{52\sqrt{2}}{3}$$

61.4 Line Integrals with Respect to a Specific Variable

We are going to take another look at line integrals, but from the perspective of just one variable and how this affects the formulas we know. For the next explanation, we will use only functions of two variables to keep the examples simple.

61.4.1 Over Scalar Field with Respect to x or y

Imagining how this works is not especially difficult. Imagine taking a line integral where we do not move along the y-axis and only care about the plane xz. Here, our line integral only changes in the fact that in the norm of our curve, the component that was responsible for y will be set to 0, and the ds changes to the variable we are integrating with respect to. In this case, dx.

$$\int_c f(x, y)dx = \int_c f(g(t), h(t))\sqrt{g'(t)^2}dt$$

61.4.2 Over Vector Field with Respect to x or y

In the case of a field, the story is different in the fact that now our output vectors change depending on the variable we are choosing to leave out.

Now, given a field $\vec{F}(x,y) = M(x,y)\imath + N(x,y)\jmath$ and the curve $\vec{r}(t) = g(t)\imath + h(t)\jmath$, the line integral of $\vec{F}$ along C is:

$$\int_\gamma \left\langle \nabla\vec{F}, \frac{\vec{r(t)}}{dt} \right\rangle dt$$

$$\int_a^b M(x,y)g'(t)dt + \int_a^b N(x,y)h'(t)dt$$

Here $g'(t)dt = dx$ and $h'(t)dt = dy$.

$$\int_a^b Mdx + \int_a^b Ndy$$

These two integrals give us both the integral with respect to x and y.

Example:

Curve C is a portion of the parabola $y = x^2$ that goes through the points $(1,1)$ to $(2,4)$.

Compute: $\int_a^b \frac{x}{y}dy$

Step 1: Parameterize C

The trick for these cases, where the curve is the graph of a function, is to set $x = t$ and then solve for y in terms of x.

In this case, we get

$$\vec{r}(t) = t\imath + t^2\jmath \text{ with } 1 \leq t \leq 2$$

Step 2: Substitute into the integral

$$h'(t)dt = 2tdt$$

$$\int_a^b \frac{t}{t^2}2tdt = \int_1^2 \frac{1}{t}2tdt = 2$$

61.5 Flow and Circulation

For velocity $\vec{F} = M(x,y)\hat{i} + N(x,y)\hat{j}$ and a smooth curve C parameterized by $\vec{r}(t)$

$$\text{Flow} = \int_C \langle \vec{F}, \vec{T} \rangle ds$$

If the curve C is closed, then it is called *Circulation*.

$$\text{Circulation} = \oint_C \langle \vec{F}, \vec{T} \rangle ds$$

Flow is a measurement of the degree to which a curve goes along the vector field.

61.5.1 Circulation Density

The *circulation density* of a vector field $\vec{F}(x,y) = M(x,y)\hat{i} + N(x,y)\hat{j}$ is given by

$$\text{CD} = \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$$

It derives from imagining having a square as our curve, where from our starting point (x,y) we move to $(x+\Delta x, y), (x+\Delta x, y+\Delta y), (x, y+\Delta y)$, generating four different curves. Therefore, our total circulation is given by

$$\oint_C \langle \vec{F}, d\vec{r}\rangle = \sum_{i=1}^{4} \int_{C_i} \langle \vec{F}, \vec{T}\rangle d\vec{r}$$

The dot products are given by

- Bottom: $\langle \vec{F}(x,y)\Delta x\hat{i}\rangle = M(x,y)\Delta x$.
- Right: $\langle \vec{F}(x+\Delta x, y)\Delta y\hat{j}\rangle = N(x+\Delta x, y)\Delta y$.
- Top: $\langle \vec{F}(x, y+\Delta y) - \Delta x\hat{i}\rangle = -M(x, y+\Delta y)\Delta x$.
- Left: $\langle \vec{F}(x,y) - \Delta y\hat{j}\rangle = N(x,y)\Delta y$.

Then the top minus the bottom is $-(M(x,y+\Delta y) - M(x,y)\Delta x)$, and notice that inside the parentheses we have something similar to the partial

$$\frac{\partial M}{\partial y} = \lim_{\Delta y \to 0} \frac{M(x, y+\Delta y) - M(x,y)}{\Delta y}.$$

Thus, by multiplying by a clever one $\frac{\Delta y}{\Delta y}$, we get $-\frac{\partial M}{\partial y}\Delta y\Delta x$. Similarly, we get for the right minus left case $\frac{\partial N}{\partial x}\Delta y\Delta x$.

Finally, by putting it all together

$$\oint_C \langle \vec{F}, d\vec{r}\rangle = -\frac{\partial M}{\partial y}\Delta y\Delta x + \frac{\partial N}{\partial x}\Delta y\Delta x = \left(-\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x}\right)\Delta y\Delta x$$

$\Delta y\Delta x$ is the area of the rectangle.

61.6 Green's Theorem Circulation-Curl Form

For C, a smooth, simple, closed curve enclosing R, $\vec{F} = M\hat{i} + N\hat{j}$ having M, N continuous first partials in an open region containing R

$$\oint_C \langle \vec{F}, d\vec{r}\rangle = \int\int_R \left(-\frac{\partial M}{\partial y} + \frac{\partial N}{\partial x}\right) dxdy$$

61.7 Flux

For a continuous field $\vec{F} = M(x,y)\hat{i} + N(x,y)\hat{j}$, a smooth curve with no intersections C and $\vec{n}$ the outward normal

$$Flux = \oint_C \langle \vec{F}, \vec{n}\rangle ds$$

The derivation of the formula for actually computing the *flux* goes as follows: First, we find ourselves in 2 dimensions and we want to find a normal vector to the curve. We can pretend that we are in 3 dimensions and define a vector

$$\vec{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

This gives us $\vec{n} = \vec{T} \times \vec{k}$. Now, either by using the determinant trick or formally computing the cross product, we get

$$\vec{T} \times \vec{k} = \vec{n} = \begin{bmatrix} \frac{dy}{ds} \\ \frac{dx}{ds} \\ 0 \end{bmatrix}$$

The final formula is then

$$Flux = \oint_C \langle \vec{F}, \vec{n} \rangle ds = \oint_C \left\langle \vec{F}, \begin{bmatrix} \frac{dy}{ds} \\ \frac{dx}{ds} \\ 0 \end{bmatrix} \right\rangle ds = \int_a^b M dy - \int_a^b N dx$$

One extra note about the derivation is that if the curve goes counterclockwise, we use $\vec{n} = \vec{T} \times \vec{k}$, but if it goes clockwise, $\vec{n} = \vec{k} \times \vec{T}$, because of the right-hand rule.

61.7.1 Flux Density

Just like for the *Circulation Density*, we will use the small square to examine the divergence in a small region. Following a similar process with the formula for flux, we can get a similar result.

$$FD = \left(-\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right)$$

61.8 Green's Theorem Divergence-Flux Form

For C, a smooth, simple, closed curve enclosing R, $\vec{F} = M\hat{i} + N\hat{j}$ having M, N continuous first partials in an open region containing R

$$\oint_C \langle \vec{F}, \vec{n} \rangle ds = \int \int_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dxdy$$

61.9 Conservative Vector Fields

A field $\vec{F}$ is *conservative* on an open domain if

$$\int_C \langle \vec{F}, d\vec{r} \rangle$$

is the same for all paths C between points A and B in the domain. In other words, it only depends on the endpoints.

61.10 The Potential Function

A vector field $\vec{F} : \mathbb{R}^n \to \mathbb{R}^n$ is called *conservative* if there exists a scalar function $f : \mathbb{R}^n \to \mathbb{R}$ such that:

$$\vec{F} = \nabla f$$

The function f is then called a *potential function* of $\vec{F}$ and also

$$\int_C \langle \vec{F}, d\vec{r} \rangle = f(B) - f(A)$$

Proof:

$$\int_C \langle \vec{F}, d\vec{r} \rangle = \int_a^b \left\langle \vec{F}(\vec{r}(t)), \frac{d\vec{r}(t)}{dt} \right\rangle dt$$

By assumption

$$= \int_a^b \left\langle \nabla f(\vec{r}(t)), \frac{d\vec{r}(t)}{dt} \right\rangle dt$$

This is like the chain rule. Therefore,

$$\left\langle \nabla f(\vec{r}(t)), \frac{d\vec{r}(t)}{dt} \right\rangle = \frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial t} + \cdots \frac{\partial f}{\partial x_n}\frac{\partial x_n}{\partial t} = \frac{d}{dt} f(\vec{r}(t))$$

$$= \int_a^b \frac{d}{dt}\phi(\vec{r}(t))dt$$

By the fundamental theorem of calculus

$$= f(B) - f(A)$$

$\mathfrak{QED}$

61.11 Check for Conservative Vector Fields

Using the definition $\vec{F} = \nabla f$ and by writing

$$\begin{aligned} \vec{F} &= f_1(x_1, x_2, \dots)\hat{i}_1 + f_2(x_1, x_2, \dots)\hat{i}_2 + \cdots + f_n(x_1, x_2, \dots)\hat{i}_n \\ &= \frac{\partial f}{\partial x_1}\hat{i}_1 + \frac{\partial f}{\partial x_2}\hat{i}_2 + \cdots \frac{\partial f}{\partial x_n}\hat{i}_n \end{aligned}$$

We get the following by reducing to 3 variables and dimensions for the sake of simplicity:

$$\frac{\partial f_1}{\partial y} = \frac{\partial}{\partial y}\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\frac{\partial f}{\partial y} = \frac{\partial f_2}{\partial x}$$

We can also keep making pairs like this, but checking one is enough.

61.11.1 Finding the Potential Function of a Vector Field

To find the potential function $f(x, y)$ of a vector field $\vec{F}(x, y) = (P(x, y), Q(x, y))$, we must check if the field is conservative and then integrate accordingly. A vector field is conservative if there exists a scalar potential function f such that:

$$\vec{F} = \nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Given:

$$\vec{F}(x,y) = \begin{bmatrix} 2xy + e^x \\ x^2 + \frac{1}{2\sqrt{y}} \end{bmatrix}$$

Let $P(x,y) = 2xy + e^x$, and $Q(x,y) = x^2 + \frac{1}{2\sqrt{y}}$.

First, check if the field is conservative by verifying:

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(2xy + e^x) = 2x$$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}\left(x^2 + \frac{1}{2\sqrt{y}}\right) = 2x$$

Since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, the field is conservative.

Step 1: Integrate $P(x,y)$ with respect to x:

$$f(x,y) = \int (2xy + e^x)\,dx = x^2y + e^x + h(y)$$

Here, $h(y)$ is a function of y only.

Step 2: Differentiate $f(x,y)$ with respect to y:

$$\frac{\partial f}{\partial y} = x^2 + h'(y)$$

Set this equal to $Q(x,y) = x^2 + \frac{1}{2\sqrt{y}}$, so:

$$x^2 + h'(y) = x^2 + \frac{1}{2\sqrt{y}} \Rightarrow h'(y) = \frac{1}{2\sqrt{y}}$$

Step 3: Integrate to find $h(y)$:

$$h(y) = \int \frac{1}{2\sqrt{y}}\,dy = \sqrt{y} + C$$

Therefore, the potential function is:

$$f(x,y) = x^2y + e^x + \sqrt{y} + C$$

61.11.2 Double Integral Over a Rectangular Region

Let $A = [x_0, x_1] \times [y_0, y_1]$ be a rectangular region in $\mathbb{R}^2$. The double integral of a function $f(x,y)$ over A is:

$$\iint_A f(x,y)\,dA = \int_{x_0}^{x_1} \int_{y_0}^{y_1} f(x,y)\,dy\,dx$$

For rectangular domains, the order of integration does not affect the result:

$$\int_{x_0}^{x_1} \int_{y_0}^{y_1} f(x,y)\,dy\,dx = \int_{y_0}^{y_1} \int_{x_0}^{x_1} f(x,y)\,dx\,dy$$

61.12 Steps for Finding the Region of Integration

When integrals are not defined over regions defined by constants, finding the bounds of integration can become a tricky task depending on the region we want to integrate.

The general process for setting up iterated integrals over a non-rectangular region is as follows:

1. **Sketch the region if possible:** Begin by sketching the region of integration. A visual representation helps you understand the boundaries, the shape of the region, and the relationships between variables.

2. **Choose the order of integration:** Decide the order in which you'll integrate the variables. Remember that integration proceeds from the inside out, so the inner integral corresponds to the variable you integrate first.

3. **Determine the bounds for each variable:** For the chosen order of integration, identify the limits of integration for each variable. The bounds for the inner variable are often functions of the outer variable. For example, if the region is described by the inequality $x + y < 1$ and you choose to integrate with respect to y first, you must express y's bounds in terms of x. This means solving the inequality for y:

$$y < 1 - x \quad \Rightarrow \quad y \in [c(x), 1 - x]$$

where $c(x)$ is the lower bound for y, possibly another function of x. Then the full double integral would look like:

$$\int_a^b \int_{c(x)}^{1-x} f(x, y)\, dy\, dx$$

Repeat this process for all variables involved. If you've sketched the region, you can often find these bounds by "slicing" the region—holding one variable constant and observing how the others vary within the region.

61.13 Integration Over Curvilinear Domains

If the region $A \subset \mathbb{R}^2$ is not rectangular, it is divided into subregions ΔA_k, and the integral is defined as the limit:

$$\iint_A f(x, y)\, dA = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k, y_k)\, \Delta A_k$$

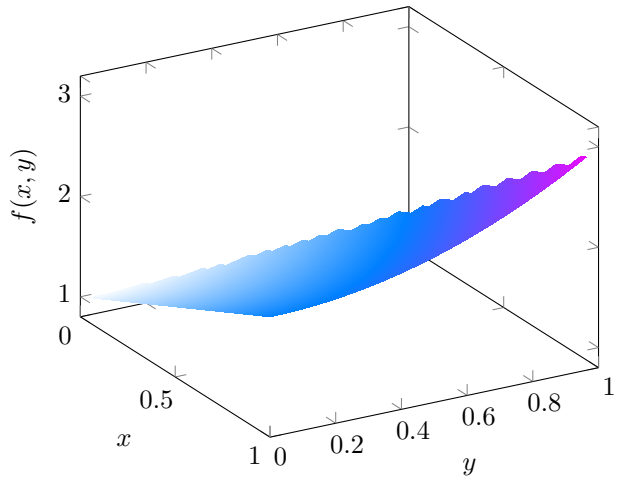
In practice, for regions bounded by curves, we integrate in the form:

$$\int_a^b \int_{u(x)}^{o(x)} f(x, y)\, dy\, dx$$

Here, $y \in [u(x), o(x)]$ describes the vertical bounds as functions of x.

Example:

Compute the volume under $f(x, y) = 1 + x + y^2$ bounded by the x-axis, $x = 1$, and the y-axis $y = \sqrt{x}$.



First we are going to imagine slicing the area under this sector vertically along the x-axis from 0 up to 1, which is the volume of an area under a curve that is represented by going along the y-axis.

$$\int_0^1 \int_0^{\sqrt{x}} 1 + x + y^2 dy dx$$

$$\int_0^1 \left| y + xy + \frac{y^3}{3} \right|_0^{\sqrt{x}} dx$$

$$\int_0^1 \left| \sqrt{x} + x\sqrt{x} + \frac{\sqrt{x}^3}{3} \right|_0^{\sqrt{x}} dx$$

$$= \left| \frac{2}{3} x^{\frac{3}{2}} + \frac{4}{3}\frac{2}{5} x^{\frac{5}{2}} \right|_0^1 = \frac{6}{5}$$

If we choose the other way, then we have an area with a fixed y on the inside that goes from y^2 to 1 with respect to x.

$$\int_0^1 \int_{y^2}^1 1 + x + y^2 dx dy$$

61.13.1 Changing the Order of Integration to Solve Tricky Integrals

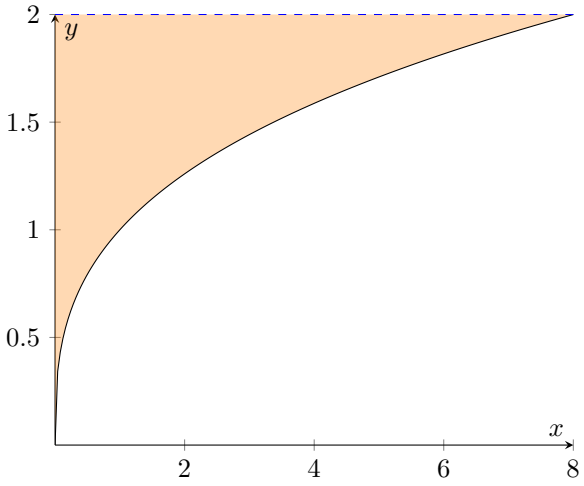
Given $\int_0^8 \int_{\sqrt[3]{x}}^2 \frac{1}{y^4+1} dy dx$ which is very hard to integrate, let us change the order to make it easier.

$$\int_0^2 \int_0^{y^3} \frac{1}{y^4 + 1} dx dy$$

The steps are the same as in the previous example.

1. Identify the area you want to integrate; what are the bounds with respect to x and y?
2. Write the same expression without bounds and change the order of the differentials
3. Imagine slicing the area underneath vertically for x and horizontally for y with their respective bounds corresponding to the perspective.
4. Write the new integral with the opposite order of integration

Visualization of the area underneath the surface



Here, the original bounds are: x goes from 0 to 8, and y goes from $\sqrt[3]{x}$ to 2. As you can see in the diagram, by slicing vertically the value of x is fixed, and we move towards the upper bound of y, which is 2. Therefore, the smallest x can get is 0 and the biggest is 2. While for y, by slicing vertically, we get that x goes from 0 to $\sqrt[3]{x}$, but we do not want to leave x, so we solve for y and get y^3. Now we can put all of this together and integrate.

In general, changing the order of integration is only about using logic to set the bounds of integration, but starting from a different perspective.

61.14 The Jacobian

The *Jacobian* matrix is a fundamental concept in multivariable calculus. The Jacobian is the best approximation of a linear transformation at a point.

For the intuition of why we use partial derivatives, just think about how we use them as these tiny changes in the components of the original basis vectors.

Given a vector-valued function

$$f : \mathbb{R}^n \to \mathbb{R}^m, \quad f(x_1, \ldots, x_n) = \begin{bmatrix} f_1(x_1, \ldots, x_n) \\ f_2(x_1, \ldots, x_n) \\ \vdots \\ f_m(x_1, \ldots, x_n) \end{bmatrix},$$

the *Jacobian matrix* of f is the $m \times n$ matrix of all first-order partial derivatives:

$$J_f(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

61.15 Local Linearity

While performing non-linear transformations on a space, we can see that in a tiny neighborhood around a point, the transformation appears to be linear, and the matrix that encodes that transformation is the *Jacobian*.

For intuition, just imagine that in the tiny neighborhood the transformation is composed of small steps in the $x_1, \ldots, x_n$ directions, which give the partial derivatives that represent these small steps.

61.16 Determinant

When $m = n$, i.e., the function maps from $\mathbb{R}^n$ to $\mathbb{R}^n$, the Jacobian matrix is square. In this case, the *Jacobian determinant* is defined as:

$$\det(J_f(x)),$$

and provides important information about the local behavior of the function. For instance:

- If $\det(J_f(x)) \neq 0$, the function is locally invertible at x by the inverse function theorem.
- The sign and magnitude of the determinant gives us information about orientation and volume scaling of the transformation.

When we change our system of coordinates, it is important that our differential makes sense in the original system we were in. Because of that, we take the determinant with respect to the new coordinates to tell us the scaling factor that we have to use for our result to make sense.

Use Cases:

The Jacobian matrix and its determinant appear in many areas, including:

- **Change of variables in integrals:** The absolute value of the Jacobian determinant is used to adjust the measure when changing coordinates in multiple integrals.
- **Nonlinear optimization:** The Jacobian is crucial for computing gradients and performing techniques like Newton's method in multiple dimensions.
- **Differential equations:** Stability analysis of dynamical systems often involves evaluating the Jacobian at equilibrium points.

61.17 List of Common Jacobians

- *Cartesian Coordinates*:
  - Definition: The position of a point is given by (x, y, z) in a 3D space along orthogonal axes.
  - Jacobian:

$$J_{\text{Cartesian}} = 1dx$$

- *Polar Coordinates (2D)*:
  - Definition: The position of a point in a plane is given by (r, θ), where r is the radial distance and θ is the angle from the positive x-axis.
  - Jacobian:

$$J_{\text{Polar}} = r\,dr\,d\theta$$

- *Cylindrical Coordinates*:
  - Definition: The position of a point in 3D space is given by (r, θ, z), where (r, θ) are the polar coordinates in the xy-plane and z is the height along the z-axis.
  - Jacobian:

$$J_{\text{Cylindrical}} = r\,dr\,d\theta\,dz$$

- *Spherical Coordinates*:
  - Definition: The position of a point in 3D space is given by (ρ, θ, ϕ), where:
    - ρ is the distance from the origin,

  - θ is the azimuthal angle (in the xy-plane from the positive x-axis),
  - ϕ is the polar angle (from the positive z-axis).

- Jacobian:

$$J_{\text{Spherical}} = \rho^2 \sin(\phi)\, d\rho\, d\theta\, d\phi$$

61.18 Integration in Polar Coordinates

For radially symmetric regions or functions, switching to polar coordinates simplifies the computation.

Transformation:

$$x = r\cos(\varphi), \quad y = r\sin(\varphi).$$

The Jacobian of the transformation gives the area element:

$$dA = r\, dr\, d\varphi$$

Integral:

$$\iint_D f(x,y)\, dA = \int_{\varphi_0}^{\varphi_1} \int_0^{R(\varphi)} f(r\cos(\varphi), r\sin(\varphi))\, r\, dr\, d\varphi$$

61.18.1 Origin of the Area Formula

When we divide our circle into sections, we get both a difference in the angle and the radius, which generates another area. The difference $\Delta A =$ Area of big wedge $-$ Area of small wedge

Radius: $r_k + \frac{\Delta r}{2}$ for the big wedge and $r_k - \frac{\Delta r}{2}$ for the small wedge

Angle: $\frac{\Delta\theta}{2\pi}$, which is the angle of the fraction of the circle because the formula for the area is $r^2\pi$ and the whole circle would be 2π

Final Formula for the difference in the Area

$$\Delta A = \frac{\Delta\theta}{2\pi}\left(\left[r_k + \frac{\Delta r}{2}\right]^2 - \left[r_k - \frac{\Delta r}{2}\right]^2\right)$$

$$= r_k \Delta r \Delta\theta$$

Multiplying by the function value and taking the limit of these sections, we get the volume:

$$V = \int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} f(r,\theta) r \Delta r \Delta\theta$$

Example:

$f(r,\theta) = r$ above the cardioid $r = 1 - \sin\theta$.

Then

$$V = \int_0^{2\pi} \int_0^{1-\sin\theta} r r dr d\theta$$

$$= \int_0^{2\pi} \frac{r^3}{3}|_0^{1-\sin\theta} d\theta = \int_0^{2\pi} \frac{(1-\sin\theta)^3}{3} d\theta = \frac{5\pi}{3}$$

61.19 Improper Integrals Over Unbounded Regions

If the domain of integration is unbounded (e.g., the entire plane $\mathbb{R}^2$), the double integral is defined via a limit process:

$$\iint_{\mathbb{R}^2} f(x,y)\,dx\,dy := \lim_{M_1,M_2\to\infty} \lim_{N_1,N_2\to\infty} \int_{M_1}^{M_2} \int_{N_1}^{N_2} f(x,y)\,dy\,dx$$

Often, using polar coordinates simplifies such cases, reducing multiple limits to one:

$$\iint_{\mathbb{R}^2} f(x,y)\,dx\,dy = \int_0^{2\pi} \int_0^{\infty} f(r\cos\varphi, r\sin\varphi)\,r\,dr\,d\varphi$$

61.19.1 Gaussian Integral

We are going to find the integral of the *unbounded integral* e^{-x^2} using a combination of techniques.

$$\begin{aligned}
S &= \int_{-\infty}^{\infty} e^{-x^2} dx \\
S^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)\left(\int_{-\infty}^{\infty} e^{-x^2} dx\right) \\
S^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2} dy\right)
\end{aligned}$$

Now let us translate to polar coordinates

$$\begin{aligned}
S^2 &= \left(\int_0^{2\pi} \int_0^{\infty} re^{-(r\cos\theta)^2} re^{-(r\sin\theta)^2} drd\theta\right) \\
S^2 &= \left(\int_0^{2\pi} \int_0^{\infty} re^{-(r\cos\theta)^2+(r\sin\theta)^2} drd\theta\right) \\
S^2 &= \left(\int_0^{2\pi} \int_0^{\infty} re^{-(r^2)} drd\theta\right) \\
S^2 &= \left(\int_0^{2\pi} \int_0^{\infty} \frac{1}{2}e^{-(s)} dsd\theta\right) \\
S^2 &= \left(\int_0^{2\pi} d\theta\right) \\
S^2 &= \pi \\
S &= \sqrt{\pi}
\end{aligned}$$

$\mathfrak{QED}$

61.20 Triple Integrals

The triple integral allows the computation of volume or mass in three-dimensional space.

Let $V \subset \mathbb{R}^3$ be a bounded region, then:

$$\iiint_V f(x,y,z)\,dV = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k,y_k,z_k)\,\Delta V_k,$$

where ΔV_k are small sub-volumes approximating V.

In practice, evaluate:

$$\iiint_V f(x,y,z)\,dz\,dy\,dx,$$

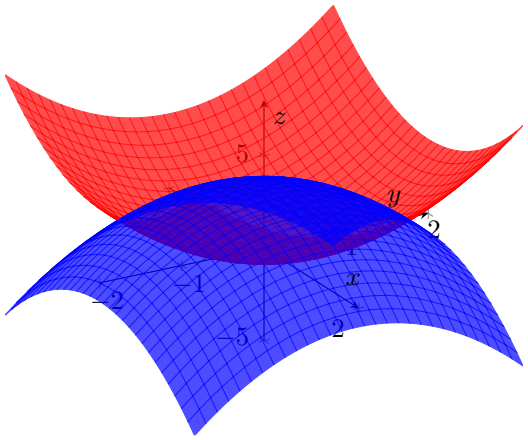
with limits determined by the geometry of the volume. The order of integration may be rearranged to simplify computation.

$$\int_{x=a}^{x=b}\int_{y=g_1(x)}^{y=g_2(x)}\int_{f_1(x,y)}^{f_2(x,y)} dzdydx$$

61.20.1 Triple Integrals in Cartesian Coordinates

In this short section, we will look at an example of how to calculate the volume enclosed between two functions using triple integrals.

Given are the functions $f_1(x,y) = x^2 + y^2$ and $f_2(x,y) = 3 - x^2 - y^2$.



Let us find the intersection

$$x^2 + y^2 = 3 - x^2 - y^2$$

$$x^2 + y^2 = \frac{3}{2}$$

Now we have to find the limits of integration for

$$\int_{x=a}^{x=b}\int_{y=g_1(x)}^{y=g_2(x)}\int_{f_1(x,y)}^{f_2(x,y)} dzdydx,$$

using our constraint and our original functions.

$$\int_{x=a}^{x=b}\int_{y=-\sqrt{\frac{3}{2}-x^2}}^{\sqrt{\frac{3}{2}-x^2}}\int_{x^2+y^3}^{3-x^2-y^2} dzdydx$$

Here, z means the height, which is just the output of the original functions.

The bounds of y come just by looking at our constraint and solving for it, and x only depends on the concrete limit of integration.

61.20.2 Coordinate Transformation and the Jacobian Determinant

Let a coordinate transformation be defined by:

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w)$$

Then the triple integral transforms as:

$$\iiint_{(x,y,z)} f(x, y, z)\, dx\, dy\, dz = \iiint_{(u,v,w)} f(x(u, v, w), y(u, v, w), z(u, v, w)) \cdot |\det J|\ du\, dv\, dw$$

where J is the *Jacobian matrix* of the transformation:

$$J = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{bmatrix}$$

Example: Transformation to Cylindrical Coordinates

The standard cylindrical coordinate transformation is:

$$x = r\cos\varphi, \quad y = r\sin\varphi, \quad z = z$$

The Jacobian determinant is:

$$|\det J| = \begin{vmatrix} \cos\varphi & -r\sin\varphi & 0 \\ \sin\varphi & r\cos\varphi & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

Thus, the triple integral in cylindrical coordinates becomes:

$$\iiint f(x, y, z)\, dx\, dy\, dz = \iiint f(r\cos\varphi, r\sin\varphi, z)\, r\, dr\, d\varphi\, dz$$

61.20.3 Calculating the Center of Mass with Triple Integrals

We can make use of triple integration to calculate the center of mass of a body with respect to x, y or z with the following formulas:

$$x_s = \iiint_V xp(x, z, y)dz\ dy\ dx$$
$$y_s = \iiint_V yp(x, z, y)dz\ dy\ dx$$
$$z_s = \iiint_V zp(x, z, y)dz\ dy\ dx$$

Where $p(x, y, z)$ is our mass function.

61.21 Surfaces

In vector calculus, we sometimes want to examine the *surface* generated by a function. One approach to analyzing these surfaces is via parameterization, and another is using explicit and implicit forms.

61.21.1 Explicit and Implicit Forms

For a 3D surface, the *implicit form* is given by

$$F(x, y, z) = c$$

and the *explicit form* by

$$z = f(x, y)$$

61.21.2 Parametric Form

It is given by

$$\vec{r}(u, v) = f(u, v)\hat{i} + g(u, v)\hat{j} + h(u, v)\hat{k}$$

where $a \leq u \leq b$ and $c \leq v \leq d$.

Example:

For the example, we take the cone formed by the function $z = \sqrt{x^2 + y^2}$, which is in the explicit form.

The implicit form is then $0 = z - \sqrt{x^2 + y^2}$, and the parametric form can be described very naturally by

$$\vec{r}(r, \theta) = r\cos(\theta)\hat{i} + r\sin(\theta)\hat{j} + r\hat{k}$$

using something similar to cylindrical coordinates with $r = \sqrt{x^2 + y^2}$.

61.21.3 Surface Area Formula

Continuing with the surfaces in 3D, we can compute the area using the following formula:

$$A = \iint_D \left\| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right\| du\, dv.$$

Notice that we are using the fact that the length of a cross product is the area of the parallelogram formed by the vectors it is orthogonal to. Using this on a small scale with the partials of u and v and then adding up results in the formula.

Example I: Area of the sphere

We want to compute the surface area of a sphere of radius a and show that it equals $4\pi a^2$. Let us proceed step by step.

The equation of a sphere of radius a centered at the origin is given implicitly by

$$x^2 + y^2 + z^2 = a^2.$$

To parametrize the sphere, one possible way is to solve for z:

$$z = \pm\sqrt{a^2 - x^2 - y^2}, \quad x^2 + y^2 \leq a^2.$$

This representation works, but it splits the sphere into two surfaces (upper and lower hemispheres). This makes the parametrization less convenient for computing surface area.

A more natural parametrization comes from spherical coordinates. Any point on the sphere can be written as

$$\begin{cases} x = a\sin\theta\cos\varphi, \\ y = a\sin\theta\sin\varphi, \\ z = a\cos\theta, \end{cases}$$

where the parameters range over

$$0 \le \theta \le \pi, \quad 0 \le \varphi < 2\pi.$$

Here, θ is the polar angle (measured from the z-axis) and φ is the azimuthal angle in the xy-plane.

The surface area of a parametrized surface $\vec{r}(u, v)$ is given by

$$A = \iint_D \left\| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right\| du\, dv.$$

For the parametrization $\vec{r}(\theta, \varphi) = (a\sin\theta\cos\varphi,\ a\sin\theta\sin\varphi,\ a\cos\theta)$, we compute:

$$\frac{\partial \vec{r}}{\partial \theta} = (a\cos\theta\cos\varphi,\ a\cos\theta\sin\varphi,\ -a\sin\theta),$$

$$\frac{\partial \vec{r}}{\partial \varphi} = (-a\sin\theta\sin\varphi,\ a\sin\theta\cos\varphi,\ 0).$$

Their cross product is

$$\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \varphi} = (a^2\sin^2\theta\cos\varphi,\ a^2\sin^2\theta\sin\varphi,\ a^2\sin\theta\cos\theta).$$

The magnitude is

$$\left\| \frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \varphi} \right\| = a^2\sin\theta.$$

Therefore, the surface area is

$$A = \int_0^{2\pi} \int_0^{\pi} a^2\sin\theta\, d\theta\, d\varphi.$$

Compute the integrals:

$$A = a^2 \left(\int_0^{2\pi} d\varphi \right) \left(\int_0^{\pi} \sin\theta\, d\theta \right).$$

$$A = a^2(2\pi)(2) = 4\pi a^2.$$

Thus, the surface area of the sphere is

$$4\pi a^2$$

$\mathfrak{QED}$

Example II:

Compute the surface area of the intersection of the plane $z = -x$ and the cylinder $x^2 + y^2 = 4$.

We will use polar coordinates because the intersection is circular.

$$\vec{r}(r,\theta) = \begin{bmatrix} r\cos(\theta) \\ r\sin(\theta) \\ -r\cos(\theta) \end{bmatrix}$$

Take the partials

$$\vec{r}_r = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \\ -\cos(\theta) \end{bmatrix}$$

$$\vec{r}_\theta = \begin{bmatrix} -r\sin(\theta) \\ r\cos(\theta) \\ r\sin(\theta) \end{bmatrix}$$

Compute the cross product:

$$\vec{r}_r \times \vec{r}_\theta = \begin{bmatrix} r \\ 0 \\ r \end{bmatrix}, \qquad \left\|\vec{r}_r \times \vec{r}_\theta\right\| = \sqrt{r^2 + r^2} = \sqrt{2}\, r$$

Now integrate over the disk $0 \le r \le 2,\ 0 \le \theta \le 2\pi$:

$$\text{Area} = \int_0^{2\pi}\int_0^2 \sqrt{2}\, r\, dr\, d\theta = \int_0^{2\pi} \left[\tfrac{\sqrt{2}}{2} r^2\right]_0^2 d\theta = \int_0^{2\pi} 2\sqrt{2}\, d\theta = 4\pi\sqrt{2}.$$

The final result is

$$4\pi\sqrt{2}$$

61.22 Implicit Surface Area

Given a smooth, continuous surface $F(x,y,z) = c$ where $\langle \nabla F, \vec{p}\rangle \neq 0$, the surface area is given by

$$SA = \iint_R \frac{\|\nabla F\|}{|\langle \nabla F, \vec{p}\rangle|} dA$$

$\vec{p}$ can be $\hat{i}, \hat{j}$ or $\hat{k}$.

61.23 Explicit Surface Area

Given a smooth, continuous surface $F(x,y,z) = f(x,y) - z = 0$ where $\nabla F = f_x\hat{i} + f_y\hat{j} - \hat{k}$, $|\langle \nabla F, \hat{k}\rangle| = |-1| = 1$ and $\|\nabla F\| = \sqrt{f_x^2 + f_y^2 + 1}$

$$SA = \iint_R \sqrt{f_x^2 + f_y^2 + 1} dA$$

61.24 Surface Integrals

We define a function $G(x, y, z)$ defined on a smooth surface S with an area $d\sigma$, whose integral is

$$SI = \iint_R G(x, y, z) d\sigma$$

The function G associates each point on the surface to a value in a higher dimension.

Surface integral in different forms:

Parametric

$$SI = \iint_R G(f(u, v), g(u, v), h(u, v)) \left\| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right\| du\, dv.$$

Explicit

$$SI = \iint_R G(x, y, z) \sqrt{f_x^2 + f_y^2 + 1} dA$$

Implicit

$$SI = \iint_R G(x, y, f(x, y)) \frac{\|\nabla F\|}{|\langle \nabla F, \vec{p} \rangle|} dA$$

61.25 The Orientation of a Surface

A smooth surface is *orientable* if there exists a continuous field of unit normal vectors to the surface.

61.26 The Flux of a Surface

For a continuous field $\vec{F}$ on a smooth surface S oriented with normal unit vectors $\vec{n}$,

$$Flux = \iint_S \langle \vec{F}, \vec{n} \rangle d\sigma$$

where σ is the surface area in the corresponding form, and the normal unit vector $\vec{n}$ for parametric forms is

$$\vec{n} = \frac{\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}}{\left\| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right\|}$$

and for implicit surfaces

$$\vec{n} = \frac{\nabla g}{\|\nabla g\|}$$

61.27 Curl

Let $\vec{F} = (F_1, F_2, F_3) : \mathbb{R}^3 \to \mathbb{R}^3$ be a vector field.

$$\operatorname{curl} \vec{F} = \nabla \times \vec{F} = \begin{bmatrix} \frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3} \\ \frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1} \\ \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2} \end{bmatrix}$$

These describe how the field spreads or rotates around a point. As a side note, the *curl* of the gradient of f is 0 if f has continuous mixed second derivatives.

There exists a connection between the curl and conservative vector fields. Notice that the components of the curl vector are similar to the conditions for a vector field to be conservative. Therefore, a conservative vector field has no curl, because if the conditions are true, then the curl is the zero vector.

A point where curl $\vec{f} \neq 0$ is called a whirlpool.

61.28 Stokes's Theorem

For S a piecewise smooth oriented surface with piecewise smooth boundary C, and $\vec{F}$ a field with continuous first partials for each component on an open region containing S. The circulation around the curve is equal to

$$C = \oint_C \langle \vec{F}, d\vec{r} \rangle = \iint_S \langle (\nabla \times F), \vec{n} \rangle d\sigma$$

61.29 Divergence

We define *divergence* as

$$\operatorname{div} \vec{F} = \langle \nabla, \vec{F} \rangle = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} + \frac{\partial F_3}{\partial x_3}$$

A point where div $\vec{f} > 0$ is called a source.

61.30 The Divergence Theorem

For S a piecewise smooth oriented, closed surface with piecewise smooth boundary C, and $\vec{F}$ a field with continuous first partials for each component on an open region containing S. The flux around the curve is equal to

$$\iint_S \langle \vec{F}, \vec{n} \rangle ds = \iiint_D \langle \nabla, \vec{F} \rangle dV$$

61.31 Unified View of Vector Calculus

Integrating a differential operator acting on a field over a domain is the same as adding the field components along the boundary.

Formally, the integral of a differential form over a boundary $\partial\Omega$ of some orientable manifold Ω is equal to the integral of its exterior derivative over the whole of Ω.

$$\int_{\partial\Omega} w = \int_{\Omega} dw$$

Chapter 62

Differential Equations

A Differential Equation is an equation that relates a function (the dependent variable) with a variable or multiple variables (the independent variables) and its derivative.

They can be categorized into *Ordinary* and *Partial* Differential Equations.

62.1 Ordinary Differential Equations

An equation of the form

$$y^n = f(x, y, x', y', \ldots, y^{n-1})$$

is called an *Ordinary Explicit Differential Equation of order n.*

If $y^n = 0$, then it is *implicit.*

If the exponent of the dependent variable is 1, and it is built like a linear equation, then it is categorized as *Linear.*

62.2 Initial Value Problems and Solutions

The specification of an explicit differential equation and the values

$$x_0, \quad y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \ldots, \quad y^{(n-1)}(x_0) = y_{n-1}$$

is called an *initial value problem (IVP).*

An n-times differentiable function $y(x)$ that satisfies the explicit differential equation is called the *general solution* of the differential equation. If it also satisfies the conditions of the initial value problem from the previous definition, it is called the *particular solution* of the IVP.

62.3 Volterra Integral

The function $y(x)$ is a solution of the IVP $y' = f(x, y), \quad y(x_0) = y_0$ if and only if

$$y(x) = y_0 + \int_{x_0}^{x} f(t, y(t))dt$$

Proof:

$$\int_{x_0}^{x} f(t, y(t))dt = \int_{x_0}^{x} y'$$

$$y(x) = y(x_0)$$

The other way would be

$$\left(\int_{x_0}^{x} f(t, y(t))dt + y_0 \right)' = f(x, y(x)) = y'$$

QED

62.4 The Picard-Lindelöf Iteration Method

We can approximate the desired solution using the Volterra integral equation as follows. Start with an arbitrary initial function, for example a constant — preferably the initial value:

$$g_0(x) = y_0$$

Then form the next approximation to the desired function $y(x)$ as:

$$g_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, g_n(t))\, dt$$

62.5 Integration Case

To solve an ODE of the form $y' = f(x)$, it is as easy as integrating $f(x)$.

62.6 Separation of Variables

An equation $F(x, y, y', \dots) = 0$ is called an equation of *separable variables* if it can be written in one of the following forms:

- $\frac{dy}{dx} = f(x)g(y)$
- $\frac{dy}{dx} = \frac{f(x)}{g(y)}$
- $\frac{dy}{dx} = \frac{g(y)}{f(x)}$

62.6.1 Steps to Solve an ODE of Separable Variables

1. Solve for the derivative
2. Check that it is indeed an ODE of separable variables
3. Separate the variables
4. Integrate with respect to the independent variable

Example:

Given is $(y - 2)y' = 3x - 5$.

It is an ODE of separable variables.

$$y' = \frac{3x - 5}{y - 2}$$

Now we separate the variables and integrate.

$$\int (y - 2)y'dx = \int 3x - 5dx$$

Note that $y' = \frac{dy}{dx}$ and $\frac{dy}{dx}dx = dy$ (only as a convenient fiction; in reality we are changing the variables), thus,

$$\int (y-2)dy = \int 3x - 5dx$$

$$\frac{y^2}{2} - 2y + c = \frac{3x^2}{2} - 5x + c$$

The solution we get is an implicit solution. To make it look nice, solve for y if you can.

62.6.2 Understanding the Fiction

The previous fiction we used can be explained in the following manner.

Given $f(y)\frac{dy}{dx}dx = \int f(y)dy$. We want to prove that this equality holds via the *chain rule* $\frac{du}{dx} = \frac{du}{dy}\frac{dy}{dx}$.

$$\begin{aligned}\frac{d}{dx}\left(\int f(y)dy\right) &= \frac{d}{dx}\left(\int f(y)dy\right)\frac{dy}{dx} \\ &= f(y)dy\frac{dy}{dx} \\ \implies \int f(y)\frac{dy}{dx} &= \int f(y)dy\end{aligned}$$

$\mathfrak{QED}$

62.7 Geometry of ODE

You can imagine the plot of an ODE as a *Slope field* where the slope (derivative) of a function obeys a certain condition, namely the differential equation $\frac{dy}{dx} = expr$.

A point on this field is called an *Initial Condition*, and the curve that goes through that point in the slope field is called an *Integral curve*, which is a solution to the differential equation with respect to that initial condition.

62.8 Existence and Uniqueness

Let $f(x, y)$ be continuous on a rectangle $[a, b] \times \mathbb{R}$. Then the differential equation

$$y' = f(x, y)$$

has at least one solution. A unique solution is only guaranteed under additional conditions.

A function $f : \mathbb{R}^2 \to \mathbb{R}$ is called *Lipschitz continuous with respect to* y if there exists a constant $L > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2| \quad \text{for all } (x, y_1), (x, y_2) \in D.$$

Let $x_0 \in [a, b]$, and let $f : [a, b] \times \mathbb{R} \to \mathbb{R}$ be continuous and bounded, and Lipschitz continuous in y. Then the initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0$$

has a unique solution on $[a, b]$.

Or in easier words: If f and $\frac{\partial f}{\partial y}$ are continuous near (x_0, y_0) then there is a unique solution on an interval $\alpha < x_0 < \beta$ to the IVP

$$y' = f(x, y), \quad y(x_0) = y_0$$

62.9 How to Deal with Initial Conditions

After solving a differential equation, we may have the problem that certain initial conditions were set at the start, thus making our general solution not specific.

Solving this problem is pretty easy, because we just have to solve for the constant terms in our general solution with respect to our initial conditions. For example, given $y(0) = 1$, we would plug in 0 for all x's in our general solution and set the whole equation equal to 1.

62.10 Solving DE with Substitution

In some cases, a complex function can be simplified via a substitution.

62.10.1 Case I

$$y' = f(ax + by(x) + c)$$

With a substitution like $z(x) = ax + by(x) + c$, the right-hand side will become $f(z)$ and for the left-hand side we get $y(x) = \frac{z(x) - ax - c}{b}$.

Now we can substitute in the original equation, and we get

$$y'(x) = \frac{1}{b}(z'(x) - a) = f(z)$$

$$z'(x) = a + bf(z)$$

Now we can integrate and find the solution and then return to our original variables.

Example:

$$y' = (x + y(x))^2$$

Here $a = 1$, $b = 1$, $c = 0$ and $f(z) = z^2$.

1. Substitute

$$z(x) = x + y(x)$$
$$y(x) = z(x) - x$$
$$y'(x) = z'(x) - 1$$

2. Plug in the DE

$$y' = (x + y(x))^2$$
$$z'(x) - 1 = z(x)^2$$

3. Integrate

$$z'(x) = 1 + z(x)^2$$
$$\frac{z'(x)}{1 + z(x)^2}$$

$$\int \frac{1}{1+z^2} dz = \int 1 dx$$

$$\arctan(z) = x + c$$

$$z = \tan(x + c)$$

4. Return to the original variables

$$x + y(x) = \tan(x + c)$$

$$y(x) = \tan(x + c) - x$$

62.10.2 Case II

$$y' = f(\frac{y}{x})$$

For this to be valid with $z(x) = \frac{y}{x}$, our $y(x)$ needs to look like $y(x) = z(x)x$ and so $y' = z(x) + z'(x)x$.

Now we substitute in the differential equation using $f(z) = z(x) + z'(x)x$.

$$z'(x) = \frac{f(z) - z(x)}{x}$$

Now we can integrate and then return to our original variables. $y(x) = z(x)x$, where $z(x)$ is our result of the integration.

Example:

$$y' = \frac{y(x)}{x} + 1$$

This is converted to $y' = f(z) = z + 1$.

1. Substitute

$$z(x) = \frac{y(x)}{x}$$

$$y(x) = z(x)x$$

$$y'(x) = z(x) + z'(x)x$$

2. Plug in the DE

$$z(x) + z'(x)x = z(x) + 1$$

3. Integrate

$$z'(x)x = 1$$

$$\int z' dz = \int \frac{1}{x} dx$$

$$z = (\ln(x) + c)x$$

62.11 Linear Differential Equations

An ODE is called linear if all occurrences of the dependent variable and itsderivatives have an exponent of at most 1.

$$a_n(x)y^n + a_{n-1}y^{n-1} + \cdots + a_1(x)y' + a_0(x)y = b(x)$$

If such an equation also has the property $b(x) = 0$, it is called *Homogeneous*.

62.12 How to Solve a Linear Differential Equation

In this case, we will focus on first-order ODEs.

Write it in the standard form $y' + p(x)y = f(x)$.

Now use the *Integrating Factor Method*.

62.13 Integrating Factor Method

We are going to multiply our expression by a function $r(x)$

$$r(x)y' + r(x)p(x)y = r(x)f(x)$$

Now we would like to write the left side as $\frac{d}{dx}yr(x)$ because, if we were to integrate, it would take us to the expression on the right.

When we evaluate this new expression, we get that

$$\frac{d}{dx}yr(x) = y'r(x) + r'(x)y$$

which is not quite what we actually have, but it is close. Now note that we can say

$$r'(x) = r(x)p(x)$$

This is a differential equation of separable variables, thus we can

$$\frac{r'(x)}{r(x)} = p(x)$$

$$\int \frac{r'(x)}{r(x)}dx = \int p(x)dx$$

$$\ln(r(x)) = \int p(x)dx$$

after using e

$$r(x) = e^{\int p(x)dx}$$

Now we have found an $r(x)$, and now that we know such a function exists we can return to our initial problem and say

$$\frac{d}{dx}r(x)y = r(x)f(x)$$

$$y = \frac{1}{r(x)} \int r(x) f(x) dx$$

and

$$r(x) = e^{\int p(x) dx}$$

To summarize, given $y' + f(x)y = g(x)$ the steps are:

1. Find $c(x) = e^{\int f(x) dx}$.
2. Multiply the original expression with $c(x)$.

$$y'c(x) + f(x)c(x)y = g(x)c(x)$$

3. Use the power rule of differentiation to simplify.

$$y'c(x) + f(x)c(x)y = \frac{d}{dx}\left(c(x), y\right)$$

4. Integrate.

$$\begin{aligned} \int \frac{d}{dx}\left(c(x), y\right) dx &= \int g(x)c(x) dx \\ c(x)y &= \int g(x)c(x) dx \\ y &= \frac{\int g(x)c(x) dx}{c(x)} \end{aligned}$$

Example:

Given are $y' + 4y = e^{-x}$ and $y(0) = \frac{4}{3}$.

Here $4 = p(x)$ and $e^{-x} = f(x)$.

Now let us use the formula for $r(x) = e^{\int p(x)}$. This gives us $r(x) = e^{\int 4dx} = e^{4x}$.

Now recall what we saw earlier.

$$e^{4x}y' + e^{4x}4y = e^{4x}e^{-x} = e^{3x}$$

Our left side is just $\frac{d}{dx}e^{4x}y$.

Now let us complete the exercise with our last formula by integrating both sides.

$$\begin{aligned} \frac{d}{dx}e^{4x}y &= e^{3x} \\ y &= \frac{1}{e^4 x} \int e^{3x} \\ y &= \frac{1}{e^4 x}\frac{1}{3}e^{3x} + c \end{aligned}$$

This is our general solution.

To find our specific solution, we just plug 0 into our general solution, which gives us $\frac{1}{3} + c = \frac{4}{3}$, thus $c = 1$.

62.14 Homogeneous Differential Equations

A differential equation

$$M(x,y)dx + N(x,y)dy = 0$$

is considered *Homogeneous* if and only if $M(x)$ and $N(x)$ are *homogeneous*, and they have the same degree.

The equation $y' + f(x)y = 0$ is called a *homogeneous linear differential equation of first order*.

62.14.1 Homogeneous Equation

A function $f(x,y)$ is called *homogeneous* if and only if it can be written as

$$f(xt,yt) = t^n f(x,y)$$

Example:

$$f(x,y) = x^3 + y^3$$
$$f(tx,ty) = (tx)^3 + (ty)^3$$
$$t^3(x^3 + y^3) = t^3 f(x,y)$$

It is important to note that the key is that each term has a total degree of n. For example: $x^2y + x3y^2$

62.14.2 Homogeneous Functions Theorem

If $f(x,y)$ is homogeneous of degree 0 in x and y, then f is a function of $\frac{y}{x}$.

Or $f(x,y,y')$ is homogeneous if it can be written as $y' = f(\frac{x}{y})$ or $y' = f(\frac{y}{x})$.

62.14.3 Solving Homogeneous DE I

1. Write in the form $M(x,y)dx + N(x,y)dy = 0$.
2. Check if it is homogeneous.
3. Change the variables to $y = ux$ or $x = uy$ depending on the situation.
4. Solve using the method of separable variables.
5. Rewrite the answer in terms of the original variables again.

Example:

Given is the function $(x-y)dx = -x\,dy$

Step 1. We write it in the desired form $(x-y)dx + xdy = 0$.

Step 2. We notice that both are of degree 1. This time we skip the rigorous check.

Step 3. We are going to choose the simpler function to do the change of variable depending on the *differential!*

The simpler function is xdy, thus we are to change y

$$y = ux$$
$$dy = ux' + xu'$$

and

$$u = \frac{y}{x}$$

Now substitute

$$\begin{aligned}(x - ux)dx + x(udx + xdu) &= 0\\ xdx - uxdx + xudx + x^2du &= 0\\ (x - ux + xu)dx + x^2du &= 0\\ xdx + x^2du &= 0\\ xdx &= -x^2du\\ dx &= -xdu\end{aligned}$$

Step 4. Now we can integrate both sides

$$\int \frac{1}{x}dx = \int du$$
$$\ln x + c = -u + c$$

Step 5. Now we use $u = \frac{y}{x}$ to return to our original variables

$$\ln x + c = -\frac{y}{x} + c$$
$$-x \ln x - xc = y$$

62.14.4 Solving Homogeneous DE II

Another method for solving this kind of equation $y' + f(x)y = 0$ is to follow the next steps.

Step 1: Separate the variables

$$\frac{dy}{dx} = -f(x)y$$

Step 2: Use the separation of variables

$$\int \frac{\frac{dy}{dx}}{y} = \int -f(x)dx$$

$$\int \frac{dy}{y} = \int -f(x)dx$$

$$\ln|y| = \int -f(x)dx$$

$$y = Ce^{\int -f(x)dx}$$

This method works more directly.

62.15 Inhomogeneous Equation

The equation $y' + f(x)y = g(x)$ is called a *linear inhomogeneous differential equation of first order*; $g(x)$ is called the *distortion function*. The corresponding IVP is called a *linear* IVP.

62.16 Solutions of LDE

Every linear IVP with continuous functions $f(x)$ and $g(x)$ plus a bounded $f(x)$ has exactly one solution.

Proof:

$$\begin{aligned} y' = -f(x)y + g(x) \text{ therefore, } | - f(x)y_1 + g(x) + -f(x)y_2 + g(x)| &= |f(x)||y_1 - y_2| \\ &\leq M|y_1 - y_2| \end{aligned}$$

Therefore, the function is Lipschitz continuous, thus, a solution is possible.

$\mathfrak{QED}$

62.17 Variation of the Constants

Recall that for the homogeneous case $y' + f(x)y = 0$ we use $y = ce^{\int -f(x)dx}$, but now we have $y' + f(x)y = g(x)$, thus, let us use the following approach

$$y = c(x)e^{\int -f(x)dx}$$

Now use the product rule to differentiate

$$\begin{aligned} y' &= c'(x)e^{\int -f(x)dx} + c(x)e^{\int -f(x)dx}(-f(x)) \\ &= c'(x)e^{\int -f(x)dx} - f(x)y \end{aligned}$$

thus,

$$y' + f(x)y = c'(x)e^{\int -f(x)dx}$$

therefore, $g(x) = c'(x)e^{\int -f(x)dx}$

Then we can get a solution for $c(x)$ from

$$\begin{aligned} g(x) &= c'(x)e^{\int -f(x)dx} \\ g(x) &= c'(x)e^{-\int f(x)dx} \\ c'(x) &= \frac{g(x)}{e^{-\int f(x)dx}} \\ c'(x) &= g(x)e^{\int f(x)dx} \end{aligned}$$

Example:

Given is $y' - \cos(x)y = xe^{\sin(x)}$ with $y(0) = 3$.

1. Solve as if it were homogeneous

$$\begin{aligned} y' - \cos(x)y &= 0 \\ y' &= \cos(x)y \\ \frac{y'}{y} &= \cos(x) \\ \int \frac{1}{y}dy &= \int \cos(x)dx \\ \ln|y| + c &= \sin(x) + c \\ y &= e^{\sin(x)}c \end{aligned}$$

2. Use $y = c(x)e^{\sin(x)dx}$

$$\begin{aligned} y' &= c(x)\cos(x)e^{\sin(x)} + c'(x)e^{\sin(x)} \\ y' &= \cos(x)y + c'(x)e^{\sin(x)} \\ y' - \cos(x)y &= c'(x)e^{\sin(x)} \end{aligned}$$

3. Compare with the original equation

$$\begin{aligned} y' - \cos(x)y &= xe^{\sin(x)} \\ y' - \cos(x)y &= c'(x)e^{\sin(x)} \end{aligned}$$

4. Solve for $c'(x)$

$$\begin{aligned} xe^{\sin(x)} &= c'(x)e^{\sin(x)} \\ x &= c'(x) \end{aligned}$$

5. Integrate

$$\begin{aligned} \int xdx &= \int c'(x)dx \\ \frac{x^2}{2} + c &= c(x) \end{aligned}$$

6. Substitute in our approach $y = c(x)e^{\sin(x)}$ **from earlier**

$$y = \left(\frac{x^2}{2} + c\right)e^{\sin(x)}$$

7. Solve for c **using our initial condition if asked**

$$\begin{aligned} y(0) &= 3 \\ 3 &= \left(\frac{0^2}{2} + c\right)e^{\sin(0)} \\ 3 &= c \end{aligned}$$

62.18 Existence and Uniqueness of Differential Equations

For the IVP

$$y^n + p_{n-1}y^{n-1} + \cdots + p_0(x)y = f(x)$$

$$y(a) = b_0, y'(a) = b_1, \ldots, y^{n-1}(a) = b_{n-1}$$

If all p_i and f are continuous on the interval I about a, then there exists a unique solution on I.

62.19 Theorem of Superposition and the General Solution

Suppose $y_1, \ldots, y_n$ solve

$$y^n + p_{n-1}y^{n-1} + \cdots + p_0(x)y = 0$$

Then $y = c_1y_1 + \cdots + c_ny_n$ also solves the equation, and it is the *General Solution* if and only if the *Wronskian* $W(t) \neq 0$ for some t_0.

62.20 Undetermined Coefficients

Given an equation $y' + ay = g(x)$, notice that a does not depend on x. The homogeneous equation to this case will be $y' + ay = 0$ with a solution y_h and some partial solution y_p. Now remember from *Linear Algebra* that *general = particular + homogeneous*, more precisely $y = y_p + y_h$; therefore, adding a particular solution to a homogeneous one does not change the general solution of our DE.

Now let us substitute in our differential equation.

$$\begin{aligned}(y_h + y_p)' + a(y_h + y_p) &= y_h' + y_p' + ay_h + ay_p \\ &= (y_h' + ay_h) + (y_p' + ay_p) = 0 + g(x) = g(x)\end{aligned}$$

62.21 Steps for Solving Constant Coefficients Problems

1. Find the *homogeneous* solution for $y' + ay = 0$. y_h is going to have a free parameter.
2. Find a partial solution for $y' + ay = g(x)$
3. Use $y = y_h + y_p$
4. If you have an IVP solve for the free parameter in y_h

Now we have to find a way to solve for y_p and y_h.

62.22 Find the Partial Solution

We will use a method based on guessing the type of the function on the right side.

Example:

Given is the following IVP: $y' + 2y = 2x + 13$ with $y(0) = 8$.

By looking carefully, we see that on the right-hand side we just have a normal polynomial; thus, we can substitute $y = bx + c$ and differentiate to get $y' = b$. In the case when we have a higher order derivative we would differentiate for each of them.

Now substitute

$$b + 2(bx + c) = 2x + 13$$

$$2bx + b + 2c = 2x + 13$$

Now by comparing the coefficients with the other side

$$2b = 2$$
$$b + 2c = 13$$

We get: $b = 1$, $c = 6$. And we have found a *particular* solution for our DE, $y_p = x + 6$

Now we have to find a *homogeneous* solution with the formula we already know, $y_h = ce^{\int f(x)dx}$; in our case $y_h = ce^{\int 2dx} = ce^{-2x}$

$$y = y_h + y_p = ce^{-2x} + x + 6$$

Now considering the condition $y(0) = 8$, this gives:

$$y(0) = c + 6 = 8 \Rightarrow c = 2$$

and we obtain the particular solution:

$$y = 2e^{-2x} + x + 6$$

62.23 Table of Reference

Type of Forcing Function	**Disturbance Function** $g(x)$	**Approach for** y_p
Constant	k_0	c_0
Linear	$k_0 + k_1 x$	$c_0 + c_1 x$
Polynomial	$\sum_{i=0}^{n} k_i x^i$	$\sum_{i=0}^{n} c_i x^i$
Exponential	$ke^{bx},\ b \neq a$ ke^{ax}	$c_0 e^{bx}$ $c_0 x e^{ax}$
Trigonometric	$k\sin(bx) + l\cos(bx)$	$c_0 \sin(bx) + c_1 \cos(bx)$

It is important to point out that sometimes it is necessary to combine different types of functions to get the solution.

Example:

$$y'' - 2x'3y = t^2 + 3e^{-t}\cos(4t)$$

Here the approach would be

$$y_p = (A + Bt + Ct^2) + D(e^{-t}\cos(4t)) + E(e^{-t}\sin(4t))$$

The best way is to tackle each of the types of functions separately.

An alternative way of using this method is to

1. Find the homogeneous solution.
2. Multiply both sides of the equation by it.
3. Integrate

This will give you the exact same result and is faster.

62.24 Bernoulli Equation

This is used for equations of the form

$$y' + P(x)y = Q(x)y^n$$

We cannot use the methods we already know for this equation because it is not linear, but with a clever substitution we can turn it into a linear differential equation.

Let us use a substitution

$$u = y^{1-n}$$

then

$$u' = (1-n)y^{-n}y'$$

Now this looks kind of similar to the original equation; multiply by y^{-n}, which looks like

$$y^{-n}y' + P(x)y^{-n}y = Q(x)y^{-n}yn$$

$$y^{-n}y' + P(x)y^{1-n} = Q(x)$$

Notice that $y^{-n} = \frac{u'}{(1-n)y'}$, thus, after substitution

$$\frac{1}{1-n}u' + P(x)u = Q(x)$$

Now our equation is *linear*, and we can use the *integrating factor* method.

Example:

Given is $y' - 5y = \frac{-5}{2}xy^3$.

$$u = y^{-2} \text{ and } u' = -2y^{-3}y'$$

$$y^{-3} = \frac{u'}{-2y'}$$

And our equation after multiplying by y^{-3}

$$y^{-3}y' - 5yy^{-3} = -\frac{5}{2}x$$

$$y^{-3}y' - 5y^{-2} = -\frac{5}{2}x$$

$$\frac{u'}{-2y'}y' - 5u = -\frac{5}{2}x$$

Now we can use the *Integrating Factor Method*. First bring it to the standard form. $y' + p(x)y = f(x)$

$$u' + 10u = 5x$$

Remember that $r(x) = e^{\int p(x)dx} = e^{\int 10dx} = e^{10x}$ therefore, we have

$$e^{10x}u' + 10ue^{10x} = e^{10x}5x$$

Now let us integrate

$$\frac{d}{dx}\left(e^{10x}u\right) = e^{10x}5x$$

$$Ce^{10x}u = \int e^{10x}5xdx = 5x\frac{e^{10x}}{10} - \int e^{10x}5dx$$

$$Ce^{10x}u = \int e^{10x}5xdx = 5x\frac{e^{10x}}{10} - \frac{5}{10}\int e^{10x}dx$$

$$Ce^{10x}u = \frac{xe^{10x}}{2} - \frac{5}{100}e^{10x} + c$$

$$Ce^{10x} = \frac{x}{2} - \frac{1}{20} + \frac{c}{e^{10x}} = y^{-2}$$

62.25 Autonomous Equations

An *Autonomous Differential Equation* only depends on the dependent variable y.

Example: $\frac{dy}{dt} = (1+y)(1-y)$.

62.26 Equilibrium Solutions

Values where $f(y) = 0$ are *Equilibrium Solutions*.

62.27 Asymptotically Stable

An equilibrium solution $y = a$ is *asymptotically stable* if solutions that start near a tend towards a as $t \to \infty$.

62.28 Asymptotically Unstable

An equilibrium solution $y = a$ is *asymptotically unstable* if solutions that start near a leave as $t \to \infty$.

62.29 Linear Inhomogeneous DEs with Non-Constant Coefficients

The method of *variation of constants* is often used alongside the *superposition principle*. In this method, the integration constant in the function $c(x)$ is omitted as it is part of the particular solution.

The superposition principle also holds for linear inhomogeneous differential equations with non-constant coefficients:

$$\begin{aligned} y_p' + f(x)y_p &= g(x) \quad \text{(particular solution)} \\ y_h' + f(x)y_h &= 0 \quad \text{(homogeneous solution)} \end{aligned}$$

For the total solution $y = y_h + y_p$, we get:

$$\begin{aligned} y' + f(x)y &= (y_h + y_p)' + f(x)(y_h + y_p) \\ &= y_h' + y_p' + f(x)y_h + f(x)y_p \\ &= g(x) \end{aligned}$$

Thus, the solution can be decomposed into the general solution of the homogeneous DE and a particular solution based on the form of the right-hand side.

Example:

$$y' = -\frac{y}{x} + 1$$

The homogeneous DE is $y' = -\frac{y}{x}$, which can be solved by separating the variables: $y_h = \frac{c}{x}$

Now let us use $y_p = \frac{c(x)}{x}$.

$$y_p' = \frac{c'(x)x - c(x)}{x^2}$$

Let us break down the fraction to find where to put the homogeneous solution we have already found

$$\frac{c'(x)}{x} - \frac{c(x)}{x^2} = \frac{c'(x)}{x} + \frac{1}{x}\frac{c}{x} = \frac{c'(x)}{x} + \frac{1}{x}y$$

and we get

$$y_p' = -\frac{y}{x} + \frac{c'(x)}{x}$$

Here we compare with the inhomogeneous part of the original equation and get

$$\frac{c'(x)}{x} = 1 \implies c'(x) = x$$

then $c(x)$ is equal to $\frac{x^2}{2}$, and therefore $y_p = \frac{x^2}{x}\frac{1}{x} = \frac{x}{2}$

As our final result, we get $y = \frac{c}{x} + \frac{x}{2}$.

62.29.1 Power Series Approach

Another method to solve a certain type of differential equation is to find the function given initial conditions. This can be better understood with an example.

$$y' = y$$
$$y(0) = 1$$

We say $y = \sum_{n=0}^{\infty} a_n x^n$ and $y(0) = a_0 = 1$.

Now let us differentiate our y

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 x^0 + a_1 x^1 + \cdots = 1 + a_1 x^1 + a_2 x^2 + \cdots$$
$$y' = 0 + a_1 + 2a_2 x + 3a_3 x^2 + \cdots$$

Thus, we can rewrite the sum as

$$y' = \sum_{n=1}^{\infty} a_n n x^{n-1}$$

Starting from 1 because 0 does not contribute to the sum. After shifting the index, we get

$$y' = \sum_{n=0}^{\infty} a_{n+1}(n+1)x^n$$

Now we build our original DE with our new definitions for y' and y.

$$\sum_{n=0}^{\infty} a_{n+1}(n+1)x^n = \sum_{n=0}^{\infty} a_n x^n$$

If we compare the coefficients, we get

$$a_n = a_{n+1}(n+1)$$

$$\frac{a_n}{(n+1)} = a_{n+1}$$

After iterating backwards, we get

$$a_{n+1} = \frac{a_n}{n+1} = \frac{\frac{a_{n-1}}{(n-1+1)}}{n+1} = \frac{\frac{\frac{a_{n-2}}{(n-1)}}{(n-1+1)}}{n+1}$$
$$a_n = \frac{a_0}{(n+1)!} = \frac{1}{(n+1)!} = 1$$

therefore, $a_n = \frac{1}{n!}$, which gives us the solution

$$\sum_{n=0}^{\infty} \frac{1}{n!} x^n = e^x$$

Theorem for the Power Series of DE

For a differential equation of the form:

$$A(x)y^n + \cdots + B(x)y' + C(x)y = 0$$

more specifically, after dividing by $A(x)$

$$y^n + \cdots + P(x)y' + Q(x)y = 0,$$

with $x = a$ as an *Ordinary Point* if $\ldots, P(x)$ and $Q(x)$ are *Analytic* at $x = a$. Otherwise, a *Singular Point*.

Then this equation has n linearly independent solutions of the form

$$y(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$$

The radius of convergence is at least as large as the distance to the nearest *Singular Point*.

62.29.2 Exact Differential Equations

We will take a short look at functions with two inputs via implicit differentiation $F(x, y(x)) = 0$. With $\frac{\partial F}{\partial x} = P$ and $\frac{\partial F}{\partial y} = Q$.

$$P(x,y)dx + Q(x,y)dy = 0$$

after dividing by dx

$$P(x,y) + Q(x,y)y' = 0$$

We define this kind of differential equation as *exact*. And $P_y = Q_x$ is called the *Integrability-Condition*. By finding the *potential*, we can solve this kind of differential equation.

Example:

Step 1: Solve for y'

$$\begin{aligned}(12xy+3)dx + 6x^2dy &= 0\\ (12xy+3) + 6x^2y' &= 0\\ y' &= -\frac{12xy+3}{6x^2}\end{aligned}$$

Step 2: Test the Integrability-Condition

$$\begin{aligned}P(x,y) &= 12xy+3\\ Q(x,y) &= 6x^2\\ P_y &= 12x\\ Q_x &= 12x\end{aligned}$$

Step 3: Integrate P with respect to x and add $h(y)$

$$\begin{aligned}F(x,y) = \int P(x,y)dx &= \int (12xy+3)dx\\ &= 6x^2y + 3x + h(y)\end{aligned}$$

Step 4: Differentiate F with respect to y and set equal to Q

$$\frac{\partial}{\partial y}\left(6x^2y + 3x + h(y)\right) = 6x^2 + c'(y)$$

Step 5: Integrate with respect to y

$$\begin{aligned}6x^2 + c'(y) &= 6x^2\\ c'(y) &= 0\\ \int c'(y) &= \int 0 = C\end{aligned}$$

But $C = 0$ because C in Q is zero. Therefore, $F(x,y) = 6x^2y + 3x$.

62.30 Euler Multipliers

Sometimes a seemingly *exact* differential equation is not possible to solve due to the partial derivatives of $P(x,y)$ and $Q(x,y)$ not being equal. An approach to this problem is the so-called *Euler Multipliers*, which are factors $\mu(x)$ or $\mu(y)$ that, when found, can help solve the differential equation.

$$P(x,y)\mu(x) + Q(x,y)y'\mu(x) = 0$$

or

$$P(x,y)\mu(y) + Q(x,y)y'\mu(y) = 0$$

The steps for solving this kind of problem are

1. Choose either $\mu(x)$ or $\mu(y)$ and multiply the whole equation by it.
2. Find the new partial derivatives of P and Q.
3. Set both partials equal and solve for μ in the newly created differential equation using the necessary method. Also make sure that the solution has no zeroes. This is necessary because otherwise we would have multiplied our original equation by zero.
4. Substitute μ with the solution. If both partials are now equal, then you can solve the exact differential equation. If not, try with another μ using the other variable. And if this also does not work, then the differential equation cannot be solved. More specifically, if $\mu(x)$ depends on y and $\mu(y)$ on x, then there is no way to make the differential equation exact.

Example:

$$3y^2 + 2xyy' = 0, x > 0$$

Here we have $P(x, y) = 3y^2$ and $Q(x, y) = 2xy$, whose partial derivatives are different

$$\frac{\partial P}{\partial y} = 6y \neq 2y = \frac{\partial Q}{\partial x}$$

Now we try to use a multiplier $\mu(x)$.

$$P_2(x, y) = 3y^2\mu(x) \text{ and } Q_2(x, y) = 2xy\mu(x)$$

Now by taking the partials again

$$\begin{aligned}\frac{\partial P_2}{\partial y} &= 6y\mu(x)\\ \frac{\partial Q_2}{\partial x} &= 2y(\mu(x) + x\mu'(x))\end{aligned}$$

Now by setting them equal $6y\mu(x) = 2y(\mu(x) + x\mu'(x))$ we can solve a new differential equation to find $\mu(x)$ and $\mu'(x)$.

$$\begin{aligned}6y\mu(x) &= 2y(\mu(x) + x\mu'(x))\\ 3\mu(x) &= \mu(x) + x\mu'(x)\\ 2\mu(x) &= x\mu'(x)\\ 2\mu &= x\frac{d\mu}{dx}\\ \int \frac{2}{x}dx &= \int \frac{d\mu}{\mu}\\ 2\ln(x) &= \ln(\mu)\\ \mu &= x^2\end{aligned}$$

Now finally we check if now the partials are equal, which in this case is true.

$$\begin{aligned}\frac{\partial P_2}{\partial y} &= 6yx^2\\ \frac{\partial Q_2}{\partial x} &= 6yx^2\end{aligned}$$

Now the equation can be solved like normal.

62.31 Constant Coefficients ODE

Consider the second-order linear differential equation with constant coefficients:

$$ay'' + by' + cy = 0,$$

where $a, b, c \in \mathbb{R}$, and $a \neq 0$.

We look for solutions of the form $y = e^{rt}$. Substituting into the equation:

$$\begin{aligned} y &= e^{rt}, \\ y' &= re^{rt} \\ y'' &= r^2 e^{rt} \end{aligned}$$

Substituting into the original equation:

$$ar^2 e^{rt} + bre^{rt} + ce^{rt} = 0$$

Factoring out e^{rt} (which is never zero):

$$e^{rt}(ar^2 + br + c) = 0 \quad \Rightarrow \quad ar^2 + br + c = 0.$$

This is a quadratic equation in r. Solving using the quadratic formula:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Suppose there are two linearly independent solution y_1 and y_2 to

$$y'' + p(x)y' + q(x)y = 0,$$

then the general solution is $y = c_1 y_1 + c_2 y_2$.

We analyze the nature of the roots r based on the discriminant $D = b^2 - 4ac$. Also, for dealing with initial conditions of the form $y^n = a, \ldots$ we only have to plug the values into our solutions $y_h, y_h', \ldots$ to get the constants.

62.31.1 Case 1: Two Distinct Real Roots $(D > 0)$

There is not much to do here; we just plug in the solutions of the root.

$$y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

62.31.2 Case 2: Repeated Real Root $(D = 0)$

For this case we need to use a trick.

$$y = c_1 e^{rt} + c_2 te^{rt}$$

Note that this could be counterintuitive, but te^{rt} is a solution and is linearly independent. For more repeated roots, you can use higher powers of t: $t, t^2, t^3, \ldots$.

62.31.3 Case 3: Complex Conjugate Roots ($D < 0$)

When the discriminant $D = b^2 - 4ac < 0$, the roots of the characteristic equation are complex:

$$r = \alpha \pm i\beta,$$

where $\alpha = -\frac{b}{2a}$ and $\beta = \frac{\sqrt{4ac-b^2}}{2a}$.

The general solution to the differential equation is:

$$y(t) = c_1 e^{(\alpha+i\beta)t} + c_2 e^{(\alpha-i\beta)t}.$$

To express this in real form, we use Euler's formula:

$$e^{i\beta t} = \cos(\beta t) + i\sin(\beta t).$$

Now rewrite each exponential term:

$$\begin{aligned} e^{(\alpha+i\beta)t} &= e^{\alpha t} \cdot e^{i\beta t} = e^{\alpha t}(\cos(\beta t) + i\sin(\beta t)), \\ e^{(\alpha-i\beta)t} &= e^{\alpha t} \cdot e^{-i\beta t} = e^{\alpha t}(\cos(\beta t) - i\sin(\beta t)). \end{aligned}$$

Now, to get rid of the imaginary part, we can do the following trick

$$\begin{aligned} \frac{y_1 + y_2}{2} &= e^{\alpha t}\cos\beta t \\ \frac{y_1 - y_2}{2i} &= e^{\alpha t}\sin\beta t \end{aligned}$$

This might feel wrong, but these are in fact linear combinations of our original solution, and they are also linearly independent; because of that, we can just use them for our general solution.

$$y = c_1 e^{\alpha t}\cos\beta t + c_2 e^{\alpha t}\sin\beta t$$

Example:

Given is $y''' + y' = 0$.

$$y = e^{rt}$$

$$\begin{aligned} r^3 + r &= 0 \\ r(r^2+1) &= 0 \\ r(r+i)(r-i) &= 0 \end{aligned}$$

Now $r_1 = 0 \quad r_2 = -i \quad r_3 = i$.

$$y = c_1 e^{0t} + c_2 e^{-it} + c_3 e^{it}$$

$$y = c_1 + c_2\cos t + c_3\sin t$$

62.31.4 Inhomogeneous Case

For the inhomogeneous case, we apply the method of *Undetermined Coefficients*, differentiating two or more times depending on the order of the differential equation, and then add the homogeneous solution to the particular solution we obtain.

Example:

Solve the differential equation:

$$y' + \frac{1}{x}y = \sin(x)$$

The integrating factor is:

$$\mu(x) = e^{\int \frac{1}{x}dx} = e^{\ln|x|} = |x|$$

Assuming $x > 0$, we take $\mu(x) = x$. Multiply through by the integrating factor:

$$xy' + y = x\sin(x)$$

The left-hand side is the derivative of xy:

$$\frac{d}{dx}(xy) = x\sin(x)$$

$$\int \frac{d}{dx}(xy)\,dx = \int x\sin(x)\,dx$$

Use integration by parts on the right:

$$\int x\sin(x)dx = -x\cos(x) + \int \cos(x)\,dx = -x\cos(x) + \sin(x)$$

So:

$$xy = -x\cos(x) + \sin(x) + C$$

$$y = -\cos(x) + \frac{\sin(x)}{x} + \frac{C}{x}$$

62.32 The Wronskian

Let $y_1, y_2, \ldots, y_n$ be n functions that are at least $(n-1)$-times differentiable on some interval. The *Wronskian* of these functions is defined as:

$$W(y_1, y_2, \ldots, y_n)(x) = \begin{vmatrix} y_1(x) & y_2(x) & \cdots & y_n(x) \\ y_1'(x) & y_2'(x) & \cdots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \cdots & y_n^{(n-1)}(x) \end{vmatrix}$$

If $W(A) \neq 0$, then they are linearly independent.

62.33 Linearity Property

Let $y_p^{(1)}$ be a particular solution of

$$y'' + ay' + by = g_1(x)$$

and $y_p^{(2)}$ a particular solution of

$$y'' + ay' + by = g_2(x),$$

then

$$y_p = y_p^{(1)} + y_p^{(2)}$$

is a particular solution of

$$y'' + ay' + by = g_1(x) + g_2(x).$$

62.34 Variation of Parameters

For a linear differential equation of the form

$$y'' + p(x)y' + q(x)y = f(x)$$

The homogeneous equation has a solution consisting of the linearly independent y_1 and y_2, of the form $y = u_1y_1 + u_2y_2$. We can guess the functions y_1 and y_2.

So, for finding a solution of these types of equations, we use the following formulas.

$$u_1 = -\int \frac{y_2 g}{y_1y_2' - y_2y_1'}dx$$

$$u_2 = -\int \frac{y_1 g}{y_1y_2' - y_2y_1'}dx$$

These formulas come from:

Differentiating two times:

$$y = u_1y_1 + u_2y_2$$

$$y' = u_1y_1' + y_1u_1' + u_2y_2' + y2u_2'$$

Here we can add another constraint that $y_1u_1' + y2u_2' = 0$. This allows us to simplify the expression to:

$$y' = u_1y_1' + u_2y_2'$$

Now we can differentiate one more time:

$$y'' = u_1y_1'' + y_1'u_1' + u_2y_2'' + y_2'u_2'$$

Substitute in the original equation

$$y'' + p(x)y' + q(x)y = f(x)$$

$$u_1y_1'' + y_1'u_1' + u_2y_2'' + y_2'u_2' + p(x)(u_1y_1' + u_2y_2') + q(x)(u_1y_1 + u_2y_2) = g(x)$$

$$u_1y_1'' + y_1'u_1' + u_2y_2'' + y_2'u_2' + p(x)(u_1y_1') + p(x)(u_2y_2') + q(x)(u_1y_1) + q(x)(u_2y_2) = g(x)$$

$$u_1y_1'' + y_1'u_1' + u_2y_2'' + y_2'u_2' + p(x)(u_1y_1') + p(x)(u_2y_2') + q(x)(u_1y_1) + q(x)(u_2y_2) = g(x)$$

With non-generic functions, a lot of the terms would cancel out. After that, we multiply one of the rows by y_1 and the other by y_2 so that we can cancel them in a row operation. This will leave us with only two integrals to solve.

62.35 Which Method to Use Depending on the Type of Differential Equations

First Order

- Separable ⇒ Separation of Variables.
- Form $y' = f(ax + by(x) + c)$ ⇒ Substitution $z = ax + by(x) + c$ and $f(z)$.
- Form $y' = \frac{y}{x}$ ⇒ Substitution $z = \frac{y}{x}$
- Linear ⇒ Integrating Factor or Variation of Constants.
- Bernoulli (Non-linear) ⇒ Bernoulli Substitution.
- Autonomous ⇒ Equilibrium Analysis.
- Non-homogeneous ⇒ Undetermined Coefficients or integrating factor.

62.35.1 Second Order and Higher

- Constant Coefficients ⇒ Guess e^{rx}.
- Non-homogeneous ⇒ Undetermined Coefficients.
- Linear ⇒ Series Solutions.

62.35.2 Exact

Using the method for exact differential equations.

62.36 Linear Systems of Differential Equations

Let us start by looking at a system of first-order homogeneous differential equations

$$y_1' = ay_1 + by_2$$
$$y_2' = cy_1 + dy_2$$

This can be written as

$$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$Ax = \vec{y}'$$

Then, if we use the guess $y(t) = \vec{v}e^{\lambda t}$ and differentiate with respect to t, we get

$$\lambda \vec{v} e^{\lambda t} = A\vec{v}e^{\lambda t}$$

$$\lambda \vec{v} = A\vec{v},$$

which is our *Eigenvalue* problem. This can be solved via

$$\det(A - \lambda I) = 0$$

$$\lambda^2 - (a+c)\lambda + ad - bc = 0$$

And this solution can lead to the three cases we already know: distinct real roots, complex conjugate roots, and repeated roots.

We can also use substitution to solve for one of the y_i, which are functions of x, like

$$\begin{aligned} y_1' &= ay_1 + by_2 \\ y_2' &= cy_1 + dy_2 \\ by_2 &= y_1' - ay_1 \\ y_2 &\frac{y_1' - ay_1}{b} \end{aligned}$$

Then we can differentiate our first equation with respect to x

$$\begin{aligned} by_2 &= y_1' - ay_1 \\ by_2' &= y_1'' - ay_1' \\ &= b(cy_2 + dy_1) \\ &= bcy_2 + bdy_1 \\ &= c(y_1' - ay_1) + bdy_1 \\ y_1'' - (a+c)y_1' + (ac - bd)y_1 + bdy_1 &= 0 \end{aligned}$$

Example:

We are given the system of differential equations:

$$\begin{cases} y' = y + z \\ z' = -2y + 3z \end{cases}$$

We write this system in matrix form:

$$u' = Au, \quad \text{where } u = \begin{bmatrix} y \\ z \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix}$$

To solve, we find the eigenvalues of A by solving the characteristic equation:

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 1 \\ -2 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) + 2 = \lambda^2 - 4\lambda + 5 = 0$$

Solving the quadratic equation:

$$\lambda = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(5)}}{2} = \frac{4 \pm \sqrt{-4}}{2} = \frac{4 \pm 2i}{2} = 2 \pm i$$

Now we find an eigenvector corresponding to $\lambda = 2 + i$ by solving:

$$(A - (2+i)I)v = 0$$

$$A - (2+i)I = \begin{bmatrix} 1-(2+i) & 1 \\ -2 & 3-(2+i) \end{bmatrix} = \begin{bmatrix} -1-i & 1 \\ -2 & 1-i \end{bmatrix}$$

From the first row of the system:

$$(-1-i)v_1 + v_2 = 0 \Rightarrow v_2 = (1+i)v_1$$

So, an eigenvector corresponding to $\lambda = 2 + i$ is:

$$v = \begin{bmatrix} 1 \\ 1+i \end{bmatrix}$$

After repeating the previous computation for $\lambda = 2 - i$, we get the complex solution to the system:

$$u(t) = c_1 e^{(2+i)t} \begin{bmatrix} 1 \\ 1+i \end{bmatrix} + c_2 e^{(2-i)t} \begin{bmatrix} 1 \\ 1-i \end{bmatrix}$$

Chapter 63

Mathematical Optimization

Optimization associates a cost function $f : \mathbb{R}^n \to \mathbb{R}^n$ with a singular or multiple optimal solution $x \in \mathbb{R}^n$. They can also be associated with some constraints.

Optimization relates closely to derivatives and iterative numerical methods for finding the solution x.

63.1 Global and Local Minima

Let $f \in \mathbb{R}^n \to \mathbb{R},\ \tilde{x} \in \mathbb{R}^n$, then $\tilde{x}$ is a *global minimum* if

$$f(\tilde{x}) \leq f(x), \quad \forall x \in \mathbb{R}^n$$

$\tilde{x}$ is a *local minimum* if $\exists\ \varepsilon > 0$ such that

$$f(\tilde{x}) \leq f(x) \quad \forall x \in \mathbb{R}^n \quad \|x - \tilde{x}\| < \varepsilon$$

63.2 Extrema and Critical Points Test of Single Variable Calculus

A function f has a critical point at $x = c$ if:

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

To determine the nature of the critical point:

- *First Derivative Test:* Check sign changes of f' around c
- *Second Derivative Test:*

$$f''(c) > 0 \Rightarrow \text{local minimum} f''(c) < 0 \Rightarrow \text{local maximum}$$

  This works because the second derivative tells us about the concavity of the function. If it is positive, then we will have a function which is concave up, a minimum; and if negative, concave down, a maximum.

63.3 Relative Extrema in $\mathbb{R}^n$

To determine local extrema for $f : \mathbb{R}^n \to \mathbb{R}$ without constraints, follow this procedure:

1. Find critical points by solving:

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = 0.$$

   This must hold for each of the equations. This process is called *finding the candidates*, which we denote as $\tilde{x}$.

2. Compute the Hessian matrix:

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(x)}{\partial x_n \partial x_n} \end{bmatrix}.$$

3. Analyze the Hessian at critical points:

   For the $\mathbb{R}^2$ case we use:

   If $H_f(x)$ is *positive definite* $f_{xx}(x_0, y_0) > 0$, $\det(H) > 0$, then x_0, y_0 is a local minimum.

   If $H_f(x)$ is *negative definite* $f_{xx}(x_0, y_0) < 0$, $\det(H) > 0$, then it is a local maximum.

   If $H_f(x)$ has mixed signs, *indefinite*, $\det(H) < 0$, then it is a saddle point.

   If $\det(H) = 0$, then the next derivative decides.

   For the general case $\mathbb{R}^n$ we use the eigenvalues.

   Positive definite if $\lambda_i > 0$, $\forall \lambda_i$, which implies a local minimum.

   Negative definite if $\lambda_i < 0$, $\forall \lambda_i$, which implies a local maximum.

   Indefinite for mixed signs, which implies a saddle point.

In real life cases, the Hessian is not widely used due to the high computational cost. Other methods are used to approximate the extreme points. However, if the Hessian shows definite behavior via eigenvectors, we can guarantee that our problem is a convex optimization problem.

Example:

Find the minima and maxima of $f(x, y) = x^2 + y^2$.

$$\nabla f = \begin{bmatrix} 2x \\ 2y \end{bmatrix} \Rightarrow \text{Critical point at } (0, 0)$$

$$H_f = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \text{positive definite} \Rightarrow (0, 0) \text{ is a local minimum}$$

63.3.1 Non-Differentiable Points

If the function f is not differentiable at some point $\vec{x}_0$, but defined there, this point may also be a candidate for extremum and should be examined separately using *limit-based analysis* or directional behavior, i.e., taking the limit at that point when analyzing the derivatives.

63.4 Lagrange Multipliers

Let $f : \mathbb{R}^n \to \mathbb{R}$ be the function to optimize, subject to the constraint $g(\vec{x}) = 0$, where $g : \mathbb{R}^n \to \mathbb{R}$.

If we look at the constraint intersection with the original function we are going to notice that along the intersection there are maxima and minima with their respective tangent lines and gradients that are normal to them. We also are going to see that both the gradient of f and the gradient of g are just scalar versions of each other.

We start with the *Lagrange system*:

$$\begin{cases} \nabla f(\vec{x}) = \lambda \nabla g(\vec{x}) \\ g(\vec{x}) = 0 \end{cases}$$

For more conditions we can use $g_1, \ldots, g_n$ and $\lambda_1, \ldots, \lambda_n$.

$$\begin{cases} \nabla f(\vec{x}) = \lambda_1 \nabla g_1(\vec{x}) + \cdots + \lambda_n \nabla g_n(\vec{c}) \\ g_1(\vec{x}) = 0 \\ \vdots \\ g_n(\vec{x}) = 0 \end{cases}$$

We can also compact these equations in the *Lagrangian* as

$$\mathcal{L}(x, y, \ldots, \lambda_1, \ldots, \lambda_n) = f(x, y, \ldots) + \lambda_1 g_1(x, y, \ldots) + \cdots + \lambda_n g_n(x, y, \ldots).$$

63.5 Determinant Method

In certain cases, this is the way to solve a problem of this topic, hence, here are the steps to follow.

1. Build the *Lagrangian*.
2. $\nabla \mathcal{L} = 0$ and maybe compute the determinant to generate another equation to solve.
3. Get rid of λ via factorization, substitution, addition of rows, etc.
4. Solve the system without λ.

63.6 The Bordered Hessian

The Bordered Hessian matrix is defined as

$$H_B = \begin{bmatrix} 0 & (\nabla g(x_1, \ldots, x_n))^T \\ \nabla g(x_1, \ldots, x_n) & \nabla^2 \mathcal{L}(x, \ldots, x_n, \lambda_1, \ldots, \lambda_n) \end{bmatrix}$$

where

- $\nabla g(x_1, \ldots, x_n)$ is the $m \times n$ Jacobian matrix of the constraints.
- $\nabla^2 \mathcal{L}(x, \ldots, x_n, \lambda_1, \ldots, \lambda_n)$ is the $n \times n$ Hessian of the Lagrangian with respect to x.

Example I:

Find the point on the circle $x^2 + y^2 = 4$ that is the closest to the point $(3, 4)$.

The function $f(x, y)$ is in this case the Euclidean norm:

$$f(x, y) = \sqrt{(x-3)^2 + (y-4)^2}$$

subject to the constraint

$$g(x, y) = x^2 + y^2 - 4 = 0.$$

The gradient of the constraint is:

$$\nabla g = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

Since $\nabla g \neq 0$, the rank condition is fulfilled, and no further critical points need to be analyzed.

Build the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda g(x, y).$$

In this case we can use a property of the root function to our advantage. Notice that the minimum of the root function is the minimum of the term under the root; therefore, we can simplify some work by writing $f(x, y)$ without the root.

$$\mathcal{L}(x, y, \lambda) = (x-3)^2 + (y-4)^2 + \lambda(x^2 + y^2 - 4)$$

We solve

$$\begin{aligned}
\textbf{I}\ \frac{\partial \mathcal{L}}{\partial x} &= 2(x-3) + 2\lambda x = 0 \\
\textbf{II}\ \frac{\partial \mathcal{L}}{\partial y} &= 2(y-4) + 2\lambda y = 0 \\
\textbf{III}\ \frac{\partial \mathcal{L}}{\partial \lambda} &= x^2 + y^2 - 4 = 0.
\end{aligned}$$

Now, this is the difficult part. The best approach in this case is to use substitution with a top-bottom approach:

$$x = \frac{3}{1+\lambda}$$

$$y = \frac{4}{1+\lambda}$$

Solving for λ is also an option, but in these kinds of systems of equations, where both the partial with respect to x and y are very similar, it is a good idea to solve for them because they tend to have a similar form when solving for them.

$$\begin{aligned}
\left(\frac{3}{1+\lambda}\right)^2 + \left(\frac{4}{1+\lambda}\right)^2 - 4 &= 0 \\
\frac{9}{(1+\lambda)^2} + \frac{16}{(1+\lambda)^2} - 4 &= 0 \\
\frac{25}{(1+\lambda)^2} &= 4 \\
25 &= 4(1+\lambda)^2 \\
\sqrt{\frac{25}{4}} &= 1+\lambda \\
\lambda &= -1 \pm \frac{5}{2}
\end{aligned}$$

$$\lambda_1 = \frac{-7}{2} \quad \lambda_2 = \frac{3}{2}$$

Now, let us substitute backwards for λ_1:

$$x = \frac{3}{1 - \frac{7}{2}} = \frac{6}{-5},$$

$$y = \frac{4}{1 - \frac{7}{2}} = \frac{8}{-5}$$

Now, for λ_2:

$$x = \frac{3}{1 + \frac{3}{2}} = \frac{6}{5}$$

and

$$y = \frac{4}{1 + \frac{3}{2}} = \frac{8}{5}$$

We construct the Hessian with constraints:

$$H = \begin{bmatrix} 2 + 2\lambda & 0 & 2x \\ 0 & 2 + 2\lambda & 2 \\ 2x & 2 & 0 \end{bmatrix}$$

For λ_1 we get $\det(H) = \frac{244}{5} > 0$, a minimum and for λ_2 $\det(H) = \frac{-244}{5}$ a maximum.

Example II:

Optimize the distance to the origin $f(x, y, z)$ with respect to the constraints $g_1(x, y, z) = z^2 - x^2 - y^2 = 0$ and $g_2(x, y, z) = x - 2z - 3 = 0$.

$$\begin{aligned} \nabla f &= \lambda \nabla g_1 + \mu \nabla g_2 \\ g_1 &= 0 \\ g_2 &= 0 \end{aligned}$$

We know that $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$. Like with other minimization problems, we can just find the minimum of the part under the root; thus, $f(x, y, z) = x^2 + y^2 + z^2$. Now we can build our system of equations:

$$\begin{aligned} \textbf{I}\ & 2x = \lambda(-2x) + \mu \\ \textbf{II}\ & 2y = \lambda(-2y) \\ \textbf{III}\ & 2z = \lambda(2z) + \mu(-2) \\ \textbf{IV}\ & z^2 = x^2 + y^2 \\ \textbf{V}\ & x = 2z + 3. \end{aligned}$$

For the equation **II**, we can already see that the equation is equal to zero only if $\lambda = -1$ or $y = 0$. Thus, we have our first numbers to work with.

Case $\lambda = -1$:

$$\begin{aligned} \textbf{I}\ & 2x = -1(-2x) + \mu \implies \mu = 0 \\ \textbf{III}\ & 2z = -1(2z) + 0(-2) \implies z = 0 \\ \textbf{IV}\ & 0^2 = x^2 + y^2 \implies x = y = 0 \\ \textbf{V}\ & 0 = 0 + 3 \implies 0 = 3 \end{aligned}$$

With the last contradiction we know that the whole case $\lambda = -1$ is not valid.

Case $y = 0$:

$$\begin{aligned}
&\textbf{IV } z^2 = x^2 + y^2 \implies z^2 = x^2 \implies z = \pm x \\
&\textbf{V } x = 2x + 3 \implies x = -3 (z = x) \\
&\textbf{V } x = -2x + 3 \implies x = 1 (z = -x)
\end{aligned}$$

Now with this new information we can get two different points $P_1(1, 0, -1)$ and $P_2(-3, 0, -3)$. We do not need to write the Hessian in this case because with geometric intuition the problem can be solved.

$$f(P_1) = 1^2 + 0^2 + (-1)^2 = 2$$

So the distance is $\sqrt{2}$.

$$f(P_2) = -3^2 + 0^2 - 3^2 = 18$$

Thus, the distance is $3\sqrt{2}$.

Our final answer is that P_1 is the closest point to the intersection of the two planes with the function f.

63.7 Convexity

A function f is *convex* if $\|x - y\| \in X, \ \forall x, y \in X$, where X is the *epigraph* of the function. Visually, in 2D and 3D, it is cup-shaped.

Convexity delivers the property that local extrema become global extrema. In the context of optimization, *convexity* generalizes linearity.

63.8 Penalty Functions

A way to get rid of constraints in an optimization problem is to add the constraint equality or inequality as a part of our target function f, accompanied by a scalar for the linear case and more exotic terms for non-linear cases. This technique allows for the maintenance of continuity.

63.9 The Primal and Dual Problem

Given a convex optimization problem with a target function $f(X)$, equality constraints $h_i(X)$, and inequality constraints $g_i(X)$, we define the *primal problem* as

$$\min_{X \in \mathbb{R}^n} f(X) + \max_{v_i, u_i \geq 0} \sum_i v_i h_i(X) + \sum_j u_j g_j(X)$$

with $u_i > 0$ and $v_i \neq 0$. We can build as the *dual problem* as

$$\max_{v_i, u_i \geq 0} \min_{X \in \mathbb{R}^n} f(X) + \sum_i v_i h_i(X) + \sum_j u_j g_j(X),$$

this problem can be easier to solve than the primal problem, and using the Karush-Kuhn-Tucker conditions, we get a much simpler problem.

63.10 Gradient Descent

The *gradient descent* is an iterative method to find the local minimum of a function f by moving in the direction of the negative gradient. The update rule is given by:

$$x_{k+1} = x_k - \alpha \nabla f(x_k)$$

where α is the learning rate, which controls the step size. The choice of α is crucial for convergence.

It is important to note that gradient descent can converge to local minima, and the choice of the initial point x_0 can significantly affect the outcome.

Chapter 64

Operations Research

Operations research is the discipline of applying advanced analytical methods to help make better decisions. It is used in logistics, finance, healthcare, manufacturing, etc.

64.1 Linear Programming

Linear programing si the process od optimizing a linear objective function subsect to a set of linear functional constraints. Formally, given $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$, a linear programming problem can be written as:

$$\text{maximize } c^T x \text{ subject to } Ax \leq b,\ x \geq 0.$$

We denote our optimal solution as

$$x^* = \text{argmax}\{c^T x \mid x \geq 0, Ax \leq b\}.$$

64.1.1 Standard Form

A linear program is in the *standard form* if $\forall b_i \in b,\ b_i \geq 0$. This means that we can manipulate our expresions to get b to the standard form if needed.

We also can modify our problem in the following way

$$g(x) \geq b \rightarrow -g(x) \leq -b$$

and

$$g(x) = b \rightarrow g(x) \leq b \wedge g(x) \geq b$$

This also allows to get an optimal solution from a minimization problem

$$\min(f(x)) = \max(-f(x)).$$

When modelling a problem as a linear program, **always** use the standard form.

Example:

Given

$$z = 3x_1 + 2x_2$$

which we want to maximaze subject to

$$\begin{aligned} -x_1 - x_2 &\leq -3 \\ 2x_1 + x_2 &\leq 8 \\ -x_1 + x_2 &\leq 1 \\ x_1 + 2x_2 &\leq 8 \end{aligned}$$

and the non-negativity constraints $x_1, x_2 \geq 0$. The *standard form* of this problem is given by

$$Ax = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 3 \\ 8 \\ 1 \\ 8 \end{bmatrix} = b.$$

With

$$c = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

64.2 Vertex Enumeration

Vertex enumeration is the process of listing all vertices (extreme points) of a convex polyhedron defined by a set of linear inequalities.

Formally, given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, we want to find all $x \in \mathbb{R}^n$ such that:

$$Ax \leq b.$$

The vertices of this polyhedron are the points that cannot be expressed as a convex combination of other points in the polyhedron.

It is computationally challenging and expensive, especially for high-dimensional polyhedra.

64.3 Families of Methods for Solving Linear Programs

64.3.1 Simplex Method

The *simplex method* is an iterative algorithm which "moves" along the edges of the feasible region defined by the linear constraints, from one vertex to another, in search of the optimal solution.

Formally, starting from an initial feasible vertex x_0, the simplex method iteratively moves to an adjacent vertex x_{k+1} such that $c^T x_{k+1} > c^T x_k$, until no further improvement is possible, at which point the current vertex is the optimal solution x^*.

The simplex method is widely used in practice due to its efficiency on many real-world problems, despite its worst-case exponential time complexity.

64.3.2 Exterior Point Methods

Exterior point methods are a class of algorithms for solving linear and nonlinear convex optimization problems. Unlike the simplex method, which traverses the boundary of the feasible region, exterior point methods move through the interior and exterior of the feasible region. They are not mainstream.

64.3.3 Intetior Point Methods

Interior point methods are a class of algorithms for solving linear and nonlinear convex optimization problems. They operate by traversing the interior of the feasible region, rather than its boundary, and iteratively improve the solution until convergence to the optimal point.

A common example of an interior point method is the *primal-dual interior point method*, which simultaneously considers the primal and dual formulations of the optimization problem to guide the search for the optimal solution.

64.4 The Simplex Method

The *simplex method* uses the corner-points of the feasable region to find the best solution. To accomplish this we solve a system of equations induced by a subset of inequalities of our LP; the **unique solutions** of this systems are the cornerpoints of our corner points. Also note, that LPs are system of inequelities, not of equalities threfore, we need to **modify** our LP get equalities.

We use projections to a higher dimensional space by adding extra *slack variables* so that our inequalities become equalities like for example:

$$x_1 \leq 4,\, x_1 \geq 0 \to x_1 + x_3 = 4,\, x_1, x_3 \geq 0.$$

We do this for all of our functional constraints, giving us the *augmented for of a linear program*.

From this augmented system there are subsets which will have unique solution and hence will give us the corner points.

64.4.1 Basic Feasable Solution

A *basic solution* is a solution which cames from a system of our augmented form which also feasable if and only if all of the decision variables and slack variables are non-negative.

The variables of our subs-system of equations are called *basic* variables while the remaning are called *non-basic*.

Adjacency

Convexity

64.4.2 The Simplex Algorithm

Note that we use the value of the objective function

$$Z = a_1x_1 + \cdots + a_nx_n$$

as a variable hence,

$$Z - a_1x_1 - \cdots - a_nx_n = 0.$$

1. Find basic feasable solution to start by using the slack variables as a basis for the first system.

$$\begin{bmatrix} x_{n+1} \\ x_{n+2} \\ \vdots \\ x_{n+m} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

2. Can a current **non-basic** variable improve the objective function? Check coefficients in row 0. If no, terminate.

3. Pick a non-basic variable with the greatest potential to improbe the objective function as the entering variable. st potential to improbe the objective function as the entering variable.

$$x_i = \max(\{c_i \mid c_i \text{ is coefficient of the target function (also after modification)}\})$$

4. Use the minimum ratio test to identify the leaving variable. Using the system of equation gained from the basic solution check which one has the smallest fraction when solving for the x_i. That variable x_j is the one which is going to hit zero first. This is the *minimum ratio test*

$$x_j = \min\left(\frac{b_i}{a_{ij}} \mid a_{ij} > 0\right)$$

5. Update the system for the new basic feasible solution using Gaussian elimination. This means we need to update the column of the variable x_i for all coefficients to be zero, except for the row **owned** by the variable the variable x_j identified by the minimum ratio test as the leaving variable.
6. Read the solution by noting the solution from the target variables and the slack variables which did not turn 0.

If a our system of equations fails to have a unique solution, then our problem is either infisible or it has infinitely many solutions. In both cases our simplex is not affected since for the first case we can not start the algorithm and for the second case, it will just find a solution and then stop.

Example:

Given $3_x1 + 5x_2 = Z$ subject to

$$\begin{aligned} x_1 &\leq 4 \\ x_2 &\leq 12 \\ 3x_1 + 2x_2 &\leq 18 \end{aligned}$$

We first bring the problem to the standard form

$$\begin{aligned} Z - 3_x1 - 5x_2 = 0 \quad x_1 + x_3 &= 4 \\ x_2 + x_4 &= 12 \\ 3x_1 + 2x_2 + x_5 &= 18 \end{aligned}$$

First, we know that there is a basic solution

$$\begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

with $x_1 = 0 = x_2$.

Now, we know that x_2 has the biggest impact on the target function since it has weight 5. Therefore, we need to take x_2 into consideration for our subsystem of equations and solve

$$\begin{aligned} x_3 &= 4 \\ x_4 &= 12 - 2x_2 \\ x_5 &= 18 - 2x_2 \end{aligned}$$

Here since $\frac{12}{2} < \frac{18}{2}$ we need need to decrease x_4 to zero first.

$$\left[\begin{array}{ccccc|c} -3 & -5 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 4 \\ 0 & 2 & 0 & 1 & 0 & 12 \\ 3 & 2 & 0 & 0 & 1 & 18 \end{array}\right]$$

Row 3 becomes

$$\left[\begin{array}{ccccc|c} -3 & -5 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & \frac{1}{2} & 0 & 6 \\ 3 & 2 & 0 & 0 & 1 & 18 \end{array}\right]$$

Now we need to get ri of the -5 in row 1 and the 2 in row 4.

$$\left[\begin{array}{ccccc|c} -3 & 0 & 0 & 2.5 & 0 & 30 \\ 1 & 0 & 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & \frac{1}{2} & 0 & 6 \\ 3 & 0 & 0 & -1 & 1 & 6 \end{array}\right]$$

This completes the first iteration of the algorithm. Note that the non-basic variable x_1 still has potential and that the coefficient of x_4 is already non-negative.

We now contine with x_1:

$$x_3 = 4 - x_1$$
$$x_5 = 6 - 3x_1$$

.

Since $4 > 2$ we know that x_5 has to leave. Hence we will perform Gaussian elimination again in the row of the leaving variable x_5:

Row 3,

$$\left[\begin{array}{ccccc|c} -3 & 0 & 0 & 2.5 & 0 & 30 \\ 1 & 0 & 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & \frac{1}{2} & 0 & 6 \\ 1 & 0 & 0 & \frac{-1}{3} & \frac{1}{3} & 2 \end{array}\right]$$

For the other rows =:

$$\left[\begin{array}{ccccc|c} 0 & 0 & 0 & \frac{3}{2} & 1 & 36 \\ 0 & 0 & 1 & \frac{1}{3} & \frac{-1}{3} & 2 \\ 0 & 1 & 0 & \frac{1}{2} & 0 & 6 \\ 1 & 0 & 0 & \frac{-1}{3} & \frac{1}{3} & 2 \end{array}\right]$$

Since our slack variables except for x_3 are zero we know that

$$Z + 0x_1 + 0x_2 + 0x_3 + \frac{3}{2}x_4 + 1x_5 = 36$$
$$0x_1 + 0x_2 + x_3 + 0x_4 + 0x_5 = 2$$
$$0x_1 + x_2 + 0x_3 + 0x_4 + 0x_5 = 6$$
$$x_1 + 0x_2 + 0x_3 + 0x_4 + 0x_5 = 2$$

which corresponds to

$$\begin{aligned} Z &= 36 \\ x_3 &= 2 \\ x_2 &= 6 \\ x_1 &= 2 \\ x_4 &= 0 \\ x_5 &= 0 \end{aligned}$$

Giving us

$$\begin{bmatrix} 2 \\ 6 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

and the maximum of our objective function 36.

Chapter 65

Functional Analysis

65.1 Contractions and Banach

65.1.1 Local Convergence

Given $x_k \in X \subseteq \mathbb{R}^n$, we define *local convergence* for the iteration $x_{k+1} = \Phi(x_k)$ as

$$\lim_{k\to\infty} x_k = x \ \forall x_0 \in X.$$

65.1.2 Contraction

Φ is considered a *contraction* according to $\|\cdot\|$ on $X \subseteq \mathbb{R}^n$ if

$$\Phi : X \to X$$

and there is an $\alpha < 1$ independent of x, y such that

$$\|\Phi(x) - \Phi(y)\| \leq \alpha \|x - y\| \ \ \forall x, y \in X.$$

To check whether a function is a contraction on a certain set or interval, we need to check the following properties:

- Check that $f : X \to X$, i.e., that the function at its local extremes and endpoints remains in the interval. This is accomplished by the derivative test or, if the function is monotone, by checking the function at the endpoints of the interval.
- Estimate α using the definition of the contraction and approximations $\geq$. Alternatively, we can use the *Mean Value Theorem* with $\max_{\xi\in X} \|f'(x)\| \leq \alpha$ via a derivative test using ξ as our variable.

Example I:

Check if $f(x) = \sqrt{x}$ is a contraction on $[1, \infty)$ with the Euclidean metric.

Since $f(x)$ is monotone, we just need to check the function at the endpoints of the interval.

$$f(1) = 1 \quad \text{and} \quad f(\infty) = \infty.$$

Thus, our first property is satisfied. Now we check the second property:

$$\begin{aligned} \left\|\sqrt{x} - \sqrt{y}\right\| &= \frac{\|x - y\|}{\left\|\sqrt{x} + \sqrt{y}\right\|} \\ &\leq \frac{\|x - y\|}{\|1 + 1\|} = \frac{1}{2}\|x - y\| \end{aligned}$$

Hence $\alpha = \frac{1}{2}$.

Example II:

$\Phi(x) = x^3 - 3x^2 + 3x$ on $X = [\frac{1}{2}, \frac{3}{2}]$.

We take the first and second derivatives:

$$\Phi'(x) = 3x^2 - 6x + 3 \quad \text{and} \quad \Phi''(x) = 6x - 6.$$

Then we perform our tests:

$\Phi'(x) = 0$:

$$\Phi'(x) = 0 \iff 3x^2 - 6x + 3 = 0 \iff x = 1 > 0$$

$\Phi''(x) \neq 0$:

$$\Phi''(1) = 6 - 6 = 0$$

We do not have any local extremes, hence we check the function at the endpoints of the interval:

$$\Phi\left(\frac{1}{2}\right) = \frac{11}{8} \in X$$

$$\Phi\left(\frac{3}{2}\right) = \frac{11}{8} \in X$$

Hence $\Phi(X) \subseteq X$. Now we check the second property:

$$\alpha \leq \max_{x \in X} \|\Phi'(x)\| .$$

65.1.3 The Fix Point Theorem of Banach

Given a closed set $X \subseteq \mathbb{R}^n$ with the contraction $\Phi : X \to X$ and

$$\|\Phi(x) - \Phi(y)\| \leq \alpha \|x - y\| \ \ \forall x, y \in X$$

such that $\alpha < 1$ is independent. Then Φ has exactly one *fix-point* $x = \Phi(x)$ in X. Our *Picard iteration* converges for all $x_0 \in X$ towards x. Additionally, the following approximations hold:

$$\|x - x_k\| \leq \frac{\alpha^k}{1 - \alpha} \|x_1 - x_0\| \quad \text{a-priori}$$

$$\|x - x_k\| \leq \frac{\alpha}{1 - \alpha} \|x_k - x_{k-1}\| \quad \text{a-posteriori}$$

Proof:

Let X be a complete space and Φ be a contraction:

$$\|\Phi(x) - \Phi(y)\| < \alpha \|x - y\|$$

We will show that x_k is a Cauchy sequence:

$$\|x_{k+1} - x_k\| = \|\Phi(x_k) - \Phi(x_{k-1})\| \leq \alpha \|x_k - x_{k-1}\|$$

Because of the contraction property and our recursive definition, we can find the following:

$$\begin{aligned}\|f(x_2) - f(x_1)\| &< \alpha \|x_2 - x_1\| \\ \|x_3 - x_2\| &< \alpha \|x_2 - x_1\| \\ \|x_4 - x_3\| &< \alpha \|x_3 - x_2\|\end{aligned}$$

Using the previous inequality, we get

$$\|x_3 - x_2\| < \alpha \|x_2 - x_1\| \iff \alpha \|x_3 - x_2\| < \alpha^2 \|x_2 - x_1\|$$

Now, by applying it multiple times, we get

$$\begin{aligned}\|x_4 - x_3\| < \alpha \|x_3 - x_2\| &< \alpha^2 \|x_2 - x_1\| \\ &\dots \\ \|f(x_{k+1}) - f(x_k)\| &< \alpha^k \|x_2 - x_1\|\end{aligned}$$

We use the following approximations for $n > m$:

$$\begin{aligned}\|x_n - x_m\| = \|x_n + x_{n-1} - x_{n-1} + \cdots + x_{m-1} - x_{m-1} - x_m\| &\leq \|x_n - x_{n-1}\| + \cdots + \|x_{m+1} - x_m\| \\ &\leq \alpha^{n-1} \|x_2 - x_1\| + \cdots + \alpha^m \|x_2 - x_1\| \\ &= (\alpha^{n-1} + \cdots + \alpha^m) \|x_2 - x_1\| \\ &= \alpha^m \sum_{i=0}^{n-m-1} \alpha^i \|x_2 - x_1\| \\ &= \alpha^m \frac{1 - \alpha^{n-m}}{1 - \alpha} \|x_2 - x_1\| \\ &< \frac{\alpha^m}{1 - \alpha} \|x_2 - x_1\|\end{aligned}$$

Hence, our sequence is Cauchy, and because X is complete, we know that $\lim_{k\to\infty} x_k = p$ for some $p \in X$.

We take the limit:

$$\lim_{k\to\infty} \|f(x_{k+1}) - f(x_k)\| < \alpha^k \|x_2 - x_1\| = 0.$$

Because X is complete, we know that $\lim_{k\to\infty} f(x_k) = p$, hence $f(p) = p$.

$\mathfrak{QED}$

65.1.4 Linear Convergence

For the iteration $\Phi(x_k) = x_{k+1}$, we define *linear convergence* as

$$\|e_{k+1}\| \leq c \|e_k\| \ \forall k, \ 0 \leq c < 1.$$

The iteration is linearly convergent if such a constant c exists.

65.1.5 Convergence With Order m

Convergence With Order $m \geq 1$ is given for an iteration $\Phi(x_k) = x_{k+1}$ if $\Phi(x)$ is linearly convergent, $0 \leq c$ is independent of k, and

$$\lim_{k\to\infty} \|e_K\| = 0 \quad \text{and} \quad \|e_{k+1}\| \leq c\,\|e_K\|^m \ \ \forall k.$$

65.1.6 Convergence of the Differentiable Iteration

If given an $m \geq 2$, a fix-point x of $\Phi : \mathbb{R} \to \mathbb{R}$ such that Φ is m-times differentiable on x with the property

$$\Phi'(x) = \Phi''(x) = \cdots = \Phi^{(m-1)}(x) = 0,$$

then there exists a closed interval $X = [x - \delta, x + \delta], \delta > 0$ such that the iteration

$$x_{k+1} = \Phi(x_k)$$

converges for all $x_0 \in X$ with at least order m.

This means that if we want to check the convergence of an iteration, we can check the derivatives of the function at the fix-point, and if they are zero up to a certain order, then we can conclude that the iteration converges with at least that order.

Chapter 66

Combinatorics

Combinatorics is an area of mathematics concerned with counting, arrangements, and combinations of objects.

66.1 Lists

A *list* is an ordered sequence of elements.

$$(a_1, a_2, \ldots, a_n)$$

Unlike sets, where the order of elements does not matter, in lists the order of elements is important. Hence, two lists with the same elements but in different orders are considered different.

66.1.1 Length

A list can have a *length*, which is the number of elements it contains.

$$\text{length}((a_1, a_2, \ldots, a_n)) = n.$$

66.1.2 Strings

A *string* is a list where the elements are not separated by commas or parenthesis.

$$a_1 a_2 \ldots a_n$$

66.2 The Three Principles of Combinatorics

The three principles of combinatorics are *Addition*, *Multiplication* and *subtraction*.

66.2.1 Principle of Multiplication

Suppose in making a list of length n there a_1 possible choices for the first entry, a_2 for the second choicesa, and so on. Then, the total number of possible lists is the product of the number of choices for each entry:

$$a_1 \cdot a_2 \cdot \cdots \cdot a_n.$$

If repetition is allowed, then each entry can be chosen independently from the same set of possibilities, and the total number of possible lists is still given by the product formula:

$$a^n$$

66.2.2 Principle of Addition

Suppose a finite set X can be decomposed as a union $X = X_1 \cup X_2 \cup \cdots \cup X_n$ where $X_i \cap X_j = \emptyset$ whenever $i \neq j$. Then, the total number of elements in X is the sum of the number of elements in each X_i:

$$|X| = |X_1| + |X_2| + \cdots + |X_n|.$$

Example:

An example of this principle is given the following task: given the number from 0 to 9, how many strings including the 6 exactly one time are possible if this words are of length 3.

We have the following cases: $X_1 = 6, _, _$, $X_2 = _, 6, _$, $X_3 = _, _, 6$. Therefore,

$$|X_1| + |X_2| + |X_3| = 9^2 + 9^2 + 9^2 = 243$$

Note that if the 6 could occur more than once, we would have to consider additional cases where the 6 appears in multiple positions, which would mean that we would use the inclusion-exclusion principle.

66.2.3 Principle of Subtraction

If X is a subset of a finite set U, then $|\bar{X}| = |U| - |X|$. In other words,

$$X \subseteq U \implies |U \setminus X| = |U| - |X|.$$

This principle is used when we have overlapping sets, and we want to count the number of elements in their union accurately by correcting for the overcounting of elements in the intersections.

Example:

Let U be the set of all 3-digit strings formed from the digits 0 to 9, and let X be the set of strings that contain the digit 6 exactly once. Then, by the principle of subtraction, we have

$$|U| = 10^3, \quad |X| = 243$$

Therefore,

$$|\bar{X}| = |U| - |X| = 10^3 - 243 = 1000 - 243 = 757$$

66.3 Inclusion-Exclusion Principle

Using the principles above, we can know state our *principle of inclusion-exclusion.*

$$\left|\bigcup_{i=1}^{n} X_i\right| = \sum_{i=1}^{n} |X_i| - \sum_{1 \leq i < j \leq n} |X_i \cap X_j| + \sum_{1 \leq i < j < k \leq n} |X_i \cap X_j \cap X_k| - \cdots + (-1)^{n+1} |X_1 \cap X_2 \cap \cdots \cap X_n|.$$

66.4 The Basic Counting Principle

For n outcomes of event A and m of B, there are $n \cdot m$ outcomes for event A and B.

In combinatorics we can model a lot of problems as having n elements from which we can take one at a time a certain number of times, and then we take into consideration if we allow repetition or the order matters or not.

Example:

A restaurant has 8 waiters and 5 types of steaks. How many ways of ordering a steak are there.

$$8 \cdot 5 = 40$$

66.5 Permutation Without Repetition

The *permutation* counts the number of ways of ordering n distinct elements. In other words, the number of tuples of n distinct elements. Of course, the order matters.

$$n!$$

Example:

5 men and 4 women are ranked by heigjht. If the men are ordered among themselves and women as well, how many rankins are possible?

$$5! \cdot 4! = 120 \cdot 24 = 2880$$

For reach ranking, we consider the men and women separately, and then multiply the number of ways to order each group.

66.5.1 Permutation With Repetition

The *permutation with repetition* counts the number of ways of ordering n non-distinct elements. Here $m_1, \ldots, m_n$ is the number of times a specific item is repeated in the original set. By dividing by the respective amount of the repeated non-distinct element, we eliminate the permutations where the identical elements swap their place.

$$\frac{n!}{m_1! \cdots m_n!}$$

This formula comes from the *multinomial coefficient* given by

$$\binom{n}{n_1 n_2 \ldots n_m} = \frac{n!}{n_1! \cdots n_m!},$$

which comes from

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n-n_1-\cdots n_{m-1}}{n_m}.$$

This is also a kind of *partition problem*, because it can be formulated as if you have n different elements, and you want to divide them into m bins with capacities $m_1, m_2, \cdots, m_m$, how many possible ways are there for these partitions.

In this case, the order matters, but only for the different elements.

Example:

66.6 Multisets

A *multiset* is similar to a set, but we allow multiple occurrences of an elements, but the order does not matter unlike lists. We will use this multiset for combinations with repetition.

As an example:

$$[1, 1, 2, 2, 2, 3]$$

Example:

Consider the multiset $\{a, a, b, b, b, c\}$. The number of distinct permutations of this multiset is given by

$$\frac{6!}{2! \cdot 3! \cdot 1!} = \frac{720}{12} = 60.$$

66.7 Combination With Repetition

We can think of the *combination* as having n types of objects, and we want to take with repetition until we fill our k slots.

$$\binom{n+k-1}{n-1}$$

For understanding, we have n types of objects, and because we want a combination of those objects, the order does not matter. Thus, we can put them together as contiguous groups. We have k slots and n types, thus we can separate our groups with $n-1$ separators $|$. The question becomes how many ways are there to put $n-1$ separators into $n+k-1$ slots. Note that we have $n+k-1$ positions if we count the actual slots and the separator slots in between.

$$*|***|**, \quad n = 6, \quad k = 2, \quad \text{positions in between } = 7$$

This called the *stars and bars* method. For this method we make the assumption that identical are next to each other, forming contiguous groups.

Example

We have 10 balls and 3 boxes, but we do not know how many ways are there to distribute the balls into the boxes. We use the bars and stars method to represent the distribution of balls into boxes. Here, the balls are represented by stars and the separations between boxes are represented by bars.

We have 10 balls and 3 boxes, so we need 2 bars to separate the boxes. The total number of positions is $10 + 2 = 12$, and we need to choose 2 positions for the bars. Thus, the number of ways to distribute the balls is

$$\binom{10+2}{2} = \binom{12}{2} = 66.$$

66.8 Variation

The *variation* counts the ways of putting n objects in k slots with repetition. Here, the order does not matter.

$$n^k$$

66.9 Variation Without Repetition

This kind of *variation* counts the ways of putting n objects in k slots without repetition. Number of tuples with k elements which can be selected from n elements, without repetition. The order matters.

$$\frac{n!}{(n-k)!}$$

66.10 Combination Without Repetition

The *combination* can be understood as choosing k objects of n elements. Or putting k elements from n into k slots where the order does not matter.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Note that like in the variation we use $\frac{n!}{(n-k)!}$ to limit our slots to k. Then for the k objects we choose we can arrange them in $k!$ ways but because the order does not matter we divide by $k!$ to get rid of the repeated combinations.

66.11 Multinomial Coefficients

When using the binomial coefficient repeatedly, one after the other, we can abbreviate these consecutive uses as

$$\binom{n}{k_1, \ldots, k_n}$$

This is like from n choose k_1, then from $n - k_1$ choose k_2, and so on.

66.11.1 Multinomial Theorem

Given $x_1, \ldots, x_r \in \mathbb{R}$ and $\mathbb{N}$, then

$$(x_1 + \cdots + x_r)^n = \sum_{k_1+\cdots+k_r=n} \binom{n}{k_1, \ldots, k_r} x_1^{k_1} \cdots x_r^{k_r}$$

where $k_1, \ldots, k_p$ are less or equal to n.

This can be better understood with an example.

Example:

Let us use the formula for $(x + y + z)^3$.

$$\sum_{i,j,k,i+j+k=4}^{4} \binom{4}{i, j, k} x^i y^j z^k$$

Proof:

Let $r = 1$ be our *base case*, then $x_1 = x_1$.

We assume as our *inductive hypothesis* that

$$(x_1 + \cdots + x_r)^n = \sum_{k_1+\cdots+k_r=n} \binom{n}{k_1, \ldots, k_r} x_1^{k_1} \cdots x_r^{k_r}$$

Now for our *inductive step* we take $n+1$

$$\begin{aligned}(x_1+\cdots+x_n+x_{r+1})^{n+1} &= ((x_1+\cdots+x_n)+x_{r+1})^{n+1}\\ &= \sum_{k=0}^{n}\binom{n}{k}(x_1+\cdots+x_n+x_n)^k x_{r+1}^{n-k}\\ &= \sum_{k=0}^{n}\binom{n}{k}\left(\sum_{k_1+\cdots+k_r=n}\binom{n}{k_1,\ldots,k_r}x_1^{k_1}\cdots x_r^{k_r}\right)x_{r+1}^{n-k}\\ &= \sum_{k_1+\cdots+k_{r+1}=n}\binom{n}{k_1,\ldots,k_r+k_{r+1}}x_1^{k_1}\cdots x_r^{k_r}x_{r+1}^{k_{r+1}}\end{aligned}$$

Note that we used both the definition of the multinomial coefficient and the binomial theorem to get this result.

$\mathfrak{QED}$

66.12 The Division Principle

Suppose n objets are placed into k boxes. Then at least one box contains $\lceil \frac{n}{k} \rceil$ or more objects and at least one box contains $\lfloor \frac{n}{k} \rfloor$ or fewer objects. This is called the *division principle*.

66.13 The Pigeonhole Principle

The *pigeonhole principle* states that if n objects are placed into k boxes and $n > k$, then at least one box contains more than one objects. If $n < k$, then at least one box is empty. This is a direct consecuence from the division principle.

66.14 Example Problems

Defected Machines

A new delivery with 10 machines from which 2 are not working properly has to be tested. For that purpose a man in charge selects 5 of the machines randomly, and then he proceeds to make a decision based on the number of faulty machines found in the delivery. If among the 5 samples only 1 machine does not work it will be accepted otherwise it will be rejected. What is the probability of a delivery being accepted?

Answer:

Total cases: $\binom{10}{5}$

$\mathbb{P}(\text{accepted}) = \frac{\binom{8}{5}}{\binom{10}{5}} + \frac{\binom{2}{1}\binom{8}{4}}{\binom{10}{5}}$

$\binom{8}{5}$ represents no faulty machine in the choice; $\binom{2}{1}\binom{8}{4}$ represents that only one machine does not work.

66.15 Disarray

A *disarray* is the number of permutations in which no object ends in the same initial spot. It is also known as the *sub-factorial*.

$$!n = n!\sum_{k=0}^{n}\left(\frac{(-1)^k}{k!}\right)$$

66.16 Bell Numbers

The *Bell numbers* count the number of ways of grouping n objects in an arbitrary number of slots.

$$B(N) = \sum_{k=0}^{N-1} \binom{N}{K} B(K),$$

with $B(1) = 1$ and $B(2) = 2$.

66.17 Ramanujan Numbers

The *Ramanujan numbers* are the same as the Bell numbers, but all objects are equal. As an example, the number of ways to decompose a number.

$$R(N) \approx \frac{1}{4N\sqrt{3}} e^{2\pi\sqrt{\frac{N}{6}}}$$

66.18 Stirling Numbers I

The ways of permuting N items with K exchanges are given by the *Stirling Numbers I*.

$$\begin{bmatrix} N+1 \\ K \end{bmatrix} = N \begin{bmatrix} N \\ K \end{bmatrix} + \begin{bmatrix} N \\ K-1 \end{bmatrix}$$

66.19 Stirling Numbers II

The ways of grouping n items in k slots are given by the *Stirling Numbers II*.

$$\begin{Bmatrix} N \\ K \end{Bmatrix} = \frac{1}{K!} \sum_{j=0}^{K} (-1)^{K-j} \binom{K}{j} j^N$$

66.20 Lah Numbers

The *Lah numbers* represent the ways of building K lists with a set of N elements.

$$\begin{bmatrix} N \\ K \end{bmatrix} = \binom{N-1}{K-1} \frac{N!}{K!} = \sum_{j=0}^{K} \begin{bmatrix} N \\ K \end{bmatrix} \begin{Bmatrix} N \\ K \end{Bmatrix}$$

66.21 Euler's Numbers

The *Euler's numbers* are the number of permutations of a set of N elements of different size in which K elements are bigger than their previous element.

$$\left\langle \begin{matrix} N \\ K \end{matrix} \right\rangle = \sum_{j=0}^{K} (-1)^j \binom{N+1}{j} (K+1-j)^N.$$

66.22 Catalan Numbers

The *Catalan numbers* have different applications like the number of ways of the triangulations of a polygon or the number of trees with N leaves.

$$\mathfrak{C}(N) = \binom{2N}{N} - \binom{2N}{N+1}$$

or

$$\mathfrak{C}(N) = \sum_{k=1}^{n} (k-1)\mathfrak{C}(N-k).$$

Chapter 67

Probability

67.1 Probability Experiments

A *probability experiment* is considered ideal if and only if it satisfies the following properties:

- We know the possible outcomes.
- It can be repeated multiple times under the same conditions.
- It will be executed under strict conditions.

67.1.1 The Birthday Paradox

This is a small question that says: what is the probability of at least two people having their birthday on the same day in a group of x persons?

$$P(x) = 1 - \prod_{n=0}^{x} \frac{365 - n}{365}$$

This formula gives us the probability of the total minus all persons having it on different days, which is what we are looking for.

67.2 The Sample Space

The *sample space* is the set of all possible outcomes of an experiment. It is denoted as Ω, and its cardinality is the number of possible outcomes.

For experiments with Ω as the result set of one repetition, we define

$$\Omega = \Omega \times \cdots \times \Omega$$

as the *sample space* of the experiment.

Finally, each subset of Ω is called an *outcome*. Each subset with only one element in it is called an *elemental outcome*. The set of all possible outcomes is $\mathcal{P}(\Omega)$.

67.3 Laplace Distribution and Experiment

A *Laplace distribution* is a probability experiment with finite outcomes where each elemental outcome has the same probability of occurring.

$$P(\omega) = \frac{\omega}{|\Omega|}$$

Then an experiment with this property is considered a *Laplace experiment.*

$$P(E) = \frac{E}{|\Omega|}$$

67.4 Sigma Algebra

Given a non-empty set Ω and $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ a set of subsets of Ω. $\mathcal{A}$ is a *σ-Algebra* if and only if

I. $\Omega \in \mathcal{A}$

II. $A \in \mathcal{A} \implies A^c \in \mathcal{A}$

III. $A_1, A_2, A_3, \cdots \in \mathcal{A} \implies \bigcup_{n\in\mathbb{N}} A_n \in \mathcal{A}$

67.4.1 The Power-Set Is a Sigma Algebra

This proof is pretty straightforward.

For the first axiom, we know that $\Omega \in \mathcal{P}$.

For the second axiom, if A is a subset of the power-set, then $\Omega \setminus A$ is also in the power-set.

And lastly, for a sequence of sets in the power-set, their union is also in the power-set, because the power-set contains all subsets of Ω.

$\mathfrak{QED}$

67.4.2 Properties of Sigma Algebras

If $\mathcal{A}$ is a sigma algebra then

- $\emptyset \in \mathcal{A}$
- $A_1, A_2, \cdots \in \mathcal{A} \implies \bigcap_{n\in\mathbb{N}} A_n \in \mathcal{A}$
- $A, B \in \mathcal{A} \implies A \setminus B \in \mathcal{A}$

Proof:

For the first, because $\mathcal{A}$ is a subset of the power-set of Ω and the empty set is always a subset of a collection, this is trivial.

Given a number of sets in $\mathcal{A}$, using the laws of De Morgan we get

$$\bigcap_{n\in\mathbb{N}} A_n = \bigcup A_n^c$$

and we know that the complements are part of $mathcal A$ by definition.

For the last part, $A \setminus B = A \cap B^c$, because $B^c \in \mathcal{A}$ we can apply De Morgan $(A^c \cup (B^c)^c)^c = (A^c \cup B)^c$. Now, because this is the union of two sets in the sigma algebra, we know by definition that their union is also in the set.

$\mathfrak{QED}$

67.5 Probability Measurement

Given a non-empty set Ω and $\mathcal{A} \subseteq \mathcal{P}$ a sigma algebra. We define the function $P : \mathcal{A} \to \mathbb{R}$ as a *probability measurement* if

I. $\forall A \in \mathcal{A} : P(A) \geq 0$.

II. $P(\Omega) = 1$.

III. $A_n \in \mathcal{A}, n \in \mathbb{N}$ disjunctive in pairs $\implies P\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} P(A_n)$.

Also, the triplet $\Omega, \mathcal{A}, P$ is called a *probability space*.

67.5.1 Probability Space of the Power-Set

We want to prove that the probability space with the power-set of omega and $P(A) = \frac{|A|}{|\Omega|}$ is in fact a probability space.

- $P(A) = \frac{|A|}{|\Omega|}$ is always greater or equal to zero.
- $P(\Omega) = \frac{|\Omega|}{|\Omega|} = 1$.
- If all of our subsets in the power-set are pairwise disjunctive, this means that their intersection is the empty-set. In other words, they are therefore independent of each other, which means that their probabilities are also independent.

67.5.2 Discrete Probability Spaces

A probability space $\Omega, \mathcal{P}(\Omega), P$ is considered *discrete* if Ω is finite or infinite countable. In that case, $\Omega = \{\omega_i : i \in I\}$ for $I = \{1, \dots, n\}$.

For $A \in \mathcal{A}$

$$P(A) = P\left(\bigcup_{i \in I : \omega_i \in A} \{\omega_i\}\right) = \sum_{i \in I : \omega_i \in A} P(\{\omega_i\}) = \sum_{i \in I : \omega_i \in A} p(\omega_i)$$

The function $P(\{\omega_i\}) = p(\omega_i)$ is called a *probability function*.

To test if a specific function or distribution is actually a probability one, we have to check:

- $\forall p(\omega) \geq 0 \mid \omega \in \Omega$.
- $\sum_{\omega \in \Omega} p(\omega) = 1$.

67.5.3 Properties of Probability-Measurements

Given $(\Omega, \mathcal{A}, P)$ and $A, B \in \mathcal{A}$, the following applies:

- $P(\emptyset) = 0$.
- $A_1, \dots, A_n \in \mathcal{A}$ pairwise disjoint: $P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$.
- $P(B \setminus A) = P(B) - P(A \cap B)$.
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- $P(A^c) = 1 - P(A)$.
- If $A \subset B$, then $P(A) \leq P(B)$.
- $0 \leq P(A) \leq 1$.
- $A_1, \dots, A_n \in \mathcal{A}$: $P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$.
- $A_1, A_2, \dots \in \mathcal{A}$: $P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i)$.

Proof:

- Define $A_n = \emptyset$ and by the properties of the probability spaces we know that $A_n \in \mathcal{A}, n \in \mathbb{N}$ disjunctive in pairs $\implies P\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} P(A_n)$.

  Therefore, $P(\emptyset) = \sum_{i=1}^{n} P(A_i)$ and the only real number with the property $x + x = x$ is 0.
- $B = (A \cap B) \cup B \setminus A$ and by the definition of the addition of probabilities

$$P(B) = P(B \setminus A) - P(A \cap B).$$

- Using the addition property and by writing the union as $(A \setminus B) \cup (B \setminus A) \cup (A \cap B)$ we get that

$$P(A \cup B) = P(B) - P(A \cap B) + P(A) - P(A \cap B) + P(A \cup B)$$

  which results in $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- $\Omega = A^c \cup A$ and $1 = P(\Omega)$, thus by the addition $P(A) + P(A^c) = 1$.
- Given that if $A = B$, then the equality holds. The case where A is a subset of B comes from the fact that if it is a strict subset of the other, then its cardinality is smaller.
- By the definition that $P(\Omega) = 1$ and the property $P(\emptyset) = 0$ we are given an upper and lower bound for our probability distribution. Therefore, every subset of Ω will have a probability less or equal to it. For the lower bound, the logic is that it has no other subset with fewer elements than it, but it is part of sets with more elements.
- Let us define $B_n = A_n \setminus \bigcap i = 1_{n-1} A_i$ which are pairwise disjoint.

$$P\left(\bigcup A_n\right) = P\left(\bigcup B_n\right) = \sum P\left(B_i\right) \leq \sum P\left(A_i\right).$$

  Due to the fact that the B_i are pairwise disjoint.
- This is practically the same but for an infinite case.

$\mathfrak{QED}$

67.6 Inclusion-Exclusion Formulas for Sets

Given M_i, the cardinality of their union is given by

$$\left|\bigcup_{i=1}^{n} M_i\right| = \sum_{I \subseteq \{1,\ldots,n\}, I \neq \emptyset} (-1)^{|I|-1} \left|\bigcap_{i \in I} M_i\right|.$$

67.6.1 Inclusion-Exclusion Formula for Two Non-Disjoint Sets

Let $|A \cap B| = k$, then:

$$|A \cup B| = |A| + |B| - |A \cap B| = n + m - k \quad \text{(since we do not count the intersection twice)}$$

67.6.2 Inclusion-Exclusion Formula for Three Non-Disjoint Sets

$$\begin{aligned}
|A \cup B \cup C| &= |(A \cup B) \cup C| \\
&= |A \cup B| + |C| - |(A \cup B) \cap C| \\
&= |A| + |B| - |A \cap B| + |C| - |(A \cap C) \cup (B \cap C)| \\
&= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|
\end{aligned}$$

67.7 Stochastic Independence

Two results $A, B \in \mathcal{A}$ are considered *independent* if

$$P(A \mid B) = P(A)$$
$$P(B \mid A) = P(B),$$

then

$$P(A \cap B) = P(A) \cdot P(B).$$

Example:

When throwing a die two times, we look at the probability of throwing an even number during the first throw and, when throwing the second die, the sum giving us an even number.

$$\Omega = \{1, \ldots, 6\}^2 \quad A = \{2, 4, 6\} \times \{1, \ldots, 6\} \quad B = \{2, 4, 6\} \cup \{1, 3, 5\}^2$$

Therefore,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{18}{36} = \frac{1}{2}$$

$$P(B) = \frac{|B|}{|\Omega|} = \frac{18}{36} = \frac{1}{2}$$

And

$$P(A \cap B) = \frac{|\{2, 4, 6\}^2|}{|\Omega|} = \frac{1}{4}$$

This is just $P(A) \cdot P(B)$. Therefore, they are independent.

$\mathfrak{QED}$

67.7.1 Independence for Multiple Results

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2) \cdots P(A_n)$$

For all pairs A_i, A_j where $i \leq j$.

67.7.2 Independence in Laplace Distribution

Given a Laplace distribution $P = \frac{|A|}{|\Omega|}$ with two independent results A and B.

$$\frac{|A \cap B|}{|A|} = \frac{B}{|\Omega|}.$$

This means that B is as probable in A as in the greater picture.

67.8 Complement Property

Given the probability space $(\Omega, \mathcal{A}, P)$ and $A, B \in \mathcal{A}$ as two independent results. Then A^c and B are also independent.

Proof:

The sets $A \cap B$ and $A^c \cap B$ are disjoint and their union is

$$B = (A \cap B) \cup (A^c \cap B).$$

If we calculate the probability

$$P(B) = P(A \cap B) + P(A^c \cap B).$$

This gives us

$$P(A^c \cap B) = P(A) - P(A \cap B).$$

Using the independence of A and B

$$P(A^c \cap B) = P(B) - P(A)P(B) = P(B)(1 - P(A))$$

$$P(B) = P(B)(1 - P(A)) = P(B) \cdot P(A^c)$$

$\mathfrak{QED}$

67.9 Independence of Blocks

Given $A_1, \ldots, A_n$ independent events, divided into two blocks $A_1, \ldots, A_k$ and $A_{k+1}, \ldots, A_n$. Let B and C be events resulting from the union, intersection, or complement of the two blocks. Then these results are independent.

67.10 The Product Space

Given two or more probability spaces with their respective tuples of two independent experiments. We define their combined probability space as the *product space* where

$$\Omega := \Omega_1 \times \cdots \times \Omega_n$$

and

$$p(\omega_1, \ldots, \omega_n) = p(\omega_1) \cdot \ldots \cdot \ldots \cdot p(\omega_n)$$

An example is throwing two dice at the same time.

67.10.1 The Product Space of Discrete Probability Spaces

Given the product space $(\Omega, \mathcal{P}(\Omega), P)$ of the spaces $(\Omega_i, \mathcal{P}(\Omega_i), P_i)$ and $A_i \in \mathcal{P}(\Omega_i)$. We define

$$\begin{aligned} A_i^{'} &:= \{\omega \in \Omega : \omega_i \in A_i\} \\ &= \{(w_1, \ldots, w_n) : \omega_j \in \Omega_j, \text{ for } j \neq i \text{ and } \omega_i \in A_i\} \\ &= \Omega_1 \times \cdots \times \Omega_{i-1} \times A_i \times \Omega_{i+1} \times \cdots \times \Omega_n \end{aligned}$$

For $i = 1, \ldots, n$. Then it holds that $P(A_i^{'}) = P_i(A_i)$ and the events $A_i^{'}$ are independent.

67.11 Bernoulli-Experiments

An experiment where $\Omega = \{0, 1\}, P(\{1\}) = p$ and $P(\{0\}) = 1 - p$ for $p \in (0, 1)$ is called a *Bernoulli-experiment.*

67.11.1 Bernoulli Schema

If we perform a Bernoulli-experiment n times with a success probability of p, where $P(A_i) = p$, we define

$$C_k = \bigcup_{I \in \{1,\ldots,n\} \wedge |I|=k} \left(\bigcap_{i \in I} A_i \cap \bigcap_{i \in I} A_i^c \right).$$

For the probability we get

$$P(C_k) = \sum_{I \in \{1,\ldots,n\} \wedge |I|=k} P\left(\bigcap_{i \in I} A_i \cap \bigcap_{i \notin I} A_i^c \right) = \sum_{I \in \{1,\ldots,n\} \wedge |I|=k} \prod_{i \in I} P(A_i) = \prod_{i \notin I} = \binom{n}{k} p^k (1-p)^{n-k}.$$

67.12 Conditional Probability

Given the events A and B where $P(A) > 0$. The *conditional probability* of B under the condition A is defined as

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

Example:

In words, what is the probability of B if we limit ourselves only to the case where A happens? Also, what is the ratio between $A \cap B$ and A? From this definition, we can also derive these two statements, which are also equivalent.

$$P(B \mid A) = P(B) \iff A \text{ and } B \text{ are independent}$$

Proof:

$$\begin{aligned} P(B \cap A) = P(A)P(B) &= P(A \cap B)P(A) \\ &= P(B) \end{aligned}$$

$\mathfrak{QED}$

67.12.1 Conditional Probability of a Subset

Given the events A and B where $P(A) > 0$. If $A \subset B$ then $P(B \mid A) = 1$.

Proof:

$A \subset B \implies A \cap B = A$, therefore

$$P(B \mid A) = \frac{P(A)}{P(A)} = 1.$$

$\mathfrak{QED}$

67.12.2 Probability Measurement of Conditional Probability

Given the probability space $(\Omega, \mathcal{A}, P)$ and $A \in \mathcal{A}$ where $P(A) > 0$. The map $B \to P(B \mid A)$ is a probability measurement with the following properties.

- $\forall \in \mathcal{A} : P(B \mid A) \geq 0$.
- $P(\Omega \mid A) = 1$.

- $B_n \in A, n \in \mathbb{N}$ pairwise disjoint $\implies P\left(\bigcup_{n\in\mathbb{N}} B_n \mid A\right) = \sum_{n\in\mathbb{N}} P(B_n \mid A)$.

We also get

$$P(\emptyset \mid A) = 0$$

and

$$P(B^c \mid A) = 1 - P(B \mid A).$$

Proof:

- Because our sets do not have negative cardinalities and the lowest we can reach is the probability of the empty-set, we cannot get below 0.
- Similar to another proof, we get $\frac{P(A)}{P(A)}$.
- By the definition of probability space.

$\mathfrak{QED}$

67.13 Multiplication Theorem

Given $A_1, \ldots, A_n$ events with $P(A_1 \cap \cdots \cap A_{n-1}) \neq 0$. Then

$$P(A_1 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot \cdots \cdot (A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

67.14 Theorem of the Total Probability

Given our standard probability space, the disjoint partitions of Ω, $B_1, \ldots, B_n$ where $B_1 \cup \cdots \cup B_n = \Omega$, $B_i \cap B_j = \emptyset$ for $i \neq j$ and $P(B_i) > 0$, then

$$P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i) \, \forall A \in \mathcal{A}.$$

Example:

You are given three bags

- Bag A has 2 red balls and 3 green balls.
- Bag B has 3 red balls and 2 green balls.
- Bag C has 1 red ball and 4 green balls.

What is the probability of a ball being red?

The answer is given by

$$P(R) = P(R \cap A) + P(R \cap B) + P(R \cap C).$$

Which gives us

$$P(R) = P(R \mid A)P(A) + P(R \mid B)P(B) + P(R \mid C)P(C).$$

Finally, we plug in the individual numbers

$$P(R) = \frac{1}{3}\left(\frac{2}{5} + \frac{3}{2} + \frac{1}{4}\right) = \frac{6}{15}.$$

67.15 Multistage Random Experiments

Let $\Omega_1, \ldots, \Omega_n$ be discrete sample spaces of n sub-experiments, and let $\Omega = \Omega_1 \times \cdots \times \Omega_n$. Furthermore, let an initial distribution be given by the probability function p_1 with

$$\sum_{\omega_1 \in \Omega_1} p_1(\omega_1) = 1$$

and transition probabilities p_j for $j = 2, \ldots, n$ satisfying

$$\sum_{\omega_j \in \Omega_j} p_j(\omega_j \mid \omega_1, \ldots, \omega_{j-1}) = 1,$$

for $(\omega_1, \ldots, \omega_{j-1}) \in \Omega_1 \times \cdots \times \Omega_{j-1}$.

This is used in the case when we model our experiment as a tree of probabilities.

For the multistage random experiment, the following two *path rules* apply:

1. Let $(\omega_1, \ldots, \omega_n) \in \Omega$, then we have:

$$p(\omega_1, \ldots, \omega_n) = p_1(\omega_1) \cdot p_2(\omega_2 \mid \omega_1) \cdot \ldots \cdot p_n(\omega_n \mid \omega_1, \ldots, \omega_{n-1}).$$

2. For $A \subset \Omega$, it holds that
$$P(A) = \sum_{(\omega_1, \ldots, \omega_n) \in A} p(\omega_1, \ldots, \omega_n).$$

Note that for independent sub-experiments, we obtain the product space with

$$p(\omega_1, \ldots, \omega_n) = p_1(\omega_1) \cdot p_2(\omega_2) \cdot \ldots \cdot p_n(\omega_n).$$

67.16 Bayes Theorem

Let $(\Omega, \mathcal{A}, P)$ be a probability space. Furthermore, let $B_1, \ldots, B_n$ be a disjoint partition of the sample space Ω with $P(B_i) > 0$ for all $i = 1, \ldots, n$. Then, for every event $A \in \mathcal{A}$, the following holds:

$$P(B_i \mid A) = \frac{P(A \mid B_i)\, P(B_i)}{P(A)} = \frac{P(A \mid B_i)\, P(B_i)}{\sum_{k=1}^{n} P(A \mid B_k) P(B_k)}.$$

In words and by changing A with E of evidence and B with H of hypothesis, we want to know how much the actual evidence changes our initial belief. The process can be described as: you have a hypothesis, you have observed some evidence and finally, you want to know the probability that your hypothesis is true given the observed evidence.

$$P(H \mid E) = \frac{P(E \mid H)P(H)}{P(H)P(E \mid H) + P(\neg H)P(E \mid \neg H)}$$

- $P(H)$ is called the *prior*. Our initial hypothesis, without seeing the evidence.
- $P(E \mid H)$ is called the *likelihood*. This is the probability that we will see our hypothesis reflected in the evidence, given that our hypothesis is true.
- $P(E \mid \neg H)$ represents the probability that our hypothesis will be reflected in the sample, given that our hypothesis is false.

With this knowledge, breaking down the formula is very intuitive.

- $P(H \mid E)$ is the *posterior*. It describes the following question: how often is H true among the cases where H is true?

- $P(H)P(E \mid H)$ is the proportion of our hypothesis in the sample being true times the actual proportion of the fitting samples inside our hypothesis.
- $P(H)P(E \mid H) + P(\neg H)P(E \mid \neg H)$ consists of our expression above plus the proportion of the counter hypothesis times the proportion of the representatives of our hypothesis which will show up in the evidence in the counter group.

Example:

A disease affects 1 percent of the population, and you were tested positive with a test which has a 99.9 percent accuracy. What is the probability of you truly being infected?

$$P(I \mid +) = \frac{P(I)P(+ \mid I)}{P(I)P(+ \mid I) + P(\neg I)P(+|\neg I)}$$

Our numbers are:

- $P(I) = 0.001$
- $P(+ \mid I) = 0.99$
- $P(\neg I) = 0.999$
- $P(+ \mid \neg I) = 0.1$

Thus, our formula gives us a 9 percent chance of actually being infected.

67.17 Weak Law of Large Numbers

Given a sequence of independent random variables $X_1, X_2, \ldots, X_n$ with the same mean μ and the same σ^2. The sample mean $Z_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ will converge to μ as $n \to \infty$.

Formally $P(|Z_n - \mu| > \epsilon)_{n\to\infty} \to 0$.

Proof:

Since our random variables are independent

$$\mathrm{E}[Z_n] = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[X_i] = \mu$$

$$\mathrm{var}[Z_n] = \frac{1}{n}\sum_{i=1}^{n} \mathrm{var}[X_i] = \frac{\sigma^2}{n}$$

as $n \to \infty, \mathrm{var}(Z_n) \to 0$.

Formally,

$$\begin{aligned} P(|Z_n - \mu| \geq \varepsilon) &\leq \frac{\mathrm{var}(Z_n)}{\varepsilon^2} \\ &= \frac{\sigma^2}{n\varepsilon^2} \to 0 \text{ as n goes to infinity} \end{aligned}$$

67.18 Strong Law of Large Numbers

Given $X_1, X_2, \ldots$ independent random variables with the same distribution and $|\mathrm{E}[|X_1|]| < \infty$ then

$$Z_n \to \mathrm{E}[X_1]$$

where $Z_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

67.19 Central Limit Theorem

Given a sequence of independent random variables $X_1, X_2, \ldots, X_n$ with the same mean μ and the same σ^2, the sample mean

$$Z_n = \frac{1}{n}\sum_{i=1}^{n} X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

or equivalently

$$\lim_{n\to\infty} P\left(\sqrt{n}\frac{Z_n - \mu}{\sigma} \leq x\right) = \Phi(x).$$

Example:

Approximate the probability of a tricked coin landing heads with $n = 10000$ throws and the number of heads between $4900 < X < 5100$.

Our variable is binomial distributed

$$\mu = \mathrm{E}[X] = np = 5000$$

$$\sigma = 50$$

Then for

$$P\left(Z_1 \leq \sqrt{n}\frac{Z_n - \mu}{\sigma} \leq Z_2\right)$$

with $Z_1 = \frac{4900-5000}{50}, Z_2 = \frac{5100-5000}{50}$ We get

$$\Phi(Z_2) - \Phi(Z_1) = 0.9545$$

67.19.1 Approximating the Binomial Distribution With the Normal Distribution

Given a binomial distribution with parameters n and p. For large n the distribution can be approximated by a normal distribution with parameters $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.

Example:

Given $n = 75$ and $p = 0.6$, approximate the probability $P(X \geq 52)$

$\mu = np = 45$, $\sigma^2 = np(1-p) = 18$

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 45}{\sqrt{18}}$$

Therefore,

$$P\left(Z \geq \frac{52 - \mu}{\sqrt{18}}\right)$$

$$= 1 - \Phi\left(\frac{52 - 45}{\sqrt{18}}\right) = 0.0582$$

For $P(50 \leq X \leq 52)$ we have

$$P\left(\frac{50-\mu}{\sqrt{18}} \leq Z \leq \frac{52-\mu}{\sqrt{18}}\right) = \Phi\left(\frac{52-45}{\sqrt{18}}\right) - \Phi\left(\frac{50-45}{\sqrt{18}}\right).$$

67.20 Theorem of Moivre-Laplace

Given $X_1, X_2, \ldots, X_n$ Bernoulli distributed random variables and $S_n = \sum_{i=1}^{n} X_i$ then the number of successes is given by

$$\lim_{n\to\infty} P\left(\frac{S_n - np}{\sqrt{np(1-p)} \leq x}\right) = \Phi(X).$$

For the case that we have a region with two bounds $a < S_n < b$, we get

$$P\left(\frac{S_n - np}{\sqrt{np(1-p)} \leq Z_2}\right) - P\left(\frac{S_n - np}{\sqrt{np(1-p)} \leq Z_1}\right).$$

If a normal approximation is used for a discrete distribution, we have to apply the continuity correction.

$$P(a < S_n < b) \approx \Phi\left(\frac{b + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{a - 0.5 - np}{\sqrt{np(1-p)}}\right).$$

67.20.1 Continuity Correction

An approximation is useful if $np(1-p) \geq 9$. This is a heuristic method.

Example:

Approximate the probability of getting between $4900 < X < 5100$ heads when throwing a tricked coin $n = 10000$ times with $p = 0.5$.

$$P\left(\frac{S_n - np}{\sqrt{np(1-p)} \leq Z_2}\right) - P\left(\frac{S_n - np}{\sqrt{np(1-p)} \leq Z_1}\right)$$

with $Z_1 = \frac{4900-5000}{50}, Z_2 = \frac{5100-5000}{50}$.

$$\Phi(Z_2) - \Phi(Z_1) = 0.9545$$

The complete formula for this case looks like this

$$P(4900 < S_n < 5100) \approx \Phi\left(\frac{5100 + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{4900 - 0.5 - np}{\sqrt{np(1-p)}}\right).$$

Chapter 68

Single Variable Distributions

68.1 Random Variables

Given a probability space $(\Omega, \mathcal{A}, \mathbb{P})$, we call a function $f : \Omega \to \mathbb{R}$ *measurable* if for all $z \in \mathbb{R}$, the following applies

$$f^{-1}((-\infty, z]) = \{\omega \in \Omega : f(\omega) \leq z\} \in \mathcal{A}$$

This measurable function $X : \Omega \to \mathbb{R}$ is a *random variable.*

Note that for discrete probability spaces $\{\omega : X(\omega) \leq z\} \subseteq \Omega \in \mathcal{P}$.

Also, it is considered discrete if it has only a finite or an infinite countable number of values. Finally, we define the probability for random variables as

$$\mathbb{P}(X = k) := \mathbb{P}(\{X = k\})$$

68.1.1 Operations With Random Variables

Given two R.V. X,Y and some number $\alpha \in \mathbb{R}$, then

- $X + Y$.
- $X - Y$.
- $X \cdot Y$.
- $\max(X, Y)$.
- $\min(X, Y)$.
- αX or αY.

The result of these operations results in a random variable as well.

Example:

A company has fixed costs of 250 euros per month. It produces toys for a price of 6 euros with a production price of 3 euros. On average, it sells 1000, $\mu = 1000$ toys with a $\sigma = \sqrt{500}$. Determine the random variables of the costs, the sales volume and the profit.

Costs: $X_c = 250 + 3X$ with $\mu_c = 250 + 3\mu = 3250$ and $\sigma^2 = 9 \cdot 500 = 1500$

Sales volume: $X_s = 6X$ with $\mu_c = 6\mu = 6000$ and $\sigma^2 = 6 \cdot 500 = 3000$

Profit: $X_p = X_s - X_c = 250 + 6X - 3X$ with $\mu_s = 3X - 250 = 2750$ and $\sigma^2 = 3 \cdot 500 = 1500$

68.2 Indicator Function

Given $A \subseteq \Omega$, we define the *indicator function* as

$$\mathbb{1}_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A \\ 0, & \text{if } \omega \notin A \end{cases}$$

We also get the following rules:

- $\mathbb{1}_\emptyset = 0$
- $\mathbb{1}_A^2 = \mathbb{1}_A$
- $\mathbb{1}_{A^c} = 1 - \mathbb{1}_A$
- $\mathbb{1}_{A \cap B} = \mathbb{1}_A \mathbb{1}_B$
- $\mathbb{1}_{A \cup B} = \mathbb{1}_A + \mathbb{1}_B - \mathbb{1}_{A \cap B}$
- $\mathbb{1}_{A+B} = \mathbb{1}_A + \mathbb{1}_B$

68.2.1 Indicator Sum

Given the events $A_1, \dots A_n \in \Omega$, we define the random variable

$$X := \sum_{j=1}^{n} \mathbb{1} - A_j$$

as the *indicator sum* or *counting variable*.

The realization of X gives us how many A_j have occurred.

$$\{X = k\} = \sum_{T \subseteq \{1,\dots,n\} : |T| = k} \left(\bigcap_{j \in T} A_j \cap \bigcap_{j \notin T} A_j^c \right), k \in \{0, 1, \dots, n\}.$$

68.3 Distributions

Given our probability space and a R.V. X, the probability measurement $\mathbb{P}_X$ to $\mathbb{R}$ defined by

$$\mathbb{P}_X(A) := \mathbb{P}(\{\omega : X(w) \in A\}), \text{ for } A \subseteq \mathbb{R}$$

is called the *distribution* of X. Also, we write $\mathbb{P}(X \in A)$ as an abbreviation for $\mathbb{P}(\{\omega : X(w) \in A\})$.

68.4 Distribution Function

Given a probability space, a probability variable and a distribution $\mathbb{P}_X$, we define the function

$$\mathbb{F}_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(\{\omega : X(w) \leq x\})$$

as the *distribution function*.

68.4.1 Properties of Distribution Functions

- $\mathbb{F}_X$ is monotonically increasing.
- It is right-sided continuous.
- $\lim_{x \to -\infty} \mathbb{F}_X(x) = 0$ and $\lim_{x \to \infty} \mathbb{F}_X = 1$.

68.4.2 Cumulative Probability Calculations

- $\mathbb{P}(X = a) = \mathbb{P}(X = a)$
- $\mathbb{P}(X \leq a) = \mathbb{P}(X \leq a)$
- $\mathbb{P}(X < a) = \mathbb{P}(X \leq a - 1)$
- $\mathbb{P}(X > a) = 1 - \mathbb{P}(X \leq a)$
- $\mathbb{P}(X \geq a) = 1 - \mathbb{P}(X \leq -1)$
- $\mathbb{P}(a \leq X \leq b) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a)$
- $\mathbb{P}(|X - a| < b) = \mathbb{P}(-b < |X - a| < b) = \mathbb{P}(b) - \mathbb{P}(-b)$

68.5 The Expected Value for Discrete Random Variables

Given $\Omega, \mathcal{A}, \mathbb{P}$, a probability space and X a discrete random variable, we define the *expected result* as

$$\mathrm{E}[X] = \sum_{x \in X(\Omega)} x\mathbb{P}(X = x)$$

which exists only if

$$\sum_{x \in X(\Omega)} |x|\mathbb{P}(X = x) < \infty$$

where x is an event and $\mathbb{P}(X = x)$ its respective probability. It represents the mean of the random variable.

68.5.1 Transformation of the Expected Value

Given $u : X(\Omega) \to \mathbb{R}$ where $\sum_{x \in X(\Omega)} |u(x)|p(x) < \infty$, then

$$\mathrm{E}[u(X)] = \sum_{x \in X(\Omega)} u(x)\mathbb{P}(X = x)$$

This represents the expectation of the function.

68.5.2 Triangle Inequality for Expected Values

$$|\mathrm{E}[X]| \leq \mathrm{E}[|X|]$$

Proof:

$$\begin{aligned} |\mathrm{E}[X]| = \left| \sum_{x \in X(\Omega)} x\mathbb{P}(X = x) \right| &\leq \sum_{x \in X(\Omega)} |x\mathbb{P}(X = x)| \\ &= \sum_{x \in X(\Omega)} |x|\mathbb{P}(X = x) = \mathrm{E}[|X|] \end{aligned}$$

$\mathfrak{QED}$

68.5.3 Properties of Expected Values

Given two random variables X and Y, if both have an expected value, then

- $\mathrm{E}[\alpha X] = \alpha \mathrm{E}[X]$
- $\mathrm{E}[X + Y] = \mathrm{E}[X] + \mathrm{E}[Y]$
- $\mathrm{E}[b] = b$
- $X \leq Y = \mathrm{E}[X] \leq \mathrm{E}[Y]$

68.6 Variance

Given a random variable X for which $\mathrm{E}[(X - \mathrm{E}[X])^2]$ exists, we define the *variance* of X as

$$\mathrm{var}(X) = \mathrm{E}[(X - \mathrm{E}[X])^2] = \sum_{x \in X} (x - \mathrm{E}[X]^2)\mathbb{P}(X = x).$$

Its root is the *standard deviation*.

68.6.1 Properties of the Variance

Let X be a (discrete) random variable for which the variance exists. Then the following hold:

- $\mathrm{var}(aX + b) = a^2 \,\mathrm{var}(X)$, for $a, b \in \mathbb{R}$
- $\mathrm{var}(X) = \mathrm{E}[X^2] - (\mathrm{E}[X])^2$
- For $a \in \mathbb{R}$, we have $\mathrm{E}[(X - a)^2] = \mathrm{var}(X) + (\mathrm{E}[X] - a)^2$, and in particular, $\mathrm{E}[(X - a)^2] \geq \mathrm{var}(X)$. Equality holds if and only if $a = \mathrm{E}[X]$.

Example:

Given $\mathrm{E}(X) = 1.2$ and $\mathbb{P}(X = 0) = 0.16, \quad \mathbb{P}(X = 1) = 0.48, \mathbb{P}(X = 2) = 0.36.$

$$\mathrm{E}[X^2] = 0^2(0.16) + 1^2(0.48) + 2^2(0.36) = 1.92$$

$$\mathrm{var}(X) = \mathrm{E}[(X - \mathrm{E})^2] = (0 - 1.2)^2(0.16) + (1 - 1.2)^2(0.48) + (2 - 1.2)^2(0.36)$$

68.7 Moments

For a random variable X, we define the k-th *moment* as $\mathrm{E}[X^k]$.

68.8 Markov Inequality

Given a random variable X, $a \in \mathbb{R}$ and $h : D \to [0, \infty)$ a monotonically increasing function with the properties $h(a) > 0$ and $X(\Omega) \subseteq D$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathrm{E}[h(X)]}{h(a)}.$$

Simpler,

$$\mathbb{P}(X \geq a) \leq \frac{\mathrm{E}[X]}{a}$$

There is a special case for $Y = |X|, a > 0$ and $h(x) = x$

$$\mathbb{P}(X \geq a) \leq \frac{\mathrm{E}[h(X)]}{a}.$$

68.9 Chebychev Inequality

Given a random variable X with a variance and $a > 0$, then

$$\mathbb{P}(|X - \mathrm{E}[X]| \geq a) \leq \frac{\mathrm{var}(X)}{a^2}$$

This states that the probability of X being at least a away from its mean cannot be too big.

68.10 Probability Density

A Lebesgue-integrable function $f : \mathbb{R} \to [0, \infty)$ is called the (probability) *density* or *density function* of the random variable X or of the distribution $\mathbb{P}_X$ if, for all $a, b \in \mathbb{R}$ with $a \leq b$, the following holds:

$$\mathbb{P}(a < X \leq b) = \mathbb{P}_X((a, b]) = \int_a^b f(x)\, dx.$$

Distributions that possess a probability density are called *continuous distributions*.

68.10.1 Properties of Continuous Distributions

- $\mathbb{F}_X$ is continuous at x if and only if $\mathbb{P}(X = x) = 0$.
- $$\int_{-\infty}^{\infty} f_X(x)\, dx = \mathbb{F}_X(\infty) = \lim_{x \to \infty} P(X \leq x) = 1.$$
- If X is a continuous random variable, then F_X is continuous.

How to Test if F Function Works as a Density Function

To test if a given function can be used as a density function, we test:

- $f(x) \geq 0 \forall x \in \mathbb{R}$
- $\int_{-\infty}^{\infty} f(x)dx = 1$

68.11 Zero Property

Given a continuous random variable X.

For $a, b \in \mathbb{R}$:

$$\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(x)dx = \mathbb{P}(a < X \leq b).$$

Also $A \in \mathcal{B}(\mathbb{R}) : \mathbb{P}(X \in A) = \int_A f_X(x)dx$.

And $A \in \mathcal{B}(\mathbb{R})$ (discrete) $: A = \{a_i\}_{i \in I}, I \in \mathbb{N}$:

$$\mathbb{P}(X \in A) = \sum_{i \in I} \mathbb{P}(X = a_i) = 0.$$

68.12 Expected Value for Continuous Random Variables

Given a continuous random variable X with a density f_X, the *expected value* of X exists if $\int |x| f_X(x)dx, \infty$. In that case, we define it as

$$\mathrm{E}[X] = \int x f_X(x)dx.$$

We also get all of its properties like for the discrete case.

68.13 Transformation Lemma for Continuous Random Variables

Given a continuous random variable X with a density f and a measurable function $u : \mathbb{R} \to \mathbb{R}$ with $\int |u(x)| f(x)dx < \infty$, then the following applies

$$\mathrm{E}[u(X)] = \int u(x) f(x)dx.$$

68.14 Transformation Theorem for Densities

Given a random variable X with values in the open interval $I \subseteq \mathbb{R}$ and the density f_X. If $j \subseteq \mathbb{R}$ is an open interval and $u : I \to J$ is a diffeomorphism, then $Y = u(X)$ has the density

$$f_Y(y) = f_X(u^{-1}(y))|(u^{-1})'(y)|.$$

For $y \in J$ and $f_Y(y) = 0$ for $y \in \mathbb{R} \setminus J$.

68.15 Quantile

Let X be a random variable with distribution F. For $p \in (0, 1)$, a value q_p is called a *p-quantile* of X (or of F) if:

$$\mathbb{P}(X \leq q_p) \geq p \text{ and } \mathbb{P}(q_p \leq X) \geq 1 - p,$$

or equivalently,

$$F(q_p) \geq p \text{ and } \lim_{x \nearrow q_p} F(x) \leq p.$$

In general, the p-quantile is not unique. If F is strictly increasing and continuous, then q_p is unique. If q_p is not unique, we can make it "unique" by defining the smallest quantile as

$$F^{-1}(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}.$$

68.16 The Moment Generating Function

The n-th *moment* of a distribution is given by $\mathrm{E}[X^n]$. We also know that the first moment is the mean and the variance is closely related to the second moment. The moments *uniquely* determine the distribution.

The *moment generating function* is defined as

$$M(t) = \begin{cases} \sum_x e^{tx}\mathbb{P}(X = x) & \text{if } X \text{ discrete} \\ \int_{-\infty}^{\infty} e^{tx}\mathbb{P}(X = x)dx & \text{if } X \text{ continuous} \end{cases}.$$

$M(t)$ uniquely determines the cumulative probability distribution of $\mathbb{P}(X = x)$.

68.16.1 Derivative of the Moment Generating Function

The $n-th$ derivative of the *moment generating function* has the following property

$$M^n(0) = \mathrm{E}[X^n].$$

Proof (Informal):

We will handle only the continuous case, but the same holds for the discrete case

$$\frac{d}{dt}\int_{-\infty}^{\infty} e^{tx}\mathbb{P}(X=x)dx = \int_{-\infty}^{\infty} xe^{tx}\mathbb{P}(X=x)dx$$

$$\frac{d}{dt}\int_{-\infty}^{\infty} xe^{tx}\mathbb{P}(X=x)dx = \int_{-\infty}^{\infty} x^2e^{tx}\mathbb{P}(X=x)dx$$

And the pattern repeats. This just gives us the definition of each of the moments.

$\mathfrak{QED}$

68.16.2 The Additive Property of the Moment Generating Function

Let $X_1, X_2, \ldots, X_n$ be random variables with their corresponding moment generating functions. Let $Z = \sum_i X_i$ then

$$M_Z(t) = \prod_i M_{X_i}(t)$$

68.17 Discrete Uniform Distribution

A random variable $X \in \{1, 2, \ldots, n\}$ with $\mathbb{P}(X = i) = \frac{1}{n}$ for $i = 1, \ldots, n$ is uniformly distributed. The distribution $\mathbb{P}_X$ is called the *discrete uniform distribution*. Noted as $\mathcal{U}(n)$.

$$\mathbb{P}(X=i) = \begin{cases} \frac{1}{n}, & i = 1, \ldots, n \\ 0, & \text{else} \end{cases}$$

The expected value is given by

$$\mathrm{E}(X) = \frac{n+1}{2}$$

and the standard deviation is given by

$$\sigma = \frac{n^2-1}{12}$$

Finally, its cumulative distribution is given by

$$\mathbb{F}_X(x) = \sum_{i=1}^{x} \frac{i}{n}$$

68.18 Continuous Uniform Distribution

A random variable $X \in \{1, 2, \ldots, n\}$ with $\mathbb{P}(X = i) = \frac{1}{n}$ for $i =\in [a, b]$ is uniformly distributed. The distribution is called the *continuous uniform distribution*. Noted as $\mathcal{U}(a, b)$.

$$\mathbb{P}(X = i) = \begin{cases} \frac{1}{b-a}, & i \in [a, b] \\ 0, & \text{else} \end{cases}$$

The expected value is given by

$$\mathrm{E}(X) = \frac{a + b}{2}$$

and the standard deviation is given by

$$\sigma = \frac{(b - a)^2}{12}.$$

Finally, its cumulative distribution is given by

$$\mathbb{F}_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & x \in [a, b] \\ 1, & x > b \end{cases}$$

68.19 Bernoulli Distribution

The *Bernoulli distribution* is used for modeling binary events with only one trial, noted as $\mathcal{B}(p)$.

$$\mathbb{P}(X = x) = \begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \end{cases}$$

The expected value, variance and cumulative distribution function are given by

$$\mathrm{E}(X) = p$$

$$\mathrm{var}(X) = p(1 - p)$$

$$\mathbb{F}_X = \begin{cases} 0, & x < 0 \\ 1 - p, & 0 \geq x < 1 \\ 1, & x \geq 1 \end{cases}$$

68.20 Binomial Distribution

From the Bernoulli distribution we can derive the *binomial distribution* $\mathrm{Bin}(n, p)$ as having a number of consecutive independent Bernoulli experiments.

This distribution is for discrete binary experiments. Either an event occurs or it does not.

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

With

$$\mathrm{E}[X] = np$$

$$\mathrm{var}(X) = \mathrm{E}[X](1-p)$$

The cumulative distribution function is given by

$$\mathbb{F}_X = \sum_{i=0}^{k} \mathbb{P}(X = i)$$

If we let $n \to \infty$ with np being not too small, then the limit of this distribution becomes the normal distribution.

68.21 Normal Distribution

The *normal distribution* $\mathcal{N}(\mu, \sigma^2)$ is the limit of the *binomial distribution* under certain conditions, and it is given by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}e^{\frac{1}{2}\left(\frac{x-u}{\sigma}\right)^2}}.$$

It also has

$$\mathrm{E}[X] = np = \mu,$$

$$\mathrm{var}(X) = \mathrm{E}[X](1-p).$$

To compute its cumulative probability, we need to use special tables, due to the fact that the function cannot be analytically integrated. To get a specific value, we use tables with pre-calculated values computed numerically using the so-called Φ function. This function is used in the following manner:

We plug the values in the expression, our expected value and the standard deviation

$$Z = \frac{X - \mu}{\sigma}.$$

For $\Phi(Z)$, you use the portion before the comma as the row and after the comma as the column to read the value from.

68.21.1 Standard Normal Distribution

The *standard* normal distribution has the parameters $\mu = 0$ and $\sigma^2 = \mathrm{var} = 1$.

To transform a normal distributed random variable X you can define a new random variable

$$Z = \frac{X - \mu}{\sigma}.$$

Example I:

Given $X \sim N(10, 4)$. What is $\mathbb{P}(8 \leq X \leq 12)$?

We define $Z = \frac{X-10}{2}$. Then we get

$$\mathbb{P}(8 \leq X \leq 12) = \mathbb{P}\left(\frac{8-10}{2} \leq Z \leq \frac{12-10}{2}\right) = \mathbb{P}(-1 \leq Z \leq 1)$$

Using the standard normal distribution table we get

$$\mathbb{P}(-1 \leq Z \leq 1) = \mathbb{F}_Z(1) - \mathbb{F}_Z(-1) = 0.8413 - 0.1587 = 0.6826$$

Example II:

The diameter of mass-produced balls is normally distributed. The mean and standard deviation can be estimated using the following experiment:

Of two sieves, sieve A has holes with a diameter of 69 mm, and sieve B has holes with a diameter of 72 mm. It is observed that 30.85 percent of the balls pass through sieve A, while 15.87 percent of the balls are retained by sieve B.

Determine estimates for the mean and standard deviation of the ball diameters based on these experimental results.

First we have:

$$\mathbb{P}(X \leq 69) = 0.3085 \quad \mathbb{P}(X \geq 72) = 0.1587 \implies \mathbb{P}(X \leq 72) = 0.8413$$

If we look at the Z-table we get the z-values -0.5 and 1

This means

$$\frac{69-\mu}{\sigma} = -0.5 \quad \frac{72-\mu}{\sigma} = 1$$

Now we can solve the system of equations and the results will be:

$$\mu = 70 \quad \sigma = 2$$

68.22 Geometric Distribution

A random variable $X \in \mathbb{N}_0$ with $\mathbb{P}(X = n) = p(1-p)^n$ for some $p \in (0,1)$ and $n \in \mathbb{N}_0$ is geometrically distributed. The distribution $\mathbb{P}_X$ is called the *geometric distribution*.

Similar to the binomial distribution, the *geometric* $\mathrm{Geo}(n,p)$ distribution is used for an experiment with two outcomes, and we want to know the probability of $k-1$ non-successes until a success happens.

$$\mathbb{P}(X = k) = (1-p)^{k-1}p.$$

We also get

$$\mathrm{E}[X] = \mu = \frac{1-p}{p}$$

$$\mathrm{var}(X) = \frac{1-p}{p^2}$$

The cumulative distribution function is given by

$$\mathbb{F}_X = \begin{cases} 0, & x < 1 \\ 1-(p-1)^{\lfloor x \rfloor}, & x \geq 1 \end{cases}$$

68.23 Poisson Distribution

A random variable $X \in \mathbb{N}_0$ with

$$\mathbb{P}(X = n) = \frac{\lambda^n}{n!} e^{-\lambda}, \quad \text{for some } \lambda > 0,$$

We also define the distribution function

$$\mathbb{F}_X(n) = \sum_{i=0}^{n} \frac{\lambda^i}{i!} i^{-\lambda}$$

is Poisson distributed. The distribution $\mathbb{P}_X$ is called the *Poisson distribution* $\text{Poi}(\lambda)$. In simple words, it is the number of events x, in a specific interval. This is a limit of the binomial distribution for large n and small p.

We also get our expected result and variance

$$\text{E}[X] = \lambda = np$$

and

$$\text{var}(X) = \lambda$$

The derivation is the following, let $\lambda = np$

$$\begin{aligned}
&\lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k} \\
&\quad = \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
&\quad = \frac{n(n-1)\dots(n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
&\quad = \frac{n(n-1)\dots(n-k+1)}{k!} \frac{\lambda^k}{n^k} \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
&\quad = \frac{\lambda^k}{k!} \frac{n(n-1)\dots(n-k+1)}{n^k} \left(1 - \frac{\lambda}{n}\right)^{n} \left(1 - \frac{\lambda}{n}\right)^{-k} \\
&\quad = \frac{\lambda^k}{k!} \cdot 1 \cdot e^{-\lambda} \cdot 1 \\
&\quad = \frac{\lambda^k}{k!} \cdot e^{-\lambda}
\end{aligned}$$

68.24 Hyper-Geometric Distribution

The *hyper-geometric distribution* is similar to the binomial distribution, but we keep track of the changes after each choice. In other words, binomial distribution without repetition. We notate this as $\text{HG}(M, N, n)$.

$$\mathbb{P}(X = k) = \frac{\binom{M}{k}\binom{N-M}{n-k}}{\binom{N}{n}}.$$

Nomenclature:

- N: Total number of elements

- M: Elements with the trait A
- $N - M$: Elements without the trait A
- k: Number of elements to take
- n: Number of iterations

Each part of the formula has a different meaning. $\binom{N}{n}$ represents the total number of ways to choose the elements from the total. $\binom{M}{k}$ gives us the ways of choose k elements with the trait from its total M and finally, $\binom{N-M}{n-k}$ represents the remaining ways of choosing elements we are not interested in.

We also get

$$\mathrm{E}[X] = n\frac{M}{N}$$

$$\mathrm{var}(X) = \mathrm{E}[X]\left(1 - \frac{M}{N}\right)\left(\frac{N-n}{N-1}\right)$$

The cumulative distribution function is given by

$$\mathbb{F}_X(x) = \sum_{i=1}^{x} \mathbb{P}(X = i)$$

68.25 Exponential Distribution

We define the *exponential distribution* $\mathcal{E}(\lambda)$ as a real-valued random variable, used mostly for the duration of time intervals.

$$f(n) = \begin{cases} \lambda e^{-\lambda n} & , n \geq 0 \\ 0 & , \text{ else} \end{cases}$$

Where *lambda* is our *rate* parameter.

$$\lambda = \frac{1}{\text{average}}$$

We also define the distribution function

$$\mathbb{F}_X(X = n) = \int_{-\infty}^{n} \mathbb{P}(X = n) = \begin{cases} 1 - e^{-\lambda n} & , n \geq 0 \\ 0 & , \text{ else} \end{cases}$$

We also have

$$\mathrm{E}(X) = \frac{1}{\lambda}$$

$$\mathrm{var}(X) = \frac{1}{\lambda^2}$$

68.25.1 Memory-less Property of the Exponential Distribution

This distribution is *memory-less*. This means that the future probabilities of an event are independent of the previous ones.

$$\mathbb{P}(X \geq x + y \mid X > y) = \mathbb{P}(X \geq x)$$

68.25.2 The Connection Between the Exponential Distribution and the Poisson Process

A neat connection between these two concepts is that given an exponential distribution over some time t then the number of events which occurred during that interval is given by a Poisson distribution $\text{Poi}(\lambda t)$.

Example:

Emails come at a rate of 5 per minute ($\lambda = 5$). What is the probability of waiting t until the next email?

The probability of getting an email after t waiting is given by the following exponential distribution

$$\mathbb{P}(T > t) = e^{-5}t$$

What is the probability of getting no emails in the next t minutes?

This is given by the following Poisson process

$$\mathbb{P}(k = 0) = \frac{5^0 e^{-5t}}{0!} = e^{-5t}$$

68.26 Gamma Distribution

This distribution comes directly from the connection between the exponential distribution and the Poisson process. Given events $w_0, \ldots, w_n$ with an exponential distribution. The *gamma* distribution gives us the waiting time for the r-th event.

$$T_r = \sum_{k=0}^{r} w_k$$

We denote this as $T_r \sim \text{gamma}(r, \lambda)$. The probability density function is given by

$$\mathbb{P}(T_r > t) = \mathbb{P}(N_t \geq r - 1) = \sum_{k=0}^{r-1} \frac{e^{-\lambda t}(\lambda t)^k}{k!} = \cdots = e^{-\lambda t}\frac{\lambda^r t^{r-1}}{(r-1)!}$$

with N_t being the number of Poisson events in t.

68.27 Chi-Square Distribution

The square of the *normal distribution* is called the *Chi-Square Distribution*, noted as $\mathcal{N}(a\mu + b, a^2\sigma^2)$ or χ^2_n with $n \in \mathbb{N}$ degrees of freedom.

Given a random variable X with density

$$f(x) = \frac{1}{2^{\frac{n}{2}}\Gamma\left(\frac{n}{2}\right)} x^{\frac{n}{2-1}} e^{\frac{-x}{2}}$$

where Γ is the gamma function defined as

$$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}dt$$

For 1 degree of freedom, we get

$$f(x) = \frac{1}{\sqrt{2\pi}} y^{-\frac{1}{2}} e^{-\frac{y}{2}}$$

We also have

$$\mathrm{E}[X] = n$$

$$\mathrm{var}(X) = \mathrm{E}[X]2n$$

68.27.1 Derivation

The probability density function is reached by taking the cumulative density function of the normal distribution which is an Integral, square it, and then we take its derivative. We proceed as follows:

Given $X \sim \mathcal{N}(0,1)$ we define $Y = X^2$. Then we get

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

This gives us

$\Phi(\sqrt{y}) - \Phi(-\sqrt{y})$ Now we take the derivative to get the density function

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy}\left(\Phi(\sqrt{y}) - \Phi(-\sqrt{y})\right) \\ &= \Phi'(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + \Phi'(-\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} \\ &= y^{-\frac{1}{2}} f_X(\sqrt{y}) \\ &= \frac{1}{\sqrt{2\pi}} y^{-\frac{1}{2}} e^{-\frac{y}{2}} \end{aligned}$$

68.28 Student's t-Distribution

This is a distribution mostly used in statistics for hypothesis testing.

$$f_n(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}$$

This is used to test the mean $\overline{X_n}$ against the true μ when σ^2 is unknown.

68.29 Activity Examples

68.29.1 Night Guard

A night guard has 2 methods for finding the correct key out of 10. The first one, A, consists of just testing each key without repetition. The second one, B, consists of choosing one random key and then if it was not the correct one it just shuffles the keys and repeats the process.

What are the distributions of both the methods?

The distribution for A is the uniform with $n = 10$, for B it is the geometric distribution $P(X = k) = \left(\frac{9}{10}\right)^{k-1} \frac{1}{10}$

The guard is drunk 1/3 of the time. When he is drunk, he uses method B, otherwise A. What is the conditional probability of him being caught drunk during work after failing 8 times?

$$\mathbb{P}(\text{drunk} \mid X > 8) = \frac{\mathbb{P}(\text{drunk})\mathbb{P}(X > 8 \mid \text{not drunk})}{\mathbb{P}(X > 8)}$$

This is

$$\mathbb{P}(B \mid X \leq 8) = \frac{\frac{1}{3} \cdot \left(1 - \left(\frac{9}{10}\right)^{8-1} \frac{1}{10}\right)}{\frac{1}{3} \cdot \left(1 - \left(\frac{9}{10}\right)^{8-1} \frac{1}{10}\right) + \frac{2}{3} \cdot \left(1 - \frac{8}{10}\right)}.$$

Chapter 69

Multivariable Distributions

Multivariable or *joint distributions* are a way of combining two or more random variables.

69.1 Measurability of Random Variables and Random Vectors

Let $X_1, X_2, \ldots, X_n$ be random variables on $(\Omega, \mathcal{A}, P)$. Then each $X_i : \Omega \to \mathbb{R}$ is measurable.

Measurability in $\mathbb{R}$.

$$X_i^{-1}((-\infty, z]) \in \mathcal{A} \quad \text{for all } z \in \mathbb{R}$$

Alternative Representation.

Define $X : \Omega \to \mathbb{R}^n$ by

$$X(\omega) = (X_1(\omega), \ldots, X_n(\omega))$$

It can be shown (from measure theory) that X is measurable.

Measurability in $\mathbb{R}^n$.

$$X^{-1}(\mathbb{R}) \in \mathcal{A}, \quad \text{where } \mathbb{R} = (a_1, b_1] \times (a_2, b_2] \times \cdots \times (a_n, b_n]$$

for arbitrary $a_1, \ldots, a_n, b_1, \ldots, b_n \in \mathbb{R}$.

From this (by measure theory) it follows that for any set $A \subseteq \mathbb{R}^n$,

$$X^{-1}(A) \in \mathcal{A}$$

Hence, the probability

$$\mathbb{P}(X \in A) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in A\})$$

is well-defined for all $A \subseteq \mathbb{R}^n$.

69.2 Joint Distribution and Random Vectors

Let $X_1, \ldots, X_n$ be random variables on a probability space $(\Omega, \mathcal{A}, P)$. Then P_X (respectively $P_{X_1,\ldots,X_n}$) is defined as

$$\mathbb{P}_X(A) = P((X_1, \ldots, X_n) \in A), \quad A \subseteq \mathbb{R}^n$$

and is called the *joint distribution* of $X_1, \ldots, X_n$. The vector $X = (X_1, \ldots, X_n)$ is called a *random vector*.

This is just the probability of n events happening simultaneously.

69.3 Joint Probability of an Event

For any random variables:

$$\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) = \mathbb{P}(X_1 = x_1)\mathbb{P}(X_2 = x_2 \mid X_1 = x_1) \cdots \mathbb{P}(X_n = x_n \mid X_1 = x_1, \ldots, X_{n-1} = x_{n-1})$$

If they are independent, then

$$\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) = \mathbb{P}(X_1 = x_1)\mathbb{P}(X_2 = x_2) \cdots \mathbb{P}(X_n = x_n)$$

69.4 Joint Probability Function

Let $X_1, \ldots, X_n$ be discrete random variables. The function p_X, defined as

$$\mathbb{P}_X(x) = \mathbb{P}(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)$$

for $x \in X(\Omega) \subseteq \mathbb{R}^n$, is the *joint probability function* of $X_1, \ldots, X_n$, or equivalently, of the random vector X.

As before, the distribution P_X is uniquely determined by the probability function p_X:

$$\mathbb{P}_X(A) = \mathbb{P}\big((X_1, \ldots, X_n) \in A\big) = \sum_{x \in A} \mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) = \sum_{x \in A} p_X(x)$$

Furthermore, for probability functions $p : \mathbb{R}^n \to \mathbb{R}$, the following holds:

$$\sum_{(x_1,\ldots,x_n) \in X(\Omega)} p(x_1, \ldots, x_n) = 1$$

$$p(x_1, \ldots, x_n) \geq 0$$

Example:

Let X, Y be the numbers shown when rolling two dice. The joint probability function is

$$p_{X,Y}(i, j) = \frac{1}{36}$$

for $i, j \in \{1, 2, \ldots, 6\}$.

Their combined probability function is

$$p_{X,Y}(i, j) = \begin{cases} \frac{1}{6}, & \text{if } i = j \\ 0, & \text{otherwise.} \end{cases}$$

69.5 Marginal Distributions of Discrete Random Variables

Let X, Y be two discrete random variables. The *marginal probability functions* of X and Y are given by

$$p_X(x) = \sum_y p_{X,Y}(x,y) \quad \text{and} \quad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

This means that we fix one variable and sum over all possible values of the other variable. For the 2d case, if we want a marginal distribution of X, we sum over all rows for a fixed column.

More generally, let $X_1, \dots, X_n$ be random variables and $1 \leq i_1 < \cdots < i_k \leq n$.

The joint distribution of $X_{i_1}, \dots, X_{i_k}$ is a *k-dimensional marginal distribution* of $X_1, \dots, X_n$.

For discrete random variables, the joint probability function of $X_{i_1}, \dots, X_{i_k}$ is called the *marginal probability function*.

Example:

Let X_1, X_2 be the outcomes of two dice rolls, and let $Y = X_1 + X_2$. The joint distribution of X_1 and Y is given by:

$X_1 \backslash Y$	2	3	4	5	6	7	8	9	10	11	12	$\mathbb{P}(X_1 = x)$
1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0	0	$\frac{1}{6}$
2	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	0	$\frac{1}{6}$
3	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	0	$\frac{1}{6}$
4	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	0	$\frac{1}{6}$
5	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	0	$\frac{1}{6}$
6	0	0	0	0	0	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{6}$
$\mathbb{P}(Y = y)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	

In the last column and row, we get the marginal distributions of X_1 and Y, respectively.

69.5.1 Marginal Probability Function for Multiple Discrete Variables

Let $X_1, \dots, X_n$ be discrete random variables with joint probability function $p(x_1, \dots, x_n)$, and let $1 \leq i_1 < \cdots < i_k \leq n$. For the *marginal probability function* of $X_{i_1}, \dots, X_{i_k}$, the following holds:

$$p_{X_{i_1},\dots,X_{i_k}}(x_{i_1}, \dots, x_{i_k}) = \sum_{(x_j)_{j \notin \{i_1,\dots,i_k\}}} p(x_1, \dots, x_n)$$

This is just notation for summing over all variables that are not of interest.

69.6 Multivariate Expected Value for Discrete Random Variables

Let $X_1, \dots, X_n$ be discrete random variables with joint probability function p, and let $u : \mathbb{R}^n \to \mathbb{R}$.

Then, if the expectation exists, the following holds:

$$\mathrm{E}[u(X_1, \dots, X_n)] = \sum_{(x_1,\dots,x_n) \in X(\Omega)} u(x_1, \dots, x_n)\, p(x_1, \dots, x_n)$$

Example:

Let X and Y be random variables with the following joint probability function p:

$Y \backslash X$	1	2	3
3	$\frac{1}{6}$	0	$\frac{1}{4}$
4	$\frac{1}{8}$	$\frac{1}{3}$	$\frac{1}{8}$

Let $u(X,Y) = xy$, then our expected value is given by:

$$\mathrm{E}[XY] = \sum_{(x,y)\in(X,Y)(\Omega)} xy\, p(x,y) = 1\cdot 3\cdot\frac{1}{6} + 2\cdot 3\cdot 0 + 3\cdot 3\cdot\frac{1}{4} + 1\cdot 4\cdot\frac{1}{8} + 2\cdot 4\cdot\frac{1}{3} + 3\cdot 4\cdot\frac{1}{8} \approx 7.42$$

69.7 Expected Value as a Special Case of the Transformation Formula

Let X, Y be discrete random variables with joint probability function p. With $u(x,y) = x$, the transformation formula gives:

$$\mathrm{E}[X] = \sum_{(x,y)\in(X,Y)(\Omega)} x\, p(x,y)$$

69.8 Joint Probability Density Function

An integrable function $f : \mathbb{R}^n \to [0,\infty)$ is called the *joint (probability) density* of the random variables $X_1, \ldots, X_n$ if, for all rectangles

$$R = (a_1, b_1] \times \cdots \times (a_n, b_n] \subseteq \mathbb{R}^n,$$

the following holds:

$$\mathbb{P}((X_1, \ldots, X_n) \in R) = \int_R f(x_1, \ldots, x_n)\, dx_1 \cdots dx_n.$$

Example I:

Find the probability of $\mathbb{P}(X \leq \frac{2}{3}, Y \leq \frac{1}{2})$, with the constraint $\Delta = \{(x,y) \in [0,1]^2 : 0 \leq y \leq x \leq 1\}$.

$$\begin{aligned}
&\int_0^{\frac{2}{3}} \int_0^{\frac{1}{2}} f(x,y)dydx \\
&\qquad = \int_0^{\frac{2}{3}} \int_0^{\frac{1}{2}} \mathbb{1}_\Delta dydx \\
&\qquad = 2\int_0^{\frac{2}{3}} \int_0^{\frac{1}{2}} 1 dydx \\
&\qquad = 2\int_0^{\frac{2}{3}} \int_0^{\min(x,\frac{1}{2})} 1 dydx \\
&\qquad = 2\left(\int_0^{\frac{1}{2}} x dx + \int_{\frac{1}{w}}^{\frac{2}{3}} \frac{1}{2} dx\right) = \frac{5}{12}
\end{aligned}$$

Example II:

Find the densities of X and Y

$$\int_0^1 \int_0^1 6\cdot x^2 \cdot y\, dy\, dx = 1$$

Density of X:

$$f_X(x) = \int_0^1 6x^2 y\, dy = 6x^2 \left[\frac{y^2}{2}\right]_0^1 = 3x^2 \quad \text{for } 0 \leq x \leq$$

$$f_X(x) = \begin{cases} 3x^2 & \text{if } 0 \le x \le 1 \\ 0 & \text{else} \end{cases}$$

Density of Y:

$$f_Y(y) = \int_0^1 6x^2 y \, dx = 6y \left[\frac{x^3}{3} \right]_0^1 = 2y \quad \text{for } 0 \le y \le 1$$

$$f_Y(y) = \begin{cases} 2y & \text{if } 0 \le y \le 1 \\ 0 & \text{else} \end{cases}$$

69.9 Multivariate Expected Value for Continuous Random Variables

Let $X_1, \ldots, X_n$ be random variables with joint density $f(x_1, \ldots, x_n)$, or equivalently, with joint distribution $P_{X_1,\ldots,X_n}$.

Let $u : \mathbb{R}^n \to \mathbb{R}$ be a measurable function. Then, if the expectation exists, the following holds:

$$\mathrm{E}[u(X_1, \ldots, X_n)] = \int_{\mathbb{R}^n} u(x_1, \ldots, x_n) f(x_1, \ldots, x_n) \, dx_1 \cdots dx_n$$

69.10 Uniform Distribution on a Measurable Set

Let $I \subseteq \mathbb{R}^n$ be a measurable set. Random variables $(X_1, \ldots, X_n) \in \mathbb{R}^n$ with joint density

$$f(x_1, \ldots, x_n) = \frac{1}{|I|} \mathbb{1}_I(x_1, \ldots, x_n)$$

are called *uniformly distributed*, where $|I|$ denotes the volume (Lebesgue measure) of the set I. The distribution $P_{X_1,\ldots,X_n}$ is called the *uniform distribution*.

69.11 Probability Over Measurable Sets

Let $X_1, \ldots, X_n$ be random variables with joint density $f(x_1, \ldots, x_n)$. For every measurable set $R \subseteq \mathbb{R}^n$, the following holds:

$$\mathbb{P}\big((X_1, \ldots, X_n) \in R\big) = \int_R f(x_1, \ldots, x_n) \, dx_1 \cdots dx_n$$

69.12 Joint Distribution Function for Multiple Variables

Let $X_1, \ldots, X_n$ be random variables. The function $F : \mathbb{R}^n \to \mathbb{R}$, defined by

$$F(x_1, \ldots, x_n) = \mathbb{P}(X_1 \le x_1, \ldots, X_n \le x_n),$$

is called the *joint distribution function* of $X_1, \ldots, X_n$.

69.13 Marginal Densities

Let $X_1, \ldots, X_n$ be random variables with joint density $f(x_1, \ldots, x_n)$. The density of X_1 is given by

$$f_{X_1}(x_1) = \int \cdots \int f(x_1, \ldots, x_n)\, dx_2 \cdots dx_n.$$

For random variables X, Y with joint density $f(x, y)$, the following holds:

$$f_X(x) = \int f(x, y)\, dy$$

$$f_Y(x) = \int f(x, y)\, dx$$

The idea for higher dimensions is to eliminate all variables we are not interested in.

69.14 Expectation With Respect to a Joint Density

Let $X_1, \ldots, X_n$ be random variables with joint density $f(x_1, \ldots, x_n)$, and let $u : \mathbb{R}^n \to \mathbb{R}$ be a measurable function. Then the following holds:

$$\mathrm{E}[u(X_1, \ldots, X_n)] = \int u(x_1, \ldots, x_n) f(x_1, \ldots, x_n)\, dx_1 \cdots dx_n$$

69.15 Stochastic Independence of Random Variables

The random variables $X_1, \ldots, X_n$ are *stochastically independent* if, for all intervals $I_1, \ldots, I_n \subseteq \mathbb{R}$, the following holds:

$$\mathbb{P}(X_1 \in I_1, \ldots, X_n \in I_n) = \prod_{k=1}^{n} \mathbb{P}(X_k \in I_k)$$

69.16 Independence of Densities

Let X, Y be discrete random variables with joint probability function $p_{X,Y}(x, y)$ and marginal probability functions $p_X(x)$ and $p_Y(y)$. Then X and Y are *independent if and only if*

$$p_{X,Y}(x, y) = p_X(x) p_Y(y)$$

Let X and Y be discrete random variables with joint probability function $p_{X,Y}(x, y)$ and marginal probability functions $p_X(x)$ and $p_Y(y)$. The random variables X and Y are said to be *independent* if and only if

$$p_{X,Y}(x, y) = p_X(x) p_Y(y)$$

Also, if $X_1, \ldots, X_n$ are independent random variables and

$$f_1, \cdots, f_n : \mathbb{R} \to \mathbb{R}$$

Then $f_1(X_1), \ldots, f_n(X_n)$ are also independent random variables.

Example:

Given

$$\int_0^1 \int_0^1 6 \cdot x^2 \cdot y \, dy \, dx = 1$$

with

$$f_X(x) = \int_0^1 6x^2 y \, dy = 6x^2 \left[\frac{y^2}{2} \right]_0^1 = 3x^2 \quad \text{for } 0 \leq x \leq$$

$$f_Y(y) = \int_0^1 6x^2 y \, dx = 6y \left[\frac{x^3}{3} \right]_0^1 = 2y \quad \text{for } 0 \leq y \leq 1$$

We test $f_{X,Y}(x,y) \stackrel{?}{=} f_X(x) \cdot f_Y(y)$

$$f_X(x) \cdot f_Y(y) = 3x^2 \cdot 2y = 6x^2 y = f_{X,Y}(x,y)$$

69.17 Variance for Independent Random Variables

Given the independent random variables $X_1, \ldots, X_n$ then

$$\text{var}\left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{var}(X_i)$$

Proof:

For $i \neq j$, we have, due to independence,

$$\text{E}[(X_i - \text{E}[X_i])](X_j - \text{E}[X_j]) = \text{E}[X_i - \text{E}[X_i]] \cdot \text{E}[X_j - \text{E}[X_j]] = 0$$

This implies that

$$\begin{aligned}
\text{var}\left(\sum_{i=1}^n X_i \right) &= \text{E}\left[\left(\sum_{i=1}^n X_i - \text{E}[X_i] \right)^2 \right] \\
&= \text{E}\left[\sum_{i=1}^n (X_i - \text{E}[X_i])^2 + \sum_{i \neq j} (X_i - \text{E}[X_i])(X_j - \text{E}[X_j]) \right] \\
&= \sum_{i=1}^n \text{E}[(X_i - \text{E}[X_i])^2] + \sum_{i \neq j} \text{E}[(X_i - \text{E}[X_i])(X_j - \text{E}[X_j])] \\
&= \sum_{i=1}^n \text{var}(X_i)
\end{aligned}$$

69.18 Conditional Distribution of Discrete Random Variables

Let X, Y be discrete random variables and $x \in \mathbb{R}$ such that $\mathbb{P}(X = x) > 0$. The mapping

$$A \mapsto \mathbb{P}(Y \in A \mid X = x)$$

for measurable $A \subseteq \mathbb{R}$, defines a probability distribution on $\mathbb{R}$ and is called the *conditional distribution of Y given $X = x$*.

The corresponding *conditional probability function* of Y given $X = x$ is defined by

$$p_{Y|X}(y \mid x) = \mathbb{P}(Y = y \mid X = x) = \frac{p_{X,Y}(x,y)}{p_X(x)}.$$

Two conditional discrete random variables X and Y are independent if the conditional probability of Y given $X = x$ is not dependent on x, i.e.

Proof:

X, Y independent $\iff p_{X,Y}(x,y) = p_X(x)p_Y(y)$

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)} = \frac{p_X(x)p_Y(y)}{p_X(x)} = p_Y(y)$$

Otherwise, if $p_{Y|X}(y \mid x) = \mathbb{P}(Y = y \mid X = x)$ is not dependent on x then

$$p_{Y|X}(y|x) = \mathbb{P}(Y = y) = p_Y(y)$$

Then $p_{X,Y}(x,y) = p_X(x)p_{Y|X}(y \mid x) = p_X(x)p_Y(y)$

69.19 Conditional Density for Continuous Random Variables

Let X, Y be continuous random variables with joint density $f(x,y)$. The conditional density $f_{Y|X}(y \mid x)$ of Y given $X = x$ is defined by

$$f_{Y|X}(y \mid x) = \begin{cases} \frac{f(x,y)}{f_X(x)}, & \text{if } f_X(x) > 0, \\ 0, & \text{if } f_X(x) = 0. \end{cases}$$

The conditional distribution of Y given $X = x$ is the distribution with density $f_{Y|X}(y \mid x)$. Thus, for any measurable set $A \subseteq \mathbb{R}$,

$$\mathbb{P}(Y \in A \mid X = x) = \int_A f_{Y|X}(y \mid x)\, dy.$$

69.20 Covariance and Correlation of Random Variables

Let X, Y be random variables with existing variances. The *covariance*, joint dependence, is defined by

$$\operatorname{cov}(X,Y) = E[(X - \mathrm{E}[X])(Y - \mathrm{E}[Y])] = \sum_{i=1}^{n}\sum_{j=1}^{m}(x_i - \mathrm{E}[X])(y_j - \mathrm{E}[Y])\,\mathbb{P}(X = x_i, Y = y_j)$$

If the variances are positive, the *correlation coefficient* $\rho_{X,Y}$ is defined as

$$\rho_{X,Y} = \frac{\operatorname{cov}(X,Y)}{\sqrt{\operatorname{var}(X)\operatorname{var}(Y)}}.$$

If $\rho_{X,Y} = 0$, the random variables X and Y are called *uncorrelated*.

69.20.1 Properties of Covariance

- $\operatorname{cov}(X,X) = \operatorname{var}(X)$
- $\operatorname{cov}(X,Y) = \mathrm{E}[XY] - \mathrm{E}[X]\mathrm{E}[Y]$
- $\operatorname{cov}(X_1 + X_2, Y) = \operatorname{cov}(X_1, Y) + \operatorname{cov}(X_2, Y)$
- $\operatorname{var}(X+Y) = \operatorname{var}(X) + \operatorname{var}(Y) + 2\operatorname{cov}(X,Y)$
- If X and Y are independent, then $\operatorname{cov}(X,Y) = 0$

69.20.2 Correlation Between Minus and Plus One

Let X, Y be random variables with existing variances. Then the following holds:

$$\rho_{X,Y} = 1 \iff Y = aX + b \text{ with } a > 0$$

$$\rho_{X,Y} = -1 \iff Y = aX + b \text{ with } a < 0$$

69.21 Variance of the Sum of Random Variables

Given the random variables $X_1, \ldots, X_n$ then

$$\operatorname{var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{var}(X_i) + 2\sum_{i \neq j} \operatorname{cov}(X_i, X_j)$$

If the random variables are independent then

$$\operatorname{var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{var}(X_i)$$

Special case: $n = 2$: $\operatorname{var}(X + Y) = \operatorname{var}(X) + \operatorname{var}(Y) + 2\operatorname{cov}(X, Y)$

69.22 Cauchy-Schwarz Inequality for Random Variables

Let X, Y be random variables with $\mathrm{E}[X^2]$ and $\mathrm{E}[Y^2]$ finite. Then the following holds:

$$|\mathrm{E}[X, Y]| \leq \sqrt{\mathrm{E}[X^2]\mathrm{E}[Y^2]}$$

It follows that

$$|\operatorname{cov}(X, Y)| \leq \sqrt{\operatorname{var}(X)\operatorname{var}(Y)}$$

69.23 Multivariable Normal Distribution

Given $X_1, \ldots, X_n$ independent $\mathcal{N}(0, 1)$, the density function of their joint distribution is given by

$$f(x_1, \ldots, x_n) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{\left(-\frac{1}{2}x_i^2\right)} = \frac{1}{(2\pi)^{n/2}} e^{\left(-\frac{1}{2}X^T \cdot X\right)} \sim \mathcal{N}(\vec{0}, I_n).$$

Thus, for $\mu \neq \vec{0} \in \mathbb{R}^n$ and $\Sigma \in \mathbb{R}^{n \times n}$ positive definite,

$$f(x_1, \ldots, x_n) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(X-\mu)^T \Sigma^{-1} (X-\mu)} \sim \mathcal{N}(\mu, \Sigma)$$

With d being the dimension of the space and $|\Sigma|$ being the determinant of the covariance matrix

$$\Sigma = \begin{bmatrix} \operatorname{cov}(X_1, X_1) & \cdots & \operatorname{cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \operatorname{cov}(X_n, X_1) & \cdots & \operatorname{cov}(X_n, X_n) \end{bmatrix}$$

where $\mu_i = \mathrm{E}[X_i]$ and $(X - \mu)^T \Sigma^{-1} (X - \mu)$ is a generalization of $\frac{(x-\mu)^2}{\sigma^2}$.

69.24 Computing Expected Values and Variances of Linear Combinations

69.24.1 Expected Value of Linear Combinations

For random variables $X_1, \ldots, X_n$ (not necessarily independent) with $\mathrm{E}[X_i] = \mu_i$ and constants $a_1, \ldots, a_n, b \in \mathbb{R}$, the expected value of a linear combination is always:

$$\mathrm{E}\left[b + \sum_{i=1}^{n} a_i X_i\right] = b + \sum_{i=1}^{n} a_i \mu_i$$

This holds regardless of independence or correlation structure.

69.24.2 Variance of Linear Combinations

The variance formula depends on whether the variables are independent:

Case 1: Independent Variables

If $X_1, \ldots, X_n$ are mutually independent with $\mathrm{var}(X_i) = \sigma_i^2$, then:

$$\mathrm{var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \sigma_i^2$$

Case 2: Dependent Variables (General Case)

If $X_1, \ldots, X_n$ are not independent, we must account for covariances:

$$\mathrm{var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \sigma_i^2 + 2\sum_{i<j} a_i a_j \mathrm{cov}(X_i, X_j)$$

Equivalently, in matrix notation with $a = (a_1, \ldots, a_n)^T$:

$$\mathrm{var}\left(\sum_{i=1}^{n} a_i X_i\right) = a^T \Sigma a$$

Special Cases:

- If $\mathrm{cov}(X_i, X_j) = 0$ for all $i \neq j$, we recover the independent case
- If $a_i = 1$ for all i (simple sum), then:

$$\mathrm{var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \sigma_i^2 + 2\sum_{i<j} \mathrm{cov}(X_i, X_j)$$

- Adding a constant b does not affect variance: $\mathrm{var}(b + Y) = \mathrm{var}(Y)$

69.24.3 Example: Bivariate Normal Distribution

Consider the random vector

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2(\mu, \Sigma),$$

with mean vector

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix},$$

and covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\,\sigma_1\sigma_2 \\ \rho\,\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix},$$

where $-1 < \rho < 1$ is the correlation coefficient.

Density:

The density of X is

$$f_X(x) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right),$$

where

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\,\sigma_1\sigma_2 \\ \rho\,\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

$$|\Sigma| = \sigma_1^2\sigma_2^2(1-\rho^2),$$

and

$$\Sigma^{-1} = \frac{1}{\sigma_1^2\sigma_2^2(1-\rho^2)} \begin{pmatrix} \sigma_2^2 & -\rho\,\sigma_1\sigma_2 \\ -\rho\,\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}.$$

Detailed Example: Spare Parts Delivery

Consider a delivery containing:

- One box of ball bearings: $X_1 \sim \mathcal{N}(125, 1)$
- Two boxes of gears: $Y_1, Y_2 \sim \mathcal{N}(84, 4)$
- Three boxes of screws: $Z_1, Z_2, Z_3 \sim \mathcal{N}(65, 3)$

All variables are independent. The total weight is:

$$G = X_1 + Y_1 + Y_2 + Z_1 + Z_2 + Z_3$$

Step 1: Expected Total Weight

Using linearity of expectation:

$$\begin{aligned} \mathrm{E}[G] &= \mathrm{E}[X_1] + \mathrm{E}[Y_1] + \mathrm{E}[Y_2] + \mathrm{E}[Z_1] + \mathrm{E}[Z_2] + \mathrm{E}[Z_3] \\ &= 125 + 84 + 84 + 65 + 65 + 65 \\ &= 488 \text{ kg} \end{aligned}$$

Step 2: Variance of Total Weight

Since all variables are independent:

$$\begin{aligned} \mathrm{var}(G) &= \mathrm{var}(X_1) + \mathrm{var}(Y_1) + \mathrm{var}(Y_2) + \mathrm{var}(Z_1) + \mathrm{var}(Z_2) + \mathrm{var}(Z_3) \\ &= 1 + 4 + 4 + 3 + 3 + 3 \\ &= 18 \text{ kg}^2 \end{aligned}$$

Therefore, $G \sim \mathcal{N}(488, 18)$ with standard deviation $\sigma_G = \sqrt{18} = 3\sqrt{2} \approx 4.243$ kg.

Step 3: Probability Calculations

Question: What is the probability that the delivery weighs more than 495 kg?

Standardize: $Z = \frac{G-488}{\sqrt{18}} \sim \mathcal{N}(0,1)$

$$\begin{aligned} P(G > 495) &= P\left(\frac{G-488}{\sqrt{18}} > \frac{495-488}{\sqrt{18}}\right) \\ &= P\left(Z > \frac{7}{\sqrt{18}}\right) \\ &= P(Z > 1.650) \\ &= 1 - \Phi(1.650) \\ &\approx 1 - 0.9505 = 0.0495 \end{aligned}$$

Thus, there is approximately a 4.95% probability that the delivery weighs more than 495 kg.

Alternative Scenario: Dependent Variables

Suppose instead that Y_1 and Y_2 are correlated with $\text{cov}(Y_1, Y_2) = 2$ (perhaps they're from the same production batch). Then:

$$\begin{aligned} \text{var}(G) &= 1 + 4 + 4 + 3 + 3 + 3 + 2\text{cov}(Y_1, Y_2) \\ &= 18 + 2(2) \\ &= 22 \text{ kg}^2 \end{aligned}$$

This increases the standard deviation to $\sqrt{22} \approx 4.690$ kg, making extreme values more likely.

69.25 Independence of Random Variables Under the Joint Normal Distribution

Given $X_1, \ldots, X_n$ random variables with a joint normal distribution. If $\text{cov}(X_i, X_j) = 0$ for all $i \neq j$, then they are mutually independent.

Importantly, this equivalence between zero correlation and independence is special to the multivariate normal distribution and does not hold in general.

69.26 Linear Combinations of the Joint Normal Distribution

Given $X_1, \ldots, X_n$ random variables with a joint normal distribution $\mathcal{N}(\mu, \Sigma)$. For any sequence of coefficients $a_1, \ldots, a_n \in \mathbb{R}$, the linear combination $\sum_{i=1}^{n} a_i X_i$ is also normally distributed.

Specifically, if $X \sim \mathcal{N}(\mu, \Sigma)$ and $a \in \mathbb{R}^n$, then:

$$a^T X \sim \mathcal{N}(a^T \mu, a^T \Sigma a).$$

If the random variables are independent with $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then:

$$b + \sum_{i=1}^{n} a_i X_i \sim \mathcal{N}\left(b + \sum_{i=1}^{n} a_i \mu_i, \sum_{i=1}^{n} a_i^2 \sigma_i^2\right).$$

Chapter 70

Statistics

In *probability*, we had models, and we assumed that they have their ideal form. In *Statistics*, we try to estimate the model from samples. We will approximate its parameters and test hypotheses about the model.

70.1 Basic Nomenclature

- *Population*: Set of all objects about which we make a statement.
- *Sample*: Finite set of the Population.
- *Sample Perimeter*: Number of samples, like the number of participants.
- *Trait*: Special characteristic we are looking for. They can be discrete, continuous, non-numerical (can be converted to numerical).
- *Trait-quantifier*: Amount of correspondence to a trait.

70.2 Random Sampling

A *random sample* of size n is a sequence of n random variables $X_1, X_2, \ldots, X_n$ which are independent and identically distributed (i.i.d.) with the same distribution as the random variable X.

70.3 Population Statistics

Let $x_1, x_2, \ldots, x_N$ be the elements of the population.

- The *population mean* (or expectation) of X is defined as

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i.$$

- The *population variance* of X is defined as

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2.$$

- The *population standard deviation* of X is defined as

$$\sigma = \sqrt{\sigma^2}.$$

70.4 The 5 Number Summary

In the *descriptive statistics* we call the *5 number summary* of a data set the list of numbers consisting of the: minimum, lower quantile, median, upper quantile and the maximum.

These numbers can be plotted in a box-plot.

70.5 Sample Statistics

Let $X_1, X_2, \ldots, X_n$ be a random sample of size n.

- The *sample mean* is defined as

$$\overline{X_n} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

  It has the property that $\overline{X_n} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$.

- The *sample variance* is defined as

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X_n})^2.$$

- The *sample standard deviation* is defined as

$$s = \sqrt{s^2}.$$

70.6 Finite n-Correction

$$\operatorname{var}(\overline{X_n}) = \frac{\sigma^2}{n} \cdot \frac{n-1}{N-1}$$

for very large N.

Proof:

$$\begin{aligned}
\operatorname{var}(\overline{X}) &= \operatorname{cov}(\overline{X}, \overline{X}) \\
&= \operatorname{cov}\left(\frac{1}{N}\sum_{i=1}^{n}\overline{X_i}, \frac{1}{N}\sum_{j=1}^{n}\overline{X_j}\right) \\
&= \frac{1}{n^2}\left(\sum_{i=1}^{n}\sum_{j=1}^{n}\operatorname{cov}(\overline{X_i}, \overline{X_j})\right) \\
&= \frac{1}{n^2}\left(\sum_{i=1}^{n}\operatorname{cov}(\overline{X_i}, \overline{X_j}) + \sum_{i=1}^{n}\operatorname{cov}(\overline{X_i}, \overline{X_j})\right) \\
&= \frac{1}{n^2}n\sigma^2 - \frac{n(n-1)}{N-1}\sigma^2 \\
&= \frac{\sigma^2}{n} \cdot \frac{n-1}{N-1}
\end{aligned}$$

$\mathfrak{QED}$

70.7 Frequency

Given a trait X with some quantifiers $(a_i)_{i\in I}$ for a discrete set I. Given a sample of observations $x_1, x_2, \ldots, x_n$ of X. The *absolute frequency* h_i of a_i is defined as

$$h_i = \sum_{j=1}^{n} \mathbb{1}_{x_j=a_i} i \in I.$$

The *relative frequency* is defined as

$$r_i = \frac{h_i}{n}, i \in I.$$

70.8 Order Statistic

Given $X_1, \ldots, X_n$ random variables. The random variables $X_{(1)}, \ldots, X_{(n)}$ are called *order Statistic* of the variables $X_1, \ldots, X_n$ if for each $\omega \in \Omega$ the following holds:

$$X_{(1)}(\omega) \leq X_{(2)}(\omega) \leq \cdots \leq X_{(n)}(\omega)$$

and for each i there exists k_i such that $X_(i)(\omega) = X_{k_i}(\omega)$.

Note

$$X_{(1)} = \min\{X_1, \ldots, X_n\}$$
$$X_{(n)} = \max\{X_1, \ldots, X_n\}$$

Example:

Given $5, 1, 3, 4, 2$ we get

$$x_{(1)} = 1, x_{(2)} = 2, \ldots, x_{(5)} = 5.$$

70.9 Empirical Median

Given $X_1, \ldots, X_n$ random variables with their corresponding order statistic. Their *empiric median* is defined as

$$\operatorname{med}(X_1, \ldots, X_n) = \begin{cases} X_{\left(\frac{n+1}{2}\right)} & n \text{ n is odd} \\ \frac{1}{2}\left(X_{X_{n/2}} + X_{n/2+1}\right) & n \text{ n is odd} \end{cases}$$

The median is equivariant for addition

$$\operatorname{med}(X_1 + a, \ldots, X_n + a) = \operatorname{med}(X_1, X_2, \ldots, X_n) + a.$$

Example:

Given $3.80, 4.33, 4.96, 3.71, 3.40$. What is the median?

Order statistic: $3.40, 3.71, 3.80, 4.33, 4.96$.

Median: $n = 5$, thus $\frac{n+1}{2} = 3$, and our median is $\operatorname{med}(X) = X_{(3)} = 3.80$.

70.10 Empirical Quantile

Given $X_1, \ldots, X_n$ random variables with their corresponding order statistic. The *empirical p-quantile* is defined as

$$q_p = \begin{cases} X_{\lfloor n \cdot p+1 \rfloor} & \text{if } np \notin \mathbb{N} \\ \frac{1}{2}(X_{np} + X_{np+1}) & \text{if } np \notin \mathbb{N} \end{cases}.$$

Note

$$\text{med}(X_1, \ldots, X_n) = q_{0.5}.$$

70.11 Modus

The most recurrent value in a set of samples for a particular feature is called the *modus*.

Given $x_1, \ldots, x_n$ samples of discrete classes $c_1, \ldots, c_n$. Define h_i as the recurrence of the class c_i in the sample set.

$$h_i = |\{x_j \in c_i : j = 1, \ldots, n\}|$$

for $i = 1, \ldots, d$. Then the class c_k is a modus if $h_k = \max_{i=1}^{d} h_i$.

70.12 Empirical Variance

Let $X_1, \ldots, X_n$ be random variables with mean $\overline{X}_n$. The *empirical variance* is defined as

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X_n})^2.$$

s is called the *empirical standard deviation*.

As a side note, if our samples are distributed uniformly then

$$\text{var}(X) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \overline{x_n})^2.$$

70.13 Sigma Rules

Given $X \sim \mathcal{N}(\mu, \sigma^2)$

- $\mathbb{P}(\mu - \sigma \leq x\mu + \sigma) \approx 68,3\%$
- $\mathbb{P}(\mu - 2\sigma \leq x\mu + 2\sigma) \approx 95,4\%$
- $\mathbb{P}(\mu - 3\sigma \leq x\mu + 3\sigma) \approx 99,7\%$

70.14 Absolute Deviation and Range

Let $X_1, \ldots, X_n$ be random variables with mean $\overline{X}_n$, minimum $X_{(1)}$ and maximum $X_{(n)}$.

The range *absolute deviation* is defined as

$$\text{MAD} = \frac{1}{n} \sum_{i=1}^{n} |X_i - \overline{X_n}|.$$

The sample *range* is defined as

$$r = X_{(n)} - X_{(1)} = \max\{X_1, \dots, X_n\} - \min\{X_1, \dots, X_n\}.$$

70.15 Interquartile Range

Let $X_1, \dots, X_n$ be random variables with 25% and 75% quantiles $q_{0.25}$ and $q_{0.75}$. The *interquartile range* is defined as

$$\text{IQR} = q_{0.75} - q_{0.25}.$$

70.16 Coefficient of Variation

Let $X_1, \dots, X_n$ be random variables with mean $\overline{X}_n$ and empirical variance s^2. The *empirical coefficient of variation* is defined as

$$V = \frac{s}{\overline{X_n}}.$$

70.17 Covariance

Covariance represents how much two values are dependent on each other

$$\text{cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X_n})(Y_i - \overline{Y_n}).$$

If the result is greater than zero, they are 'moving' in the same direction and the slope is positive, less 'opposite' with negative slope and equal to zero means that they are independent, and the slope is zero.

70.18 Pearson Correlation Coefficient

The *Pearson correlation coefficient* describes the degree to which two features are linearly correlated, and it is given by:

$$\rho_p = \frac{\text{cov}(X, Y)}{\sigma(X)\sigma(Y)}, \quad -1 < r < 1.$$

The closer the result, the closer the relationship between the values. Zero means independence.

Classification of the values:

$$\rho_p \begin{cases} 1 & \text{the line has a positive slope} \\ -1 & \text{the line has a negative slope} \\ 0 & \text{the line is constant} \end{cases}$$

The individual interpretation of these values and the slope of the line is dependent on the context.

70.18.1 Invariance Under Linear Transformation

Given random variables X, Y, we can transform them linearly

$$\tilde{X} = a_x X - b_x, \quad \tilde{Y} = a_y Y + b_y$$

and get the following result when applying the Pearson correlation coefficient

$$\rho_p(\tilde{X}, \tilde{Y}) = |\rho_p(X, Y)|$$

70.19 Spearman Correlation Coefficient

Given an order statistic $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ from the sample $x_1, \ldots, x_n$, we define the *rank* as:

$$\text{rank}(x_i) = k, \quad x_i = x_{(k)}.$$

We now define the *Spearman correlation coefficient* as

$$\rho_s = \frac{\sum_{i=1}^{n} \left(\text{rank}(x_i) - \frac{n+1}{2}\right)\left(\text{rank}(y_i) - \frac{n+1}{2}\right)}{\sqrt{\sum_{i=1}^{n} \left(\text{rank}(x_i) - \frac{n+1}{2}\right)^2 \sum_{i=1}^{n} \left(\text{rank}(y_i) - \frac{n+1}{2}\right)^2}}$$

Or in simpler terms, we compute the Pearson correlation of the ranks.

This coefficient is used to measure non-linear, monotone correlation.

70.19.1 Invariance Strictly Monotone Transformations

Given random variables X, Y, we can transform them using strictly monotone functions

$$\tilde{X} = g(X), \quad \tilde{Y} = h(Y).$$

The following holds for the Spearman correlation coefficient

$$\rho_s(\tilde{X}, \tilde{Y}) = \begin{cases} \rho_s(X, Y), & \text{if both } g \text{ and } h \text{ are either decreasing or increasing} \\ -\rho_s(X, Y), & \text{else} \end{cases}$$

70.20 Empirical Distribution Function

Let $X_1, \ldots, X_n$ be random variables. The function

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{X_i \leq x}$$

is the *empirical distribution* function

$$F_n(a_i) = \frac{1}{n} \sum_{j=1}^{n} \mathbb{1}_{X_j \leq a_i}$$

If the traits are sorted $a_1 < a_2 < \ldots$ then

$$F_n(a_i) = \frac{1}{n} \sum_{j=1}^{n} \mathbb{1}_{X_j \leq a_i} = \sum_{k=1}^{i} \frac{1}{n} \sum_{j=1}^{n} \mathbb{1}_{x_j = a_k} = \sum_{k=1}^{i} r_k$$

where r_k is the relative frequency of k.

Example:

Given the samples $3, 2, 4, 3, 1, 4, 3$ we have the traits $1, 2, 3, 4$ with relative frequencies

- $r_1 = \frac{1}{7}$
- $r_2 = \frac{1}{7}$

- $r_3 = \frac{3}{7}$
- $r_4 = \frac{2}{7}$

Then the empirical distribution function is given by

- $F_n(1) = r_1 = \frac{1}{7}$
- $F_n(2) = r_1 + r_2 = \frac{2}{7}$
- $F_n(3) = r_1 + r_2 + r_3 = \frac{5}{7}$
- $F_n(4) = r_1 + r_2 + r_3 + r_4 = 1$

70.21 Absolute Frequency Sum

$$nF_n(a_i) = n\sum_{k=1}^{i} r_k = \sum_{k=1}^{i} h_k$$

where h_k is the absolute frequency of k.

70.22 Grouped Data

When dealing with a lot of data, sometimes it is easier to put it into *classes* $A_1, \ldots, A_k$ which represent a certain range $[a, b], [a, b)$, etc. with a corresponding *absolute frequency* h_i, *relative frequency* r_i, *class middle* $\frac{a_{i+1}-a_i}{2} = \alpha_i$.

For the next definitions, the following example table will be used.

i	**Duration**	$\mathbf{h_i}$	$\mathbf{r_i}$	$\mathbf{F_i}$
1	$[0, 2)$	1650	0.3300	0.3300
2	$[2, 4)$	1111	0.2222	0.5522
3	$[4, 6)$	720	0.1440	0.6962
4	$[6, 8)$	508	0.1016	0.7978
5	$[8, 10)$	332	0.0664	0.8642
6	$[10, 15)$	427	0.0854	0.9496
7	$[15, 30]$	252	0.0504	1.0000

$F_i = \sum_{k=1}^{i} r_k$.

Empirical Distribution Function:

$$F_n(x) \approx \mathbb{P}(X \leq x)$$

Mean:

$$\overline{x} \approx \frac{1}{n}\sum_{i=i}^{k} h_i\alpha_i = \sum_{i=1}^{k} r_i\alpha_i$$

Empirical Variance:

$$s^2 \approx \frac{1}{n-1}\sum_{i=1}^{k} h_i(\alpha_i - \overline{x_n})^2$$

Empirical Quantile:

For the quantile q_p, find $F_i \geq p$ and $F_{i-1} < p$; this will give you a class $A(a_i, a_{i+1}]$. Then plug into the formula.

$$q_p = a_i + \frac{p - F_{i-1}}{F_i - F_{i-1}}(a_{i+1} - a_i)$$

Example:

$i = 2$, $A_2 = [2, 4)$ with $q = 0.5$

$$2 + \frac{0.5 - 0.33}{0.5522 - 0.33}(4 - 2) = 3.53$$

Mode Class

With $A_i = \text{interval}(a_i, a_{i+1})$

$$A_i \text{ with } h_i = \max_{j=1}^{k} h_j$$

70.23 Samples via Random Variables

Let $\mathcal{X} \subset \mathbb{R}^n$ and $X : \Omega \to \mathcal{X}$ be a random variable. A *sample is a realization* of $X = (X_1, \ldots, X_n)^T$, i.e., $X(\omega)$ for some $\omega \in \Omega$.

- The set $\mathcal{X}$ is called the *sample set*.
- The samples $x_1, \ldots, , x_n$ correspond to the coordinates of the vector

$$\begin{bmatrix} X_1(\omega) \\ \vdots \\ X_n\omega \end{bmatrix} = X(\omega)$$

70.24 Statistical Models

Let $\mathcal{X} \subset \mathbb{R}^n$ and $X : \Omega \to \mathcal{X}$ be a random variable.

- A *statistical model* is a family of probability measures P on $\mathcal{X}$.
- If $P = \{P_\theta : \theta \in \Theta\}$, then Θ is called the parameter space of P, and $\theta \in \Theta$ is referred to as a parameter.

Example:

n-fold coin toss

- Sample space: $\mathcal{X} = \{0, 1\}^n$.
- $X_i \sim \text{Ber}(p)$ (with values in $\{0, 1\}$).
- $X = (X_1, \ldots, X_n)^T$ is a random variable in $\{0, 1\}^n$.
- Sample: $X(\omega) = (x_1, \ldots, x_n) \in \{0, 1\}^n$.
- Informally: "Let $x_1, \ldots, x_n$ be a sample..."
- Marginal distribution of the independent tosses: $\mathbb{P}_p(X_1 = 1) = p$.
- Joint distribution: $\mathbb{P}_p^n$.
- Statistical model for the coin toss:

$$P = \{\mathbb{P}_p^n : p \in (0, 1)\}.$$

70.25 Estimators

Let $X : \Omega \to \mathcal{X}$ be a random variable with statistical model $P = \{P_\theta : \theta \in \Theta\}$.

1. A mapping $T_n : \mathcal{X} \to \Theta$, or the random variable $T_n(X)$, is called an *estimator* for θ.
2. An estimate is a realization of the estimator, i.e., $T_n(X)(\omega)$ for some $\omega \in \Omega$.
3. If for the true distribution of X we have $\mathbb{P}_X = \mathbb{P}_{\theta_0}$, then θ_0 is called the true parameter.

In simpler terms, an estimator is a function that takes in our samples and gives us an estimate of the parameter we are looking for.

70.26 Consistence of an Estimate

If $\hat{\theta}_n$ is an estimate of θ based on the sample size n, then $\hat{\theta}_n$ is consistent if

$$\mathbb{P}(|\hat{\theta}_n - \theta| < \varepsilon) \to 0 \quad asn \to \infty.$$

This means that $\hat{\theta}_n$ is an *unbiased estimate.*

70.27 Distortion

Let $X : \Omega \to \mathcal{X}$ be a random variable with statistical model $P = \{P_\theta : \theta \in \Theta\}$. For $\theta \in \Theta$, let E_θ denote the expectation with respect to P_θ, and let var_θ denote the corresponding variance.

- The *bias* of an estimator T_n is
$$\mathrm{Bias}_{T_n}(\theta) = \mathrm{E}_\theta[T_n(X)] - \theta.$$
- An estimator is called *unbiased* if $\mathrm{Bias}_{T_n}(\theta) = 0$ for all $\theta \in \Theta$, i.e.,
$$\mathrm{E}_\theta[T_n(X)] = \theta.$$
- A sequence of estimators $(T_n)_{n\in\mathbb{Z}}$ is called consistent if for all $\varepsilon > 0$ and all $\theta \in \Theta$,
$$\lim_{n\to\infty} \mathbb{P}_\theta(|T_n(X) - \theta| > \varepsilon) = 0.$$

70.28 Method of Moments Estimators

Let $X : \Omega \to \mathcal{X}$ be a random variable with $X = (X_1, \ldots, X_n)^T$ for independent, identically distributed random variables $X_1, \ldots, X_n$.

The *method of moments* estimates the target distribution parameters based on the moments of the sample. The first moment is the expectation, the second moment is the variance, etc.

Based on the first and second moments, we have the following *unbiased and consistent estimators*:

Given $X_1, \ldots, X_n$ random variables from a population with mean μ and variance σ^2. Then for some $iin1, \ldots, n$

$$\mathrm{E}[X_i] = \mu$$

$$\mathrm{var}[X_i] = \sigma^2$$

$$\sigma[X_i] = \sigma$$

70.29 Expected Value, Standard Deviation and Variance of the Sample Mean

$$\mathrm{E}[\overline{X_n}] = \mu$$

$$\sigma(\overline{X_n}) = \frac{\sigma}{\sqrt{n}}$$

$$\mathrm{var}(\overline{X_n}) = \frac{\sigma^2}{n}$$

70.30 Errors in the Method of Moments

Approximating the error when estimating the mean with the sample mean or variance can be challenging depending on the size of the population and the kind of distribution of the data.

Sometimes we can use our methods for unbiased and consistent estimators, but they might not be accurate enough.

Another method is bootstrapping, which consists of resampling with replacement from the sample and calculating the estimator.

70.31 Sample Variance as Estimator

The *sample variance* is given by

$$\hat{\sigma}_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X_n})^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i^2\right) - \overline{X_n^2}$$

and

$$\mathrm{E}[\hat{\sigma}^2] = \frac{\sigma^2}{n}\cdot\frac{N(n-1)}{N-1}.$$

Proof:

$$\mathrm{E}[X_i^2] = \mathrm{var}(X_i) + \mathrm{E}[X_i]^2 = \sigma^2 + \mu^2$$

From this fact we can derive that

$$\begin{aligned}
\mathrm{E}[\hat{\sigma}^2] &= \mathrm{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \overline{X_n^2}\right] \\
&= \frac{1}{n}\sum_{i=1}^{n}\mathrm{E}\left[X_i^2 - \overline{X_n^2}\right] \\
&= \frac{1}{n}\sum\left[(\mathrm{var}(X_i) + \mathrm{E}[X_i]^2 - \mathrm{var}(X_i) - \mathrm{E}[\overline{X_n}]^2\right] \quad \text{apply the n-correction result} \\
&= \frac{1}{n}\sum_{i=1}^{n}\sigma^2 + \mu^2 - \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right) - \mu^2 \\
&= \frac{\sigma^2}{n}\left[n - \left(1 - \frac{n-1}{N-1}\right)\right] \\
&= \frac{\sigma^2}{n}\cdot\frac{N(n-1)}{N-1}
\end{aligned}$$

$\mathfrak{QED}$

70.32 Normal Approximation to Distribution of the Sample Mean

Let $X_1, \ldots, X_n$ be random variables from a population with mean μ and variance σ^2. The *sample mean* $\overline{X_n}$ is approximately normally distributed with mean μ and variance $\frac{\sigma^2}{n}$ for large n.

Note that this fact comes from the *Central Limit Theorem*, which only applies to independent random variables. In our case, because our population is very large, we can assume independence due to the covariance between samples being very small.

This approximation gives us the ability to calculate probabilities about the sample mean using the normal distribution.

$$\mathbb{P}\left(\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \leq z\right) \approx \Phi(z)$$

70.33 Linear Regression

Given samples $(x_1, y_1), \ldots, (x_n, y_n)$ of random variables X and Y. The *linear regression* of Y on X is an approximation using the linear function

$$Y = aX + b$$

where we minimize *the sum of squared errors*.

$$Q(a, b) = \sum_{i=1}^{n} (y_i - (ax_i + b))^2.$$

by differentiating with respect to a and b.

$$\frac{\partial Q}{\partial a} = \sum_{i=1}^{n} ax_i^2 + bx_i - y_i x_i = 0$$

$$\frac{\partial Q}{\partial b} = \sum_{i=1}^{n} ax_i + b - y_i = 0$$

We proceed by defining the following terms

$$S = \sum_{i=1}^{n} x_i^w$$

$$R = \sum_{i=1}^{n} y_i x_i$$

Then

$$\frac{\partial Q}{\partial a} = \sum_{i=1}^{n} ax_i^2 + bx_i - y_i x_i = aS + bn\overline{X_n} - R$$

$$\frac{\partial Q}{\partial b} = \sum_{i=1}^{n} ax_i + b - y_i = an\overline{X_n} + nB - \overline{Y_n}.$$

The Hessian is given then by

$$H = \begin{bmatrix} S & n\overline{X_n} \\ n\overline{X_n} & n \end{bmatrix}$$

With $\det(H_f) = n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2 > 0$

In general, this comes from

$$\begin{bmatrix} S & n\overline{X_n} \\ n\overline{X_n} & n \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} R \\ n\overline{Y_n} \end{bmatrix}$$

by solving for a and b we get using the inverse of the Hessian

$$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{nS - n^2\overline{X_n}^2} \begin{bmatrix} n & -n\overline{X_n} \\ -n\overline{X_n} & S \end{bmatrix} \begin{bmatrix} R \\ n\overline{Y_n} \end{bmatrix}$$

Thus, our square error is minimized when $S > 0, \det(H_f) > 0$.

With a simple calculation we get

$$b = \frac{s_{XY}}{s_X^2} \quad a = \overline{Y_n} - b\overline{X_n}$$

where the *variance*

$$s_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{X_n})^2$$

and the *covariance*

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{X_n})(y_i - \overline{Y_n}).$$

70.33.1 Simplified Linear Regression

Ordinary least square with $y = b_0 + b_1 x + \cdots$ or for the simple case $\hat{y} = mx + b$.

- $m = b_1 = \frac{\sum_{i=1}^{n}(X_i - \overline{X_n})(Y_i - \overline{Y_n})}{\sum_{i=1}^{n}(X_i - \overline{X_n})^2}$
- $b = b_0 = \overline{Y_n} - b_1\overline{X_n}$

70.34 Likelihood

The *Likelihood* answers the following question: "Given the data observed, which parameter values make that data most plausible?".

$$\mathbb{P}_\theta(x_1, x_2, \ldots, x_n \mid \theta)$$

If we choose just some random parameters, we may get a probability distribution, but our observations could be so rare that we could misinterpret our data.

Assuming that our data is independent then we get the following function:

$$L(\theta) = \mathbb{P}_\theta(x_1, x_2, \ldots, x_n \mid \theta) = \mathbb{P}(x_1 \mid \theta)\mathbb{P}(x_2|\theta) \cdot \cdots \cdot \mathbb{P}(x_n \mid \theta).$$

Note that we choose to make a product out of this measurement we want to do, because for higher probabilities our product becomes bigger. Maximizing this function is interpreted as: maximizing the probability of observing the whole data given a variable parameter θ.

Example:

A coin has been flipped 4 times and we observed the following data: H H H H. Our intuition tells us that the coin is biased; thus we want to find the probability $p \in [0, 1]$ of getting heads, with counter probability $1 - p$ of getting tails.

Our sequence is given by

$$\mathbb{P}(X = HHHH) = p^4.$$

As we increase p the likelihood of observing our data increases. Thus, the maximum likelihood is achieved when $p = 1$.

70.34.1 Maximum Likelihood

Let $X = (X_1, \ldots, X_n)^T : \Omega \to \mathcal{X}$ be independent random variables with statistical model $P = \{P_\theta^n : \theta \in \Theta\}$. Furthermore, let $x_1, \ldots, x_n$ be a sample.

The function $L : \Theta \to [0, 1]$, defined by

$$L(\theta) = \begin{cases} \prod_{i=1}^n \mathbb{P}_\theta(X_i = x_i) & \text{for disc. random vars.} \\ \prod_{i=1}^n f_\theta(x_i) & \text{for cont. random vars.} \end{cases}$$

is called the *likelihood function*.

The parameter $\hat{\theta}$ that maximizes $L(\theta)$ is called the *maximum-likelihood estimate*.

$$\hat{\theta} = \text{argmax}_\theta L(\theta).$$

The mapping $\hat{\theta}(x_1, \ldots, x_n)$ is called the *maximum-likelihood estimator*.

70.34.2 The Log-Likelihood-Estimator

The likelihood function can be pretty challenging to optimize; therefore, the less complicated *Log-Likelihood-function* is often used for maximization problems.

$$l(\theta) = \begin{cases} \sum_{i=1}^n \log(\mathbb{P}_\theta(X_i = x_i)) & \text{for disc. random vars.} \\ \sum_{i=1}^n \log(f_\theta(x_i)) & \text{for cont. random vars.} \end{cases}$$

Example:

Given $X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$

$$f(x_1, \ldots, x_n|\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

We can calculate the log-likelihood function with $\theta = \lambda$ as

$$
\begin{aligned}
l(\lambda) &= \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \\
&\iff \ln\left(\prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}\right) \\
&= \sum_{i=1}^{n} \ln\left(\frac{\lambda^{x_i} e^{-\lambda}}{x_i!}\right) \\
&= \sum_{i=1}^{n} \ln(\lambda^{x_i} e^{-\lambda}) - \ln(x_i!) \\
&= \sum_{i=1}^{n} \ln(\lambda^{x_i}) + \ln(e^{-\lambda}) - \ln(x_i!) \\
&= \sum_{i=1}^{n} x_i \ln(\lambda) - \lambda - \ln(x_i!)
\end{aligned}
$$

Now we can derive with respect to λ and set to zero to find the maximum.

$$\frac{\partial l(\lambda)}{\partial \lambda} = \sum_{i=1}^{n} \frac{x_i}{\lambda} - 1 = 0.$$

The second derivative is

$$\frac{d\partial^2 l(\lambda)}{\partial \lambda^2} = \sum_{i=1}^{n} -\frac{x_i}{\lambda^2} < 0.$$

Always less than zero, thus we have a maximum.

The first derivative gives us

$$
\begin{aligned}
\sum_{i=1}^{n} \frac{x_i}{\lambda} - 1 &= 0 \\
\frac{1}{\lambda} \sum_{i=1}^{n} [x_i] - n &= 0 \\
\frac{1}{\lambda} \sum_{i=1}^{n} [x_i] &= n \\
\lambda = \frac{1}{n} \sum_{i=1}^{n} x_i &= \overline{X_n}
\end{aligned}
$$

We get the sample mean as the maximum likelihood estimator for λ in a Poisson distribution.

Properties of the Maximum Likelihood Estimator

Given

$$l(\theta) = \sum \ln(\mathbb{P}(x_i \mid \theta))$$

where $\hat{\theta} = \text{argmax}_\theta l(\theta)$. We get the following properties:

- $\hat{\theta}$ is consistent, $\hat{\theta} \to \theta_{\text{true}}$ for $n \to \infty$.

- $\hat{\theta} \sim \mathcal{N}\left(\theta_{\text{true}}, \frac{1}{nI(\theta_{\text{true}})}\right)$ where $I(\theta) = \mathrm{E}\left[\frac{\partial}{\partial\theta}\ln(\mathbb{P}(x \mid \theta))\right]^2 = -\mathrm{E}\left[\frac{\partial^2}{\partial\theta^2}\ln(\mathbb{P}(x \mid \theta))\right]$.
- *Cramér-Rao Inequality:* Given $X_1, \ldots, X_n$ i.i.d. random variables with the unbiased estimator $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$

$$\mathrm{var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

for all smooth $\mathbb{P}(X \mid \theta)$.

70.35 Bayesian Maximum A-Posteriori Estimation

Recall the MLE, but we model the probability for the observations of the samples as $\mathbb{P}(x_i \mid \theta)$ and treat them as real conditional probabilities instead of as a notational trick for the probability of x with the parameter θ, then we get

$$\mathbb{P}(\theta \mid x) = \frac{\mathbb{P}(x \mid \theta)\mathbb{P}(\theta)}{\mathbb{P}(x)}$$

If we optimize θ we get

$$MAP = \max \log(\mathbb{P}(x \mid \theta)\mathbb{P}(\theta)).$$

This is the *maximum a-posteriori estimation*.

70.36 Confidence Intervals

A *confidence interval* p for μ is a random interval centered at $\overline{X_n}$ that contains the true mean μ with probability p.

$$\mathbb{P}\left(-\epsilon \leq \overline{X_n} - \mu \leq \epsilon\right) = p = 1 - \alpha.$$

Given $\Phi(z) = 1 - \alpha$, if we standardize the expression we get

$$\mathbb{P}\left(-Z_{\frac{\alpha}{2}} \leq \frac{\overline{X_n} - \mu}{\sigma/\sqrt{n}} \leq Z_{\frac{\alpha}{2}}\right) = p = 1 - \alpha$$

This allows us to find the confidence interval given an α by:

Finding the critical value $Z_{\frac{\alpha}{2}}$ from the $Z-$table.

This is done by looking for $1 - \frac{\alpha}{2}$ in the body of the table and finding the corresponding z value.

In general,

$$(x, y) = \overline{X_n} \pm Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}.$$

Example:

Given *Confidence*$= 0.95$, $\alpha = 0.05$, $\overline{X_n} = 8978814.2$

The critical probability is $1 - \alpha/2 = 0.975$, which leads us to look at the $Z-$table, which gives us 1.96.

Next we have $\sigma - 12471455.74$ and $n = 15$

giving the result

$$(2667401.97, 15290226.43) = \overline{X_n} \pm Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}$$

70.36.1 Confidence Interval Formal Definition

Let $X = (X_1, \ldots, X_n)^T : \Omega \to \mathcal{X}$ be a random variable with statistical model $P = \{\mathbb{P}_\theta^n : \theta \in \Theta\}$. Furthermore, let $x_1, \ldots, x_n$ be a sample and let $\alpha \in [0, 1]$.

A mapping I that assigns to each sample $x = (x_1, \ldots, x_n)^T \in X$ a subset $I(x) \subset \Theta$ is called a $(1-\alpha)$-confidence region for θ if

$$\mathbb{P}_\theta\big(\theta \in I(X_1, \ldots, X_n)\big) \geq 1 - \alpha$$

for all $\theta \in \Theta$.

If $\Theta \subset \mathbb{R}$ and $I(x)$ is an interval for all $x \in X$, then I is called a $(1-\alpha)$-*confidence interval*.

70.37 t_{n-1}

The t_{n-1} distribution is used when the standard deviation σ is not known. It is defined as

$$t_{n-1} = \frac{\overline{X_n} - \mu}{s_n/\sqrt{n}}.$$

This distribution is symmetric about the y-axis and approximates the standard normal distribution for large n.

70.38 Confidence Intervals for the Expected Value and Standard Deviation of Normal Distributions

70.38.1 Confidence Interval for μ With Known σ

Given $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. We can build a confidence interval for μ as follows:

$$\left[\overline{X_n} - \frac{q\sigma}{\sqrt{n}}, \overline{X_n} + \frac{q\sigma}{\sqrt{n}}\right]$$

where q is the $(1-\alpha/2)$-quantile of the standard normal distribution. This is computed using the formula

$$q = \Phi^{-1}(1 - \alpha/2)$$

If σ^2 is not known, we can use s_n^2.

70.38.2 Confidence Interval for μ With Unknown σ

Given $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. We can build a confidence interval for μ as follows:

$$\left[\overline{X_n} - \frac{t_{n-1,\alpha/2}s_n}{\sqrt{n}}, \overline{X_n} + \frac{t_{n-1,\alpha/2}s_n}{\sqrt{n}}\right]$$

where $t_{n-1,\alpha/2}$ is the $(1-\alpha/2)$-quantile of the t-distribution with $n-1$ degrees of freedom.

70.38.3 Confidence Interval for σ With Known μ

Given $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. We can build a confidence interval for σ as follows:

$$\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{n,1-\alpha/2}}, \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{n,\alpha/2}}\right]$$

70.38.4 Confidence Interval for σ With Unknown μ

Given $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$. We can build a confidence interval for σ as follows:

$$\left[\frac{(n-1)s_n^2}{\chi^2_{n-1,1-\alpha/2}}, \frac{(n-1)s_n^2}{\chi^2_{n-1,\alpha/2}}\right].$$

70.39 Degrees of Freedom

The *degrees of freedom* of a random variable is the number of independent values or quantities which can be assigned to a statistical distribution.

For example, if we have n samples, and we know the mean, then we have $n-1$ degrees of freedom because the last value is determined by the mean and the other $n-1$ values.

70.40 Hypothesis Testing

Let $X : \Omega \to \mathcal{X}$ be a random variable with statistical model $P = \{\mathbb{P}_\theta : \theta \in \Theta\}$. A subset Θ_i of Θ with $H_1 \in \Theta_i$ is called a *hypothesis*.

The *null-hypothesis* is denoted by $H_0 : \theta \in \Theta_0$ and the *alternative hypothesis* by $H_1 : \theta \in \Theta_1$. The *hypothesis test* is a decision rule to decide whether to reject H_0 or not based on the sample.

Example:

We want to test if a coin is fair.

- $\Theta = [0, 1]$
- $\Theta_0 = \{0.5\}, H_0 : p = 0.5$
- $\Theta_1 = [0, 1] \setminus \{0.5\}, H_1 : p \neq 0.5$

70.41 Test Statistic

Let $X : \Omega \to \mathcal{X}$ be a random variable with statistical model $P = \{\mathbb{P}_\theta : \theta \in \Theta\}$. A *test statistic* is a mapping $\varphi : \mathcal{X} \to \{0, 1\}$ where $\varphi(x) = 1$ means that we reject H_0 and $\varphi(x) = 0$ means that we do not reject H_0.

70.41.1 Right Assertions

- *Type I Assertion:* Accepting H_0 when it is true.
- *Type II Assertion:* Rejecting H_0 when it is false.

70.41.2 Error Types

- *Type I Error:* Rejecting H_0 when it is true. The probability of this error is called the *significance level* α.
- *Type II Error:* Not rejecting H_0 when it is false. The probability of this error is called β.

We can sadly not minimize both errors at the same time, therefore, we choose a significance level α and try to minimize β.

70.42 Quality, Power and Level

Let $X : \Omega \to \mathcal{X}$ be a random variable with statistical model $P = \{\mathbb{P}_\theta : \theta \in \Theta\}$. For the test $\varphi : \mathcal{X} \to \{0, 1\}$ of the hypothesis $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$:

70.42.1 Quality Function

The function $\beta_\varphi \theta \to [0,1]$ defined by

$$\beta_\varphi(\theta) = \mathbb{P}_\theta(\varphi(X) = 1)$$

is called the *quality function* of the test φ.

Properties of the quality function:

- The probability of a Type I error is given by $\beta_\varphi(\theta) = \mathbb{P}_\theta(\varphi(X) = 1)$ for $\theta \in \Theta_0$.
- The probability of a Type II error is given by $1 - \beta_\varphi(\theta) = \mathbb{P}_\theta(\varphi(X) = 0)$ for $\theta \in \Theta_1$.

70.42.2 Significance Level

For $\alpha \in [0,1]$ the test φ is called a *significance level* α test if

$$\sup_{\theta \in \Theta_0} \beta_\varphi(\theta) \leq \alpha$$

$\sup_{\theta \in \Theta_0} \beta_\varphi(\theta)$ is called the *level* of the test.

70.42.3 Power

For $\theta \in \Theta_1$ is $\beta_\varphi(\theta)$ called the *power* of the test φ at θ.

Note the following:

- In practice, we choose a significance level α and look for a test φ with level $\leq \alpha$ that maximizes the power $\beta_\varphi(\theta)$ for $\theta \in \Theta_1$.
- A good test should have, for $n \to \infty$, $\beta_\varphi(\theta) \to 1$.

70.42.4 Ideal Matrix

	H_0 true	H_0 false
Reject H_0	$1 - \beta_\varphi(\theta) \geq 1 - \alpha$	$1 - \beta_\varphi(\theta)$
Do not reject H_0	$\beta_\varphi(\theta) \leq \alpha$	$\beta_\varphi(\theta)$

70.43 Critical Value

Let $X : \Omega \to \mathcal{X}$ be a random variable with statistical model $P = \{\mathbb{P}_\theta : \theta \in \Theta\}$. For the test $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$ and test φ with level α. A value $c_\alpha \in \mathbb{R}$ is called a *critical value* if

$$\varphi(x) = \begin{cases} 1 & \text{if } t(x) > c_\alpha \\ 0 & \text{else} \end{cases}$$

70.44 Test-Statistic

Let $X : \Omega \to \mathcal{X}$ be a random variable with statistical model $P = \{\mathbb{P}_\theta : \theta \in \Theta\}$. $t : \mathcal{X} \to \mathbb{R}$ is called a *test-statistic* for the test $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$ and c_α the critical value for the choice rule

$$\text{Reject } H_0 \text{ if } t(x) > c_\alpha$$

of test φ with level α.

70.45 p-Value

The *p-value* of a test-statistic t for the observation x is defined as

$$p = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(t(X) \geq t(x)) = 1 - \mathbb{P}_\theta(t(X) < t(x))$$

where θ is a parameter under the null-hypothesis H_0, $t(x)$ is the value of the test-statistic for the observation x and X is the random variable.

The p-value gives us the probability of observing a test-statistic value as extreme or more extreme than the one observed, assuming that the null hypothesis H_0 is true.

- A small p-value means the observed data would be rare under H_0.
- A large p-value means the data are consistent with H_0.

Example:

We use the number of heads in n coin tosses to test if the coin is fair. $X \sim \text{Bin}(n, p)$

Our hypotheses are

- $H_0 : p = 0.5$
- $H_1 : p \neq 0.5$

We define our test φ as follows:

$$\varphi(x) = \begin{cases} 1 & \text{if } |X - \frac{n}{2}| > k \\ 0 & \text{else} \end{cases}$$

$\frac{n}{2}$ is the expected value for our random variable X under H_0.

Our quality function is given by

$$\begin{aligned} \beta_\varphi(p) = \mathbb{P}_p(\varphi(X) = 1) &= \mathbb{P}_p(|X - \frac{n}{2}| > k) \\ &= 1 - \mathbb{P}_p(\left|X - \frac{n}{2}\right| \leq k) \\ &= 1 - \mathbb{P}_p(\frac{n}{2} - k \leq X \leq \frac{n}{2} + k) \\ &= \sum_{i=\lceil \frac{n}{2} \rceil - k}^{\lfloor \frac{n}{2} \rfloor + k} \binom{n}{i} p^i (1-p)^{n-i} \end{aligned}$$

To find the level of the test we compute for $n = 100$ and $k = 13$, and get $\beta_\varphi\left(\frac{1}{2}\right) = 0.0066$. Which is less than 0.05 thus our test is a level 0.05 test.

In summary, we have:

- **Significance Level:** $\alpha = 0.01$.
- **Sample:** $x = (x_1, \ldots, x_{100}) \in \{0, 1\}^{100}$.
- **Test-Statistic:** $t(x) = \left|\sum_{i=1}^{100} x_i - 50\right|$.
- **Critical Value:** $c_\alpha = 13$ If we get heads 33 times then H_0 will be rejected since $t(x) = |33 - 50| = 17 > 13 = c_\alpha$.
- **p-Value:**

$$p = \mathbb{P}_{1/2}(t(X) \geq 17) = \frac{1}{2^{100}} \sum_{i=0}^{33} \binom{100}{i} + \sum_{i=67}^{100} \binom{100}{i} \approx 0.0087$$

Due to $p = 0.0087 < \alpha = 0.01$ our observation seems to be rare under H_0 thus we reject H_0.

70.46 Standard Error

The *standard error* (SE) of a statistic is the standard deviation of its sampling distribution or an estimate of that standard deviation. It is often used to measure the precision of a sample mean by estimating the variability between sample means that one would obtain if multiple samples were drawn from the same population.

$$SE = \frac{\sigma}{\sqrt{n}}$$

70.47 General Procedure for Hypothesis Testing

You are given a sample $X = (X_1, \ldots, X_n)$ and you want to test some hypothesis about the distribution of the data. Mostly, you will have n, the sample size, α the significance level and some information about the distribution like the mean or variance.

The general procedure is as follows:

- Compute the missing information about the distribution (mean, variance, standard deviation, etc.) as needed.
- State the null and alternative hypothesis H_0 and H_1. This will influence the test statistic and rejection region.
- Determine the test statistic and the rejection region based on the hypotheses and significance level (Z-test, Chi-Test, t-Test, etc.).
- Compute the test statistic value from the sample data.
- Using the significance level α and the test statistic distribution, find the critical value(s) for the rejection region.
- Compute the p-value for the test statistic value if needed.
- Compute the quality function if needed.

We will take a look at the kind of tests that can be performed.

70.48 Hypothesis Testing on Normal Distributed Random Variables

Given $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d. random variables. We would want to test the following hypotheses:

- *Two-sided*: $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$.
- *Right-sided*: $H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$.
- *Left-sided*: $H_0 : \mu \geq \mu_0$ vs. $H_1 : \mu < \mu_0$.

If σ^2 is known we use the *Gauss-Test*, otherwise we use the *t-Test*.

70.49 Gauss-Test (Z-Test)

Given $X = (X_1, \ldots, X_n) \in \mathcal{X}$. We denote with z_α the $\alpha-$quantile of $\mathcal{N}(0, 1)$. We also have the test-statistic

$$Z = \frac{\overline{X_n} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

if $\mu = \mu_0 : Z \sim \mathcal{N}(0,1)$

Test Type	Rejection Region	Critical Value	p-Value
Two-Sided	$\|Z\| > z_{1-\alpha/2}$	$z_{1-\alpha/2}$	$2(1-\Phi(\|z\|))$
Right-Sided	$Z > z_{1-\alpha}$	$z_{1-\alpha}$	$1-\Phi(z)$
Left-Sided	$Z < -z_{1-\alpha}$	$-z_{1-\alpha}$	$\Phi(z)$

Use Case: This test is used when we want to know if the mean of a normal distribution is equal to some value μ_0 when the variance σ^2 is known.

70.49.1 Error Types in Gauss-Test

[h!]	Error Type I	Error Type II
$H_0 : \mu \leq \mu_0$	$1 - \Phi\left(z_{1-\alpha} + \sqrt{n}\frac{\mu_0-\mu}{\sigma}\right)$	$\Phi\left(z_{1-\alpha}\sqrt{n}\frac{\mu_0-\mu}{\sigma}\right)$
$H_0 : \mu \geq \mu_0$	$\Phi\left(z_{\alpha} + \sqrt{n}\frac{\mu_0-\mu}{\sigma}\right)$	$1 - \Phi\left(z_{\alpha} + \sqrt{n}\frac{\mu_0-\mu}{\sigma}\right)$
$H_0 : \mu = \mu_0$	$2 - \xi$	$\xi - 1$

with $\xi = \Phi\left(z_{1-\frac{\alpha}{2}} + \sqrt{n}\frac{\mu_0-\mu}{\sigma}\right) + \Phi\left(z_{1-\frac{\alpha}{2}} - \sqrt{n}\frac{\mu_0-\mu}{\sigma}\right)$

Example:

Given a sample of size $n = 400$, $\mu_0 = 1,74m$ and $\sigma_0^2 = 0.1m$ from height measurements of men in a city. We get a sample mean of $\overline{X_n} = 1.72m$. We want to test if the mean height is less than $1.74m$ with a significance level of $\alpha = 0.05$.

Following hypotheses are stated:

- $H_0 : \mu \geq 1.74m$
- $H_1 : \mu < 1.74m$

We compute the test-statistic

$$Z = \frac{\overline{X_n} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{1.72 - 1.74}{\frac{\sqrt{0.1}}{\sqrt{400}}} = -4$$

The critical value is given by $z_{0.05} = -1.645$. Since $-4 < -1.645$, we reject H_0 and conclude that the mean height is less than $1.74m$.

70.50 t-Test

Given $X = (X_1, \ldots, X_n) \in \mathcal{X}$. We denote with $t_{n-1,\alpha}$ the $\alpha-$quantile of the t_{n-1} distribution. We also have the test-statistic

$$T = \sqrt{n}\frac{\overline{X_n} - \mu_0}{s_n}$$

if $\mu = \mu_0 : T \sim t_{n-1}$

[h!] Test Type	Rejection Region	Critical Value	p-Value
Two-Sided	$\|T\| > t_{n-1,1-\alpha/2}$	$t_{n-1,1-\alpha/2}$	$2(1-F_{t_{n-1}}(\|t\|))$
Right-Sided	$T > t_{n-1,1-\alpha}$	$t_{n-1,1-\alpha}$	$1-F_{t_{n-1}}(t)$
Left-Sided	$T < -t_{n-1,1-\alpha}$	$-t_{n-1,1-\alpha}$	$F_{t_{n-1}}(t)$

Use Case: This test is used when we want to know if the mean of a normal distribution is equal to some value μ_0 when the variance σ^2 is unknown.

Example:

Given a sample of size $n = 25$ $\mu_0 = 100$, and we observed the following data:

[h!] 102	98	96	100	101
99	97	100	102	100
98	97	96	101	99
100	98	97	101	99
100	102	98	97	101

We want to test if the mean is different from 100 with a significance level of $\alpha = 0.05$.

We compute the sample mean $\overline{X_n} = 99.32$ and the sample standard deviation $s_n = 1.683$.

The test-statistic gives us

$$T = \sqrt{n}\frac{\overline{X_n} - \mu_0}{s_n} = \sqrt{25}\frac{99.32 - 100}{1.683} = -1.618$$

The critical value is given by $t_{24,0.975} = 2.064$. Since $|-1.618| < 2.064$, we do not reject H_0 and conclude that the mean is not significantly different from 100.

70.51 Between-Sample t-Tests

Given two independent samples $X_1, \ldots, X_n \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y_1, \ldots, Y_m \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$.

Define $D_i := X_i - Y_i \sim \mathcal{N}(\mu_X - \mu_Y, \sigma_{X+Y}^2)$.

Use Case: The null hypothesis is given by $H_0 : \mu_X = \mu_Y$ vs. $H_1 : \mu_X \neq \mu_Y$.

We get the test-statistic

$$T = \frac{\overline{X_n} - \overline{Y_m}}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim \mathcal{N}\left(\frac{\mu_X - \mu_Y}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}, 1\right)$$

We reject H_0 if $|T| > z_{1-\alpha/2}$.

70.51.1 Case With Unknown Variances

If the σ_X^2 and σ_Y^2 are unknown we use the test-statistic

$$T = \frac{\overline{X_n} - \overline{Y_m}}{\sqrt{\frac{m+n}{mn}s^2}}$$

with

$$s^2 = \frac{1}{n+m-2}((n-1)s_X^2 + (m-1)s_Y^2)$$

where s_X^2 and s_Y^2 are the sample variances of X and Y.

Use Case: Reject H_0 if $|T| > t_{n+m-2,1-\alpha/2}$.

70.52 Chi-Square-Test

Given $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d. random variables.

Use Case: We want to test $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 \neq \sigma_0^2$.

If μ is known we use the test-statistic

$$T_0 := \frac{1}{\sigma_0^2} \sum_{i=1}^{n} (X_i - \mu)^2 \sim \chi_n^2$$

Reject H_0 if $|T_0| > \chi^2_{n,1-\alpha/2}$ or $|T_0| < \chi^2_{n,\alpha/2}$.

If μ is unknown we use the test-statistic

$$T_1 := \frac{(n-1)\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X_n})^2}{\sigma_0^2} \sim \chi^2_{n-1}$$

Reject H_0 if $|T_1| > \chi^2_{n-1,1-\alpha/2}$ or $|T_1| < \chi^2_{n-1,\alpha/2}$.

Example:

Given $n = 18$, $\overline{x} = 109$, $s = 13,5$, $\mu = 100$ and $\sigma = 15$. We want to test if the standard deviation is less than 15 with a significance level of $\alpha = 0.05$.

We state the hypotheses:

- $H_0 : \sigma^2 \geq 15^2$
- $H_1 : \sigma^2 < 15^2$

We compute the test-statistic

$$T_1 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{17 \cdot 13.5^2}{15^2} = 16.38$$

The critical value is given by $\chi^2_{17,0.05} = 8.672$. Since $16.38 > 8.672$, we do not reject H_0 and conclude that the standard deviation is not significantly less than 15.

70.53 Bayesian Inference

The big idea is

$$\mathbb{P}(X \mid \theta) \to \mathbb{P}(\theta \mid X) = \frac{\mathbb{P}(X \mid \theta)\mathbb{P}(\theta)}{\mathbb{P}(X)}$$

where $\mathbb{P}(X \mid \theta)$ is the *likelihood* and $\mathbb{P}(\theta)$ is the *prior*.

We want to derive a statistical model for some collected data with its corresponding parameters using prior beliefs.

70.53.1 Bayesian Update

Iterative *Bayesian Update Process*

$$\mathbb{P}_{k+1}(\theta \mid X) = \frac{\mathbb{P}_k(X \mid \theta)\mathbb{P}_k(\theta)}{\mathbb{P}_k(X)}$$

with $\mathbb{P}_{k+1} = \mathbb{P}_k(\theta \mid X)$ and its base value is our defined prior.

Chapter 71

Information Theory

Information theory is a branch of applied mathematics and electrical engineering involving the quantification of information.

71.1 Core Concepts

An important note: depending on the type of random variable, discrete or continuous, the formulas use the logarithm base 2 or the natural logarithm.

71.1.1 Surprise

The *surprise* of an event $X = x$ is defined as

$$S(x) = \log\left(\frac{1}{P(X = x)}\right) = -\log(P(X = x)).$$

The intuition behind this definition is the inverse relationship between a high probability and the amount of information it delivers. An event with a high probability is not "surprising", while an improbable event is.

The logarithm is used to get rid of discontinuities and simplify work.

71.1.2 Entropy

Entropy is defined as the **expected surprise** of an event of a random variable X

$$H(X) = -\sum_{x \in X} P(X = x)S(x) = -\sum_{x \in X} P(X = x)\log(P(X = x)).$$

Or in a simpler way

$$H(X) = \sum p(x)\log\left(\frac{1}{p(x)}\right).$$

71.1.3 Conditional Entropy

Conditional Entropy is the expected surprise of an event of random variable X given the realization of another event of a random variable Y. It is defined as

$$H(X|Y) = -\sum_{y \in Y} P(Y = y)\sum_{x \in X} P(X = x|Y = y)\log(P(X = x|Y = y)).$$

For the intuition, given an event y, what is our average surprise under such conditions?

71.2 Mutual Information

Mutual Information measures how much information one random variable X contains about another random variable Y. It is defined as

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} P(X = x, Y = y) \log \left[\frac{P(X = x, Y = y)}{P(X = x)P(Y = y)} \right].$$

If one of the random variables is continuous, then we can make bins for different ranges of values and then proceed to work with the bins as the discrete classes.

$$I(X;Y) = H(X) - H(X|Y).$$

For the intuition, given a certain entropy (uncertainty) of X, after you observe Y you have a new entropy $H(X|Y)$, and the difference between the two is the mutual information.

Imagine that after observing Y, the amount of entropy of X is reduced by half; then Y contains a lot of information about X.

71.3 Cross Entropy

Cross Entropy is a measure of the difference between two probability distributions for a given random variable or set of events.

$$H(P, Q) = -\sum_{x \in X} P(X = x) \log(Q(X = x)).$$

where P is the true distribution and Q is the estimated distribution (model).

When there is no discrepancy between both distributions, the cross entropy is equal to the entropy of P.

It also has the following properties:

- *Lower Bound:*
$$H(P, Q) \geq H(P)$$
- *Asymmetry:*
$$H(P, Q) \neq H(Q, P)$$

71.4 KL-Divergence

Kullback-Leibler Divergence is a measure of how one probability distribution Q diverges from a second expected probability distribution P.

$$D_{KL}(P\|Q) = \sum_{x \in X} P(X = x) \log \left(\frac{P(X = x)}{Q(X = x)} \right).$$

It measures surprise from using Q to approximate P. It is the reverse of the cross entropy, where we use P to approximate Q.

We can also write it as

$$D_{KL}(P\|Q) = H(P, Q) - H(P).$$

71.5 Information Gain

Information Gain measures how much information a feature A gives us about the class label Y. It is defined as

$$IG(S, A) = H(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} H(S_v).$$

where $H(S)$ is the entropy of the original set S, S_v is the subset of S for which feature A has value v, and $|S|$ and $|S_v|$ are the sizes of sets S and S_v respectively.

In the context of Trees we can think about information gain as

$$IG = H(\text{parent}) - \sum_i w_i H(\text{child}_i).$$

where $w_i = \frac{|\text{parent}|}{|\text{child}_i|}$.

Example:

Suppose we have a dataset S with 10 samples, where 6 belong to class A and 4 belong to class B. The entropy of the original set S is calculated as:

$$H(S) = -\left(\frac{6}{10}\log_2\left(\frac{6}{10}\right) + \frac{4}{10}\log_2\left(\frac{4}{10}\right)\right) \approx 0.970$$

Now, let's say we split the dataset based on a feature A that can take two values: v_1 and v_2. After the split, we have:

Subset S_{v_1} with 4 samples (3 from class A and 1 from class B) and subset S_{v_2} with 6 samples (3 from class A and 3 from class B).

The entropy for each subset is calculated as:

$$H(S_{v_1}) = -\left(\frac{3}{4}\log_2\left(\frac{3}{4}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right)\right) \approx 0.811$$

$$H(S_{v_2}) = -\left(\frac{3}{6}\log_2\left(\frac{3}{6}\right) + \frac{3}{6}\log_2\left(\frac{3}{6}\right)\right) = 1.0$$

The weighted average entropy after the split is:

$$\sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} H(S_v) = \frac{4}{10} \cdot H(S_{v_1}) + \frac{6}{10} \cdot H(S_{v_2}) \approx 0.4 \cdot 0.811 + 0.6 \cdot 1.0 \approx 0.924$$

Therefore, the information gain from splitting on feature A is:

$$IG(S, A) = H(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} H(S_v) \approx 0.970 - 0.924 = 0.046$$

71.6 Gini Impurity

Gini Impurity is a measure of how often a randomly chosen element from the set would be incorrectly labeled if it were randomly labeled according to the distribution of labels in the subset. It is calculated as

$$Gini(S) = 1 - \sum_{i=1}^{C} p_i^2.$$

where C is the number of classes and p_i is the proportion of instances in class i in the set S.

The closer the Gini impurity is to 0, the more pure the set is.

Chapter 72

Machine Learning and Data Science

Machine Learning is an area of computer science which focuses on making accurate predictions based on gathered data and mathematical models. It heavily uses *data science*, which is another area of computer science with the goal of analyzing data, identifying patterns and performing statistical analysis.

72.1 Core Vocabulary

Our main terminologies and notation are:

- Historical Data which refers to the data on which we are going to base our model.
- Feature/Predictor vector X: This is a vector whose components represent the different numeric values of the characteristics of some object.
- Function f: This is called the *target function* $f : \mathcal{X} \to \mathcal{Y}$, which makes a perfect approximation.
- Label y: This is the image of some $x \in \mathcal{X}$.
- Data Sample D: this is a set of pairs (x_i, y_i)....

Our goal is to find a function $g : \mathcal{X} \to \mathcal{Y}$ which approximates f.

72.1.1 Classification and Regression

Classification predicts categories based on discrete labels, i.e., classes or groups, while *regression* predicts continuous values, i.e., identifying trends and patterns.

Common classification functions with example models:

- Binary cross entropy in logistic regression, ironically.
- Categorical cross entropy.
- Hinge loss for support vector machines.

Common regression functions with example models:

- Mean square error in logistic regression.
- Information gain in decision tree regressors.

72.1.2 Solution Strategy

- The learning algorithm $\mathcal{A}$ uses data to select, from a set of candidate functions, the function g that best approximates the target function f.
- $\mathcal{H}$: Set of candidate functions (hypothesis set). (Example: linear functions)
- $g \in \mathcal{H}$ is also called the *final hypothesis*, which the algorithm selects at the end.

72.2 Central Learning Problems in Machine Learning

72.2.1 Supervised Learning

- Sought: Unknown target function.
- Data: Input-output pairs.
- Approximate the unknown target function f_{target} with a hypothesis h such that $h \approx f_{\text{target}}$.

72.2.2 Unsupervised Learning

- Sought: Unknown function that describes the data.
- Data: Only input data (no labels).
- Find a function f that describes the structure or distribution of the data well.

72.2.3 Reinforcement Learning

- Sought: Unknown strategy S that maximizes a reward.
- Data: (1) Input (2) Possible output (3) Evaluation of output (reward).
- Find a function f that approximates the optimal strategy S^* that maximizes the expected reward.

72.3 Data Preparation

The data set $\mathcal{D}$ will be split into *training* and *test* data to minimize our margin of error.

If a model is trained using only the same data again and again, it will become less reliable with new data which differs from its training set, making out-of-sample errors difficult to spot.

Another way of improving the quality of the data set is to add a certain degree of noise to our sample data, thus converting y into a probability variable dependent on x. Therefore, $f \to P(y \mid x)$.

72.3.1 Data Preparation Techniques

- *Train-Test-Split*: Splitting the data set into the data used for training and validation.
- *Encoding Non-numerical Values*: Categorical data and other non-numerical features need to be encoded or discarded, depending on the context.
- *Dropping/Replacing NaN-Values*: Either we discard columns or rows which contain incomplete data or we fill the gaps using some placeholder values, which should have a neutral effect during the training process, like 0.

72.3.2 Explorative Data Analysis

Explorative data analysis is the process of analyzing, visualizing and exploring the data using software, statistical methods, intuition, etc., with the goal of establishing hypotheses about the data for further analysis. This step is to be done extensively before any real training of the data.

72.3.3 Feature Engineering

Feature engineering is the process of selecting the right features/predictors using domain-knowledge. This procedure is also accompanied by the selection of the right mathematical model for the desired goal.

The main difference from the explorative data analysis is that the goal is to create data-driven models, i.e., mathematical models from our data to make predictions instead of just exploring and analyzing the data.

72.3.4 Feature Scaling

Feature scaling is a technique used to standardize the range of independent variables or features or predictors of data. In machine learning, it is crucial because many algorithms are sensitive to the scale of the input data.

Common methods for feature scaling include:

- *Min-Max Scaling (Normalization)*: This technique scales the data to a fixed range, usually 0 to 1. The formula is:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}.$$

  By adding $(b - a)$ to the formula we can scale it to any range $[a, b]$:

$$X' = a + \frac{(X - X_{min})(b - a)}{X_{max} - X_{min}}.$$

- *Standardization (Z-score Normalization)*: This method transforms the data to have a mean of 0 and a standard deviation of 1. The formula is:

$$Z = \frac{X - \mu}{\sigma},$$

  where μ is the mean and σ is the standard deviation of the feature.

- *Robust Scaling*: This method uses the median and the interquartile range (IQR) for scaling, making it robust to outliers. The formula is:

$$X' = \frac{X - \text{median}}{\text{IQR}}.$$

72.4 Hyper Parameters

A *hyper-parameter* is a parameter used by a model/algorithm which cannot be approximated directly during the training process. For example, the k in K-Means is given at the start.

72.4.1 Hyper-Parameter Tuning

Hyper-parameter tuning is the process of improving the hyper-parameters via validation methods. In such methods not only is the model tested against other models with the same hyper-parameters, but also against the same model with different hyper-parameters. We do this via cross-validation among other techniques.

72.5 Accuracy Metrics for Classification

A *confusion matrix* is a table that is often used to describe the performance of a classification model (or classifier) on a set of test data for which the true values are known. It allows the visualization of the performance of an algorithm.

For example, in a binary classification problem, the confusion matrix is a 2×2 table that summarizes the counts of true positive TP, true negative TN, false positive FP, and false negative FN predictions made by the model.

	Positive	Negative
Positive	TP	FN
Negative	FP	TN

72.5.1 Accuracy

The *accuracy* is then given by

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Portion of correctly predicted observations to the total observations. Ideally high.

72.5.2 Precision

The *precision* is then given by

$$\text{Precision} = \frac{TP}{TP + FP}.$$

Ratio of the correctly predicted positive observations to the total predicted positive observations. Ideally high.

72.5.3 Recall

The *recall* or true positive rate (TPR) is then given by

$$\text{Recall} = \frac{TP}{TP + FN}.$$

Ratio of real positives to the total of positives. Ideally high.

72.5.4 Specificity

The *specificity* or true negative rate (TNR) is then given by

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{TN}{N}.$$

Number of correctly classified negative observations in relation to the correctly and incorrectly classified negative observations. Ideally high.

72.5.5 The F1 Metric

The *F1 metric* is a measure which combines two of the previous metrics into one:

$$\text{F1} = 2\frac{\text{Precision Recall}}{\text{Precision} + \text{Recall}}.$$

72.5.6 Receiver Operating Characteristic (ROC) Curves

A *ROC curve* is a graphical representation of a set of points (y_i, x_i) with $y_i = \text{Recall}_\theta$, $x_i = \text{Specificity}_\theta$, where θ is the threshold for a classification problem.

When we read a value like $(0.8, 0.2)$, we can say that 80 percent were correctly classified and 20 percent were misclassified.

The higher the curve, the better our classification model. This can be compared using the area under the ROC curve of different curves.

72.6 Accuracy Metrics for Regression

72.6.1 R-Squared

R-squared provides a measurement of how much variation in the response the model explains.

$$R^2 = \frac{\text{var(fitted)}}{\text{var(response)}}, \quad 0 \leq R^2 \leq 1.$$

with

$$\text{var(response)} = \text{var(fitted)} + \text{var(residuals)}.$$

This metric is not available for all models. Linear regressions, among others, support this metric.

A problem of this metric is that it increases when more predictors are added, which means that even irrelevant coefficients are taken into consideration.

72.6.2 Adjusted R-Squared

This is a penalized version of R^2 to avoid taking irrelevant predictors.

$$\text{adjusted } R^2 = 1 - (1 - R^2)\frac{N-1}{N-p},$$

with N as the sample size and p as the number of coefficients in our linear model.

72.6.3 Mean Squared Error

This is the classical function for our linear regression given by

$$\text{MSE} = \sum_{i=1}^{N}(y_i - \tilde{y_i})^2.$$

72.7 Validation

Validation refers to the process of evaluating our models to approximate the out-of-sample error, compare them, and find *hyperparameters*.

The hypothesis gained from a model which is going to be validated is marked as $\overline{g_n}$.

The types of data which arise from the splitting of the data are:

- *Training Data* $\mathcal{D}_T$: Data used for training the model.
- *Validation Data* $\mathcal{D}_V$: Used for testing our model against other models in situations like hyper-parameter tuning. We normally use a size of K datapoints where $K = N/5$.

Quote From the Statistical Learn-Theory

The more often you use a data-set for training a model, taking decisions, etc., the harder it is to approximate the capacity of the data-set to approximate the out-of-sample error.

72.7.1 Leave One Out Cross Validation

Leave One Out Cross Validation works by always leaving **one data point out**, instead of a fold. This is the same as an N-fold cross validation.

The nomenclature:

- $e_n := E_{\text{val}}(\overline{g_n}) = e(\overline{g_n}(x_n), y_n)$, represents the point-wise error, which is the number of misclassified samples.
- *Cross Validation Error* $E_{\text{CV}} = \frac{1}{N}\sum_{n=1}^{N} e_n$

72.7.2 V-Cross Validation

V-Cross Validation is a technique to improve a model's accuracy by training the model on different splits of the data.

The procedure consists of splitting the data-set into V different sub-sets of equal size called *folds*. Then, we train our model V-times, each time using $V-1$ disjoint sub-sets for training and the remaining one for testing. The final out-of-sample error is given by the average of all the out-of-sample errors obtained from each iteration. We will have in total V models.

The nomenclature:

- *Number of data points in the training set* $N_{\text{train}} = \frac{V-1}{V}N$
- *Number of data points in the validation set* $N_{\text{val}} = \frac{1}{V}N$
- $e_n := E_{\text{val}}(\overline{g_n}) = e(\overline{g_n}(x_n), y_n)$, which, again, is the number of misclassified samples.
- *Cross Validation Error* $E_{\text{V-fold CV}} = \frac{1}{V}\sum_{n=1}^{V} E_v$

Cross validation is mainly used for *model selection*, which is the process of comparing different machine learning algorithms, and *hyper-parameter-tuning*, not to directly deliver a model.

72.7.3 Hold-Out Validation

Hold-Out validation works by just splitting the data set into training and validation data. We only get one model.

72.7.4 Sensitivity and Specificity

Sensitivity measures the proportion of actual positives in the data that are correctly identified as such (true positive rate). It is calculated as:

$$\text{Sensitivity} = \frac{TP}{TP+FN}.$$

The denominator represents the sum of the column corresponding to actual positive cases, while the numerator represents the true positive cases.

Specificity measures the proportion of actual negatives in the data that are correctly identified as such (true negative rate). It is calculated as:

$$\text{Specificity} = \frac{TN}{TN+FP}.$$

The numerator is the sum of all cells which are not in row or column of positive cases, while the denominator is the sum of the row corresponding to actual negative cases plus our previous numerator.

72.8 Learning Principles

Ockham

The simplest model which approximates the data is also the most plausible. Hence, we should aim for the simplest model (least parameters possible) which approximates the data well enough.

Corrupted Samples

If the samples are corrupted, biased or contain errors, then the model will also not be neutral and will give false approximations.

Data Snooping

If a data-set affects the learning process, then its capacity to approximate the out-of-sample error was reduced. Therefore, we should not mix test and training data.

72.9 Feature Selection

This is the process of choosing the "right" features to use in a model. This can be done manually using statistical testing, examining correlation, and domain knowledge or automatically via algorithms.

The automatic variants do deliver accurate models on paper, but they may not reflect the reality of the function we are trying to approximate, or they do not take into account certain relationships in the data. For these reasons, regularization methods are mostly preferred, but feature selection is still useful if done correctly.

72.9.1 Best Subset Selection

Best subset selection consists of testing all 2^p possibilities of combining p features and then perform cross validation with all possibilities to get the best features.

This is obviously very expensive.

72.9.2 Forward Stepwise Selection

Adding variables one at a time, choosing a variable which delivers the best result.

72.9.3 Backward Stepwise Selection

Removing variables one at a time, choosing the one which contributes the least to the error function.

72.10 Errors

72.10.1 In Sample Errors

The *in-sample-error* is the mean of the amount of samples which were incorrectly labeled.

$$E_{\text{in}}(h) = \frac{1}{N}\sum_{i=1}^{N} W(h(x_i) \neq f(x_i))$$

where

$$W(\text{Expr}) = \begin{cases} 1 & \text{Expr} = \text{True} \\ 0 & \text{Expr} = \text{False} \end{cases}$$

It is the median of the errors made on the training data.

72.10.2 Out of Sample Errors

This refers to all errors which can occur due to the lack of data. Obviously, we do not have all possible samples which will exist. But we can estimate this error using the split of the training and validation data by computing the accuracy of the model.

$$E_{\text{out}}(h) = P[h(x) \neq f(x)]$$

These errors have a randomness factor, or in other words an unknown probability distribution. They are harder to calculate due to their nature.

72.11 Approximation and Generalization

When trying to approximate the target function f we encounter the problem that our data may not give us the exact function because of the lack of data. We want to minimize the in-sample error but also make our approximation as general as possible.

Approximation: Small in-sample-error

Generalization: $E_{\text{out}} \approx E_{\text{in}}$.

72.11.1 Bias and Variance

Fitting the training data well, but making poor predictions, is called the *bias-variance* trade-off.

The *bias* measures how much our model differs from our training data (ground truth), while *variance* tells us how well our model can make future predictions. A high bias leads to **underfitting** while a high variance to **overfitting**.

With this in mind we now define

$$E_{\text{out}} = \text{Bias} + \text{Variance}.$$

Example:

Our target is $\sin(\pi x)$, and our hypothesis-sets are $\mathcal{H}_0$ the set of all horizontal lines $h(x) = b$ and $\mathcal{H}_1$ the set of all lines $h(x) = ax + b$. The training-sets are two data-points of the interval $[-1, +1]$.

72.11.2 Learning Curves

Learning curves plot E_{in} and E_{out} depending on the number of the data-points.

72.11.3 Overfitting

Overfitting refers to the situation when a model adapts itself too much to a given data-set. Precisely, it is the process in which hypotheses h continuously reduce the in-sample error, but leading to a bigger out-of-sample error.

We define *overfitting* for a hypothesis h, if for another hypothesis h' the following is true

$$E_{\text{in}}(h) < E_{\text{in}}(h') \,\text{but}\, E_{\text{out}}(h) > E_{\text{out}}(h')$$

In summary, noise in the data (stochastic or deterministic) has a negative effect on the learning process and can lead to overfitting. To address this problem we will use *regularization*.

72.11.4 Underfitting

Underfitting refers to the case when the model has not been trained with enough data to make accurate approximations.

Underfitting can be handled by: using more data for the training, reducing regularization, feature selection, etc.

72.11.5 The Augmented Error

We define the *augmented error* as

$$E_{\text{aug}}(w) = E_{\text{in}}(w) + \frac{\lambda}{N} w^T w,$$

where λ is the parameter for how much we should regularize, $w^T w$ is the penalty term based on the weight of the vector and N is the number of data points.

72.12 Regularization

Regularization is the process of making the algorithm less sensitive to out-of-sample data. It helps the learning algorithm to minimize the out-of-sample error. This is accomplished by different methods like adding error terms of a *regularization vector*

$$\beta = [\beta_0, \beta_1, \dots]$$

with one or multiple parameters λ.

72.12.1 LASSO Regression L1

This is similar to Ridge regression in the sense that we are adding a penalty term times λ at the end of our optimization problem, but instead of summing the squares we add the absolute value

$$\text{LASSO} : h(x_i) + \lambda \sum_{i=1}^{p} |\beta_i|.$$

It is important to note that both this and Ridge regularization are **very sensitive** to the scaling of our features.

Note that adding the absolute values is equivalent to defining a multidimensional **polygon**.

72.12.2 Ridge Regression L2

This is a regularization technique used in linear models to prevent overfitting by discouraging the model from assigning excessively large values to its parameters.

To define the region in which we want our parameters to be we use a **hyper-sphere** as region by setting a sum of squares as a constraint.

$$\text{Ridge} : h(x_i) + \lambda \sum_{i=1}^{p} \beta_i^2.$$

Example:

We use linear regression for this derivation:

$$\begin{aligned}
E_{\text{in}}(h) &= \frac{1}{N} \sum_{n=1}^{N} (h(\phi(x_n)) - y_n)^2 \\
E_{\text{in}}(h) &= \frac{1}{N} \sum_{n=1}^{N} (w^T z_n - y_n)^2 \\
E_{\text{in}}(h) &= \frac{1}{N} (Zw - y_n)^T (Zw - y)
\end{aligned}$$

And

$$Z := \begin{bmatrix} z_1^T \\ z_2^T \\ \vdots \\ z_n^T. \end{bmatrix}$$

Our augmented error is given by

$$\begin{aligned} E_{\text{aug}}(w) &= E_{\text{in}} + \frac{\lambda}{N} w^T w \\ &= \frac{1}{N}(Zw - y_n)^T (Zw - y) + \frac{\lambda}{N} w^T w \end{aligned}$$

To minimize it we want to set its gradient to 0 which gives us

$$\begin{aligned} \nabla_w E_{\text{aug}}(w) = 0 \implies & Z^T(Zw - y) + \lambda w = 0 \\ & Z^T Z w - Z^T y + \lambda w = 0 \\ & (Z^T Z - Z^T I)w + Z^T y = 0 \\ & w_{\text{reg}} = (Z^T Z + \lambda I)^{-1} Z^T y. \end{aligned}$$

Finally, as a remainder the formula without regularization

$$w_{\text{lin}} = (Z^T Z)^{-1} Z^T y.$$

72.13 The Perceptron

The perceptron is our first model in machine learning. It consists of having a separate vector $w \in \mathbb{R}^d$ which consists of weights which are going to be associated with the components of $x \in \mathbb{R}^d$. We also define $y \in \{-1, 1\}$, and some value b to work as our border between the categories of y. We classify our prediction based on the result

$$\sum_{i=1}^{d} w_i x_i > b \text{ accept },$$

$$\sum_{i=1}^{d} w_i x_i < b \text{ reject }.$$

More precisely

$$\begin{aligned} \sum_{i=1}^{d} w_i x_i + 1 \cdot b &> 0 \\ \sum_{i=1}^{d} w_i x_i + x_0 \cdot w_0 > 0 \mid x_0 = 1 \quad w_0 &= b \\ \sum_{i=0}^{d} w_i x_i &> 0 \\ w^T x &> 0. \end{aligned}$$

Finally, to get our respective y we define the following function

$$\text{sign}(w^T x) = \begin{cases} 1, & \sum_{i=1}^{d} w_i x_i > 0 \\ -1, & \sum_{i=1}^{d} w_i x_i < 0. \end{cases}$$

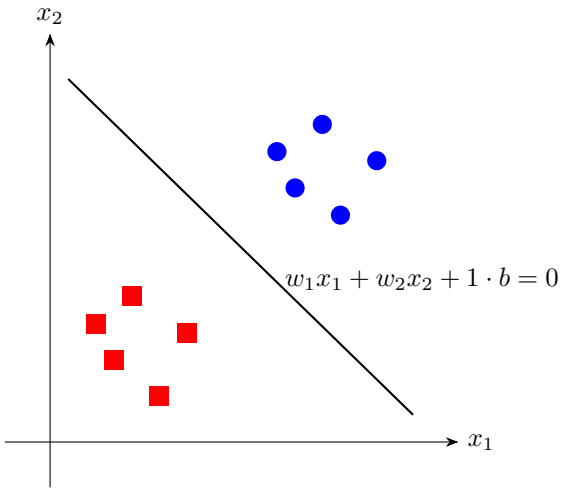
And we get our final perceptron equation

$$h(x) = \text{sign}(w^T x).$$

For the case $d = 2$, our vector w defines a *line*.

$$\begin{aligned} w_0 \cdot 1 + w_1 x_1 + w_2, x_2 = 0 \\ x_2 = -\frac{w_1}{w_2} x_1 - \frac{w_0}{w2} \\ x_2 = m x_1 + c. \end{aligned}$$

Here a visualization



Note that y is not used for the visualization, as it is just using in the sign function to as for the classification.

72.14 The Perceptron Learning Algorithm (PLA)

The *pereptron learning algorithm (PLA)* modifies w until the data are linearly separated, as long it is possible. It can be the case that in the current space it is not possible. Formally, if there exists a w^* such that $h(x_i) = y_i, \forall (x_i, y_i) \in \mathcal{D}$.

It goes as follows:

- *Data:* (x_i, y_i) with x being a vector $i = 1, \ldots, N$.
- *Assumption:* The data are linearly separable, i.e., we can separate the data into their classes using a line ($d = 2$) or a hyperplane ($d > 2$).

Let $t = 0, 1, 2, \ldots$ be the current iteration step, and let $w^{(t)}$ denote the vector w at iteration step t. Set $t = 0$ and $w^{(0)} = 0$.

1. Use the perceptron with $w^{(t)}$ and classify all data $\mathbf{x_i}, i = 1, \ldots, N$. The labels predicted by the perceptron are denoted by $y_{i,\text{pred}}$.
2. Select any pair (x_j, y_j) that was misclassified (i.e., $y_{j,\text{pred}} \neq y_j$), and denote it by $(x^{(t)}, y^{(t)})$. If no such pair exists, terminate.

3. Update the weight vector w using the following rule:

$$w^{(t+1)} = w^{(t)} + y^{(t)}x^{(t)}.$$

This just means taking the current w and adding the vector x times the corresponding label y which was expected.

4. Increase t by 1 and return to step 1.

5. After we got our final weights, we get our line or hyperplane by manipulating our expression $\sum_{i=0}^{d} x_i w_i = 0$.

72.14.1 Pocket Algorithm

The *pocket algorithm* is an extension of the perceptron learning algorithm to handle non-linear separable data-sets. This achieved by setting a maximum number iterations while also keeping track of the vector with the lowest in-sample error value.

1. Initialize w and set $w_{\text{best}} = w$.

2. Compute the current in-sample error $E_{\text{best}} = E_{\text{in}}(w)$.

3. For each iteration:

   (a) If a point (x_i, y_i) is misclassified, update:

$$w \leftarrow w + y_i \tilde{x}_i.$$

   (b) Compute the new error $E_{\text{in}}(w)$.

   (c) If $E_{\text{in}}(w) < E_{\text{best}}$, then

$$w_{\text{best}} \leftarrow w, \quad E_{\text{best}} \leftarrow E_{\text{in}}(w).$$

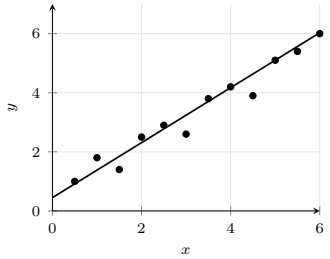
4. After all iterations, output w_{best}.

We use the following formula for error.

$$E_{\text{in}}(w) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\big(\text{sign}(w^T \tilde{x}_i) \neq y_i\big).$$

72.15 Linear Regression

Given our data set D with N samples and m features, we have the assumption that our data is linearly related, and we want to find a fitting line to approximate it $\hat{y} = wx + b$ where w is a vector of weights.



One way of doing *linear regression* is via the method of the least squares found in the linear algebra section of this book:

$$w = (X^T X)^{-1} X^T y = X^{\dagger} y.$$

Yet, this is computationally very expensive and does not gives us b. An alternative way is by minimizing the *mean squared error* function given by

$$\text{MSE} = J(w,b) = \frac{1}{N}\sum_{i=1}^{N}(y_i - (w^T x_i + b))^2$$

For this we compute the gradient of of $J(w,b)$

$$J'(w,b) = \begin{bmatrix} \frac{\partial J}{\partial w} \\ \frac{\partial J}{\partial b} \end{bmatrix} = \begin{bmatrix} \frac{1}{N}\sum -2x_i(y_i - (w^T x_i + b)) \\ \frac{1}{N}\sum -2(y_i - (w^T x_i + b)) \end{bmatrix} = \begin{bmatrix} \frac{1}{N}\sum 2x_i(\hat{y}_i - y_i)) \\ \frac{1}{N}\sum 2(\hat{y}_i - y_i) \end{bmatrix}$$

Then by applying gradient descent we can approach a minimum of our function by taking in the direction negative to the gradient multiplied by a *learning rate* $0 < \eta$, giving us the following update step.

$$w = w - \eta\frac{\partial J}{\partial w}(w,b)$$
$$b = b - \eta\frac{\partial J}{\partial b}(w,b)$$

This iterative approach is stopped after a certain number of iterations defined as parameter.

We need to make emphasis that we are having some assumptions such as that the data is **independent**, there exists a predictable trend, the variance of the residuals does not change.

72.16 Non-Linear Transformations

The two algorithms covered until now have the condition that both of them are linear, but what if our data cannot be linearly separated? We can modify the input variable x_i in a non-linear way to be able to separate the data linearly in another space and then apply our algorithms.

For this purpose, we define the new space $\mathcal{Z}$ whose elements are $z_n = \phi(x_n)$, where $\phi : \mathcal{X} \to \mathcal{Z}$ is a non-linear transformation. We also define $\hat{g}(z) = \text{sign}(\langle \hat{w}, z \rangle)$. Finally, our hypothesis set $\mathcal{H}_\phi$ contains functions of the form $h(x) = \hat{h}(\phi(x))$.

72.17 Logistic Regression

Logistic Regression is a statistical model which maps a continuous variable to a discrete number of classes using the *sigmoid* function and thresholds for the different classes.

To achieve the desired result we apply a *linear regression* and then apply the *logistic function* to the result to squish it between 0 and 1, hence, we get probabilities.

The logistic function is defined as

$$\theta(s) = \frac{e^s}{1+e^s} = \frac{1}{1+e^{-s}}$$

called the *sigmoid function*, whose domain is $(0,1)$ and with the properties

$$\theta(-s) = 1 - \theta(s), \quad \frac{\partial \theta}{\partial s} = \frac{1}{1+e^s}\frac{e^{-s}}{1+e^{-s}}.$$

Note that our original problem with $\hat{y} = w^T x + b$ is transformed into

$$h_\theta(x_i) = \frac{1}{1+e^{-(w^T x_i + b)}}.$$

Now our function returns the **probability** of a data point being assigned a corresponding label.

Instead of minimizing the mean squared error, which could be a valid strategy, we use the likelihood function which uses $p_i(x_i)^{y_i}(1-p_i(x_i))^{1-y_i}$ for penalizing errors more equally.

Let our loss function be

$$L(\theta) = \prod_{n=1}^{N} p(x_i)^{y_i}(1-p(x_i))^{1-y_i}$$

where $p(x_i)$ is the probability of x_i being assigned the label $y = 1$ and $1 - p(x_i)$ the same for $y = 0$. We choose this because for each of the samples we have a discrete *geometric distribution*.

We take that expression and modify it using $h_\theta(x_i)$:

$$L(\theta) = \prod_{n=1}^{N} (h_\theta(x_i))^{y_i}(1-h_\theta(x_i))^{1-y_i}.$$

Taking the logarithm of the likelihood function gives us the log-likelihood:

$$\log L(w,b) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(h_\theta(x_i)) + (1-y_i)\log(1-h_\theta(x_i))].$$

And this is the function we want to maximize. Note that this is the *cross entropy loss function*.

72.17.1 Gradient of the Cross Entropy Loss Function

Using the cross entropy we can define our error for logistic regression. By taking the gradient and applying gradient descent.

$$\log L(w,b) = -\frac{1}{N}\sum_{i=1}^{n}[y_i \log(h_\theta(x_i) + (1-y_i)\log(1-h_\theta(x_i))].$$

We get

$$\nabla \log L(\theta) = \begin{bmatrix} \frac{\partial \log L(w,b)}{\partial w} \\ \frac{\partial \log L(w,b)}{\partial b} \end{bmatrix} = \begin{bmatrix} \frac{1}{N}\sum_{i=1}^{N} 2x_i\left(\frac{1}{1+e^{-(w^T x_i+b)}} - y_i\right) \\ \frac{1}{N}\sum_{i=1}^{N} 2\left(\frac{1}{1+e^{-(w^T x_i+b)}} - y_i\right) \end{bmatrix} = \begin{bmatrix} \frac{1}{N}\sum_{i=1}^{N} 2x_i(h_\theta(x_i) - y_i) \\ \frac{1}{N}\sum_{i=1}^{N} 2(h_\theta(x_i) - y_i) \end{bmatrix}.$$

In vector form

$$\nabla_w \log L(w,b) = \frac{1}{N}X^T(h_\theta(X) - y),$$

$$\nabla_b \log L(w,b) = \frac{1}{N}\sum_{i=1}^{N}(h_\theta(x_i) - y_i).$$

72.17.2 Logistic Regression Algorithm

1. Initialize w and b (e.g., to zeros or small random values).
2. For each iteration until convergence or for a fixed number of iterations:

(a) Compute the predictions for all training examples:

$$y_{\text{pred}} = h_\theta(x^i) = \frac{1}{1 + e^{-(w^T x^i + b)}}.$$

(b) Compute the gradients:

$$\nabla_w \log L(w, b) = \frac{1}{N} X^T (h_\theta(X) - y).$$

$$\nabla_b \log L(w, b) = \frac{1}{N} \sum_{i=1}^{N} (h_\theta(x^i) - y^i).$$

(c) Update the parameters using gradient descent:

$$w^{(k+1)} = w^{(k)} - \alpha \nabla_w \log L(w, b).$$

$$b^{(k+1)} = b^{(k)} - \alpha \nabla_b \log L(w, b).$$

3. After all iterations, output the learned parameters w and b.
4. We predict our data by

$$y_{\text{pred}} = h_\theta(X),$$

followed by assigning each prediction a label based on single threshold or a list of ranges.

72.18 Decision Trees

Decision trees are used mainly for *regression trees* or *classification trees* which are used to partition the data into subsets based on a purity measurement.

72.18.1 Classification Trees

We can take everything from the regression tree except the cost function. For *classification trees* the cost function is given as the label of the specific classes itself.

Nomenclature:

- C is the number of classes in the classification problem.
- $p(i)$ is the probability of the class relative to the complete set.

Gini Impurity

Represents the mean probability for a data point being misclassified. $p(i) = p_i$.

This probability that a data point was misclassified is given by

$$\sum_{k \neq i}^{C} p_k = 1 - p_i.$$

The mean is then

$$E_{\text{Gini}} = \sum_{i}^{C} p_i (1 - p_i) = \sum_{i}^{C} p_i - \sum_{i}^{C} p_i^2 = 1 - \sum_{1}^{C} p_i^2.$$

Information Gain

Information gain is a measure commonly used in decision tree algorithms such as ID3, C4.5, and CART to determine which feature should be used to split the dataset at each step. It quantifies how much uncertainty about the class labels is reduced after splitting the data based on a particular feature.

$$\mathrm{IG} = \mathrm{H}(P) - \sum w_i \mathrm{H}(C_i).$$

where $H(X)$ is the entropy and $w_i = \frac{|C_i|}{|P|}$ the relative cardinality of the child with respect to the parent.

72.18.2 Regression Trees

A *regression tree* divides the feature space into disjoint sub-spaces in such a way that it brings the value of all labels y_i for $x_i \in \mathcal{D}$ as close as possible to the true label $\overline{y_i} \approx y_i$.

The process behind this goal is called *recursive binary splitting* and can be performed in different manners, based on how the "purity" of a node is measured. This is of course an optimization in which we want to maximize the purity of our node given a split.

The process is similar to the one used for classification trees, but we use *variance reduction* for our measurements.

Variance Reduction

The *variance reduction* is given by

$$\mathrm{VR} = \mathrm{var}(P) - \sum w_i \mathrm{var}(C_i),$$

where $w_i = \frac{|P|}{|C_i|}$.

72.18.3 Recursive Binary Splitting

We previously stated an optimization problem, which sadly cannot be solved efficiently; instead, we will approach it with *recursive binary splitting*, an algorithm which goes as follows:

1. Given R_0 a hyperquadrant with training data (x_i, y_i). Given a feature vector $x_i = (x_{i1}, \ldots, x_{id})$. Also, given $j \in \{1, \ldots, d\}$ as the indexes of the feature and $s \in \mathbb{R}$ a bias term.
2. Define the regions

   $$R_1(j,s) = \{x \mid (x)_j < s, x \in R_0\},$$
   $$R_2(j,s) = \{x \mid (x)_j \geq s, x \in R_0\}.$$

   Find (j, s) which minimizes the following

   $$\theta = \min \left(\sum_{i:x_i \in R_1(j,s)} (y_i - \overline{y_{R_1}})^2 + \sum_{i:x_i \in R_2(j,s)} (y_i - \overline{y_{R_2}})^2 \right).$$

3. For the next step repeat step one, $R_0 = R_1$ or $R_0 = R_2$ depending on the smallest, else terminate.

The termination criteria are for example:

- Each leaf which has to be divided has a minimal number of data points contained.
- The tree can have maximal depth.
- The tree size is limited.

72.18.4 CART Algorithm

The *CART* (Classification and Regression Trees) algorithm is a popular decision tree algorithm used for both classification and regression tasks. The CART algorithm constructs binary trees by recursively splitting the data based on feature values to minimize impurity or variance.

The steps of the CART algorithm are as follows:

1. **Select the Best Split:** For each feature, evaluate all possible split points and calculate the impurity (e.g., Gini impurity for classification or variance for regression) for each split. Choose the feature and split point that results in the lowest impurity.
2. **Create Child Nodes:** Split the dataset into two subsets based on the selected feature and split point, creating left and right child nodes.
3. **Repeat Recursively:** For each child node, repeat steps 1 and 2 until a stopping criterion is met (e.g., maximum tree depth, minimum number of samples in a node, or no further reduction in impurity).
4. **Assign Class Labels or Values:** For classification tasks, assign the most common class label to each leaf node. For regression tasks, assign the mean value of the target variable to each leaf node.

72.18.5 Variance Reduction

In regression tasks, the CART algorithm aims to minimize the variance within each node after a split. The variance reduction is calculated as follows:

$$\text{Variance Reduction} = \text{Var}(R) - \left(\frac{N_{left}}{N} \text{Var}(R_{left}) + \frac{N_{right}}{N} \text{Var}(R_{right}) \right)$$

with

$$\text{var}(Y) = \frac{1}{N-1} \sum_{i=1}^{N} (y_i - \overline{y})^2.$$

72.19 Nearest Neighbors Models

Nearest neighbors models NNM use the labels of the neighbors in the proximity to perform classification or regression.

72.19.1 Quantifying the Differences Between Features

Examples of metrics, distance measurements:

- *Euclidean Distance*: $d(x, y) = \|x - y\|$.
- *Cosine Similarity*: $\text{sim}(x, y) = \frac{x \cdot y}{\|x\| \|y\|}$.
- *Jaccard Similarity*: $\text{sim}(A, B) = \frac{|A \cap B|}{|A \cup B|} \in [0, 1]$.

72.19.2 Nearest Neighbors

Given a data-set $\mathcal{D} = \{(x_1, y_1), \ldots, (x_N, y_N)\}$ and a new point x we want to predict its label y. We choose a *distance metric* $d(x, x')$ to quantify the difference between two feature vectors.

To predict the label of x we find the k nearest neighbors of x in the data-set $\mathcal{D}$ based on the distance metric d.

Feature vectors in the training set are noted with their distance to x. $x_{[1]}$ symbolizes the nearest neighbor of x, $x_{[2]}$ the second nearest and so on.

$$d(x, x_{[1]}) \leq d(x, x_{[2]}) \leq \ldots d(x, x_{[N]}).$$

The *Nearest Neighbor Model* gives us our final hypothesis $g(x) = y_{[1]}(x)$.

72.19.3 K-Nearest Neighbors Classification

The *KNN classification* is a generalization of the *NN model*.

Given an integer $k \geq 1$ and $y \in \{-1, 1\}$ the *k-nearest neighbor* model of binary classification is given by

$$h(x) = \text{sign}\left(\sum_{i=1}^{k} y_{[i]}(x)\right).$$

Note, that for $k = 1$ we get the previous NN model. This is a majority vote model.

72.19.4 K-Nearest Neighbor Regression

The *KNN regression* determines the label of the feature based on the **mean** of the label of the neighbors and a series of thresholds.

$$h(x) = \frac{1}{k}\sum_{i=1}^{k} y_{[i]}(x).$$

72.19.5 KNN Algorithm

Given a distance metric $d(x, y)$ we classify our training data using the following steps for each data point z:

1. Compute its distance to all other data points $d(z, v_i)$.
2. Sort the distances in ascending order, by also keeping track of the indexes.
3. Based on the parameter k get the distances of the k closest points.
4. Depending on regression or classification, apply one of the following formulas

$$\text{Regression} : h(x) = \text{sign}\left(\sum_{i=1}^{k} y_{[i]}(x)\right) \quad \text{Classification} : h(x) = \frac{1}{k}\sum_{i=1}^{k} y_{[i]}(x)$$

   using the indexes of the k nearest neighbors for getting the corresponding labels.

72.20 Kernel Trick

The *kernel trick* allows us to compute the inner product in the transformed feature space without explicitly performing the transformation. For that we use so-called *kernel functions* K to use as inner products in the feature space. We are basically using mathematical elegance to use high dimensional spaces without paying the computational cost.

The kernel functions must satisfy the following properties *Mercer's Theorem*:

K is a *kernel* if the kernel matrix K is symmetric and positive semi-definite for any set of input data points.

$$x^T K x \geq 0 \quad K = K^T.$$

Common kernel-functions are:

- *Linear:* $k(x, y) = x^T y$
- *Polynomial:* $k(x, y) = (x^T y + c)$
- *RBF:* $k(x, y) = \exp(-\gamma \left\|x - y\right\|^2)$

- *Sigmoid:* $k(x, y) = \tanh(\gamma x^T y + c)$

We can also give parameters α, β for $k(\beta + \alpha x^T, y)$.

Example:

Given $\Phi(x_1, x_2) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, \sqrt{2}x_1x_2, x_2^2)$ we get for

$$\langle \Phi(x), \Phi(y) \rangle = 1^2 + \sqrt{2}x_1\sqrt{2}y_1 + \sqrt{2}x_2\sqrt{2}x_2 + x_1^2y_1^2 + \sqrt{2}x_1x_2\sqrt{2}y_1y_2 + x_2^2y_2^2.$$

The following kernel by simplifying

$$\langle \Phi(x), \Phi(y) \rangle = 1 + 2x^T y + (x^T y)^2 = (1 + x^T y)^2 = k(x, y).$$

Generalized $(d + x^T y)^2$.

72.20.1 The Radial Kernel

The *radial basis function (RBF) kernel* is defined as

$$K(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

where $\sigma > 0$ is a parameter that controls the width of the Gaussian function, and σ is the standard deviation of the Gaussian distribution.

The infinite-dimensional property comes from the fact that the RBF kernel can be expressed as an infinite series expansion,

$$K(x, y) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)^n.$$

72.21 Support Vector Machines

Support vector machines are a family of algorithms which use the perceptron as a basis for classifying points into classes using a hyperplane

$$w_0 + w_1x_1 + \cdots + w_px_p = 0.$$

But this time we also define a region defined by the normal vectors of the hyperplane and define it as our *margin*, which we will try to optimize to be as large as possible.

As starting point we take the perceptron we already defined with $b = w_0$ being the bias term

$$y_k = \text{sign}(w^T x + b) = \text{sign}(\tilde{w}^T \tilde{x}),\ y_k \in \{-1, 1\}.$$

The size of the margin is given by the normal vector w in the following way: Imagine, that we want to take a step into the direction of w from a point x on our decision boundary ($w^T x + b = 0$), such that we reach the portion of the other class. We also make the assumption that for a point of the first class $w^T x_+ + b = 1$, $w^T x_- + b = -1$.

$$\begin{aligned} w^T(x + k\frac{w}{\|w\|}) + b &= 1 \\ w^T x + w^T k\frac{w}{\|w\|} + b &= 1 \quad \text{note that } (w^T x + b = 0) \\ k\frac{w^T w}{\|w\|} &= 1 \\ k &= \frac{1}{\|w\|} \end{aligned}$$

Therefore, the length taking into consideration both directions is $\frac{2}{\|w\|}$.

Hence, for the maximization of the margin we need to minimize $\|w\|$ with the constraints that

$$\begin{aligned} y_i = 1 &\implies w^T x + b \geq 1 \iff x \in C_1 \\ y_i = -1 &\implies w^T x + b \leq -1 \iff x \in C_2. \end{aligned}$$

In general

$$y_i(w^T x_i + b) \geq 1.$$

This comes from the fact that $1(1) \geq 1$, $-1(-1) \geq 1$ given our rules. Thus, we combine both into one expression.

72.21.1 Hard Margin SVM

A *hard margin SVM* is a perceptron with maximal margin and no misclassifications allowed.

Formally, we want to solve the following optimization problem:

Minimize: $\frac{1}{2}\|w\|$ under N *constraints* $y_i(w^T x_i + b) \geq 1$.

Maximize: $y_n(w^T x_n + b) \geq 1, (n = 1, \ldots, N)$.

First, we use the technique of Lagrange multipliers giving us

$$\mathcal{L}(w, \alpha_1, \ldots, \alpha_n, b) = \frac{1}{2}\langle w, w\rangle - \sum \alpha_i[y_i(\langle w, x\rangle + b) - 1].$$

Giving us

$$\frac{\partial \mathcal{L}}{\partial w} = w - \sum \alpha_i y_i x_i = 0 \iff w = \sum \alpha_i y_i x_i$$

and

$$\frac{\partial \mathcal{L}}{\partial b} = \sum \alpha_i y_i = 0.$$

By putting our new terms in our original equation we get that

$$\mathcal{L}(w, \alpha_1, \ldots, \alpha_n, b) = \frac{1}{2}\left(\sum \alpha_i y_i x_i\right)\left(\sum \alpha_i y_i x_i\right) - \sum \alpha_i \left[y_i\left(\left\langle \sum \alpha_i y_i x_i, x\right\rangle + b\right) - 1\right].$$

Giving us

$$-\sum \alpha_i - \frac{1}{2}\sum_i \sum_j \alpha_i \alpha_j y_i y_j \langle x_i, x_j\rangle.$$

This is the so-called *dual optimization problem.*

72.21.2 Soft Margin Support Vector Machine

The idea of *soft margin SVM* is to allow misclassifications but penalize them in the optimization problem. To get the best soft margin we can use cross validation.

Formally, we want to solve the following optimization problem:

Minimize: $\frac{1}{2}w^T w + C\sum_n \xi_n$ under N *constraints*

In other words, maximize $\frac{1}{2}w^T w$, the width of the margin, minimize $\sum_n \xi_n$, the total error.

With the constraint $y_n(w^T x_n + b) \geq 1 - \xi_n, (n = 1, \ldots, N)$,

$$\xi_n = \max(0, 1 - y_n(w^T x_n + b))$$

Thus,

$$\frac{1}{2}w^T w + C\sum_n \max(0, 1 - y_n(w^T x_n + b))$$

where C is our regularization parameter.

under the conditions

$$\sum_{n=1}^{N} \alpha_n y_n = 0, 0 \leq \alpha_n \leq C\,\forall n.$$

Also,

$$y_i = \text{sign}\left(\sum_{n=1}^{N} \alpha_n y_n x_n^T x_i + b\right).$$

72.21.3 Non-Linear Support Vector Machines

We will allow non-linear feature Transformations $x \to \Phi(x)$ such that

$$x_n^T x_i \to \Phi(x_n)^T \Phi(x_i).$$

Now our y_i becomes

$$y_i = \text{sign}\left(\sum_{n=1}^{N} \alpha_n y_n \Phi(x_n)^T \Phi(x_i) + b\right).$$

Let us also define $K := \Phi(x_n)^T \Phi(x_i)$.

72.21.4 Least Squares Support Vector Machine

This is a simplified version of the soft margin SVM.

Formally, we want to minimize $\frac{1}{2}w^T w + \frac{\gamma}{2}\sum_n^N e_n^2$ under N constraints.

$$y_n(w^t x_n + b) = 1 - e_n$$

with

$$e_n^2 = y_n^2(y_n - (w^T x_n + b))2 = (y_n - (w^T x_n + b)).$$

The Lagrangian we get is

$$\mathcal{L} = \frac{1}{2}w^T w + \frac{\gamma}{2}\sum_{n}^{N} e_n^2 - \sum_{n=1}^{N}\alpha_n[y_n(w^T x_n + b) - 1 + e_n].$$

Apply $\nabla_{w,b,e,\alpha}\mathcal{L} = 0$:

$$\begin{aligned}
\textbf{I}\ \frac{\partial \mathcal{L}}{\partial w} &= w - \sum_{k=1}^{N}\alpha_k y_k x_k = 0 \iff w = \sum_{k=1}^{N}\alpha_k y_k x_k \\
\textbf{II}\ \frac{\partial \mathcal{L}}{\partial b} &= -\sum_{k=1}^{N}\alpha_k y_k = 0 \iff y^t\vec{\alpha} = 0 \\
\textbf{III}\ \frac{\partial \mathcal{L}}{\partial e} &= \gamma e_k - a_k = 0 \iff \alpha_k = \gamma e_K \iff e = \frac{1}{\gamma}\vec{\alpha} \\
\textbf{IV}\ \frac{\partial \mathcal{L}}{\partial \alpha_k} &= -y_k(w^T x_k + b) + 1 - e_k \iff y_k(w^T x_k + b) + e_k = 1 \forall k.
\end{aligned}$$

Now, let us combine our results

$$\begin{aligned}
\textbf{I}\ w &= \sum_{k=1}^{N}\alpha_k y_k x_k \\
\textbf{II}\ y^t\vec{\alpha} &= 0 \\
\textbf{III}\ e &= \frac{1}{\gamma}\vec{\alpha} \\
\textbf{IV}\ y_k(w^T x_k + b) + e_k &= 1 \forall k.
\end{aligned}$$

Substitute I and III in IV

$$\begin{aligned}
& y_k(w^T x_k + b) + e_k = 1 \forall k \\
\iff & y_k(\sum_{j=1}^{N}\alpha_j y_j x_j^T x_k + b) + \frac{1}{\gamma}\alpha_k = 1 \forall k \\
\iff & y_k b + \sum_{j=1}^{N}\alpha_j y_j x_j^T x_k y_k + \frac{1}{\gamma}\alpha_k = 1 \forall k.
\end{aligned}$$

Now, replace the dot product with the kernel function

$$\iff \text{V}\ yb + (yyT \cdot K)\vec{a} + \frac{1}{\gamma}I\vec{a} = 1.$$

The only results left to use are II and V which we can combine into

$$A = \begin{bmatrix} 0 & y^T \\ y & \Omega + \frac{1}{\gamma}I \end{bmatrix}.$$

After this process we get

$$y_i = \text{sign}(w^T x_i + b) \implies \text{sign}\left(\sum_{k=1}^{N}\alpha_k y_k x_k^T x_i + b\right).$$

We can apply our kernel function and get

$$y_i = \text{sign}\left(\sum_{k=1}^{N} \alpha_k y_k K(x_k, x_i) + b\right).$$

Least Square Support Vector Machine Implementation

1. Implement the kernel function, typically

$$K(x_k, x_i) = \exp\left(-\frac{\|x_k - x_i\|^2}{2\sigma^2}\right).$$

2. Compute the $n \times n$ matrix $\Omega = yy^T \cdot K$ with y being the label vector and K the kernel matrix.
3. Find the matrix

$$A = \begin{bmatrix} 0 & y^T \\ y & \Omega + \frac{1}{\gamma} I \end{bmatrix}.$$

4. Invert A and get b and $\vec{\alpha}$.

$$\begin{bmatrix} b \\ \vec{\alpha} \end{bmatrix} = A^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

5. Predict the labels

$$y_i = \text{sign}\left(\sum_{k=1}^{N} \alpha_k y_k K(x_k, x_i) + b\right).$$

72.22 Average Vector

Given N feature vectors with D features, the *average vector* is given by

$$\overline{x} = \frac{1}{N} \sum_{n=1}^{N} x_n.$$

This vector is used to center the data by subtracting it from each feature vector. This vector has as its components the mean of the corresponding feature across all feature vectors.

72.23 Z Scoring

Given $\overline{X}$ as the mean and σ as the standard deviation of X. The standardized feature Z (z-score) is given by

$$Z = \frac{X - \overline{X}}{\sigma}.$$

In practice, data are often organized in matrices where each column represents a feature. In this case, standardize each feature by transforming every column separately so that it has mean 0 and variance 1 (i.e., standard deviation 1).

72.24 Dimension Reduction

The *curse of dimensionality* is a concept that describes the problems which arise when dealing with high-dimensional data. A high number of features (dimensions) leads to an increase in the complexity of the model as well as the computational cost. Also, in data science it is important to be able to visualize the data, which is not possible in high dimensions. Therefore, we want to reduce the number of features while preserving as much information as possible.

To reduce dimensionality we have two categories of methods: *linear* and *non-linear*. The linear methods use the assumption that the data lies on a linear subspace of the original feature space which means that they use, mostly, the Euclidean distance as a measure of similarity. Non-linear methods take into consideration the topology of the data and use non-linear transformations to reduce the dimensionality.

In the next sections we will cover some algorithms for this purpose.

72.25 Principal Component Analysis (PCA)

PCA (Principal Component Analysis) finds new axes called components for the data, such that the data show the largest possible variance along these axes. In other words, PCA looks for directions in which the data vary the most.

Axes along which the data have little, or almost no variance can usually be ignored, because they contain little useful information.

When projecting our points onto a new axis we use the projection formula

$$x' = \text{Proj}_u(x) = \frac{\langle x, u\rangle}{\langle u, u\rangle} u = \langle x, u\rangle \, u \quad \text{with } \|u\| = 1.$$

We can use the term $\langle x, u\rangle$ as the measure of **information preserved between the two vectors** x **and** x'. This is maximal when they are **parallel** and minimal when they are **orthogonal**.

For that, we use the variance of the projections with our unit vector u:

$$\begin{aligned}
\frac{1}{N}\sum_{i=1}^{N}(u^T x_j - u^T \overline{x})^2 &= \frac{1}{N}\sum_{i=1}^{N}[u^T(x_j - \overline{x})]^2 \\
&= \frac{1}{N}\sum_{i=1}^{N} u^T(x_j - \overline{x}) u^T(x_j - \overline{x}) \\
&= \frac{1}{N}\sum_{i=1}^{N} u^T(x_j - \overline{x})(u^T(x_j - \overline{x}))^T \\
&= \frac{1}{N}\sum_{i=1}^{N} u^T(x_j - \overline{x})(x_j - \overline{x})^T u_i \\
&= u^T S u.
\end{aligned}$$

Hence, we want to maximize the sum of the inner products between the original data and the projected data onto the directions u, while minimizing the loss of information.

Note the covariance matrix inside

$$S = \frac{1}{N}\sum_i (x_i - \overline{x})(x_i - \overline{x})^T.$$

If we assume that the data is centered around the mean, then we can write

$$S = \frac{1}{N}\sum_i x_i x_i^T = X^T X.$$

For

$$X = \begin{bmatrix} - & X_1 & - \\ - & X_2 & - \\ & \vdots & \\ - & X_n & - \end{bmatrix}$$

in our optimization problem we can replace the covariance matrix and get

$$\max_{\|u\|=1} (u^T S u)$$

with the constraint

$$u_i^T u_i = 1 \iff 1 - u_i^T u_i = 0.$$

We also have the constraint that u is a unit vector, so we can write the optimization problem as

$$\max_{\|u\|=1} (u^T S u).$$

Now, using the method of Lagrange multipliers we can write the Lagrangian as

$$\mathcal{L}(u, \lambda) = u^T S u - \lambda(u^T u - 1)$$

Taking the gradient and setting it to zero gives us

$$\nabla_u \mathcal{L} = 2Su - 2\lambda u = 0 \implies Su = \lambda u$$

which is an eigenvalue problem. The solution u is an eigenvector of the covariance matrix S and λ is the corresponding eigenvalue.

Because we want to maximize $u^T S u$, we need to choose the eigenvector corresponding to the largest eigenvalue. This eigenvector is called the *first principal component*.

From

$$\begin{aligned} Su_i &= \lambda_i u_i \\ u_i^T S u_i &= \lambda_i. \end{aligned}$$

note that this is maximal when λ_i is maximal.

Depending on how many dimensions we want to reduce to, we can choose the eigenvectors corresponding to the largest eigenvalues.

72.25.1 Projection of the Data After the Analysis

Each of our principal components represents a direction in our high dimensional space. Given our data set $X \in \mathbb{R}^{n \times p}$ and our matrix of k principal components $W^T \in \mathbb{R}^{k \times p}$; the projection of our data onto the components is given by

$$Z = XW^T, \quad Z \in \mathbb{R}^{n \times k}.$$

This comes from the fact that given our projection:

$$\text{Proj}_{u_i}(x) = (u_i^T x) u_i \iff u_i^T u_i = 1.$$

When we multiply our matrices X and W we get

$$\begin{bmatrix} - & X_1 & - \\ - & X_2 & - \\ & \vdots & \\ - & X_n & - \end{bmatrix} \begin{bmatrix} - & u_1 & - \\ - & u_2 & - \\ & \vdots & \\ - & u_k & - \end{bmatrix}^T = \begin{bmatrix} \langle X_1, u_1 \rangle & \langle X_1, u_2 \rangle & \dots & \langle X_1, u_k \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle X_n, u_1 \rangle & \langle X_n, u_2 \rangle & \dots & \langle X_n, u_k \rangle \end{bmatrix}$$

which essentially describes our vector as a linear combination:

$$Z_1 = \langle X_1, u_1 \rangle u_1 + \langle X_1, u_2 \rangle u_2 + \cdots + \langle X_1, u_k \rangle u_k = \sum_{i=1}^{k} \langle X_1, u_i \rangle u_i.$$

Note that we are using the row vectors of X and the transpose of the matrix of principal components W^T to get the projection, because of the way most software libraries implement matrices as arrays of row vectors.

72.25.2 Explained Variance

If we divide each eigenvalue by the sum of all eigenvalues we get the *explained variance EV* of each principal component.

$$\text{PVE} = \frac{\lambda_k}{\sum_{j=1}^{D} \lambda_j}.$$

This tells us how much of the variance in the data is explained by each principal component. We can use this to decide how many principal components to keep based on a desired level of explained variance.

72.25.3 Covariance Optimization

For the computation of our principal components, we have been using the expression $S = X^T x$. Although, this is mathematically correct, for certain data sets with more features than samples this can become a really big matrix. To improve this problem we make use of one of the following properties of our eigenvalue problem.

$$\begin{aligned} Su_i &= \lambda_i u_i \\ X^T X u_i &= \lambda_i u_i \\ XX^T X v_i &= X \lambda_i X u_i \\ (XX^T)(Xu_i) &= \lambda_i (Xu_i) \\ (XX^T)v_i &= \lambda_i v_i \end{aligned}$$

This means that our eigenvalues and eigenvectors are the same if, instead of the *feature covariance matrix*, we use the *sample covariance matrix* XX^T.

72.25.4 PCA Algorithm

We want to reduce the number of features from D to $M < D$ or just inspect our data for the features which contribute the most variance.

The algorithm goes as following:

1. We have N feature vectors with D features each.
2. Compute the average vector $\overline{x} = \frac{1}{N} \sum_{n=1}^{N} x_n$.
3. Subtract the mean $B = X - \overline{X}$.
4. We have a $D \times D$ covariance matrix $S = B^T B$ or BB^T depending on the number of samples and features.
5. We compute the eigenvalues and eigenvectors of S.
6. We sort the eigenvalues with their corresponding eigenvectors in decreasing order and take the first M eigenvectors.
7. These eigenvectors form the new basis for the feature space, with which we can project our feature vectors.

72.26 Kernel PCA

This is an extension of PCA to non-linear feature spaces via the kernel trick.

72.26.1 Reformulation of the PCA With Non-Linear Features

The covariance matrix $S = \frac{1}{N}\sum_{n=1}^{N}(x_n - \overline{x})(x_n - \overline{x})^T$ can be reformulated with $x_i \to \Phi(x_i)$ to

$$S = \frac{1}{N}\sum_{n=1}^{N}(\Phi(x_n) - \overline{\Phi(x)})(\Phi(x_n) - \overline{\Phi(x)})^T.$$

Assume that the features vectors are centered in the feature space, i.e. $\overline{\Phi(x)} = 0$. Then we have

$$S = \frac{1}{N}\sum_{n=1}^{N}\Phi(x_n)\Phi(x_n)^T.$$

72.26.2 Eigenvector as Linear Combination of the Transformed Features

$$\begin{aligned}
Su_j &= \lambda_j u_j \\
Su_j &= \frac{1}{N}\sum_{n=1}^{N}\Phi(x_n)\Phi(x_n)^T u_j \\
&= \frac{1}{\lambda_j N}\sum_{n=1}^{N}\Phi(x_n)(\Phi(x_n)^T u_j) \\
&= \frac{1}{\lambda_j N}\sum_{n=1}^{N}\Phi(x_n)^T u_j(\Phi(x_n)) \\
&= \sum_{n=1}^{N}\left(\frac{1}{\lambda_j N}\Phi(x_n)^T u_j\right)\Phi(x_n) \\
&= \sum_{n=1}^{N} a_{ji}\Phi(x_i)
\end{aligned}$$

with $u_j = \sum_l \alpha_{jl}\Phi(x_l)$.

Because we can express the eigenvectors as a linear combination of the transformed features, we can find the coefficients α_{jl} instead of u_j.

72.26.3 Find the Eigenvectors

$$\begin{aligned}
\frac{1}{N}\sum_i \Phi(x_i)\Phi(x_i)^T\sum_l \alpha_{jl}\Phi(x_l) &= \lambda_j\sum_l \alpha_{jl}\Phi(x_l) \\
\frac{1}{N}\sum_i \Phi(x_i)\left(\sum_l \alpha_{jl}\Phi(x_i)^T\Phi(x_l)\right) &= \lambda_j\sum_l \alpha_{jl}\Phi(x_l) \\
\frac{1}{N}\sum_i \Phi(x_i)\left(\sum_l \alpha_{jl}K(x_i,x_l)\right) &= \lambda_j\sum_l \alpha_{jl}K(x_k,x_l) \\
\frac{1}{N}KK\vec{\alpha}_j &= \lambda_j K\vec{\alpha}_j \\
K\vec{\alpha}_j &= N\lambda_j K\vec{\alpha}_j
\end{aligned}$$

Therefore, we have to solve the following eigenvalue problem

$$K\vec{\alpha}_j = N\lambda_j\vec{\alpha}_j$$

which we reformulate to

$$K\vec{\alpha}_j = \hat{\lambda_j}\vec{\alpha}_j.$$

The coefficients α_j are the components of the eigenvectors of the kernel matrix K.

72.26.4 Normalization of the Eigenvectors

To normalize the eigenvectors u_j we have to impose the condition

$$u_j = \sum_l \alpha_{jl}\Phi(x_l) \implies \hat{u} = \sum_l \hat{\alpha_{jl}}\Phi(x_l) = \hat{u_j^T}u_j^T = 1\forall j.$$

This gives us

$$\sum_k \sum_l \hat{\alpha_{jk}}\hat{\alpha_{jl}}K(x_k, x_l) = 1\forall j$$

which can be reformulated to

$$\hat{\vec{\alpha}_j^T}\hat{\lambda_j}\hat{\vec{\alpha}_j} = 1\forall j \implies \|\hat{\vec{\alpha}_j}\| = \frac{1}{\sqrt{\hat{\lambda_j}}}$$

with

$$\vec{\alpha_j} \to \hat{\vec{\alpha}_j} = \frac{\vec{\alpha_j}}{\sqrt{\hat{\lambda_j}}}\frac{\vec{\alpha}_j}{\|\vec{\alpha}_j\|}.$$

72.26.5 Projection Onto the KPCA Axis

$$\begin{aligned}\Phi(x)^T u_j &= \Phi(x)^T \sum_l \alpha_{jl}\Phi(x_l) \\ &= \sum_l \alpha_{jl}\Phi(x)^T\Phi(x_l) \\ &= \sum_l \alpha_{jl}K(x, x_l)\end{aligned}$$

72.26.6 Centering of the Features

We assumed that the features were centered in the transformed feature space. Now we want to normalize them.

Given

$$\hat{\Phi}(x_i) = \Phi(x_i) - \frac{1}{N}\sum_l \Phi(x_l),$$

and the kernel function

$$\begin{aligned} K(x_i, x_j) &= \Phi(\hat{x_i})^T \Phi(\hat{x_j}) \\ &= \left(\Phi(x_i) - \frac{1}{N} \sum_l \Phi(x_l) \right)^T \left(\Phi(x_j) - \frac{1}{N} \sum_l \Phi(x_l) \right) \\ &= K(x_i, x_j) - \frac{1}{N} \sum_l K(x_i, x_l) - \frac{1}{N} \sum_k K(x_k, x_j) + \frac{1}{N^2} \sum_{k,l} K(x_k, x_l). \end{aligned}$$

Simplified

$$\hat{K} = K - 1_N K - K 1_N + 1_N K 1_N,$$

where 1_N is the $N \times N$ matrix with all entries equal to $\frac{1}{N}$.

72.26.7 Kernel PCA Algorithm

Now we can summarize the K-PCA algorithm

1. Choose a kernel function $K(x, y)$ and compute the kernel matrix K with entries $K_{ij} = K(x_i, x_j)$.
2. Center the kernel matrix to obtain $\hat{K}$ using $\hat{K} = K - 1_N K - K 1_N + 1_N K 1_N$
3. Solve the eigenvalue problem $\hat{K}\vec{\alpha}_j = \hat{\lambda}_j \vec{\alpha}_j$ to find the eigenvalues $\hat{\lambda}_j$ and eigenvectors $\vec{\alpha}_j$.
4. Normalize the eigenvectors $\vec{\alpha}_j \to \frac{1}{\sqrt{\hat{\lambda}_j}} \frac{\vec{\alpha}_j}{\|\vec{\alpha}_j\|}$
5. Project the data points onto the K-PCA axes using

$$\Phi(x)^T \hat{u_j} = \sum_l \hat{\alpha_{jl}} K(x, x_l).$$

72.27 Multidimensional Scaling MDS

Multidimensional Scaling (MDS) is a family of procedures for the visualization and dimension reduction of features. The non-Euclidean MDS variants do not have a closed form solution, they are iterative methods; hence, we will present the classical MDS with the Euclidean distance.

Given the pairwise distances of N vectors in matrix form

$$D = (d_{i,j})_{i,j}^n = \delta_{i,j}.$$

In this classical approach $d_{i,j}$ represents the distances in the low-dimensional configuration, $\delta_{i,j}$ the observed distances or dissimilarities and $\hat{d}_{i,j} = f(\delta_{i,j})$ the disparities from our data X, which we do not know and want to find X.

We take our matrix D and square its entries giving us our *Gram matrix*:

$$\tilde{D} = (d_{i,j}^2)_{i,j}^n.$$

Then, we construct our matrix B for which the following holds:

$$B = XX^T, \quad b_{i,j} = \langle x_i, x_j \rangle .$$

Using the fact that we can write our inner products as

$$\langle x_i, x_j \rangle = \frac{1}{2}(\langle x_i, x_i \rangle + \langle x_j, x_j \rangle - d_{i,j}^2).$$

we write B as

$$B = -\frac{1}{2} H \tilde{D} H, \quad H = I - \frac{1}{N} \vec{1}\vec{1}^T.$$

and H is our *double centering matrix* which subtracts both the row and column mean.

Finally, reconstruct X using the eigendecomposition with V as the matrix of eigenvectors and Λ the eigenvalue matrix:

$$B = V \Lambda V^t.$$

We get

$$X = V\sqrt{B} = \left[\sqrt{\lambda_1} v_1 \quad \dots \quad \sqrt{\lambda_k} v_1\right].$$

Depending on the number of dimensions we want to keep, we can use a certain number of sorted eigenvalues with their corresponding eigenvectors to control the dimension reduction.

The idea is similar to PCA, but instead of using correlation between samples, we use the distance/metric to group samples. Choosing the right metric for the method is part of the art. As an important note, the previously discussed method only works with the Euclidean distance; for a generalization with any metric there is an alternative algorithm.

72.27.1 Proportion of Distance Matrix Explained (PDME)

The *PDME* is given by:

$$\text{PDME} = \frac{\lambda_j}{\sum_{i=1}^{N} |\lambda_i|}.$$

72.27.2 Euclidean Constraint

The reason that our distance matrix has to be build using the Euclidean metric is due to the loss of positive definiteness. If $B = XX^T$, then

$$a^T B a = a^T X X^T a = (Xa)^T (Xa) \geq 0, \ \forall a$$

which guarantees positive eigenvalues. Negative eigenvalues lead to non-real matrices when reconstructing our data.

This is mostly not a problem in practice, since the negative eigenvalues and their associated eigenvectors are discarded during the algorithm, but this can limit the possibilities of projecting our data, meaning more loss of information.

72.28 Linear Discriminant Analysis

The goal of the *linear discriminant analysis* is to find directions $u_1, u_2, \dots, u_k$ such that they best separate our classes. It is also a dimension-reduction technique, in which we want to maximize the separability of the classes.

For the direction u_1 with the projection of some data point x_i onto that direction defined as

$$\hat{x_i} = u_1^T x_i.$$

72.28.1 Mean Vector of a Class

Let

$$m^{(k)} = \frac{1}{N_k} \sum_{n \in \mathcal{C}_k} x_n$$

be the *mean vector* of class k.

The projection onto u_1 is given by

$$\hat{m}^{(k)} = u_1^T m^{(k)}.$$

It is important to note that this is a supervised learning method.

72.28.2 Within-Class Scatter

For quantifying the separation of the points **projected onto** u_i we define the *within-class scatter* as

$$\begin{aligned} s_w^2 &:= \sum_{n \in \mathcal{C}_k} (\hat{x}_n - \hat{m}^{(k)})^2 \\ &= \sum_{n \in \mathcal{C}_k} (u_1^T x_n - u_1^T m^{(k)})^2 \\ &= u_1^T \left[\sum_{n \in \mathcal{C}_k} (x_n - m^{(k)})(x_n - m^{(k)})^T \right] u_1 \\ &= u_1^T S_{wk} u_1. \end{aligned}$$

The *total scatter* for two classes is given by

$$\begin{aligned} s_1^2 + s_2^2 &= u_1^T (S_{w1} + S_{w2}) u_1 \\ &= u_1^T S_W u_1. \end{aligned}$$

For the $n-$classes we have

$$S_W = \sum_{k=1}^{n} S_{wk} = \sum_{k=1}^{n} \sum_{n \in \mathcal{C}_k} (x_n - m^{(k)})(x_n - m^{(k)})^T.$$

72.28.3 Between-Classes Scatter

The quantification of the separation between **the middle points**, called the *between-classes-scatter* is given by

$$\begin{aligned} (\hat{m}^{(1)} - \hat{m}^{(2)})^2 \to (u_1^T m^{(1)} - u_1^T m^{(2)})^2 \\ &= u_1^T (m^{(1)} - m^{(2)})(m^{(1)} - m^{(2)})^T u_1 \\ &= u_1^T S_b u_1. \end{aligned}$$

$$S_b = (m^{(1)} - m^{(2)})(m^{(1)} - m^{(2)})^T$$

where $m^{(1)}$ and $m^{(2)}$ are the mean vectors of class 1 and class 2.

The general form is

$$S_b = \sum_{k=1}^{n} \sum_{n \in \mathcal{C}_k} (m^{(n)} - m^{(k)})(m^{(n)} - m^{(k)})^T.$$

72.28.4 Fisher Discriminant

The *Fisher Discriminant* is the ratio of the **between-class scatter** and the **within-class scatter**. In this case we will do the derivation by considering only two classes.

$$J(u_1) = \frac{u_1^T S_b u_1}{u_1^T S_W u_1} = \frac{(u_1^T (m^{(1)} - m^{(2)})(m^{(1)} - m^{(2)})^T u_1)}{u_1^T S_W u_1}.$$

We want to maximize $J(u_1)$. This means to find the direction u_1 such that the between-class scatter is maximized and the within-class scatter is minimized. To accomplish this we take the gradient and set it to zero.

$$\begin{aligned}
\frac{dJ(u_1)}{du_1} &= u_1^T S_W u_1 \frac{d}{du_1}(u_1^T S_b u_1) - u_1^T S_b u_1 \frac{d}{du_1}(u_1^T S_W u_1) \\
&= 2S_b u_1 (u_1^T S_W u_1) - 2S_W u_1 (u_1^T S_b u_1) = 0 \quad \text{divide by } 2(u_1^T S_W u_1) \\
&\iff S_b u_1 = \frac{u_1^T S_b u_1}{u_1^T S_W u_1} S_W u_1 \\
&\iff S_b u_1 = J(u_1) S_W u_1 \\
&\iff (S_W^{-1} S_b) u_1 = J(u_1) u_1.
\end{aligned}$$

We arrive at an eigenvalue problem where u_1 is the eigenvector of $S_W^{-1} S_b$ with the corresponding eigenvalue $J(u_1)$.

72.28.5 Linear Discriminant Algorithm

1. Compute the within-class scatter matrix S_W and the between-class scatter matrix S_b.

$$S_W = \sum_{k=1}^{n} \sum_{n \in \mathcal{C}_k} (x_n - m^{(k)})(x_n - m^{(k)})^T \quad S_b = \sum_{k=1}^{n} \sum_{n \in \mathcal{C}_k} (m^{(n)} - m^{(k)})(m^{(n)} - m^{(k)})^T.$$

2. Solve the eigenvalue problem $(S_W^{-1} S_b) u = \lambda u$ to find the eigenvalues λ and eigenvectors u.
3. Sort the eigenvectors by their corresponding eigenvalues in descending order.
4. Select the top k eigenvectors to form a projection matrix W.
5. Project the original data points onto the new subspace using $z_i = W^T x_i$.

72.29 Naive Bayes

The *naive Bayes* is a probabilistic classifier based on Bayes' Theorem with strong (naive) independence assumptions between the features.

Recall Bayes' theorem as

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \cdot \mathbb{P}(A)}{\mathbb{P}(B)}.$$

For our case we will use

$$\mathbb{P}(y \mid X) = \frac{\mathbb{P}(X \mid y) \cdot \mathbb{P}(y)}{\mathbb{P}(X)},$$

$$\mathbb{P}(y \mid X) = \frac{(\prod_{i=i}^{n} \mathbb{P}(x_i \mid y)) \cdot \mathbb{P}(y)}{\mathbb{P}(X)}$$

where $X = (x_1, x_2, \cdots, x_n)$ and with the assumption that the features are independent.

We want to select the class with the highest posterior probability

$$\begin{aligned} y &= \text{argmax}_y \mathbb{P}(y \mid X) = \text{argmax}_y \frac{(\prod_{i=i}^{n} \mathbb{P}(x_i \mid y)) \cdot \mathbb{P}(y)}{\mathbb{P}(X)} \\ y &= \text{argmax}_y \prod_{i=i}^{n} \mathbb{P}(x_i \mid y)\mathbb{P}(y) \\ y &= \text{argmax}_y \sum_{i=i}^{n} \ln(\mathbb{P}(x_i \mid y)) + \ln(\mathbb{P}(y)). \end{aligned}$$

We will use the following nomenclature:

- $\mathbb{P}(y)$ is the proper probability, which in this case is the frequency of each class.
- $\mathbb{P}(x_i \mid y)$ is the class conditional probability which is modeled with a normal distribution.

$$\mathbb{P}(x_i \mid y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right).$$

72.29.1 Naive Bayes Algorithm

1. Calculate the mean, variance and prior (frequency) for each class.
2. Calculate the posterior for each class with

$$y = \text{argmax}_y \sum_{i=i}^{n} \ln(\mathbb{P}(x_i \mid y)) + \ln(\mathbb{P}(y))$$

   and the normal distribution formula.
3. Choose the class with the highest probability.

72.30 Clustering

Clustering is a data analysis technique aimed at partitioning the data into mostly disjoint sets such that the elements of each set exhibit a great similarity based on the measurement chosen.

It is important to note that clustering algorithms always find clusters, no matter the data given. Therefore, it is the responsibility of the data scientist to analyze the results and determine if the clusters are meaningful or not.

72.30.1 K-Means Clustering

K-Means is an unsupervised learning iterative process that clusters the data set into k different clusters. Each sample is assigned to the cluster with the nearest mean, and then the means, *centroids*, and clusters are updated until there is no more change or the maximum number of iterations is reached.

Centroid

The *centroid* of a cluster C_i is defined as

$$\mu_i = \frac{1}{|C_i|} \sum_{x_j \in C_i} x_j.$$

It is the mean of the values inside the cluster.

72.30.2 Intra-Cluster Variance

Given $C_1, \ldots, C_n$ clusters with centroids $\mu_1, \ldots, \mu_n$, the *intra-cluster variance* is given by

$$J = \frac{1}{|C_i|} \sum_{i=1}^{k} \sum_{x_j \in C_i} \|x_j - \mu_i\|^2 .$$

The goal of the K-Means algorithm is to minimize J,

$$\text{argmin}_{C_1,\ldots,C_n} \frac{1}{|C_i|} \sum_{i=1}^{k} \sum_{x_j \in C_i} \|x_j - \mu_i\|^2 ,$$

but due to it being a rather complex optimization problem, K-Means is computed in an iterative manner.

72.30.3 K-Means Algorithm

1. Initialize centroids randomly by choosing a certain number of indexes from our data.
2. Initialize empty arrays for the number of clusters to which a centroid is going to be assigned.
3. Repeat until converged or max iterations are reached:

   Update the clusters by assigning each sample to a cluster based on the closest centroid.

   Update centroids by computing the new mean. If a cluster is empty, then a random value from the sample is used as such.

   Check convergence by checking if the old centroids are the same as the new ones.
4. We can return a label array with a label based on the cluster they ended in.

72.30.4 Elbow Heuristic

The *elbow heuristic* consists of plotting the explained variance as a function of the number of clusters and looking for an elbow point where the rate of decrease sharply changes.

- For $1, \ldots, K$ determine the clustering and the intra-cluster variance $J(k)$.
- Visualize $J(k)$ against k.
- Choose k at the elbow point where the decrease in $J(k)$ starts to slow down significantly.

72.31 Hierarchical Clustering

In contrast to the partitioning clustering like K-Means, *hierarchical clustering* creates different divisions in each iteration of the clustering process based on similarity. This can happen in two ways:

- *Top-Down*: We start with our set D and start splitting into subsets using a similarity metric, until we have sets with only one data point.
- *Bottom-Up*: This is the inverse process to the top-down approach, starting with singletons and then creating sets which, at the root level, are then merged into D.

72.31.1 Similarity Measurement

Given two sets X, Y we can measure their similarity in the following ways:

- *Single Linkage*: $D(X,Y) = \min_{x\in X, y\in Y} \|x-y\|$.
- *Complete Linkage*: $D(X,Y) = \max_{x\in X, y\in Y} \|x-y\|$.
- *Average Linkage*: $D(X,Y) = \frac{1}{|X||Y|}\sum_{x\in X}\sum_{y\in Y}\|x-y\|$.

72.32 External Cluster Validation

Under this category there are several methods, but commonly both *purity* and *mutual information* are used.

72.32.1 Purity

Given $C_1, \dots C_k$ clusters with T_1, T_k true clusters and $n_i = |C_i|$, we define the *purity of a cluster* i as

$$\text{purity}_i = \frac{1}{n_i}\max_{j=1}^{k}|C_i \cap T_j|$$

which gives 1 if the C_i contains only data points of T_j.

The *total purity* is given by

$$\text{purity} = \sum_{i=1}^{k}\frac{n_i}{n}\text{purity}_i = \frac{1}{n}\sum_{i=1}^{k}\max_{j=1}^{k}|C_i \cap T_j|$$

where n is the total number of samples.

72.32.2 Mutual Information

Mutual information is used to measure the similarity between the computed and true clusters C and T.

$$I(C,Y) = \sum_{i,j} p_{i,j}\log_2\left(\frac{p_{i,j}}{p_{C_i}p_{T_j}}\right)$$

where

- $p_{i,j} = \frac{|C_i \cap T_j|}{n}$ is the probability of a point in C_i being in T_j.
- $p_{C_i} = \frac{|C_i|}{n}$ is the probability of a point being in C_i.
- $p_{T_j} = \frac{|T_j|}{n}$ is the probability of a point being in T_j.

The bigger the mutual information, the more similar.

72.33 Internal Cluster Validation

Due to the fact that knowing the true clusters is not possible when performing explorative data analysis, we use methods such as *silhouette plots* and *prediction strength* to analyze the quality of our clusters.

72.33.1 Silhouettes

The *silhouette-index* is a measurement of how good a point was clustered using *cohesion* for the similarity to other points in the same cluster and also *separation* for measuring the differences to other clusters.

Given C_1, C_k clusters with $x_i, x_j \in C_m$ and the measurement $\|x_j - x_j\|_2$, we define *cohesion* as

$$a(x_i) = \frac{1}{|C_j| - 1} \sum_{x \in C_j, x \neq x_i} \|x_i, x\|_2 \,,$$

the mean distance from x_i to all other points in the cluster.

And *separation* as

$$b(x_i) = \min_{k \neq j} \frac{1}{|C_k|} \sum_{x \in C_j, x \neq x_i} \|x_i, x\|_2 \,,$$

the minimal mean distance from x_i to all other points in the closest cluster.

Finally, with these two concepts we can define the *silhouette-index* as

$$s(x_i) = \begin{cases} 1 - \frac{a(x_i)}{b(x_i)}, & \text{if } |C_j| > 1 \wedge a(x_i) < b(x_i) \text{ (good)} \\ 0, & \text{if } |C_j| = 1 \wedge a(x_i) = b(x_i) \text{ (not clear)} \\ \frac{b(x_i)}{a(x_i)} - 1, & \text{if } |C_j| > 1 \wedge a(x_i) > b(x_i) \text{ (bad).} \end{cases}.$$

We can also define the *average silhouette width* and *silhouette score* as

$$\text{average silhouette width}(C_j) = \overline{s_j} = \frac{1}{|C_j|} \sum_{x \in C_j} s(x),$$

$$\text{silhouette score} = \overline{s} = \frac{1}{n} \sum_{i=1}^{n} s(x_i).$$

72.33.2 Prediction Strength

The main idea of the *prediction strength* is that the quality of a clustering is good if the belonging to a specific cluster can be well predicted.

The Co-Membership Matrix and the Prediction Strength

Given the clusters $C_1, \ldots, C_k$ of $\mathcal{D}_{\text{train}}$ and $A_1, \ldots, A_k$ of $\mathcal{D}_{\text{val}}$ with size $n' = |\mathcal{D}_{\text{val}}|$; we denote our *co-membership matrix* $M \in \mathbb{R}^{n' \times n'}$.

$$M_{i,j} = \begin{cases} 1, & \text{if } \forall k : x_i, x_j \in C_k \\ 0, & \text{else} \end{cases} \quad \forall x_i, x_j \in \mathcal{D}_{\text{val}}.$$

This means that the entry is 1 if two data-points of our validation set are in the same cluster with respect to our training-centroids.

We define our *prediction strength* as

$$\text{ps}(k) = \min_{l=1,\ldots k} \frac{1}{(|A_l|)(|A_l| - 1)} \sum_{x_i, x_j \in A_l : x_i \neq x_j} M_{i,j}.$$

This means that we count the number of matches in our co-membership matrix computed from the train-labels for our current cluster l of the validation labels and divide it by the number of elements in the current cluster times the same amount minus 1.

Note that this algorithm is highly dependent on the chosen clustering strategy, here the K-Means variant was used for the clustering metric, but for example, if we do agglomerative clustering, we would use the mean Euclidean distance as our similarity measurement.

Prediction Strength Algorithm for K-Means

1. Divide the data into a training and validation set.
2. Apply the K-Means algorithm to both the training and validation sets, while storing both the labels and centroids.
3. Compute the co-membership matrix

$$M_{i,j} = \begin{cases} 1, & \text{if } \forall k : x_i, x_j \in C_k \\ 0, & \text{else} \end{cases} \quad \forall x_i, x_j \in \mathcal{D}_{\text{val}}.$$

4. Compute the prediction strength

$$\text{ps}(k) = \min_{l=1,\dots k} \frac{1}{(|A_l|)(|A_l|-1)} \sum_{x_i, x_j \in A_l : x_i \neq x_j} M_{i,j}.$$

Chapter 73

Introduction Numerical Mathematics

Numerical mathematics is the study of algorithms for the problems of continuous mathematics (Lloyd Trefethen, 1992).

73.1 Floating Point Numbers

A *floating point number* of a basis b has the representation

$$z = v \cdot m \cdot b^c,$$

where

- $b \in \mathbb{N} \setminus \{1\}$.
- $v \in \{-1, 1\}$.
- m is the *mantissa*, a b-adic fraction

$$m = m_{-k} \cdot \cdots \cdot m_{-1} m_0 m_1 \cdot \cdots \cdot m_l = m_{-k} b^k + \cdots + m_l b^{-l} = \sum_{i=-k}^{l} m_i b^{-i}$$

  with $m_i \in \{0, 1, \ldots, b-1\}$.
- $e \in \mathbb{N}$.

In this representation, numbers are not really well differentiable; hence, 0 cannot be represented properly.

Example:

$b = 2, \quad 10.11 \cdot 2^1 = (1 \cdot 2^1 + 0 \cdot 2^0 + 1 \cdot 2^{-1} + 1 \cdot 2^{-2}) \cdot 2 = \frac{11}{2}$.

73.1.1 Normalized Representation

The *normalized representation* of floating point numbers is given by

$$\pm(m_0 \cdot m_1 \cdot \cdots \cdot m_l) b^c = \pm \left(\sum_{i=0}^{l} m_i b^{-i} \right) b^c, \cdot m_0 \neq 0$$

It is well defined for numbers $\neq 0$.

73.1.2 Machine Precision

Given a floating point number in a basis b with mantissa length l,

$$\varepsilon_M = \frac{1}{2} b^{-l}$$

is called the *machine precision*.

73.1.3 Interpretation

In the floating point system numbers do not really represent one single number x but a range of numbers close to that x in the Reals. This combined with also rounding errors generated by trade-offs in range precision lead to the well-known errors when dealing with floating point numbers.

73.2 Errors

Almost all problems in numerical analysis can be described as a zero problem with a function g, input x, and output y: $g(x, y) = 0$.

For example, a linear system of equations $Ax = b$ is equivalent to

$$g(x, y) = 0, \quad x = (A, b), \quad g(x, y) = Ay - b$$

Criteria for Problems in the N.A.

The 3 criteria for problems in numerical analysis are:

1. A solution should exist. $g(x, y) = 0 \iff y = f(x)$.
2. For each x there should be exactly one y.
3. There should be a continuous dependence of y on x, i.e. $x' \to x \implies y' \to y$. Also, small changes in the input mean small changes in the output.

73.2.1 Well- and Ill-Posed Problems

Given a problem $g(x, y) = 0$ for which a solution exists, is unique, and has continuous dependence, the problem is *well-posed*; if not, it is *ill-posed*.

73.2.2 Error Measurement

Given a well-posed problem with x and

$$y = f(x) \quad f : X \to V,$$

where X and Y are vector spaces with metrics $\|\cdot\|_X$ and $\|\cdot\|_Y$. We denote the following scenarios:

- y is the exact solution.
- $\tilde{x}$ is the floating point representation of x.
- $\hat{y}$ is the result of solution via exact arithmetic with the exact data of x.
- $\tilde{y}$ is the floating point solution obtained by using $\tilde{x}$.
- f is the original function.
- $\hat{f}$ is the discrete (finite), simplification of the problem with which we replace f.
- $\tilde{f}$ is the floating point implementation of $\hat{f}$.

Example:

Given

$$f(x) = e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \quad x = \frac{5}{3}$$

With the exact solution $y = e^{\frac{5}{3}} \approx 5.29449$.

We use the substitute problem

$$\hat{f}(x) = \sum_{k=0}^{4} \frac{x^k}{k!} = \frac{10009}{1944} \approx 5.14866$$

Using three decimal places for FP, we get $\tilde{x} = 1.67$ and hence, $\tilde{y} = 5.16$.

73.2.3 Error Types

Given y and $\tilde{y}$ we get the following error types:

- $e = \tilde{y} - y$ is the *error*.
- $e_a = \|\tilde{y} - y\|_Y$ is the *absolute error*, the actual amount of deviation.
- $e_r = \frac{\|\tilde{y}-y\|_Y}{\|y\|_Y}$, for $\|y\|_Y \neq 0$ is the *relative error*, the percentage of deviation.

The relative error is in practice more important for analyzing numerical problems, but due to its definition, when the absolute error is small, it can lead to unnecessary precision constraints. For this reason, for practical applications, it is preferred to use a combination of both.

Example:

We are given $t_2 = -p + \sqrt{p^2 + q}$.

First,

$$x = (p, q), \quad y = f(x), \quad f(x_1, x_2) = -x_1 + \sqrt{x_1^2 + x_2}, \quad f : \mathbb{R}^2 \to \mathbb{R}$$

In this case $\hat{f} = f$, hence we do not need a substitute problem. We use for the metric $\|y\|_Y = |y|$. For $p = 10, q = 1$ we get

$$y = \sqrt{101} - 10 \approx 0.04988, \quad \tilde{y} = 0,$$

with

$$\begin{aligned} e &= \tilde{y} - y \approx -0.4988, \\ e_a &= |\tilde{y} - y| \approx 0.4988, \\ e_r &= \frac{|\tilde{y} - y|}{|y|} = 1 \end{aligned}$$

73.2.4 Error Criteria

Given $\varepsilon_a, \varepsilon_r$ as bounds for the absolute and relative errors, we try to find an approximation $\tilde{y}$ of y such that:

$$\|\tilde{y} - y\|_Y \leq \varepsilon_a + \|y\|_Y \varepsilon_r.$$

We call this a *mixed error criterion*.

Note that if $\|y\|_Y$ is small, then ε_a is dominant and

$$e_a = \|\tilde{y} - y\|_Y \lessapprox \varepsilon_a.$$

Else, the dominant term is ε_r, resulting in

$$\|\tilde{y} - y\|_Y \lessapprox \|y\|_Y \varepsilon_r$$

i.e.,

$$e_r = \frac{\|\tilde{y} - y\|_Y}{\|y\|_Y} \lessapprox \varepsilon_r.$$

Example:

Let us take the last example and add $\varepsilon_a = \varepsilon_r = 10^{-2}$.

$$y = \sqrt{101} - 10 \approx 0.04988, \quad \tilde{y} = 0$$

Then,

$$\|\tilde{y} - y\| \approx 0.04988, \quad \varepsilon_a + |y|\varepsilon_r \approx 1.04988 \cdot 10^{-2}$$

Therefore,

$$\|\tilde{y} - y\|_Y > \|y\|_Y \varepsilon_r.$$

Hence, we reject the approximation $\tilde{y}$.

73.2.5 Error Kinds

Starting from the error $\tilde{y} - y = \tilde{f}(\tilde{x}) - f(x)$, let us expand by adding zero.

$$\tilde{y} - y = \tilde{f}(\tilde{x}) - + \hat{f}(\tilde{x}) + \hat{f}(\tilde{x}) - f(\tilde{x}) + f(\tilde{x}) - f(x)$$

From this we define the following error kinds:

- *Input Error*: $f(\tilde{x}) - f(x)$. Caused by the conversion to the floating point format. It describes the sensitivity of the original problem in correlation to changes in the input data. It cannot really be influenced; the perspective highly depends on our choices.
- *Procedural Error*: $\hat{f}(\tilde{x}) - f(\tilde{x})$. Caused by the quality of the substitute problem. It gives a hint about the quality of the substitute problem. Can be reduced by using a better substitute problem.
- *Accumulation Error*: $\tilde{f}(\tilde{x}) - \hat{f}(\tilde{x})$. Accumulation of the rounding errors with FP. Can be handled by taking the properties of FP into consideration by ordering our operations accordingly and/or using more stable representations for decimal numbers.

In practice, all error kinds should be considered.

73.2.6 Input Error

Due to our problem being well-posed, we know that

$$\|\tilde{x} - x\|_X \implies \|f(\tilde{x}) - f(x)\|_Y < \varepsilon$$

We want to study how the conversion to the floating point format affects the results and how we can solve our problems with, if possible, small divergences.

73.3 Condition

Given $f : X \to Y$, the smallest constants c_a, c_r such that:

$$\|f(x') - f(x)\|_Y \leq c_a \|x' - x\|_X \iff c_a = \lim_{\Delta x \to 0} \frac{\|\Delta y\|}{\|\Delta x\|}, \quad c_a = \|J_f(x)\|$$

$$\frac{\|f(x') - f(x)\|_Y}{\|f(x)\|} \leq c_r \frac{\|x' - x\|_X}{\|x\|_X} \iff c_r = \frac{\|x\| \, \|f'(x)\|}{\|f(x)\|}, \quad c_r = \frac{\|x\|}{\|f(x)\|} \|J_f(x)\|$$

with $\|x' - x\|_X \leq \delta$; are called *absolute condition* and *relative condition* of f.

The interpretation of c_a is closely related to the Lipschitz continuity, meaning that a small variation in the input data due to conversion problems does not affect the results drastically.

For c_r, it is useful for when a user wants to limit the relative error by some amount ε_r, then

$$c_r \leq \frac{\varepsilon_r}{\varepsilon_M} \implies e_r(y) \leq \varepsilon_r$$

For c_a, it can be deduced that f is a Lipschitz continuous function with a smallest Lipschitz constant c_a. Hence,

$$\left| \frac{f'(x) - f(x)}{x' - x} \right| \leq c_a$$

and for $x' \to x$, we get $|f'(x)| \leq c_a$.

73.3.1 Interpretation

- $c \approx 1$, well-conditioned.
- $c >> 1$, ill-conditioned.
- $c = \infty$, ill-posed.

73.3.2 Approximations

For a continuous, differentiable f the following approximations are valid:

$$c_a \leq \max_{|\xi - x| \leq \delta} |f'(\xi)|, \quad k_r \leq \frac{|x|}{|f(x|} \max_{|\xi - x| \leq \delta} |f'(\xi))|$$

where $\xi \in \{z \mid z = x + t(x' - x), t \in [0, 1]\{$ from the mean value theorem

$$\frac{f(x') - f(x)}{x' - x} = f'(\xi)$$

Example:

Given $f(x) = 1 + \sqrt{x - 1}$ for $x > 1$.

The function is differentiable

$$f'(x) = \frac{1}{2\sqrt{x - 1}}.$$

$|\xi - x| \leq \delta$ means $\xi \in [x - \delta, x + \delta]$, therefore:

$$\max_{|\xi - x| \leq \delta} \frac{1}{2\sqrt{\xi - 1}} = \frac{1}{2\sqrt{x - \delta - 1}}$$

For $x = 2, \delta = 10^{-8}$ we get

$$\max_{|\xi - x| \leq \delta} |f'(\xi)| = \frac{1}{2\sqrt{1 - 10^{-8}}} \leq \frac{1}{2\sqrt{\frac{1}{2}}} = \sqrt{\frac{1}{2}} < 1$$

Hence $c_a < 1$ and $c_r < \frac{|2|}{|f(2)|} = 1$.

73.4 Error Propagation Formulas of the Differential Error Analysis

The approximations

$$c_a \approx |f'(x)|, \quad c_r \approx \frac{|x|}{|f(x)|}|f'(x)|$$

are considered the *error propagation formulas of the error analysis.*

For the **general case** $f : \mathbb{R}^n \to \mathbb{R}^m$, we take the generalized mean value theorem

$$f(x') = f(x) + \left(\int_0^1 f'((x + t(x' - x)))dt \right) (x' - x),$$

where

$$f'(x) = \begin{bmatrix} \partial_{x_1} f_1(x) & \dots & \partial_{x_n} f_1(x) \\ \dots & \dots & \dots \\ \partial_{x_1} f_m(x) & \dots & \partial_{x_n} f_m(x) \end{bmatrix}$$

We apply the metric

$$\|f(x') - f(x)\|_X \leq \left\| \int_0^1 f'((x + t(x' - x)))dt \right\|_{X,Y} \|x' - x\|_X$$
$$\leq \max_{\xi \in [x-\delta, x+\delta]} \|f'(\xi)\|_{X,Y} \|x' - x\|_X$$

For all x' holds that $\|x' - x\|_X \leq \delta$, then

$$c_a \leq \max_{\|\xi - x\|_X \leq \delta} \|f'(\xi)\|_{X,Y}$$

and

$$c_r \leq \frac{\|x\|_X}{\|f(x)\|_Y} \max_{\|\xi - x\|_X \leq \delta} \|f'(\xi)\|_{X,Y}$$

Example:

We observe $f : \mathbb{R}^2 \to \mathbb{R} \mapsto x_1 + x_2$ with the metrics

$$\|x\|_X = \|x\|_\infty = \max(\|x_1\|, \|x_2\|), \quad \|y\|_Y = \|y\|_\infty = |y|$$

For the Jacobian we get

$$f'(x) = [\partial_{x_1} f(x), \partial_{x_2} f(x)] = [1, 1]$$

Due to our chosen vector metrics we use the infinite metric for matrices

$$\|f'(x)\|_{X,Y} = \|f'(x)\|_\infty = \|[1,1]\| = 2$$

Therefore, $c_a \approx 2$ and $c_r \approx 2\frac{\|x\|_\infty}{|f(x)|}$.

73.5 Procedural Error

In contrast to the input error, this is much simpler and depends on the class of problem. For a linear system of equations using just matrix decomposition the difference $\hat{f}(\tilde{(x)}) - f(\tilde{x}) = 0$, while for an infinite sum it can vary depending on the number of terms used of the infinite sum.

73.6 Accumulation Error

This error occurs due to the floating point arithmetic not being associative for multiplication and addition. In contrast, they mostly preserve the commutativity. They are a commutative *monoid* with identity for each operation $(F,+),(F,\cdot)$.

73.7 Error Analysis for Linear Systems

We have a square system $Ax = b$ as the well-posed problem

$$f(A,b) = A^{-1}b = x.$$

We analyze the problem with the condition, i.e., the sensitivity of x to changes in the input, and denote this as $\tilde{A}\tilde{x} = \tilde{b}$. Then

$$\begin{aligned}\Delta A &= \tilde{A} - A\\ \Delta b &= \tilde{b} - b\\ \Delta x &= \tilde{x} - x\end{aligned}$$

Under these conditions we can write

$$\begin{aligned}x - \tilde{x} &= A^{-1}b - A^{-1}\tilde{b}\\ \Delta x &= A^{-1}(\Delta b)\end{aligned}$$

Hence, if we take the metric and divide by $\|x\|$

$$\begin{aligned}\frac{\|\Delta x\|}{\|x\|} &= \frac{\|A^{-1}(\Delta b)\|}{\|x\|}\\ &\leq \frac{\|b\|}{\|x\|}\|A^{-1}\|\frac{\|\Delta b\|}{\|b\|}\\ &= \frac{\|Ax\|}{\|x\|}\|A^{-1}\|\frac{\|\Delta b\|}{\|b\|}\\ &\leq \frac{\|A\|\|x\|}{\|x\|}\|A^{-1}\|\frac{\|\Delta b\|}{\|b\|}\\ &= \|A\|\|A^{-1}\|\frac{\|\Delta b\|}{\|b\|}\end{aligned}$$

We used two important facts; first, $Ax = b$ and second, $\|Ax\| \leq \|A\|\|x\|$.

73.7.1 Condition of a Matrix

Given a metric $\|\cdot\|$ on $\mathbb{R}^n$ with A being regular. We define

$$\kappa(A) = \|A\|\|A^{-1}\|$$

as the *condition number* of the matrix.

It has the following properties:

- κ depends highly on the metric.
- If $\alpha\|B\|' \leq \|B\| \leq \beta\|B\|'$, then $\alpha^2\kappa'(B) \leq \kappa(B) \leq \beta^2\kappa'(B)$.
- $\kappa(A) \geq 1 \implies \|I\| = \|AA^{-1}\| \leq \|A\|\|A^{-1}\| = \kappa(A)$.
- $\kappa(A) = \kappa(A^{-1})$.
- $\kappa(AB) \leq \kappa(A)\kappa(B)$.

73.8 Extended Condition

Our condition is until now only defined for regular square matrices, which means that they are invertible. We will expand this concept in the following way.

Start with

$$\|A^{-1}\| = \sup_{x\neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{z\neq 0} \frac{\|z\|}{\|Az\|}$$

because $x = Az$, and because our matrix is a bijective map, we assume that we never get the zero vector, but all other real numbers.

$$\begin{aligned}\sup_{z\neq 0} \frac{\|z\|}{\|x\|} &= \sup_{z\neq 0} \frac{1}{\left\|A\frac{z}{\|z\|}\right\|} \\ &= \sup_{\|z\|=1} \frac{1}{\|Az\|} \\ &= \max_{\|z\|=1} \frac{1}{\|Az\|} = \frac{1}{\min_{\|z\|=1}\|Az\|}\end{aligned}$$

By the fact that we are in a compact space, we assume the equality of the supremum and maximum, and now we can define the condition for any real matrix as follows:

$$\kappa(A) = \|A\|\|A^{-1}\| = \frac{\max_{\|y\|=1}\|Ay\|}{\min_{\|z\|=1}\|Az\|}$$

73.8.1 Inequality for Error Analysis in Linear Systems

Given a metric $\|\cdot\|$ on $\mathbb{R}^n$ with A being regular and $\|\Delta A\| < \frac{1}{\|A^{-1}\|}$, where $\|\cdot\|$ is an induced metric. Then, $\tilde{A}$ is regular and the following holds:

$$\frac{\|\Delta x\|}{\|x\|} \leq \frac{\kappa(A)}{1-\kappa(A)\frac{\|\Delta A\|}{\|A\|}}\left(\frac{\|\Delta A\|}{\|A\|} - \frac{\|\Delta b\|}{\|b\|}\right).$$

This is the case for an error of type $(A+\Delta A)(x+\Delta x) = (b+\Delta b)$.

There are also the cases

- $A(x + \Delta x) = (b + \Delta b)$:

$$\frac{\|\Delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\Delta b\|}{\|b\|}$$

- $(A + \Delta A)(x + \Delta x) = b$:

$$\frac{\|\Delta x\|}{\|x\|} \leq \frac{\kappa(A)}{1 - \kappa(A)} \frac{\|\Delta b\|}{\|b\|}$$

Example:

Given are

$$A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} -0.4 & 0.2 \\ 0.2 & 0.1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \ \|\Delta b\|_1 = 10^{-5}, \ \|\Delta A\|_1 = 0$$

We are looking for an upper bound for $\|\Delta x\|_1$.

In this case we are handling the case $A(x + \Delta x) = (b + \Delta b)$:

$$\begin{aligned} \kappa(A) &= \|A\|_1 \cdot \|A^{-1}\|_1 = \frac{42}{10} \\ \|b\|_1 &= 8 \\ \|x\|_1 &= 2 \end{aligned}$$

Resulting into

$$\begin{aligned} \frac{\|\Delta x\|}{\|x\|} &\leq \kappa(A) \frac{\|\Delta b\|}{\|b\|} \\ &\leq \frac{42}{10} \cdot \frac{10^{-5}}{8} 2 = 105 \cdot 10^{-5} \end{aligned}$$

Chapter 74

Numerical Linear Algebra

74.1 QR-Decomposition for Numerical Mathematics

We want to find $A = QR$, where R is an upper-triangular matrix and Q is just a matrix.

74.1.1 Givens Matrices

This algorithm uses rotations for the elimination of the elements underneath the main diagonal. We use matrices of the form:

$$Q_{i,j} = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & c & & s & & \\ & & & \ddots & & & \\ & & -s & & c & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix}$$

where, given an element $a_{k,l}$ to eliminate, we have

$$\begin{aligned} q_{k-1,k-1} &= c \\ q_{k-1,k} &= s, \\ q_{k,k-1} &= -s, \\ q_{k,k} &= c \end{aligned}$$

and

$$s = \frac{a_{k,l}}{\sqrt{a_{k-1,l}^2 + a_{k,l}^2}}, \quad c = \frac{a_{k-1,l}}{\sqrt{a_{k-1,l}^2 + a_{k,l}^2}}$$

When chained, this gives us

$$Q_{n-1} \cdot \cdots \cdot Q_1 A = R$$

By the orthonormality of the Q_i, we have

$$A = Q_1^T \cdot \cdots \cdot Q_{n-1}^T R = QR$$

Givens Algorithm

We have our system $Ax = b$ in $\mathbb{R}^{n\times n}$ and are going to store the Q and R in the same matrix. We start with $Q = I$ and $R = A$.

1. For each element $a_{i,j}$ underneath the main diagonal we compute:

$$s = \frac{a_{i,j}}{\sqrt{a_{i,j}^2 + a_{k,j}^2}}, \quad c = \frac{a_{k,j}}{\sqrt{a_{i,j}^2 + a_{k,j}^2}}$$

   where k is the **index of the first non-zero entry in the current column above** $a_{i,j}$.

2. Create the matrix $Q_{i,j}$.
3. Multiply $Q_{i,j}$ with A and store at the position of the generated zero the value s and at the position of c the value c.
4. Continue with the next element in the column until all elements are eliminated. Note that when applying the multiplication for the new A, we handle the places where we have stored the $\rho_{i-1,j-1}$ as if they were zero, because they are not used for the next steps.
5. Then, continue with the next column in the new sub-matrix and so on until we have an upper triangular matrix.
6. Now, compute $y = Q^T b = Q_{n-1} \cdot \cdots \cdot Q_1 b$. One matrix at a time. We use the stored coefficient $\rho_{i,j} = s$ to then compute $s = \sqrt{1 - s^2}$ for the multiplication.
7. Finally, solve $Rx = Q_{n-1} \cdot \cdots \cdot Q_1 Ax = y$ for x using back substitution.

It is important to note that this algorithm is four times more expensive than the LU-Decomposition, but it is more stable.

Example:

Find the QR-Decomposition of A and solve $Ax = b$.

$$A = \begin{bmatrix} 12 & -40 & 46 \\ -9 & 30 & 153 \\ 20 & 100 & 135 \end{bmatrix}, b = \begin{bmatrix} -16 \\ 12 \\ 140 \end{bmatrix}$$

We start by eliminating $a_{2,1}$:

$$c = \frac{12}{\sqrt{12^2 + (-9)^2}} = \frac{4}{5}, \; s = \frac{-9}{\sqrt{12^2 + (-9)^2}} = -\frac{3}{5}$$

Therefore

$$Q_{2,1} = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} & 0 \\ \frac{3}{5} & \frac{4}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Then

$$Q_{2,1}A = \begin{bmatrix} 15 & -50 & -55 \\ 0 & 0 & 150 \\ 20 & 100 & 135 \end{bmatrix}$$

We store $\rho_{2,1} = s = -\frac{3}{5}$ at the generated 0 at $a_{2,1}$.

$$\begin{bmatrix} 15 & -50 & -55 \\ -\frac{3}{5} & 0 & 150 \\ 20 & 100 & 135 \end{bmatrix}$$

Now, we eliminate the element $a_{3,1}$:

$$c = \frac{15}{\sqrt{15^2 + 20^2}} = \frac{3}{5},\ s = \frac{20}{\sqrt{15^2 + 20^2}} = \frac{4}{5}$$

Now we get

$$Q_{3,1} = \begin{bmatrix} \frac{3}{5} & 0 & \frac{4}{5} \\ 0 & 1 & 0 \\ -\frac{4}{5} & 0 & \frac{3}{5} \end{bmatrix}$$

Applying the multiplication to our previously computed matrix $Q_{2,1}A$

$$Q_{3,1}Q_{2,1}A = \begin{bmatrix} 25 & 50 & 75 \\ 0 & 0 & 150 \\ 0 & 100 & 125 \end{bmatrix}$$

We store $\rho_{3,1} = s = \frac{4}{5}$ at $a_{3,1}$.

$$\begin{bmatrix} 25 & 50 & 75 \\ -\frac{3}{5} & 0 & 150 \\ \frac{4}{5} & 100 & 125 \end{bmatrix}$$

Continuing with $a_{3,2}$:

$$c = \frac{0}{\sqrt{0^2 + 100^2}} = 0,\ s = \frac{100}{\sqrt{0^2 + 100^2}} = 1$$

$$Q_{3,2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

Multiply

$$Q_{3,2}Q_{3,1}Q_{2,1}A = \begin{bmatrix} 25 & 50 & 75 \\ 0 & 100 & 125 \\ 0 & 0 & -150 \end{bmatrix}$$

Now, we store $\rho_{3,1} = s = 1$ at $a_{3,2}$.

$$\begin{bmatrix} 25 & 50 & 75 \\ -\frac{3}{5} & 100 & 125 \\ \frac{4}{5} & 1 & -150 \end{bmatrix}$$

Now we solve $Qy = b \iff Q^T b = y$ for $Rx = y$ using

$$Q = (Q_{3,2}Q_{3,1}Q_{2,1})^T = \frac{1}{25}\begin{bmatrix} 12 & -16 & -15 \\ -9 & 12 & -20 \\ 20 & 15 & 0 \end{bmatrix}$$

and

$$R = Q_{3,2}Q_{3,1}Q_{2,1}A = 25\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & -6 \end{bmatrix}$$

For each of the matrices in $Q^T b$, we reconstruct the $Q_{i,j}$ matrix and perform the multiplication.

Using $\rho_{2,1} = -\frac{3}{5}, c = \sqrt{1-s^2} = \frac{4}{5}$:

$$Q_{2,1}b = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} & 0 \\ \frac{3}{5} & \frac{4}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -16 \\ 12 \\ 140 \end{bmatrix} = \begin{bmatrix} -20 \\ 0 \\ 140 \end{bmatrix}$$

Then repeat for the others, and multiply the new b by the reconstructed matrix, and we get

$$y = \begin{bmatrix} 100 \\ 100 \\ 0 \end{bmatrix}$$

Finally, for $Rx = y$ we have

$$25 \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & -6 \end{bmatrix} x = y$$

which gives us $x = [2, 1, 0]^T$.

74.1.2 Householder Algorithm

The *Householder algorithm*, similarly to the Givens algorithm, will give us a decomposition of a matrix A such that $A = QR$ with Q being orthonormal and R being upper triangular. The difference lies in the method of the transformations, which, instead of being rotations, are reflections.

Householder Transformation

Given a vector $v \in \mathbb{R}^n$ we denote

$$H_v = I - 2\frac{vv^T}{v^T v}$$

as the *Householder transformation* for v. Equivalently

$$H_v = I - 2\frac{vv^T}{\|v\|_2^2}.$$

If v is normalized, then $H_v = I - 2vv^T$, which is a reflection across the hyperplane orthogonal to v.

Example:

Given $v = [1, 2]^T$ we have

$$H_v = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2\frac{\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}}{\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2\frac{1}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} -3 & -4 \\ -4 & 3 \end{bmatrix}$$

74.1.3 Effect of a Householder Transformation

Given $x \in \mathbb{R}^n$ when multiplying H_k, which is the k-th Householder transformation, with x we get

$$H_k x = H_k \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \\ x_k \\ x_{k+1} \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \\ ? \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The vector to be used for the transformation is defined as

$$v = x \pm \|x\|_2 \, e_k$$

The sign is not really important, but it is usually chosen to avoid cancellation errors.

The k-th Householder transformation is then defined as $\tilde{x}$, where all elements before the k-th element are zero, i.e. if we have to eliminate all elements below the first element, then we have $k = 1$ and $\tilde{x} = x$, but if we have to eliminate all elements below the second element, then we have $k = 2$ and $\tilde{x} = [0, x_2, x_3, \ldots, x_n]^T$ and so on. Note that the resulting transformation will still be used with x.

Householder Algorithm

Given $A \in \mathbb{R}^{n \times n}$ we start with $Q = I$ and $R = A$. Similar to the LU-Decomposition we will start with the first column and then focus on the sub-matrix and so on.

1. Eliminate the elements below the pivot element $a_{i,i}$ by applying the Householder transformation H_i to the i-th column of the current matrix.
2. Multiply H_i with the current matrix and store the resulting matrix which will be used for the next step.
3. Continue with the next column and so on until we have an upper triangular matrix. The result should be $H_k \cdot H_{k-1} \cdot \cdots \cdot H_1 \cdot A = R$ and $Q = H_1 \cdot H_2 \cdot \cdots \cdot H_k$.

Note that the Householder matrices are self-inverse, i.e. $H_i^{-1} = H_i$ and therefore $A = QR$.

Example

Given is

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

We start with the first column and compute the Householder transformation for the first column. We have

$$v = \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \sqrt{10} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 + \sqrt{10} \\ 3 \end{bmatrix}$$

Then, we compute the Householder transformation

$$H_1 = I - 2\frac{vv^T}{v^T v} = \begin{bmatrix} -\frac{1}{5} & -\frac{3}{5} \\ -\frac{3}{5} & \frac{1}{5} \end{bmatrix}$$

And we apply it to A

$$H_1 A = \begin{bmatrix} -\frac{1}{5} & -\frac{3}{5} \\ -\frac{3}{5} & \frac{1}{5} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -\sqrt{10} & -\frac{4}{5}\sqrt{10} \\ 0 & \frac{2}{5}\sqrt{10} \end{bmatrix} = R$$

Now, we have an upper triangular matrix and we can compute Q as H_1 and R as $H_1 A$, and we have $A = QR$.

74.2 Iterative Methods for Linear Systems

An *iterative method*

$$x^{(k+1)} = \phi_k(x^{(k)}, x^{(k-1)}, \ldots, x^{(0)})$$

is convergent if $x^{(k)} \to x$ for all $x^{(0)} \in \mathbb{R}^n$. It also has the *fixed point property* $x = \phi(x) \iff Ax = b$.

The *error* $e^{(k)}$ and the *residuum* $r^{(k)}$ are defined as

$$e^{(k)} = x - x^{(k)}, \quad r^{(k)} = b - Ax^{(k)}$$

If, for the computation of $x^{(k+1)}$, only the direct predecessor $x^{(k)}$ is used, then

$$x^{(k+1)} = \phi(x^{(k)})$$

the process is considered with *one step* $\phi_k : \mathbb{R}^n \to \mathbb{R}^n$, else it is considered with *multi step* $\phi_k : (\mathbb{R}^n)^{k+1} \to \mathbb{R}^n$.

If $\phi_k = \phi$ for all k then the process is called *stationary*, else it is called *non-stationary*.

74.2.1 Single Step Stationary Iteration Method

A *single step stationary iteration method* is given by

$$x^{(k+1)} = \phi(x^{(k)}) = Bx + c,$$

where B is the *iteration matrix* and c is the *iteration vector*.

74.2.2 Error of Single Step Stationary Iteration Method

If the single step stationary iteration method has the fixed point property, then for $e^{(k)} = x - x^{(k)}, r^{(k)} = b - Ax^{(k)}$ we have

$$Ae^{(k)} = r^{(k)}$$

Proof:

$$Ae^{(k)} = A(x - x^{(k)}) = Ax - Ax^{(k)} = b - Ax^{(k)} = r^{(k)}$$

$\mathfrak{QED}$

74.2.3 Convergence

Analyzing our regular linear system $Ax = b$ with the single step iteration

$$x^{(k+1)} = \phi(x^{(k)}) = Bx^{(k)} + c$$

with the fixed point property $x = \phi(x) = Bx + c \iff Ax = b$, we get

For the error $e^{(k)} = x^{(k)} - x$ we have

$$e^{(k+1)} = x^{(k+1)} - x = Bx^{(k)} + c - (Bx + c) = Be^{(k)}$$

Therefore,

$$e^{(k)} = B^k e^{(0)}$$

Then

$$e^{(k)} \to 0 \quad \text{as} \quad k \to \infty$$

for an arbitrary $e^{(0)}$ if and only if $B^k \to 0 \quad \text{as} \quad k \to \infty$.

74.2.4 The Spectral Radius

The *spectral radius* of a matrix B is defined as

$$\rho(B) = \max\{|\lambda| \mid \lambda \in \sigma(B)\},$$

where $\sigma(B)$ is the *spectrum* of B, i.e. the set of all eigenvalues of B.

Properties of the Spectral Radius

- $\rho(B) \leq \|B\|_I$ for all induced metrics.
- $\forall \varepsilon > 0$ there exists an induced metric $\|\cdot\|_I$ from a vector metric $\|\cdot\|_V$ such that:

$$\|B\|_I = \sup_{w \neq 0} \frac{\|Bw\|_w}{\|w\|_V}$$

  with $\|B\|_I \leq \rho(B) + \varepsilon$.

74.2.5 Gelfand

For any matrix metric $\|\cdot\|$ the following holds:

$$\rho(B) = \lim_{k \to \infty} \sqrt[k]{\|B^k\|}.$$

Also for an induced matrix metric $\|\cdot\|_I$ $\|B\|_I^{\frac{1}{k}}$ converges towards $\rho(B)$.

74.2.6 Splitting Method for Linear Systems

Given a regular matrix $A \in \mathbb{R}^{n\times n}$ in a system $Ax = b$, we can split A into

$$A = M - N$$

where M is regular and N is arbitrary. Then, we can write

$$Mx = Nx + b$$

and we can define the iteration

$$x^{(k+1)} = M^{-1}Nx^{(k)} + M^{-1}b,$$

and we set $x^{(0)} = \vec{0} \in \mathbb{R}^n$ as the initial guess.

Example:

Given the system

$$\begin{bmatrix} 4 & -1 & 0 & 0 \\ -1 & 4 & -1 & 0 \\ 0 & -1 & 4 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} x = \begin{bmatrix} 15 \\ 10 \\ 10 \\ 10 \end{bmatrix}$$

Then we can split A into

$$M = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}, \quad N = \begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

Which results in the iteration

$$x^{(k+1)} = \begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 15 \\ 10 \\ 10 \\ 10 \end{bmatrix} = \begin{bmatrix} 3.75 \\ 2.5 \\ 2.5 \\ 3.333\ldots \end{bmatrix}$$

We can then continue the process using our resulting vector as the new $x^{(k)}$ and we will get closer to the solution of the system.

Triple Splitting

We can also define the splitting

$$A = D - L - U$$

where D is the diagonal of A, L is the lower triangular part of A and U is the upper triangular part of A. We then derive from that

$$\begin{aligned} Ax = b &\iff (D - L - U)x = b \\ &\iff Dx = (L + U)x + b \end{aligned}$$

The iteration is then given by

$$x^{(k+1)} = D^{-1}(L + U)x^{(k)} + D^{-1}b$$

74.2.7 The Residuum as Stopping Criterion

The *residuum* is defined as

$$r^{(k)} = b - Ax^{(k)}$$

When $\left\|r^{(k)}\right\| \leq \varepsilon \|b\|$ for a given $\varepsilon > 0$ then we can stop the iteration because we are close enough to the solution.

74.2.8 Convergence for Semi Positive Definite Matrices

Given semi positive definite matrices A, M; if $2M - A$ is also semi positive definite, then

$$\phi(x) = Bx + d,\ B = I - M^{-1}A,\ d = M^{-1}b$$

converges and also $\|B\|_A = \|B\|_M = \rho(B) < 1$.

Also, the splitting $M + M^T - A$ is also spd. and $\rho(B) \leq \|B\|_A < 1$.

74.2.9 Strictly Diagonal-Dominant

A is strictly diagonal-dominant if

$$|a_{i,i}| > \sum_{j \neq i} |a_{i,j}|,\ \forall i = 1, \ldots, n$$

74.2.10 The Jacobi Method for Linear Systems

The Jacobi method is a single step stationary iteration method with the splitting $A = D - (L + U)$ and the iteration:

$$x^{(k+1)} = D^{-1}(L + U)x^{(k)} + D^{-1}b$$

Where the *iteration-matrix* B is given by:

$$B = D^{-1}(L + U).$$

The Jacobi method converges if A is strictly diagonal-dominant or if $\rho(B) < 1$.

Example:

Given the system

$$\begin{bmatrix} 4 & -1 & 0 & 0 \\ -1 & 4 & -1 & 0 \\ 0 & -1 & 4 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} b = \begin{bmatrix} 15 \\ 10 \\ 10 \\ 10 \end{bmatrix}$$

We can apply the Jacobi method with the splitting $A = D - (L + U)$ where

$$D = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}, \quad L + U = \begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

Then, we can compute the iteration

$$x^{(k+1)} = \begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 15 \\ 10 \\ 10 \\ 10 \end{bmatrix}$$

Giving us

$$x^{(1)} = \begin{bmatrix} 3.75 \\ 2.5 \\ 2.5 \\ 3.333\ldots \end{bmatrix}$$

74.2.11 Gauss-Seidel Method

The Gauss-Seidel method is a single step stationary iteration method with the splitting $A = (D-L)-U$ and the iteration

$$x^{(k+1)} = (D-L)^{-1}Ux^{(k)} + (D-L)^{-1}b$$

Where the *iteration-matrix* B is given by:

$$B = (D-L)^{-1}U.$$

The Gauss-Seidel method converges if $\rho(B) < 1$.

74.2.12 After Iteration

This process refers to using an iterative method after using a mathematical algorithm for solving a certain problem to mitigate rounding errors. In our case we will present such a process with the LU-Decomposition.

While performing the LU-Decomposition we will get $\tilde{L}\tilde{U} = M$ which are the L, U matrices, but with rounding errors. We use the stationary linear iteration to improve the solution $\tilde{x}$ for which $A\tilde{x} = b$.

1. Perform the LU-Decomposition to get the triangular matrices and the approximate solution with errors.
2. Set $x_0 = \tilde{x}$.
3. Repeat the following steps as long as $\frac{\|p_k\|}{\|x_{k+1}\|} < \varepsilon$:
   - $r_k = b - Ax_k$.
   - $p_k = M^{-1}r_k$, in other words $\tilde{L}\tilde{U}p_k = r_k$. Also

     ```
     p_k = np.linalg.solve(A, r_k)
     ```
   - $x_{k+1} = x_k + p_k$.

74.2.13 Gradient Descent

Given our stationary iteration

$$x^{(k+1)} = x^{(k)} + M^{-1}r^{(k)}$$

We add the parameter $\alpha^{(k)}$

$$x^{(k+1)} = x^{(k)} + \alpha^{(k)}M^{-1}r^{(k)}$$

The goal in each step is to find an $\alpha^{(k)}$ which gives us the "best" $x^{(k+1)}$. This means

$$\begin{aligned} e^{(k+1)} &= x - x^{(k+1)} \\ &= x - x^{(k)} + \alpha^{(k)} M^{-1} r^{(k)} \\ &= e^{(k)} - \alpha^{(k)} p^{(k)} \end{aligned}$$

Geometrically, $e^{(k+1)}$ is on a straight line determined by $e^{(k)}$ in the direction $p^{(k)}$. This means that $\left\|e^{(k+1)}\right\|$ will be minimal when $p_{(k)}$ is orthogonal to $e^{(k+1)}$.

$$\begin{aligned} \left\langle e^{(k+1)}, p^{(k)} \right\rangle &= \left\langle p^{(k)}, e^{(k)} - \alpha^{(k)} p^{(k)} \right\rangle \\ &= \left\langle p^{(k)}, e^{(k)} \right\rangle - \alpha^{(k)} \left\langle p^{(k)}, p^{(k)} \right\rangle \\ \alpha^{(k)} &= \frac{\left\langle p^{(k)}, e^{(k)} \right\rangle}{\left\langle p^{(k)}, p^{(k)} \right\rangle} \end{aligned}$$

Hence, we have constructed the following iterative algorithm

$$\begin{aligned} r^{(k)} &= b - Ax^{(k)} \\ p^{(k)} &= M^{-1} r^{(k)} \\ \alpha^{(k)} &= \frac{\left\langle p^{(k)}, e^{(k)} \right\rangle}{\left\langle p^{(k)}, p^{(k)} \right\rangle} \\ x^{(k+1)} &= x^k + \alpha^{(k)} p^{(k)} \end{aligned}$$

The problem is that technically we do not know the exact solution x, but this can be solved using a suitable inner product with metric

$$\|x\| = \|x\|_A = \sqrt{x^T A x}$$

Giving us

$$\alpha^{(k)} = \frac{\left\langle p^{(k)}, e^{(k)} \right\rangle_A}{\left\langle p^{(k)}, p^{(k)} \right\rangle_A} = \frac{\left\langle p^{(k)}, Ae^{(k)} \right\rangle}{\left\langle p^{(k)}, Ap^{(k)} \right\rangle} = \frac{\left\langle p^{(k)}, r^{(k)} \right\rangle}{\left\langle p^{(k)}, Ap^{(k)} \right\rangle}$$

74.3 Iterative Methods for Eigenvalue Problems

74.3.1 The Vector Iteration

Given a normal matrix $A \in \mathbb{C}^{n \times n}$, which is unitarily diagonalizable, i.e. there exists an orthonormal basis of eigenvectors, $u_1, \ldots, u_n$.

Each vector $x \in \mathbb{C}^n$ can be written as $x = \sum_i x_i u_i$ and

$$Ax = \sum_i x_i \lambda_i u_i, \quad A^k x = \sum_i x_i \lambda_i^k u_i$$

If $|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|$ then the dominant eigenvalue is λ_1. This means that by applying A to x multiple times, we get closer to the dominant eigenvector u_1 and the dominant eigenvalue λ_1.

Algorithm

1. Choose a $x^{(0)}$ such that $\left\|x^{(0)}\right\|_2 = 1$ and $\left\langle x^{(0)}, u_1 \right\rangle \neq 0$.
2. Repeat the following steps until convergence:

$$
\begin{aligned}
y^{(k+1)} &= Ax^{(k)} \\
x^{(k+1)} &= \frac{y^{(k+1)}}{\left\|y^{(k+1)}\right\|_2} \\
\mu^{(k+1)} &= \left\langle x^{(k+1)}, Ax^{(k+1)} \right\rangle = \frac{\left\langle x^{(k+1)}, y^{(k+1)} \right\rangle}{\left\|x^{(k)}\right\|_2^2}
\end{aligned}
$$

74.3.2 The Jacobi Method for Eigenvalues

The *Jacobi method* is an iterative method for computing the eigenvalues of a symmetric matrix A by applying a sequence of orthogonal transformations to A to make it diagonal.

The eigenvalues of A are then given by the diagonal elements of the resulting matrix.

The derivation will come from applying orthonormal matrices repeatedly to our original matrix.

$$A^{(0)} = A, \quad A^{(k)} = Q_k^T A^{(k-1)} Q_k$$

Using our Givens-Rotations

$$
Q_k = \begin{bmatrix}
1 & & & & & & \\
& \ddots & & & & & \\
& & q_{k-1,k-1} = c & & q_{k-1,k} = s & & \\
& & & 1 & & & \\
& & & & \ddots & & \\
& & q_{k,k-1} = -s & & q_{k,k} = c & & \\
& & & & & 1 & \\
& & & & & & \ddots \\
& & & & & & & 1
\end{bmatrix}
$$

where

$$
c = \cos(\alpha), \quad s = \cos(\alpha), \alpha = \begin{cases} \frac{\pi}{4} & a_{i_0,i_0} = a_{j_0,j_0} \\ \frac{1}{2} \arctan\left(\frac{2a_{i_0,j_0}}{a_{j_0,j_0} - a_{i_0,i_0}}\right) & \end{cases}
$$

And we use the iteration

$$Q^{(0)} = I, \quad Q^{(k)} = Q^{(k-1)} Q_k$$

Stop Criterion

We define the square sum of the diagonal elements of the matrices $A^{(k)}$ as

$$N(A^{(k)}) = \sum_{i \neq j} \left(a_{i,j}^{(k)}\right)^2.$$

It holds that

$$N(A^{(k+1)}) \leq \alpha N(A^{(k)}), \quad \alpha \in [0,1)$$

converges *strictly monotonously decreasing* towards zero. Thus, this is our convergence criterion given an upper bound.

74.4 The QR Method

We take a matrix $A \in \mathbb{R}^{n \times n}$ with n start vectors whose dimension is also n.

Choose $X^{(0)} = [x_1^{(0)}, \ldots, x_n^{(0)}]$, where the vectors form an orthonormal basis.

For $k = 0, 1, \ldots$ repeat the following steps:

$$\begin{aligned} Y^{(k+1)} &= AX^{(k)} \\ \tilde{Q^{(k+1)}} R^{(k+1)} &= Y^{(k+1)} \\ X^{(k+1)} &= \tilde{Q^{(k+1)}} \\ T^{(k+1)} &= {X^{(k)}}^T Y^{(k+1)} = {X^{(k)}}^T A X^{(k)} \end{aligned}$$

Then for

$$\begin{aligned} T^{(k)} &= {X^{(k-1)}}^T A X^{(k-1)} \\ &= {X^{(k-1)}}^T Y^{(k)} \\ &= {X^{(k-1)}}^T \tilde{Q^{(k)}} R^{(k)} \\ &= ({X^{(k-1)}}^T X^{(k)}) R^{(k)} \\ &= Q^{(k)} R^{(k)} \end{aligned}$$

We can also use $Q^{(k)} = {X^{(k-1)}}^T X^{(k)}$. And because the vectors of $X^{(k)}$ are orthonormal, we also have that

$$\begin{aligned} T^{(k+1)} &= {X^{(k)}}^T A X^{(k)} \\ &= {X^{(k)}}^T A X^{(k-1)} {X^{(k-1)}}^T X^{(k)} \\ &= {X^{(k)}}^T Y^{(K)} {X^{(k-1)}}^T X^{(k)} \\ &= {X^{(k)}}^T \tilde{Q^{(k)}} R^{(k)} {X^{(k-1)}}^T X^{(k)} \\ &= {X^{(k)}}^T X^{(k)} R^{(k)} {X^{(k-1)}}^T X^{(k)} \\ &= R^{(k)} ({X^{(k-1)}}^T X^{(k)}) \\ &= R^{(k)} Q^{(k)} \end{aligned}$$

This means that our iteration can be simplified to

$$A^{(k)} = Q^{(k)} R^{(k)}, \quad A^{(k+1)} = R^{(k)} Q^{(k)}$$

Because A is in the real Schur-form $A = QTQ^T$, under certain circumstances we get:

$$A^{(k)} \to^{k \to \infty} T, \quad \hat{Q}^{(k)} \to^{k \to \infty} T$$

with $\hat{Q}^{(k)} = Q^{(0)} \cdot \ldots \cdot Q^{(k-1)}$.

74.5 Roots of Polynomials

When dealing with different kinds of situations and models, most problems can be interpreted as looking for the roots of a non-linear polynomial. For this, there are plenty of theories and methods.

74.5.1 The Bisection Algorithm

Given a continuous function $f : \mathbb{R} \to \mathbb{R}$, with $a_0, b_0 \in \mathbb{R}$ such that

$$f(a_0) < 0, f(b_0) > 0 \quad \text{or} \quad f(a_0) > 0, f(b_0) < 0$$

that is $f(a_0)f(b_0) < 0$, then there must be a root in between (Intermediate Value Theorem).

We define our iteration as

$$x_k = \frac{a_k + b_k}{2}$$

and apply the following rules:

If $f(x_k)f(a_k) < 0$ then

$$a_{k+1} = a_k, \quad b_{k+1} = x_k$$

Else

$$a_{k+1} = x_k, \quad b_{k+1} = b_k$$

then we repeat the process. The termination criterion is the number of iterations set for the computation.

74.5.2 The Regula-Falsi Algorithm

This is a variation of the *bisection algorithm* which improves the convergence velocity of the algorithm.

We start as before:

Given a continuous function $f : \mathbb{R} \to \mathbb{R}$, with $a_0, b_0 \in \mathbb{R}$ such that

$$f(a_0) < 0, f(b_0) > 0 \quad \text{or} \quad f(a_0) > 0, f(b_0) < 0$$

that is, $f(a_0)f(b_0) < 0$ implies there must be a zero in between.

We define our iteration as

$$x_k = \frac{a_k + b_k}{2}$$

Now we use the information of $(a_k, f(a_k))$ and $(b_k, f(b_k))$ to define a line

$$s(x) = f(a_k) + \frac{f(b_k) - f(a_k)}{b_k - a_k}(x - a_k)$$

We set it to $s(x) = 0 = ...$, which gives us

$$s(x_k) = 0 \iff x_k = -\frac{b_k - a_k}{f(b_k) - f(a_k)} f(a_k) = \frac{a_k f(b_k) - b_k f(a_k)}{b_k - a_k}$$

We use the point at which our secant crosses the x-axis as our new estimation, giving us the new iteration rules:

$$x_k = \frac{a_k f(b_k) - b_k f(a_k)}{b_k - a_k}$$

If $f(x_k)f(a_k) < 0$ then

$$a_{k+1} = a_k, \quad b_{k+1} = x_k$$

Else

$$a_{k+1} = x_k, \quad b_{k+1} = b_k$$

74.5.3 The Secant Algorithm

This is again a slight modification of the previous algorithm given x_{-1}, x_0.

$$x_{k+1} = x_k - \frac{f(x_k)}{\frac{f(x_k)-f(x_{k-1})}{x_k - x_{k-1}}}$$

This time we want to find a root, and for that we define different secants which will approximate us to that specific point.

74.6 The Newton Algorithm

Taking the Taylor series of our function f and only considering the first linear term, we get

$$f(y) = f(x_k) + (y - x_k)f'(x_k), \quad y = x_k - \frac{f(x_k)}{f'(x_k)}$$

The *Newton algorithm* defines the following iteration:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

If we were to stop at the quadratic term, we would get:

$$x_{k+1} = x_k - \frac{f'(x_k) \pm \sqrt{(f'(x_k))^2 - 2f(x_k)f''(x_k)}}{f''(x_k)}.$$

The geometric interpretation is that we are approximating the zero by the tangent at the point $(x_k, f(x_k))$.

This algorithm comes from the idea that when taking a derivative if its slope is negative, then we need to move in the positive x direction: $x_{k+1} = x_k + \alpha f'(x_k)$ and if the slope is positive, then we move in the negative direction $x_{k+1} = x_k - \alpha f'(x_k)$.

74.6.1 The Newton Algorithm Under Banach

Given the function $f(x)$ we define our Newton iteration as

$$\Phi(x) = x - \frac{f(x)}{f'(x)}$$

with

$$\Phi'(x) = \frac{f(x)f''(x)}{(f'(x))^2}$$

$$\Phi''(x) = \frac{((f'(x))^3 f''(x) + f(x)f'(x))^2 f'''(x) - 2f(x)f'(x)(f''(x))^2}{(f'(x))^4}$$

We can show that our new function is a contraction by showing:

$$\|\Phi'(x)\| \leq \alpha < 1$$

For this, we must find an upper bound for $\|f(x)\|, \|f''(x)\|$ or a lower bound for $\|f'(x)\|^2$.

Example:

Show that the Newton algorithm is a contraction for $f(x) = x + \ln(x) - 2$ on $X = [1, 2]$.

$$f'(x) = 1 + \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2}$$

This returns

$$\Phi'(x) = \frac{f(x)f''(x)}{(f'(x))^2} = \frac{-\frac{1}{x^2}(x + \ln(x) - 2)}{\left(1 + \frac{1}{x}\right)^2} = \frac{-(x + \ln(x) - 2)}{(x+1)^2}$$

Now with $|\cdot|$

$$|\Phi'(x)| = \frac{|(x + \ln(x) - 2)|}{(x+1)^2}$$

We use the derivatives of $f(x)$ to analyze the way we are going to find our bounds.

$f'(x)$ is monotonically increasing, therefore

$$f'(x) = 1 + \frac{1}{x} > 0, \quad \text{for} \quad x \in [1, 2] \implies f(x) \in [f(1), f(2)]$$

For our denominator we know that it becomes minimal when x is minimal, therefore

$$|\Phi'(x)| = \frac{|f(x)|}{(x+1)^2} \leq \frac{1}{4} = \alpha$$

74.6.2 Convergence of the Newton Algorithm

If $f : \mathbb{R} \to \mathbb{R}$ is two times continuously differentiable with $f(x) = 0, f'(x) \neq 0$, i.e. x is a root of f. If $x_0 \in [x - \varepsilon, x + \varepsilon]$ where ε is infinitesimally small, then the Newton algorithm converges *quadratically.*

74.6.3 Taylor Expansion for Zero Roots of the Newton Algorithm

If we have the case that x is a zero root of $f(x), f'(x), \ldots$ or even leads to undefined expressions, then we can write f as a Taylor polynomial

$$f(y) = (y - x)^q g(y)$$

where $q - 1$ is the degree of the last derivative such that $f'(x) = 0, f''(x) = 0, \ldots, f^{(q-1)} = 0, f^{(q)} \neq 0$.

This gives us

$$f'(x) = q(y - x)^{q-1} g(y) + (y - x)^q g'(x)$$

Then

$$\frac{f(y)}{f'(x)} = \frac{(y-x)^q g(y)}{q(y-x)^{q-1}g(y) + (y-x)^q g'(x)} = \frac{(y-x)g(x)}{qg(x) + (y-x)g'(x)}$$

Resulting in

$$\lim_{y \to x} \frac{f(y)}{f'(y)} = 0.$$

Chapter 75

Polynomial Interpolation

Polynomial interpolation is a family of procedures to find a polynomial given a certain number of data points $(x_0, y_0), \ldots, (x_n, y_n)$ such that

$$f(x_i) = y_i, \quad \forall i = 0, 1, \ldots, n.$$

This results in finding a polynomial of degree at most n, with $n+1$ terms and n constant terms c_i

$$f(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n.$$

75.1 Set of Polynomials of Degree n

We define the set of polynomials of degree n as P_n.

75.1.1 Maximal Number of Zeros for a Polynomial

A polynomial $p \in P_n$ has at most n zeros or is identically zero.

75.1.2 Uniqueness of the Polynomial Interpolation

Given a set of points $(x_0, y_0), \ldots, (x_n, y_n)$, there exists only one polynomial of degree at most n such that $f(x_0) = y_0, \ldots$.

Proof:

Uniqueness: Let us assume that such a polynomial $p(x)$ exists, and also $q(x) \neq p(x)$ such that both interpolate our points.

Using the fact that a polynomial $p \in P_n$ has at most n zeros or is identically zero.

If we perform the following operation

$$p(x) - q(x) = 0, \quad \forall x \in \{x \mid (x, y),\ y = 0\}$$

We have a polynomial $p(x) - q(x) \in P_n$ but it has zeros at $n+1$ places, hence $p(x) - q(x) = 0 \implies p(x) = q(x)$.

75.2 Vandermonde Systems

For our interpolation $p(x)$ we define the following system

$$Vc = \begin{bmatrix} 1 & x_0 & \dots & x_0^n \\ \vdots & \vdots & & \vdots \\ 1 & x_n & \dots & x_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} y_0 \\ \vdots \\ y_n \end{bmatrix} = y$$

$Vc = y$ as our *Vandermonde system* with V as the *Vandermonde matrix*.

Now our problem of finding the constants in the vector c has been transformed into finding the inverse of V.

75.2.1 Invertibility of the Vandermonde Matrix

There exists a V^{-1} for the corresponding V.

Proof:

$\exists V^{-1} \iff \det(V) \neq 0$, therefore we need to compute the determinant in the following way.

Remember that row/column operations do not change the determinant

$$\det(A) = \det(A') = \det(C_i A) = \det(C_i)\det(A) = 1 \cdot \det(A)$$

For

$$V = \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix}$$

We add from the first column times $-x_0$, then the second $x_0 x_1$ and so on for each column

$$\begin{aligned}
\det(V) &= \det \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \\
&= \det \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \\
&= \det \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & x_1 - x_0 & x_1^2 - x_0 x_1 & \dots & x_1^n - x_0 x_1^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n - x_0 & x_n^2 - x_0 x_n & \dots & x_n^n - x_0 x_n^{n-1} \end{bmatrix} \\
&= \det \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & x_1 - x_0 & x_1(x_1^2 - x_0) & \dots & x_1^{n-1}(x_1 - x_0) \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n - x_0 & x_n(x_n - x_0) & \dots & x_n^{n-1}(x_n - x_0) \end{bmatrix} \\
&= (x_1 - x_0)(x_2 - x_0) \cdot \dots \cdot (x_n - x_0) \det \begin{bmatrix} 1 & x_1 & \dots & x_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & x_n & \dots & x_n^{n-1} \end{bmatrix}
\end{aligned}$$

By repeatedly applying the column subtraction, we get a transformed matrix from which we can extract the scalar factors from each column and then using the Laplace method for determinants, repeat the process for the inner sub-matrix.

Giving us an inductive procedure and the following results

$$\det(V) = \prod_{0 \leq i < j \leq n} (x_j - x_i) \neq 0 \iff x_i \neq x_j \ \forall i \neq j.$$

75.3 Lagrange Polynomials

Lagrange polynomials are used to interpolate polynomials by using the following property

$$L_j(x_i) = \begin{cases} 1, & i = j \\ 0, \end{cases}$$

This is done by the definition

$$L_j(x) = \prod_{i=0}^{j-1} \frac{x - x_i}{x_j - x_i} \prod_{i=1}^{n} \frac{x - x_{j+i}}{x_j - x_{j+i}} = \prod_{i=0, i \neq j}^{n} \frac{x - x_i}{x_j - x_i}$$

This comes from the fact that, given our points $x_0, x_1, \ldots, x_n$, we want for x_n that $L_m(x_m) = 1$ and zero elsewhere, so we get

$$(x - x_1)(x - x_2) \cdot \cdots \cdot (x - x_{m-1})(x - x_{m+1}) \cdot \cdots \cdot \dot{(x - x_n)} \neq 1, \quad \text{but zero for every other } x_k$$

To get it to be 1, we just need to normalize the term

$$\frac{(x - x_1)(x - x_2) \cdot \cdots \cdot (x - x_{m-1})(x - x_{m+1}) \cdot \cdots \cdot \dot{(x - x_n)}}{(x_m - x_1)(x_m - x_2) \cdot \cdots \cdot (x_m - x_{m-1})(x_m - x_{m+1}) \cdot \cdots \cdot \dot{(x_m - x_n)}}$$

Our Lagrange polynomial is defined as

$$p(x) = \sum_{i=0}^{n} y_i L_i(x_i) = \sum_{i=0}^{n} y_i \left[\prod_{i=0, i \neq j}^{n} \frac{x - x_i}{x_j - x_i} \right]$$

Example:

Given are the points $(0, 3), (1, 2), (3, 6)$ find the interpolating polynomial.

$$L_0(x) = \frac{(x-1)(x-3)}{(0-1)(0-3)} = \frac{(x-1)(x-3)}{3}$$

$$L_1(x) = \frac{(x-0)(x-3)}{(1-0)(1-3)} = \frac{(x)(x-3)}{-2}$$

$$L_2(x) = \frac{(x-0)(x-1)}{(3-0)(3-1)} = \frac{(x)(x-1)}{6}$$

Resulting in the following interpolating polynomial

$$p(x) = 3 \cdot \frac{(x-1)(x-3)}{3} + 2 \cdot \frac{(x)(x-3)}{-2} + 6 \cdot \frac{(x)(x-1)}{6} = x^2 - x + 3$$

75.4 Newton Polynomials

The interpolation polynomial in the *Newtonian form* is defined as

$$p(x) = y_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n \prod_{i=0}^{n-1}(x - x_i)$$

Note that for $p(x) = y_0 + a_1(x - x_0)$, if we put $p(x_0) = y_0$, which is an equality we want. If we set our polynomial to y_1, we also get the first *divided difference*

$$y_1 = y_0 + a_1(x_1 - x_0) \iff a_1 = \frac{y_1 - y_0}{x_1 - x_0} = f[x_0, x_1]$$

where $f[x_i, x_j]$ is the *divided difference* defined as

$$f[x_i, x_j] = \frac{f[x_i, j-1] - f[x_{i-1}, x_{j-1}]}{x_i - x_{i-j}}$$

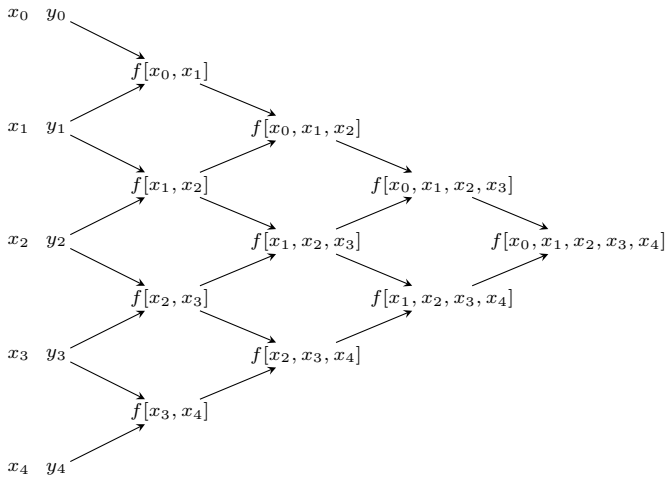
with

$$f[x_i, 0] = y_i$$

We can keep the step we did without a recurrence relation for the next higher degree polynomials and get all of our differences.

75.4.1 Recurrence Relation Table

When working out an exercise, it is useful to write down the divided differences in a table as follows:



Although the table looks complicated, when using the actual values for x_i and x_j we follow a simple pattern to fill in the table by going diagonally down and up. For example, for

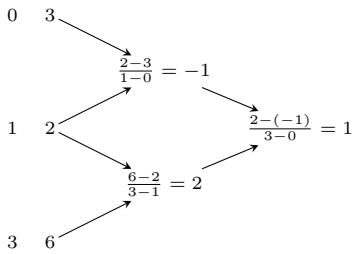
$$f[x_0, x_1] = \frac{y_1 \text{ (start)} \ - y_0 \text{ (up)}}{x_1 \text{ (diagonally down)} \ - x_0 \text{ (diagonally up)}}$$

It is important to note that the first column of the table is just the y_i values; for the next columns we use the previous column to compute the numerator, and the denominator is still defined by the x_i values.

Example:

Given are the points $(0, 3), (1, 2), (3, 6)$ find the interpolating polynomial.

We start by filling in the first column of the table with the y_i values



The resulting interpolating polynomial is

$$p(x) = 3 + (-1)(x-0) + 1(x-0)(x-1) = x^2 - x + 3$$

75.5 Polynomial Spline Interpolation

For $x_0 < x_1 < \cdots < x_n$ we denote $\Omega_n = \{x_0, \ldots, x_n\}$ the *subdivision of the interval* $[x_0, x_n]$. A function s_k such that

$$s_k \mid_{[x_i, x_{i+1}]} \text{ is a polynomial of degree } c \leq k,$$

$$s_k \text{ is } C^{k-1} \text{ for } [x_0, x_n],$$

where C^{k-1} means differentiable to degree at most $k-1$; is denoted as *polynomial spline* of order k with respect to Ω_n.

Polynomial splines are a way of interpolating a polynomial to fit a continuous function in a certain interval, and then combining those piecewise functions into one continuous function. This can be used when an interpolation method delivers errors in certain regions of the domain. To address the discontinuities, we use constraints using the derivatives of our polynomials.

75.5.1 Cubic Splines

A *cubic polynomial spline* of degree 3 has the form

$$s_3 \mid_{[x_i, x_{i+1}]} (x_i) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3.$$

Note that s_3 is two times differentiable.

We get our main conditions from

$$\begin{aligned} s_3(x_1) &= y_i \\ &\vdots \\ s_3(x_i) &= y_i \end{aligned}$$

and our smoothness conditions from

$$\begin{aligned} C^0 &: s_3(x_j)_- = s_3(x_j)_+ \\ C^1 &: s_3'(x_j)_- = s_3'(x_j)_+ \\ C^2 &: s_3''(x_i)_- = s_3''(x_j)_+ \end{aligned}$$

They just mean that our function will be continuous on the points where two splines meet.

Other types of conditions are for example:

the *natural spline*

$$s_3''(x_0) = s_3''(x_n) = 0,$$

the *periodic spline*

$$s_3'(x_0) = s_3'(x_n) \wedge s_3''(x_0) = s_3''(x_n),$$

the *hermitian spline*

$$s_3'(x_0) = us_3'(x_n) = v,$$

given both u, v.

Direct Computation

Given the derivatives of s_3

$$\begin{aligned} s_3(x) &= a_i + b_i x + +c_i x^2 + d_i x^3 \\ s_3'(x) &= b_i + 2c_i x + 3d_i x^2 \\ s_3''(x) &= 2c_i + 6d_i x, \end{aligned}$$

the interpolation and continuity properties are fulfilled if

$$\begin{aligned} a_i + b_i x_i + +c_i x_i^2 + d_i x_i^3 &= y_i \\ a_i + b_i x_{i+1} + +c_i x_{i+1}^2 + d_i x_{i+1}^3 &= y_{i+1}, \end{aligned}$$

also the continuity of the first derivative delivers the following property

$$b_{i-1} + 2c_{i-1}x_i + 3d_{i_1}x_i^2 = b_i + 2c_i x_i + 3d_i x_i^2$$

and the second derivative the following

$$2c_{i-1} + 6_{i-1}x_i = 2c_i + 6d_i x_i$$

for $i = 0, \ldots, n-1$. This provides us with

$$\begin{aligned} 2_c 0 + 6d_0 x_0 &= 0 \\ 2c_{n-1} + 6_{n-1}x_i &= 0 \end{aligned}$$

This will deliver a system of equations of $4n \times 4n$.

Example:

For four variables we would get the following cubic splines S_i with the main conditions:

$$
\begin{aligned}
S_1(x_1) &= y_1 \\
S_1(x_2) &= y_2 \\
S_2(x_2) &= y_2 \\
S_2(x_3) &= y_3 \\
S_3(x_3) &= y_3 \\
S_3(x_4) &= y_4 \\
S_4(x_4) &= y_4
\end{aligned}
$$

the first derivative conditions for the intersection points

$$
\begin{aligned}
S_1'(x_2) &= S_2'(x_2) \\
S_2'(x_3) &= S_3'(x_3)
\end{aligned}
$$

and lastly the second derivative conditions

$$
\begin{aligned}
S_1''(x_2) &= S_2''(x_2) \\
S_2''(x_3) &= S_2''(x_3)
\end{aligned}
$$

Computation Over Moments

We define a *moment* as

$$\beta_i = s_3''(x_i), \quad i = 0, \ldots, n.$$

We also define $\alpha_i := s_3'(x_i)$.

Given that our first and second derivatives are both defined, we can define our second derivative in the form

$$s_3''(x) = \beta_i + \frac{x - x_i}{x_{i+1} - x_i}(\beta_{i+1} - \beta_i) = \frac{\beta_{i+1} - \beta_i}{h_i}(x - x_i)$$

with $h_i = x_{i+1} - x_i$.

If we integrate over the region $[x_i, x]$

$$s_3'(x) = \int_{x_i}^{x} \frac{\beta_{i+1} - \beta_i}{h_i}(x - x_i) = \beta_i(x - x_i) + \frac{b_{i+1} - \beta_i}{2h_i}(x - x_i)^2$$

Giving the following line

$$s_3'(x) = \alpha_i + \beta_i(x - x_i) + \frac{b_{i+1} - \beta_i}{2h_i}(x - x_i)^2$$

Then, repeating the integration over the same integral delivers

$$s_3(x) = y_i + \alpha_i(x - x_i) + \frac{\beta_i}{2}(x - x_i)^3 + \frac{\beta_{i+1} - \beta_i}{6h_i}(x - x_i)^3$$

By substituting $x = x_{i+1}$ in $s_3'(x)$ we get:

$$\begin{aligned}
\alpha_{i+1} &= s_3'(x_{i+1}) \\
\alpha_{i+1} &= \alpha_i + \beta_i(x_{i+1} - x_i) + \frac{b_{i+1} - \beta_i}{2h_i}(x_{i+1} - x_i)^2 \\
\alpha_{i+1} - \alpha_i &= \frac{b_i + \beta_{i+1}}{2}
\end{aligned}$$

Continuing, by substituting $x = x_{i+1}$ into $s_3(x_i)$ we get

$$\begin{aligned}
s_3(x_{i+1}) = y_{i+1} &= y_i + \alpha_i(x_{i+1} - x_i) + \frac{\beta_i}{2}(x_{i+1} - x_i)^3 + \frac{\beta_{i+1} - \beta_i}{6h_i}(x_{i+1} - x_i)^3 \\
&= u_i + \alpha_i h_i + \frac{\beta_i}{2}h_i^2 + \frac{\beta_{i+1} - \beta_i}{6}h_i^3 \\
\frac{y_{i+1} - y_i}{h_i} &= \alpha_i + \frac{\beta_i}{3}h_i + \frac{\beta_{i+1}}{6}h_i
\end{aligned}$$

By evaluating on the inner points of an interval $[x_i, x_{i+1}]$, $[x_{i-1}, x_i]$ we get

$$\begin{aligned}
\frac{y_{i+1} - y_i}{h_i} &= \alpha_i + \frac{\beta_i}{3}h_i + \frac{\beta_{i+1}}{6}h_i \\
\frac{y_i - y_{i-1}}{h_{i-1}} &= \alpha_{i-1} + \frac{\beta_{i-1}}{3}h_{i-1} + \frac{\beta_i}{6}h_{i-1}
\end{aligned}$$

Now, by subtracting the two equations, we get

$$\begin{aligned}
\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}} &= \alpha_i - \alpha_{i-1} + \frac{\beta_i}{3}h_i + \frac{\beta_{i+1}}{6}h_i - \frac{\beta_{i-1}}{3}h_{i-1} - \frac{\beta_i}{6}h_{i-1} \\
&= \frac{\beta_i + \beta_{i+1}}{2}h_{i-1} + \frac{\beta_i}{3}h_i + \frac{\beta_{i+1}}{6}h_i - \frac{\beta_{i-1}}{3}h_{i-1} - \frac{\beta_i}{6}h_{i-1} \\
&= \frac{h_{i-1}}{6}\beta_{i-1} + \frac{h_{i-1} + h_i}{3}\beta_i + \frac{h_i}{6}\beta_{i+1}
\end{aligned}$$

More specifically, we get the following system of equations

$$h_{i-1}\beta_{i-1} + 2(h_{i-1} + h_i)\beta_i + h_i\beta_{i+1} = 6\left(\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\right)$$

for $i = 1, \ldots, n-1$. This is a tridiagonal system of equations which can be solved in $O(n)$ time.

$$\begin{bmatrix}
2(h_0 + h_1) & h_1 & 0 & \dots & 0 \\
h_1 & 2(h_1 + h_2) & h_2 & \dots & 0 \\
0 & h_2 & 2(h_2 + h_3) & \dots & 0 \\
\vdots & \vdots & \vdots & & \vdots \\
0 & 0 & 0 & \dots & 2(h_{n-2} + h_{n-1})
\end{bmatrix}
\begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \vdots \\ \beta_{n-1} \end{bmatrix} =
\begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \vdots \\ \gamma_{n-1} \end{bmatrix}$$

where

$$\gamma_i = 6\left(\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\right), \quad h_i = x_{i+1} - x_i$$

Everything but the α_i is known, but they can be computed by

$$\alpha_i = \frac{y_{i+1} - y_i}{h_i} - \frac{\beta_i}{3} h_i - \frac{\beta_{i+1}}{6} h_i$$

Hence, we get the following algorithm for computing the cubic spline interpolation.

75.5.2 Algorithm for Cubic Spline Interpolation

Given the points $(x_0, y_0), \ldots, (x_n, y_n)$, we can compute the cubic spline interpolation by the following steps:

1. Compute h_i and γ_i for $i = 1, \ldots, n-1$.
2. Solve the tridiagonal system of equations $A\beta = \gamma$.
3. Compute all α_i
4. Substitute the values of α_i and β_i into the formula for $s_3(x)$ to get the cubic spline interpolation.

Example:

We are given the following table of values

i	0	1	2	3	4
x_i	-3	-1	0	1	3
y_i	$\frac{3}{5}$	1	$\frac{3}{2}$	1	$\frac{3}{5}$

for the function $\frac{1}{2} + \frac{1}{x^2+1}$, compute the cubic interpolation.

We use our formula for $h_i = x_{i+1} - x_i$ to compute our differences

i	0	1	2	3	4
h_i	2	1	1	2	-

We substitute into our system

$$\begin{bmatrix} 2(2+1) & 1 & 0 \\ 1 & 2(1+1) & 1 \\ 0 & 1 & 2(1+2) \end{bmatrix} \vec{\beta} = \begin{bmatrix} 6\left(\frac{\frac{3}{2}-1}{1} - \frac{1-\frac{3}{5}}{2}\right) \\ 6\left(\frac{1-\frac{3}{2}}{1} - \frac{\frac{3}{2}-1}{2}\right) \\ 6\left(\frac{\frac{3}{2}-1}{2} - \frac{1-\frac{3}{2}}{1}\right) \end{bmatrix}$$

using the condition $\beta_0 = \beta_n = 0$, we get

i	0	1	2	3	4
β_i	0	$\frac{3}{5}$	$-\frac{9}{5}$	$\frac{3}{5}$	0
α_i	0	$\frac{3}{5}$	0	$-\frac{3}{5}$	-

Hence, our splines using the formula

$$s_3(x) = y_i + \alpha_i(x - x_i) + \frac{\beta_i}{2}(x - x_i)^3 + \frac{\beta_{i+1} - \beta_i}{6h_i}(x - x_i)^3$$

are:

$$s_3(x) = \begin{cases} \frac{x^3}{20} + \frac{9x^2}{20} + \frac{27x}{20} + \frac{39}{20} & x < -1 \\ -\frac{2x^3}{5} - \frac{9x^2}{10} + \frac{3}{2} & x < 0 \\ \frac{2x^3}{5} - \frac{9x^2}{10} + \frac{3}{2} & x < 1 \\ -\frac{x^3}{20} + \frac{9x^2}{20} - \frac{27x}{20} + \frac{39}{20} & \text{else} \end{cases}$$

75.6 Interpolation Curves

We define a continuous function

$$\gamma : [a, b] \subseteq \mathbb{R} \to \mathbb{R}^m, \quad t \to \gamma(t)$$

as *curve* with a parameter $t \in [a, b]$.

Now, given $(t_i, z_i), t_i \in [a, b], z_i \in \mathbb{R}^m$, we want to find $f : [a, b] \to \mathbb{R}^m$ such that $f(t_i) = z_i \ \forall i$. We will use *polynomial curves*

$$f(t) = \begin{bmatrix} p_1(t) \\ \vdots \\ p_m(t) \end{bmatrix}$$

where $p_j(t)$ is a polynomial of degree n.

One of the methods for computing such polynomials is Newtonian divided differences, but instead of using individual y_i we use the vectors with the corresponding values from x_i for the differences.

Chapter 76

Approximations

76.1 Polynomial Linear Regression

Given data-points $(x_i, y_i) \in \mathbb{R}^2,\ i = 1, \ldots, m$, we define the *mean square error* as

$$\tilde{e} = \sqrt{\sum_{i=1}^{m} \left(f(x_i) - y_i\right)^2} = \sqrt{\sum_{i=1}^{m} e_i^2} = \|e\|_2, \quad e = \begin{bmatrix} e_1 \\ \vdots \\ e_m \end{bmatrix}.$$

Now, instead of using a linear term, we define a polynomial of degree $n-1$:

$$f(x) = p_1 + p_2 x + \cdots + p_m x^{n-1}.$$

Hence, we want to minimize the expression

$$\|e\|_2 = \left\| \begin{bmatrix} f(x_1) - y_1 \\ \vdots \\ f(x_m) - y_m \end{bmatrix} \right\|_2$$

which delivers the following system

$$\left\| \begin{bmatrix} 1 & x_1 & \cdots & x_1^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & x_m & \cdots & x_m^{n-1} \end{bmatrix} \begin{bmatrix} p_1 \\ \vdots \\ p_n \end{bmatrix} - \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} \right\|_2 = \|Ap - y\|_2 .$$

Thus, given $A \in \mathbb{R}^{m \times n}$ such that $\operatorname{rank}(A) = n$ and $m \geq n$, we have the minimization problem

$$\min_{x \in \mathbb{R}^n} \|Ap - y\|_2$$

whose solution $\tilde{x} \in \mathbb{R}^n$ is also the solution of the *normal equation*

$$A^T A \tilde{x} = A^T b$$

where $A^T A \in \mathbb{R}^{n \times n}$ is regular.

Proof:

We first want to show that $A^T A$ is regular; hence, we do the following:

If $A^T A x = 0$, then

$$0 = x^T A^T Ax = (Ax)^T Ax = \|Ax\|_2^2 \implies Ax = 0.$$

Because A has maximal rank, $x = 0$; hence, $A^T A$ is regular.

Now we can show that $\tilde{x} = (A^T A)^{-1} A^T b$ minimizes $\|Ax - b\|_2^2$:

$$\begin{aligned} J(x) = \|Ax - b\|_2 &= (Ax - b)^T (Ax - b) \\ &= ((Ax)^T - b^T)(Ax - b) \\ &= x^T (A^T Ax - Ab^T) + b^T (b - Ax) \end{aligned}$$

Hence, $J(\tilde{x}) = \tilde{x}^T((A^T)(A^T A)^{-1} A^T b - A^T b) + b^T(b - A\tilde{x}) = b^T(b - A\tilde{x})$.

If we assume there exists another solution $\tilde{y} = \tilde{x} + v,\ v \in \mathbb{R}^n,\ v \neq 0$, then

$$\begin{aligned} J(\tilde{y}) &= (\tilde{x} + v)^T (A^T A(\tilde{x} + v) - A^T b) + b^T (b - A(\tilde{x} + v)) \\ &= (\tilde{x} + v)^T (A^T A\tilde{x} + A^T Av - A^T b) + b^T (b - A\tilde{x} + Av)) \\ &= (\tilde{x} + v)^T A^T A\tilde{x} + A^T Av + b^T (b - A\tilde{x}) - b^T A^v \\ &= v^T A^T A(\tilde{x} + v)^T - v^T A^T b + b^T (b - A\tilde{x}) \\ &= v^T (A^T A\tilde{x}(\tilde{x} - A^T b)^T + v^T A^T b + b^T (b - A\tilde{x}) \\ &= \|Av\|_2^2 + J(\tilde{x}) \end{aligned}$$

Because $v \neq 0$ and A has the maximal rank, $Av \neq 0$ and $\|Av\|_2^2 > 0$; hence

$$J(\tilde{y}) = \|A(\tilde{y} - \tilde{x})\|_2^2 + J(\tilde{x}) > J(\tilde{x}).$$

Therefore, $\tilde{x}$ is the solution with the minimal distance.

$\mathfrak{QED}$

Example:

Given the points $(0, 1), (2, 2), (3, 2)$ find the minimal solution to the minimization problem using $f(x) = p_1 + p_2 x$, $(n = 2)$. Notice that our basis is 1, x; hence:

$$A = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad x = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}$$

Hence for $A^T Ax = A^T b$ we get

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 5 & 13 \end{bmatrix}$$

and

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

Hence by solving

$$\begin{bmatrix} 3 & 5 \\ 5 & 13 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

using Gaussian elimination we get $p_1 = \frac{15}{14}$ and $p_2 = \frac{5}{14}$, giving us the final polynomial

$$f(x) = \frac{5}{14} + \frac{15}{14}x.$$

76.2 Pseudo Inverse

Given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, there is exactly one solution to the minimization problem $x \in \mathbb{R}^n$, which is dependent on the matrix $A^+ \in \mathbb{R}^{n \times m}$ with the property

$$A^+ b = x, \ \forall b \in \mathbb{R}^m.$$

This matrix is the *pseudo inverse of* A and obeys the following properties:

- $(A^+ A)^T = A^+ A$
- $(AA^+)^T = AA^+$
- $A^+ AA^+ = A^+$
- $AA^+ A = A$

76.3 Singular Value Decomposition

Given $A \in \mathbb{R}^{m \times n}$, then there exist matrices $U \in \mathbb{R}^{m \times m}, \ V \in \mathbb{R}^{n \times n}$ such that

$$A = U \Sigma V^T$$

where

$$\Sigma \in \mathbb{R}^{m \times n}, \ \Sigma = \text{diag}(\sigma_1, \ldots, \sigma_p), \ p = \min(m, n)$$

and

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0.$$

This is the *singular value decomposition of* A.

For the case $A \in \mathbb{R}^{n \times n}$, this is just a diagonalisation.

Properties of the SVD

- $V \Sigma^T U^T$ is SVD of A^T and has the same *singular values* as A.
- If $\sigma_1 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_p = 0$, then $\text{rank}(A) = r$.
- $\sigma_1^2, \ldots \sigma_p^2$ are the eigenvalues of $AA^T, \ A^T A$.
- The columns of $U, \ V$ are the eigenvectors of $AA^T, \ A^T A$.
- If $r = \text{rank}(A), \ A = U \Sigma V^T$ with $\Sigma = \text{diag}(\sigma_1, \ldots, \sigma_r, 0, \ldots, 0) \in \mathbb{R}^{m \times n}$, then

$$A^+ = V \Sigma^+ U^T, \quad \Sigma^+ = \text{diag}(\frac{1}{\sigma_1}, \ldots, \frac{1}{\sigma_r}, 0, \ldots, 0,) \in \mathbb{R}^{n \times m}$$

Hence, to compute the singular value decomposition, we need to compute the eigenvalues and eigenvectors of AA^T or $A^T A$ and take the square root of its eigenvalues to get the *singular values*.

76.4 SVD Regularization

Given $A \in \mathbb{R}^{m\times n}$ with a SVD $A = U\Sigma V^T$ with

$$\Sigma = \operatorname{diag}(\frac{1}{\sigma_1}, \dots, \frac{1}{\sigma_r}, 0, \dots, 0,) \in \mathbb{R}^{m\times n}.$$

For $x = A^+ b$, $\Delta x = A^+ \Delta b$, the following approximations are valid:

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \frac{\sigma_1}{\sigma_r} \frac{\|\Delta b\|_2}{\|P_r U^T b\|_2},$$

with $P_r = \operatorname{diag}(1_1, \dots, 1_r, 0, \dots, 0) \in \mathbb{R}^{m\times m}$. There is always a b and Δb such that

$$\frac{\|\Delta x\|_2}{\|x\|_2} = \frac{\sigma_1}{\sigma_r} \frac{\|\Delta b\|_2}{\|b\|_2}.$$

We also define the *condition* of our matrix as $\kappa_2(A) = \frac{\sigma_1}{\sigma_r}$ if all of our singular values are greater than zero.

76.4.1 Truncated SVD (TSVD)

Given $A \in \mathbb{R}^{n\times n}$ with a SVD $A = U\Sigma V^T$,

$$\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0,) \in \mathbb{R}^{m\times n},\ \sigma_1 \geq \cdots \geq \sigma_r > 0.$$

For $\alpha > 0$, $A_\alpha = U\Sigma_\alpha V^T$ with

$$\Sigma_\alpha = \operatorname{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0,) \in \mathbb{R}^{m\times n},\ \frac{\sigma_k}{\sigma_1} \geq \sqrt{\alpha} \geq \frac{\sigma_{k+1}}{\sigma_1}.$$

is the *truncated singular value decomposition* of A.

From this definition, we also get the fact that $\kappa_2(A_\alpha) \leq \frac{1}{\sqrt{\alpha}}$.

76.5 Tikhonov Regularization

When computing $x = A^+ b$, we get the equivalent minimization problem

$$\min \|Ax - b\|_2, \quad \min \|x\|.$$

By combining them in an expression, we get

$$\begin{aligned} J_\alpha(x) &= \|Ax - b\|_2^2 + \alpha \|x\|_2^2 \\ &= \left\| \begin{bmatrix} Ax \\ \sqrt{\alpha} x \end{bmatrix} - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|_2^2 \\ &= \left\| \begin{bmatrix} A \\ \sqrt{\alpha} I \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|_2^2 \\ &= \|A_\alpha x - b_\alpha\|_2^2 \end{aligned}$$

The rank is preserved by $\alpha > 0$, hence we can minimize the normal equation

$$A_\alpha^T A_\alpha x_\alpha = A_\alpha^T b_\alpha.$$

We can rewrite as

$$[A^T, \alpha I] \begin{bmatrix} A \\ \alpha I \end{bmatrix} x_\alpha = [A^T, \alpha I] b_\alpha$$

giving us

$$x_\alpha = (A^T A + \alpha I)^{-1} A^T b$$

Formally, for $A \in \mathbb{R}^{m \times n},\ \alpha > 0$,

$$x_\alpha = (A^T A + \alpha I)^{-1} A^T b$$

is the *Tikhonov regularization* of $x = A^+ b$ with parameter α.

Chapter 77

Numerical Integration

The main idea of numerical integration is to find a function $g(x)$ which is easier to integrate, instead of integrating $f(x)$ directly.

77.1 Newton Cotes

Given $-\infty < a < b < \infty$ and $f \in [a, b]$, we want to approximate

$$\int_a^b f(x)dx$$

with the use of equidistant points $x_k = a(b-a)\frac{k}{n} \in [a, b]$, k being the number of points. Hence, we use polynomial interpolation which gives us:

$$\int_a^b g(x)dx = (b-a)\sum_{k=0}^{n} \beta_k f(x_k)$$

where β_k are the *weights* given by our Newton interpolation algorithm.

It is also preferred to divide the target interval into sub-intervals and then apply the approximation on those intervals to avoid using high degree polynomials.

$$\int_a^b f(x)dx = \sum_{i=0}^{m-1} \int_{a_i}^{b_i} g(x)$$

with $a = a_0 < b_0 = a_1 < b_1 \ldots a_{m-1} < b_{m-1} = b$.

In general, this is the template for the coming methods.

77.2 The Midpoint Rule

The *midpoint rule* builds rectangles by taking the middle point of the sub-intervals as the height of our rectangles.

$$M_n = \sum_{i=1}^{n} f\left(\frac{x_{i-1} - x_i}{2}\right) h, \quad h - \frac{b-a}{2}.$$

With the error

$$E \leq \frac{(b-a)^3}{24n^2} \max_{a \leq x \leq b} |f''(x)|.$$

77.3 Trapezoidal Sum

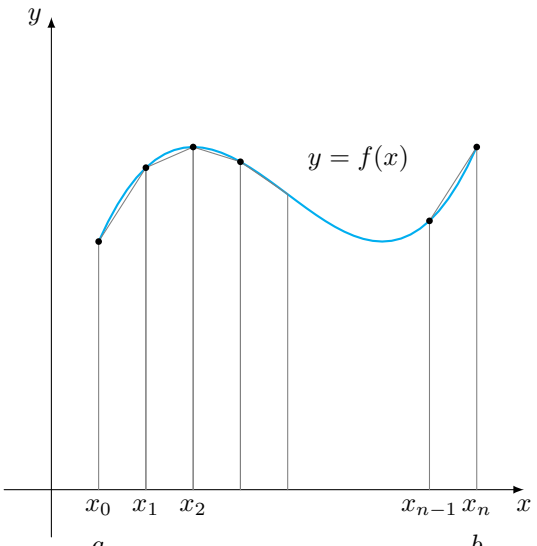
One of the most common approaches for approximating integrals which is very easy to compute is the *trapezoidal sum* which consists of dividing our interval $[a, b]$ into n trapezoids whose area is given by $\frac{h}{2}(a + b)$.

$$T_n = \frac{h}{2}\left(f(a) + 2\sum_{i=1}^{n-1} f(a + ih) + f(b)\right), \quad h = \frac{b-a}{n}.$$

With the error

$$E \leq \frac{(b-a)^3}{12n^2} \max_{a \leq x \leq b} |f''(x)|.$$

Visually it looks like:



(C)

The simple trapezoidal rule is a special case of the trapezoidal sum with $n = 1$.

$$T_1 = \frac{b-a}{2}(f(a) + f(b)).$$

77.4 Simpsons Rule

The *Simpsons rule* is an improvement of the trapezoidal rule which uses parabolas using two intervals in order to improve the approximation of the sum.

$$S_n = \frac{b-a}{3n}\left(f(a) + \sum_{i=1}^{n-1} s(i) f(a + ih) + f(b)\right)$$

where

$$s(i) = \begin{cases} 4, & \text{if } i \text{ odd} \\ 2, & \text{else} \end{cases}.$$

With error

$$E \leq \frac{(b-a)^5}{180n^4} \max_{a \leq x \leq b} |f^{(4)}(x)|.$$

The simple Simpsons rule is a special case of the Simpsons rule with $n = 1$.

$$S_1 = \frac{b-a}{6}(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)).$$

77.5 Accuracy

We can use the error formulas and solve for n to find the number of sub-intervals to use with our approximation given an initial precision.

Example:

Given $\int_{-2}^{1}(-x^4 - 2x^3 + 12x^2 + 2x - 5dx)$, find the n for a result using the Simpsons rule with accuracy of 0.1.

Using our formula

$$E \leq \frac{(b-a)^5}{180n^4} \max_{a \leq x \leq b} |f^{(4)}(x)|.$$

Assuming we already know our maximum, we would get:

$$\begin{aligned} \frac{(1-(-2))^5}{180n^4} \max_{a \leq x \leq b} |f^{(4)}(x)| &\leq \frac{1}{10} \\ \frac{(1-(-2))^5}{180n^4} \max_{a \leq x \leq b} 24 &\leq \frac{1}{10} \\ n &\leq 4.24 \end{aligned}$$

Hence, we need to use an $n \geq 4.24$.

77.6 Lagrange Integration for $[0, 1]$

For the interval $[0, 1]$ we get the equidistant points $x_k = \frac{k}{n}$ giving us the interpolation

$$g(x) = \sum_{k=0}^{n} f(x_k) L_k(x),$$

where

$$L_k(x) = \prod_{i=0, i \neq k} \frac{x - x_i}{x_k - x_i}$$

is our *Lagrange polynomial*.

Using them gives us the approximation

$$\int_1^0 g(x) = \sum_{k=0}^{n} f(x_k) \int_0^1 L_k(x)dx = \sum_{k=0}^{n} \beta_k f(x_k)$$

with $\beta_k = \int_0^1 L_k(x)dx$.

77.6.1 Generalization for $[a, b]$

We can extend the previous idea to any real, closed interval

$$\int_a^b f(x)dx = (b-a)\int_0^1 f(a+(b-a)y)dy \approx (b-a)\sum_{k=0}^{n}\beta_k f\left(a+(b-a)\frac{k}{n}\right).$$

77.7 Summation for Error Mitigation

$$\int_a^b f(x)dx = (b-a)\sum_{i=0}^{m-1}\int_{a_i}^{b_i} f(x)dx \approx \sum_{i=0}^{m-1}(b_i-a_i)\sum_{k=0}^{n}\beta_k f(x_{k,i})$$

with $a = a_0 < b_0 = a_1 < b_1 \ldots a_{m-1} < b_{m-1} = b$. $x_{k,i} = a_i + (b_i - a_i)\frac{k}{n}$.

For the case when the intervals have the same size then our intervals boundaries are given by

$$h = \frac{b-a}{m} = b_i - a_i, \quad a_i = a + ih,\ b_i = a + (i+1)h.$$

77.7.1 Bernoulli Polynomials and the Bernoulli Numbers

The *Bernoulli polynomials* b_k are defined by

$$b_0(x) = 1, \quad b_k'(x) = k, \ \int_0^1 b_k(x)dx = 0,\ k \geq 1.$$

The *Bernoulli numbers* are given by

$$B_0 = b_0(0) = b_0(1), \quad B_1 = b_1(0) = -b_1(1), \quad B_k = b_k(0) = b_k(1),\ k \geq 2.$$

These numbers are used in the next theorem.

77.8 Euler-Maclaurin Summation

If $f \in C^{2n}([a, b])$, then the following formula holds:

$$\sum_{n=a}^{b} f(n) = \int_a^b f(x)dx + \frac{f(a)-f(b)}{2} + \sum_{n=1}^{\infty}\frac{B_{2n}}{(2n)!}(f^{(2n-1)}(b) - f^{(2n-1)}(a)).$$

This formula is used to approximate integrals without computing a lot of terms.

Chapter 78

Computability

78.1 Church Turing Thesis

The intuitive computable functions are exactly all functions which can be computed by a *Turing Machine*. In other words, every computational model can be expressed as such.

78.2 Textual Substitution

For a given condition of the form $\{E[x := R]\}$, we perform the substitution in the following way:

$$(2x + y)[x := 4y] = 8y + y = 9y$$

If there are multiple replacements of the form $[\dots], \dots, [\dots]$, we perform those from left to right.

78.3 Leibniz Inference Rule

To clarify, the notation we will be using from now on looks like a fraction, but it means that assuming the statement above is true, then the following (below) is also true.

$$\frac{X = Y}{E[z := X] = E[z := Y]}$$

78.4 Correctness of Algorithms

The *correctness* of an algorithm is determined by the sum of the following properties: the *partial correctness*, which means the algorithm at each step does the right thing, and *termination*, which, like the name, means that the algorithm terminates.

78.4.1 Partial Correctness

If before running P the condition φ is true, then after the termination of P, ψ should be true; these are the pre- and post-conditions.

Example:

$$< x = 0 > x := x + 1 < x > 0 >$$

78.4.2 Hoare Calculus

To read the following expressions, which look like fractions but are not, use the following:

- *Above*: if this is true
- *Below*: then this is also true

Assignation Rule

If **x** is a variable and **t** is an expression without side effects, then φx/t is ψ with all **x** substituted for **t**.

$$< \varphi[\mathrm{x/t}] > \ \mathrm{x} = \mathrm{t}; \ < \psi >$$

Note that this rule only has the part below, which means that this rule has no precondition.

Example I:

$< \mathrm{y} \ = 5 > \mathrm{x} = 5; < \mathrm{x} \ = \ \mathrm{y} >$

Example II:

$< 5 > 4 > \mathrm{x} = 5; < \mathrm{x} \ > 4 >$

Consequence Rule (Strong pre-condition)

Here α is a stronger condition than ϕ. This way, our condition ϕ has an extra layer of protection.

$$\frac{< \varphi > P < \psi > \quad \alpha \implies \varphi}{< \alpha > P < \psi >}$$

Example:

$$\frac{< 5 = 5 > \mathrm{x} \ = 5; < \mathrm{x} \ = 5 > \ \mathrm{true} \implies 5 = 5}{< \mathrm{true} > x = 5; < \mathrm{x} \ = 5 >}$$

In code:

```
<true>
<5=5>
x=5;
<x=5>
```

Consequence Rule (Weak post-condition)

This lets us make the post-condition weaker.

$$\frac{< \varphi > P < \psi > \quad \psi \implies \beta}{< \varphi > P < \beta >}$$

Example:

$$\frac{< 5 = 5 > \mathrm{x} \ = 5; < \mathrm{x} \ = 5 > \ \mathrm{x} \ = 5 \implies x \geq 5}{< \mathrm{true} > x = 5; < \mathrm{x} \ \geq 5 >}$$

Note that two guards can only be consecutive if the one below is implied by the one above.

Sequence Rule

This rule allows us to compress our conditions.

$$\frac{< \psi > P < \psi > \quad < \psi > Q < \beta >}{< \psi > PQ < \beta >}$$

Example:

```
<true>
x = 5;
res = x * x + 6;
<res = 31>
```

Condition Rule I

This rule enforces that, no matter which branch of the program is taken, the post-condition is true.

$$\frac{< \phi \wedge B > P < \psi > \quad \phi \wedge \neg B \implies \psi}{< \phi > \text{ if } (B)\{P\} < \psi >}$$

Example:

```
<true>
<y=y>
res=y
<res=y>

if (x > y) {
    <res = y \land x > y>
    <x = max(x,y)>
    res = x
    <res = max(x,y)>
}
<res = max(x,y)>
```

Condition Rule II

Similar to the previous case, we want to make sure that, independent of the branch, we have a correct post-condition.

$$\frac{< \phi \wedge B > P < \psi > \quad < \phi \wedge \neg B > Q < \psi >}{< \phi > \text{ if } (B) \text{ else } \{Q\} < \psi >}$$

Example:

```
<true>
<x=x>
res=x
<res=x>

if (x < 0) {
    <true \land x < 0>
    <-x = |x|>
    res = -x
    <res = |x|>
} else {
    <true \land -x < 0 >
    <x = |x|>
    res = x
    <res = |x>
}
<res = |x|>
```

Loop Rule

In this rule, we look for some *invariant* ϕ which is true before and after the loop.

$$\frac{< \phi \wedge B > P < \phi >}{< \phi > \text{ while } (B)\{Q\} < \phi \wedge \neg B >}$$

Note that from the guard before the loop $< \alpha >$, we have $\alpha \implies \phi$. Finding the invariant can be tricky, but it commonly involves finding an integer expression, or some formula which holds for the entire loop.

Example:

In this case, the invariant is $i!res = n!$.

```
<true>
i = n;
res = 1;
<i = n \land res = 1>
<i! res = n!>
while (i > 1) {
<i! res = n! \land i > 1>
<(i - 1)! res i = n!>
    res = res * i;
<(i - 1)! res = n!>
    i = i - 1;
<i! res = n!>
}
<\phi \land \neg i > 1>
<res = n!>
```

78.4.3 Termination

For each loop **while** $(B)\{P\}$, we have to find some **int** expression V called the *variant* such that

$$B \implies V \geq 0 \text{ and } < V = m \land B > P < V < m >> .$$

In simpler words, we look for an integer expression which continuously decreases.

Example:

```
<i = m \land i > 1>
while ( i > q) {

<i - 1 < m >
    res  = res * i;
<i -1 < m>
    i = i - 1;;
<i < m>
}
```

Chapter 79

Complexity Theory

Asymptotic notations describe how algorithms perform as input size (n) grows.

They help analyze:

- *Time Complexity*: How runtime grows.
- *Space Complexity*: How memory usage grows.

79.1 Definitions

To clarify, all of these are sets of functions.

79.1.1 *Big O* (O)

- Upper Bound.
- Describes the *worst-case* growth rate.
- $f(n) \in O(g(n))$ means $\exists c, n_0$ such that $\forall n > n_0, f(n) \leq c \cdot g(n)$. In other words, at some point our function f is always going to be less than g.

79.1.2 *Omega* (Ω)

- Lower Bound.
- Describes the *best-case* growth rate.
- $f(n) \in \Omega(g(n))$ means $\exists c, n_0$ such that $\forall n > n_0, f(n) \geq c \cdot g(n)$. In other words, at some point our function f is always going to be greater than g.

79.1.3 *Theta* (Θ)

- Tight Bound.
- Describes the *average-case* or exact growth rate.
- $f(n) \in \Theta(g(n))$ means $f(n) \in O(g(n))$ and $f(n) \in \Omega(g(n))$. That is, at some point our function f is always going to be in between the two bounds.

Example I:

$$f(n) = 4n^2 + 8n + 15 \in O(n^2)$$

$$\begin{aligned} f(n) &\leq cn^2 \\ 4n^2 + 8n + 15 &\leq cn^2 \\ 4n^2 + 8n + 15 &\leq 4n^2 + 8n^2 + 15n^2 = 27n^2 \end{aligned}$$

Thus, we have found a constant $c = 27$, from which our function is always less than $g = cn^2$. Therefore, $f \in O(g)$.

QED

Example II:

$$n^3 \notin O(n^2)$$

Suppose the opposite; then

$$n^3 \in O(n^2) \implies n^3 \leq cn^2 \implies n \leq c \forall n \geq N, c \in R.$$

Choose $n = \lceil n_0 + c + 1 \rceil$ which is a natural number.

$$\lceil n_0 + c + 1 \rceil \leq c.$$

This is a contradiction. Therefore, $n^3 \notin O(n^2)$.

QED

79.2 Properties of the Complexity Classes

Linearity:

If $g(n) = \alpha f(n) + \beta$ where $\alpha, \beta \in R^+$ and $f \in \Omega(1)$, then

$$g \in O(f)$$

Addition:

$$f + g \in O(\max(f, g)) = \begin{cases} O(g) & \text{if } f \in O(g) \\ O(f) & \text{if } g \in O(f) \end{cases}$$

Multiplication:

$$a \in O(f) \wedge b \in O(g) \implies ab \in O(fg)$$

Limit:

If the limit

$$\lim_{n \to \infty} \frac{g(n)}{f(n)}$$

exists, then $g \in O(f)$. The reverse does not hold.

79.3 Master's Theorem

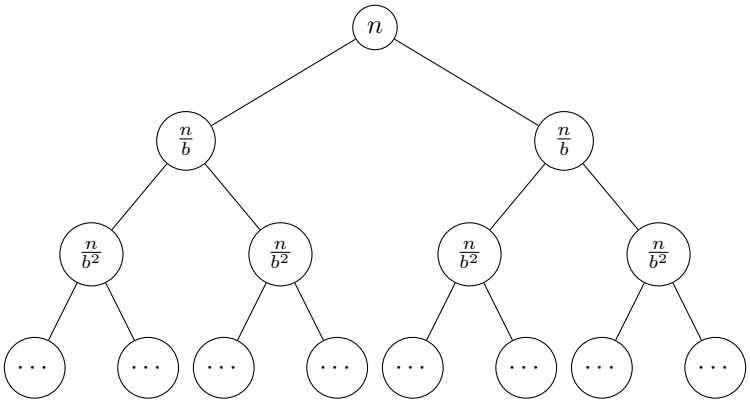
This helps us find the time complexity of divide-and-conquer recurrences:

$$T(n) = aT(\left\lceil \frac{b}{n} \right\rceil) + c(n^d)$$

Where:

- $a \geq 1$ is the number of sub-problems
- $b > 1$ is the factor by which the problem size shrinks
- $c(n^d), d \geq 1$ is the cost of the work outside recursion (e.g., combining results)

Imagine a tree generated each time we divide the problem, until some point where solving it becomes trivial $\Theta(1)$. This tree is going to have a depth of $\log_b n$, which is the number of times we can divide the problem, and a maximum width of $n^{\log_b a}$, which is the number of leaves at the bottom.



Work at each level: $n, n, n, \ldots, n \Rightarrow \log n$ levels

Total work: $n \log n$

Therefore, our total work is going to be

$$\begin{aligned} T(n) &= c(n) + ac(n/b) + a^2c(b/n^2) + \cdots + \Theta(n^{log_b a}) \\ &= \Theta(n^{log_b a}) + \sum_{j=0}^{\log_b(n-1)} a^j c(n/b^j) \end{aligned}$$

79.3.1 Cases of the Master Theorem

When looking at the tree generated by the divide-and-conquer, we can get three cases:

$$T(n) \in \begin{cases} \text{The problem gets easier} & O(n^d) \text{ if } a < b^d \\ \text{The hardness is evenly distributed} & O(n^d \log n) \text{ if } a = b^d \\ \text{The problem gets harder} & O(n^{\log_b a}) \text{ if } a > b^d \end{cases}$$

Example:

$$T(n) = 4T(\frac{n}{2}) + c(n)$$

Here $a = 4$, $b = 2$ and $d = 1$. Since $d < \log_b a$ $1 < \log_2 4 = 2$

$$T(n) \in O(n^{log_b a} = O(n^2)$$

79.4 How to Identify Time Complexity of a Function

The process can be summarized into 5 steps:

1. Write the definition of the bound we are looking for.
2. Identify the most dominant term in $f(n)$; the one that grows the fastest.
3. Set up the inequality and solve for c.
4. Take the limit of $\frac{f(n)}{g(n)}$.
5. Choose a c using the knowledge of the limit and find n_0.

Example:

Find the time complexity $O(n)$, the constant, and the starting term for the inequality of $\sqrt{4n}$.

Step 1: Definition

$$f(n) \in O(g(n)) \iff f(n) \leq cg(n)$$

Step 2: Dominant Term

The dominant term is $\sqrt{n}$.

Step 3: Inequality

We are going to assume that it is part of $O(\sqrt{n})$.

$$\sqrt{n} \leq c\sqrt{n}$$

Step 4: Limit

$$\lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n}} = 1 \leq c$$

Step 5: Choose c

Let us choose $c = 2$ which gives us $n_0 = 1$.

79.5 How to Find Time Complexity in Code

1. **Count Primitive Operations**
   - Additions, multiplications, comparisons, etc.
2. **Identify Loops**
   - Single loop $\Rightarrow O(n)$
   - Nested loop $\Rightarrow$ Multiply the complexities
3. **Check Recursive Patterns**
   - Use recurrence relations (often solve with Master's Theorem)
4. **Ignore Constants**
   - Drop lower-order terms and constants (e.g., $O(n^2 + 3n + 5) \implies O(n^2)$)

79.5.1 Examples

Example I: Simple Loop

```
for i in range(n):
    print(i)
```

$\Rightarrow O(n)$

Example II: Nested Loop

```
for i in range(n):
    for j in range(n):
        print(i, j)
```

$\Rightarrow O(n^2)$

79.5.2 Common Time Complexities

Complexity	Name	Example Algorithms
$O(1)$	Constant	Array access
$O(\log n)$	Logarithmic	Binary search
$O(n)$	Linear	Linear search
$O(n \log n)$	Linearithmic	Merge Sort, Heap Sort
$O(n^2)$	Quadratic	Bubble Sort, Selection Sort
$O(n^3)$	Cubic	Naive matrix multiplication
$O(2^n)$	Exponential	Recursive Fibonacci
$O(n!)$	Factorial	Traveling Salesman (brute force)

Chapter 80

State Machines and Languages

80.1 Languages and Alphabets

A *language* L is a subset of words of an *alphabet* Σ, which is a finite, non-empty set of symbols that always includes the ε-word, the empty word.

80.1.1 Concatenation

Concatenation is an operation that can be performed with members of an alphabet, and it is like basic addition. Therefore, it is associative, and when ε, the zero of alphabets, is added, it does not affect the words or their length.

$$\text{“}a\text{”} + \text{“}b\text{”} = ab$$

80.2 Inductive Definitions

In the field of languages and alphabets, an inductive function $f : \Sigma^* \to X$ will be defined in the following way.

1. *The base case*: $f(\varepsilon)$.
2. *The induction step*: where the actual behavior of the function is defined. Helper functions can also be defined using the same structure.

It also has the following properties:

- $|\varepsilon| = 0$
- For $w = v + a$ then $|w| = |v| + 1$ for $a \neq \varepsilon$.

Example:

Define a function that counts the occurrences of ue in a word with letters of the alphabet $\Sigma = \{a, \ldots, z\}$.

$$f(\varepsilon) = 0$$
$$f(w \cdot x) = \begin{cases} f(w) & \text{if } x \neq e \\ \text{help}(w) & \text{if } x = e \end{cases}$$

The auxiliary function $\text{help} : \Sigma^* \to \mathbb{N}$ is defined as

$$\text{help}(w \cdot x) = \begin{cases} f(w) + 1 & \text{if } x = u \\ f(w) & \text{otherwise} \end{cases}$$

That is, $\text{help}(w) = f(w) + 1$ if the last letter of w is u, and $f(w)$ otherwise.

80.3 Head and Tail

Head returns the first symbol of a word and is defined by

$$\begin{aligned} \text{head}(\varepsilon) &= \varepsilon \\ \text{head}(\varepsilon + a) &= a \\ \text{head}(v + a) &= \text{head}(v) \end{aligned}$$

Tail removes the first symbol of a word and is defined by

$$\begin{aligned} \text{tail}(\varepsilon) &= \varepsilon \\ \text{tail}(\varepsilon + a) &= \varepsilon \\ \text{tail}(v + a) &= \text{tail}(v) + a \end{aligned}$$

80.4 Fixes

Let Σ be an alphabet and let $v, w \in \Sigma^*$ be words.

- v is called a *substring* of w iff $\exists s, t \in \Sigma^* : w = svt$.
- v is called a *suffix* of w iff $\exists s \in \Sigma^* : w = sv$ - we write $v \sqsupseteq w$.
- v is called a *prefix* of w iff $\exists t \in \Sigma^* : w = vt$ - we write $v \sqsubseteq w$.
- v is called a *proper substring (prefix/suffix)* of w iff $v \neq \varepsilon \wedge v \neq w$ and v is a substring (prefix/suffix) of w. We then write $v \sqsubset w$ or $v \sqsupset w$.

ε is a prefix, suffix, and substring of all words.

80.5 The Canonical Order

Let $\Sigma = \{a_1, \ldots, a_n\}$ be an alphabet with $n \geq 1$, and let $<\subseteq \Sigma \times \Sigma$ be an order on Σ with $a_1 < a_2 < \cdots < a_n$.

We define the *canonical order* on Σ^* as follows:

$$u < v \quad \text{iff} \quad |u| < |v| \ \vee \ \big(|u| = |v| \ \wedge \ u = w \cdot a_i \cdot u' \ \wedge \ v = w \cdot a_j \cdot v' \ \wedge \ i < j\big)$$

for arbitrary $w, u, u', v, v' \in \Sigma^*$.

80.6 Languages

Let Σ be an alphabet. A *language* L over Σ is a subset of Σ^* ($L \subseteq \Sigma^*$).

$$\mathcal{P}(\Sigma^*) = \{L \mid L \subseteq \Sigma^*\}.$$

The power set of Σ^* is the set of all languages over Σ.

Furthermore, $L_\emptyset = \emptyset$ and $L_\varepsilon = \{\varepsilon\}$.

Other parts of a language are the *syntax*, which defines the words in the language, and the *semantics*, which determines the meaning of words.

80.7 Pumping Lemma

For a language L which can be recognized by a state machine, there is a natural number n such that for each word in L with length greater than or equal to n, the word w can be decomposed into three subwords x, y, z.

- $|xy| \leq n$
- $y \neq \varepsilon$
- $xy^i z \in L$ for all $i \geq 0$

Example:

Let $L = \{a^n b^n \mid n \geq 0\}$. This language is **not** regular, and the Pumping Lemma can be used to prove this by contradiction.

80.7.1 Pumping Lemma Algorithm

Given two alphabets Σ_1, Σ_2 and $\mathcal{D} \subseteq \Sigma_1^*$, an *algorithm* is a total function

$$A : \mathcal{D} \to \Sigma_2^*.$$

Two algorithms are equivalent if and only if

$$\forall x \in \mathcal{D} : A(x) = B(x).$$

80.8 The Decision Problem

The *decision problem* $\langle \Sigma, L \rangle$ for a given alphabet Σ and a language $L \subseteq \Sigma^*$ is to decide, for each $w \in \Sigma^*$, whether

$$w \in L \quad \text{or} \quad w \notin L.$$

Therefore, one also speaks of the *word problem* $\langle \Sigma, L \rangle$.

An algorithm $A\colon \Sigma^* \to \{0, 1\}$ *solves* the decision problem $\langle \Sigma, L \rangle$ if, for all $w \in \Sigma^*$, we have

$$A(w) = \begin{cases} 1 & \text{if } w \in L, \\ 0 & \text{if } w \notin L. \end{cases}$$

We also say: *A recognizes L.*

80.9 Recursive Languages

Let Σ be an alphabet and let $L \subseteq \Sigma^*$ be a language. L is called *recursive* if and only if there exists an algorithm A that solves the decision problem $\langle \Sigma, L \rangle$.

80.10 Enumeration Problem

Let Σ be an alphabet, $L \subseteq \Sigma^*$ a language, and $n \in \mathbb{N}$. The *enumeration problem* $\langle \Sigma, L, n \rangle$ for L consists of listing the canonical first n words of L.

An algorithm $A\colon N \to \mathcal{P}(L)$ is an *enumeration algorithm* for L if, for every input $n \in N$, A outputs the canonical first n words $w_0, w_1, \ldots, w_{n-1}$ of the language L.

80.11 Grammars

Let

$$G = (V, \Sigma, R, S)$$

where:

- V is a finite set of *variables* (also called non-terminal symbols).
- Σ is a finite set of *terminal symbols*, such that $V \cap \Sigma = \emptyset$.
- R is a finite set of *production rules* of the form

  $$A \to \alpha,$$

  where $A \in V$ and $\alpha \in (V \cup \Sigma)^*$.
- $S \in V$ is the **start symbol**.

Thus, a *grammar* G defines a *formal language* $L(G)$ as

$$L(G) = \{\, w \in \Sigma^* \mid S \Rightarrow^* w \,\}$$

where $\Rightarrow^*$ denotes the reflexive and transitive closure of the derivation relation defined by R.

Context Free Grammars

Chapter 81

State Machines

A *state machine* is a computational model used to design algorithms, systems, or software by breaking them down into a finite number of *states*, *transitions*, *inputs*, and *outputs*.

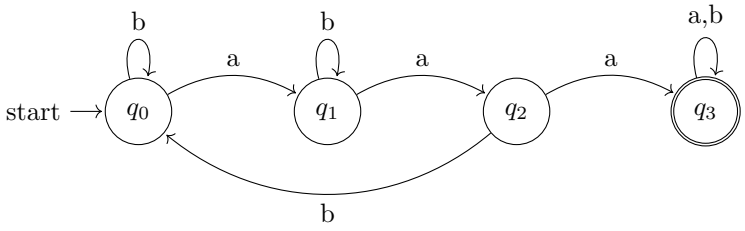
81.1 Deterministic Finite Automaton (DFA)

DFAs are defined by a quintuple $(Q, \Sigma, \delta, q_0, F)$ which consists of Q states, an input alphabet Σ, a transition function $\delta : Q \times \Sigma \to Q$, a start state $q_0 \in Q$, and a set of termination states $F \subset Q$.

This kind of automaton has the particularity that each state has *exactly one transition* for each possible input. Therefore, *deterministic*.

Example:

State	a	b
q_0	q_1	q_0
q_1	q_2	q_1
q_2	q_3	q_0
q_3	q_3	q_3



Here, our language is

$$L = \{w \in \{a, b\}^* \mid \text{the sequence of transitions on } w \text{ leads to state } q_3\}$$

81.2 Non-Deterministic Finite Automaton (NFA)

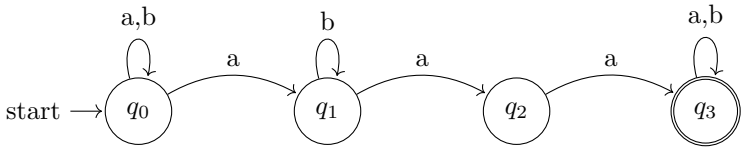
An *NFA* is a state machine defined by a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where: Q is a finite set of states, Σ is the input alphabet, $\delta : Q \times \Sigma \to \mathcal{P}(Q)$ is the transition function (mapping to a set of states), $q_0 \in Q$ is the start state, and $F \subset Q$ is the set of accepting states.

Unlike a DFA, an NFA allows:

- Multiple possible next states for a given input.
- No transition at all.

Example:

State	a	b
q_0	$\{q_0, q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_1\}$
q_2	$\{q_3\}$	$\emptyset$
q_3	$\{q_3\}$	$\{q_3\}$



Here our language is still:

$$L = \{w \in \{a, b\}^* \mid \text{some computation path leads to } q_3\}$$

81.3 Epsilon-NFA (ε-NFA)

An *ε-NFA* is a non-deterministic finite automaton that allows transitions on the empty string ε, meaning the automaton can change states without consuming input.

It is defined as a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, with the transition function:

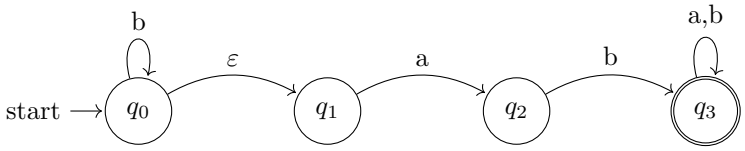
$$\delta : Q \times (\Sigma \cup \{\varepsilon\}) \to \mathcal{P}(Q)$$

This means the automaton has the following properties:

- It can transition normally on $a \in \Sigma$ or move to other states "for free" via ε-transitions.
- *Epsilon-Closure:* It is the set of all states reachable via ε from a particular state, plus itself.

Example:

State	a	b	ε
q_0	$\emptyset$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\emptyset$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$	$\emptyset$
q_3	$\{q_3\}$	$\{q_3\}$	$\emptyset$



The language is:

$$L = \{w \in \{a, b\}^* \mid w \text{ contains the substring } ab\}$$

via the path:

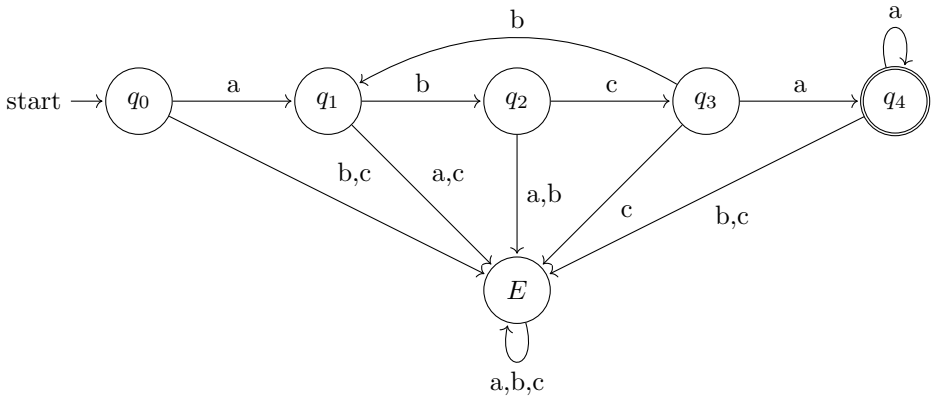
$$q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{a} q_2 \xrightarrow{b} q_3$$

81.4 How to Construct a State Machine

To construct a state machine, it is necessary that an alphabet and a language are given. Then the process revolves around identifying which words are part of the language and how they can be expressed as a list of states and transitions.

Example:

Given are the alphabet $\Sigma = \{a, b, c\}$ and the language $L = \{w \in \Sigma^* \mid w = a(bc)^m a^n \text{ and } \}$. Construct a DFA.



81.5 Regular Expressions

Regular expressions (regex) are a practical application of finite automata (DFA/NFA). They define search patterns for strings and are processed internally using state machines.

Example:

Regex: `a(b|c)*d`

- *Start State*: Awaiting input.
- *Transition on 'a'*: Move to a state expecting b or c.
- *Loop State*: Repeats b or c any number of times.
- *Final Transition*: Ends with a d.

Regular expressions describe sets of strings using a small set of operations:

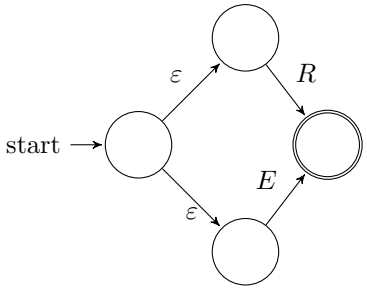
- *Concatenation* $(R.S)$: A string from R followed by a string from S.
- *Union* $(R + S)$: A string from either R or S.
- *Kleene star* (R^*): Zero or more repetitions of strings from R.

Special symbols:

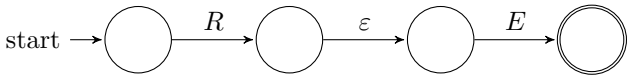
- $\emptyset$: Matches no string.
- ε: Matches the empty string.
- Literal characters (e.g., a, b, c): Match themselves.

Below are ε-NFAs (nondeterministic automata with ε-transitions) for some regular expressions:

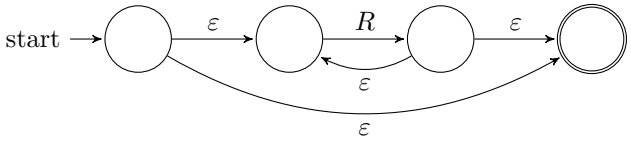
R + E



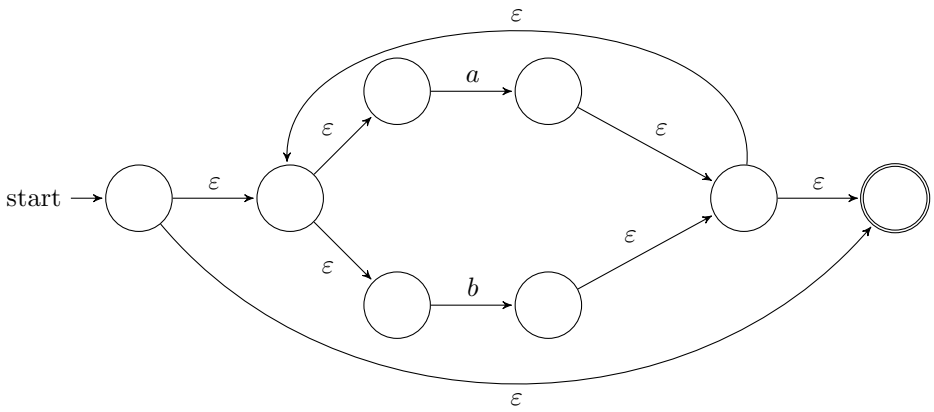
R.E



R*



Example: `(a+b)*`



81.6 State Machine Conversions and Minimization

81.6.1 Non-Deterministic Finite Automaton (NFA) to Deterministic Finite Automaton (DFA)

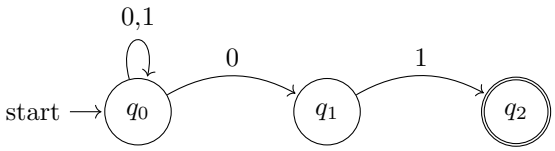
The process for the conversion is the so-called *Power-Set-Construction-Method* or *Subset-Construction-Method*.

Steps:

1. Write the transition table of the given automaton.
2. Write a similar table with the transition as columns.
3. Start with q_0 and find which states can be reached with the transitions. If from our current state we can reach more than one state with our current transition, write the possible states as a set, then proceed to write the set as the next entry to check (write in the next row).
4. We terminate the algorithm when, after evaluating our states, we only get states we have already visited.

Example:

Convert the NFA to a DFA



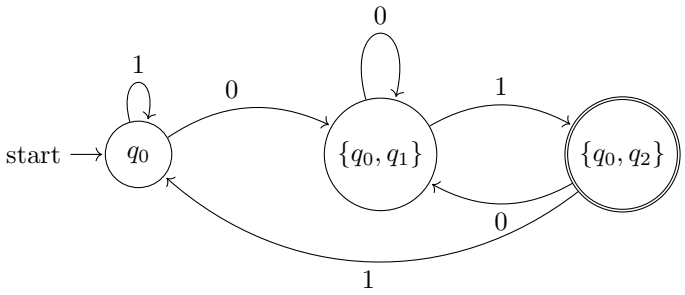
Step 1: Transition Table

State	0	1
q_0	$\{q_0, q_1\}$	q_0
q_1	$\emptyset$	q_2
q_2	$\emptyset$	$\emptyset$

Step 2: Table of new states

State	0	1
q_0	$\{q_0, q_1\}$	q_0
$\{q_0, q_1\}$	$\{\{q_0, q_1\}, \emptyset\}$	$\{q_0, q_2\}$
$\{q_0, q_2\}$	$\{\{q_0, q_1\}, \emptyset\}$	$\{q_0, \emptyset\}$

Step 3: Construct DFA



81.6.2 Epsilon-NFA to DFA

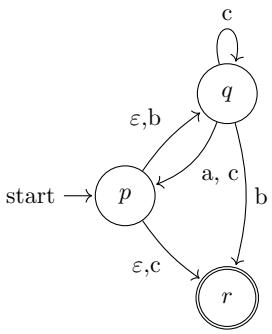
The process is very similar to the previous one. The catch is that we have to keep in mind that our initial state is not only q_0 but also all states reachable from q_0 via epsilon.

Steps:

1. Write a transition table.
2. Write the new initial state which is going to be a set of states with the original initial state and all states reachable via ε from it. This also includes states reachable via multiple epsilon transitions. This is the epsilon closure of the initial state.
3. Find the states reachable via the first transition of this epsilon closure.
4. Then find the epsilon closure of the previously found set of states. This becomes a new state in the DFA.
5. Repeat the process with each new created state and transition until you get no new epsilon closure.

Example:

Convert to DFA



Step 1: Transition Table

State	a	b	c	ε
p	$\emptyset$	q	r	$\{q, r\}$
q	p	r	p, q	$\emptyset$
r	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Step 2: Epsilon Closure and Transitions

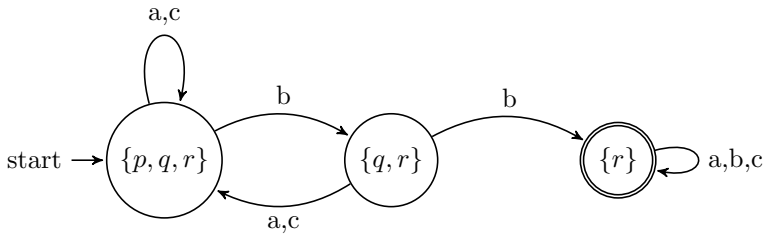
The initial state is $A = \varepsilon(p) = \{p, q, r\}$.

Using the transition a we get the set $\{p\}$ with $\varepsilon(p) = \{p, q, r\}$. This state is just A again. Using the transition b we get the set $\{q, r\}$ with $B = \varepsilon(\{q, r\}) = \{q, r\}$. And finally with c we get the set $\{p, q, r\}$ with $\varepsilon(\{p, q, r\}) = \{p, q, r\}$.

Repeating this process with the rest of state will give us:

State	a	b	c
$\{p, q, r\}$	$\{p, q, r\}$	$\{q, r\}$	$\{p, q, r\}$
$\{q, r\}$	$\{p, q, r\}$	r	$\{p, q, r\}$
r	$\emptyset$	$\emptyset$	$\emptyset$

Step 3: Construct the Automaton



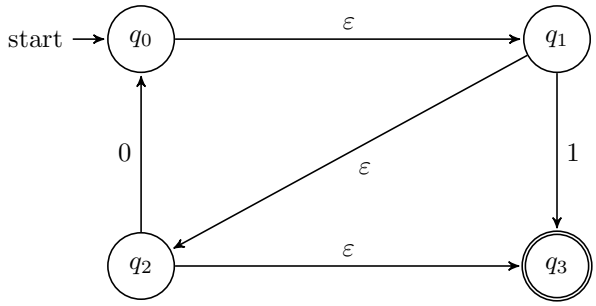
81.6.3 Epsilon-NFA to NFA

Here the key is to eliminate the ε transitions from the automaton. It can be done with the following steps:

1. Write an empty transition table of the automaton but without the epsilon transition, and then list all epsilon closures of each state
2. For each state and transition, take the ε-closure of the state, find where these states go with the current transition, and then take their epsilon closure. The final closure goes into the corresponding transition column of the current state.
3. Look for new final states. A final state will be a state that can be reached only using epsilon transitions.
4. Finally, write the new automaton using the information from the transition table with the new final states. It is important to note that the automaton is not minimal.

Example:

Transform to a nondeterministic automaton



Step 1: Set up table and closures

State	0	1
q_0		
q_1		
q_2		
q_3		

ε-Closures

$$q_0 = \{q_0, q_1, q_2, q_3\}$$
$$q_1 = \{q_1, q_2, q_3\}$$
$$q_2 = \{q_2, q_3\}$$
$$q_3 = \{q_3\}$$

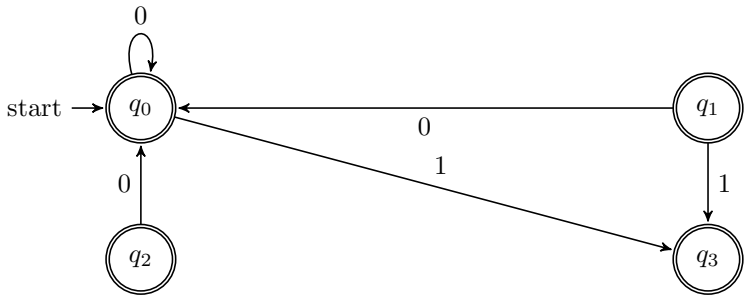
Step 2: Filling the table using the transition and closures

State	0	1
q_0	q_3	q_0
q_1	q_3	q_0
q_2	$\emptyset$	q_0
q_3	$\emptyset$	$\emptyset$

Step 3: Identify Final States

Here, every state is a final state, because every state contains the final state in its epsilon closure.

Step 4: Draw the automaton



81.6.4 NFA to Regex

Converting a regular expression into an NFA is not really difficult, but the other direction is not so trivial. The most common approach to tackle this type of conversion is via a system of equations.

You are given some NFA with n states and m transitions. Set your system as follows:

1. Write an equation L_i for each state q_i. Each equation L_i represents the set of strings (language) accepted when starting from state q_i. These equations will be defined recursively in terms of transitions to other states.
2. For each L_i, describe all transitions from state q_i: For every possible outgoing transition from q_i to some q_j on symbol(s) w, add a term wL_j to the equation. If q_i is an accepting state, also include ε (the empty string).

   Example: If from state 1 there is a transition $a \to 1$ and $b \to 2$, and state 1 is accepting, then

   $$L_1 = aL_1 \cup bL_2 \cup \varepsilon$$

3. Rewrite the equations using regular expression notation. Replace set notation with regex syntax: $\cup \to +$, concatenation is written without an operator (e.g., aL_1), and replace sets of symbols (e.g., $\{a, b\}$) with grouped regex (e.g., $(a + b)$).

   Example:

   $$L_1 = aL_1 + bL_2 + \varepsilon$$

4. Substitute equations recursively to reduce the system. There are two cases:
   - If an equation L_i does not reference itself (i.e., no recursion), substitute it into the other equations directly.
   - If an equation L_i does reference itself (e.g., $L_i = rL_i + s$), apply *Arden's Lemma*:

     $$\text{If } L = YL \cup X, \text{ then } L = Y^*X$$

   Example:

   (a) Write for each state the equation L_i.

   (b) Now treat each state as a start state and look for the sets of words this state accepts that lead to a terminating state, then combine them with $\cup$. If the current state is also a terminating state, then ε must also be added. Example: $L_1 = \{ab\}L_1 \cup bL_2$.

   A trick is to write the table with the transition and for each row (state) you write $L_{row} = transition_1 L_{state_x} + transition_2 L_{state_y} + \cdots$. Where the transition is a set of letters of the alphabet.

   (c) Repeat step 2 for all L_i, and then replace the set of words with regular expressions and the $\cup$ with $+$. Example: $L_1 = (a + b)L_1 + bL_2$.

   (d) Start substituting equations into other equations. Here there are two cases: either the equation $L_x, \ldots$ depends only on other equations different from itself, or $L_x, \ldots$ also depends on itself, which causes infinite recursion. For the first case, the problem solves itself by just doing the substitution until we have only one equation, which is going to be our regex; but for the other case, we will use *Arden's Lemma*.

   $$L = (a + b)L + b \Rightarrow L = (a + b)^*b$$

   Continue substituting until all equations are resolved. The expression for the initial state gives the regular expression for the NFA.

81.7 Arden's Lemma

Given X, Y as formal languages with $\varepsilon \notin Y$, the following holds:

$$L = YL \cup X \text{ has exactly one solution: } L = Y^*X$$

And for the special case $X = \emptyset$:

$$L = YL + \emptyset = Y^*\emptyset = \emptyset$$

Proof:

To show that $L = Y^*X$ is a solution and, more specifically, the only solution.

$$\begin{aligned} L &= YL \cup X \\ Y^*X &= YY^*X \cup X \text{ This shows that it is indeed a solution} \end{aligned}$$

For the second part, we can argue that if L is a solution for the equation, then

$$Y^*X \subseteq L \implies X \cup YX \cup YYX \cup \cdots \subseteq L$$

Here, we always took longer words, which are also part of the language, and united them. For the second part, we are going to use shorter words.

Assume that

$$L \subsetneq Y^*X$$

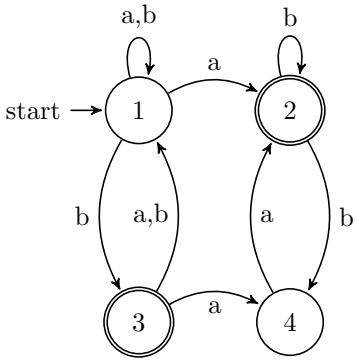
Then there is the shortest word w in L which is not in Y^*X. But because $L = YL \cup X$, w must be either a word in X, which is not possible because $w \notin Y^*X$, or in YL. This implies that w can be decomposed into uv with $u \in Y, u \neq \varepsilon, v \in L$. But this is a contradiction, since then w would be a word of the form Y^*X, which is not true, and this would imply that there is an even shorter word than the shortest word, and so on. Therefore

$$L \subseteq Y^*X$$

QED

Example:

Convert to Regex



Step 1: Initial Equations

$$\begin{aligned} L_1 &= \{a,b\}L_1 \cup aL_2 \cup bL_3 \\ L_2 &= \varepsilon \cup bL_2 \cup bL_4 \\ L_3 &= \{a,b\}L1 \cup \varepsilon \cup aL4 \\ L4 &= aL_2 \end{aligned}$$

Step 2: Regex Equations

$$
\begin{aligned}
L_1 &= (a+b)L_1 + aL_2 + bL_3 \\
L_2 &= \varepsilon + bL_2 + bL_4 \\
L_3 &= (a+b)L1 + \varepsilon + aL4 \\
L4 &= aL_2
\end{aligned}
$$

Step 3: Substitution

L_4 in L_3 and L_2

$$L_3 = (a+b)L1 + \varepsilon + aaL_2$$

$$L_2 = \varepsilon + bL_2 + baL_2$$

Now we can solve L_2 using the lemma.

$$
\begin{aligned}
L_2 &= \varepsilon + bL_2 + baL_2 \\
&= \varepsilon + (b+ba)L_2 \\
&= (b+ba)^*\varepsilon \\
&= (b+ba)^*
\end{aligned}
$$

L_2 in L_3 and L_1

$$
\begin{aligned}
L_1 &= (a+b)L_1 + a(b+ba)^* + bL_3 \\
L_3 &= (a+b)L_1 + \varepsilon + aa(b+ba)^*
\end{aligned}
$$

L_3 in L_1

$$
\begin{aligned}
L_1 &= (a+b)L_1 + a(b+ba)^* + b((a+b)L1 + \varepsilon + aa(b+ba)^*) \\
&= (a+b)L_1 + a(b+ba)^* + b(a+b)bL1 + b + baa(b+ba)^* \\
&= (a+b)L_1 + b(a+b)L_1 + a(b+ba)^* + b + baa(b+ba)^* \\
&= ((a+b) + b(a+b))L_1 + a(b+ba)^* + b + baa(b+ba)^* \\
&= ((a+b) + b(a+b))L_1 + a(b+ba)^* + b + baa(b+ba)^* \\
&= ((a+b) + b(a+b))^*(a(b+ba)^* + b + baa(b+ba)^*)
\end{aligned}
$$

This is the answer but we can simplify. Here $((a+b) + b(a+b))$ is a word of $(a+b)^*$

$$
\begin{aligned}
L_1 &= ((a+b) + b(a+b)^*(a(b+ba)^* + b + baa(b+ba)^*) \\
&= (a+b)^*(a(b+ba)^* + b + baa(b+ba)^*)
\end{aligned}
$$

81.8 Minimizing State Machines

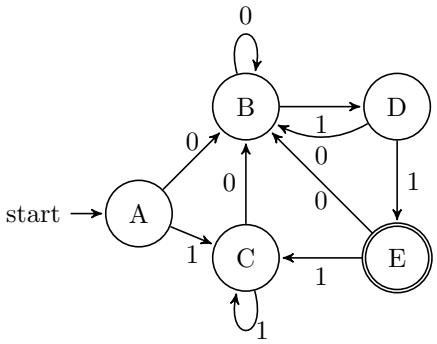
Minimization aims to create an equivalent DFA with the *fewest number of states*. One widely used method is the *table-filling algorithm*.

81.8.1 Table Method for DFA Minimization

Steps:

1. Put all non-accepting states in an equivalence class α, and the accepting states in another called Ω. Write a table with the following columns: state, transition 1, transition 2, ..., transition n. On the first row put α and in the state column the states of the class. Then do the same with Ω. You can also optionally make a table of transition for the states.
2. Start evaluating each state with each transition, and write the equivalence class reached by that transition. This means that you have to write the name of the set (class) the reached state is in. For example if we evaluate a with transition 0, and it leads to b and b is in the same set as a then for the row a with the transition 0 we write the name of that set.
3. For each state in the current class which has a tuple of classes different from the rest, create a new class, which is going to be included in the next iteration of the table. Then repeat step 2.
4. Terminate the process if no more new class tuples appear.
5. Build a new DFA with merged equivalent states and updated transitions.

Example:



Step 1 and 2: Table

Class	State	0	1
α	A B C D		
Ω	E		

Step 3: Evaluation

First iteration

Class	State	0	1
α	A	α	α
	B	α	α
	C	α	α
	D	α	Ω
Ω	E	α	α

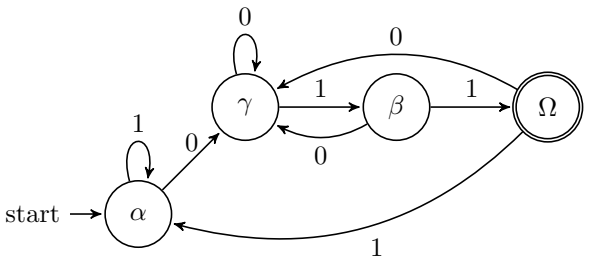
Second iteration

Class	State	0	1
α	A	α	α
	B	α	β
	C	α	α
β	D	α	Ω
Ω	E	α	α

Third and final iteration

Class	State	0	1
α	A	γ	α
	C	γ	α
β	D	γ	Ω
γ	B	γ	β
Ω	E	γ	α

Step 4: Draw the automaton



Chapter 82

Regular Expressions (RegEx)

RegEx is a system which focuses on the pattern of text by specifying a syntax used to represent a state machine for one or more patterns. In the real world, it is used, for example, for searching and validating, among other operations.

82.1 Basic Characters

- `a` : Matches character 'a' exactly.
- `abc` : Matches the string "abc" exactly.

82.2 Meta Characters (Special Characters)

- `.` : Matches any single character except a newline.
- `\` : Escape character (used to escape special characters, e.g., `\.` to match a literal period).
- `^` : Matches the start of a string.
- `$` : Matches the end of a string.
- `|` : Alternation (logical OR, e.g., `cat|dog` matches either "cat" or "dog").

82.3 Character Classes

- `[abc]` : Matches any one of 'a', 'b', or 'c'.
- `[^abc]` : Matches any character except 'a', 'b', or 'c'.
- `[a-z]` : Matches any lowercase letter.
- `[A-Z]` : Matches any uppercase letter.
- `[0-9]` : Matches any digit.
- `[a-zA-Z0-9]` : Matches any alphanumeric character.
- `.` : Matches any character except newline.
- `\d` : Matches any digit (equivalent to `[0-9]`).
- `\D` : Matches any non-digit character.
- `\w` : Matches any word character (letters, digits, underscore).
- `\W` : Matches any non-word character.
- `\s` : Matches any whitespace character (spaces, tabs, newlines).

- `\S` : Matches any non-whitespace character.

82.4 Quantifiers

- `*` : Matches 0 or more occurrences of the preceding character (e.g., `a*` matches "", "a", "aa", etc.).
- `+` : Matches 1 or more occurrences (e.g., `a+` matches "a", "aa", etc.).
- `?` : Matches 0 or 1 occurrence (e.g., `a?` matches "" or "a").
- `{n}` : Matches exactly `n` occurrences (e.g., `a{3}` matches "aaa").
- `{n,}` : Matches at least `n` occurrences (e.g., `a{2,}` matches "aa", "aaa", etc.).
- `{n,m}` : Matches between `n` and `m` occurrences (e.g., `a{2,4}` matches "aa", "aaa", or "aaaa").

82.5 Grouping and Capturing

- `(abc)` : Capturing group, treats "abc" as a single unit.
- `(?:abc)` : Non-capturing group, groups "abc" without capturing.
- `\n` : Refers to the nth captured group (e.g., `(\w)\1` matches doubled letters like "oo").

82.6 Assertions and Anchors

- `^` : Matches the start of a string.
- `$` : Matches the end of a string.
- `\b` : Matches a word boundary (e.g., `\bword\b` matches "word" but not "sword").
- `\B` : Matches a non-word boundary.
- `(?=...)` : Positive look ahead (e.g., `a(?=b)` matches 'a' only if followed by 'b').
- `(?!...)` : Negative look ahead (e.g., `a(?!b)` matches 'a' only if NOT followed by 'b').
- `(?<=...)` : Positive look behind (e.g., `(?<=a)b` matches 'b' only if preceded by 'a').
- `(?<!...)` : Negative look behind (e.g., `(?<!a)b` matches 'b' only if NOT preceded by 'a').

82.7 Advanced Constructs

- `(?(condition)yes|no)` : Conditional expression.
- `(?P<name>...)` : Named capturing group (e.g., `(?P<word>\w+)`).
- `(?P=name)` : Refers to a named capturing group.
- `(?#...)` : Comment within regex for readability.
- `(*UTF8)` : Enforces UTF-8 mode for international text processing.
- `(?>...)` : Atomic group (prevents backtracking within the group).

82.8 Examples

Example: Email validation

```
^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$
```

Matches valid email addresses.

Example: Phone number validation

```
^\(?\d{3}\)?[-.]?\d{3}[-.]?\d{4}$
```

Matches US phone numbers like (123) 456-7890, 123-456-7890, or 123.456.7890.

Example: URL validation

```
^https?:\/\/(www\.)?[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}(/\S*)?$
```

Matches URLs like http://example.com or https://www.example.com/path.

Example: Hex color code validation

```
^#([A-Fa-f0-9]{6}|[A-Fa-f0-9]{3})$
```

Matches hex color codes like #FFF or #A1B2C3.

Example: Date format (YYYY-MM-DD) validation

```
^\d{4}-\d{2}-\d{2}$
```

Matches dates formatted as 2024-03-01.

Chapter 83

Graph Theory

A *Graph* $G = \langle V, E \rangle$ consists of a finite set of vertices V and a relation $E \subseteq V \times V$ over the set. Depending on the traits of E, it can be categorized into

- *Undirected*: E is symmetric.
- *Directed*: E is not symmetric.
- *Complete*: $E = V \times V$ for directed and $|E| \approx |V|(|V| - 1)/2$ for undirected graphs.
- *Loop-free*: $\forall (v, v) \in V \times V : (v, v) \notin E$
- *Weighted*: $G = \langle V, E, \phi \rangle$, where $\phi : E \to R$ and R is an arbitrary set. It mostly refers to adding a cost to each edge.
- *Dense*: $|E| \approx |V|^2$ for directed and $|E| \approx |V|(|V| - 1)/2$ for undirected graphs.
- *Sparse*: $|E| << |V|^2$ for directed and $|E| << |V|(|V| - 1)/2$ for undirected graphs.

In computer science, the density of a graph is a main criterion when modeling a graph.

83.1 Connected Graphs

Let $G = (V, E)$ be an undirected graph. Then it is called *connected* if, for all pairs $(v_i, v_j) \subseteq V \times V$, there exists a path from v_i to v_j. We define E_S as the symmetric relation.

$$(v_1, v_2) \in E \iff (v_1, v_2) \in E_S \wedge (v_2, v_1) \in E_S$$

If G is directed, then it is connected if and only if its undirected version is connected.

83.2 Paths

Given a graph G, we call a *path* a set of vertices if

$$\forall i \in \{1, \ldots, k - 1\} : (w_i, w_{i+1}) \in E$$

The most common types of paths are

- *Circle*: $w_1 = w_k$
- *Eulerian Circuit*: Each edge will be visited one time. $\forall (s, t) \in E : \exists_1 i : s = w_i \wedge t = w_{i+1}$
- *Hamiltonian Circuit*: Each vertex will be visited one time $\forall v \in V : \exists_1 i : v = w_i$
- *Loop-free*: $\forall (v, v) \in V \times V : (v, v) \notin E$

83.2.1 Path Cost

We define the *path cost* for a weighted graph as

$$\phi(p) = \sum_{i=1}^{k-1} \phi(w_i, w_{i+1})$$

83.3 The Adjacency Matrix

For a graph G with vertices $V = \{v_1, \ldots, v_n\}$, we define the *adjacency matrix* $A = (a_{ij})$ as a $n \times n$ matrix with

$$a_{ij} = \begin{cases} 1 & if (v_i, v_j) \in E \\ 0 & else \end{cases}$$

83.4 The Relax Operation

In graph theory, when referring to the action of updating the cost of visiting a vertex in a weighted graph, we call this operation *relaxation*.

$$\text{relax}(u, v, \text{weight}) = \begin{cases} d[u] + \text{weight}(u, v) & \text{if} d[u] + \text{weight}(u, v) < d[v] \\ d[u] & \text{else} \end{cases}$$